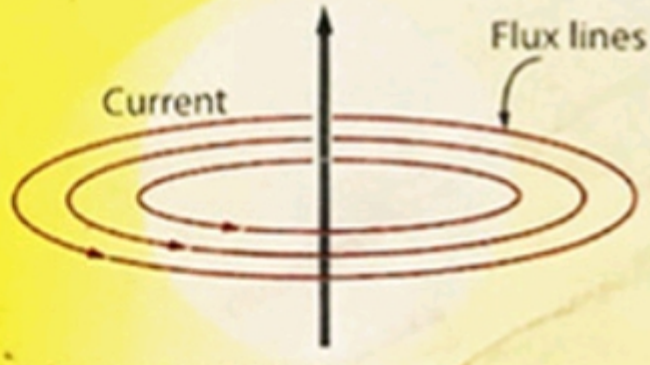
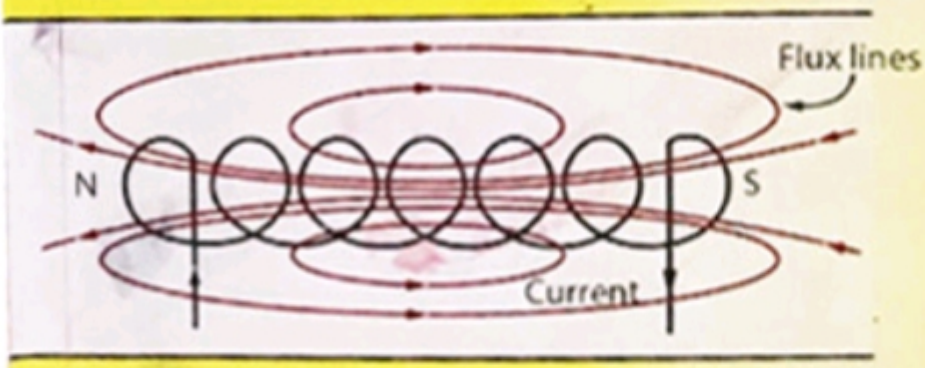
Extreme Series

PHYSICS

GRADE 9-10





Based on Common Currently Used
National and International Curriculum

Extreme Series

2012 E.C

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The Greek Alphabet

Name	Capital	Small	Name	Capital	Small
Alpha	A	α	Nu	N	ν
Beta	B	β	Xi	≡	ξ
Gamma	Γ	γ	Omicron	O	o
Delta	Δ	δ	Pi	Π	π
Epsilon	E	ϵ	Rho	P	ρ
Zeta	Z	ζ	Sigma	Σ	σ
Eta	H	η	Tau	T	τ
Theta	θ	Θ	Upsilon	Y	υ
Iota	I	ι	Phi	Φ	ϕ
Kappa	K	κ	Chi	X	x
Lambda	Λ	λ	Psi	Ψ	ψ
Mu	M	μ	Omega	Ω	ω

Prefixes for Powers of 10

Power of ten	Prefix	Power of ten	Prefix
10^{-24}	Yocto (y)	10^{24}	Yotta (Y)
10^{-21}	Zepto (Z)	10^{21}	Zetta (Z)
10^{-18}	Atto (a)	10^{18}	Eta (E)
10^{-15}	Femto (f)	10^{15}	Peta (P)
10^{-12}	Pico (P)	10^{12}	Tera (T)
10^{-9}	Nano (n)	10^{9}	Giga (G)
10^{-6}	Micro (N)	10^{6}	Mega (M)
10^{-3}	Milli (m)	10^{3}	Kilo (K)
10^{-2}	Centi (c)		

The key to Succeed in Physics

Your success is your decision. If you have the burning desire to succeed in physics, develop positive attitude towards the subject and be committed to understand the basic concepts and principles.

How to Understand Concepts and Principles in Physics?

Be persistent and practice these time tested and powerful strategies.

- Attend class, be focused, take notes and ask questions if you feel so.
- Set up a daily study schedule and make an in depth reading
- Avoid rote memorization of facts and formulas. Devote to understand the basic concepts of ideas so as to solve problems.

How to solve Physics Problems?

Understand the basic concepts of algebra, geometry & trigonometry to develop mathematical skill and practice the following extremely important logical steps in problem solving.

First, read the problem carefully & identify given and required variables. Next, state any assumptions & approximations, draw diagrams where needed. Then, identify the governing concept or principle & select suitable equation & solve for the unknown. Finally, evaluate your answer. Check if it makes sense. If it does not, go back and revise.

How to Score Best in Physics Exams?

Remember, "there is no free lunch." Work hard in advance and turn confusion to confidence. At exam time, be punctual, disciplined and show your work in net & shown your work in next & logical manner.

☞ This book is designed on the new curriculum and aims to enhance students; analysis, creativity, reasoning and problem solving skills. For this purpose, so many conceptual questions with their explanations and a variety of problems with their solutions are embedded at every section of the book. What is more, about ten years of EGSCE exams including the 2007 have been solved and included at the end of each unit.

And, remember, "you will harvest what you sow" so work hard and practice. Practice will make you smart. It is the master key to succeed.

≈ Good Luck ≈

Unit one

Vectors

Airplane pilots, ship captains, car drivers, engineers and all of us make use of vectors in our everyday life.

Physical quantities: Are either scalars or vectors. For instance, if you say the time is now 3:30 AM, the mass of this book is 1.5 kg or the room temperature is 22°C, all of this information makes sense. No need of saying 3:30 AM to the East or 1.5 kg to the South or 22°C to the North. Such quantities that are fully expressed by magnitude only are called *scalar quantities*.

> **Scalar quantity**: Is a quantity that is completely described by its magnitude.

Examples of scalar quantities include: time, mass, length, speed, temperature, power, density, work, energy, pressure, potential difference, electric current, area, volume, charge, amount of substance, luminous intensity etc.

But, if you say your home is 2 km away from your school or a car is moving at 50 km/h or a man is pushing a cart with a force of 100N, all of these information is incomplete unless specified with particular direction as, 2 km away due East, or moving at 50 km/h towards South or pushing a cart with a force of 100N towards West. Such quantities that require both magnitude and direction for their complete description are called *vectors*.

> **Vector quantity**: Is a quantity that is completely described by both, its magnitude and direction.

Examples of vector quantities include; displacement, velocity, acceleration, force, weight, torque, electric field strength, magnetic field strength etc.

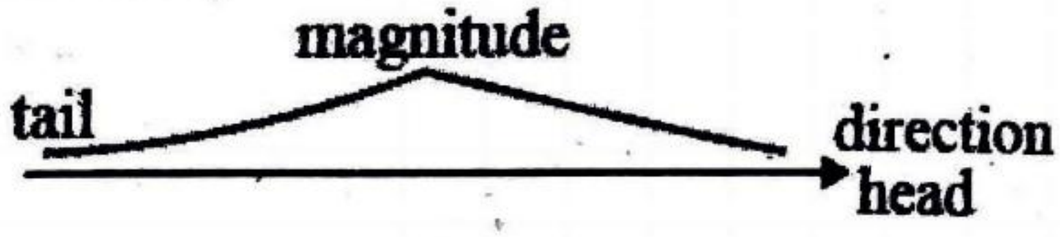
Since vectors are different from scalars, they do have their own notation algebraically or graphically.

1.1 Representation of Vectors

1) **Algebraically**, vectors are written with a bold face letter or a litter with an arrow over its head. For example, a force vect

may be written as **F** or $\vec{F}$. The magnitude of the vector is denoted by a light face letter or a letter inside an absolute value. Thus, the magnitude of the force vector may be written as F or |F|.

2) **Graphically**, a vector is represented by an arrow. The length of the arrow between its tail and head gives the magnitude of the vector and the orientation of the arrow head tells us the direction of the vector. That is,



☞ **Important points to remember**

- **Physical quantity**: Is any number that is used to describe a physical phenomenon quantitatively.
- Physical quantities may be categorized as; *basic* or *drived*, *scalar* or *vector*.
  - → **Basic (Fundamental quantities):** Are quantities that are defined based on some standard. **Example**; time, mass, length, temperature, amount of substance, electric current and luminous intensity.
  - → **Drived quantities**: Are quantities that are obtained by multiplying or dividing the basic quantities. **Example**: speed, volume, acceleration, force, pressure, work, charge, heat etc.
  - → **Magnitude:** Is size of given physical quantity. It tells us "how much" or "how big" something measures.
    Magnitude is expressed using a number and a unit. **Example**; 2 kg, where 2 is number and kg is unit.
  - → **Scalars** can be positive, negative or zero.
    There is no direction associated with scalars. That is, it would be nonsense to say 5 kg due North etc.
  - → **All vector** quantities have a direction associated with them. The magnitude of a vector is always positive. But, vectors can have positive or negative direction.

Conceptual Examples

1. Suppose a given physical quantity has both magnitude and direction. Does it mean that it is necessarily a vector quantity? Explain

   Explanation: Not necessarily. A quantity to be a vector, it should have both magnitude and direction and it should obey the laws of

vector addition. For example, electric current has both magnitude and direction but it is a scalar quantity since it does not obey the law of vector addition.

2. Do you think that all basic physical quantities are scalar quantities? Explain

   Explanation: Yes! All the seven physical quantities are scalars since they all do not obey vector addition.

3. Do you think that all derived physical quantities are vectors? Explain

   Explanation: No! not all drived physical quantities are vectors. For example, speed, work, volume, pressure, density etc are drived quantities, but none of them are vectors.

4. Can the magnitude of a vector be ever be negative? Explain

   Explanation: Never! The magnitude of a vector cannot be negative. But it may point in the negative axis. For example, the displacements $\Delta s = 2m$ and $\Delta s = -2m$ do have the same magnitude of 2 but are pointing in opposite directions.

1.2 Addition and Subtraction of vectors

Scalar quantities can be added or subtracted in the usual way using ordinary arithmetic. For example; 4 kg + 4 kg = 8 kg. However, when adding a 4N force and a 4N force, it is possible to get a value that ranges for`8N to 0N. That is, the rules of ordinary algebra cannot be used to combine vector quantities. Instead, vector quantities are added or subtracted using vector rules either graphically or analytically.

Combining parallel or antiparallel vectors

When two or more vectors are combined, they give another single vector called the *resultant* whose effect is the same as the individual vectors acting together.

> **Resultant**: Is a single vector whose effect is the same as the individual vectors acting together.

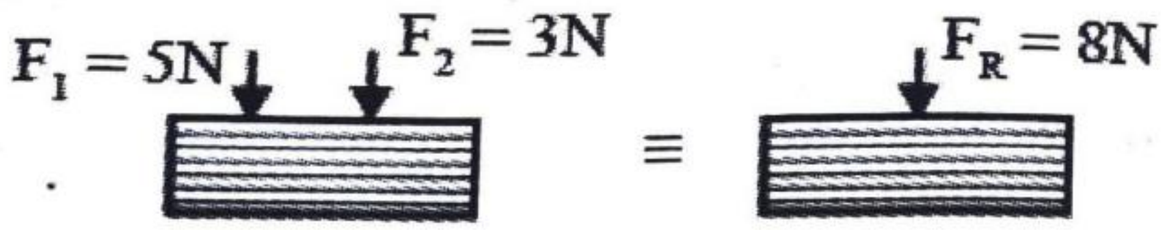
For instant, if a body moves 3m East and then 5m East, then its resultant displacement is 8m East. And, if it moves 3m east and then 5m West, now its resultant is 2m West.

☞ **Bear in mind the following important points**

> The magnitude of the resultant of two parallel vectors, that is vectors acting in the same direction is the sum of the magnitude of the two vectors and its direction is the same as the directions of the two vectors.

That is; if $F_1 = 5N$ South and $F_2 = 3N$ South, then the resultant force, F_R is;

$$F_R = F_1 + F_2 = 8N \text{ South}$$

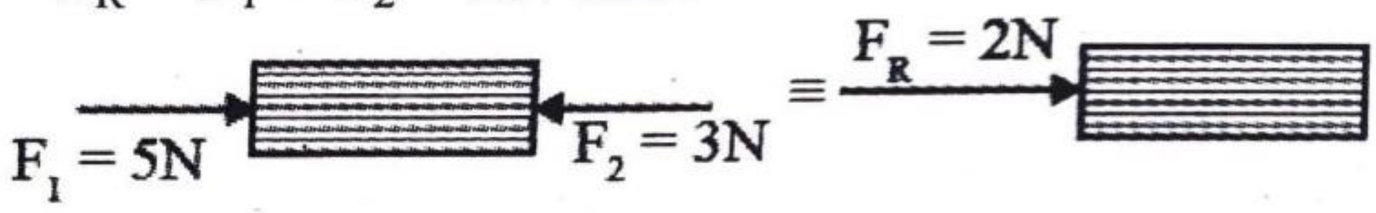


And;

> The magnitude of the resultant of two anti-parallel vectors, that is, vectors acting in opposite directions is the difference of the magnitude of the vectors, and its direction is the same as the direction of the bigger vector.

That is; if $F_1 = 5N$ East and $F_2 = 3N$ West, then the resultant force F_R is;

$$F_R = F_1 + F_2 = 2N \text{ East}$$



Conceptual Example

5. If two vectors are equal in magnitude but opposite in direction, what can you say about the magnitude and direction of their resultant?

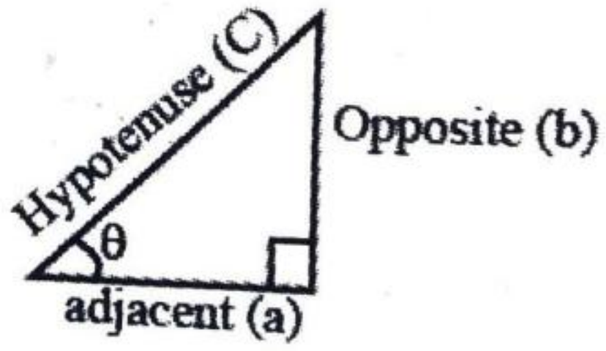
   Explanation: Magnitude zero and direction arbitrary. Such vector is called null vector.

Combining perpendicular vectors

Perpendicular vectors can be combined using Pythagoras's theorem and the direction of the resultant is determined using basic trigonometric functions.

> **Pythagoras's theorem:** "the square of the hypotenuse of a right angled triangle is equal to the sum of the squares of the other two sides."

That is; for the given triangle,



Pythagoras's theorem; $\Rightarrow \boxed{c^2 = a^2 + b^2}$

The three basic trigonometric functions are defined by:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{b}{c}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{a}{c}$$

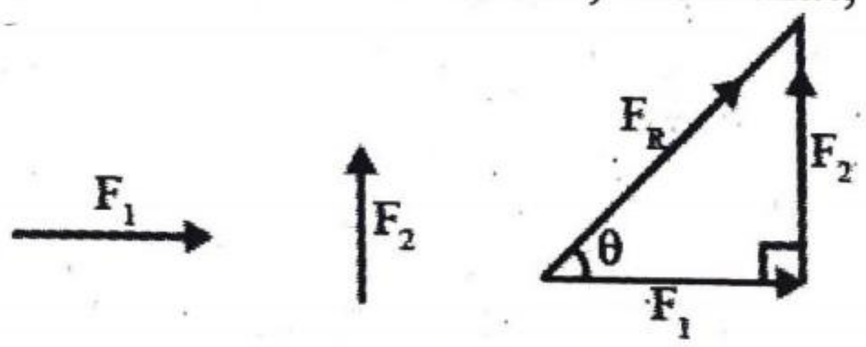
$$\tan\theta = \frac{\text{adjacent}}{\text{adjacent}} = \frac{b}{a}$$

☞ To find the angle θ in degree, we rearranging the trigonometric function as $\theta = tan^{-1}\left(\frac{b}{a}\right) = cos^{-1}\left(\frac{a}{c}\right) = sin^{-1}\left(\frac{b}{c}\right)$ where the operator ($\tan^{-1}$) is called "*tan inverse*" or "*arc tan*". The same naming holds for the others.

It is worth noting to remember that:

The magnitude of the resultant of two perpendicular vectors is obtained using Pythagoras's theorem and its direction is determined using trigonometry function.

That is, if we have two forces, such that,



$\Rightarrow F_R^2 = F_1^2 + F_2^2 \Rightarrow \boxed{F_R = \sqrt{F_1^2 + F_2^2}}$, and its direction say from F_1 is obtained by;

$$\boxed{\tan\theta = \frac{F_2}{F_1}}$$

Conceptual Example

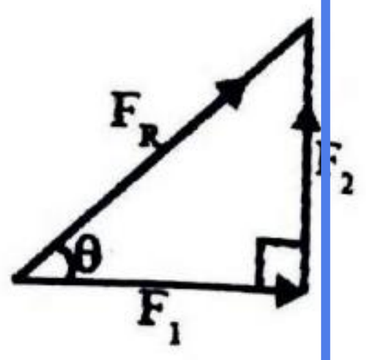
6. If the magnitude of the sum and difference of two nonzero vectors $\vec{A}$ and $\vec{B}$ is equal, what can you say about the two vectors?

Explanation: If $|A + B| = |A - B|$, then two vectors that is vector A and vector B are mutually perpendicular.

Illustrative Examples

7. Two force vectors 5N and 12N act on a body in such a way that they make an angle of 90° with each other. Find the resultant of the two forces.

Solution: Since the force vectors that we are given are perpendicular.



The resultant is obtained using Pythagoras's theorem.

$$F_R = \sqrt{F_1^2 + F_2^2} = \sqrt{5+12} = 13N$$

Its direction is; $\tan\theta = \frac{F_2}{F_1} = \frac{12}{5} \Rightarrow \theta = \tan^{-1}\left(\frac{12}{5}\right) = 17°$

8. Two mutually perpendicular vectors $\vec{A}$ & $\vec{B}$ are given such that $\vec{A} = 9.6$ units and $|A + B| = 17.98$. Find vector $\vec{B}$?

Solution: Since the two vectors are perpendicular, we have,

$$R = |A + B| = \sqrt{A^2 + B^2}$$

$$\Rightarrow B^2 = R^2 - A^2 = (17.98)^2 - (9.6)^2$$

$$B = 15.2 \text{ units}$$

Equal vectors: Two vectors are equal if they have the same magnitude and the same direction.

Zero \Null\ vector & Unit vector

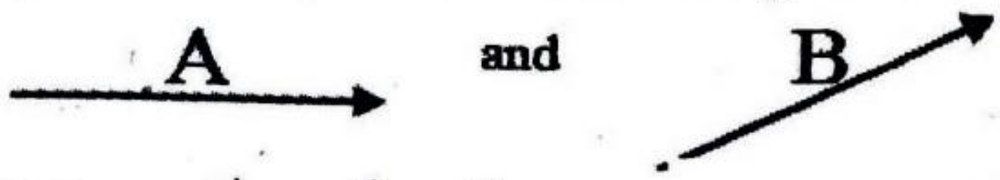
A vector with magnitude zero is called **zero vector**. A vector with magnitude **1** is called **a unit vector**.

Non parallel and non-perpendicular vectors

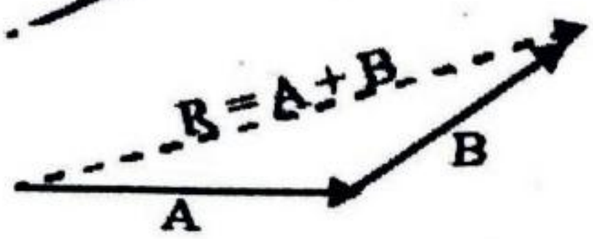
Non parallel and non-perpendicular vectors can be added using the *triangle law* or the *parallelogram law*.

1. **Triangle law**: states that:
"Join the two given vectors *head* to *tail* and the resultant is the vector that is drawn from the *tail* of the first to the *head* of the second vector."

That is, if we have two vectors say $\vec{A}$ and $\vec{B}$



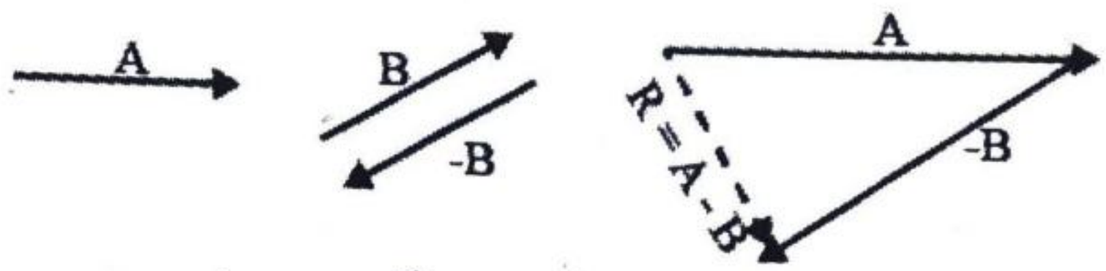
Then, $\vec{R} = \vec{A} + \vec{B} \equiv$



Subtraction of vectors

Let there are two vectors $\vec{A}$ and $\vec{B}$, then the difference $\vec{A}-\vec{B}$ is obtained by adding $-\vec{B}$ to $\vec{A}$. Hence, $\boxed{\vec{A}-\vec{B}=\vec{A}+(-\vec{B})}$.

Then, $\vec{R}=\vec{A}-\vec{B}=\vec{A}+(-\vec{B})$

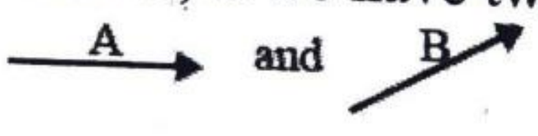


That is, if we have two vectors $\vec{A}$ and $\vec{B}$

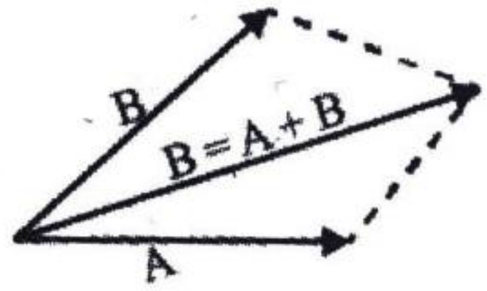
Where, $-\vec{B}$ is a vector that is equal in magnitude to vector $\vec{B}$ but pointing in opposite direction.

2. Parallelogram law: sates that;
"Join the two given vectors *tail* to *tail* and the resultant is the *diagonal* of the parallelogram so formed."

That is, if we have two vectors $\vec{A}$ and $\vec{B}$, such that



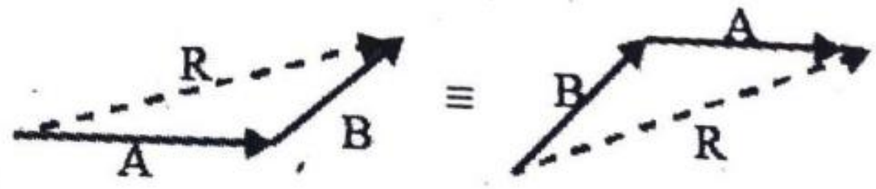
Then $\vec{R}=\vec{A}+\vec{B}\equiv$



☛ Important points to remember

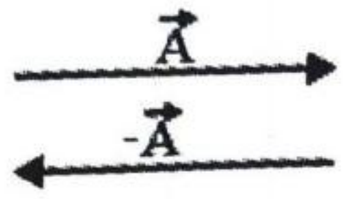
- Vector addition obeyes *commutative law*. That is the order of vector addition doesn't make any difference in the resultant.
  So that, $\vec{A}+\vec{B}=\vec{B}+\vec{A}$

  Graphically,



- While adding vectors graphically, use appropriate scales, such as 1 cm to 5 km or 1 cm to 5N or 1 cm to 5 m/s. Thus, a 3 cm value on paper will correspond to a 15 km or 15N or 15 m/s on actual context and the direction of each vector can be determined using a protractor.

- The negative of a vector is another vector whose magnitude is equal to the given vector but senses in opposite direction. That is, if vector $\vec{A}$ happens to be;

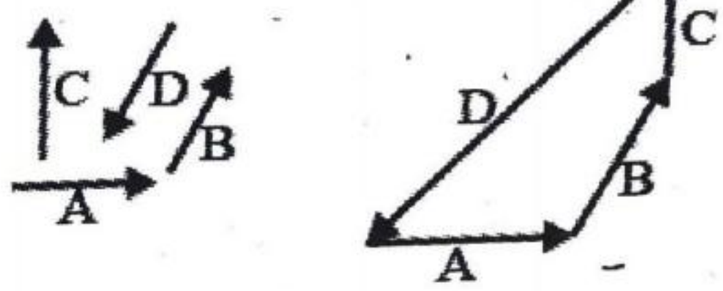


is, then the negative of vector A

Conceptual Example

9. Assume you are adding more than two vectors using the "head to tail" method, if the head of the last vector coincide with the tail of the first vector, what can you say about the magnitude of the resultant of the vectors.

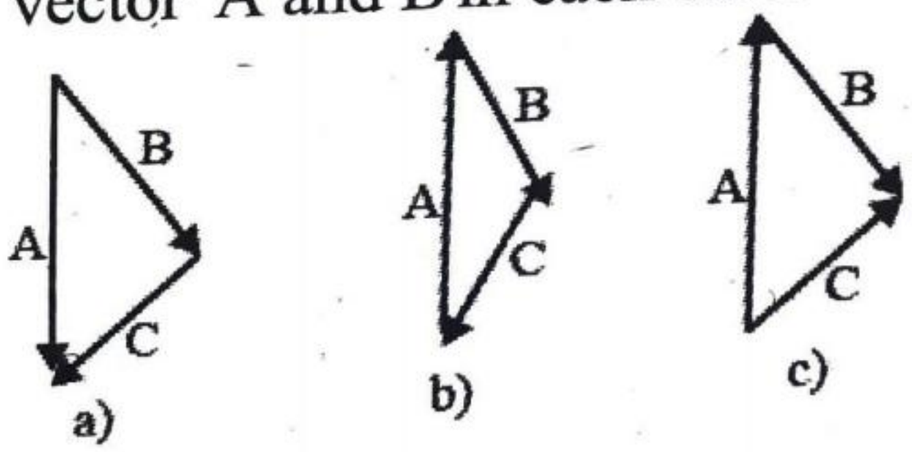
Explanation: Assume we have four vectors with the orientation given, then $\vec{R} = A + B + C + D$, using head to tail.



If this is the case, then the resultant is zero, since it a closed loop.

Illustrative Example

10. If you are given the following, express vector $\vec{C}$ in terms of vector $\vec{A}$ and $\vec{B}$ in each case.



Solution:

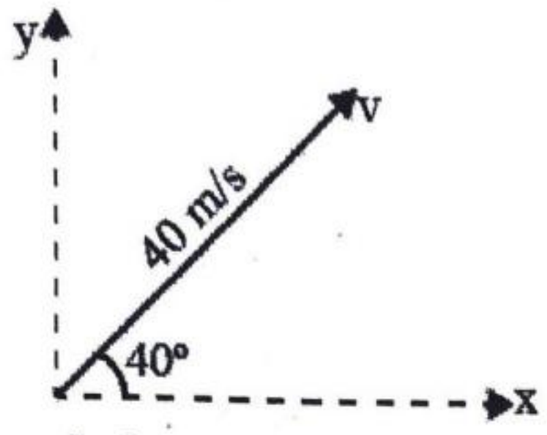
a) $\vec{B} + \vec{C} = \vec{A} \Rightarrow \vec{C} = \vec{A} - \vec{B}$

b) $\vec{A} + \vec{B} + \vec{C} = 0 \Rightarrow \vec{C} = -(\vec{A} + \vec{B})$

c) $\vec{A} + \vec{B} = \vec{C}$

11. Represent a velocity of 40 m/s directed at an angle of 40° East to North.

Solution: First assume, a scale such as; 1 cm = 10 m/s thus 40 m/s = 4 cm and the angle is measured using a protractor starting from the positive x axis, so we have;

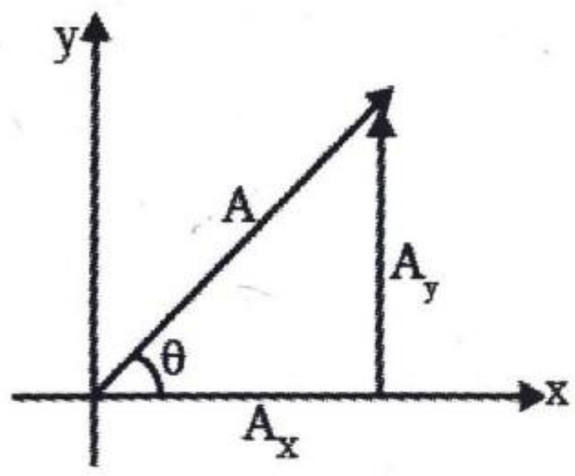


1.3 Resolving vectors

To make vector addition, subtraction or even multiplication simple, vectors are resolved.

Resolution: Is the process of splitting a vector into components.

Assume we have vector $\vec{A}$ in the xy plane making an angle θ from the positive x axis.



Then, the components of vector $\vec{A}$ can be obtained using trigonometry relations as;

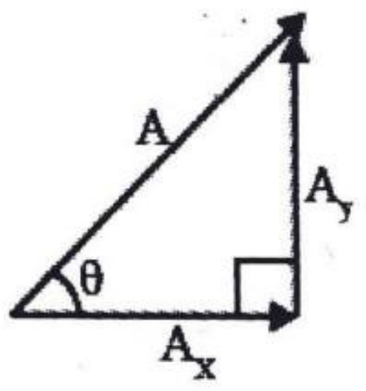
$$\cos\theta = \frac{A_x}{A} \Rightarrow \boxed{A_x = A\cos\theta}$$

$$\sin\theta = \frac{A_y}{A} \Rightarrow \boxed{A_y = A\cos\theta}$$ where,

A_x is the component of vector $\vec{A}$ along the x axis, and

A_y is the component of vector $\vec{A}$ along the y axis.

If the components A_x and A_y are given, then the magnitude of vector A can be obtained using the Pythagoras's theorem.



$$A^2 = A_x^2 + A_y^2$$

$$\boxed{A = \sqrt{(A_x)^2 + (A_y)^2}}$$

The angle that vector $\vec{A}$ makes from, say the positive x axis is obtained from the definition of tangent as;

$$\boxed{\tan\theta = \frac{A_y}{A_x}}$$

The sum of two vectors, say $\vec{A}$ and $\vec{B}$ via the component method is computed as;

$$\vec{R} = \vec{A} + \vec{B} = (A_x + A_y) + (B_x + B_y)$$

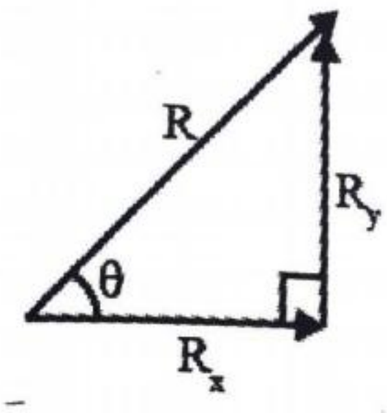
Next, adding components' of the same kind, we have;

$$\vec{R} = (A_x + B_x) + (A_y + B_y)$$

$$\Rightarrow \boxed{\vec{R} = \vec{R}_x + \vec{R}_y}$$

Where, $\vec{R}_x = \vec{A}_x + \vec{B}_x$ and $\vec{R}_y = \vec{A}_y + \vec{B}_y$

The magnitude and direction of the resultant vector can be calculated by;



$$\boxed{R = \sqrt{(R_x)^2 + (R_y)^2}} \text{ and, } \boxed{\tan\theta = \frac{R_y}{R_x}}$$

In the same manner, the difference between vectors $\vec{A}$ and $\vec{B}$ can be obtained by;

$$\vec{R} = \vec{A} - \vec{B} = (A_x + A_y) - (B_x + B_y)$$

$$\vec{R} = (A_x - B_x) + (A_y - B_y)$$

$$\Rightarrow \boxed{\vec{R} = \vec{R}_x + \vec{R}_y}$$

Where, $\vec{R}_x = \vec{A}_x - \vec{B}_x$, and $\vec{R}_y = \vec{A}_y - \vec{B}_y$

☛ **Important points to remember**

→ To add or subtract physical quantities, they must represents the same type of physical quantity. That is, it would be meaningless to add a *velocity* to a *force*. In the same manner, there is no point to add *time* with *mass*.

→ Components of a vector are parts of the vector obtained after the process of resolution.

→ The resultant of more than two vectors in component from can be obtained in the same manner as we combined two vectors. That is, if we have;

$$\vec{R} = \vec{A} + \vec{B} + \vec{C} + \ldots$$

$$\vec{R}_x = A_x + B_x + C_x + \ldots$$

$$\vec{R}_y = A_y + B_y + C_y + \ldots$$

Then, $\vec{R} = \vec{R}_x + \vec{R}_y$

Conceptual Examples

12. If the sum of two vectors is zero, what can you conclude about their rectangular components.

Explanation: Assume we have two nonzero vectors. $\vec{A}$ and $\vec{B}$. If $\vec{R} = \vec{A} + \vec{B} = (A_x + B_x) + (A_y + B_y) = 0$ then, $A_x + B_x = 0 \Rightarrow A_x = -B_x$ and $A_y + B_y = 0 \Rightarrow A_y = -B_y$

Which means that, the two vectors do have equal but opposite components?

13. Assume two vectors $\vec{A}$ and $\vec{B}$. What must be the relation of $\vec{A}$ to $\vec{B}$ if the magnitude of their sum is:

a) A + B b) Zero

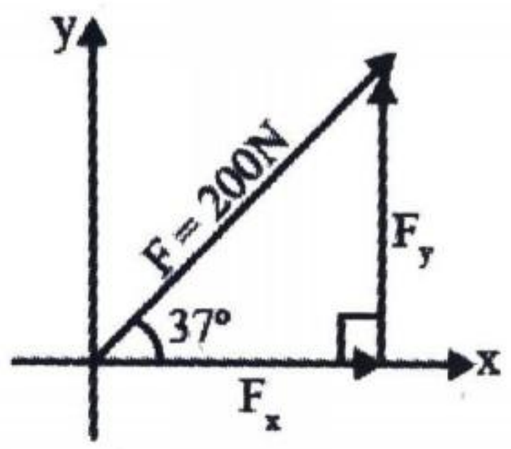
Explanation:

a) A is equal to B

b) A is equal but opposite to B

Illustrative Examples

14. A force vector of magnitude 200N is acting at 37° North to East. Resolve this vector into its x and y components.



Solution: We can draw the given vector in the xy plane as;

Then we have;

$F_x = F \cos\theta = (200)(\cos 37)$

$= (200)(0.8) = 160N$

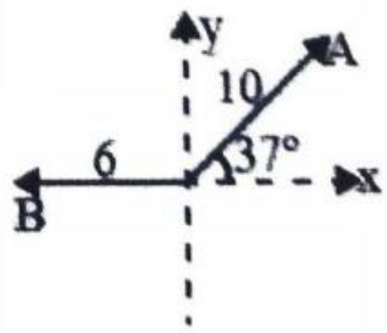
$F_y = F \sin\theta = (200)(\sin 37) = (200)(0.6) = 120N$

15. The magnitude of vectors $\vec{A}$ and $\vec{B}$ are 3 and 4 respectively. What are the largest and smallest possible values of the magnitudes of their resultant?

Solution: The largest value, is when they are parallel. Thus, $R_{max} = |A + B| = 3 + 4 = 7$.

The smallest value is when they are antiparallel. Thus, $R_{min} = |A - B| = |3 - 4| = 1$.

16. From the given arrangement, find



a) A + B b) A – B

Solution:

a) First of all resolve vector A

$\Rightarrow A_x = A \cos 37° = (10)(0.8) = 8$ and

$A_x = A \sin 37° = (10)(0.6) = 6$, but

$B_x = -6$ and $B_y = 0$

$R = A + B = (A_x + B_x) + (A_y + B_y) \Rightarrow R_x = A_x + B_x$, $R_y = A_y + B_y$

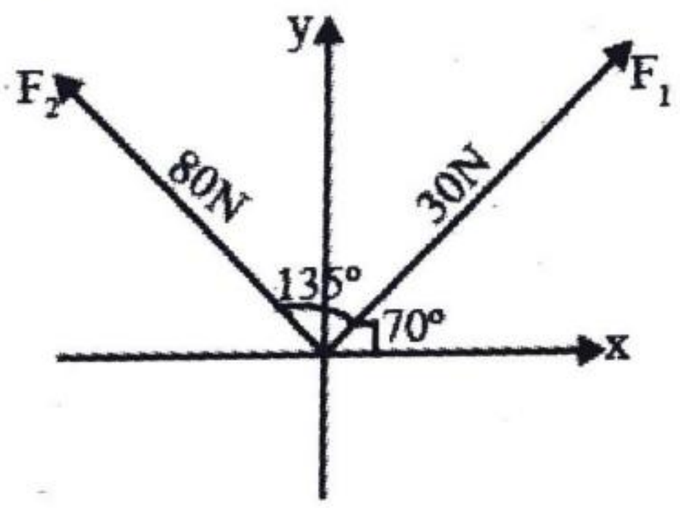
Hence, $R_x = (8 + (-6)) = 2$ and $R_y = (6 + 0) = 6$

Finally, $R = \sqrt{R_x^2 + R_y^2} = \sqrt{(2)^2 + (6)^2} = \sqrt{40} = 6.3$.

b) $R = A - B = (A_x - B_x) + (A_y - B_y)$
$R_x = (A_x - B_x) = (8 - (-6)) = 14$ and
$R_y = (A_y - B_y) = (6 - 0) = 6$
$\Rightarrow R = \sqrt{R_x^2 + R_y^2} = \sqrt{(14)^2 + (6)^2} = 15$

17. What third force must be added to the two forces, 30N at 70° and 80N at 135°??

Solution: Let $F_1 = 30N$ and $F_2 = 80N$. So we have;



$F_{1x} = F_1 \cos \theta_1 = 30 \cos 70° = (30)(0.3) = 9N$
$F_{1y} = F_1 \sin \theta_1 = 30 \sin 70° = (30)(0.9) = 27N$
$F_{2x} = F_2 \cos \theta_2 = 80 \cos (135) = (80)(-0.7) = -56N$
$F_{2y} = F_2 \sin \theta_2 = 80 \sin (135) = (80)(0.7) = 56N$

Since, $F_R = 0$

$F_{Rx} = 0 \Rightarrow F_{1x} + F_{2x} + F_{3x} = 0 \Rightarrow F_{3x} = -(F_{1x} + F_{2x})$
$\Rightarrow F_{3x} = -(9 - 56) = 47N$
And, $F_{Ry} = 0 \Rightarrow F_{1y} = F_{2y} = F_{3y} = 0 \Rightarrow F_{3y} = -(F_{1y} + F_{2y})$
$\Rightarrow F_{3y} = -(27 + 56) = -83N$

Therefore, $F_3 = \sqrt{(F_{3x})^2 + (F_{3y})} = \sqrt{(47)^2 + (-83)^2} = 95\,N$ and

$$\tan\theta = \frac{F_{3y}}{F_{3x}} = \frac{-83}{47} \Rightarrow \theta = \tan^{-1}\left(\frac{-83}{47}\right) = 300° \text{ from +ve x axis.}$$

18. The x and y components of two vectors $\vec{P}$ and $\vec{Q}$ are given to be; $P_x = 6$, $P_y = 8$, $Q_x = 3$, $Q_y = 4$. Find the magnitude and direction of their sum.

Solution: $\vec{R} = \vec{R}_x + \vec{R}_y = (P_x + Q_x) + (P_y + Q_y)$

$R_x = (6 + 3) = 9$ & $R_y = (8 + 4) = 12$

Thus, $R = \sqrt{R_x^2 + R_y^2} = \sqrt{(9)^2 + (12)^2} = 15$ and

$$\tan\theta = \frac{R_y}{R_x} = \frac{12}{9} \Rightarrow \theta = \tan^{-1}(1.33) = 53°$$

Hence, their resultant is R = 15 units at 53°.

19. Calculate the magnitude and direction of a vector whose x and y components are given to be 30 and 40 units respectively.

Solution: Let the vector be denoted by letter $\vec{A}$ so we have, $\vec{A} = \vec{A}_x + \vec{A}_y$, its magnitude,

$$|A| = \sqrt{(A_x)^2 + (A_y)^2} = \sqrt{(30)^2 + (40)^2} = 50 \text{ units}$$

Its direction, $\tan\theta = \frac{A_y}{A_x} = \frac{40}{30} \Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right) = 53°$

20. Two boys are pushing a table on a smooth horizontal surface with forces of 200N and 300N respectively. What will be the magnitude of the resultant force on the table when the boys push it

a) In the same direction

b) In opposite directions

Solution:

a) $F_R = |F_1 + F_2| = 200N + 300N = 500N$

b) $F_R = |F_1 + F_2| = |200 - 300| = 100N$, in the 300N force directions.

21. Calculate the resultant displacement of a boy if he walks 500m towards East and 1200m towards.

a) East b) West c) South

Solution: Let displacement be denoted by letters

a) $S = S_1 + S_2 = 500 + 1200 = 1700m$ towards East

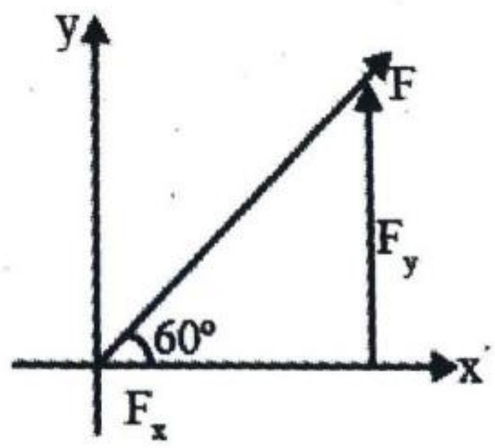
b) $S = S_1 - S_2 = 500 - 1200 = -700m$ East $= 700m$ West

c) $S = \sqrt{(S_1)^2 + (S_2)^2} = \sqrt{(500)^2 + (1200)^2} = 1300m$ and

$$\tan\theta = \frac{S_2}{S_1} = \frac{1200}{500} \Rightarrow \theta = \tan^{-1}\left(\frac{12}{5}\right) = 67°$$

22. Calculate the component of a force of 200N at a direction of 60° to the force.

Solution: Let the force arrangement be as shown; then the component whose shadow lies 60° from the force is F_x.

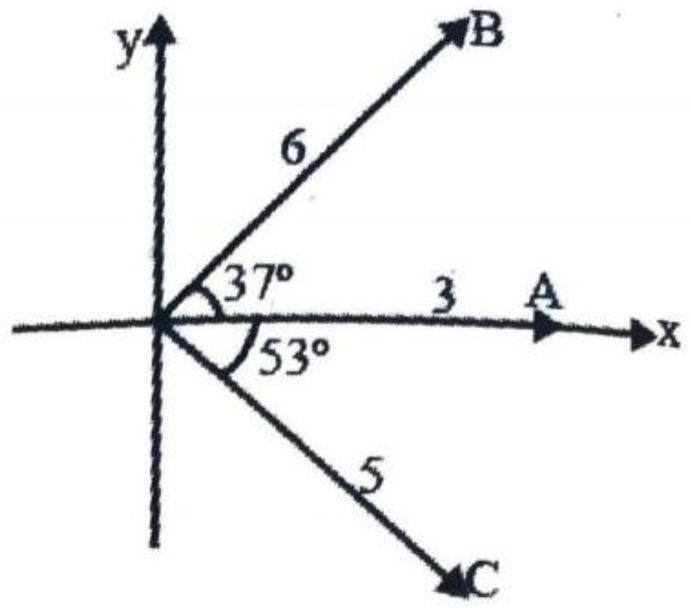


Thus, $F_x = F\cos\theta = (200)(\cos 60°)$

$= (200)(0.5)$

$F_x = 100N$

23. For the vectors shown, find the resultant of:



a) $\vec{A}+\vec{B}+\vec{C}$

b) $\vec{A}+\vec{B}-\vec{C}$

c) $\vec{A}-2\vec{C}$

d) $\vec{B}-(\vec{A}+\vec{C})$

Solution:

Let's first decompose each vector

Vector	x component	Y component
$A = 3$	$A_x = 3$	$A_y = 0$
$B = 6$	$B_x = B\cos 37 = 6(0.8) = 4.8$	$B_y = B\sin 37 = 6(0.6) = 3.6$
$C = 5$	$C_x = C\cos 53 = 5(0.6) = 3$	$C_y = -C\sin 53 = -5(0.8) = -4$

a) $\vec{R} = \vec{R}_x + \vec{R}_y \Rightarrow R_x = A_x + B_x + C_x = (3 + 4.8 + 3) = 10.8$

$R_y = A_y + B_y + C_y = (0 + 3.6 - 4) = -0.4$

Thus, $R = A + B + C = \sqrt{(10.8)^2 + (-0.4)^2} = 10.8$

$$\tan\theta = \frac{R_y}{R_x} = \frac{-0.4}{10.8} \Rightarrow \theta = \tan^{-1}\left(\frac{-0.4}{10.8}\right) = 358° \text{ from +ve x axis}$$

b) $\Rightarrow R_x = A_x + B_x - C_x = 3 + 4.8 - 3 = 4.8$

$R_y = A_y + B_y - C_y = 0 + 3.6 - (-4) = 7.6$

Thus, $R = \sqrt{R_x^2 + R_y^2} = \sqrt{(4.8)^2 + (7.6)^2} = 9$

$$\tan\theta = \frac{R_y}{R_y} = \frac{7.6}{4.8} \Rightarrow \theta = \tan\left(\frac{7.6}{4.8}\right) = 58°$$

c) $\vec{R} = \vec{A} - 2\vec{C} \Rightarrow R_x = A_x - 2C_x = 3 - 2(3) = -3$

$R_y = A_y - 2C_y = 0 - 2(-4) = 8$

Thus, $R = \sqrt{(-3)^2 + (8)^2} = 8.5$

$$\tan\theta = \frac{R_y}{R_x} = \frac{8}{-3} \Rightarrow \theta = \tan^{-1}\left(\frac{-8}{3}\right) = 110.6° \text{ from +ve x axis}$$

d) $\vec{R} = \vec{B} - (\vec{A} + \vec{C}) \Rightarrow R_x = B_x - (A_x + C_x) = 4.8 - (3 - 4) = 5.8$

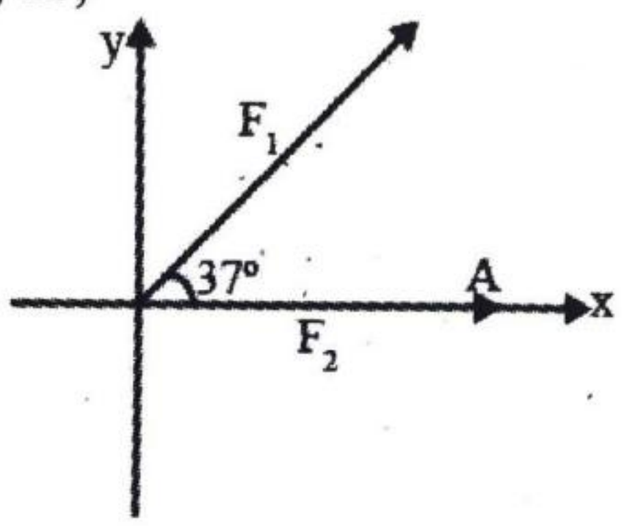
$R_y = B_y - (A_y + C_y) = 3.6 - (0 - 4) = 7.6$

Thus, $R = \sqrt{(5.8)^2 + (7.6)^2} = 9.6$

$$\tan\theta = \frac{R_y}{R_x} = \frac{7.6}{5.8} \Rightarrow \theta = \tan^{-1}\left(\frac{7.6}{5.8}\right) = 52.7°$$

24. Calculate the resultant of two forces each 5N acting at a point at an angle of 37° with each other.

Solution: Let the force vectors be acting as;



$F_{1x} = F_1 \cos 37° = 5(0.8) = 4N$

$F_{1y} = F_1 \sin 37° = 5(0.6) = 3N$

$F_{2x} = 5N$ and $F_{2y} = 0$

Then, $\vec{F}_R = \vec{F}_1 + \vec{F}_2 = \vec{F}_{Rx} + \vec{F}_{Ry}$

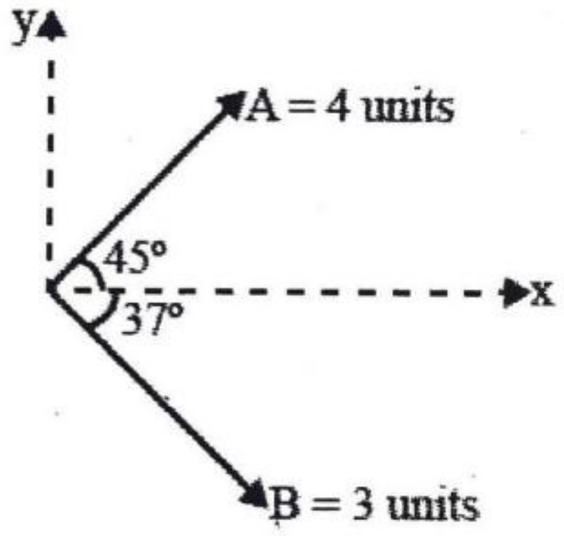
$F_{Rx} = (F_{1x} + F_{2x}) = (4 + 5) = 9N$

$F_{Ry} = (F_{1y} + F_{2y}) = (3 + 0) = 3N$

Thus, $F_R = \sqrt{(F_{Rx}) + (F_{Ry})} = \sqrt{(9)^2 + (3)^2} = 9.5N$

$$\tan\theta = \frac{F_{Ry}}{F_{Rx}} = \frac{3}{9} \Rightarrow \theta = \tan^{-1}\left(\frac{1}{3}\right) = 18°$$

25. For the vectors A and B in the figure, determine the magnitude of their vector sum.



Solution: Let's first resolve the vectors.

$A_x = A \cos 45 = (4)(0.7) = 2.8$ units

$A_y = A \sin 45 = (4)(0.7) = 2.8$ units

$B_x = B \cos 37° = (3)(0.8) = 2.4$ units

$B_y = B \sin 37° = -(3)(0.6) = -1.8$ units

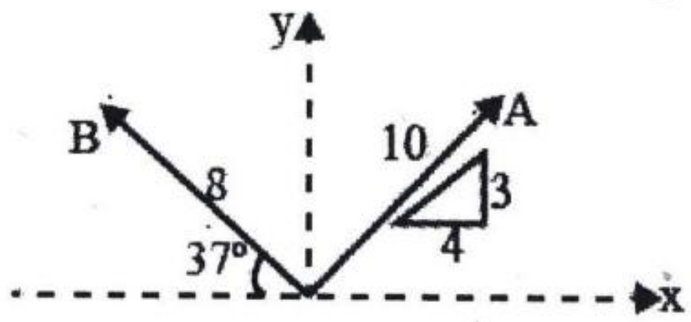
Then, $\vec{R} = \vec{A} + \vec{B} = \vec{R}_x + \vec{R}_y$

$R_x = (A_x + B_x) = 2.8 + 2.4 = 5.2$ units

$R_y = (A_y + B_y) = 2.8 - 1.8 = 1$ unit

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(5.2)^2 + (1)^2} = 5.59 \text{ units}$$

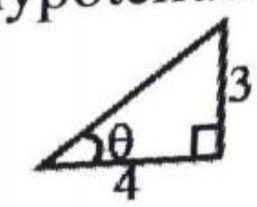
26. Determine the magnitude of the vector sum $\vec{R} = \vec{A} + \vec{B}$ and the direction which the resultant makes from the positive x axis.



Solution: First of all, let's find the components of each vector.

$A_x = A\cos\theta$, but from, the hypotenuse is 5.

$$= 10\left(\frac{4}{5}\right) = 8 \text{ units}$$



$$\cos\theta = \frac{4}{5}$$

$$A_y = A\sin\theta = 10\left(\frac{3}{5}\right) = 6 \text{ units}$$

$$\sin\theta = \frac{3}{5}$$

$B_x = -B\cos 37° = -8(0.8) = -6.4$ units

$B_y = B\sin 37° = 8(0.6) = 4.8$ units

Then, $\vec{R} = \vec{R}_x + \vec{R}_y$, $R_x = A_x + B_x = 8 + (-6.4) = 1.6$ units

$R_y = A_y + B_y = 6 + 4.8 = 10.8$ units

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(1.6)^2 + (10.8)^2} = 10.9$$

$$\tan\theta = \frac{R_y}{R_x} = \frac{10.8}{1.6} \Rightarrow \theta = \tan^{-1}\left(\frac{10.8}{1.6}\right) = 81.6°$$

27. The x and y components of vector $\vec{A}$ are 4 and 6 respectively. The x and y components of vector $(\vec{A} + \vec{B})$ are 10 and 9 respectively. Find the components of vector $\vec{B}$?

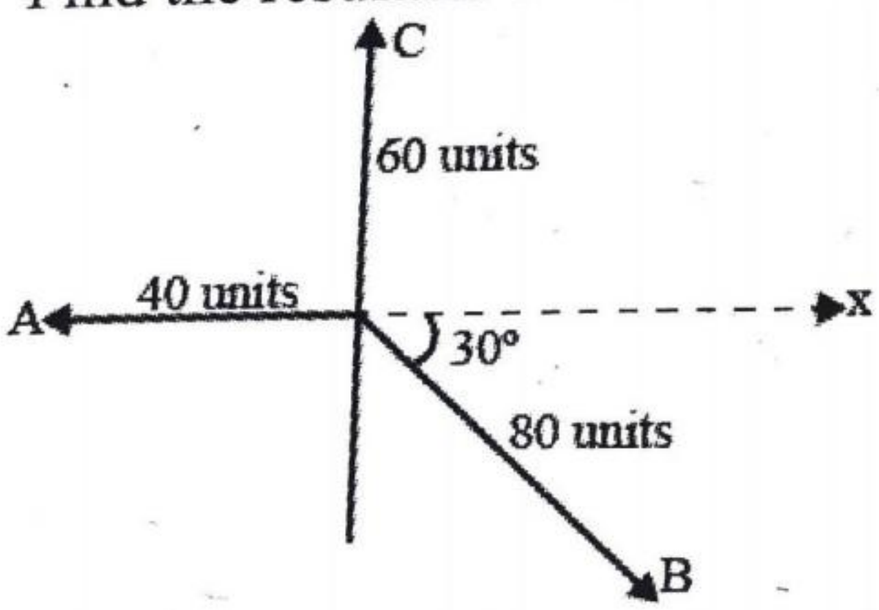
Solution: We have, $A_x = 4$, $A_y = 6$ and $A_x + B_x = 10$

$\Rightarrow B_x = 10 - A_x = 10 - 4 = 6$

$A_y + B_y = 9 \Rightarrow B_y = 9 - A_y = 9 - 6 = 3$

Thus, the components of vector B are; $B_x = 6$ & $B_y = 3$.

28. Find the resultant of the vectors shown below.



Solution: $R = A + B + C$, $R_x = -40 + 80\cos 30° + 0 = 29.3$ units

$R_y = 0 - 80\sin 30° + 60 = 20$ units

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{29.3^2 + 20^2} = 35.4 \text{ units}$$

$$\tan\theta = \frac{R_y}{R_x} = \frac{20}{29.3} \Rightarrow \theta = \tan^{-1}\left(\frac{20}{29.3}\right) = 34.3^\circ$$

General Secondary education certificate exam

29. Two forces 10N and 40N CAN NOT be combined to give a resultant of (EGSECE).

A. 10N B. 40N C. 50N D. 35N

Solution: The magnitude of the resultant of two vectors say $\vec{A}$ & $\vec{B}$ is always in the range $|A - B| \leq R \leq |A + B|$ in our case, $|10 - 40| \leq R \leq |10 + 50| \Rightarrow 30 \leq R \leq 60$. Thus, the resultant can have a magnitude ranging from 30 to 60. **Answer: A**

30. Which of the following differences between two points is a vector quantity? (EGSECE)

A. Pressure B. potential C.electric filed D. temperature

Answer: C

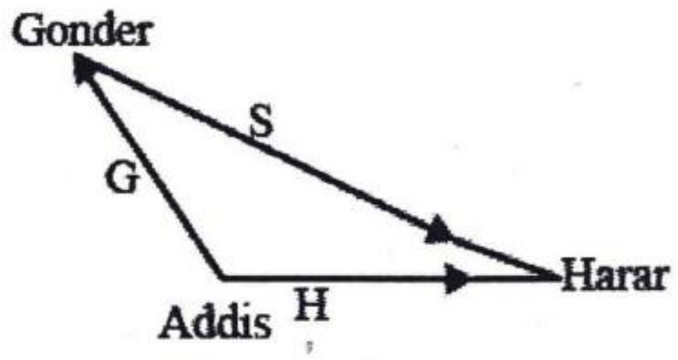
31. A vector of magnitude 20 is added to a vector of magnitude 25. The magnitude of their sum might be? (EGSECE)

A. 12 B. 3 C. 0 D. 47

Solution: The magnitude of the resultant of two vector A and B is always in the range. $|A - B| \leq R \leq |A + B|$ in our case, $|20 - 25| \leq R \leq |20 + 25| \Rightarrow 5 \leq R \leq 45$.

Answer: A

32. A pilot of a tourist plane would like to know the a real displacement from Gonder to Harar as shown. He can determine this displacement between the two tourist attraction sights by calculating. (EGSECE)



A. $H + G$ B. $G + H$ C. $G - H$ D. $H - G$

Solution: From triangle law, $\vec{G} + \vec{S} = \vec{H}$

$\Rightarrow \vec{S} = \vec{H} - \vec{G}$ **Answer: D**

33. If the tail of vector B is placed at the tip of A and the tail of C is placed at the tip of B, the sum of the three vectors is drawn from the:

A. Tail of A to tip of B
B. Tail of A to tip of C
C. Tail of B to tip of C
D. Tip of C to tail of B

Answer: B

Unit One: Conceptual Questions

1. Suppose that $\vec{A}$ and $\vec{B}$ have non zero magnitude. Is it possible for $\vec{A} + \vec{B}$ to be zero?
2. Given that $\vec{A} + \vec{B} = 0$. What can you conclude about the two given vectors.
3. Can a component of a vector be greater than the vector's magnitude?
4. If each component of a vector is doubled, how does the magnitude and direction of the vector changes.
5. Can a vector with zero magnitude have one or more components that are nonzero? Explain
6. Suppose vector $\vec{A}$ has x and y components of equal magnitude. What can you say about the possible direction of $\vec{A}$?
7. The components of vector $\vec{A}$ satisfy $A_x < 0$ and $A_y < 0$. What is the range of possible angle of $\vec{A}$?
8. The components of a vector $\vec{A}$ satisfy $A_x > 0$ and $A_y < 0$. What s the range of possible angles for $\vec{A}$?
9. The components of a vector $\vec{A}$ satisfy $A_x = -A_y$. What are the possible directions of $\vec{A}$?
10. Use a sketch to show that two vectors of unequal magnitude cannot add to zero, but that three vectors of unequal magnitude can.
11. Two forces of 50N and 80N act upon a body. What are the maximum and minimum possible values of the resultant force?
12. What is the difference between vector sum and algebraic sum?
13. If A and B are any two nonzero vectors, is it possible that A – B = B – A.
14. Is it possible to add a vector quantity to a scalar quantity?

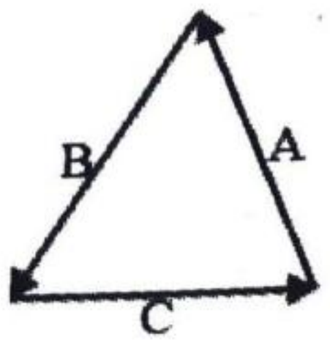
Unit One: Problems

Representation of vectors

15. Represent the following vectors graphically.
    a) 6 m/s north of east
    b) 20m, 37° south of west

Addition and subtraction of vectors

16. A force of $F_1 = 5N$ is applied to a block in a horizontal direction. A second force $F_2 = 4N$



is applied to the object at an angle of 30° above the horizontal. Use graphical method to find the resultant force.

17. Three vectors A, B and C have the same length L and form an equilateral triangle. Find the magnitude and direction of the vectors.

a) $A + B$ b) $A + B + C$

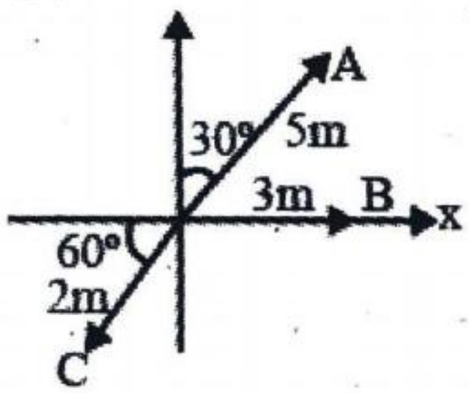
c) $A + B - C$

18. Vector $\vec{A}$ points in the negative y direction and has a magnitude of 5 units. Vector B has twice the magnitude and points in the positive x direction. Find the magnitude of:

A. $\vec{A} + \vec{B}$ B. $\vec{A} - \vec{B}$

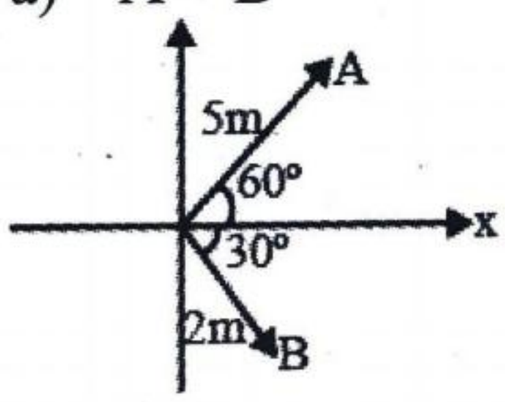
Resolving vectors

19. Compute the x and y components of each of the vectors A, B and C.

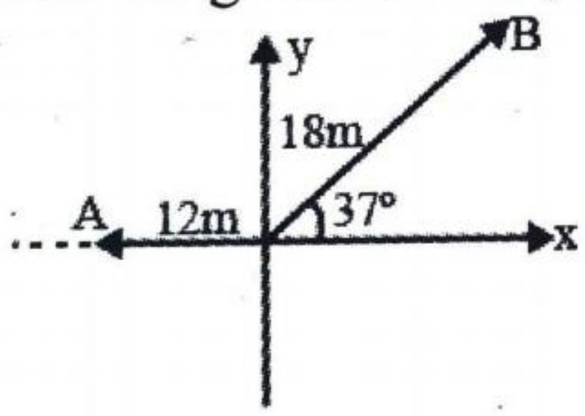


20. For the given vectors find, the component of:

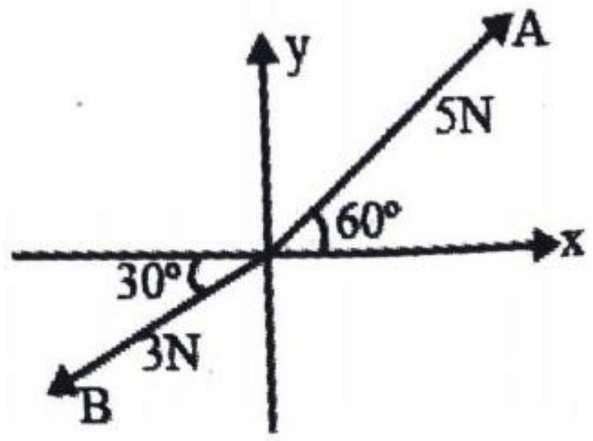
a) $A + B$ b) $A - B$



21. For the given vectors, find the magnitude of $\vec{A} + \vec{B}$.

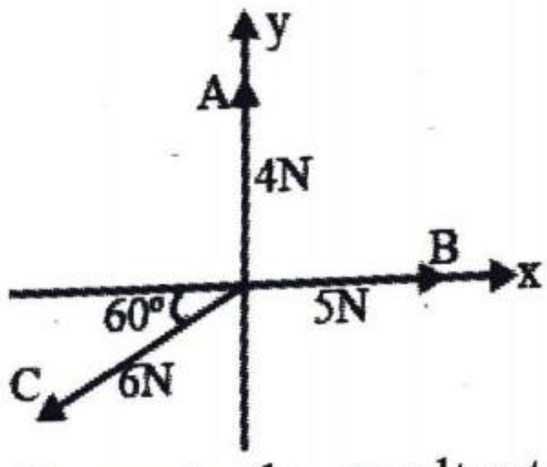


22. You are given vectors $\vec{A}$ and $\vec{B}$ as shown, find $|3\vec{A} - 2\vec{B}|$.

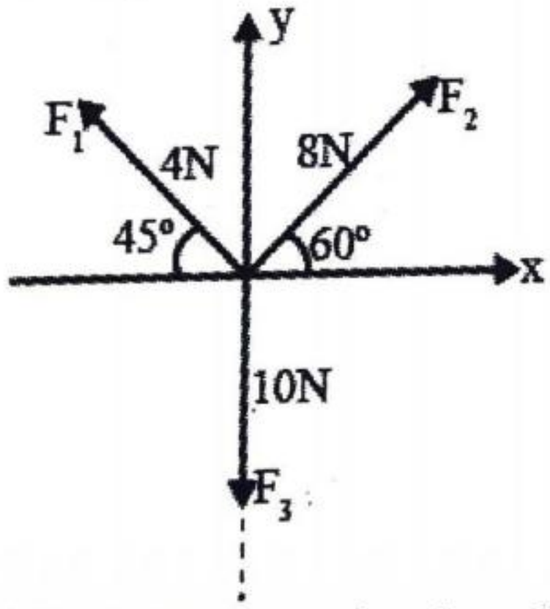


23. If you are given the x and y component of two vectors $\vec{P}$ and $\vec{Q}$ as; $P_x = 3$, $P_y = 6$ and $Q_x = 8$, $Q_y = 10$. Find $|\vec{P} + \vec{Q}|$

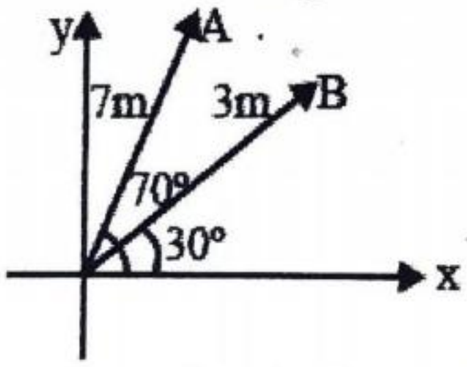
24. Find the resultant of the three force vectors acting on a particle as shown below.



25. Compute the resultant of the forces given below.



26. Find the magnitude of A – B for



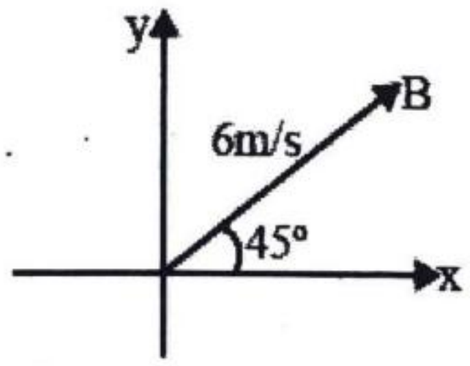
Unit One: Explanation to Questions

1. Yes. $\vec{A} + \vec{B} = 0$, when the vectors are equal and opposite.
2. Either they both are zero vectors or the vectors are equal and opposite.
3. No! It may be case for a resultant.
4. The magnitude of the vector also doubled, but its direction is unchanged.
5. No! If we are considering rectangular components.

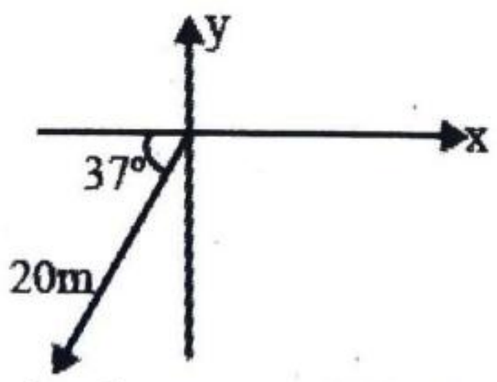
6. If $\vec{A}$ is zero vector, its direction will be arbitrary. And if it is nonzero, one possible direction is 45°.
7. The direction angle for $\vec{A}$ is between 180° and 270°.
8. The direction for $\vec{A}$ is between 270° and 360°.
9. The possible directions of $\vec{A}$ are between 90° and 180°, and between 270° and 360°.
10. Suppose we have two unequal vectors $\vec{A}$ and $\vec{B}$ then, $\vec{A}+\vec{B}$ can never be zero. But, three vectors of unequal magnitude can add to zero if they form a triangle.
11. The resultant of any two vectors is always within the range $|A - B| \leq R \leq |A + B|$ for our case, let A = 50N, B = 80N. So we have; $|50 - 80| \leq R \leq |50 + 80|$. Thus the maximum is 130N and the minimum is 30N.
12. Algebraic sum does not require direction. Scalar can be added using it. But vector sum is direction sensitive, vectors are combined using it.
13. No! Vector subtraction is not commutative. That is $A - B \neq B - A$.
14. No! it does not make sense to add for example, time with force. Only quantities of the same kind can be added or subtracted.

Unit One: Solution to problems

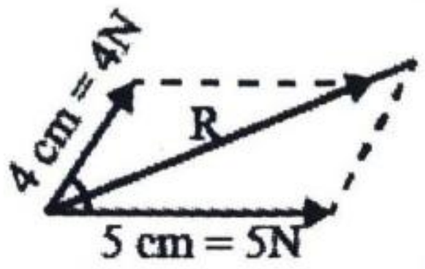
15. a) Assume 1 cm = 2 m/s, then 6 m/s ≡ 3 cm, so we have; 45° is measured using protractor.



b) Assume the scale 1 cm = 5m, then 20m ≡ 4 cm, next measure 37° from the negative x axis anti clockwise so you have.



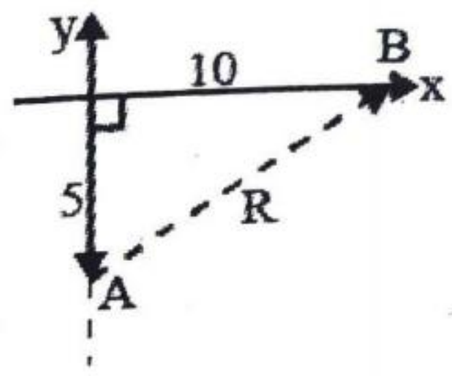
16. Assume the scale 1 cm = 1N, for all forces you are given and draw them tail to tail,



complete the parallelogram then the diagram will be either resultants where R = 8.7 cm ⇒ R = 8.7 N.

17. a) $A + B = -C = -L$ b) $A + B + C = 0$

c) $A + B - C = L$



18. $\vec{A} = 5$ units, $\vec{B} = 10$ units.

$$R = \sqrt{A^2 + B^2} = \sqrt{5+10} = 11.2 \text{ units}$$

19. A, $A_x = A\cos 60° = 5 \times 0.5 = 2.5m$

$A_y = A \sin 60° = 5 \times 0.87 = 4.3m$

B, $B_x = 3m$, $B_y = 0$

C, $C_x = -C\cos 60° = -2 \times 0.5 = -1 = 1m$ to nègative x axis

$C_y = -C \sin 60° = -2 \times 0.87 = -1.74 = 1.74$ to –ve y axis

20.

Vector	**x component**	**y component**
A = 5m	2.5m	4.3m
B = 2m	1.74m	-1m

a) $A + B = R_x + R_y \Rightarrow R_x = (2.5 + 1.74) = 4.2m$, $R_y = 3.3m$

b) $A - B \Rightarrow R_x = (2.5 - 1.74) = 0.76m$, $R_y = (4.3 - (-1)) = 5.3m$

21.

Vector	**x component**	**y component**
A = 12	$A_x = -12m$	$A_y = 0$
B = 18	$B_x = 14.4m$	$B_y = 10.8m$

$R = A + B = R_x + R_y \Rightarrow R_x = (-12 + 14.4) = R_x = 2.4m$ and

$R_y = (0 + 10.8) = 10.8m$, then $R = \sqrt{R_x^2 + R_y^2}$

$$R = \sqrt{(2.4)^2 + (10.8)^2} = 11.1m$$

22.

Vector	**Vector**	**x component**	**y component**
A = 5N	3A = 15N	$3A_x = 7.5N$	$3A_y = 13N$
B = 3N	2B = 6N	$2B_x = -5.2N$	$2B_y = -3N$

$R = 3A - 2B \Rightarrow R_x = (3A_x - 2B_x) = 7.5 - (-5.2) = 12.7N$, and

$$|3A - 2B| = R = \sqrt{R_x^2 + R_y^2} = \sqrt{(12.7)^2 + (16)^2} = 20.4N$$

23. $R_x = P_x + Q_x = 3 + 8 = 11$ units, $Ry = P_y + Q_y = 6 + 10 = 16$ units

Then $R = |P + Q| = \sqrt{R_x^2 + R_y^2} = \sqrt{(11)^2 + (16)^2} = 19.4$ units

24.

Vector	x component	y component
A = 4N	$A_x = 0$	$A_y = 4N$
B = 5N	$B_x = 5N$	$B_y = 0$
C = 6N	$C_x = -3N$	$C_y = -5.2N$

$R = A + B \Rightarrow R_x = (0 + 5 - 3) = 2N,\ R_y = (4 + 0 - 5.2) = -1.2N$

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(2)^2 + (-1.2)^2} = 2.33N,\ \tan\theta = \frac{R_y}{R_x} \Rightarrow \frac{-1.2}{2} = 329°$$

25.

Vectors	x component	y component
$F_1 = 4N$	$F_1x = -2.8N$	$F_1y = 2.8N$
$F_2 = 8N$	$F_2x = 4N$	$F_2y = 6.9N$
$F_3 = 10N$	$F_3x = 0$	$F_3y = -10N$

$F_{Rx} = (-2.8 + 4 + 0) = 1.2N$ and $F_{Ry} = (2.8 + 6.9 + 0) = 9.7N$

$$F_R = \sqrt{(F_{Rx}) + (F_{Ry})} = \sqrt{(1.2)^2 + (9.7)^2} = 9.8N,\ \tan\theta = \frac{F_{Ry}}{F_{Rx}} \Rightarrow \theta = 83°$$

26.

Vectors	x component	y component
A = 7,	$A_x = 2.4m$	$A_y = 6.6m$
B = 3m	$B_x = 2.6m$	$B_y = 1.5m$

$A - B \Rightarrow R_x = (2.4 - 2.6) = -0.2;\ R_y = (6.6 - 1.5) = 5.1$

$$\Rightarrow R = \sqrt{(0.2)^2 + (5.1)^2} = 5.1m$$

Unit Two

Motion in a Straight Line

Everything in the universe is in a state of motion. The earth itself spins on its axis at over 450 m/s and it is traveling around the sun with a speed of 30000 m/s. What is more, the earth is also filled with moving things. A dog chasing a cat, a horse racing in the field, a train moving along the railway, a car moving along the road, an airplane moving in the air, a bird flying in the air, a soccer ball kicked to the goal, are all in motion. While we are walking to school, to work or to shop, we are in motion. In one or the other way, motion affects our daily lives.

2.1 Uniform Motion

Assume that a boy takes one second to walk one meter on this way from home to school. Thus, if he continues walking at this rate, we can say, his speed is 1m/s. This means that he walked 1m in the first second, another 1m in the second, and so on. Thus, say for example after 50 seconds, he will be 50m away from home. His distance increases by 1m every 1 second. Since the body's speed is constant, such a motion is said to be *uniform*.

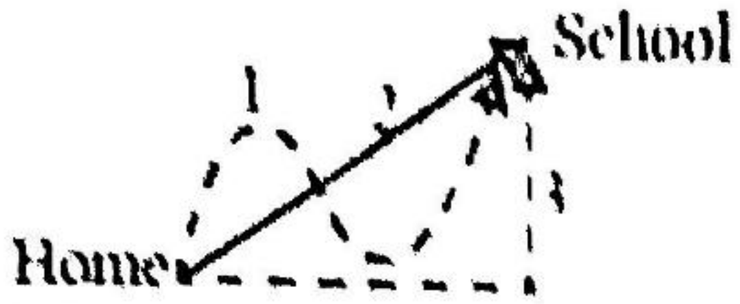
> **Uniform motion:** Is the motion of a body with constant speed in a straight line.

When a body moves in uniform motion, it covers equal distances in equal intervals of time.

In describing motion, we often use terms such as *distance*, *displacement*, *speed*, *velocity*, *acceleration* and so on.

Distance and displacement

In our daily life, distance and displacement are used interchangeably as if they mean the same thing. Yet they are different quantities having different definitions and meanings. Imagine there are three alternative paths that lead you from your home to school as shown below.



Then, the shortest path that will lead you to school is path 2, and we call it *displacement*. However, path 1 and path 3 are not short cuts and hence, they cannot be your displacement rather, they can be taken as the distance covered by you while you go from home to school.

Distance (s): Is the actual path length travelled by a moving body.

Distance is a scalar quantity and its SI unit is meter (m).

Displacement ($\vec{s}$): Is the shortest path length covered by a body between two points.

Displacement is a vector quantity and its SI unit is meter (m).

☞ **Important points to remember**

Motion: Is the continuous change in position of a moving body, with respect to some frame of reference.

Frame of reference: Is a certain point or place about which motion of a body can be explained. For instance, the earth can be taken as a frame of reference.

Uniform motion: Is the motion of a body with steady speed in a straight line.

Displacement: Is the shortest path length covered by a body between its starting and end points. It tells us how far a body is located from its starting point.

Displacement = distance + direction

Conceptual Examples

1. If the moving body is speeding up, slowing down or changing direction, can we say the body is in uniform motion?

   Explanation: No, we cannot say, if the body is speeding up, slowing down or changing direction, we say the body is under non uniform or accelerated motion.

2. Can the distance covered by a body be ever be less than its displacement?

   Explanation: Never! Distance can never be less than the displacement of the moving body.

3. Under what condition will the distance covered by a body be equal to its displacement?

Explanation: Distance covered by a body can be equal to the displacement of the body when the body is moving under uniform motion in a straight line.

4. Can the distance covered by a body ever be negative? What about its displacement.

Explanation: Distance can never be negative since it represents the actual path length of a moving body. However, displacement can be negative while the body is moving in the negative direction.

5. Can the displacement of a body be zero and at the same time its distance be nonzero?

Explanation: Yes! This happens when the moving body returns to its original position or starting point.

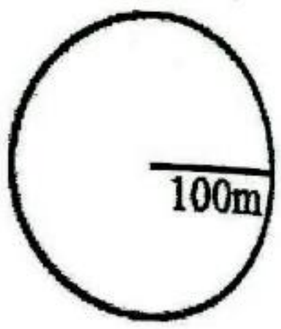
6. For the two conditions shown below find:

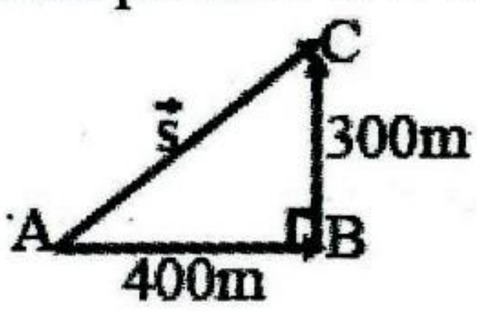
a) The displacement

b) The distance covered by an athlete

i) While running around a circular track of radius 100m once.

ii) While walking from point A to B and then to C.





Explanation: i) a) $\vec{s}$ = zero

b) $s = 2\pi r = 2 \times \pi \times 100 = 200\pi m$

ii) a) $\vec{s} = \sqrt{(300)^2 + (400)^2} = 500m$

b) $s = 300 + 400 = 700m$

7. If a body is thrown vertically upward to height h and returns to its original position. What will be its?

a) Distance b) Displacement

Explanation: a) Distance = h + h = 2h

b) Displacement = h – h = 0

8. An Athlet is running around a circular track of radius 100m. After he completes 3 revolutions, what will be

a) His displacement b) his distance

Explanation: a) $\vec{s} = 0$

b) $s = 3(2\pi r) = 3(2\pi \times 100) = 600\pi m$

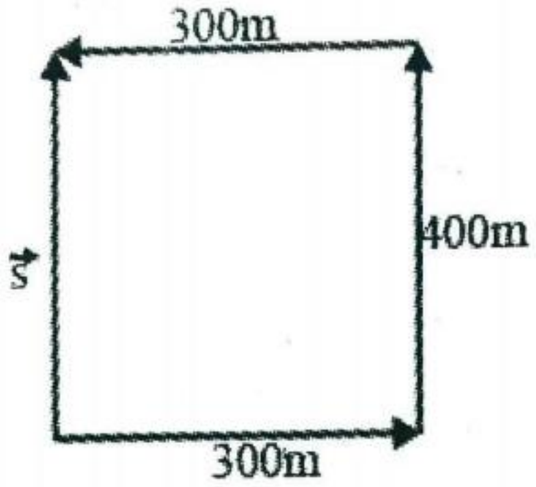
Illustrative Examples

9. A student walks 300m East, 400m north and then 300m West. Calculate
   a) The distance travelled, and
   b) The displacement of the student from his starting point.

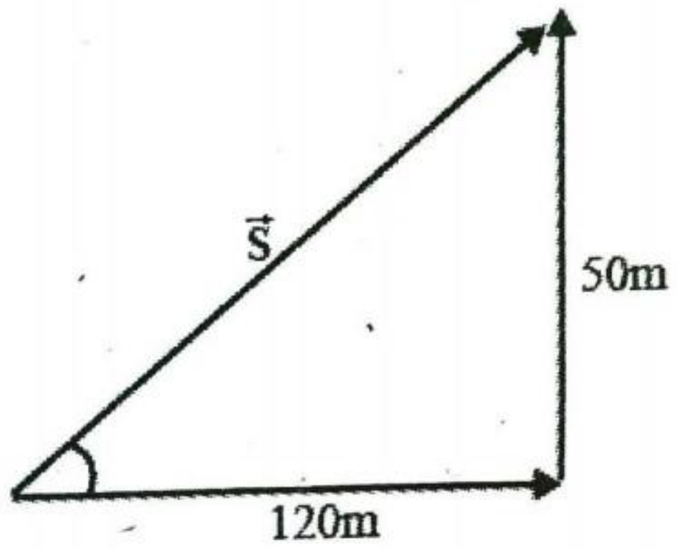
Solution: a) Distance $\Rightarrow$ s = 300m + 400m + 300m = 1000

b) Displacement $\Rightarrow \vec{s}$ = 400m North

That is



10. A boy walks 120m east and then 50m north. Calculate his
    a) Distance travelled
    b) Displacement



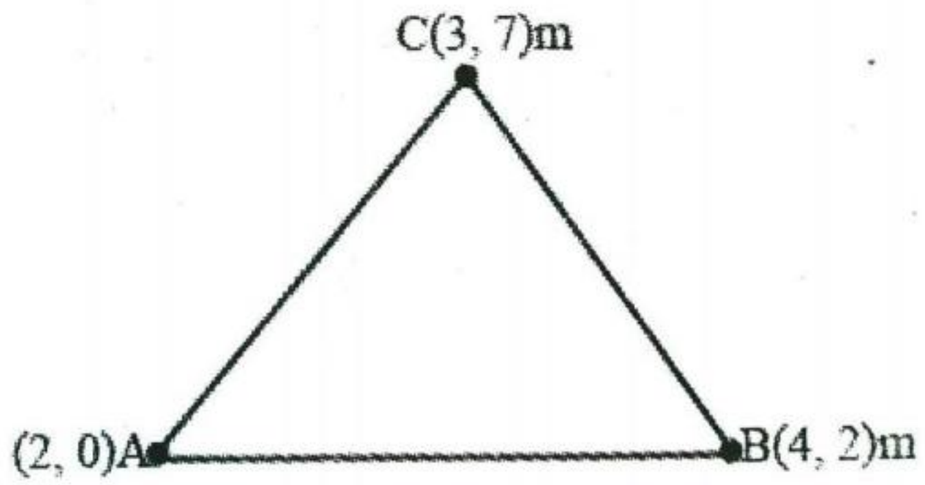
Solution:

a) Distance $\Rightarrow$s = 120m + 50m = 170 m

b) Displacement $\Rightarrow \vec{s} = \sqrt{(20)^2 + (50)^2} = 130\text{m}$ and, its direction is;

$$\tan\theta = \frac{50}{120} \Rightarrow \tan\theta = \frac{5}{12} \Rightarrow \theta = \tan\left(\frac{5}{12}\right) = 23^\circ$$

11. A trained dog runs starting from A once along a triangle whose vertices given below what is
    a) The distance travelled by the dog.
    b) Its displacement.



Solution:

First we need to find the side lengths of the triangle.

$$AB = \sqrt{(4-2)^2 + (2-0)^2} = 2.83\,\text{m}$$

$$BC = \sqrt{(3-4)^2 + (7-2)^2} = 5.1\,m$$

$$AC = \sqrt{(3-2)^2 + (7-0)^2} = 7.1\,m$$

a) $S = AB + BC + AC = 2.83 + 5.1 + 7.1 = 15m$

b) $\Delta s = 0$, since it has returned to its initial position.

Speed and Velocity

Even if speed and velocity describe how fast a body is moving, they are different quantities. If you say a car is moving at 50 km/h, you are telling us its speed. But if you say the car is moving at 50 km/h towards East, now you are telling us the velocity of the car.

Speed: Is distance travelled by a moving body divided by the time taken.

That is, $\text{speed} = \dfrac{\text{distance travelled}}{\text{time taken}}$

$$\boxed{v = \frac{s}{t}}$$

Speed is a scalar quantity and its SI unit is m/s.

Velocity: Is displacement of the moving body divided by the time taken.

That is, $\text{velocity} = \dfrac{\text{displacement}}{\text{time taken}}$

$$\boxed{\vec{v} = \frac{\Delta\vec{s}}{\Delta\vec{t}}}$$

Where $\Delta\vec{s} = \vec{s}_2 - \vec{s}_1$

The Greek letter Δ(delta) tell us change.

Velocity is a vector quantity and its SI unit is m/s. The magnitude of velocity is speed. And thus, we can write.

Velocity = speed + direction

Average speed and velocity

In real life situations, the motion of a body may not be uniform for many reasons. For example, if you critically examined your motion from home to school or vice versa, then you may find yourself moving faster at a time, slower at another time or you may even stop moving at some

other time. Thus, your speed or velocity is not uniform as you are moving. In such cases, we consider the average speed or velocity.

Average speed: Is the total distance travelled by a moving body divided by the total time taken.

That is, Average speed $= \dfrac{\text{total distance travelled}}{\text{total time taken}}$

$$\boxed{\vec{V}_{av} = \frac{\Delta\vec{s}}{\Delta\vec{t}}}$$

Where: $\Delta\vec{s} = \vec{s}_2 - \vec{s}_1$

$\Delta t = t_2 - t_1$

Average velocity tells us not only how fast a body is moving on average but also in which direction it is moving. However, one thing is certain that. Average speed or average velocity of a body gives us a general information on how fast a body moves with in a given time interval or during the whole journey. But the fact is that at different moments, the body might have been moving faster or moving slower or it might have been changed direction or stopped moving at different times in its journey history. Thus the average values do not give us the actual speed or velocity of the moving body at a particular instant of time or at a specific position of its motion. For such cases we use the concept of *instantaneous speed* or *velocity* to describe the nature of motion of the body at an instant of time or specific position.

Instantaneous speed: Is the speed of a body at a given instant of time or particular position.
Instantaneous velocity: Is the velocity of a body at a given instant of time or at a specific point.

☞ **Important points to remember**

Speed: Is a scalar quantity and has magnitude only. However, velocity is a vector quantity and has magnitude as well as direction.
Speed tells us how fast a body is moving but, **velocity** tells us how fast and in which direction the body is moving.

Conceptual Examples

12. Can a body have average speed without having average velocity? Explain

Explanation: Yes it can have! Think of a ball thrown upwards and returning to the point of throw.

13. Can the speed of a moving body be ever be negative? What about its velocity?
Explanation: Speed of a moving body can never be negative. But velocity can be negative when the body is moving in the negative axis.

14. Under what circumstance will, the average speed of a body be equal to its instantaneous speed and the average velocity be equal to its instantaneous velocity.
Explanation: When the body is in uniform motion.

15. In uniform motion in a straight line, we say speed is constant and velocity is constant. Does this mean that speed and velocity are equal in such cases?
Explanation: When speed is constant, it means, the body covers equal distances in equal time intervals. But when velocity is constant, it means, its magnitude is constant and the body is moving in a fixed direction. Hence in uniform motion in a straight line speed and velocity do have the same magnitude.

16. Can a decreasing speed be considered as a negative speed?
Explanation: No! a decreasing speed is still positive. Speed can never be negative.

17. Can a body have a zero displacement and a nonzero average velocity? A nonzero velocity?
Explanation: If the displacement of a body is zero, then its average velocity is also zero. But, the body can have a nonzero instantaneous velocity.

18. After a tennis match, the players dash to the net to congratulate one another. If they both run with a speed of 3 m/s, are their velocities equal?
Explanation: No! since the players are moving in opposite directions, their velocities are different.

19. Is it possible for an athlete to run around a circular track with constant velocity?
Explanation: No! when a body moves around a circular track, its direction continuously changes and hence, the body's velocity can never be constant.

20. If the position of an object is zero, does its speed have to be zero? Explain

Explanation: No! speed does not depend on position of a body, instead it depends on the actual distance travelled by the body.

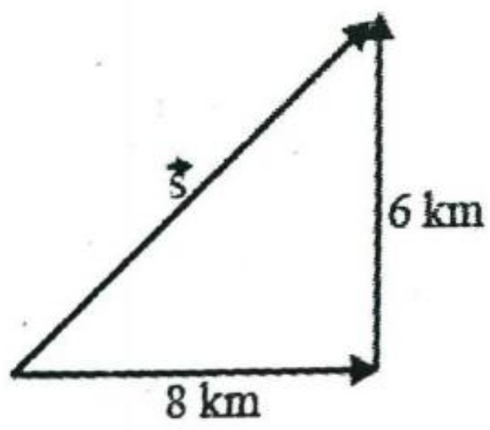
21. If the average velocity of a body is positive, is there any chance for the body to have a negative instantaneous velocity.

Explanation: Yes, even if the average velocity of a body is positive, it is possible for the body to have zero or negative instantaneous velocity during its trip.

22. A car moves 8 km due East and then 6 km due North, if it takes 2 hours to cover the trip. Find,

a) The speed

b) The velocity of the car



Solution:

a) $v = \frac{s}{t} = \frac{8+6}{2} = 7 \text{ km/hr}$

b) $\vec{v} = \frac{\vec{s}}{\vec{t}} = \frac{\sqrt{(8)^2 + (6)^2}}{2} = 5 \text{ km/hr}$

23. A car travels a distance of 45m in 3 seconds. What is the speed of the car in.

a) m/s b) km/h

Solution:

a) $v = \frac{s}{t} = \frac{45m}{3\,sec} = 15 \text{ m/s}$

b) $v = 15 \text{ m/s}$, but, $1m = \frac{1}{1000} \text{ km}$ and $1 \text{ sec} = \frac{1}{3600} \text{ hr}$

$$\Rightarrow v = 15 \times \frac{\frac{1}{1000} \text{km}}{\frac{1}{3600} \text{hr}}$$

$$v = 15 \times \frac{3600 \text{km}}{1000 \text{hr}} = 15 \times 3.6 \text{ km/hr}$$

$v = 54 \text{ km/hr}$

24. A boy runs at 10 m/s for 4 sec, then walks at 8 m/s for 6 sec and rides a bicycle at 20 m/s for 12 sec. What is the average speed of the boy for the entire trip?

Solution:

$$V_{av} = \frac{s_T}{t_T} = \frac{s_1 + s_2 + s_3}{t_1 + t_2 + t_3} = \frac{v_1 t_1 + v_2 t_2 + v_3 t_3}{t_1 + t_2 + t_3}$$

$$V_{av} = \frac{(10 \times 4) + (8 \times 6) + (20 \times 12)}{4 + 6 + 12} = 14.9 \text{m/s}$$

25. A car is moving towards east in such a way that its position is 20m at time t = 2 sec and 80m at time t = 6 sec. What is the average velocity of the car during this time interval?

Solution: $\vec{V}_{av} = \frac{\Delta \vec{s}}{\Delta \vec{t}} = \frac{\vec{s}_2 - \vec{s}_1}{t_2 - t_1} = \frac{80 - 20}{6 - 2} = 15 \text{m/s}$

26. One day you goes to school and returned to home along the same path after 60 minutes. If the distance between the school and your home is 1200m. What is your?

a) Average speed

b) Average velocity

Solution: a) $V_{av} = \frac{s_T}{t_T} = \frac{s_1 + s_2}{t_T} = \frac{1200 + 1200}{60 \times 60} = 0.67 \text{m/s}$

b) $\vec{V}_{av} = \frac{\Delta s}{\Delta t} = \frac{s - s}{\Delta t} = 0 \text{m/s}$

27. The velocity of a car moving along a straight line increase from 5 m/s to 20 m/s in 4 second. What is the displacement of the car during this time interval?

Solution: $V_{av} = \frac{\Delta s}{\Delta t} \Rightarrow \Delta s = (V_{av})t = \left(\frac{v + u}{2}\right)(t)$

$$\Delta s = \left(\frac{5 + 20}{2}\right)(4) = 50 \text{m}$$

28. A boy walked one third of his journey at a speed of 2 m/s and ran the rest of the journey at 3 m/s. What was the average speed of the boy?

Solution: $V_{av} = \frac{s_T}{t_T} = \frac{s_1 + s_2}{t_1 + t_2} = \frac{\frac{1}{3}s + \frac{2}{3}s}{\frac{\frac{1}{3}s}{v_1} + \frac{\frac{2}{3}s}{v_2}}$

$$t_1 = \frac{s_1}{V_1} = \frac{\frac{s}{3}}{v_1} = \frac{s}{3v_1}$$

$$t_2 = \frac{s_2}{V_2} = \frac{\frac{2}{3}s}{v_2} = \frac{2s}{3v_2}$$

$$V_{av} = \frac{s}{\frac{s}{3v_1} + \frac{2s}{3v_2}} = \frac{3v_1v_2}{2v_1 + v_2} = \frac{3 \times 2 \times 3}{2 \times 2 + 3}$$

$V_{av} = 2.57$ m/s

29. During a hard sneeze, your eyes might shut for 0.5 sec. If you are driving a car at 90 km/h, how far does it move during that time?

Solution: $s = vt = \frac{90}{3.6} \times 0.5 = 2.5m$

30. A car travels up a hill at a constant speed of 40km/h and returns down the hill at a constant speed of 60 km/h. Calculate the average speed for the round trip.

Solution: $V_{av} = \frac{s_T}{t_T} = \frac{s_2 + s_2}{t_1 + t_2} = \frac{s + s}{\frac{s}{v_1} + \frac{s}{v_2}} = \frac{2v_1v_2}{v_1 + v_2}$

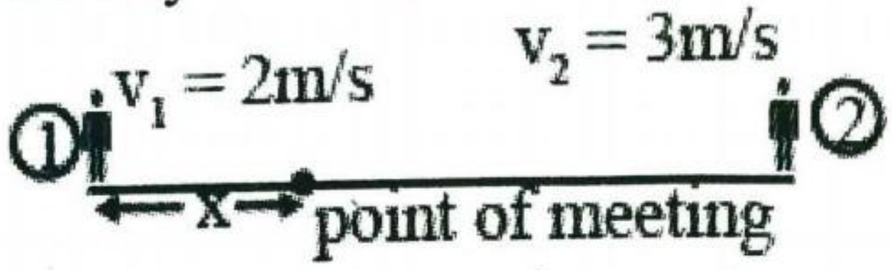
$$V_{av} = \frac{2 \times 40 \times 60}{40 + 60} = 48m/s$$

31. It was a dark and stormy night, when suddenly you saw a flash of lightning. Three and a half seconds later, you heard the thunder. If the speed of sound in air is about 340 m/s, how far away was the lighting produced?

Solution: $s = vt = 340 \times 3.5 = 1190m$

32. Two girls start running straight toward each other from two points 200m apart. One runs with a speed of 2 m/s and the other at 3 m/s. how close are they to the slower one's starting point when they meet?

Solution: The time they run to each other is same.



Since $t_1 = t_2$, let x be distance moved by girl one (slower girl).

$$\frac{x}{v_1} = \frac{200 - x}{v_2}$$

$$\Rightarrow \frac{x}{2} = \frac{200 - x}{3} \Rightarrow 3x = 400 - 2x \Rightarrow 5x = 400$$

$x = 80m$

33. A moving object increases its velocity uniformly from 20 m/s to 40 m/s in 2 min. What is;

a) Average velocity

b) Displacement of the object during the 2 min.

Solution: a) $\vec{v}_{av} = \frac{v + u}{2} = \frac{40 + 20}{2} = 30m/s$

b) $\vec{s} = (v_{av})(t) = 30 \times (2 \times 60) = 3600m$

34. A car travels along a u shaped curve for a distance of 400m in 20 sec. Its final location however, is only 40m from the starting point. What;

a) The average speed

b) The magnitude of the average velocity

Solution: a) $v_{av} = \frac{s_T}{t_T} = \frac{400}{20} = 20m/s$

b) $\vec{v}_{av} = \frac{\Delta s}{\Delta t} = \frac{40}{20} = 2m/s$

35. A marble rolls up an inclined plane a distance of 5m and then stops and returns to a point 5m below its starting point. If the entire trip took only 2 sec, what was

a) The average speed, and

b) The average velocity of the marble

Solution: a) $v_{av} = \frac{s_T}{t_T} = \frac{5+5+5}{2} = 7.5\text{m/s}$

b) $\vec{v}_{av} = \frac{\Delta s}{\Delta t} = \frac{-5}{2} = 2.5\text{m/s}$

36. A man walks for 1 min at a speed of 1 m/s and then runs for 1 min at 3 m/s along a straight track. What is the average speed of the man?

Solution: $v_{av} = \frac{s_T}{t_T} = \frac{s_1 + s_2}{t_1 + t_2} = \frac{v_1 t_1 + v_2 t_2}{t_1 + t_2}$

$$= \frac{1\times 60 + 3\times 60}{60 + 60} = 2\text{m/s}$$

37. A train travels from one station to another at a speed of 40 m/s and returns to the first station at a speed of 60 m/s. Calculate the
   a) Average speed
   b) Average velocity of the train

Solution: a) $v_{av} = \frac{s_T}{t_T} = \frac{s_1 + s_2}{t_1 + t_2}$

In this case, $s_1 = s_2 = s$ and $t_1 = \frac{s_1}{v_1} = \frac{s}{v_1}$ and $t_2 = \frac{s_2}{v_2} = \frac{s}{v_2}$

$$\Rightarrow v_{av} = \frac{s + s}{\frac{s}{v_1} + \frac{s}{v_2}} = \frac{2v_1 v_2}{v_1 + v_2} = \frac{2\times 40\times 60}{40 + 60} = 48\text{m/s}$$

b) $\vec{v}_{av} = \frac{\Delta s}{\Delta t}$, but $\Delta s = 0$

$$\Rightarrow \vec{v}_{av} = 0$$

38. A toy car travel from point A to point B at a constant speed of 20 km/h and without stopping at point B it returns to point A at a constant speed v. If the average speed of the car is 24 km/h, calculate the value of v.

Solution: $V_{av} = \frac{s_T}{t_T} = \frac{s+s}{t_1+t_2} = \frac{2s}{\frac{s}{v_1}+\frac{s}{v}}$

$$\Rightarrow V_{av} = \frac{2v_1 v}{v+v_1} \Rightarrow 24 = \frac{2\times 20\times v}{20+v}$$

$20 \times 24 + 24v = 2 \times 24 \times v$

$\Rightarrow v = 20$ km/h

39. A body moves in a straight line along the x axis. Its distance from the origin is given by $x = 8t - 3t^2$. Calculate the average speed of the body during the time interval $t = 0$ sec to $t = 1$ sec.

Solution: $V_{av} = \frac{s_T}{t_T} = \frac{s_1+s_2}{t_1+t_2}$

Where, $s_1 = 8(0) - 3(0)^2 = 0$, when $t = 0$

$s_2 = 8(1) - 3(1)^2 = 5$, when $t = 1$

$$\Rightarrow V_{av} = \frac{0+5}{0+1} = 5\text{m/s}$$

40. A man walked for 4 hr and bicycled for 3 hr covering a total distance of 45 km. Later he walked for 2 hr and bicycle for 4 hr covering a total distance of 50 km. What was his average speed of walking?

Solution: In the first phase; $4v_w + 3v_b = 45$ . . . *

and, in the second phase, $2v_w + 4v_b = 50$ . . . **

solving the two equations at a time,

$v_w = 3$ km/hr

41. Assume Kennenisa can run 4.5 km in 4.5min and MohFarah takes 6 min to finish this distance. If they compete for this distance, how far apart will they be at the finish of the race?

Solution:

In 4.5 min Kennenisa runs 4.5 km and, the speed of Mohfarah is $\frac{4.5}{6}$ km/min thus, at 4.5 min, he can runs a distance of $\left(\frac{4.5}{6}\right)(4.5) = 3.75$ km. Which means, when Kinennisa runs 4.5 km, Mohfarah can run 3.375 km. Thus, the difference in their

distance is $\Delta s = s_k - s_m = 4.5 - 3.375 = 1.125$km. That is, at the finish of the race, Kenneisa will be 1.125 km ahead of Mohfarah.

42. Haile and Portergate can run at 5 m/s and 4 m/s respectively. If Haile starts 10 min after Portergate started running, how far will they be from the starting point when Haile overtakes Portergate.

Solution: It is clear that, when they pass each other, they have covered equal distances.

$s_H = s_P$, but $t_P = t_H + 600$ sec

$v_H t_H = v_P t_P$

$5t_H = 4(t_H + 600) \Rightarrow t_H = 2400$ sec

Thus, $s_H = 5 \times 2400 = 12000$ m $= 12$km

43. Abdu travelled a distance of 265 m. He drove at 40 km/hr during the first part of the trip and at 35 km/hr during the remaining. If the whole trip takes him 7h. How long did he travel at 40 km/h?

Solution: $s = s_1 + s_2 = v_1 t_1 + v_2 t_2$, $t_2 = 7 - t_1$

$265 = 40t_1 + 35(7 - t_1) \Rightarrow t_1 = 4$hr

44. The velocity of an electric train between two points is $v_1 = 80$ km/h. The average velocity over the entire path is $v_2 = 60$ km/h. If one hour was spent when the train stops between stations, calculate the distance between the two points.

Solution: $v_1 = \frac{s}{t_1}, v_2 = \frac{s}{t_2} = \frac{s}{(t_1 + 1)}, t_2 = t_1 + 1$

$$\frac{v_1}{v_2} = \frac{t_1 + 1}{t_1} \Rightarrow \frac{80}{60} = \frac{t_1 + 1}{t_1} \Rightarrow t_1 = 3\text{hr}$$

Thus, $s = v_1 t_1 = 80 \times 3 = 240$ km

2.2 Uniform accelerated Motion

When a moving body changes its velocity with time, that is, if the body gets faster, gets slower or changes direction, we say it is accelerating.

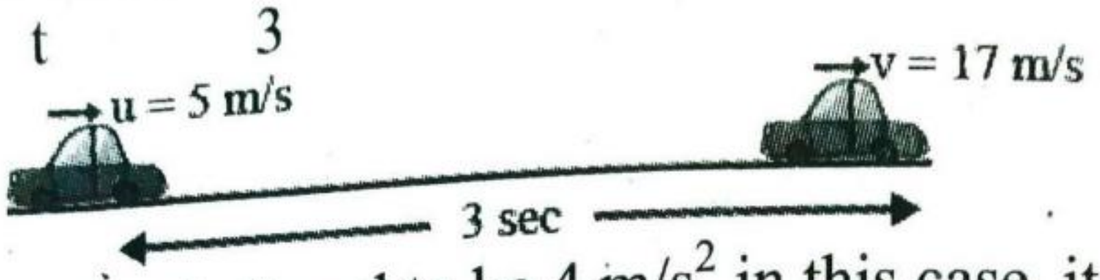
Acceleration: Is the rate of change of velocity

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t} = \frac{v - u}{\Delta t}.$$

Acceleration is a vector quantity and its SI unit is m/s^2.

For a body moving in a straight line with a velocity that is changing at a constant rate, we call the body to be under uniform acceleration or constant acceleration. As an example consider a car moving with a constant acceleration. If the car takes 3 seconds to change its velocity

from 5 m/s to 17 m/s, then its acceleration is obtained by: $\vec{a} = \frac{\Delta\vec{v}}{\Delta\vec{t}} = \frac{v-u}{t} = \frac{17-5}{3} = 4 m/s^2$.



Since the acceleration is found to be 4 m/s^2 in this case, it means that the velocity of the car is changing by 4 m/s every second. Which means that after one second, the velocity of the car will be 9 m/s after two seconds, it will be moving at 13 m/s and after three seconds its velocity become 17 m/s.

☞Important points to remember

- A body is said to be accelerating whenever its velocity changes in magnitude, that is, when it gets faster or slower or when it changes its direction.
- The direction of the acceleration is always in the direction of the change in velocity. That is, the acceleration and the velocity point in the same direction when the body *speeds up* and they point in opposite direction when the body *slows down*.
- When the body slows down, it is said to be *decelerating*.
- A body is said to be moving with uniform acceleration if its velocity changes by equal amount in equal intervals of time. In other words uniform acceleration means the same acceleration at every instant of time.

Conceptual Examples

45. Can a body have a nonzero velocity and at the same time a zero acceleration.

Explanation: Yes! This happens when the velocity is constant.

46. Is it possible to have a zero velocity and a nonzero acceleration? Explain

Explanation: Yes! Think of a ball thrown up ward, at the maximum height, the velocity of the ball is zero but the ball is still accelerating at gravitational acceleration.

47. Under what circumstances will the velocity and acceleration of a body have opposite signs?

Explanation: When the body slows down. That is, when the velocity of the body decreases with time.

48. When a particle's speed decreases, it is often said to be decelerating. But, can you conclude that declaration means a negative acceleration? Explain.
Explanation: Not necessarily! Deceleration can be caused by a positive or a negative acceleration. All that is required is that, the velocity and acceleration have opposite signs, that is, they point in opposite directions.

49. Is it possible for a body to have an acceleration without its speed is changing?
Explanation: Yes! Think of a body moving around a circular track at constant speed. Even if its speed is constant, its direction is changing continuously, hence it is accelerating.

50. If a car is traveling eastward, can its acceleration be westward? Explain
Explanation: Yes! This happens when the car slows down.

51. What does it means, when we say the acceleration of a body is
a) $2\ m/s^2$ b) $-2\ m/s^2$
Explanation:
a) $2\ m/s^2$ means the velocity of the moving body increases by 2 m/s every second.
b) $-2m/s^2$ means the velocity of the moving body decrease by 2 m/s every second.

52. Can the acceleration of a body be;
a) Positive c) zero
b) Negative
Explanation: Yes! a body can have a positive or a negative or even a zero acceleration.

Illustrative Examples

53. An air craft decelerates at $0.5\ m/s^2$. After 8 minutes, its velocity has dropped to 160 m/s. Find its initial velocity.

Solution: From, $a = \frac{v-u}{t} \Rightarrow v = u + at$, but $a = -0.5$ m/s

$\Rightarrow u = v + at = 160 + 0.5(8 \times 60) = 400$ m/s

54. A car accelerates uniformly from an initial velocity of 2 m/s to a final velocity of 10 m/s in 8 seconds. It then slows down uniformly to a velocity of 4 m/s in 6 seconds. Calculate the acceleration of the car
a) During the first 8 seconds.

b) During the last 6 seconds

Solution: a) $a = \frac{v-u}{t} = \frac{10-2}{8} = 1 m/s^2$

b) $a = \frac{v-u}{t} = \frac{4-10}{6} = -1 m/s^2$

55. A taxi moving towards east at 30 m/s comes up behind a city bus and is unable to pass. If the taxi decreases its speed to 10 m/s in 5 seconds, what is the acceleration of the taxi during this time?

Solution: $a = \frac{v-u}{t} = \frac{10-30}{5} = \frac{-20}{5} = -4 m/s^2$

It means that the taxi decreases its speed by 4 m/s every second.

56. If you are running at 6 m/s and decelerate at 2 m/s^2, how long will it take you to stop?

Solution: $a = \frac{v-u}{t} \Rightarrow t = \frac{v-u}{a}$, but v = 0 and a = -2 m/s^2

$\Rightarrow t = \frac{v-u}{a} = 3 sec$

2.3 Equations of uniformly accelerated motion

For a body moving with constant acceleration, we use the following simple but basic and powerful equations.

Acceleration, $a = \frac{\Delta v}{\Delta t} = \frac{v-u}{(t_i - t_f)}$, usually $t_i = 0$.

$\Rightarrow \boxed{v = u + at} \ldots (1)$

Average velocity, $v_{av} = \frac{\Delta s}{\Delta t} \Rightarrow \Delta s = (v_{av})(\Delta t)$

$\Rightarrow \boxed{s = \left(\frac{v+u}{2}\right)t} \ldots (2)$

Now, substituting v by $u + at$ in to equation (2), we have

$$s = \left(\frac{u+at+u}{2}\right)t = \frac{2ut+at^2}{2}$$

$$\Rightarrow \boxed{s = ut + \frac{1}{2}at^2} \ldots (3)$$

And, substituting, t by, $\frac{v-u}{a}$ into equation (2) we get

$s = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right)$ multiplying each term of the numerator and the denominator separately, we find $S = \frac{v^2 - uv + uv - u^2}{2a}$.

$$\Rightarrow \boxed{v^2 = u^2 + 2as} \ldots (4)$$

Again substituting u by $v - at$ into equation (2) we obtain.

$$s = \frac{(v + v - at)}{2}t = \frac{2vt - at^2}{2}$$

$$\Rightarrow \boxed{s = vt - \frac{1}{2}at^2} \ldots (5)$$

☞ To find the distance moved by the object in the n^{th} second, we can proceed as follows.

$$S_n^{th} = S_n - S_{n-1}$$

$$= \left(ut_n + \frac{1}{2}at_n^2\right) - \left(u(t_n - 1) + \frac{1}{2}a(t_n - 1)^2\right)$$

$$= ut_n + \frac{1}{2}at_n^2 - \left[ut_n - u + \frac{1}{2}a(t_n^2 - 2t_n + 1)\right]$$

$$= ut_n + \frac{1}{2}at_n^2 - ut_n + u - \frac{1}{2}at_n^2 + \frac{2t_n a}{2} - \frac{1}{2}a$$

$$= u + \frac{2t_n a}{2} - \frac{1}{2}a$$

$$\Rightarrow \boxed{\begin{array}{l} S_n^{th} = u + a\left(t_n - \frac{1}{2}\right) \\ \quad \text{or} \\ S_n^{th} = u + \frac{a}{2}(2t_n - 1) \end{array}}$$

☞ **Remember:** A given problem related to the concept of uniformly accelerated motion can be solve in any one of the relations obtained above. But to use the equation in a quick and efficient manner, one has to read the given problem carefully and identify the given and required variables and then decide on the straight forward equation to reach the intended solution.

☠ **Warning**! The equations formulated above can only be used for a body that is uniformly accelerating. That is, we cannot use these equations for a body that is moving at different acceleration in the given time interval.

Illustrative Examples

57. A racing car is travelling at 10 m/s North. It accelerates uniformly and covers a distance of 725m in 10 sec. Calculate the acceleration of the car.

Solution: $s = ut + \frac{1}{2}at^2 \Rightarrow 725 = 10 \times 10 + \frac{1}{2}a \times (10)^2$

$\Rightarrow a = 12.5\ m/s^2$

The racing car is accelerating at $12.5\ /s^2$ North.

58. A motorcycle, starting from rest moves in a straight line with a constant acceleration and covers a distance of 64m in 4 sec. **Calculate**:

a) Its acceleration
b) Its final velocity
c) At what time the motorcycle covered half of the total distance.
d) What distance the motorcycle covered in half of the total time?

Solution:

a) $s = ut + \frac{1}{2}at^2 \Rightarrow 64 = 0 + \frac{1}{2}a(4)^2$

$\Rightarrow a = 8\ m/s^2$ east

b) $v = u + at = 0 + 8 \times 4 = 32$ m/s east

c) $\frac{s}{2} = ut + \frac{1}{2}at^2, \frac{s}{2} = \frac{64}{2} = 32$

$$32 = 0 + t + \frac{1}{2} \times 8 \times t^2$$

$t^2 = 8 \Rightarrow t = 2.83$ sec

d) $s = u\left(\frac{t}{2}\right) + \frac{1}{2}(a)\left(\frac{t}{2}\right)^2, \frac{t}{2} = \frac{4}{2} = 2$

$$s = 0 \times 2 + \frac{1}{2} \times 7 \times (2)^2$$

$s = 16$m East

59. A moving body increases its velocity uniformly from 20 m/s to 40 m/s in 2 min. How far did it travel in the 2 min?

Solution: $s = (v_{av})t = \left(\frac{v+u}{2}\right)(t), t = 2 \times 60 = 120$ sec

$$s = \left(\frac{20+40}{2}\right)(120) = 3600\text{m}$$

60. A train traveling initially at 16 m/s is under constant acceleration of 2 m/s^2.

a) How far will it travel in 20 sec?

b) What will be its final velocity?

Solution: a) $s = ut + \frac{1}{2}at^2 = 16 \times 20 + \frac{1}{2} \times 2 \times 20^2 = 720$m

b) $v = u + at = 16 + 2 \times 20 = 56$ m/s

61. If you are running at 6 m/s and decelerate at 2 m/s^2.

a) How long will it take you to stop?

b) How far will you go while decelerating?

Solution: a) $v = u + at$, but $v = 0$, $a = -2\ m/s^2$

$0 = 6 - 2t \Rightarrow t = 3$ sec

b) $v^2 = u^2 + 2as$, $a = -2\ m/s^2$, $v = 0$

$0 = (6)^2 - 2 \times 2 \times 5$

$\Rightarrow s = 9$m

62. A frog's tongue flicks out 10 cm in 0.1 sec to cath a fly. If it accelerates at a constant rate during this time. What is its

a) Average velocity

b) Acceleration

Solution:

a) $s = (v_{av})t \Rightarrow v_{av} = \frac{s}{t} = \frac{0.1}{0.1} = 1 m/s$

b) $a = \frac{v-u}{t}$, assuming that the frog's tongue starts from rest

we have, $v_{av} = \frac{v+u}{2} = \frac{v}{2} \Rightarrow v = 2v_{av} = 2\ m/s$

Thus, $a = \frac{v-u}{t} = \frac{2-0}{0.1}$

$a = 20\ m/s^2$

63. A truck gradually starts off from rest with a uniform acceleration of 2 m/s^2 and reaches a velocity of 16 m/s. Calculate the distance it travels while it is accelerating.

Solution: $v^2 = u^2 + 2as \Rightarrow s = \frac{v^2 - u^2}{2a}$

$s = \frac{(16)^2 + (0)^2}{2 \times 2} = \frac{256}{4} = 64m$

64. A truck covers 40m in 8.5 sec while slowing down to a final speed of 2.8 m/s.

a) Find the truck's original speed.

b) Find its acceleration.

Solution:

a) $s = \frac{(v+u)}{2}t \Rightarrow u = \frac{2s}{t} - v$

$u = \frac{2 \times 40}{2} - 2.8 = 37.2 m/s$

b) $a = \frac{v-u}{t} = \frac{2.8 - 37.2}{8.5} = -4 m/s^2$

65. A boat increases its speed uniformly from 20 m/s to 30 m/s in a distance of 200m. Find
    a) The magnitude of its acceleration.
    b) The time it takes to travel this distance.

Solution:

a) $v^2 = u^2 + 2as \Rightarrow a = \dfrac{v^2 - u^2}{2s}$

$$a = \frac{(30)^2 - (20)^2}{2 \times 200} = 1.25\,\text{m/s}^2$$

b) $v = u + at \Rightarrow t = \dfrac{v - u}{a}$

$$t = \frac{30 - 10}{1.25} = 16\,\text{sec}$$

66. A racing car moving at 180 km/h brakes and stops in a distance of 200m. What is its acceleration?

Solution:

$v^2 = u^2 + 2as \Rightarrow a = \dfrac{v^2 - u^2}{2s}$, but $v = 0$

$$a = -\left(\frac{\frac{180}{3.6}}{2 \times 100}\right)^2 = -12.5\text{m/s}^2$$

67. A cyclist starts from rest and accelerates at 1 m/s^2 for 20 second. He then travels at a constant speed for 1 minute and finally decelerates at 2 m/s^2 until he stops. Find:
    a) His maximum speed.
    b) The total distance covered.

Solution: First phase; $u = 0$, $a = 1\ m/s^2$, $t = 20$ sec

$\Rightarrow v = u + at = 0 + (1)(2) = 20$ m/s

$$s_1 = ut + \frac{1}{2}at^2 = 0 \times 20 + \frac{1}{2} \times 1 \times (20)^2 = 200\text{m}$$

Second phase, $u = 20$ m/s, $t = 60$ sec

$s_2 = vt = 20 \times 60 = 1200$m

Third phase, $u = 20$ m/s, $v = 0$, $a = -2$ m/s^2

$$v^2 = u^2 + 2as_3 \Rightarrow s_3 = \frac{v^2 - u^2}{2a} = \frac{0 - (20)^2}{2 \times (-2)} = 100\text{m}$$

a) $v = 20$ m/s

b) $s = s_1 + s_2 + s_3 = 200 + 1200 + 100 = 1500$m

68. A Volkswagen and a Cadillac are driven on the same 120 km trip. The Volkswagen travels at 80 km/h all the time. The Cadillac starts at the same time, driving at 100 km/h, but the driver stops for ten minutes after he has travelled for half an hour. Which car is the first to arrive at the destination?

Solution: The Volkswagen takes, $t_v = \frac{120}{80} = 1.5\text{hours}$ and the Cadillac takes, $t_c = \frac{120}{100} = 1.2\text{hours}$ but the Cadillac was not moving for 10 min, hence the total time it takes to cover the distance is;

1.2 hours + 10 min $\approx$ 1.4 hours

Still, it is the Cadillac that arrives first at the destination.

69. A car moving with constant acceleration covers the distance between two points 60m apart in 6 sec. Its speed as it passes the second point is 15 m/s.

a) What is the speed at the first point?

b) What is the acceleration?

c) At what prior distance from the first point was the car at rest?

Solution:

a) $s = \frac{(v+u)}{2}t \Rightarrow u = \frac{2s}{t} - v$

$$u = \frac{2 \times 60}{6} - 15 = 5\text{m/s}$$

b) $a = \frac{v-u}{t} = \frac{15-5}{6} = 1.67\text{m/s}^2$

c)

$u^2 = u'^2 + 2as$, $u' = 0$

$$s = \frac{u^2}{2a} = \frac{(5)^2}{2 \times \frac{10}{6}} = 7.5\text{m}$$

70. A driver has a thinking time of 0.5 sec. He is travelling at 30 m/s when he sees a tree blocking the road 50m ahead. If it takes him 4 sec to stop the car, will he hit the tree on the road.

Solution: Total distance travelled before stopping is,
reaction time distance + breaking distance, $s = s_r + s_b$

$$\Rightarrow s = ut_r + \left(\frac{v+u}{2}\right)t_b,\ t_r = 0.5 \text{ sec};\ t_b = 4 \text{ sec}$$

$$= 30 \times 0.5 + \frac{(0+30)}{2}4$$

$S = 75\text{m} > 50\text{m}$

Since the distance moved is greater than the distance between the car and the tree blocking the road, the car hits the tree.

71. A bus initially at rest accelerates uniformly to a speed of 25 m/s in 4 seconds. What is the distance covered by the bus during the fourth second?

Solution: $a = \frac{v-u}{t} = \frac{25-0}{4} = 6.25\text{m/s}^2$

$$S_n^{th} = u + \frac{a}{2}(2t_n - 1), u = 0$$

$$S_4^{th} = 0 + \frac{6.25}{2}(2 \times 4 - 1)$$

$$S_4^{th} = 21.9\text{m}$$

72. An object moving at 13 m/s slows uniformly at the rate of 2 m/s each second for a time of 6 sec. Determine

a) Its final velocity.
b) Its average velocity.
c) The distance moved.

Solution:

a) $v = u + at,\ a = -2\ \text{m/s}^2 = 13 - 2 \times 6$
$v = 1\ \text{m/s}$

$$\frac{s}{2} = \frac{(v^1)^2 - u^2}{2a}, \; v^1 = \text{velocity at the middle of the trip}$$

Taking ratio $\frac{\frac{s}{2}}{s} = \frac{(v^1)^2}{v^2} \Rightarrow (v^1)^2 = \frac{v^2}{2}$

$v^1 = 10\sqrt{2}$ m/s

80. A taxi is moving at 8 m/s. If the driver sees a dog walking across the road 30m ahead, and applied the brakes and gives the taxi a deceleration of 1.2 m/s^2. Assuming the reaction time of the driver to be 0.2 sec, will the taxi hit the dog?

Solution: Distance traveled by the taxi before it stops is;

s = reaction distance + braking distance = $s_r + s_b$

$$= ut_r + \frac{v^2 - u^2}{2a}$$

$$= 8 \times 0.2 + \frac{0 - (8)^2}{-2 \times 1.2} = 28.27\text{m}$$

That is, the taxi stops before it reaches the dog.

81. Two cars A and B, at the same initial position accelerate uniformly from rest. After 5 second, A is 2m ahead of B. What will be their separation after 8 seconds?

Solution: $s_A = \frac{1}{2} a_A t^2$ and $s_B = \frac{1}{2} a_B t^2$

In the first case, $s_A - s_B = \frac{1}{2} t^2 (a_A - a_B)$

$$\Rightarrow 2 = \frac{1}{2} \times (4)^2 (a_A - a_B)$$

$\Rightarrow a_A - a_B = 0.25$ m/s^2

In the second case:

$$s_A - s_B = \frac{1}{2} t^2 (a_A - a_B)$$

$$s_A - s_B = \frac{1}{2} \times (8)^2 (0.25)$$

$s_A - s_B = 8\text{m}$

82. A car starting from rest accelerates at 4 m/s^2. If it travelled 40m in the last second, what is the total distance covered by the car?

Solution: $S_n^{th} = u + \frac{a}{2}(2t_n - 1)$

$$40 = 0 + \frac{4}{2}(2t_n - 1)$$

$\Rightarrow t = 10.5$ sec

Then, $s = ut + \frac{1}{2}at^2 = 0 \times 10.5 + \frac{1}{2} \times 4 \times (10.5)^2$

$s = 220.5m$

83. A careless motorist travelling at 90 km/h passes a police car travelling at 50 km/h. Immediately, the police car starts accelerating at 3 m/s^2. At what distance will the police catch the motorist?

Solution: After sometime, their distance is the same.

$$s_m = s_p \, , \, t_m = t_p = t$$

$$v_{mt} = u_p t + \frac{1}{2}at^2$$

$$\frac{90}{3.6} = \frac{50}{3.6} + \frac{1}{2} \times 3 \times t \Rightarrow t = 7.4 \text{ sec}$$

$$\Rightarrow s = v_m t = 25 \times 7.4 = 185 \text{ m}$$

84. Two sport cars start from rest and move one after the other in 1 minute interval with an acceleration of 0.4 m/s^2 each. How long after the start of the first car is the distance between them equal to 4200m?

Solution: $s_1 = \frac{1}{2}a_1 t_1^2$ and $s_2 = \frac{1}{2}a_2 t_2^2$ but $a_1 = a_2 = 0.4$ m/s^2 and $t_2 = t_1 - 60$ sec their separation is;

$$s_1 - s_2 = \frac{1}{2}at_1^2 - \frac{1}{2}at_2^2$$

$$4200 = \frac{1}{2} \times 0.4 \times t_1^2 - \frac{1}{2} \times 0.4 \times (t_1 - 60)^2$$

Solving for t_1, we get

$t_1 = 145$ sec

85. A bullet moving with speed 200 m/s can penetrate a 4 cm thick block. What should be the speed of the bullet to penetrate a 10 cm thick block of the same material?

Solution: $v^2 = u^2 - 2as$, here $v = 0$, a is negative $\Rightarrow u = \sqrt{2as}$

That is, the speed is proportional to the square root of the thickness.

$\Rightarrow u \alpha \sqrt{s}$

So we have, $\frac{u_2}{u_1} = \frac{\sqrt{s_2}}{\sqrt{s_1}} \Rightarrow u_2 = u_1\sqrt{\frac{s_2}{s_1}} = 200\sqrt{\frac{10}{4}} = 316 m/s$

86. A body travels 2m in the first two seconds and 2.2m in the next 4 seconds. What is the velocity of the body 7 seconds from the start of motion?

Solution: $s = ut + \frac{1}{2}at^2$

$$\text{At, } t = 2 \text{ sec, } 2 = 2u + \frac{1}{2}a(2)^2 \Rightarrow u + a = 1$$

$$\text{At, } t = 2 + 4, 6 \text{ sec, } 2 + 2.2 = 6u + \frac{1}{2}a(6)^2 \Rightarrow u + 3a = 0.7$$

Solving for u and a, $u = 1.15$ m/s and $a = -0.15 m/s^2$

At $t = 7$ sec, $v = u - at = 1.15 - 0.15 \times 7 = 10$ cm/s

87. A bus is beginning to move with an acceleration of 1 m/s^2. A boy who is 48m behind the bus starts running at 10 m/s. After what time will the boy be able to catch the bus?

Solution: Distance moved by the boy, $s_1 = vt = 10t$

$$\text{Distance moved by the bus, } s_2 = \frac{1}{2}at^2 = \frac{1}{2}t^2$$

The boy will catch the bus if; $s_1 = 48 + s_2$

$$10t = 48 + \frac{1}{2}t^2 \Rightarrow t^2 - 20t + 96 - 0$$

$t = 8$ sec or $t = 12$ sec

Earlier time, $t = 8$ sec

88. A car moving with a uniform acceleration of 10 m/s^2 has a velocity of 100 m/s. In what time will its velocity be doubled?

Solution: $v = u + at$, $v = 2u$

$$200 = 100 + 10t \Rightarrow t = 10 \text{ sec}$$

89. The motion of a body is given by the equation $v = mt + n$, where m is 5 m/s^2 and n is 10 m/s. How far will it travel in the first 2 seconds?

Solution: Method one: $s = \frac{(v+u)}{2}t$, but u is at $t = 0$,

$$\Rightarrow u = 5t + 10 = 0 + 10 = 10 \text{ m/s}$$

v is at $t = 2$ sec $\Rightarrow v = 5t + 10 = 5 \times 2 + 10 = 20$ m/s

$$\text{Thus, } s = \frac{(20+10)2}{2} = 30\text{m}$$

Method two: From $v = mt + n = 5t + 10$ . . . (*)

And we know, $v = u + at$ . . . (**),

comparing the two equations u = 10 m/s and a = 5 m/s^2 and using,

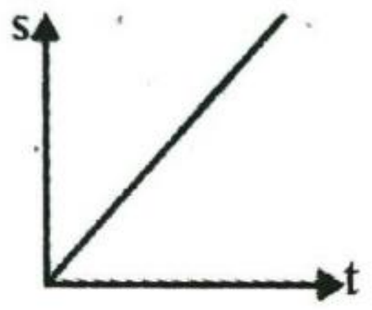
$$s = ut + \frac{1}{2}at^2 = 10 \times 2 + \frac{1}{2} \times 5 \times (2)^2 = 30\text{m}$$

2.4 Graphical description of uniformly accelerated Motion

A graphical description of motion of a body is a powerful means that help us to visualize the nature of motion of the body under consideration. To draw a motion graph, we need to plot the independent variable that is *time* along the x axis and the dependent variable such as *velocity*, *displacement* and *acceleration* along the y axis and then label these axis with appropriate scales.

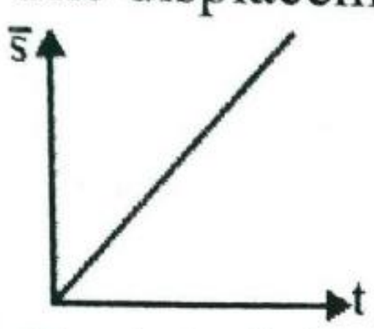
Commonly we do have distance time graph, displacement time graph, velocity time graph and acceleration time graph.

Distance time graph for a uniform motion in a straight line, s – t graph is a straight line that passes through the origin. And, the *slope* of the *s – t* graph represents the *speed* of the moving body.

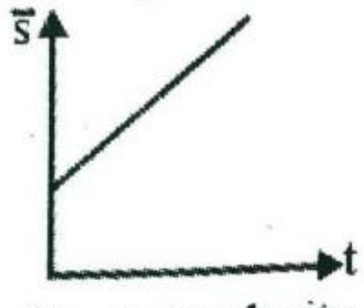


Displacement – time graph

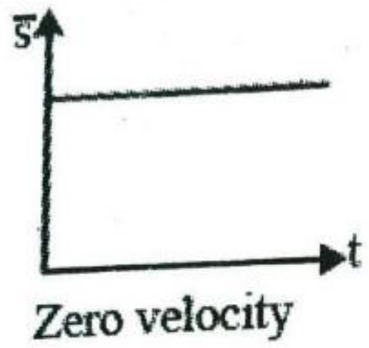
The displacement time graph do have the following common forms.



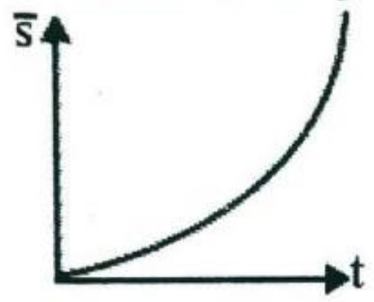
Constant velocity



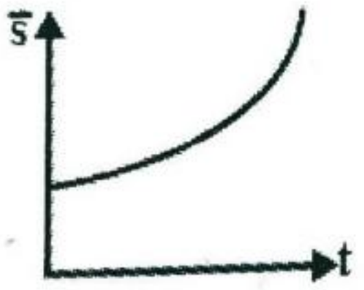
Constant velocity



Zero velocity



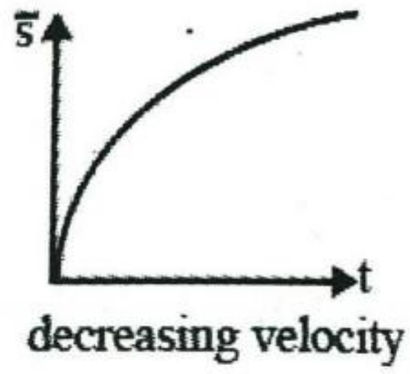
Changing velocity



Changing velocity

☞ **Remember:**

- The *slope* of *displacement time* graph represents the *velocity* of the moving body.
- For a retarded motion, that is for a decreasing velocity, the $\bar{s}$ – t graph is concave down. That is

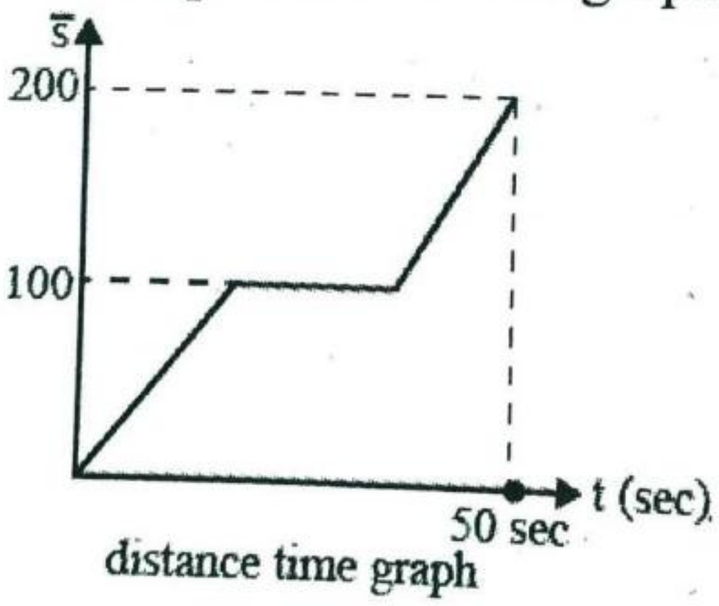


decreasing velocity

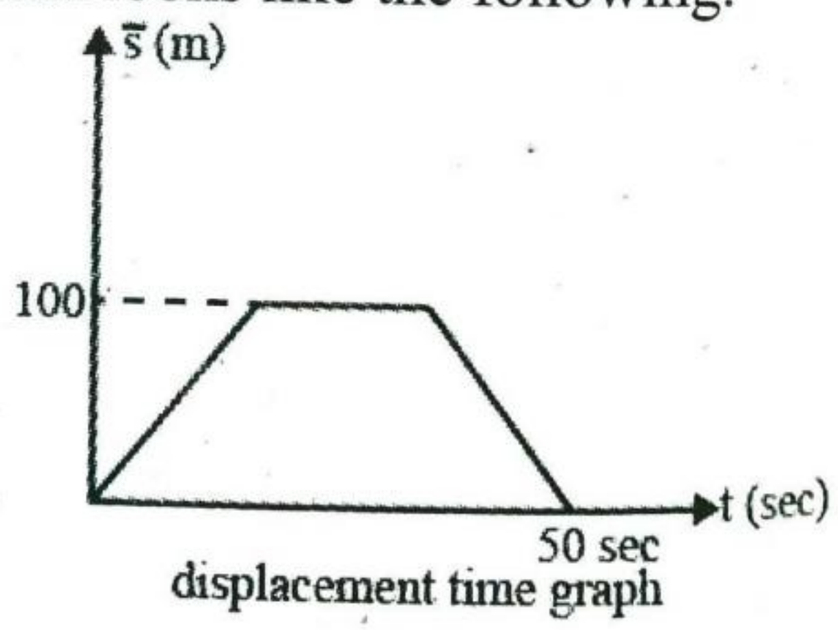
Conceptual Example

90. Can the distance time graph goes down ward. What about displacement time graph. Explain

Explanation: Never, distance time graph can go up but never down. However displacement time graph can go down that is it can have a negative slope. For example, if you walk 100m due east from your house to a shop and return home after 50 seconds along the same road, then your trip can have distance time graph and displacement time graph which looks like the following.



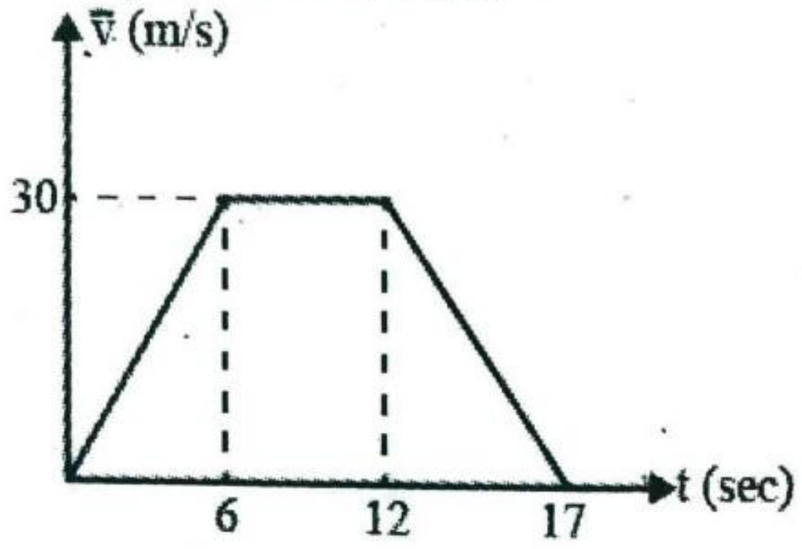
distance time graph



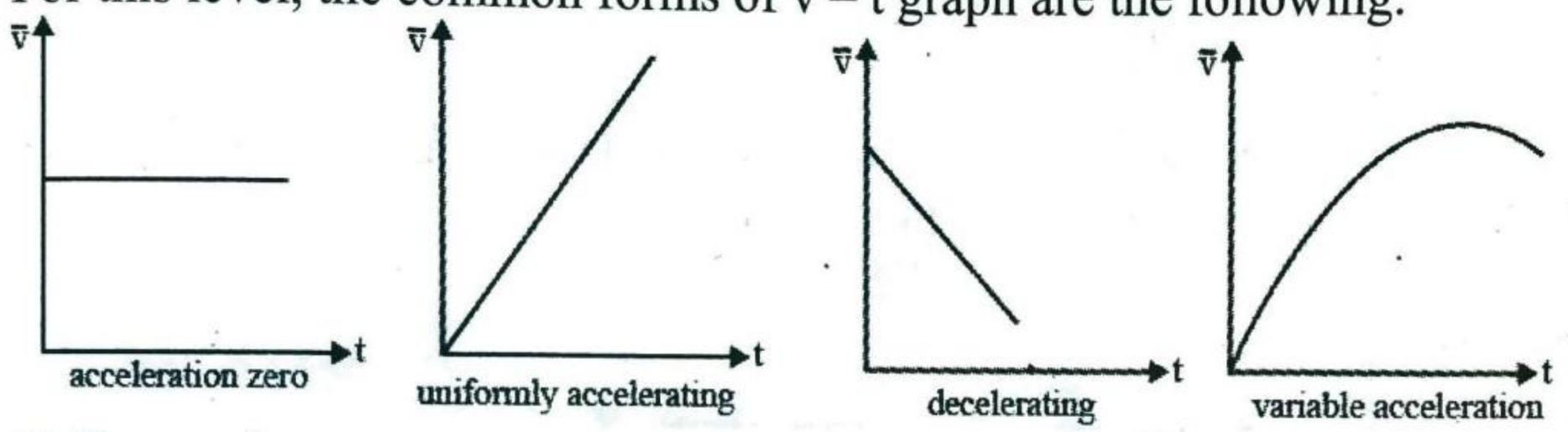
displacement time graph

Velocity – time graph

The velocity time graph shows how the velocity of a body changes as it moves along. Assume a car starts from rest and reached a velocity of 30 m/s in 6 seconds, and moves at this rate for another 6 seconds and then slows down uniformly so as to stop within 5 seconds. Graphically, this information can be described as follows.

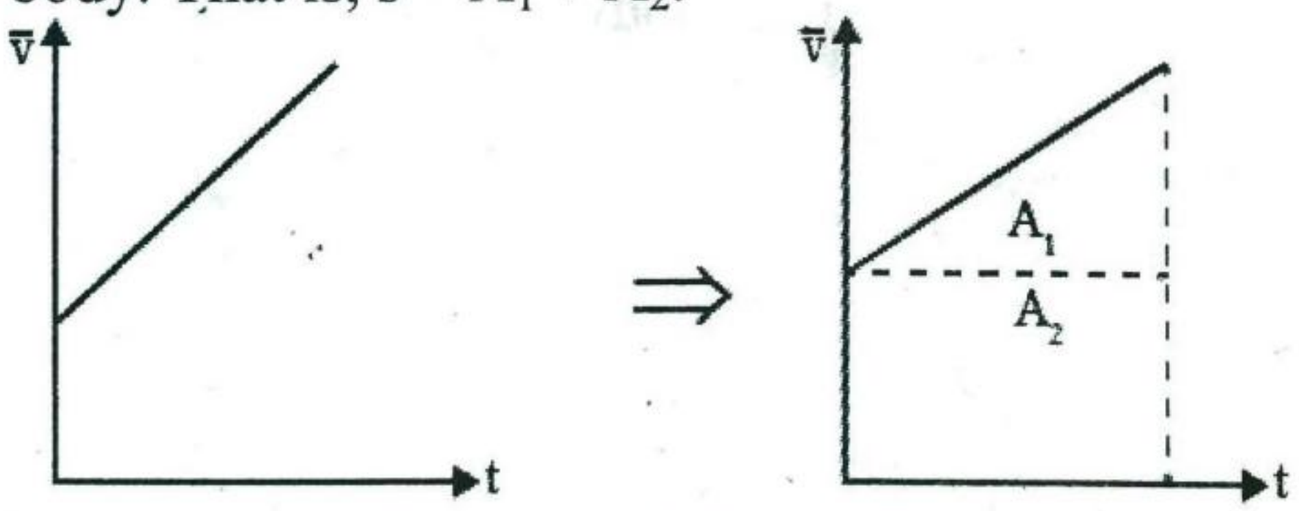


For this level, the common forms of v – t graph are the following.



☞ **Remember**

- The slope of a velocity time graph represents acceleration of the body.
- Acceleration is positive for an upward sloping $\bar{v} - t$ graph and negative for a downward sloping $\bar{v} - t$ graph.
- The area under $\bar{v} - t$ graph represents the displacement of the body. That is, $s = A_1 + A_2$.



Conceptual Example

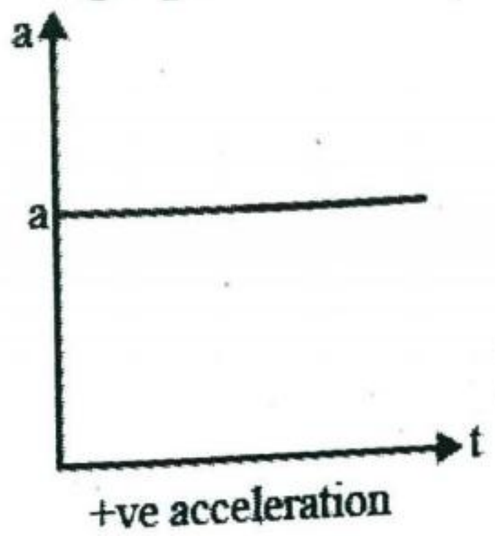
91. What can you say about the motion of the body if?
    a) The displacement time graph is perpendicular to the time axis.
    b) The velocity time graph is perpendicular to the time axis.

Explanation: a) It means the body is moving with infinite velocity. Practically speaking, it is impossible situation.

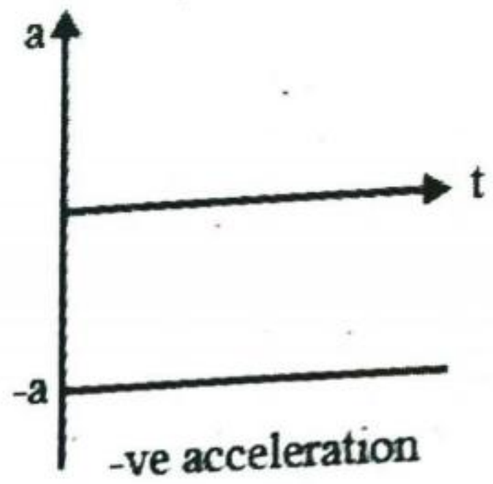
b) This means that, the body is moving with infinite acceleration. This is practically impossible.

Acceleration time graph

Commonly, we do have positive or negative uniform accelerations. So, the $\bar{a} - t$ graph has the form;



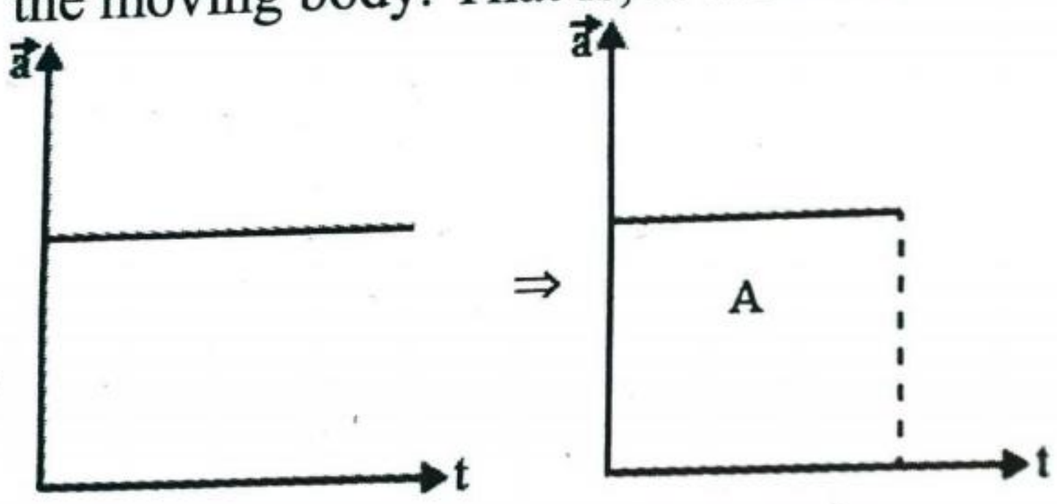
+ve acceleration



-ve acceleration

☞ **Note that;**

- The area under $\bar{a} - t$ graph represents the change in velocity of the moving body. That is, if we have



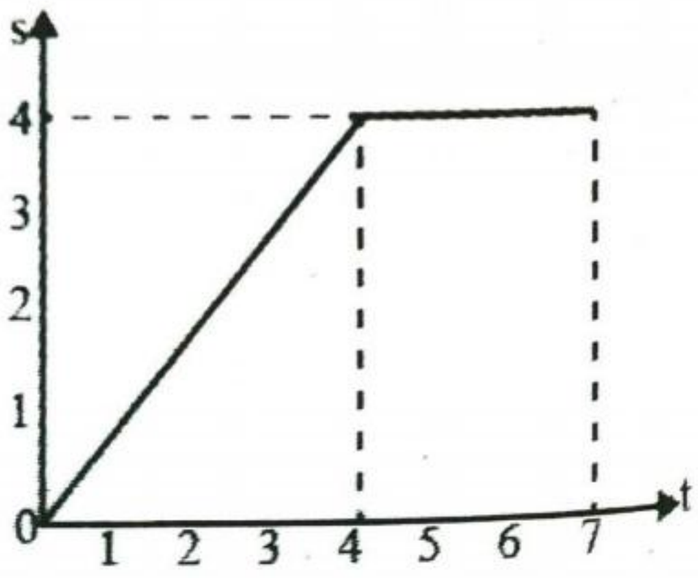
$\Delta v = A = at$

Illustrative Examples

92. From the given displacement – time graph.

a) What is the velocity of the body in the first 4 seconds?

b) What is the velocity of the body from t = 4 sec, to t = 7 sec?



Solution:

a) $v = \frac{\Delta s}{\Delta t} = \frac{4-0}{4-0} = 1 m/s$

b) $v = \frac{\Delta s}{\Delta t} = \frac{4-4}{7-4} = 0$

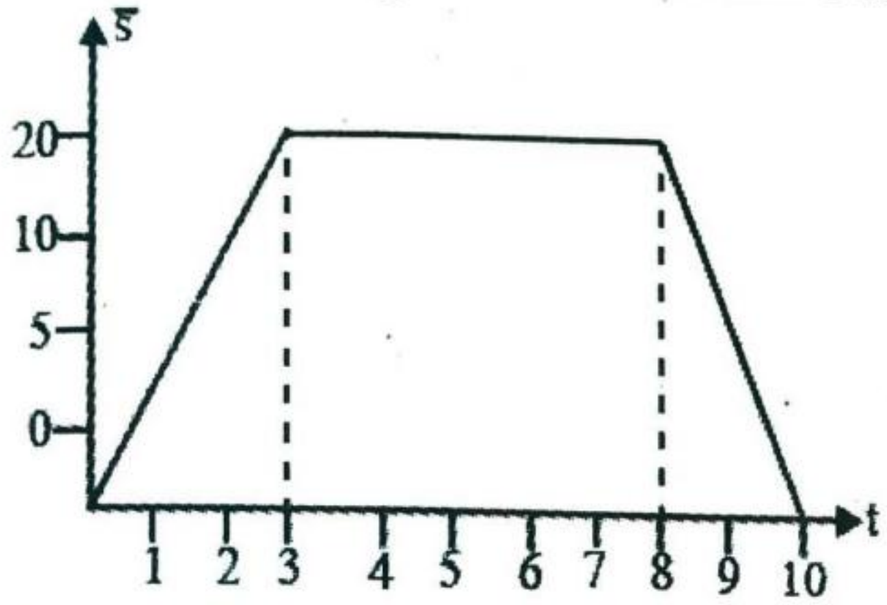
93. The displacement – time graph of a moving car is plotted below.

a) State the nature of the motion at each interval.

b) Calculate the total distance covered.

c) Calculate the velocity of the car in each interval.

d) What is the displacement of the car?



Solution:

a) From 0 to 3 sec, it is moving at constant speed. From 3 to 8 sec, it is at rest. From 8 to 10 sec it is decelerating until it stops.

b) $s_T = 20 + 0 + 20 = 40\text{m}$

c) $v_1 = \dfrac{\Delta s_1}{\Delta t_1} = \dfrac{20-0}{3-0} = \dfrac{20}{3}\text{m/s}$

$$v_2 = \frac{\Delta s_2}{\Delta t_2} = \frac{20-20}{8-3} = 0$$

$$v_3 = \frac{\Delta s_3}{\Delta t_3} = \frac{0-20}{10-8} = -10\text{m/s}$$

d) Since the body returned to its initial position its displacement is zero or $\Delta s = 20 + 0 + (-20) = 0$.

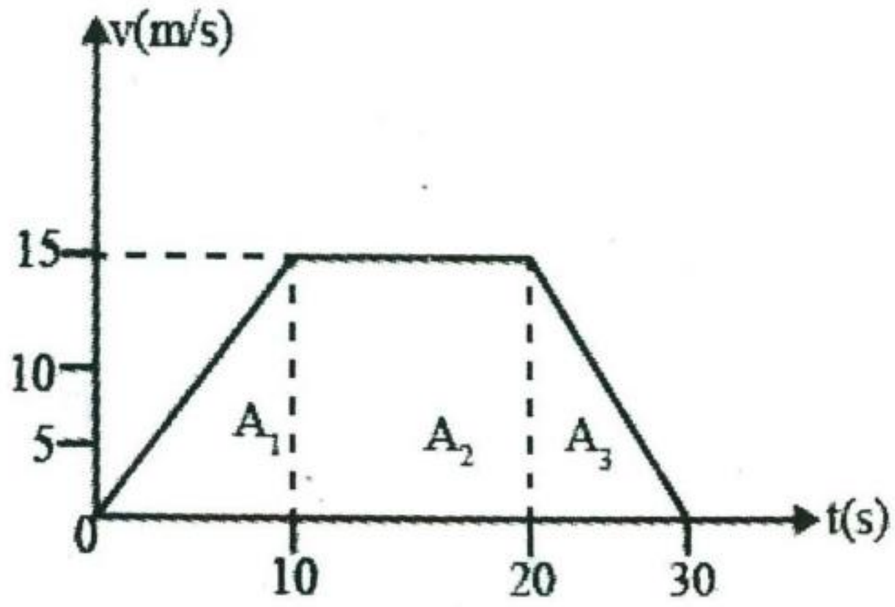
94. The velocity time graph of a body is given as follows.

a) Describe the nature of the motion for each interval.

b) Find the acceleration of the body at each interval.

c) Find the total distance covered by the body.

d) Calculate the displacement of the body.



Solution:

a) From 0 sec to 10 sec the body is accelerating at 1.5 m/s^2. From 10 sec to 20 sec the body is moving at constant velocity, acceleration is zero. From 20 sec to 30 sec, the body is decelerating at 1.5 m/s^2.

b) From 0 sec to 10 sec, $a = \frac{\Delta v}{\Delta t} = \frac{15-0}{10-0} = 1.5\,m/s^2$

From 10 sec to 20 sec, $a = \frac{\Delta v}{\Delta t} = \frac{15-15}{20-10} = 0\,m/s^2$

From 20 sec to 30 sec, $a = \frac{\Delta v}{\Delta t} = \frac{0-15}{30-20} = -1.5\,m/s^2$

c) $s_1 = A_1 = \frac{1}{2} \times 10 \times 15 = 75m$

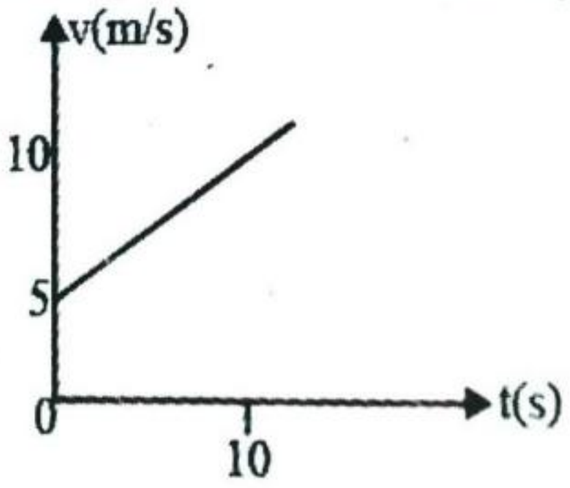
$s_2 = A_2 = 15 \times 10 = 150m$

$s_3 = A_3 = \frac{1}{2} \times 10 \times 15 = 75m$

$s = s_1 + s_2 + s_3 = 75 + 150 + 75 = 300m$

d) In this case distance and displacement are equal.

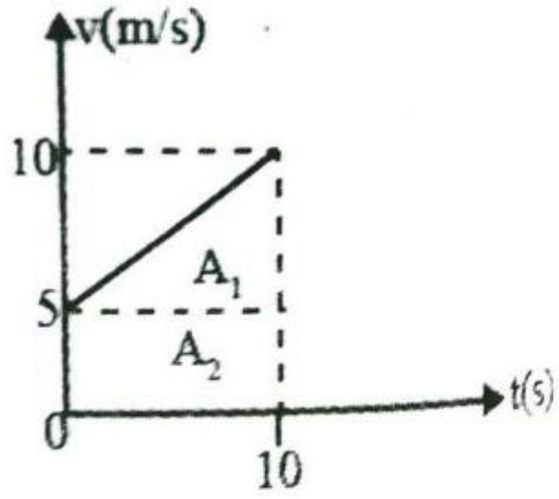
95. The following graph is the velocity time graph of a body. What is the distance covered by the body?



Solution:

$s = A_1 + A_2 = \frac{1}{2} \times 5 \times 10 + 5 \times 10$

$= 75m$



96. The velocity time graph of a truck is given below. Calculate the
a) Distance covered by the truck.
b) Displacement of the truck.

Solution:

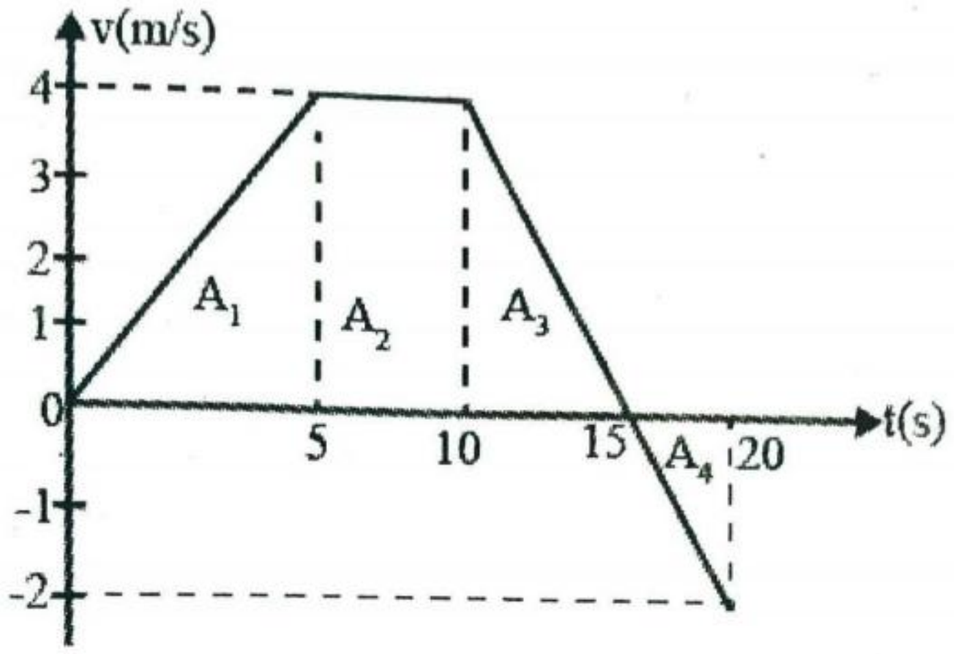
a) $s_1 = A_1 = \frac{1}{2} \times 4 \times 5 =$

10m

$s_2 = A_2 = 5 \times 4 = 20m$

$s_3 = A_3 = \frac{1}{2} \times 4 \times 5 =$

10m



$s_4 = A_4 = \frac{1}{2} \times 2 \times 5 = 5m$

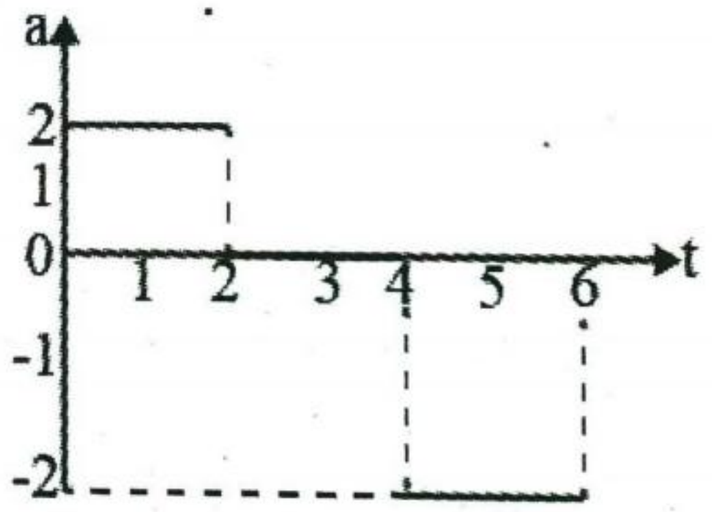
$s = s_1 + s_2 + s_3 + s_4 = 10 + 20 + 10 + 5 = 45m$

b) Since the velocity of the truck is negative in the last 5 sec, it means that the truck is returning to its original position. So we have $\Delta s = 10 + 20 + 10 - 5 = 35m$.

97. The acceleration time graph for a car starting from rest is given below.

a) Calculate the velocity of the car at each interval.

b) Draw the velocity time graph of the car.



Solution:

a) From 0 to 2 sec,

$\Delta v_1 = A_1 = 2 \times 2 = 4$ m/s

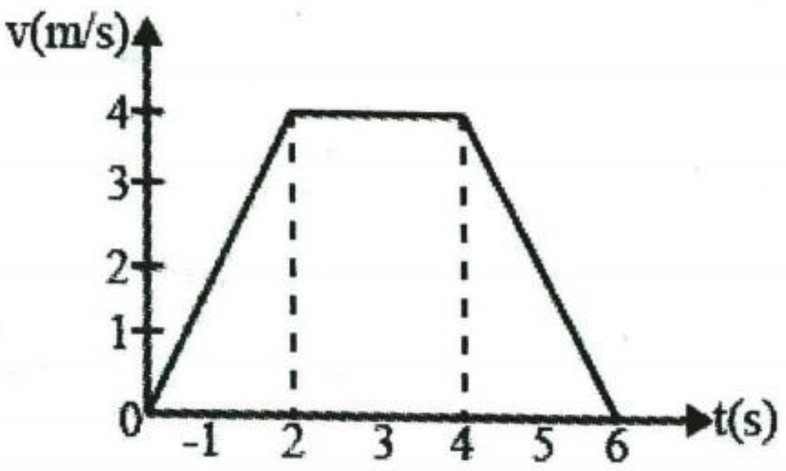
From 2 sec to 4 sec:

$\Delta v_2 = A_2 = 2 \times 0 = 0$ m/s

From 4 sec to 6 sec

$\Delta v_3 = A_3 = 2 \times -2 = -4$ m/s

b) The velocity of the car is 4 m/s at t = 2 sec, it is 0 from t = 2 sec to t = 4 sec and it decreases from 4 m/s to 0 m/s from t = 4 sec to t = 6 sec. That is,



Free Fall Motion

The most famous example of motion with constant acceleration is free fall. But, what does it mean? First of all, the word **free** in free fall means free from any effects other than gravity. In free fall, we assume that the object's motion is not influenced by any form of friction or air resistance. Next, the word **fall** in free fall does not mean the object is

necessarily moving downward. That is, if you drop a ball, it is free fall, if you throw a ball upward or downward it is in free fall as soon as it leaves your hand.

> **Free fall:** Is the motion of a body under the influence of gravity alone.

For a freely falling body, since the acceleration is constant, we use the very same equations we used for uniformly accelerated motion in a straight line with some modifications in symbols and signs. So here we use the symbol g for a and Y or h for S and in this book, we use positive sign for g when the motion is downward and negative for an upward motion.

→ For a body dropped from some height,

$$v = gt$$
$$y = \frac{1}{2}gt^2$$
$$v^2 = 2gh$$

→ For a body thrown upward,

$$v = u - gt$$
$$y = ut - \frac{1}{2}gt^2$$
$$v^2 = u^2 - 2gh$$

☛ **Important points to remember**

* An object is in free fall as soon as it is released, weather it is dropped from rest, thrown downward, or thrown upward.

* In the absence of air resistance, all bodies fall with the same acceleration regardless of their mass.

Conceptual Examples

98. If you throw a ball upward, it will rise to some might and then return back. Does it mean that the velocity of the ball become zero at the maximum height? What about its acceleration? Explain.

Explanation: The velocity of the ball is zero at the maximum height. But, it does not mean that the acceleration is zero. The fact is that, the velocity of the object that is thrown upward near the surface of the body remains constant, that is 9.81 m/s^2 even at the top of its path.

Illustrative Example

99. A ball is kicked upwards with an initial velocity of 20 m/s from the ground. How long will it take the ball to return to its initial position? Use $g = 10\ m/s^2$.

Solution: $y = ut - \frac{1}{2}gt^2$, when it return, $y = 0$

$$\Rightarrow 0 = ut - \frac{1}{2}gt^2 \Rightarrow t = \frac{2u}{g} = \frac{2\times 20}{10} = 4\,sec$$

100. At the time a ball is dropped from the top of a 20m building, a stone is thrown upward from the ground. If they both hit the ground at the same time, what was the initial velocity of the stone?

Solution: For the ball, $y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2\times 20}{10}} = 2\,sec$

For the stone, $y = ut - \frac{1}{2}gt^2 = 0 \Rightarrow u = \frac{1}{2}gt^2 = \frac{1}{2}\times 10\times (2)^2$

$u = 20\ m/s$

101. From what height must a body be dropped so that its velocity upon striking the ground is 30 m/s.

Solution: $v^2 = u^2 + 2gh$, $u = 0$

$$h = \frac{v^2}{2g} = \frac{(30)^2}{2\times 10} = 45m$$

102. A body thrown upwards with some velocity reaches a maximum height of 50m. Another body of double the mass thrown up with double the initial velocity will reach a maximum height of.

Solution: $v^2 = u^2 - 2gh \Rightarrow 0 = u^2 - 2gh \Rightarrow h = \frac{u^2}{2g}$ thus, $h \alpha u^2$.

$$\frac{h_2}{h_1} = \frac{(u_2)^2}{(u_1)^2},\ \text{but } u_2 = 2u_1$$

$$h_2 = h_1\left(\frac{2u_1}{u_1}\right)^2 = 50\times (2)^2 = 200m$$

103. A body projected upward returns to the starting point in 4 sec. If $g = 10\ m/s^2$, with what velocity was the body projected.

Solution: $y = ut - \frac{1}{2}gt^2$, but $y = 0$.

$$\Rightarrow u = \frac{gt}{2} = \frac{4\times10}{2} = 20 m/s$$

104. A stone is released from the top of 180m cliff. If one second later a ball is thrown down and both hit the ground at the same time, with what speed will the stone hit the ground?

Solution: For the stone, $y = ut - \frac{1}{2}gt_s^2, \Rightarrow t_s = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2\times180}{10}}$

$t_s = 6$ sec. For the ball, $y = ut_b + \frac{1}{2}gt_b^2$, but $t_b = t_s - 1$, $t_b = 5$ sec

$$\Rightarrow 180 = u\times5 + \frac{1}{2}\times10\times(5)^2 \Rightarrow u = 61\ m/s \text{ and}$$

$$v = u + gt = 61 + 10\times5 = 111\ m/s$$

2.5 Relative Velocity in one direction

There is no motion that is absolute. It is all relative. In its simpels case, experience can tell all of us. For instance, if you are moving by bus at 20 m/s, in the first case, if you observe the outside environment, then it seems for you as if fixed objects such as the electric pole, trees or buildings are moving in opposite direction at that speed. In the second case, if you observe a car that is moving at 20 m/s in the same direction as your bus, then its seems for you that the car is standing still. In the third case, if you observe a car that is moving at 20 m/s in opposite direction to your bus, then it seems for you that car is moving at 40 m/s. These are all relative velocity.

Relative velocity: Is the rate at which a body changes its position with respect to another body.

The relative velocity between two bodies is obtained by taking the difference of their velocities. That is, if body A is moving at velocity v_A and body B is moving at velocity v_B, then the relative velocity between the two bodies is given by:

$$\mathbf{v_{AB} = v_A - v_B}$$
$$\mathbf{v_{BA} = v_B - v_A}$$

Where, v_{AB}, is the velocity of body A with respect to body B. and v_{BA}, is the velocity of body B with respect to body A.

Illustrative Examples

105. A car moving at 20 m/s along a straight road is overtaken by another car moving at 60 m/s. What is the velocity of the second car as seen by the driver of the first car?

Solution: Let $v_A = 20$ m/s and $v_B = 60$ m/s

$v_{BA} = v_B - v_A = 60 - 20 = 40$ m/s

That is, the second car seems as if it is moving at 40 m/s when seen by an observer in the first car.

106. Two cars A and B are moving along a straight road in opposite directions with velocities of 50 km/h and 60 km/h respectively. What is the relative velocity of car A with respect to car B.

Solution: $v_{AB} = v_A - v_B$, take $v_B = -60$ km/h

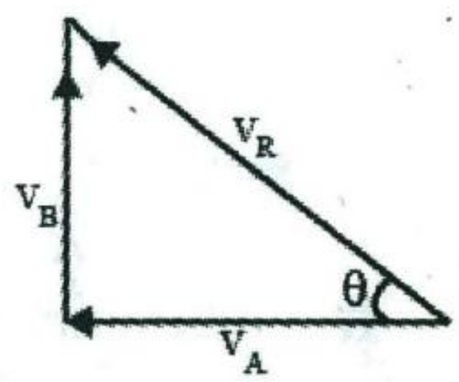
$v_{AB} = 50 - (-60) = 110$ km/h

107. A car is moving due west at 80 m/s. At the same time, another car is moving due North at 60 m/s. What is the relative velocity of the cars with respect to each other?

Solution: $v_{AB} = v_{BA} = v_R$, let $v_A = 80$ m/s, $v_B = 60$ m/s

$$v_R = \sqrt{(v_A)^2 + (v_B)^2} = \sqrt{(80)^2 + (60)^2}$$

$v_R = 100$ m/s



$$\tan\theta = \frac{v_B}{v_A} = \frac{60}{80}, \theta = 37$$

General Secondary Education Certificate Exams

108. A bird is flying due east at 10 m/s. This statement best describes the bird's constant.

A. Acceleration C. Speed

B. Direction D. Velocity

Explanation: It is speed plus direction. **Answer: D**

109. Which physical quantity is a non zero constant in a uniform motion?

A. Distance C. Velocity

B. Displacement D. Acceleration

Explanation: In uniform motion, velocity is non zero constant.

Answer: C

110. A student in a laboratory observed an insect moving along a meter stick. He recorded the position of the insect (x) on the stick and the time (t) taken as shown below.

x(m)	5	15	25	35	45	55	65	75	85	95
t(s)	0	5	10	15	20	25	30	35	40	45

The student could then describe the motion of the insect by the relation.

A. $x = 3t$ B. $x = 2t + 5$ C. $x = t + 10$ D. $x = 2t$

Answer: B

111. An insect moves along the edge of a table. It is observed that the insect advances equal distance in equal times. What do you refer from this observation?

A. The insect has constant velocity.
B. The insect is uniformly accelerating.
C. The insect changes its position slowly.
D. The insect is moving faster and faster.

Answer: A

112. If a body moves at a constant speed along a straight line, its average speed will be:

A. Exactly half its speed C. Less than its speed
B. Greater than its speed D. Equal to its speed

Answer: D

113. A student walk from home to school at a constant speed of 2 m/s. If it takes him 30 min to reach the school, how far did he travel?

A. 3.6km B. 3.6m C. 2 km D. 1.8 km

Solution: $s = vt = (2)(1800) = 3600m = 3.6$ km **Answer: A**

114. The great Ethiopian athlete Tirunesh Dibaba defended her world 10,000m crown in dramatic, historic fashion on August 25, 2007 in Osaka Japan she finished the 10,000m in 31 min and 55.41 seconds. What was her average speed?

A. 10.4 m/s B. 3.32 m/s C. 8.44 m/s D. 5.22 m/s

Solution:

$$v_{av} = \frac{s_T}{t_T} = \frac{10,000}{1915.41}, \; t_T = (31 \times 60) + 55.41 = 1915.41 \text{ sec}$$

$v_{av} = 5.22$ m/s

Answer: D

115. A uniformly moving train covered a 24 km distance in 10 minutes. How far can it travel with 1 hr and 30 minutes?

A. 144 km B. 145.5 km C. 216 km D. 375 km

Solution: $v_1 = \frac{s_1}{t_1} = \frac{24}{0.166} = 144.5\text{km/r}$

$v_2 = \frac{s_2}{t_2} \Rightarrow s_2 = v_2 \times t_2 = 144 \times 1.5 = 216\text{ km}$ **Answer: C**

116. Two soccer players are approaching each other. Initially, the players are 48m apart. If each player has a speed of 8 m/s. How much time passes before they collide?

A. 3 sc B. 4 sec C. 6 sec D. 8 sec

Solution: If the players are approaching each other, their relative velocity of approach is 16 m/s. Thus, $v = \frac{s}{t} \Rightarrow t = \frac{s}{v} = \frac{48}{16} = 3\,\text{sec}$

Answer: A

117. An engineer wanted to put a speed limit at the entrance to a train at 500m away as it comes to rest to a plat form. His plan is to give exactly 1 minute for the train to decelerate uniformly within the 500m distance. What must be the maximum speed at the entrance?

A. 15 km/hr B. 60 km/hr C. 30 km/hr D. 20 km/hr

Solution: $v = \frac{s}{t} = \frac{500}{60} = 8.33\text{m/s} \simeq 30\text{km/hr}$ **Answer: C**

118. If you are running at 6 m/s and decelerate at 2 m/s^2. How long will it take you to stop and how far will you go respectively?

A. 2 sec and 12 m C. 3 sec and 18m

B. 6 sec and 18m D. 3 sec and 9m

Solution: $t = \frac{v-u}{a} = \frac{0-6}{-2} = 3\,\text{sec}$

$s = ut - \frac{1}{2}at^2 = 6 \times 3 - \frac{1}{2} \times 2 \times 3^2 = 9\text{m}$ **Answer: D**

119. The table below shows the position and elapsed time of a car starting from rest and uniformly accelerating in a straight line.

Time (sec)	0	1	2	3
Distance (m)	0	2	8	s

What is the distance x in 3 seconds?

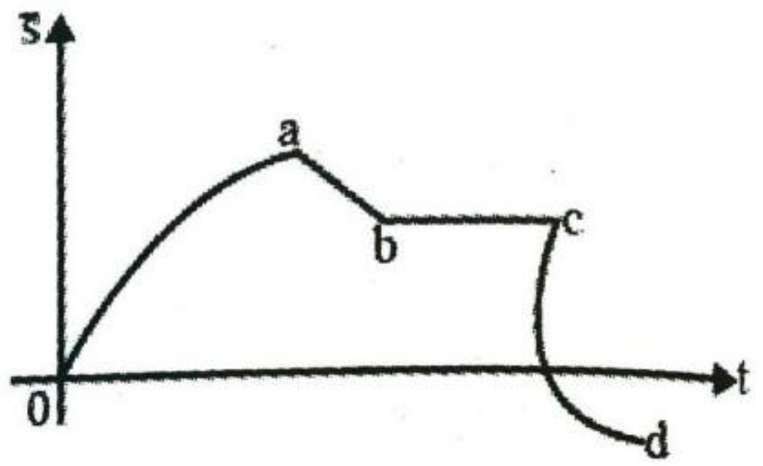
A. 12m B. 16m C. 18m D. 24m

Solution: $s = ut + \frac{1}{2}at^2 = \frac{1}{2}at^2$, $u = 0$ at $t = 2$ sec, $s = 8m$

$\Rightarrow a = \frac{2s}{t} = \frac{2\times8}{(2)^2} = 4m/s^2$ at $t = 3$ sec,

$s = \frac{1}{2}at^2 = \frac{1}{2}\times4\times(3)^2 = 18m$ **Answer: C**

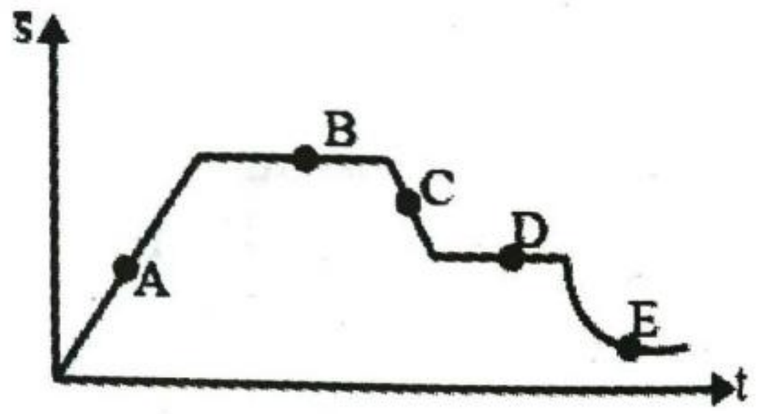
120. The following displacement time graph represents the motion of a body. In which part of the graph is the body moving uniformly with non zero velocity.



A. between b and c
B. between c and d
C. between 0 and a
D. between a and b

Solution: Between a and b, the $\overline{s} - t$ graph is a straight line sloping downward, thus the velocity is uniform and non zero between 0 and a, it is accelerating, between b and c it is at rest and between c and d it is accelerating. **Answer: D**

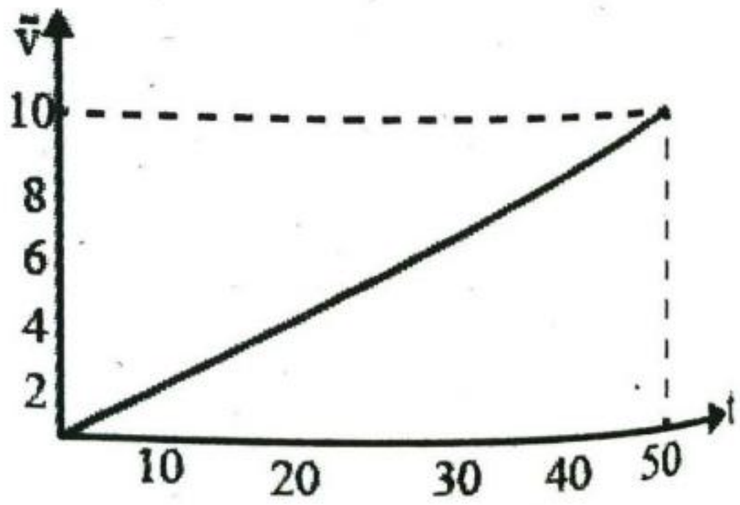
121. The following graph represents the distance at every time of the motion of a body. Where is the body moving uniformly with non zero speed.



A. At B and D
B. At A and C
C. At B and E
D. At D and E

Solution: The graph is straight line at points A and C. Hence at those parts, the body is moving uniformly with non zero speed. **Answer: B**

122. Which one of the following is the correct interpretation of the velocity time graph in the figure below?

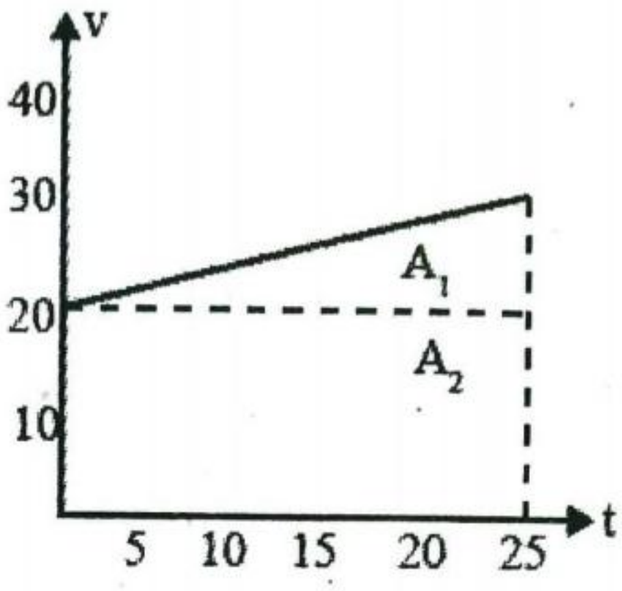


A. The average velocity is 4 m/s.
B. The total distance covered is 250m.
C. The acceleration is zero.
D. The average speed is 0.2 m/s.

Solution: $s = A = \frac{1}{2} \times 50 \times 10 = 250m$ **Answer: B**

123. The following graph is the velocity time graph of a body. The distance covered by the body in 25 seconds is:

A. 40m B. 750m C. 600m D. 500m



Solution:

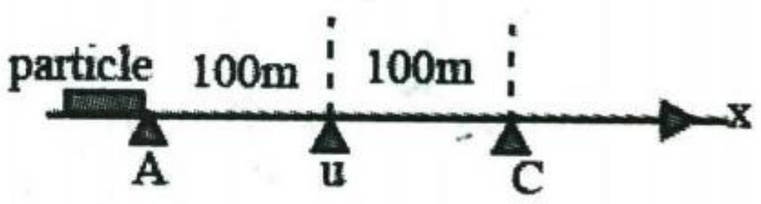
$$s = A_1 + A_2 = \frac{1}{2} \times 25 \times 20 + 20 \times 25$$

$s = 750m$ **Answer: B**

124. A particle is accelerating uniformly along the x axis. Three observers, Almaze (A), Ubang (u) and Chaltu (C) recorded the time when the particle reaches them. They measured that the time taken by the particle is to cover the 100m from A to u was 10 sec, and that from u to C was 5 sec. What is the acceleration of the particle?

A. $8\ m/s^2$ B. $\frac{10}{3}\ m/s^2$ C. $2\ m/s^2$ D. $\frac{4}{3}\ m/s^2$

Solution: For the first 100m,



$$s = \frac{(v_A + v_u)}{2} t \Rightarrow 100 = \frac{(v_A + v_u)}{2} 10 \Rightarrow v_A + v_u = 20 \ \ldots (*)$$

For the second 100m, $s = \frac{(v_u + v_c)}{2} t \Rightarrow 100 = \frac{(v_u + v_c)}{2} 5$

$\Rightarrow v_c + v_u = 40 \ldots (**)$

From Equation (*) and (**)

$v_C - v_A = 20$, and $t_T = 10 + 5 = 15$

$$a = \frac{\Delta v}{t} = \frac{20}{15} = \frac{4}{3} m/s^2$$ **Answer: D**

125. The displacement time graph of an object moving along the x axis is a straight line with a positive slope. The object has a

A. Constant velocity
B. constant non zero acceleration
C. constant displacement
D. steady increasing velocity

Answer: A

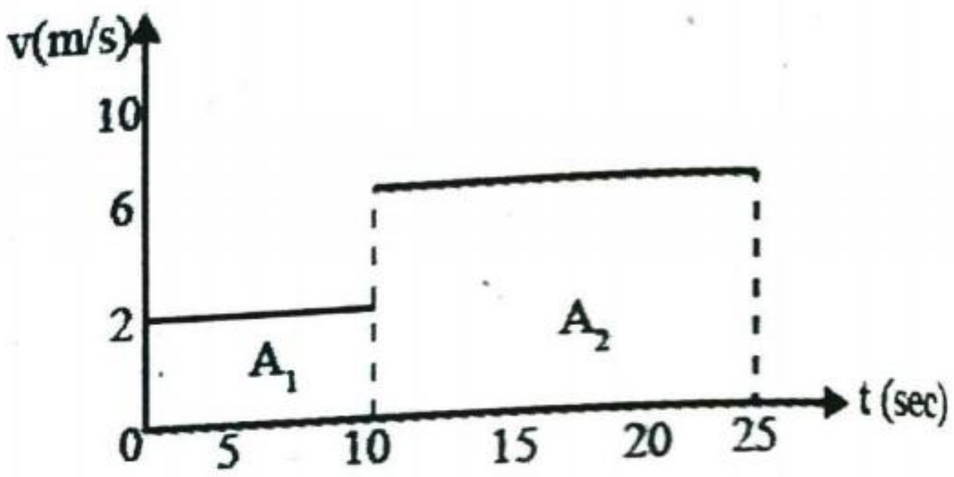
126. If a uniformly accelerated body starts from rest, its displacement will be directly proportional to:
A. The instantaneous velocity
B. The square of the time interval
C. The total time interval
D. Half of the time interval

Solution: $s = ut + \frac{1}{2}at^2$, if $u = 0$

$$s = \frac{1}{2}at^2 \Rightarrow s \alpha t^2$$

Answer: B

127. The following graph represents the motion of a particle in two regions. What is the average velocity, of the particle in 25 sec motion?



A. 2 m/s C. 3.8 m/s
B. 3.5 m/s D. 6 m/s

Solution: $s_1 = A_1 = 2 \times 10 = 20m$

$s_2 = A_2 = 5 \times 15 = 75m$

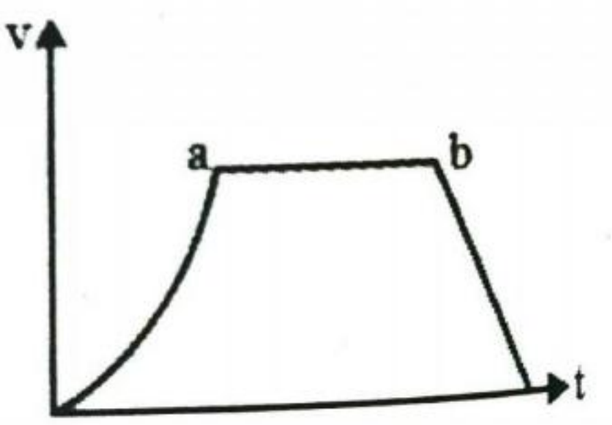
$s_T = s_1 + s_2 = 20 + 75 = 95m$

$t_T = 25$ sec

$$v_{av} = \frac{s_T}{t_T} = \frac{95}{25} = 3.8m/s$$

Answer: C

128. The following v – t graph represents motion of a moving body along a straight line. What does a horizontal line from a to b represents in this graph?



A. Uniform motion
B. Uniform declaration
C. Uniform acceleration
D. Variable acceleration

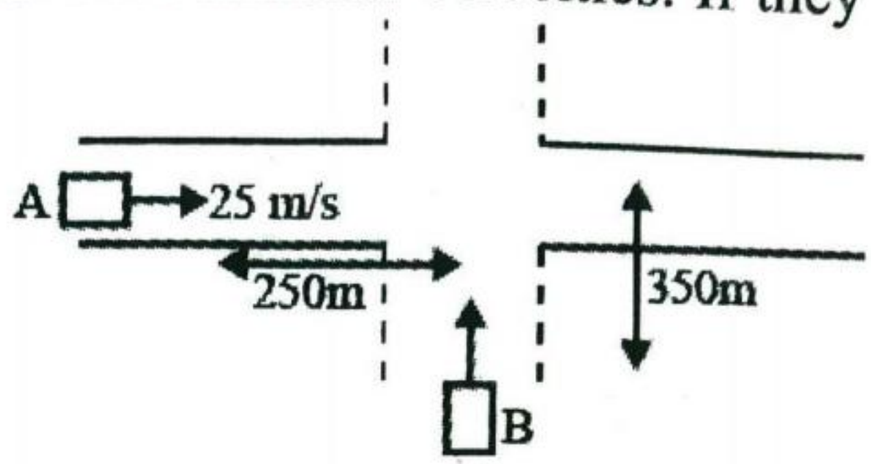
Answer: A

129. A car driver observes a motorcycle moving at 20 km/hr in the same direction relative to him. If the speed of the motorcycle relative to the ground is 70 km/h. What is the speed of the car relative to the ground?
A. 50 km/hr B. 60 km/h C. 40 km/hr D. 30 km/h

Solution: $v_{mc} = v_{mg} - v_{cg}$

$v_{cg} = v_{mg} - v_{mc} = 70 - 20 = 50$ km/h **Answer: A**

130. Two cars heading to a cross road with uniform velocities. If they collide at the center of the cross road, what must be the speed of car B?



A. 10 m/s C. 35 m/s

B. 25 m/s D. 30 m/s

Solution: When they collide, car A covers the 250m and car B covers the 350m.

$v_A = \frac{s_A}{t_A} \Rightarrow t_A = \frac{s_A}{v_A} = \frac{250}{25} = 10\,\text{sec}$ car B also travelled for 10 sec.

$v_B = \frac{s_B}{t_B} = \frac{350}{10} = 35\,\text{sec}$ **Answer: C**

131. The length of a train is 44m. Its front is 100m from a pole. it accelerates from rest at 0.5 m/s^2. At what speed do its front and rear pass the pole? Respectively.

A. 12 m/s, 44 m/s C. 10 m/s, 44 m/s

B. 10 m/s, 12 m/s D. 22 m/s, 50 m/s

Solution: Front of the train is 100m from the pole and its rear is 100m plus length of the train which is 100 + 44 = 144m from the pole. So we have;

Front, $v_f = u^2 + 2as_f = 0 + 2 \times 0.5 \times 100$, $v_f = 10$ m/s

Rear, $v_r = u^2 + 2as_r = 0 + 2 \times 0.5 \times 144$, $v_r = 12$ m/s **Answer: B**

132. If the distance x of an object is related to its speed v according to the relation x = kv, the constant k has a dimension of:

A. Acceleration B. Time C. length D. mass

Solution: From $x = kv$, $k = \frac{x}{v} = \frac{\text{meter}}{\text{meter/sec}} = \text{second}$

Thus, k has a time dimension. **Answer: B**

133. A traffic police on a motorbike spotted a car at a distance of 100m ahead of him on a straight level road. The car is moving uniformly at a velocity of 15 m/s. How long will it take the traffic police to overtake the car if he drives the bike at a velocity of 20 m/s?

A. 2.86 sec B. 20 sec C. 10 sec D. 5.56 sec

Solution: Distance moved by the car is $s_c = 100 + 15t$ distance moved by the motorbike $s_m = 20t$. Since after time t they meet, $100 + 15t = 20t \Rightarrow st = 100 \Rightarrow t = 20$ sec

Answer: B

134. An automobile driver spotted a track 50m ahead of him moving at a speed of 10 m/s. At that instant, the speed of the automobile was 20 m/s. How far should he drive before he overtakes the track?

A. 100m B. 50m C. 5m D. 150m

Solution: Distance moved by the automobile be s_a, its speed be v_a. And distance moved by the track be s_t, and its speed v_t we have;

$s_a - s_t = v_a t - v_t t = (v_a - v_t)t$

$$50 = (20 - 10)t \Rightarrow t = \frac{50}{10} = 5\,\text{sec}$$

Then $s_a = v_a t = 20 \times 5 = 100$m **Answer: A**

135. The velocity time graph of an object is shown below. Which statement is a correct sequence about the motion from t = 0 to t = 4 sec.

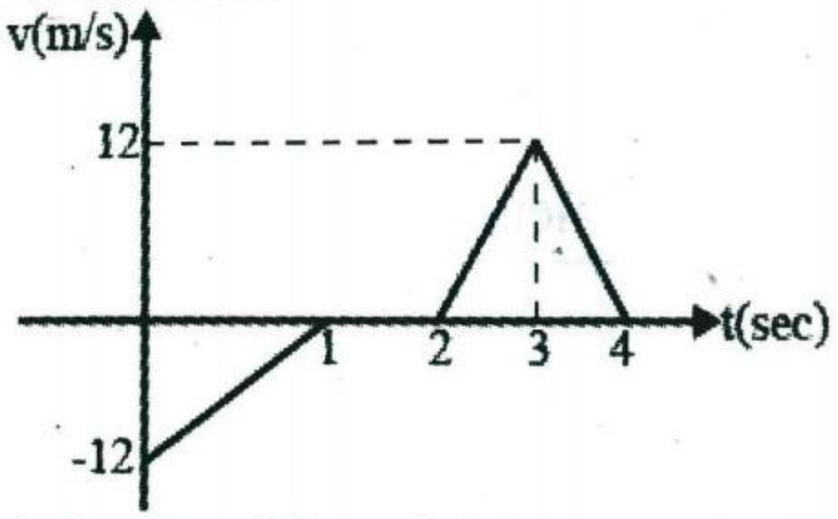
A. Accelerate, stop, accelerate, reverse direction, decelerate, stop

B. Decelerate, stop, reverse direction, accelerate, decelerate, stop

C. Decelerate, stop, accelerate, reverse direction, decelerate, stop

D. Accelerate, decelerate, accelerate, reverse direction, stop

Answer: A



136. A body with an initial speed of 1 m/s accelerates uniformly along a straight line at 2 m/s^2. What distance does it cover in the thrid second?

A. 4m B. 12m C. 8m D. 6m

Solution: $sn^{th} = u + \frac{a}{2}(2t_n - 1) = 1 + \frac{2}{2}(2 \times 3 - 1) = 6m$

Answer: D

137. A physics student shot a ball straight upward from a spring cannon. The ball's maximum height was found to be 80 cm. What was the ball's initial velocity as it left the muzzle of the cannon? (Use $g = 10\ m/s^2$)

A. 32 m/s B. 8 m/s C. 16 m/s D. 4 m/s

Solution: $v^2 = u^2 - 2gh$, $v = 0$

$u = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.8} = 4m/s$ **Answer: D**

138. An object is thrown down with initial speed of 12 m/s from a 20m high building. If the y axis is chosen to point upward, which of the following describe how the height of the object varies with time.

A. $y = 20 - 12t - \frac{1}{2}(9.8)t^2$ C. $y = 20 - 12t + \frac{1}{2}(9.8)t^2$

B. $y = 20 + 12t - \frac{1}{2}(9.8)t^2$ D. $y = 20 + (12)t + \frac{1}{2}(9.8)t^2$

Solution: $y + 12t + \frac{1}{2}(9.8)t^2 = 20$ **Answer: A**

Unit Two Conceptual Questions

1. Can a body have a constant velocity and still have a varying speed? Explain
2. Can a body have zero velocity and still be accelerated? Explain
3. Under what condition can the velocity change its direction while its acceleration is constant in magnitude.
4. Can a body have a constant speed and still have a varying velocity? Explain
5. Consider a ball that is through upward. What can you say about its acceleration at the top of its flight?
6. If a body starts from a point and returns back to the same points, what can you say about its?
   a) average speed
   b) average velocity
7. If a body moves with a uniform velocity, what can you say about its
   a) positions

b) acceleration
c) speed

8. If ball A is dropped from the top of a tower while ball B is thrown down, what can you say about the balls acceleration?
9. If the displacement of a particle is zero, then can we conclude that the distance covered by it is zero?
10. You observe a body covering distances in the direction proportional to the time elapsed. What conclusion might you draw about its acceleration?
11. A body having zero velocity will necessary have zero acceleration. Is this statement true?

Unit two: problems

Distance, displacement, average speed and average velocity

12. The earth is, on average 150 million km from the sun. Calculate its average speed in orbit.
13. A runner jogs 12 km North then turns and runs 16 km East in three hours.
    a) What is his displacement?
    b) Calculate his average speed.
    c) Calculate his average velocity.
14. A car is moving 12 km due North and then 16 km due east. What is its
    a) distance b) displacement
15. The motion of a body in a uniform motion is labeled as follows. What is the general relationship between s and t?

s(m)	0	8	16	24	32	40	48	56	64
t(sec)	0	1	2	3	4	5	6	7	8

16. The following table represents uniform motion. How long does it take for the body to move 100m?

s(m)	10	40	100
t(sec)	2	8	?

17. A car is moving with uniform velocity of 8 m/s east. How long does it take to cover 40m?
18. An athlete runs a rectangular field of side 100m by 80m in a minute. What is the average speed of the athlete?
19. The displacement of a certain body that is in motion is given by $s = 2t^2 + t + 10$. What is the average speed of the body for the first 10 seconds of its motion?

20. What is the average speed in km/h of a runner who run 1500m in 4 minutes?
21. A boy walks to school at 2.5 km/h and walks back with a constant speed of 4 km/h. Assuming the school is 6 m away, what is his average speed for the round trip?
22. A man leaves his house for a cycle ride. He comes back to his house after half an hour, after covering two km. What is his average velocity for the ride?
23. A car travels from A to B at a speed of 20 km/h and returns at a speed of 30 km/h. Find the average speed of the car for the whole journey.
24. A cheetah can run at 30 m/s, but only for about 12 sec. How far will it run in that time?

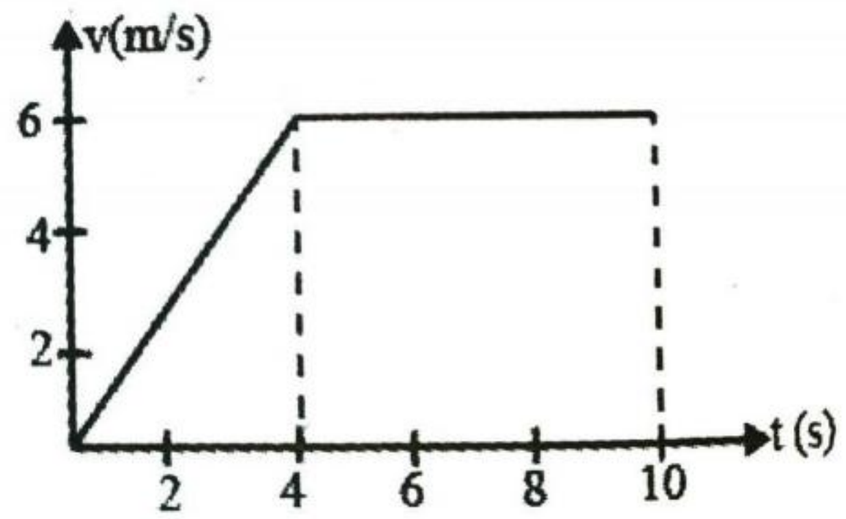
Uniform accelerated motion

25. A car accelerates from 10 m/s to 28 m/s in 6 sec. Find the average acceleration.
26. A bus accelerates from 10 m/s to 18 m/s in 3 sec. Find:
    a) The distance the bus covers.
    b) The acceleration of the bus.
27. A taxi is travelling at 15 m/s and accelerates at 3 m/s^2 for 4 sec. What is its new velocity?
28. A body initially at rest accelerates at a rate of 4 m/s^2. How long will it take the body to travel 200m?
29. A car starts from rest and accelerated at 4 m/s^2 for 5 sec. Calculate the velocity of the car in km/hr.
30. What is the distance covered by a body whose acceleration is 3 m/s^2 while its velocity changes from 20 m/s to 40 m/s?
31. How long does it takes a taxi to stop if it is initially moving at a speed of 72 m/s and then decelerating at 6 m/s^2?
32. A car starts from rest and acquires a velocity of 12 m/s. If it attains a constant acceleration of 3 m/s^2, what is the total distance it covers?
33. The speed of a car decreases uniformly from 80 m/s to 20 m/s in 6 sec. What is the distance travelled during the first 4 sec?
34. A body starts from rest and moves in a straight line with constant acceleration and covers a distance of 128 m in 4 sec. What is the distance covered by the body as it moves at 32 m/s?
35. A boat that starts from rest accelerates at 12 m/s^2. What is the distance traveled by the boat on the 8^{th} second?

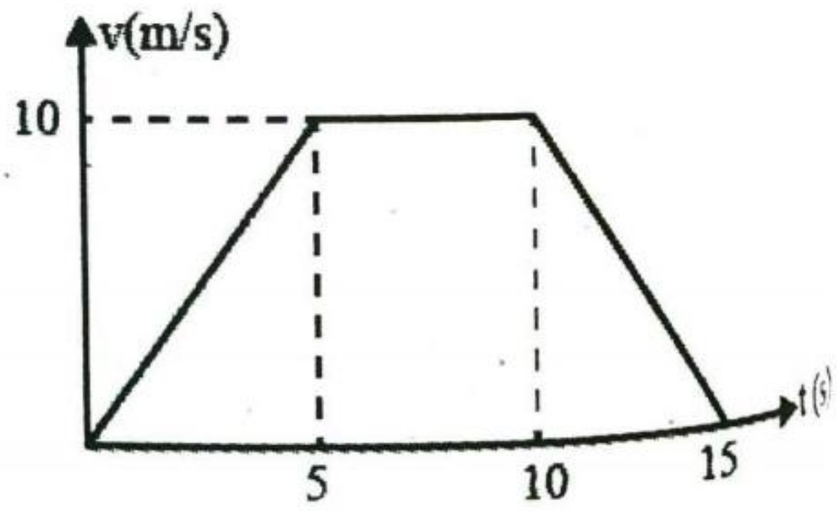
36. A body moving at a speed of 10 m/s attains a speed of 20 m/s in 100m. What is the acceleration of the body?
37. A certain body starts from rest and moves with a constant acceleration of 0.05 m/s^2. What is the distance it moved in one minute?
38. A body that was moving at a speed of 10 m/s accelerates uniformly at 0.1 m/s^2. What is the distance it moved in 100 seconds?
39. A body moving at 5 m/s^2 has an initial velocity of 50 m/s. In what time will its velocity be doubled?
40. If a body having initial velocity of zero is accelerating uniformly at 8 m/s^2, what is the distance travelled in the fifth second?
41. The velocity of a body moving with uniform acceleration of 2 m/s^2 is 10 m/s. What is its velocity after 4 seconds?
42. A trolley runs down a slope from rest with uniform acceleration. In the first second, it travels 1.6m. What is its acceleration?
43. A body travels 2m in the first 2 sec and 2.2m in the next 4 sec with uniform deceleration. What is the velocity of the body at the end of 9 seconds?
44. A body starts from rest and moves with a uniform acceleration of 2 m/s^2 in a straight line.
    a) What is its velocity after 5 sec?
    b) How far has it travelled in this time?
    c) After how long will the body be 100m from its starting point?

Graphical description of uniformly accelerated motion

45. From the given velocity – time graph, find
    a) The acceleration of the body during the first 4 seconds?
    b) The total distance it travelled.

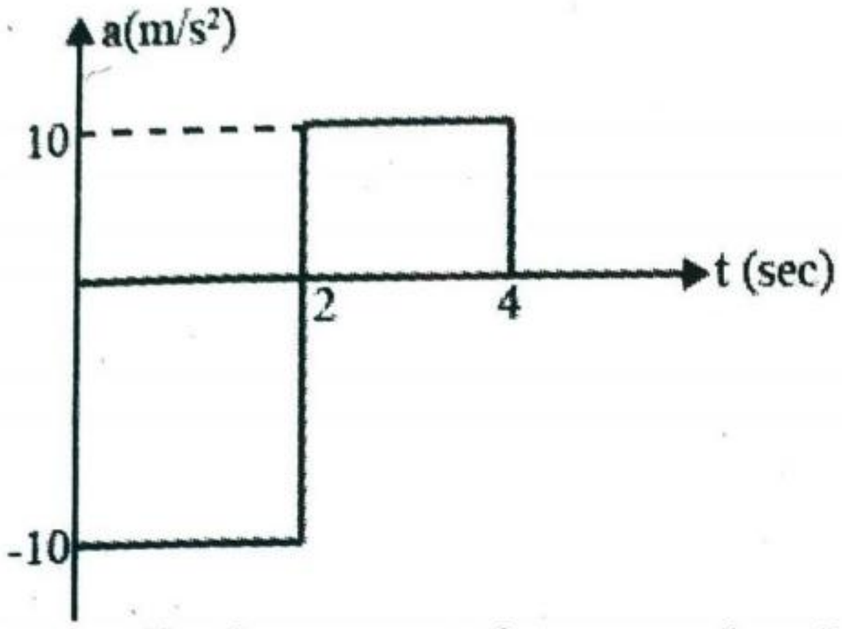


46. The velocity time graph of a train is depicted in the figure below. Find:
    a) The maximum acceleration of the train.

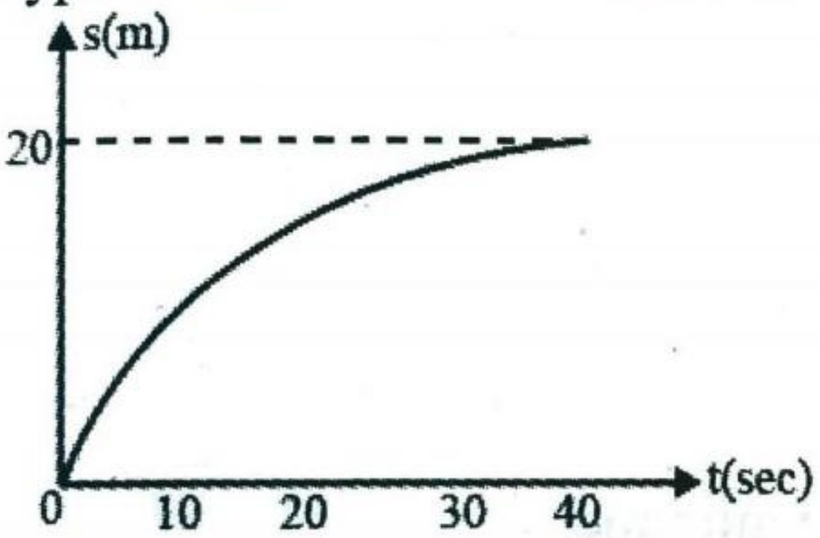


b) The total distance covered.

47. A particle starts from rest and moves at constant acceleration. At what time will the speed be maximum for more information look at the following acceleration time graph.



48. The displacement time graph of body looks the following. What type of motion does it indicate?



Free fall

49. A stone is dropped off a building. It takes 5.3 sec to hit the ground. Determine the height of the building.
50. A stone is dropped from the top of a 45m high building. How fast will it be moving when it reaches the ground?
51. A football player kicks a ball upwards giving it an initial speed of 16 m/s. What is the maximum height to which the ball will rise?
52. A body falls from a cliff 125m high. Find the time of fall. How far did body fall in half of this time?
53. A particle is dropped a height of 3.46958m in what time the particle would travel the first meter distance? (use $g = 10\ m/s^2$)

Relative velocity in one dimension

54. An air craft heads North at 320 km/h relative to the wind. The wind velocity is 80 km/h from the North. Find the velocity of the aircraft relative to the ground.

55. Two aircrafts P and Q are flying at the same speed, 300 m/s. If their directions are at right angles, find the magnitude of the velocity of aircraft p relative to aircraft Q.
56. What is the relative velocity of a car moving at 40 km/h due east with respect to itself?
57. A train is moving east at a speed of 5 m/s. A bullet fired westward with a velocity of 10 m/s crosses the train in 8 seconds. What is the length of the train?

Unit two: Explanation to conceptual Questions

1. No! When the velocity is constant, then its magnitude that is speed is constant and its direction is fixed.
2. Yes! Think of a body through upward, at its maximum height, its velocity is zero and its acceleration is constant.
3. When the body is in free fall, that is when it is thrown upward and then return or when the body is moving in circular path.
4. Yes! This is so when the body is under uniform circular motion.
5. It is constant. That is acceleration due to gravity.
6. a) Its average speed is non zero, $v_{av} = 2s/t$

   b) Its average velocity is zero, because, $\overline{v}_{av} = \frac{\Delta \overline{s}}{\Delta t}$, $\Delta s = 0$.
7. a) Its position changes with time.

   b) Its acceleration becomes zero, because, $a = \frac{\Delta v}{\Delta t}$, $\Delta v = 0$.

   c) Its speed becomes constant.
8. It is all the same. Both will have an acceleration due to gravity, $g = 9.81\ m/s^2$.
9. No! If a body travels a distance s to the right and then returns to the left along the same road, then its displacement is zero, but its distance is $s_T = 2s$.
10. If this is so, then we can say, the body is in uniform motion in straight line. Where it covers equal distances in equal time intervals hence, its acceleration is zero.
11. No! Again, think of a body in free fall.

Unit two: Solution to problems

12. $$v_{av} = \frac{s_T}{t_T} = \frac{2\pi r}{t} = \frac{2 \times \pi \times 15 \times 10}{365 \times 24 \times 60 \times 60} = 2985.8 m/s \approx 30,000 m/s$$

13. a) $\vec{s} = \sqrt{(12)^2 + (16)^2} = 20\text{km}$

b) $v_{av} = \dfrac{s_T}{t_T} = \dfrac{12\text{km} + 16\text{km}}{3\text{h}} = 9.3\text{km/h}$

c) $\vec{v}_{av} = \dfrac{\Delta s}{\Delta t} = \dfrac{20}{3} = 6.67\text{km/h}$

14. a) $s = s_1 + s_2 = 12\text{ km} + 16\text{ km} = 28\text{ km}$

b) $\overline{s} = \sqrt{s_1^2 + s_2^2} = \sqrt{(12) + (16)} = 20\text{km}$

15. $s = 8t$

16. $v = \dfrac{s}{t} = \dfrac{10}{2} = 5\text{m/s}, t = \dfrac{s}{v} = \dfrac{100}{5} = 20\,\text{sec}$

17. $s = vt \Rightarrow t = s/v = 40/8 = 5\text{ sec}$

18. $v_{av} = \dfrac{s_T}{t_T} = \dfrac{2(100+80)}{10} = 6\text{m/s}$

19. $v_{av} = \dfrac{s_T}{t_T} = \dfrac{2(10)+10+10}{10} = 22\text{m/s}$

20. $v_{av} = \dfrac{s_T}{t_T} = \dfrac{1500}{4\times 60} = 6.25\text{m/s} = 22.5\text{km/h}$

21. $v_{av} = \dfrac{s_T}{t_T} = \dfrac{6+6}{\frac{6}{2.5}+\frac{6}{4}} \approx 3\text{km/h}$

22. $v_{av} = \dfrac{\Delta s}{\Delta t} = 0$

23. $v_{av} = \dfrac{s_T}{t_T} = \dfrac{2s}{\frac{s}{20}+\frac{s}{30}} = \dfrac{2s}{\frac{3s+2s}{60}} = \dfrac{120s}{5s} = 24\text{km/h}$

24. $s = vt = 30 \times 12 = 360\text{m}$

25. $a = \dfrac{\Delta v}{\Delta t} = \dfrac{28-10}{6} = 3\text{m/s}^2$

26. a) $s = \frac{(u+v)}{2}t = \frac{(10+18)}{2}3 = 57\text{m}$

b) $a = \frac{\Delta v}{\Delta t} = \frac{18-10}{3} = 2.67\text{m/s}^2$

27. $v = u + at = 15 + 3 \times 4 = 27\ \text{m/s}$

28. $v^2 = u^2 + 2as \Rightarrow v = \sqrt{2as} = \sqrt{2 \times 4 \times 200} = 40\text{m/s}$

$a = \frac{v-u}{t} \Rightarrow t = \frac{v-u}{a} = \frac{40-0}{4} = 10\,\text{sec}$

29. $v = u + at = 0 + 4 \times 5 = 20\ \text{m/s} = 3.6 \times 20 = 72\ \text{km/h}$

30. $v^2 = u^2 = 2as \Rightarrow s = \frac{v-u}{2a} = \frac{(40)-(20)}{2 \times 3} = 200\text{m}$

31. $v = u + at \Rightarrow t = \frac{v-u}{a} = \frac{0-72}{-6} = 12\,\text{sec}$

32. $s = \frac{v^2 - u^2}{2a} = \frac{(12)^2 - 0}{2 \times 3} = 24\text{m}$

33. $a = \frac{v-u}{t} = \frac{20-80}{6} = -10\text{m/s}^2$

$s = ut + \frac{1}{2}at^2 = 80 \times 4 + \frac{1}{2} \times (-10)(4)^2 = 4\text{m}$

34. $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}a(4) = 128 \Rightarrow a = 16\text{m/s}^2$

$v^2 = u^2 + 2as \Rightarrow s = \frac{v^2 - u^2}{2a} = \frac{(32)^2}{2 \times 16} - 0 = 32\text{m}$

35. $S_n^{th} = u + \frac{a}{2}(2t_n - 1) = 0 + \frac{12}{2}(2 \times 8 - 1) = 42\text{m}$

36. $v^2 = u^2 + 2as \Rightarrow a = \frac{v^2 - u^2}{2s} = \frac{(20)^2 - (10)^2}{2 \times 100} = 1.5\text{m/s}^2$

37. $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 0.05 \times (60)^2 = 90\text{m}$

38. $s = ut + \frac{1}{2}at^2 = 10 \times 100 + \frac{1}{2} \times 0.1 \times (100)^2 = 1500\text{m}$

39. $v = u + at \Rightarrow 2(50) = 50 + 5t \Rightarrow t = 10 \text{ sec}$

40. $s = u + \frac{a}{2}(2t - 1) = 0 + \frac{8}{2}(2 \times 5 - 1) = 36\text{m}$

41. $v = u + at = 10 + 2 \times 4 = 18 \text{ m/s}$

42. $s = ut + \frac{1}{2}at^2 \Rightarrow 1.6 = 0 + \frac{1}{2}a(1)^2 \Rightarrow a = 3.2\text{m/s}^2$

43. $s = ut + \frac{1}{2}at^2$, at $t = 2$ sec, $2 = 2u + \frac{1}{2}a(2)^2 \Rightarrow u + a = 1$

at $t = 2 + 4 = 6$ sec, $2 + 2.2 = 6u + \frac{1}{2}a(6)^2 \Rightarrow u + 3a = 0.7$

From the two equations, $u = 1.15$ m/s and $a = -0.15$ m/s^2

at, $t = 9$ sec, $s = ut + \frac{1}{2}at^2 = (1.15 \times 9) - \frac{1}{2} \times 0.15 \times 9^2 = 4.3\text{m}$

44. a) $v = u + at = 0 + 2 \times 5 = 10$ m/s

b) $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times (2) \times (5)^2 = 25\text{m}$

c) $v^2 = u^2 + 2as \Rightarrow v = \sqrt{2as} = \sqrt{2 \times 2 \times 100} = 20\text{m/s}$

$$a = \frac{v - u}{t} \Rightarrow t = \frac{v - u}{a} = \frac{20 - 0}{2} = 10\,\text{sec}$$

45. a) $a = \frac{\Delta v}{\Delta t} = \frac{6}{4} = 1.5\text{m/s}^2$

b) $s = A_1 + A_2 = \frac{1}{2} \times 6 \times 4 + 6 \times 6 = 48\text{m}$

46. a) $a = \frac{\Delta v}{\Delta t} = \frac{10}{5} = 2\,\text{m/s}^2$

b) $s = A_1 + A_2 + A_3 = \frac{1}{2} \times 10 \times 5 + 5 \times 10 + \frac{1}{2} \times 10 \times 5 = 100\text{m}$

47. At, $t = 2$ sec

48. Retared motion, where the body finally stops moving.

49. $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (5.3)^2 = 140.45m$

50. $v^2 = u^2 + 2gh \Rightarrow v = \sqrt{2gh} = \sqrt{2 \times 10 \times 45} = 30m/s$

51. $v^2 = u^2 + 2gh \Rightarrow h = \frac{u^2}{2g} = \frac{(16)^2}{2 \times 10} = 12.8m$

52. $h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 125}{10}} = 5\,sec$

$$h = \frac{1}{2}g\left(\frac{t}{2}\right)^2 = \frac{1}{2} \times 10 \times \left(\frac{5}{2}\right)^2 = 31.25m$$

53. $h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 1}{10}} = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}}\,sec$

54. $v_{aw} = 320$ km/h, $v_w = 80$ km/h

55. $v_{PQ} = \sqrt{v_P^2 + v_Q^2} = \sqrt{(300)^2 + (300)^2} = 424.3m/s$

56. $v_{CC} = v_C - v_C = 40 - 40 = 0$ km/h

57. $v_{tb} = v_t + v_b = 5 + 10 = 15$ m/s
$s = vt = 15 \times 8 = 120m$

Unit Three

Force and Newton's Laws of Motion

Whether we know it or not, forces are all around us. From keeping buildings, bridges and other structures standing on the earth, to moving the earth and other planets around the sun. In sum, we experience forces of different kind every single day of our lives.

3.1 Forces in nature

Push or pull your book across your desk. The push or the pull applied on the book is called *force*.

> **Force:** Is the push or pull action of a body on another body.

Force is a vector quantity and its SI unit is Newton (N).

☞ **The four basic force in nature**

1. **Gravitational force:** Is the force of attraction between any two masses.
2. **Electromagnetic force:** Is a force that holds atoms and molecules together.
3. **Strong nuclear force:** Is the force that binds protons and neutrons together in the nucleus.
4. **Weak nuclear force:** Is the force that causes atoms to break apart, thus this force is responsible for the radioactive decay such as the Beta decay.

Contact and Non contact forces

The basic forces that exist in nature can be of *contact* or *non contact* type.

> **Contact force:** Is a force that is transmitted by touching the body while exerting force on the body.

Examples of contact force includes:

→ Writing a note on your exercise book.
→ Holding your physics book in the air.
→ Kicking a ball to your friend.
→ Pushing or pulling a table across the floor.
→ opening or closing the door etc.

Non contact force: Is a force that exerts a push or a pull effect on a body at a distance without touching the body.

Example of non contact forces includes.

→ Gravitational fórce
→ Magnetic force
→ Electrostatic force . . . etc

Effects of force

Whenever a force is applied on an object, it will cause the object to:

→ Change shape or size, which we call *deformation*
→ Change direction of motion
→ Change the state of motion of the object, that is, it will cause the body to change its speed and hence the body may move or stop.

Newton's law of motion

Newton formulated three surprisingly simple yet powerful laws of motion. We are all subjected to Newton's laws of motion. You cannot move your body, ride a bicycle or kick a ball in a way that violates his rules.

Newton's first law (Law of inertia)

According to this law, if you put your book on a table, it will be there forever, and if you kick a ball, it will move at constant velocity and keeps moving forever unless an external force acts on it.

Newton's firs law: States that; "a body at rest will remain at rest and a body in motion will continue its motion with constant speed in a straight line unless an external force acts on it."

☞ Newton's first law is also known as *law of inertia*. This is so because, the literal meaning of inertia is "**laziness**" speaking loosely, we can say, matter is "**lazy**" because, it will not change its state of motion unless forced to do so. That is, if an object is at rest, it will not start moving by itself, and if it is already moving with constant velocity it will not change its speed or direction unless an external force causes it to change.

Mass and Inertia

It is a common experience that, if you are in a car that is initially at rest, then your body will tend to fall backward when the car suddenly starts moving and your body will be thrown forward when the car stops suddenly. This is so because, all bodies do have a natural property to resist any change to their state of motion. That is, if they are at rest, they

want to be at rest forever and if they are in motion they want to be in motion forever. This inherent property of bodies is called *inertia*.

Inertia: Is the tendency of an object to resist changes to its motion.

☞ The inertia of a body depends on its mass. This can be explained as follows; It will be easy for you to stop a moving ball than to stop a car moving at the same speed. It will also be easy to kick a small stone than to kick a large rock. This is so because a body with small mass has a small inertia and requires a small force to change its state of motion, and a body with large mass has a large inertia and requires a large force to change its state of motion so we can say.

The larger the mass of the body, the greater its inertia.

☞ **Remember**

According to Newton's first law, being at rest and moving with constant velocity are practically the same.

Conceptual Examples

1. Is it possible to have motion in the absence of force? Explain

   Explanation: Yes. Motion requires no force. Newton's first law says an object in motion continues to move by itself in the absence of external force.

2. Is it possible to have force in the absence of motion?

   Explanation: Yes. Equal and opposite forces may act on a body and the body may not move.

3. When a dog gets wet, it shakes its body from head to tail to remove the water. How is this possible?

   Explanation: According to Newton's first law, the dog uses the principle of inertia to shake water from its body. As he shakes, he is applying a net force on the water particles and hence the particles start moving from its body.

4. If you kick a ball, the ball will stop moving after some time. Does this mean that Newton's first law is wrong? Explain

   Explanation: Not really. Newton's first law is valid when there is no external force acting. In the case of the ball, external forces such as air resistance and friction are acting on it and instead of moving forever, the ball slows down and finally stops.

5. If a car is traveling at 120 km/h in a straight line, the passengers in the car will be thrown forward when the car suddenly stops. Why?

   Explanation: This is because, at first, the passengers are also moving at 120 km/h and from that time on according to Newton's first law, they keep travelling at 120 km/h.

6. Why do passengers get thrown to one side when a car moving in a straight line suddenly turns around a curve? Explain

   Explanation: When the car turns, the turning force acts on the car and not on the passengers. By Newton's first law, the passengers body continues moving in a straight line at constant speed.

Hooke's Law

The effect of force on springs was studied by Robert Hooke. In his experiment, he used springs fixed at one end and applied a force to the other end of the spring. He found that, the applied force and the extension of the spring are proportional. Which means, when the force doubled, the extension is doubled and when the force is tripled, the extension is also tripled etc.

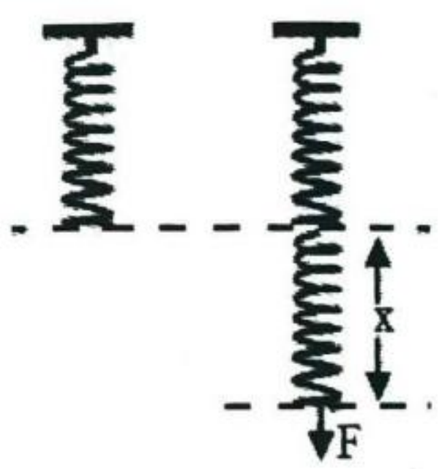
> **Hooke's Law:** The applied force to spring is directly proportional to the extension produced up to the elastic limit.

In symbols, $\boxed{F = kx}$

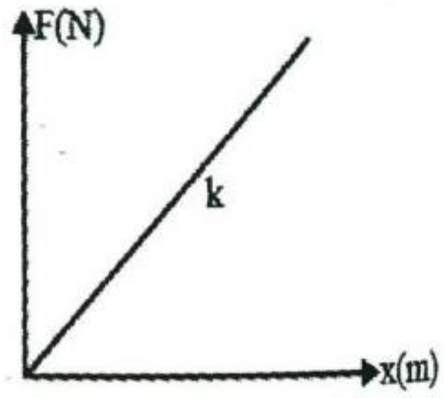
States that; Where k = spring (force) constant

x = extension (stretch)

F = force applied



Graphically, Hook's law can be visualized by a straight line graph of force – extension as follows. The gradient, that is the slope of the force – extension graph gives the spring constant (k).



The elastic limit

If we keep on applying a force on the spring, at some point in the process, the spring will break and hence will be deformed permanently.

> **Elastic Limit**: Is a point up to which a spring stretches elastically and beyond which the spring is no more elastic.

Elastic and plastic deformations

Materials may deform *elastically* or *plastically*.

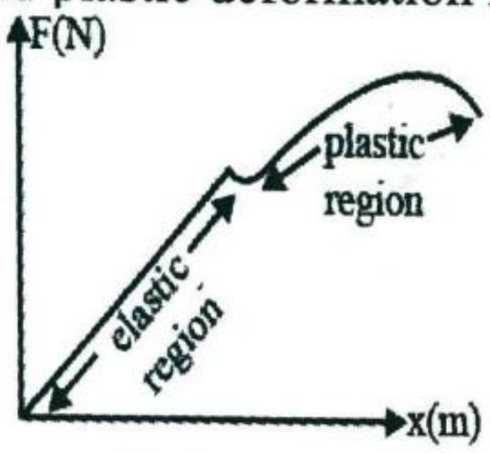
Elastic deformation: Is a deformation where the applied force is proportional to the extension and the spring returns to its original position when the applied force is removed.

☛ **Remember**

Hooke's law holds true if and only if the applied force is directly proportional to the extension.

Plastic deformation: Is a deformation where the applied force is not proportional to the extension and the spring will not return to its original position when the applied force is removed.

The regions for elastic and plastic deformation is shown below.



Springs in series and in parallel connection

a) springs in series

Two springs are said to be connected in series when they are connected one next to the other as shown below. The effective spring constant of the combination is given by; $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow \boxed{k = \frac{k_1 k_2}{k_1 + k_2}}$

Where: k, effective spring constant

k_1, spring constant of spring one

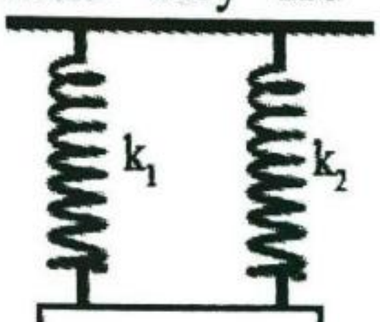
k_2, spring constant of spring two

b) springs in parallel

Two springs are said to be connected in parallel when they are connected side by side as shown below.

The effective spring constant of the combination is given by:

$$\boxed{k = k_1 + k_2}$$



Illustrative Examples

7. What force is required to stretch a spring of force constant 400 N/m by 4 cm?

Solution: $F = kx \rightarrow$ hooke's law

$F = (400)(0.04) = 16N$

8. A 6 kg mass is suspended on a vertical spring of force constant 200 N/m. By how much does the spring stretch? (Take g = 10 m/s^2)

Solution: F = kx, here F = mg

$$\Rightarrow x = \frac{F}{k} = \frac{mg}{k} = \frac{(6)(10)}{200} = 0.3m = 30cm$$

9. What is the extension of the system when a block of mass 5 kg is suspended on two identical springs of force constant 200 N/m when they are connected.

a) in series

b) in parallel (take g = 10 m/s^2)

Solution:

a) $$k = \frac{k_1 k_2}{k_1 + k_2} = \frac{(200)(200)}{200 + 200} = 100N/m$$

from F = kx, $x = \frac{F}{k}$, but F = mg

$$x = \frac{mg}{k} = \frac{(5)(10)}{100} = 0.5m = 50cm$$

b) $k = k_1 + k_2 = 200 + 200 = 400$ N/m,

from, F = kx,

$x = \frac{F}{k}$, but F = mg; $x = \frac{mg}{k} = \frac{(5)(10)}{400} = 0.125m = 12.5cm$

10. In his experiment, a grade 10 student reported the force and stretch data as follows. What is the average force constant of the spring?

F(N)	0.2	0.4	0.6	0.8
x(m)	0.01	0.025	0.03	0.042

Solution: First find the value of k for each reading

$$k_1 = \frac{F_1}{x_1} = \frac{0.2}{0.01} = 20N/m \qquad k_3 = \frac{F_3}{x_3} = \frac{0.6}{0.03} = 20N/m$$

$$k_2 = \frac{F_2}{x_2} = \frac{0.4}{0.025} = 16N/m \qquad k_4 = \frac{F_4}{x_4} = \frac{0.8}{0.042} = 19N/m$$

The average force constant is then:

$$k = \frac{k_1 + k_2 + k_3 + k_4}{4} = \frac{20 + 16 + 20 + 19}{4} = 18.75N/m$$

11. What will be the extension of the system when a force of 100N acts on two identical springs of force constant 1000 N m when they are connected.

a) in series

b) in parallel

Solution

a) in series connection, the effective force constant is,

$$k = \frac{k_1 k_2}{k_1 + k_2} = \frac{(1000)(1000)}{1000 + 1000} = 500 N/m$$

$$\Rightarrow F = kx,\ x = \frac{F}{k} = \frac{1000}{500} = 0.5m$$

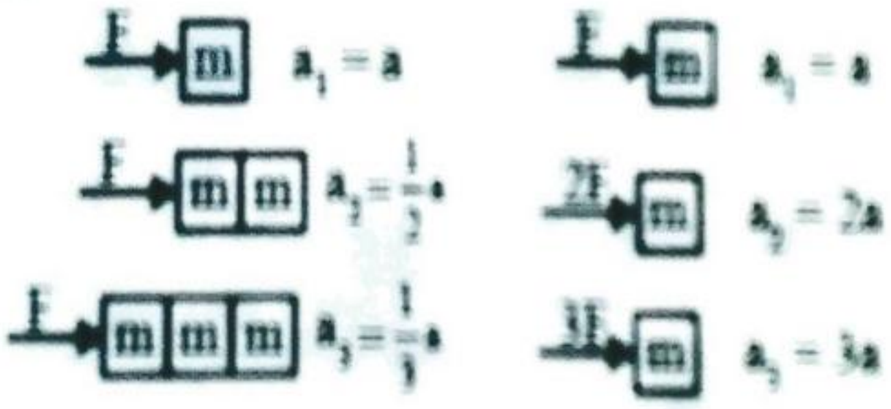
b) in parallel connection, the effective force constant is,

$$k = k_1 + k_2 = 1000 + 1000 = 2000\ N/m$$

$$\Rightarrow F = kx,\ x = \frac{F}{k} = \frac{1000}{2000} = 0.05m$$

3.2 Newton's second law (law of acceleration)

Newton found the acceleration of a body to increases with force and decreases with mass. That is keeping the mass constant, the acceleration doubled when the applied force is doubled and keeping the applied force constant, the acceleration decreased by half when the mass is doubled. That is;



That is for a constant force, the acceleration is small for a large mass and the acceleration is large for a small mass

$$M_a = {}_M a$$

This fact is explained using Newton's second law.

> **Newton's second law:** states that;
> "The acceleration of an object is directly proportional to the applied net force and inversely proportional to the mass of the object."

In symbols, $\boxed{F = ma}$

Mass and Weight

These two quantities may seem to mean the same thing and usually used interchangeably. Yet they have different meanings.

Mass: Is the amount of matter contained in a body.

Mass is a scalar quantity and its SI unit is kilogram (kg). It is constant everywhere.

Weight: Is the force of gravitational pull of the earth on the object.

Weight is a vector quantity and its SI unit is Newton (N). It varies from place to place.

Mathematically, weight is given by: $\boxed{W = mg}$

Where; W → weight

m → mass of the body

g → acceleration due to gravity

Variation of weight with location

The weight of a 10 kg body is about 100N on the earth's surface and this very same object will weight only about 16N on the surface of the moon. This variation is due to the change in the gravitational acceleration. *Acceleration due to gravity increases with increasing latitude and decreases with increasing altitude*. For instance g = 9.83 m/s^2 at the North pole, g = 9.7 m/s^2 at the equator and its is about 1.6 m/s^2 on the surface of the moon.

True Weightlessness and apparent Weightlessness

If the gravitational field strength is zero, then there would be no gravitational acceleration and hence, the body will float since its weight become zero.

True Weightlessness: Is when there is no gravitational field strength. Here, weight is zero.

The body may have apparent weightlessness if the only force acting on the body is its weight.

Apparent Weightlessness: Is when the only force acting on the body is gravity.

The apparent weight of a body

The apparent weight of a body is the resultant of the real weight and any contact force acting on it.

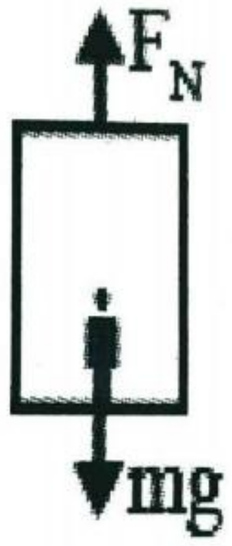
The apparent weight of a body can best be explained when the body is moving in an elevator. It is a common experience that, we feel **heavier**

when an elevator moving downward comes to rest and we feel **lighter** when an elevator moving upward comes to rest. But why?

☛ The reason is that our sensation of weight in this case is due to the force exerted on our feet by the floor of the elevator. If the force is greater than our weight, **mg**, we feel heavy and if it is less than **mg** we feel light.

Now, imagine you are in an elevator:

(1) **When the elevator moving upward with an acceleration *a***. The forces acting on you are your weight (mg) and the upward force exerted on your feet by the floor of the elevator (F_N). Now according to Newton's second law,



$F_N - mg = ma \Rightarrow F_N = mg + ma$

$\Rightarrow \boxed{F_N = m(g+a)}$

☛ Since F_N is greater than your weight mg, you feel *heavier*.

(2) **When the elevator accelerating downwards**. Here we have,

$mg - F_N = ma$

$\Rightarrow \boxed{F_N = m(g-a)}$

☛ In this case, you feel *lighter* than usual.

(3) **When the elevator is at rest or moving at constant velocity in any direction**.

$F_N - mg = ma$, but here, $a = 0$

$\Rightarrow \boxed{F_N = mg}$

☛ Here, your apparent weight is equal to your *true weight*.

(4) **When the elevator cable breaks and the elevator falls freely**.

$mg - F_N = ma$, but $a = g$

$\Rightarrow \boxed{F_N = 0}$

☛ Now you will feel a state of *weightlessness*.

Conceptual Examples

12. If a body is not accelerating, does it mean that no force is acting on it.

Explanation: No. It means, no net force is acting on it.

13. If gold were sold by weight, would you prefer to buy it in Dalule or in Ras Dashen? Why?

Explanation: In Ras Dashen. Because g is smaller there.

14. Where would an object have greater weight, at the North Pole or at the Equator? Why?

Explanation: At the North Pole. This is so because, the pole is closer to the center of the earth and gravitational force is maximum at the earth's center.

15. Is it possible for an object to be in motion and yet have zero net force acting on it? Explain

Explanation: Yes. an object with zero net force acting on it has a constant velocity.

Illustrative Examples

16. A constant force acts on a 10 kg object and increases its velocity from 8 m/s to 20 m/s in 3 sec. Find the force applied.

Solution: $F = ma$, but $a = \frac{v-u}{t} = \frac{20-8}{3} = 4m/s^2$

$\Rightarrow F = (10)(4) = 40N$

17. A certain force gives an acceleration of 2 m/s^2 to a 5 kg mass. What acceleration will this very same force give to a 20 kg mass?

Solution: $F = m_1a_1$ . . . (*), and

$F = m_2a_2$ . . . (**), dividing the two equations

$m_2a_2 = m_1a_1$

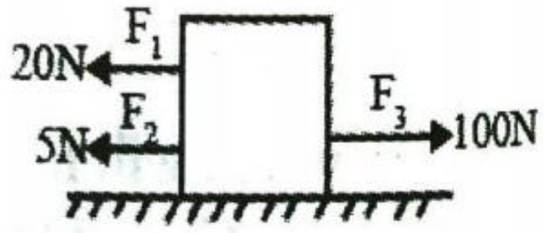
$\Rightarrow a_2 = m_1\frac{a_1}{m_2} = \frac{(5)(2)}{20} = 0.5m/s^2$

18. The net force of the given system of forces give the block resting on a smooth surface an acceleration of 3 m/s^2. Determine the mass of the block.

Solution: $F_{net} = F_3 - F_1 - F_2 = 100 - 20 - 5 = 75N$

$F = ma$

$\Rightarrow m = \frac{F}{a} = \frac{75}{3} = 25kg$



19. A force of 100N acts on a 20 kg body. **Calculate** the acceleration of the body when the

a) Force is halved
b) Force is doubled
c) mass is halved
d) mass is doubled

Solution:

a) $\frac{F}{2} = \frac{100}{2} = 50N$ and $a = \frac{F'}{m} = \frac{50}{20} = 2.5m/s^2$

b) $2F = (2)(100) = 200N$ and $a = \frac{F'}{m} = \frac{200}{20} = 10m/s^2$

c) $\frac{m}{2} = \frac{20}{2} = 10kg$ and $a = \frac{F}{m'} = \frac{100}{10} = 10m/s^2$

d) $2m = (2)(20) = 40\ kg$ and $a = \frac{F}{m'} = \frac{100}{40} = 2.5m/s^2$

20. A pick up car of mass 1000 kg accelerates at 3 m/s^2. What will be the acceleration of the car if a block of mass 200 kg is added o the top of it. Assume the force is constant.

Solution: $F = m_1a_1 \ \ldots (*)$

$F = m_2a_2 \ldots (**), m_2 = m_1 + 200 = 1200$

Taking ratio; $m_2a_2 = m_1a_1 \Rightarrow a_2 = \frac{m_1}{m_2}a_1 = \frac{(1000)}{1200}(3) = 2.5m/s^2$

21. A force of 6N acts on a 4 kg body that is initially at rest. At what time will the body attain a velocity of 25 m/s?

Solution: $F = ma \Rightarrow a = \frac{F}{m} = \frac{6}{4} = 1.5m/s^2$ and

$$a = \frac{v-u}{t} \Rightarrow t = \frac{v-u}{a} = \frac{25-0}{1.5} = 17\,sec$$

22. A certain force gives mass m_1 an acceleration of 4 m/s^2 and mass m_2 an acceleration of 0.5 m/s^2. What acceleration will this force give to:

a) $m_1 + m_2$ b) $m_2 - m_1$

Solution:

a) $F = ma, m_1a_1 = m_2a_2 \Rightarrow m_2 = \frac{a_1}{a_2}m_1$

$F = 4m_1 = 0.5m_2$

Since $a = \frac{F}{m} = \frac{4m_1}{m_1 + m_2} = \frac{4m_1}{m_1 + \frac{a_1}{a_2}m_1} = \frac{4m_1a_2}{m_1a_2 + m_1a_1}$

$$a = \frac{4a_2}{a_1 + a_2} = \frac{(4)(0.5)}{4 + 0.5} = 0.4 m/s^2$$

b) $$a = \frac{F}{m} = \frac{4m_1}{m_2 - m_1} = \frac{4m_1}{\frac{a_1}{a_2} m_1 - m_1} = \frac{4m_1 a_2}{a_1 m_1 - a_2 m_1}$$

$$a = \frac{4a_2}{a_1 - a_2} = \frac{(4)(0.5)}{4 - 0.5} = \frac{2}{3.5} = 0.6 m/s^2$$

23. If a force of 100N displaces a 6 kg body from rest to 50m in 10 second. What force is required to displace half of this body over the same distance during half this time?

Solution: $F_1 = m_1 a_1$, but $s_1 = \frac{1}{2} a t_1^2 \Rightarrow a_1 = \frac{2s}{t_1^2}$

$$F_2 = m_2 a_2,\ m_2 = \frac{m_1}{2}, s_1 = s_2 \text{ and } t = \frac{t_1}{2}$$

Taking ratio,

$$\frac{F_2}{F_1} = \frac{m_2 a_2}{m_1 a_1} = \frac{\frac{1}{2} m_1 \left(\frac{2s}{\left(\frac{t_1}{2}\right)^2} \right)}{m_1 \left(\frac{2s}{t_1^2} \right)} = \frac{\left(\frac{1}{2} m_1 \frac{2s}{\frac{t_1^2}{4}} \right)}{\left(m_1 \frac{2s}{t_1^2} \right)} = 2$$

$F_2 = 2F_1 = (2)(100) = 200N$

24. A car of mass 2000 kg starts moving with an acceleration which remains constant for 10 sec. If the car covers a distance of 500m. What is the force applied?

Solution: $F = ma,\ s = \frac{1}{2} a t^2 \Rightarrow a = \frac{2s}{t^2}$

$$F = m\left(\frac{2s}{t^2}\right) = (2000)\frac{(2)(500)}{(10)} = 20{,}000N = 20KN$$

25. A 4 kg mass is moving east at 5m/s. What will its speed after 10 seconds be if a force of 100N acts on it?

a) due east b) due west

Solution:

a) $F = ma \Rightarrow a = \frac{F}{m} = \frac{100}{4} = 25 m/s^2$

and $v = u + at = 5 + (25)(10) = 255$ m/s

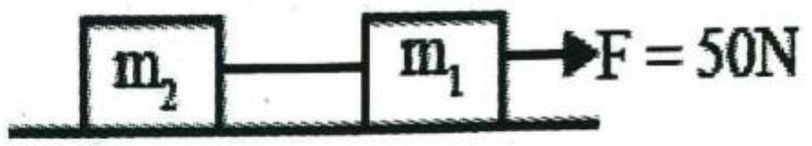
b) $a = \frac{F}{m} = \frac{-100}{4} = -25 m/s^2$

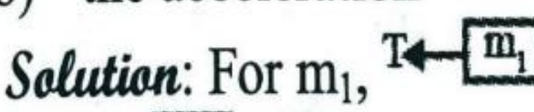
$v = u + at = 5 - (25)(10) = 245$ m/s

26. Two blocks of mass $m_1 = 2$ kg and $m_2 = 3$ kg are connected by a string and dragged horizontally by a 50N force. If the surface is smooth and they attain a constant acceleration. Find the:

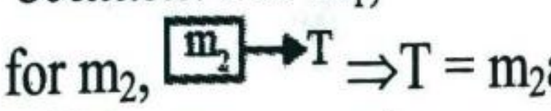
a) tension in the string

b) the acceleration



Solution: For m_1, 

$\Rightarrow F - T = m_1 a \ldots (*)$

for m_2, 

$\Rightarrow T = m_2 a \ldots$ (**)

add the two equations,

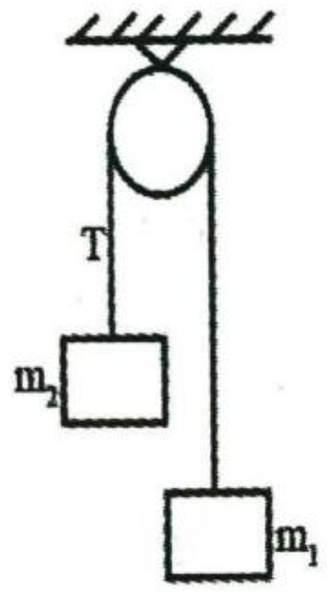
$\Rightarrow F = m_1 a + m_2 a$

$\Rightarrow a = \frac{F}{m_1 + m_2} = \frac{50}{2+3} = 10 m/s^2$ and

$T = m_2 a = (3)(10) = 30N$

Thus, a) T = 30N and b) $a = 10$ m/s^2

27. Two masses $m_1 = 5$ kg and $m_2 = 3$ kg are hang over a mass less and frictionless pulley. Find

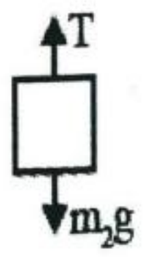


a) Tension in the string

b) acceleration of the masses (take g = 10 m/s^2)

Solution: for m_1

, $m_1 g - T = m_1 a \ldots (*)$

for m_2 

, $T - m_2 g = m_2 a \ldots (**)$

adding the two equations,

$$m_1g - m_2g = m_1a + m_2a \Rightarrow a = \frac{m_1g - m_2g}{m_1 + m_2} = \frac{10(5-3)}{5+3}$$

$\Rightarrow a = 2.5\ m/s^2$, and $T - m_2g = m_2a$

$\Rightarrow T = m_2(a + g) = 3(2.5 + 10) = 37.5N$

Thus, a) $T = 37.5N$ and b) $a = 2.5\ m/s^2$

28. The engine of a car produces an acceleration of 6 m/s^2 in the car. If this car pulls another car of the same mass then the acceleration would be?

Solution: $F = m_1a_1 \ldots (*)$

$F = m_2a_2 \ldots (**)$, but $m_2 = 2m_1$

Taking ratio of the two equations

$$m_2a_2 = m_1a_1 \Rightarrow a_2 = \frac{m_1}{m_2}a_1 = \frac{m_1}{2m_1}a_1 = \frac{6}{2} = 3m/s^2$$

29. A 2 kg body is sliding with a constant velocity of 8 m/s on a smooth surface. What force is required to keep the body moving with this velocity?

Solution: $F = ma$, but $a = \frac{\Delta v}{\Delta t} = \frac{0}{\Delta t} = 0 \Rightarrow F = 0$

Which means, no force is needed to keep the body moving at constant velocity.

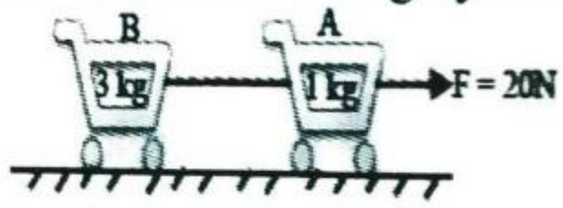
30. A car of mass 120 kg is moving with a uniform velocity of 108 km/h. The force required to stop the car in 10 second is:

Solution: $F = ma$, $a = \frac{\Delta v}{\Delta t} = \frac{0 - \frac{108}{3.6}}{10} = -3m/s^2$

$F = (120)(-3) = -360N$

The negative sign indicates, the force acts in opposite direction to the direction of motion of the car.

31. Two carts are pulled by a child on a smooth surface. What is the force exerted on the string by cart B?



Solution: For cart A: $F - T = m_Aa \ldots (*)$

For cart B: $T = m_Ba \ldots (**)$

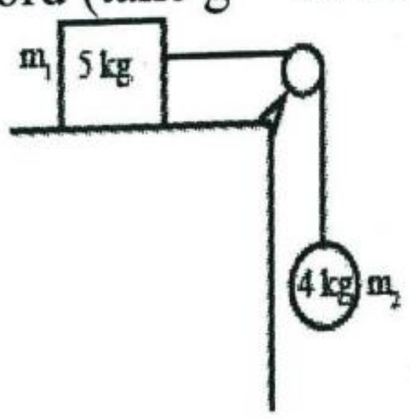
adding these two equations, $a = \frac{F}{m_A + m_B} = \frac{20}{3+1} = 5 m/s^2$

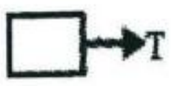
$T = m_B a = (3)(5) = 15N$

32. A 5 kg block is attached to a 4 kg body as shown below. If the system is frictionless. Find:

a) the acceleration

b) the tension in the cord (take $g = 10\ m/s^2$)



Solution: for m_1 

a) $T = m_1 a \ldots (*)$

for m_2,

$m_2 g - T = m_2 a \ldots (**)$

adding the two equations

$m_2 g = (m_1 + m_2)a$

$\Rightarrow a = \frac{m_2 g}{m_1 + m_2} = \frac{(4)(10)}{(5+4)}$

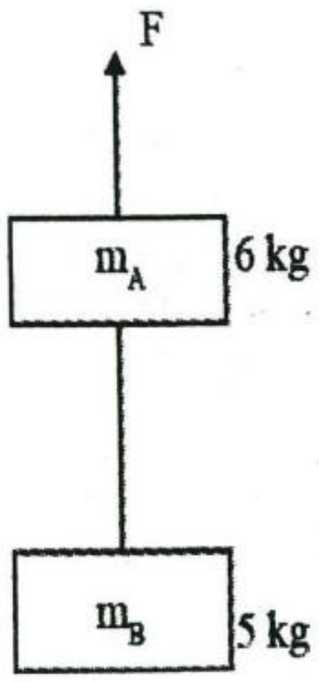
$\Rightarrow a = 4.4\ m/s^2$

b) $T = m_1 a = (5)(4.4) = 22.2N$

33. Two blocks are connected by a rope and are lifted using a force of 200N. Calculate:

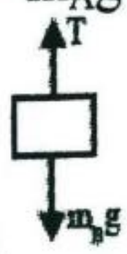
a) the acceleration of the system

b) the tension in the rope



Solution: for m_A

a) $F - T - m_A g = m_A a \ldots (*)$



for m_B

$T - m_B g = m_B a \ldots (**)$

adding the two equations,

$$F - m_A g - m_B g = m_A a + m_B a \Rightarrow a = \frac{F - g(m_A + m_B)}{m_A + m_B}$$

$$a = \frac{200 - 10(5+6)}{5+6} = \frac{90}{11} = 8.2 m/s^2$$

b) $T = m_B a + m_B g = (5)(8.2) + (5)(10) = 91N$

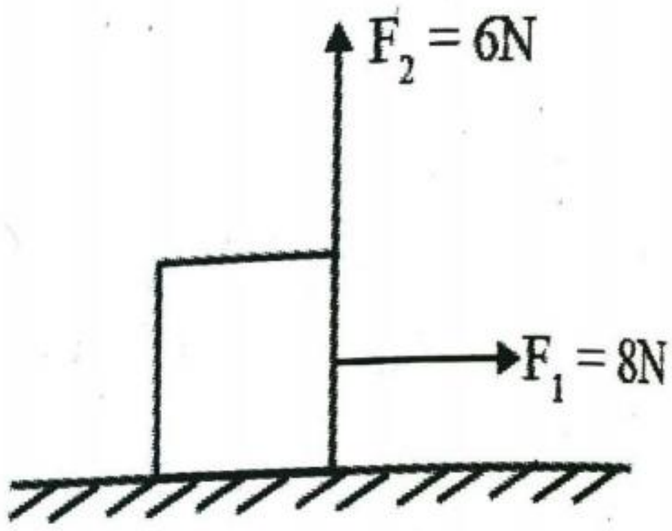
34. A 2×10^4N ford car traveling in the x direction is subjected to a 1.5×10^4N horizontal force. What is the acceleration of the car?

Solution: $F = ma$, but $m = \frac{w}{g} = \frac{2\times10^4}{10} = 2000kg$

$$\Rightarrow a = \frac{F}{m} = \frac{1.5\times10^4}{2\times10^3}$$

$a = 7.5\ m/s^2$

35. A body of mass 5 kg is acted upon by two perpendicular forces 8N and 6N. What is the magnitude and direction of the acceleration of the body?



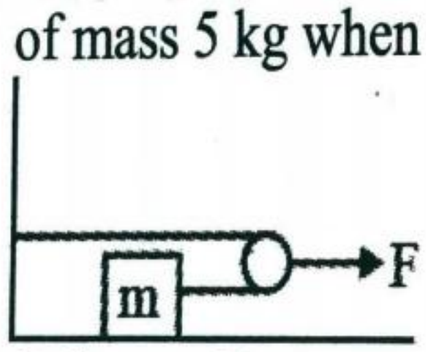
Solution:

$$F = \sqrt{F_1^2 + F_2^2} = \sqrt{(8)^2 + (6)^2} = 10N$$

From $F = ma$, $a = \frac{F}{m} = \frac{10}{5} = 2m/s^2$ and $\tan\theta = \frac{F_2}{F_1} = \frac{6}{8} = \frac{3}{4}$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right) = 37°$$

36. If the system is frictionless, what is the acceleration of the block of mass 5 kg when a force of 100N acts on it?



Solution: $T = ma$ . . . (*)

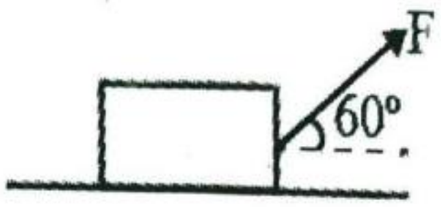
and $F - T = ma$ . . . (**)

adding the two equations,

$$F = 2ma \Rightarrow a = \frac{F}{2m} = \frac{100}{(2)(5)} = 10m/s^2$$

37. A force of 60N acts at an angle of 60° on a block of mass 6 kg and displaces it horizontally. What is the horizontal acceleration of the block if there is no friction?

Solution: The component of the force that displaces the block horizontally is,

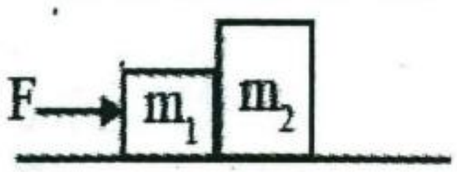


$F_x = F\cos\theta = (60)(\cos 60°)$

$F_x = (60)(0.5) = 30N$

$$a_x = \frac{F_x}{m} = \frac{30}{6} = 5m/s^2$$

38. Two blocks with masses $m_1 = 3$ kg and $m_2 = 5$ kg are in contact on a frictionless horizontal surface. If a horizontal force of 40N pushes them. What is:



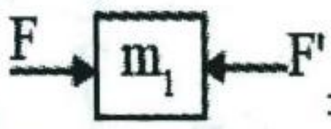
a) The acceleration of the system.

b) The force that block m_1 exerts on m_2.

Solution:

a) $F = ma$, $m = m_1 + m_2 = 3 + 5 = 8$ kg

$$a = \frac{F}{m} = \frac{40}{8} = 5m/s^2$$

b) for m_1, 

$\Rightarrow F - F' = m_1a = \ldots (*)$

for m_2,

$\Rightarrow F' = m_2a \ldots (**)$

From equation (*) or (**)

$F' = m_2a = (5)(5) = 25N$

Where F' is the contact force between m_1 and m_2.

39. A body of mass 5 kg is at rest on a frictionless surface. If a net force of 50N acts on it for 3 seconds.

a) What is the final velocity of the body?

b) How far will the body move in the 3 sec interval?

Solution:

a) $F = ma \Rightarrow a = \frac{F}{m} = \frac{50}{5} = 10m/s^2$ and $v = u + at = 0 +$

$(10)(3) = 30$ m/s

b) $s = ut + \frac{1}{2}at^2 = (0)(3) + \frac{1}{2}(10)(3)^2$

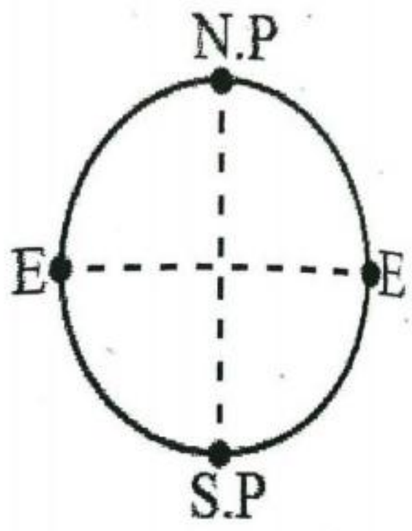
$s = 45m$

40. A block of mass 2kg is suspended on a spring of force constant 200N/m. What will be the acceleration of the block if the spring is stretched to 10 cm and released?

Solution: From Hook's & Newton's second laws $F = ma = kx$

$$\Rightarrow a = \frac{kx}{m} = \frac{(200)(0.1)}{2} = 10m/s^2$$

41. Find the weight of a 50 kg body at the North pole (N.P) where $g = 10\ m/s^2$ and at the equator (E) where $g = 9.6\ m/s^2$.



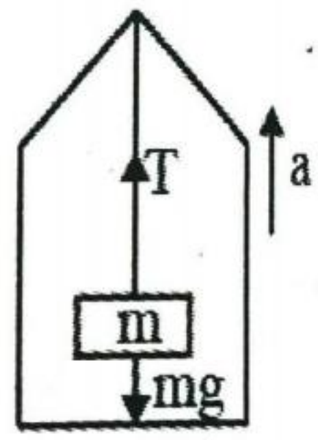
Solution: Weight at the (N.P)

$W = mg = (50)(10) = 500N$

Weight at the (E)

$W = mg = (50)(9.6) = 480N$

42. In a certain elevator, a certain body of mass 1 kg is suspended from the roof of the elevator with a string. What is the tension in the string when the elevator is moving upward with acceleration of $1\ m/s^2$ (take $g = 10\ m/s^2$)



Solution: $T - mg = ma$

$T = ma + mg = (1)(1) + (1)(10) = 11N$

43. A 4000 kg elevator is accelerating upwards. If the tension in the supporting cable is 48000N. Calculate acceleration of the elevator? (take $g = 10\ m/s^2$)

Solution: $T - mg = ma \Rightarrow T = mg + ma$

$$4800 = 4000\ (10 + a) \Rightarrow a = 2\ m/s^2$$

44. What is the apparent weight of a 60 kg mass gentle man moving in an elevator when the elevator accelerates at $3\ m/s^2$?

a) upwards

b) downwards

Solution:

a) $F_N - mg = ma \Rightarrow F_N = mg + ma$

$F_N = (60)(10) + (60)(3) = 780N$

b) $mg - F_N = ma \Rightarrow F_N = mg - ma$

$F_N = (60)(10) - (60)(3) = 420N$

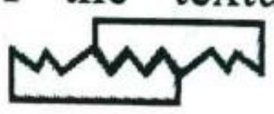
Frictional Force

Friction is a force we experience every single day. It arises when one surface tries to slide over another surface due to the microscopic hills and valleys of the surfaces interlocking together.

Friction: Is a force opposing the relative motion of two surface in contact.

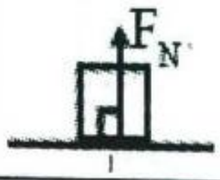
Factors affecting friction force

Perhaps the most dominant factors that affect friction force are *roughness of the surface in contact* and *the normal force.*

1. **Roughness:** Is the measure of the texture of degree of smoothness of the surface in contact. 

When the surfaces in contact are rougher, the force of friction is larger, this is due to the increase in bonding or interlocking between the contact surfaces.

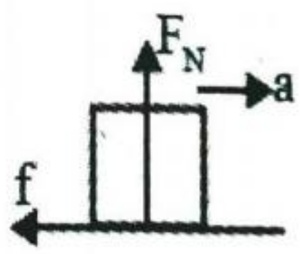
2. **Normal Force:** Is a supporting force that presses and holds two bodies together by acting perpendicularly to the surfaces in contact.



☞ A heavier object will push down and press harder on the surfaces in contact and hence lock the bumps together, as a result increases the friction force. Since the friction force is proportional to the normal force, we have;

$f \sim F_N$

$\Rightarrow \boxed{f = \mu F_N}$

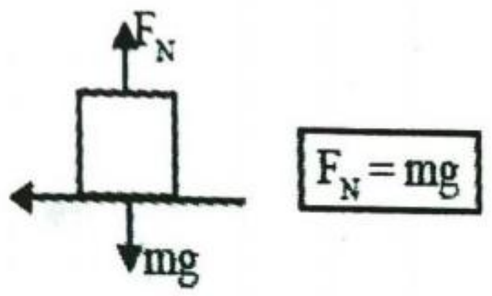


Where f = friction force

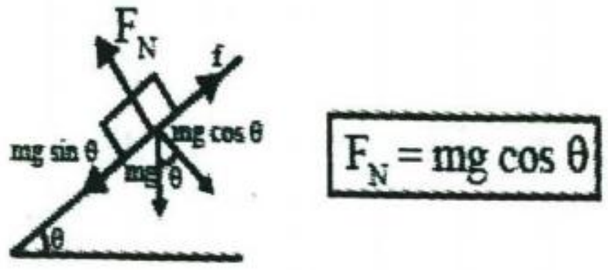
μ = coefficient of friction which depends on the surfaces in contact

F_N = Normal force

- **For a level surface**, the normal force equals the weight of the body.



- **For a body on an inclined surface**, the normal force is obtained by:



Types of friction

Depending on the relative movement between the surfaces in contact, in general terms, friction is of two types.

1. **Static friction**: Is the friction force between two surfaces that are in contact and trying to move over each other but not yet moving.

In symbols, $f_s = \mu_s F_N$

Where; f_s = static friction

μ_s = coefficient of static friction

☞ **Remember:** If you push a table gently, at first, it will not move. This is so because the force you are applying is cancelled out due to static friction. As you gradually increase the force the static friction also increases. If you continue to push harder, eventually the table will slide. This maximum value of static friction is called *limiting friction*.

Limiting Friction: Is the maximum static friction just before sliding occurs.

2. **Kinetic friction:** Is the friction force between surfaces that are in relative motion with each other.

In symbols, $f_k = \mu_k F_N$

Where; f_k = kinetic friction

μ_k = coefficient of kinetic friction

Note that:

- Once a body starts moving, it is easy to make the body maintain its motion. Thus, it is often observed that, kinetic friction is less than static friction.
- Frictional force is not affected by the area of contact of the surfaces. That is, if we change the contact area between two surfaces from small to large, or vice versa, the friction force between the given surfaces do not change.

Advantage and disadvantage of friction

Friction do have advantage and as well as disadvantage.

Advantage: Friction help us to walk, to write and cars to move etc. Without friction, our daily activities would be impossible.

Disadvantage: friction causes our shoe, cloths, car tires and parts of machinery to wear out. It causes to waste useful energy in the form of

heat energy. Friction retards motion, produce noise in machines and makes engines consume much fuel.

Conceptual Example

45. Walking on a horizontal slippery ice can be much more tiring than walking on ordinary road. Why?

Explanation: In slippery road, backward motion is the dominant one than the forward motion.

Illustrative Examples

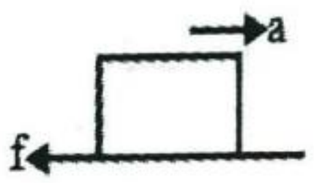
46. A body of mass 100 kg moving with a speed of 10 m/s on a horizontal surface acted on only by friction force. If the coefficient of kinetic friction between the surfaces is 0.1. What is the distance the body moves before it comes to rest?

Solution: $-f = ma$

Since $f = \mu mg \Rightarrow -\mu mg = ma \Rightarrow a = -\mu g = (-0.1)(10) = -1\ m/s^2$

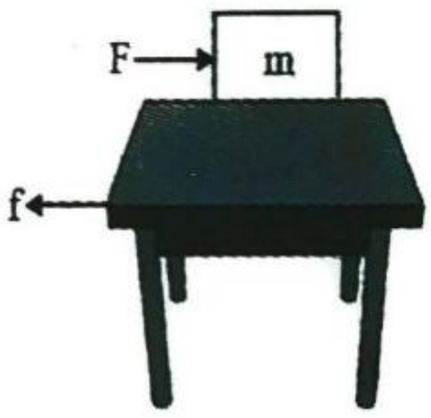
and from, $v^2 = u^2 + 2as$, $v = 0$

$-u^2 = -2as$



$$\Rightarrow s = \frac{u^2}{2a} = \frac{(10)^2}{(2)(1)} = 50m$$

47. A 10 kg block is placed on a table. If a horizontal force of 32N is applied on it, and a friction force of 7N is present between the table and the block. Calculate the acceleration of the block.



Solution: $F - f = ma$

$$\Rightarrow a = \frac{F - f}{m} = \frac{32 - 7}{10} = 2.5m/s^2$$

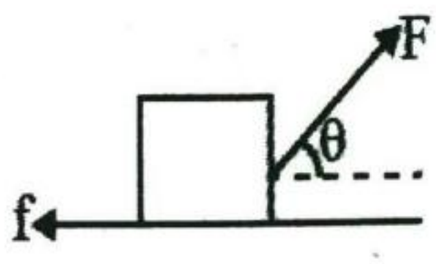
48. A man is pulling a 20 kg box with the horizontal. If he applies a force of 150N and a frictional force of 15N is present. Calculate the acceleration of the box.

Solution: $F - f = ma$

The force component that moves the box horizontally is $F_x = F \cos\theta = (150)(\cos 60) = (150)(0.5) = 75N$

$\Rightarrow F_x - f = ma$

$$a = \frac{F_x - f}{m} = \frac{75 - 15}{20} = 3m/s^2$$



49. A force of 200N acts at 30° on a block of mass 50 kg and causes it to move with constant velocity. **Calculate**:

a) the normal force
b) the coefficient of friction

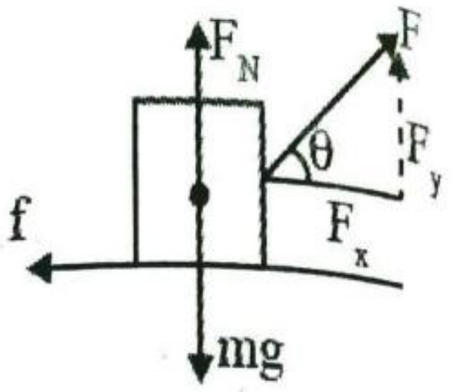
Solution:

a) $\sum F_y = 0$

$\Rightarrow F_N + F_y = mg$

$F_N = mg - F_y = (50)(10) - (200)(\sin 30)$

$F_N = 500 - (200)(0.5) = 400N$

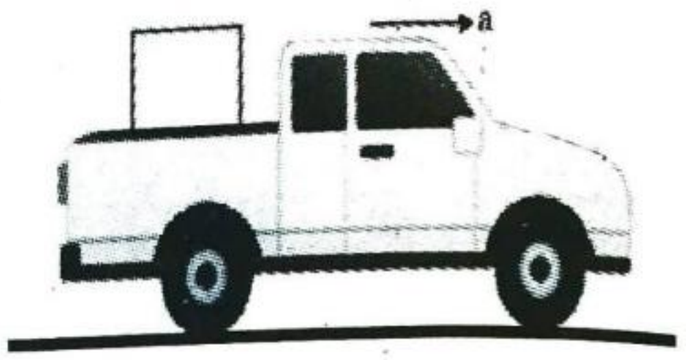


b) $\sum F_x = 0$

$\Rightarrow f = F_x \Rightarrow \mu F_N = F\cos 30$

$$\mu = \frac{F\cos 30}{F_N} = \frac{(200)(\cos 30°)}{400} = \frac{(200)(0.87)}{400} = 0.4$$

50. If the coefficient of friction between a car and a box on the car is 0.5. What is the maximum acceleration the car can have for the box not to slide?



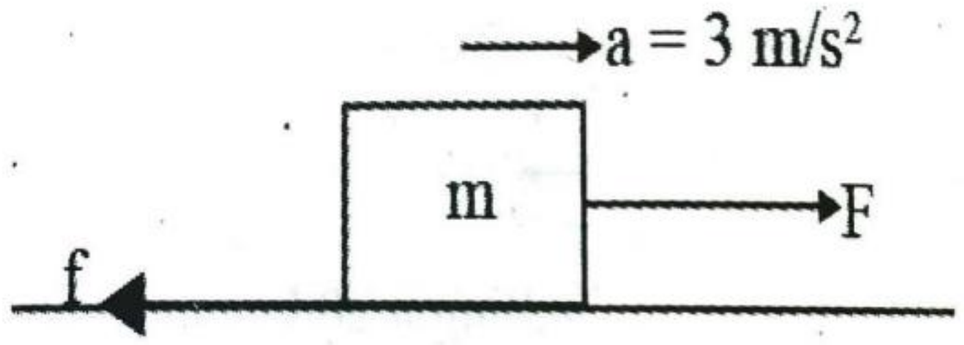
Solution:

Now, $f = ma$

$\mu mg = ma$

$a = \mu g = (0.5)(10) = 5\ m/s^2$

51. A force of 10N is applied horizontally on a block resting on a horizontal surface and accelerate it by 3 m/s^2. If the coefficient of friction is 0.2. What is the mass of the block?



Solution:

$F - f = ma$

$F - \mu mg = ma$

$\Rightarrow F = ma + \mu mg$

$$m = \frac{F}{a + \mu g} = \frac{10}{3 + (0.2)(10)} = 2kg$$

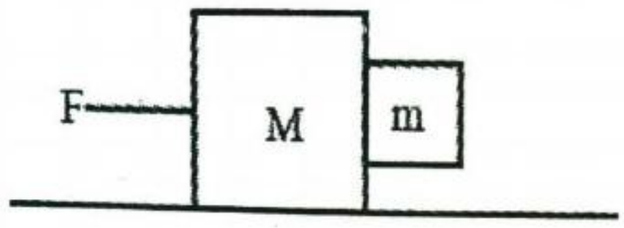
52. If the coefficient of friction between the two blocks, M = 3kg and m = 0.5 kg is 0.4, how large should be F, in order for m not to slide?

Solution: $f = mg$, but $f = \mu F_N$ and $F_N = ma$

$\Rightarrow \mu F_N = mg$

$\mu ma = mg$

$$\Rightarrow a = \frac{g}{\mu} \ldots (*)$$



and from Newton's second law.

$F = ma$, here $m' = m + M$

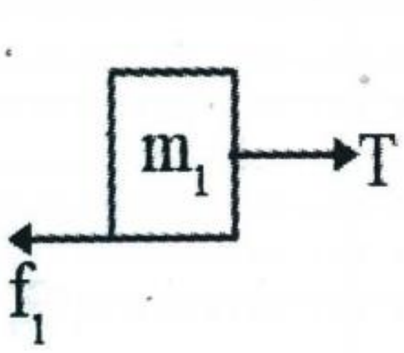
$$\Rightarrow a = \frac{F}{m'} = \frac{F}{m+M} \ldots (**)$$

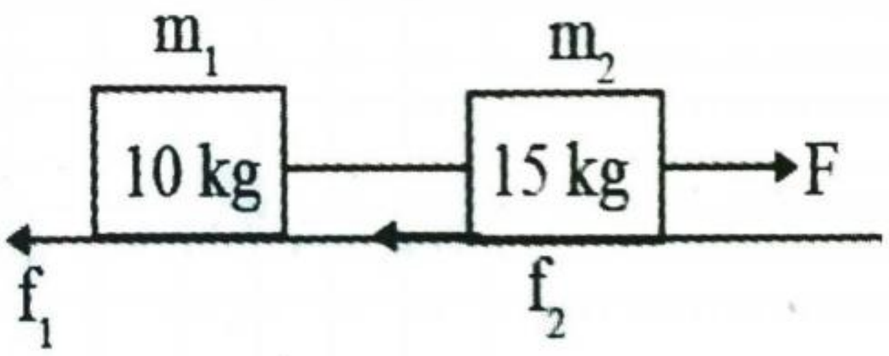
from the two equations,

$$\frac{g}{\mu} = \frac{F}{m+M} \Rightarrow F = \frac{g(m+M)}{\mu} = \frac{10(3+0.5)}{0.4} = 87.5N$$

53. Two blocks 10 kg and 15 kg, are connected by a string and a force of 500N is applied. If a 150N force of fiction acts on the 10 kg and 300N force of friction acts on the 15 kg block, and the blocks move with an acceleration of 2 m/s^2. Calculate the tension in the string.

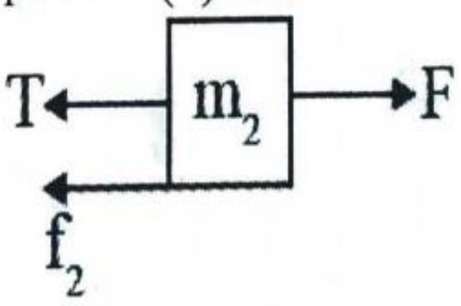
Solution: for m_1





$T - f_1 = m_1 a \ldots (*)$ and

for m_2,



$F - T - f_2 = m_2 a \ldots (**)$

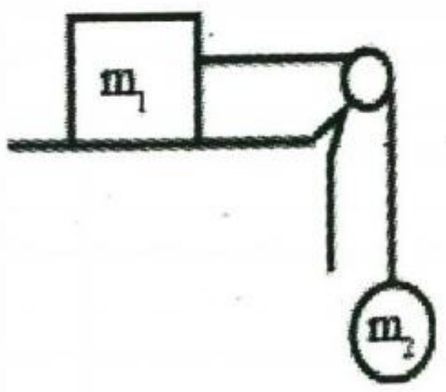
From the two equations, that is using one of the equations, say

$T - f_1 = m_1 a$

$\Rightarrow T = m_1 a + f_1 = (10)(2) + 150$

$T = 170N$

54. A block of mass $m_1 = 2$ kg rests on a table where the coefficient of friction is 0.2, and is connected to a block of mass $m_2 = 5$ kg as shown. Calculate:

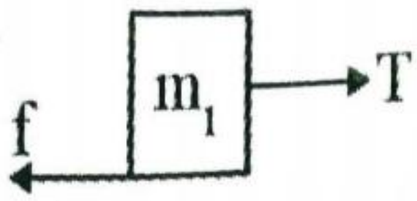


a) The acceleration of the blocks.

b) The tension in the string

Solution:

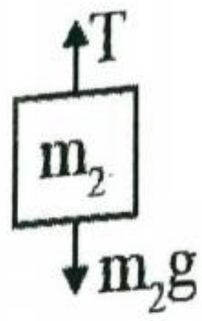
a) for m_1,



$\Rightarrow T - f = m_1 a$

$\Rightarrow T - \mu m_1 g = m_1 a \ldots (*)$

for m_2,



$\Rightarrow m_2 g - T = m_2 a \ldots (**)$

adding the two equations

$m_2 g - \mu m_1 g = m_1 a + m_2 a$

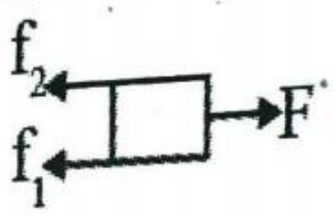
$$\Rightarrow a = \frac{g(m_2 - \mu m_1)}{m_1 + m_2} = \frac{10(5 - (0.2 \times 2)}{2+5} = 6.6 m/s^2$$

b) $T - \mu m_1 g = m_1 a$

$\Rightarrow T = m_1 a + \mu m_1 g = (2)(6.6) + (0.2)(2)(10) = 17.2N$

55. If the coefficient of friction between all the surfaces is 0.4. Find the acceleration between all the surfaces is 0.4. Find the acceleration of $m_1 = 6$ kg. Assume $m_2 = 2$ kg and $g = 10\ m/s^2$.

Solution: for m_1



$F - f_1 - f_2 = m_1 a$, where

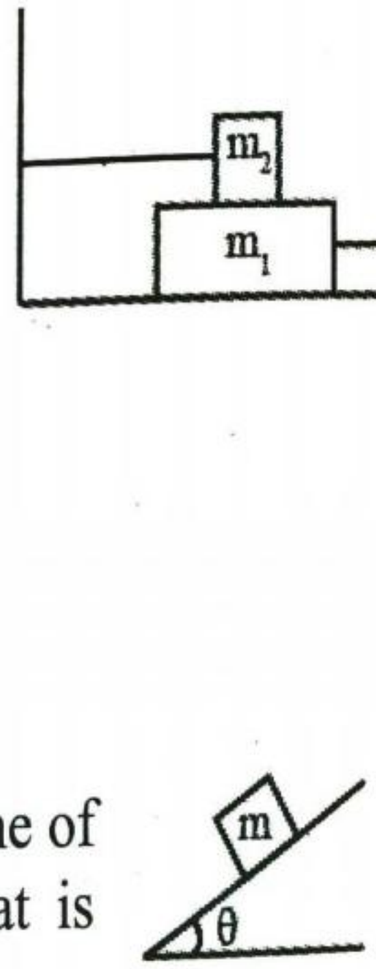
$f_1 = \mu(m_1 + m_2)g$ and $f_2 = \mu m_2 g$

so we have,

$$a = \frac{F - f_1 - f_2}{m_1} = \frac{F - \mu(m_1 + m_2)g - \mu m_2 g}{m_1}$$

$$a = \frac{49 - 0.4(6+2)10 - 0.4(2)(10)}{6}$$

$a = 1.5\ m/s^2$



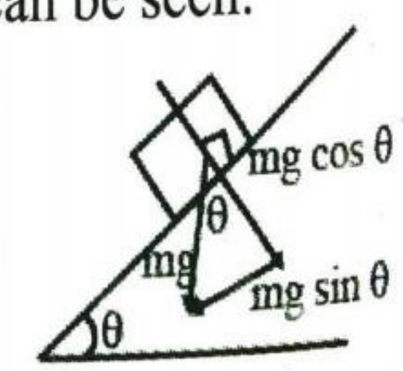
56. A block of mass 3 kg rests on an inclined plane of 30°. Find the components of the weight that is parallel and perpendicular to the inclined plane.

Solution: The components of the weight can be obtained by decomposing it horizontally and vertically as can be seen:

→ Component of the weight parallel to the inclined plane is mg sin θ.

$W_{//} = mg \sin\theta = (3)(10)(\sin 30°)$

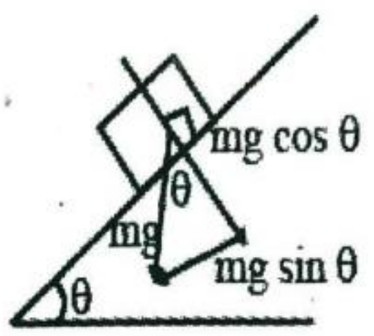
$= (3)(10)(0.5) = 15N$



→ Component of the weight perpendicular to the inclined plane is:

$W_{\perp} = mg \cos\theta = (3)(10)(\cos 30°) = (3)(10)(0.87) = 26N$

57. A 50 kg crate is placed on an inclined plane of 30° with the horizontal. If the box does not slide.



Calculate

a) The normal force

b) The frictional force

Solution:

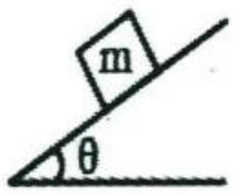
a) The normal force is equal to the component of the weight perpendicular to the inclined plane.

$\Rightarrow F_N = mg \cos\theta = (50)(10)(\cos 30°) = 500 \times 0.87 = 437N$

b) The friction force is equal and opposite to the component of the weight that is parallel to the inclined plane.

$\Rightarrow f = mg \sin\theta = (50)(10)(\sin 30) = 500 \times 0.5 = 250N$

58. A 100N object is pulled up a 37° incline with constant speed. Find the force parallel to the incline.



a) If friction is negligible.

b) If the coefficient of friction is 0.2.

Solution:

a) $F = mg \sin\theta = (100)(\sin 37) = 100 \times 0.6 = 60N$

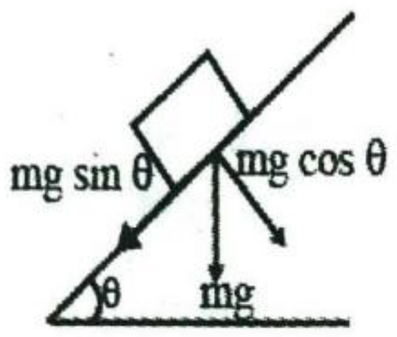
b) $F = f + mg \sin\theta$

$f = \mu F_N + mg \sin\theta, F_N = mg \cos\theta$

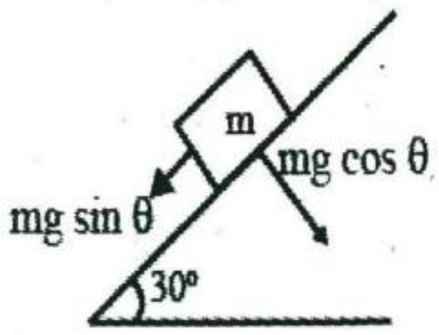
$= (0.2)(mg \cos\theta) + mg \sin\theta$

$= (0.2)(100)(\cos 37) + (100 \times \sin 37)$

$= (0.2)(100 \times 0.8) + (100 \times 0.6) = 140N$



59. What force is required to move a block of mass 10 kg on an inclined plane of 30° with an acceleration of 1 m/s^2 (take $\mu_k = 0.2$).



a) Down the plane.

b) UP the plane.

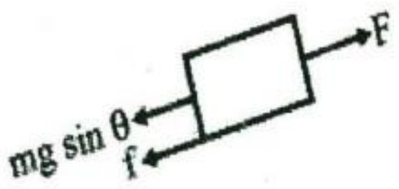
Solution:

a) $F + mg \sin\theta - f = ma, f = \mu F_N = \mu mg \cos\theta$

$F = ma + \mu mg \cos\theta - mg \sin\theta$

$F = 10 \times 1 + 0.2 \times 10 \times 10 \times 0.87 - 10 \times 10 \times 0.5$

$F = 22.6N$ (upwards)



b) $F - mg \sin\theta - f = ma$, $f = \mu mg \cos\theta$

$F = ma + mg \sin\theta + \mu\, mg \cos\theta$

$F = 10 \times 1 + 10 \times 10 \times 0.5 + 0.2 \times 10 \times 10 \times 0.86$

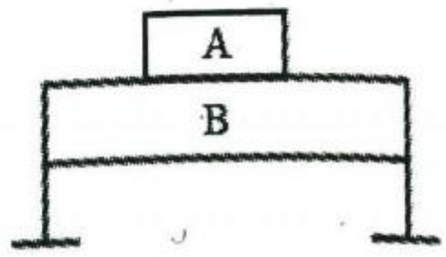
$F = 77.3N$

Newton's third law (Law of action and reaction)

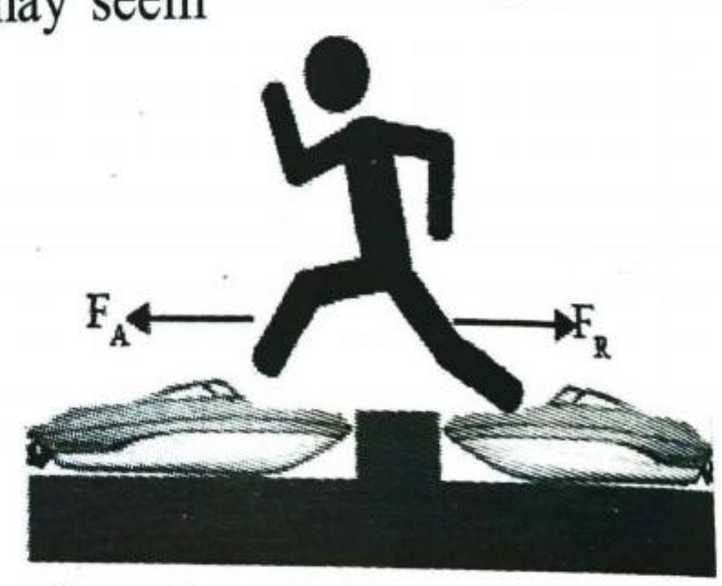
Nature never produces just one force at a time, forces always exist in pair. The pair of forces act on different bodies and are equal in magnitude and opposite in direction. This is Newton's third law.

> **Newton's third law:** If body A exerts a force on body B, then body B will exert an equal and opposite force on body A.

That is, if we have; then $\boxed{F_{AB} = -F_{BA}}$



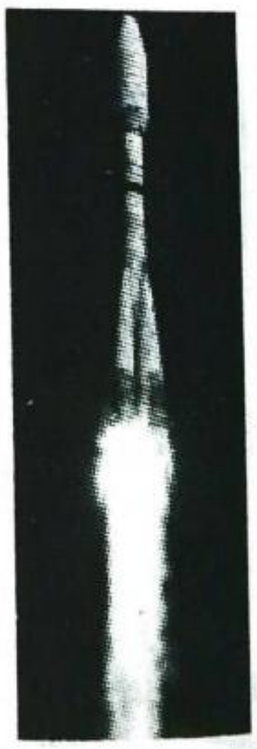
☞ According to this law, for every action, there is always an equal and opposite reaction. There is no such thing as a single isolated force. In simple terms, if you push a table by a force of say 50N, then the table will push you back by a force of exactly 50N. It may seem strange but, if you drop your pen from some height, it will fall downwards and we say gravity pulls it down, but the fact is, the pen also pulls the earth upward with the same force that acts on it. Moreover, when we walk, we push the earth backwards and at the same time, the earth pushes us forward and we accelerate.



When a gun is fired, the bullet is projected forward and the gun recoils backwards.

When a man jumps out of a boat, as he jumps forward, the reaction force causes the boat to go backwards.

When a rocket is launched, the fuel is burnt and the resulting hot gases are forced out of the back of the rocket. A reaction force drives the rocket forward.



Conceptual Examples

60. Since action and reaction forces are equal in magnitude and opposite in direction, they cancel each other.
Explanation: No even if action and reaction forces are equal and opposite, they cannot cancel each other. This is because, they act on two different bodies.

61. If a Bajage collides with a truck, which one experience a large acceleration?
Explanation: Even if the forces each experience is the same, the Bajage will get a greater acceleration due to its smaller mass.

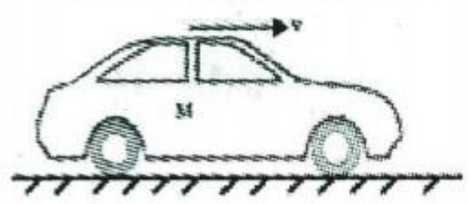
Linear Momentum and Impulse

Momentum is a measure of how hard to stop moving object. It is the "un stop ability" of the object.

It is more difficult to stop heavy truck than to stop a ball moving at the same speed as the truck. And a high speed bullet has a much stronger interaction with a tree than does a small stone thrown at the tree. This aspect of moving bodies is what we call *momentum*.

Momentum: Is the quantity of motion of a body which is obtained by taking the product of the mass and velocity of the moving body.

In symbols, $\boxed{\vec{P} = m\vec{v}}$



Momentum is a vector quantity and its SI unit is kg m/s.

☛ If a heavy truck and a ball are both moving at the same speed, the truck will have larger momentum than the ball because of its larger mass. On the other hand, a high speed bullet and a slow moving train can have equal moment.

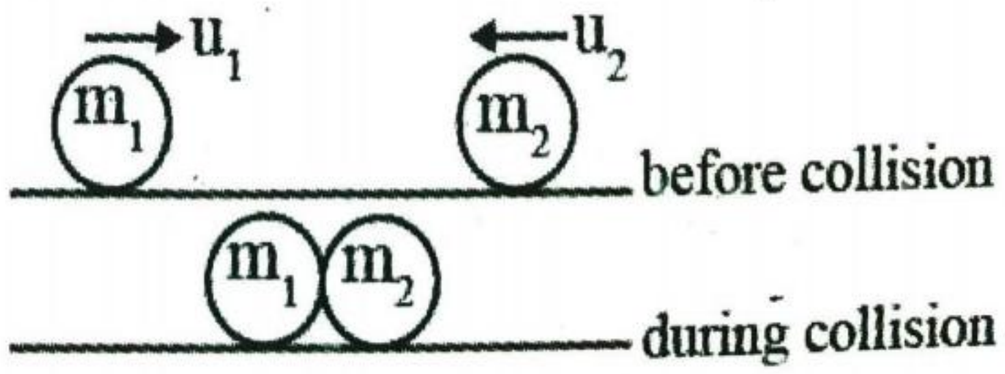
The law of conservation of linear momentum

One of the most important conservation laws in physics is the law of conservation of momentum.

Law of conservation of momentum: states
"In the absence of external force, the total linear momentum of a system remain constant."

In mathematical terms, $\boxed{\sum P_i = \sum P_f} \Rightarrow \Delta P = 0.$

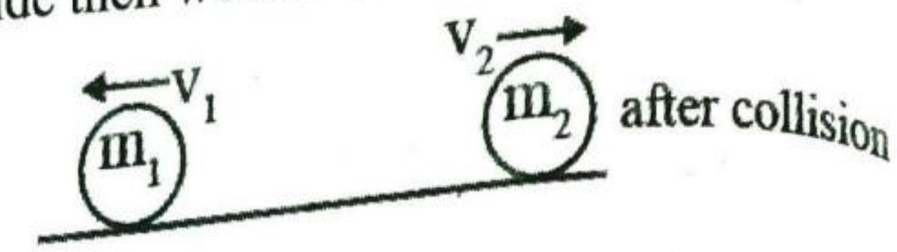
In simple words, assuming there is no external force, when two bodies collide, the



sum of their initial momentum equals the sum of their final momenta. That is, if ball 1 and ball 2 collide then we have;

$\sum P_i = P_{i1} + P_{i2} = m_1u_1 + m_2 u_2$

$\sum P_f \doteq P_{f1} + P_{f2} = m_1v_1 + m_2v_2$



Thus we have, $\boxed{\sum P_i = \sum P_f}$

$\Rightarrow \boxed{\boxed{m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2}}$

The Greek letter Σ (sigma) stands for "sum of"

Impulse

If a body of mass m is accelerated by a net force F, then according to Newton's second law, $F = ma$, but $a = \frac{v-u}{t}$

$$F = m\left(\frac{v-u}{\Delta t}\right) \Rightarrow \boxed{\boxed{F\Delta t = mv - mu = \Delta P}}$$

Impulse: Is the change in momentum and is given by the product of the force and the time for which it acts.

In symbols, $\boxed{J = F\Delta t = \Delta P}$

Where the Greek letter Δ(delta) means "change in".

Explosions

When a gun is fired, an explosion occurs and the bullet moves forward while the gun recoils. Since the bullet has small mass it attains a large velocity while because of its large mass the gun gets low velocity. Initially, the total momentum of the bullet and the gun is zero and from the conservation of linear momentum their final momentum is also zero.

$P_i = P_f = 0$

$\boxed{mv_b + Mv_g = 0}$

Collision

Collision is an event where two or more bodies make physical contact for a very short time and experience impulsive forces.

In collisions, the total momentum of the system is always conserved, but its total kinetic energy may not be conserved.

If the total kinetic energy is conserved, such as the collision of gas molecules with the walls of the container, the collision is said to be *elastic*.

Elastic collision: Is a collision where the total kinetic energy of the system is conserved.

That is, $\boxed{k_{Ei} = k_{Ef}}$

☛ Sometimes, the kinetic energy of the system may be converted to other forms, such as heat, sound and vibration in this case the kinetic energy is not conserved. As an example, this can happen when a lump of mud is thrown on to the wall and sticks on it.

Inelastic Collision: Is a collision where the total kinetic energy of the system is nto conserved. Here $\boxed{k_{Ei} \neq k_{Ef}}$

☛ **Remember:** For both, elastic and inelastic collisions, momentum is conserved.

Conceptual Examples

62. An open box slides on a frictionless surface. What happens to the speed of the box as water from a rain collected in it?

Explanation: It speed decreases.

63. A Karatis can break a bundle of bricks with the side of his bare hand. How is this possible?

Explanation: $F\Delta t = m\Delta v \Rightarrow F = \frac{m\Delta v}{\Delta t}$, as the contact time decreases the impact force increases.

Illustrative Examples

64. Calculate the momentum of a car of mass 1200 kg travelling with a velocity of 3 m/s.

Solution: $p = mv = (1200)(30) = 36000$ kgm/s

65. A car of mass 500 kg is moving at 24 m/s. A lion of mass 100 kg drops on to the roof of the car form an overhanging branch. What is the common velocity of the car and the lion?

Solution: $m_C u_C + m_L u_L = m_C v_C + m_L v_L$, but $u_L = 0$

$m_C u_C = (m_C + m_L)v$ and $v_C = v_L = v$

$$\Rightarrow v = \frac{m_C u_C}{m_C + m_L} = \frac{(500)(24)}{(500+100)} = 20 m/s$$

66. A car of mass 600 kg is moving at a speed of 20 m/s. It collides with a stationary car of mass 900 kg. If the first car bounces back at 4 m/s, at what speed does the second car move after the collision?

Solution: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$, but $u_2 = 0$

$m_1u_1 = m_1v_1 + m_2v_2$

$600)(20) = (600)(4) + 900\ v_2$

$\Rightarrow v_2 = 10.7$ m/s

67. A ball of mass 4 kg falls to the floor, it lands with a speed of 6 m/s. It bounces off with the same speed. Calculate the change in momentum of the ball.

Solution: $\Delta P = mv - mu = m(v - u)$, but $v = -u$

$\Delta P = m(2u) = 4(2 \times 6) = 48$ kgm/s

68. A bullet of mass 0.01kg is fired into a sand bag of mass 0.49 kg hanging from a tree. The sand bag with bullet embedded into it, swings away at 10 m/s. Find

a) The momentum after the collision.
b) The momentum before the collision.
c) The velocity of the bullet.

Solution:

a) $P_f = m_bv_b + m_sv_s$, but $v_b = v_s = v$

$P_f = (m_b + m_s)\ v = (0.01 + 0.49)10$

$P_f = 5$ kg m/s

b) From the conservation of linear momentum, $P_i = P_f$ $= 5$ kgm/s

Hence, $P_i = 5$ kgm/s

c) $P_i = m_bu_b + m_su_s$, but $u_s = 0$

$$P_i = m_bu_b \Rightarrow u_b = \frac{P_i}{m_b} = \frac{5}{0.1} = 500\text{m/s}$$

69. A child of mass 40 kg jumps off a wall and hits the ground at 4 m/s. He bends his knees and stops in 1 sec. Calculate the force required to slow him down.

Solution: $F\Delta t = m\Delta v \Rightarrow F = \dfrac{m\Delta v}{\Delta t} = \dfrac{(40)(4)}{1} = 160\text{N}$

70. A car of mass 1400 kg accelerating from 10 m/s to 15 m/s over 3.5 sec. Find the average force acting on it.

Solution: $F\Delta t = m\Delta v \Rightarrow F = \dfrac{m\Delta v}{\Delta t} = \dfrac{1400(15-10)}{3.5} = 2000N$

71. A resultant force of 10N acts for 2 seconds on a mass of 2 kg that is initially at rest. What is the change in its momentum?

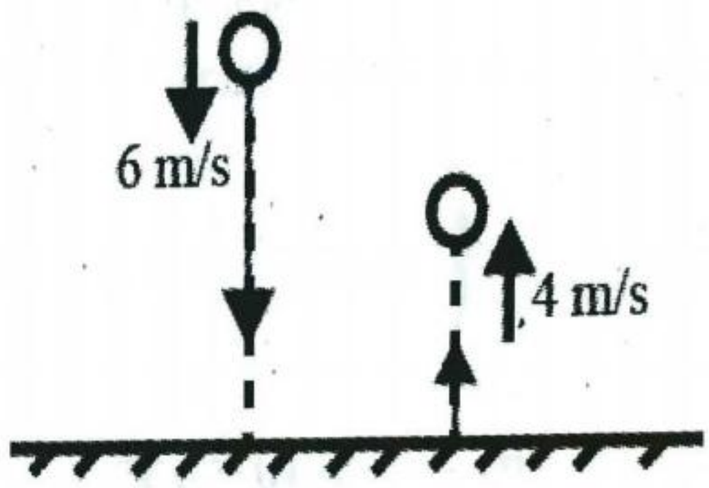
Solution: $F\Delta t = m\Delta v \Rightarrow \Delta P = F\Delta t = (10)(2) = 20Ns$

72. Two billiard balls roll towards each other. They each have a mass of 0.3 kg. ball 1 is moving at 1 m/s to the right while ball 2 is moving at 1 m/s to the left. Calculate the momentum of the system.

Solution: $\sum P = P_1 + P_1 = m_1u_1 + m_2u_2$

$$\sum P = (0.3)(1) + (0.3)(-0.8) = 0.06 \text{ kg m/s}$$

73. A rubber ball of mass 0.8 kg is dropped and strikes the floor with an initial velocity of 6 m/s. It bounces back with a final velocity of 4 m/s. Calculate the change in momentum of the rubber ball by the floor.



Solution: $\Delta P = mv - mu$

$= (0.8)(-4) - (0.8)(6)$

$\Delta P = -8$ kg m/s $= 8$ kgm/s upward

74. A 150N force acts on a 300 kg car. Calculate how long it takes this force to change the car's velocity from 2 m/s to 6 m/s.

Solution: $F\Delta t = m\Delta v \Rightarrow \Delta t = \dfrac{m\Delta v}{F} = \dfrac{m(v-u)}{F}$

$$\Delta t = \frac{300(6-2)}{150} = 8\,\text{sec}$$

75. A toy car of mass 1 kg moves westwards with a speed of 2 m/s. It collides head on with a toy train of mass 1.5 kg moving at 1.5 m/s eastwards. If the car rebounds at 3 m/s. Calculate the velocity of the train.

Solution: $m_Cu_C + m_tu_t = m_Cv_C + m_tv_t$

$(1)(-2) + (1.5)(1.5) = (1)(3) + (1.5)v_t$

$\Rightarrow v_t = 1.8$ m/s

76. A helicopter flies at a speed of 275 m/s. The pilot fires a missile forward at a speed of 700 m/s. If the respective masses of the helicopter and the missile are 500 kg and 50 kg. Calculate the

speed of the helicopter immediately after the missile had been fired.

Solution: $(m_1 + m_2)u = m_1v_1 + m_2v_2$

$(5000 + 50)275 = 5000v_1 + (50)(700)$

$\Rightarrow v_1 = 270.75$ m/s

77. A bullet of mass 50g travelling at 500 m/s strikes a stationary block of mass 2 kg resting on a smooth surface. The bullet goes through the block and comes out on the other side at 200 m/s. Calculate the speed of the block after the bullet has come out.

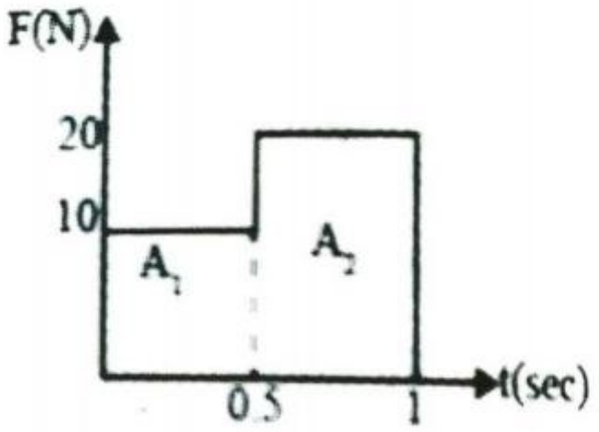
Solution: $P_i = P_f$

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$(0.05)(500) + (2)(0) = (0.05)(200) + (2)v_2$

$\Rightarrow v_2 = 7.5$ m/s

78. Force is applied on an object of mass 2 kg initially at rest on a smooth surface. What will be the velocity of the object after 1 second?



Solution: $F\Delta t = m\Delta t$ = area

$m\Delta v = A_1 + A_2$

$m(v - u) = (10)(0.5) + (20)(0.5)$

$mv = 5 + 10 = 15$

$$v = \frac{15}{m} = \frac{15}{2} = 7.5\text{m/s}$$

79. What is the mass of a car that moves with a velocity of 60 m/s in order to have the same momentum as a 6000 kg truck whose velocity is 20 m/s?

Solution: $m_Cv_C = m_Tv_T \Rightarrow m_C = \dfrac{v_T}{v_C}m_T = \dfrac{20}{60}(6000) = 2000\text{kg}$

80. A ball of mass 200 g moves towards a wall with a speed of 10 m/s. It hits the wall and bounces with 80% of its original speed along the same line. What is the magnitude of the change in momentum of the ball?

Solution: $\Delta P = mv - mu = m(v - u) = m(-0.8u - u)$

$\Delta P = 1.8mu = (1.8)(0.2)(10)$

$\Delta P = 3.6$ kgm/s

The first condition of equilibrium

If the net force on an object is zero, then the object is said to be in equilibrium. A hanging lamp, a bridge, a high rise building etc are all examples of equilibrium situations.

☛ A body in equilibrium can be at rest or moving with constant velocity.

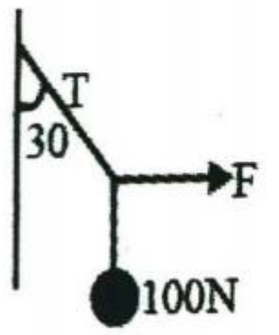
Equilibrium: A body is said to be in the first condition of equilibrium if and only if the vector sum of the forces acting on it is zero.

In symbol, $\boxed{\sum F = 0} \Rightarrow \sum F_x = 0$ and $\sum F_y = 0$

That is, when a body is in equilibrium, then the forces acting on it horizontally as well as vertically must add up to zero.

Illustrative Examples

81. A 100N ball suspended by a rope. If the rope is pulled by a horizontally force F, and held at some point, find the tension in the rope and the applied force.

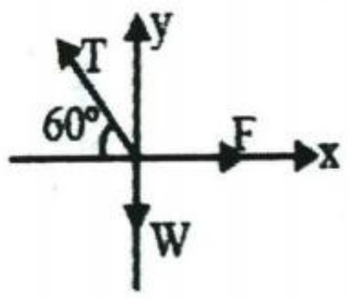


Solution: $\sum F_x = 0 \Rightarrow F - T\cos 60 = 0$

$\Rightarrow F = 0.5T \ \ldots (*)$ and $\sum F_y = 0$

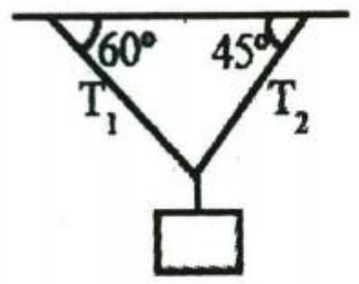
$\Rightarrow T\sin 60 = W \ldots (**)$

$$0.87T = 100 \Rightarrow T = \frac{100}{0.87} = 115N$$

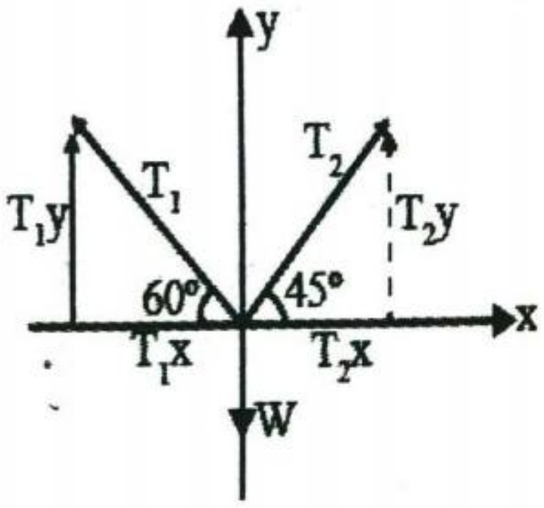


And $F = 0.5T = 0.5(115) = 57.5N$

82. A 200N block hangs from a cord. Find the tension in the ropes.



Solution: The free body diagram is,



$\sum F_x = 0 \Rightarrow T_1x = T_2x$

$T_1\cos 60 = T_2\cos 45$

$0.5T_1 = 0.7T_2 \ldots (*)$

$\sum F_y = 0 \Rightarrow T_1y + T_2y = W$

$T_1\sin 60 + T_2\sin 45 = 200$

$0.87T_1 + 0.7T_2 = 200 \ \ldots (**)$

But from $0.5T_1 = 0.72T_2$

$$\Rightarrow T_1 = \frac{0.7}{0.5}T_2 = 1.4T_2$$

Substitutions this into equation (**)

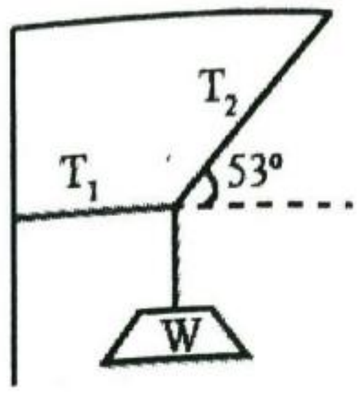
$0.87(1.4T_2) + 0.7T_2 = 200$

$1.2T_2 + 0.7T_2 = 200$

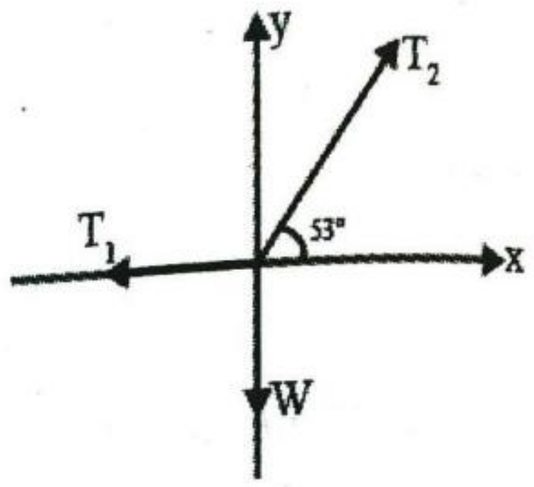
$1.9T_2 = 200 \Rightarrow T_2 = 104N$

And $T_1 = 1.4(T_2) = 1.4(104) = 146N$

83. The system shown below is in equilibrium. If the tension $T_1 = 100N$. Find T_2 and the weight of the block W.



Solution: The free body diagram is



$\sum F_x = 0 \Rightarrow T_1 = T_2 \cos 53°$

$T_1 = 0.6T_2$

$$\Rightarrow T_2 = \frac{T_1}{0.6} = \frac{100}{0.6} = 167N$$

$\sum F_y = 0 \Rightarrow T_2 \sin 53° = W$

$(167)(0.8) = W \Rightarrow W = 133N$

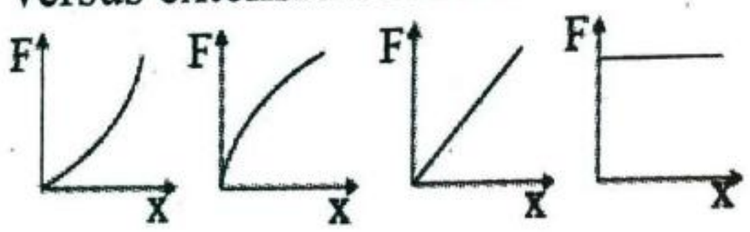
General Secondary Education Certificate Exam

84. The stretch of a spring is

A. Inversely proportional to the square of the restoring force.
B. Directly proportional to the square of the restoring force.
C. Directly proportional to the restoring force.
D. Inversely proportional to the restoring force.

Explanation: According to Hooke's law, $F_r = kx$. **Answer: C**

85. Which of the given graph represents a Hookean type of force versus extension curve?



Explanation: Hooke's law holds for a straight line graph that passes through the origin. **Answer: C**

86. How does the weight of a body vary from place to place on the surface of the earth?

A. It is maximum between the equator and the pole.
B. It decreases as you go to the equator.
C. It decreases as you go to the pole.
D. It is constant everywhere.

Answer: B

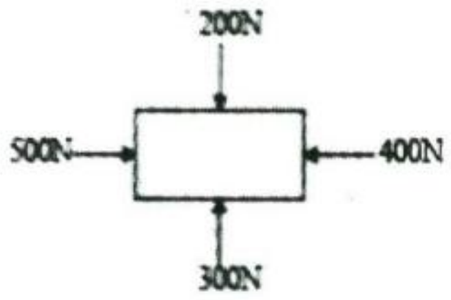
87. What happens to your weight and mass when you move from Dallol depression to Ras Dashen, which are about 120m below sea level and 4500m above sea level respectively?
A. Both weight and mass increase.
B. Both weight and mass decrease.
C. Weight increases and mass remains the same.
D. Weight decreases and mass remains the same.

Explanation: As altitude that is height, increases, the acceleration due to gravity decrease and hence weight decreases. **Answer: D**

88. Which combination of fundamental quantities can be used to express force?
A. Acceleration and mass
B. Mass, time and velocity
C. Acceleration, length and time
D. Time, length and mass

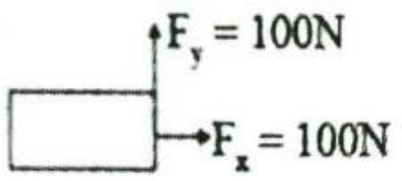
Solution: From $F = ma \Rightarrow 1N = 1\ kg\ m/s^2 \equiv \frac{ML}{T^2}$ **Answer: D**

89. Four forces act on a body as shown. What is the angle that the resultant makes with the horizontal?



A. $\tan^{-1}(1)$ C. $\tan^{-1}\left(\frac{4}{5}\right)$

B. $\tan^{-1}\left(\frac{1}{2}\right)$ D. $\tan^{-1}\left(\frac{3}{4}\right)$



Solution:
The resultant force components are;

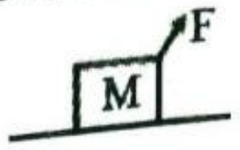
$$\Rightarrow \tan\theta = \frac{F_y}{F_x} = \frac{100}{100} \Rightarrow \theta = \tan^{-1}(-1)$$ **Answer: A**

90. Which one of the following is not correct about the effect of force?
A. Force can cause accelerated motion.
B. Force can cause deformation.
C. Force never creates change in mass.
D. Force can create change in momentum.

Solution: Change in momentum is caused by impulse and not force. **Answer: D**

91. A net force F is acting on a block of mass M as shown below. In which case is the friction force.

A.

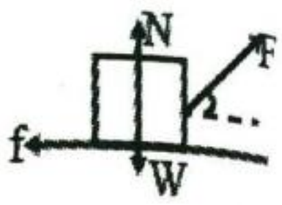
C. 

B.

D.

Explanation: $f = \mu F_N$, the friction force is larger when the normal force (F_N) is larger.. Since the normal force is a pressing force, it is large in case D. **Answer: D**

92. All of the forces acting on the block are those indicated. What can the possible explanation about the force acting on it be if the block is accelerated along the surface?



A. Force F is equal to friction.
B. The normal force N and the weight cancel each other.
C. The vertical component of F is smaller than its horizontal component.
D. The resultant is in the horizontal direction.

Solution: Since the block moves horizontally. **Answer: D**

93. How large should the vertical component of a 15N force be if its horizontal component is 9N?

A. 17.5N B. 6N C. 24N D. 12N

Explanation: $F^2 = F_x^2 + F_y^2 \Rightarrow F_y = \sqrt{F^2 - F_x^2} = \sqrt{(15)^2 - (9)^2}$

$F_y = 12N$ **Answer: C**

94. Two children are fighting over a toy. One pulls on it with a force of 50N due East, the other pulls with a force of 54N due west. What is the net force on the toy?

A. 104N West C. 104N East
B. 4N West D. 4N East

Solution: $F_R = F_W - F_E = 54 - 50 = 4N$ West **Answer: B**

95. A block of mass m is pulled at constant velocity a long a rough horizontal floor by an applied force T as shown. What is the magnitude of the friction force?

A. $\mu(mg + T\cos\theta)$ C. $T\sin\theta$
B. $\mu(mg + T\sin\theta)$ D. $T\cos\theta$

Solution: $T_x - f = ma$, but $a = 0$

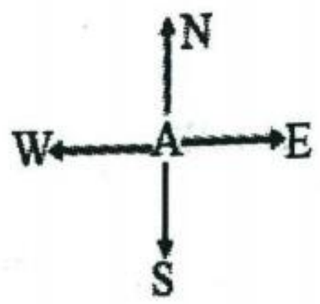
$\Rightarrow f = T\cos\theta$ **Answer: D**

96. A girl weighs 500N standing on a scale in a stationary elevator. If some time later the reading on the scale is 0N the elevator must be:

A. Moving with constant acceleration upward.
B. Moving with constant speed upward.
C. Moving with constant speed downward.
D. Falling freely vertically downward.

Solution: When the elevator moves down ward at gravitational weight is, $T = ma - mg$, $a = g = 0$ **Answer: D**

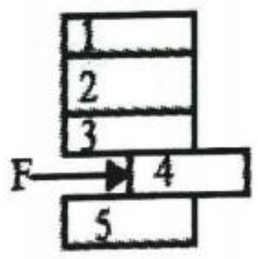
97. You stand at A near a river flowing from W to E. The land is covered by mud due to a heavy rain and become highly slippery. As you try to walk you tend to slip towards the river. Fortunately, you have an orange in your pocket. In what direction do you thrown the orange in order to safe from sliding to the river?



A. Towards E
B. Towards N
C. Towards W
D. Towards S

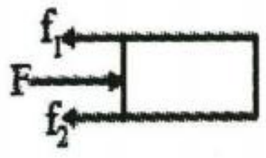
Explanation: Towards N, thus, you will be pushed towards S by the reaction force. **Answer: B**

98. There are five similar books in a stack with a weight of 3N. With what force must you push to slid the fourth book out of the stack. ($\mu = 0.1$)



A. 2.4N
B. 1.2N
C. 2.1N
D. 0.9N

Solution: Free body diagram of the fourth book. Where $f_1 = \mu m_1 g$ $= 3\mu mg$ and $f_2 = \mu m_2 g = 4\mu mg$



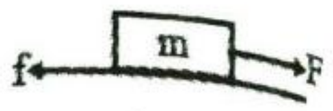
$\Rightarrow F = 3\mu mg + 4\mu mg = \mu mg\,(3 + 4)$

$= (0.1)(3)(7) = 2.1N$ **Answer: C**

99. An object of mass M moves with a constant speed on a horizontal rough surface by the action of a horizontal force of magnitude F. If the coefficient of static friction is μ. What is the magnitude of the acceleration when the force is removed?

A. g B. $\frac{1}{2}\mu g$ C. μg D. 0

Solution: $F - f = 0 \Rightarrow F = f = \mu mg$ . . . (*) and



$F = ma$ . . . (*) from equations (*) and (**)

$a = \mu g$

Answer: C

100. A block slides down a 37° incline with a constant velocity. The coefficient of friction between the surfaces is? (sin 37 = 0.6 and cos 37 = 0.8)

A. 0.21
B. 0.38
C. 0.75
D. 0.14

Solution: $F - f = 0$

$mg \sin\theta - \mu mg \cos\theta = 0$

$$\Rightarrow \mu = \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{0.6}{0.8} = 0.75$$

Answer: C

101. A block is projected up a 37° frictionless plane, with a velocity of 6 m/s. What distance the block will move along the inclined plane?

A. 6m B. 1.5m C. 3m D. 2m

Solution: $F = ma$

$mg \sin\theta = -ma$

$\Rightarrow a = -g \sin\theta = -10 \times \sin 37° = -10 \times 0.6 = -6\ m/s^2$

$$\text{And } v^2 = u^2 + 2as \Rightarrow s = \frac{v^2 - u^2}{2a}, v = 0$$

$$s = \frac{-u^2}{-2a} = \frac{(6)}{2(6)} = 3m$$

Answer: C

102. In the diagram below. What is the reaction to the pull of the hand?

A. The force the box exerts on the hand.
B. The force the rope exerts on the box.
C. The force the rope exerts on the hand.
D. The force the box exerts on the box.

Explanation: The hand pulls on the rope. By Newton's third law, the rope exerts a reaction force on the hand.

Answer: C

103. A bird flies by pushing the air downward. The air in turn pushes the bird upward. Which of the physics laws does the bird uses to fly?
   A. Pascal's law
   B. Archimedes's Principle
   C. Newton's third law
   D. Kepler's third law

   Answer: C

104. "You cannot push a table without it pushes back on you," which of the physics laws describes this statement
   A. Newton's third law
   B. Faraday's Law
   C. Newton's second law
   D. Coulomb's law

   Answer: A

105. Suppose you are carrying a bag of books. Which of the following statements is correct?
   A. The force you exert is oppositely directed to the force the books exert.
   B. The force you exert is larger and directed oppositely to that the books exert.
   C. The two forces are equal in magnitude and parallel.
   D. The force you exert on the books is larger.

   Answer: A

106. Newton's third law of motion describes about:
   A. Two bodies exerting opposite forces on each other.
   B. Two equal but opposite forces acting on a body.
   C. A force acting on and accelerating the body.
   D. The gravitational pull of the earth on a body.

   Answer: A

107. A body of mass 100g that was originally, at rest is acted on by a force of magnitude 0.1N. What is the magnitude of the velocity of the body after 10 seconds?
   A. 0.01 m/s B. 10 m/s C. 100 m/s D. 0.1 m/s

   Solution: $F\Delta t = m\Delta t = m(v-u) \Rightarrow v-u = \frac{F\Delta t}{m} \Rightarrow v = \frac{F\Delta t}{m} + u$

   $$v = \frac{(0.1)(10)}{0.1} + 0 = 10 m/s$$

   Answer: B

108. Which situation will produce the greatest change of momentum for 1 kg car?
   A. Accelerating it from rest to 3 m/s.
   B. Accelerating it from 1 m/s to 3 m/s.

C. Applying a net force of 10N for 0.5 sec.
D. Applying a net force of 5N for 2 sec.

Solution: $F\Delta t = m\Delta t = \Delta P$

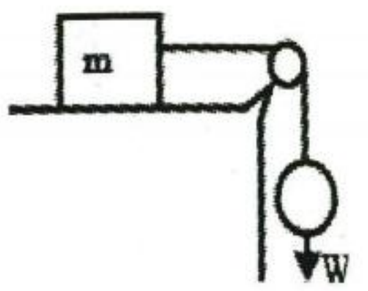
$\Delta P = F\Delta t = (5)(2) = 10$ kgm/s **Answer: D**

109. What happens to a body on which a constant net force acts?
A. Its velocity increases constantly.
B. It's acceleration increases constantly.
C. It moves with constant velocity.
D. Its acceleration becomes opposite to the force.

Explanation: A constant force gives the body a constant acceleration thus, the velocity increase constantly. **Answer: A**

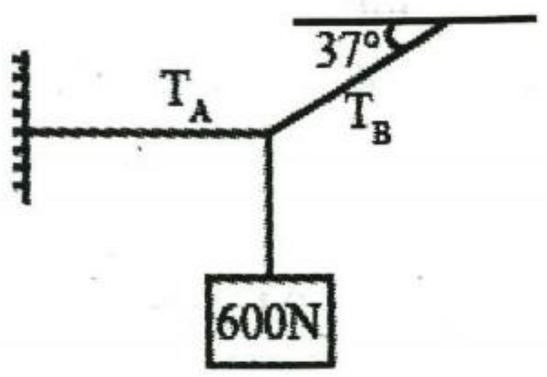
110. A student attempts to measure the mass of a brick by measuring the force required to accelerate it at 1 m/s^2. The student concluded that the brick has a mass of 2 kg. A balance shows that the mass of the brick is 1.5 kg. The experimental error is most likely due to?
A. Gravity B. Friction C. Inertia D. Work
Answer: B

111. Which force is the reacting force to the weight (W) shown in the diagram below?



A. The tension in the string.
B. The pull of the block on the string.
C. The pull of the sphere on the earth.
D. The normal force of the table.
Answer: C, as the earth pulls the body, the body also pulls the earth upwards.

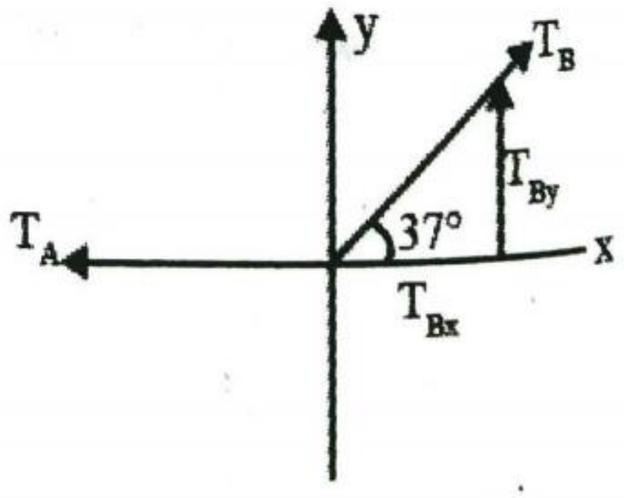
112. What are the tensions in cable A and B supporting the 600N body? (sin 37°, cos 37° = 0.8)



A. $T_A = 800N$ and $T_B = 600N$
B. $T_A = 1000N$ and $T_B = 800N$
C. $T_A = 600N$ and $T_B = 800N$
D. $T_A = 800N$ and $T_B = 1000N$

Solution: $T_Bx = T_B \cos 37$

$T_Bx = 0.8\ T_B$



From the first condition of equilibrium:

$$\sum F_x = 0 \Rightarrow T_Bx - T_A = 0 \Rightarrow 0.8T_B = T_A$$

$$\sum F_y = 0 \Rightarrow T_B y - W = 0 \Rightarrow 0.6T_B = W = 600N$$

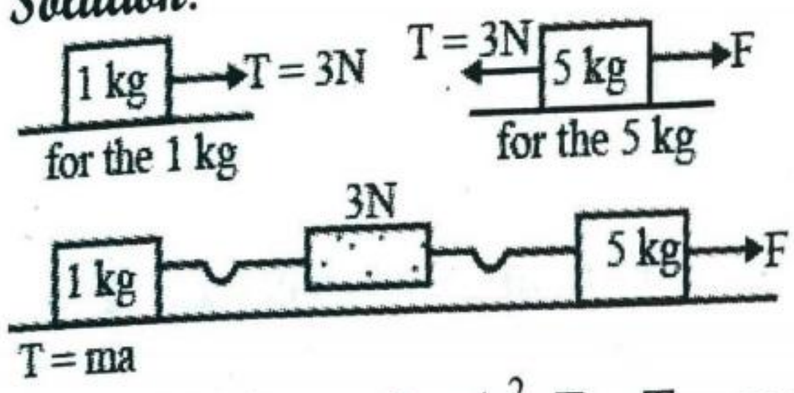
$$\Rightarrow T_B = \frac{600}{0.6} = 1000N \;\&\; T_A = 0.8T_a = (0.8)(1000) = 800N$$

Answer: D

113. Two blocks of mass 1 kg and 5 kg are connected by a string which has a tension of 3N. A force F acts in a direction shown to the right. Neglecting friction, what is the value of F?

A. 18N B. 8N C. 9N D. 1.8N

Solution:

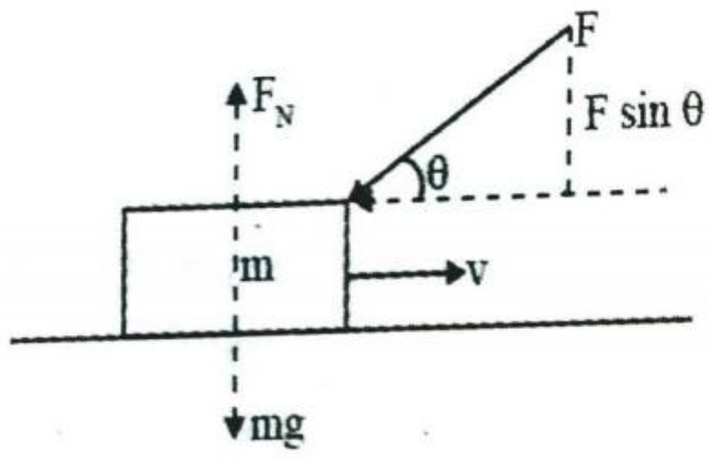


$\Rightarrow 3 = (1)(a)$, $a = 3\ m/s^2$, $F - T = ma \Rightarrow F = T + ma = 3 + (5)(3)$

$F = 18N$

Answer: A

114. A block of mass m moving to the right is subjected to a retarding force F at an angle θ as shown. What is the magnitude of the friction force acting on the block if the coefficient of friction is μ?



A. μmg

C. μ(mg – F sin θ)

B. μ(mg + f cos θ) D. μ(mg + F sin θ)

Solution:

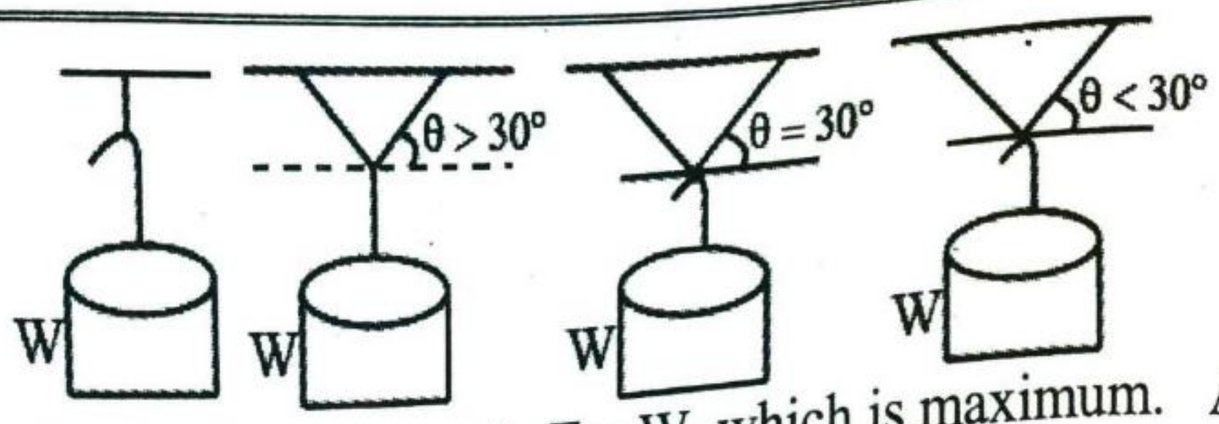
$f = \mu F_N$ but $F_N = mg + F \sin\theta \Rightarrow f = \mu(mg + F \sin\theta)$ **Answer: D**

115. Consider the deflating balloon shown. Which of the physics laws best describes the behavior of the balloon's motion?

A. Newton's third law
B. Archimedes' principle
C. Law of flotation
D. Newton's first law

Answer: A

116. Assume the objects have equal mass and the strings holding them are identical, in which case would the tension in the string be greatest? (sin 30° = 0.5)



Explanation: In case A, T = W, which is maximum. **Answer:** A

117. A spring is stretched by 5 cm when a load of 80N is suspended from its end. What load stretch the spring by 2 cm?

A. 32N B. 16N C. 94N D. 200N

Solution: From $F = kx$, $k = \frac{F}{x} = \frac{80}{0.05} = 1600N/m$

When $x = 2\ cm = 0.02m$

$F = kx = (1600)(0.02) = 32N$ **Answer:** A

118. Which combination of three concurrent forces acting on a body could NOT produce equilibrium?

A. 3N, 4N, 5N
B. 3N, 4N, 1N
C. 3N, 4N, 8N
D. 3N, 4N, 7N

Answer: C

119. A brick slides on a horizontal surface. Which of the following will increase the magnitude of the friction force on it?

A. Decreasing the mass of the brick.
B. Decreasing the surface area of contact.
C. Increasing the surface area of contact.
D. Putting a second brick on top of it.

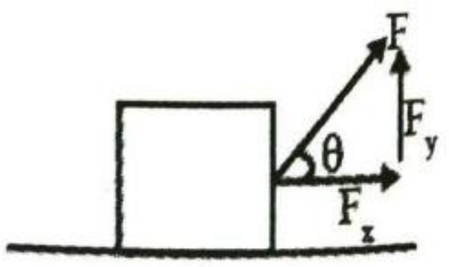
Explanation: Friction is independent of contact area but it increases as the normal force increase. **Answer: D**

120. The vertical component of a force is 20N. If the force makes an angle of 53° to the horizontal, what is the magnitude of the force? (sin 53 = 0.8)

A. 25N B. 16N C. 25N
D. 33N

Solution: $F_y = F \sin\theta$

$$\Rightarrow F = \frac{F_y}{\sin\theta} = \frac{20}{0.8} = 25N$$



Answer: C

121. Which of the following are contact forces?

A. Friction, gravity, electrostatic and kick.
B. Friction, tension, thrust, air resistance.

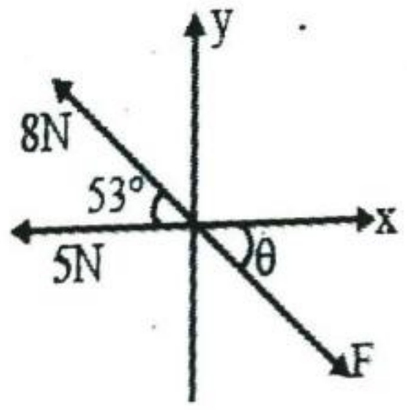
C. Electrostatic magnetic, weight, viscous.
D. Gravity, magnetic, electrostatic, buoyant.

Explanation: Thrust is an upward pushing force such as the force of the burning gas up on a rocket. **Answer: B**

122. In the diagram below three forces act concurrently on an object. If the object is in equilibrium, what must be the magnitude of component of F (sin 53° = 0.8, cos 53° = 0.6)

A. 9.8N and 6.4N
B. 5N and 8N
C. 5N and 6.4N
D. 11.4N and 6.4N

Solution:



For equilibrium $\sum F_x = 0$

$\Rightarrow 5 + 8\cos 53° = F\cos\theta$

$\Rightarrow F\cos\theta = 5 + 8(0.6) = 9.8N$

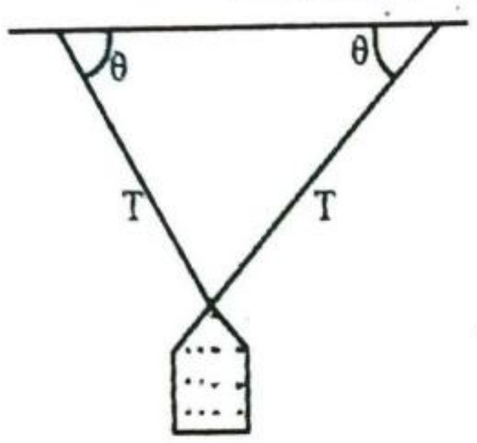
$\sum F_y = 0 \Rightarrow 8\sin 53° = F\sin\theta$

$\Rightarrow F\sin\theta = 8(0.8) = 6.4N$ **Answer: A**

123. A system is in linear equilibrium if and only if the sum of all:
A. Forces acting on the x axis is zero.
B. Forces acting on the y axis is zero.
C. Forces acting on the system is zero.
D. Torque acting on the system is zero.

Explanation: For a linear equilibrium, the necessary condition is $\sum F = 0$. **Answer: C**

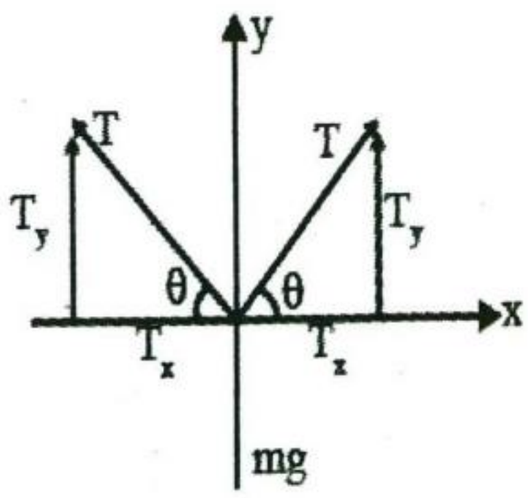
124. A concrete block of mass m hangs from two wires. The wires make the same angle with the horizontal and the tension in them is the same as shown below. If the system is in equilibrium, what will be the tension in each wire?



A. $T = \dfrac{mg}{\cos\theta}$
B. $T = \dfrac{mg}{2\sin\theta}$
C. $T = \dfrac{mg}{\sin\theta}$
D. $T = \dfrac{mg}{2\cos\theta}$

Solution: $\sum F_y = 0, T_y + T_y = mg$

$T\sin\theta + T\sin\theta = mg$



$\Rightarrow T(2\sin\theta) = mg \quad T = \dfrac{mg}{2\sin\theta}$

Answer: B

125. A 0.2 kg ball of mud falls vertically onto a 2.8 kg cart that is rolling freely at 3 m/s on a horizontal surface. What is the find speed of the cart?

A. 3 m/s B. 5.6 m/s C. 2.8 m/s D. 0.4 m/s

Solution: $P_i = P_f$

$m_m u_m + m_c u_c = (m_m + m_c)v,\ u_m = 0$

$\Rightarrow v = \dfrac{m_c u_c}{m_m + m_c} = \dfrac{(2.8)(3)}{0.2 + 2.8} = 2.8\text{m/s}$

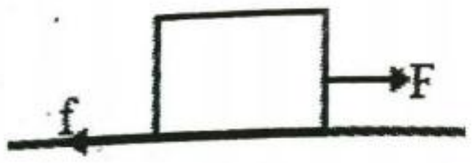
Answer: C

126. A 30N force is applied on a 50N block that is at rest on a horizontal surface. If $\mu_s = 0.7$ and $\mu_k = 0.5$. What friction force is blocking the body from sliding when the 30N force acting on it?

A. 30N static friction
B. 25N kinetic friction
C. 30N kinetic friction
D. 35N static friction

Solution:

$f_s = \mu_s F_N = (0.7)(50) = 35\text{N} > 30\text{N}$



Answer: A

127. In the laboratory to determine the force constant of a spring, a student obtained the following data.

F(N)	0	2	4
x(m)	0	0.015	0.028

What is the average value of the spring constant k?

A. 92 N/m B. 138N/m C. 133.3N/m D. 142.8N/m

Solution: $k = \dfrac{k_1 + k_2 + k_3}{3}$, and from $F = kx$

$k = \dfrac{F}{x} \Rightarrow k_1 = \dfrac{F_1}{x_1} = \infty$

$k_2 = \dfrac{F_2}{x_2} = \dfrac{2}{0.015} = 133.3\text{N/m}$

$k_3 = \dfrac{F_3}{x_3} = \dfrac{4}{0.028} = 142.9\text{N/m}$

$k = \dfrac{k_2 + k_3}{2} = \dfrac{133.3 + 142.9}{2} = 138\text{N/m}$

Answer: B

128. Suppose you are travelling up with an elevator to the 10^{th} floor of a building. When the elevator is about to stop, your weight appear to be:

A. Heavier
B. The same
C. Lighter
D. Weightlessness

Answer: A

Conceptual Questions on unit three

1. Does a body necessarily move in the direction of the net force on it? Explain
2. Can a motor cycle and a truck have the same momentum? Explain
3. Does the length of time that a force if applied to an object affect the change in the object's momentum. Explain
4. What does it mean to say that someone "weighs" 50 kilograms?
5. If you jump from a boat on the river to the sea shore, the boat moves in the opposite direction. Why?
6. As you sit in your chair, the chair exerts an upward force on you. Why don't you accelerate upward?
7. Since every force has an oppositely directed reaction force of equal magnitude, how can there ever by a net force on an object?
8. A friend thinks that Astronauts are weightless because there is no gravity, in outer space. Criticize this statement.
9. Can an object move if there is no net force on it?
10. Earth's gravity pulls a satellite toward the center of the earth. so why doesn't the satellite actually fall to the earth?
11. If surfaces are very highly polished, friction actually increases. Why might this be?
12. A coffee cup made of glass is more likely to break when it lands on a concrete floor than when it lands on a pond of water when dropped from the same might. Why?
13. A clothesline always sags a little even if nothing hangs from it. Explain why this happens.
14. A train typically requires a greater distance to come to rest for a given initial speed than does a car why?
15. Can an object be in equilibrium if it is moving? Explain
16. When rounding a corner on a bicycle or a motorcycle, the driver leans inward, toward the center of the circle. Why?
17. If you drop your pen, its momentum increases as it fall. Why is the momentum not conserved?

18. Is the impulse produced by a large force always greater than the impulse produced by a small force? Explain
19. Crash statistics show that it is safer to be riding in a heavy car in an accident than in a light car. Why?
20. Two cars collide at an interaction. If the cars do not stick together, can we conclude that their collision was elastic? Explain
21. An object at rest on a frictionless surface is struck by a second object. Is it possible that after collision both objects are at rest? Explain

Unit Three: Problems

Hooke;s Law

22. A spring is compared by 5 cm when a force of 10N is applied on it. What is its force constant?
23. What force is needed to stretch a rubber by 10 cm if its spring constant is 5000N/m?
24. A 50N force stretches a spring by 5mm. How much will this spring stretch if the applied force is 80N?
25. A spring of length 15 cm is placed on a vertical position. If it is compressed by 3 cm when a block of mass 2 kg is placed on it, what is the force constant of the spring.

Newton's Second law

26. The speed of a 100 kg car changes from rest to 80 m/s in 10 sec. What force is supplied by its engine?
27. In a grocery store, you push a 12 kg shopping cart with a force of 20N. If the cart starts from rest, how far will it move in 3 sec?
28. A 0.5 kg billiard ball is given a speed fo 10 m/s during 0.05 sec. What force acted on the wall?
29. A force of 5N gives mass m_1 an acceleration of 8 m/s^2 and mass m_2 an acceleration of 24 m/s^2. What acceleration would it give whey they are fastened together?
30. A motorcycle of mass 100 kg is accelerated by a force of 500N. If the motorcycle starts from rest.
    a) What is its acceleration?
    b) How fast will it be moving after 20 sec?
    c) How long will it take to reach a speed of 35 m/s.
    d) How far will it move from its starting point in 15 sec?

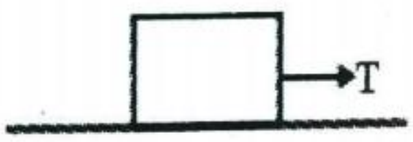
Apparent weight, mass and weight

31. An elevator is moving up with an acceleration of g/5. What is the apparent weight of 60 kg man standing in the lift?
32. At the surface of mars the acceleration due to gravity is 3.7 m/s^2. A body weighs 64N at the surface of the earth. What are its mass and weight on the surface of mars?
33. If I can barely lift a 50 kg concrete block on earth, how massive a block can I lift on the moon? ($g_m = 1.6\ m/s^2$)
34. Find the mass and weight of a body if a resultant force of 400N causes it to decrease its velocity by 4 m/s in 3 seconds?

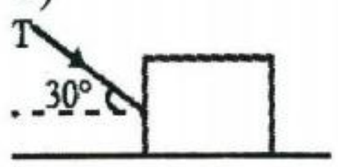
Friction force

35. A car weighing 9800N travels at 30 m/s. What braking force brings it to stop in 100m?
36. The friction force acting on a 400 kg car is found to be 750N. What acceleration will a force of 2250N produce on the car?
37. A horizontal force of 20N is required to slide a box along a level floor at a constant speed of 0.3 m/s. How large is the friction opposing the motion?
38. A horizontal force of 5000N is required to give a 1000 kg car an acceleration of 0.2 m/s^2 on a level road. How large is the friction opposing the motion?
39. Each block weighs 50N and T = 20N. Find the normal force in each case.

a)



b)



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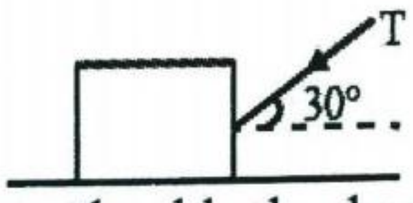
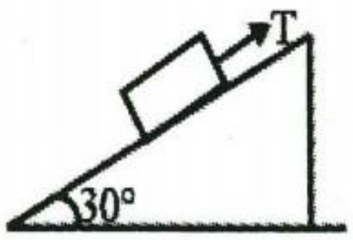
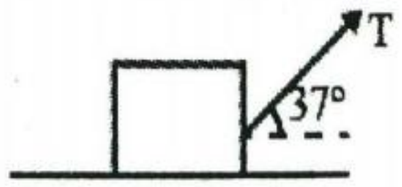


d)



40. Suppose the block shown weighs 40N and T = 50N if the coefficient of friction is 0.3. Find
    a) Friction force
    b) Acceleration of the block.



41. A 5 kg box slides down a 30° incline under the action fo gravity. If the box slides with constant speed, how large is the friction that retards its motion.

42. If the coefficient of friction between a car's tires and a roadway is 0.7. What is the least distance in which the car can accelerate from rest to 15 m/s?
43. A 50 kg woman stands inside an elevator. What does the scale read when the elevator accelerates at 3 m/s^2?
    a) Upward
    b) Downward
44. A block weighing 20N rests on a horizontal surface. If $\mu_s = 0.2$. What is:
    a) The friction force size when the block is at rest.
    b) The magnitude of friction if a horizontal force of 5N is applied on the block.
    c) The minimum force required to start the block moving.
    d) The minimum force required to keep the block in motion.
    e) If the horizontal force is 10N, what will be the friction force?
45. A 12 kg box is placed on a rough surface. If a force of 20N applied at 30° to the horizontal cannot move the box, calculate the friction force.

Liner Momentum

46. The velocity of a 4 kg mass changes from 10 m/s to 18 m/s by applying a force. What is the impulse produced by the force?
47. A force of 6N acts on an object of mass 3 kg for 10 sec, what is
    a) The change in its velocity.
    b) The change in its momentum.
48. A car of mass 1100 kg moves at 22 m/s. What braking force is needed to bring the car to a halt in 20 sec?
49. A net force of 2000N acts on a rocket of mass 1000 kg. How long does it take this force to increase the rocket's velocity from 0 m/s to 200 m/s?
50. A constant force acts on a 600 kg mass for 68 sec. The velocity of the mass is 10 m/s before the force is applied, and its final velocity is 44 m/s.
    a) What change in momentum is produced?
    b) What is the magnitude of the force?
51. A car of mass 700 kg travels at 20 m/s. The car collides with a stationary truck of mass 1400 kg. If they stick together. What is their common velocity?
52. A 900 kg car going at 20 m/s strikes a wall and stops in 3m.
    a) How long does it take the car to stop?

b) What is the stopping force?

53. An 8 kg fish moving at 3.2 m/s to the right swallows a 2 kg fish that is swimming at 6.8 m/s to the left. What is the speed of the larger fish after its lunch?

54. A 20 kg dog is running due North at 2.5 m/s while a 5 kg cat is running due East at 3 m/s. Their 70 kg owner has the same momentum as the two pets taken together. Find the magnitude of the owner's velocity.

Unit three: Explanation to Questions

1. Yes, a net force produces acceleration in its direction.
2. Yes, this is possible when the motorcycle have a very large velocity and the truck a very small velocity.
3. Yes, as time length increase, change in momentum also increase. Because, $F\Delta t = m\Delta v = \Delta p$.
4. It means the mass of the body is 50 kg. But it is not fair to use the term "weight".
5. It is due to the reaction force.
6. Because, there is no net force, $\sum F = 0$.
7. A difference in mass creates acceleration, thus net force.
8. There is an apparent weightlessness not true weightlessness.
9. Yes, the body can move at constant velocity without force.
10. Due to the action – reaction interaction.
11. It is due to viscosity that is cohesion and adhesion.
12. It is due to the difference in time of transmission of the force between the interacting surfaces. The shorter the time, the greater the force.
13. This is due to the effect of gravity.
14. This is because, the friction between the train and the railway is small and that between the car and the road is large.
15. Yes, if it is moving at constant velocity, $\sum F = 0$.
16. This is due to centripetal force.
17. Because, as it falls, a net force acts on it.
18. No. a small force can produce large impulse if it acts for a longer time.
19. Because, during collision, the larger car accelerate less than the smaller car, hence less sever.
20. No. a collision can be elastic if kinetic energy is conserved.

21. Yes. This happens when the collision is perfectly inelastic. That is when they stick together.

Unit three: solutions to problem

22. $F = kx \Rightarrow k = \frac{F}{x} = \frac{10}{0.05} = 200\text{N/m}$

23. $F = kx \Rightarrow F = (5000)(0.1) = 500\text{N}$

24. $F_1 = kx_1$ and $F_2 = kx_2 \Rightarrow \frac{F_1}{x_x} = \frac{F_2}{x_2} \Rightarrow x_2 = \frac{F_2}{F_1}(x_1)$

$$x_2 = \frac{80}{50}(0.005) = 8\,\text{mm}$$

25. $F = kx \Rightarrow mg = kx \Rightarrow k = \frac{mg}{x} = \frac{(2)(10)}{0.03} = 666.7\text{N/m}$

26. $F = ma = m\frac{(v-u)}{t} = 100\frac{(80-0)}{10} = 800\text{N}$

27. $F = m_1a_1 = m_2a_2 \Rightarrow a = \frac{F}{m} = \frac{20}{12} = 1.7\text{m/s}^2$, and

$$s = \frac{1}{2}at^2 = \frac{1}{2}(1.7)(3)^2 = 7.5\text{m}$$

28. $F = ma = m\frac{(v-u)}{t} = 0.5\frac{(10-0)}{0.05} = 100\text{N}$

29. $F = m_1a_1 = m_2a_2 \Rightarrow m_1 = \frac{a_2}{a_1}m_2 = \frac{24}{8}m_2 = 3m_2$

$$F = ma \Rightarrow a = \frac{F}{m} = \frac{m_1a_1}{m_1 + m_2} = \frac{(3m_2)(a_1)}{3m_2 + m_2} = \frac{3a_1}{4} = \frac{(3)(8)}{4}$$

$a = 6\ \text{m/s}^2$

30. a) $F = ma \Rightarrow a = \frac{F}{m} = \frac{500}{100} = 5\text{m/s}^2$

b) $v = u + at = 0 + (5)(20) = 100\ \text{m/s}$

c) $a = \frac{v-u}{t} \Rightarrow t = \frac{v-u}{a} = \frac{35-0}{5} = 7\,\text{sec}$

d) $s = ut + \frac{1}{2}at^2 = (0)(15) + \frac{1}{2}(5)(15)^2 = 562.5\text{m}$

31. $T = mg + ma = m(g + a) = m\left(g + \frac{1}{5}g\right) = 60\left(10 + \frac{10}{5}\right) = 1200\text{N}$

32. $m = \frac{w}{g} = \frac{64}{10} = 6.4\,\text{kg}$ and $W_m = mg_m = (6.4)(3.7) = 24\text{N}$

33. $W = mg = (50)(10) = (m)(1.6) \Rightarrow m = 312\text{ kg}$

34. $F = ma = m\left(\frac{\Delta v}{\Delta t}\right) \Rightarrow 400 = m\left(\frac{4}{3}\right) \Rightarrow m = 300\text{kg}$

$W = mg = (300)(10) = 3000\text{N}$

35. $F = ma = m\frac{(v^2 - u^2)}{2s} = 980\frac{(0-900)}{200} = 4410\text{N}$

36. $F - f = ma \Rightarrow 2250 - 750 = 400a \Rightarrow a = 3.75\text{ m/s}^2$

37. $F - f = 0 \Rightarrow f = F = 20\text{N}$

38. $F - f = ma \Rightarrow f = F - ma = 5000 - (1000)(0.2) = 4800\text{N}$

39. a) $F_N = mg = 50\text{N}$

b) $F_N = mg - T\sin 30° = 50 - (20 \times 0.5) = 40\text{N}$

c) $F_N = mg + T\sin 30° = 50 + (20 \times 0.5) = 60\text{N}$

d) $F_N = mg\cos 30° = (50)(0.87) = 43.5\text{N}$

40. a) $f = \mu F_N = \mu(mg - T\sin 37°) = 0.3(40 - 50 \times 0.6) = 3\text{N}$

b) $T\cos 3 + -f = ma \Rightarrow 50 \times 0.8 - 3 = 4a \Rightarrow a = 9.25\text{ m/s}^2$

41. $mg\sin\theta - f = 0 \Rightarrow f = mg\sin\theta = (5)(10)(0.5) = 25\text{N}$

42. $f = ma \Rightarrow \mu mg = m\frac{(v^2 - u^2)}{2s} \Rightarrow s = \frac{(15)^2}{(2)(0.7)(10)} = 16\,\text{m}$

43. a) $T = m(g + a) = 50(10 + 3) = 650\text{N}$

b) $T = m(g - a) = 50\,(10 - 3) = 350\text{N}$

44. a) $f = \mu_s mg = (0.4)(20) = 8\text{N}$

b) $f = F = 5\text{N}$

c) $f = F = 8\text{N}$

d) $f = \mu_k mg = (0.2)(20) = 4\text{N}$

e) since $F > f_s \Rightarrow f = f_k = 4\text{N}$

45. $F - f = 0 \Rightarrow f = F = 20\cos 30° = 20 \times 0.87 = 17.4\text{N}$

46. $\Delta P = m\Delta v = 4(18 - 20) = 32\text{ kg m/s}$

47. a) $F\Delta t = m\Delta v \Rightarrow \Delta v = \frac{F\Delta t}{m} = \frac{(6)(10)}{3} = 20 m/s$

b) $\Delta P = F\Delta t = m\Delta t = (3)(20) = 60$ kgm/s

48. $F\Delta t = m\Delta v \Rightarrow F = \frac{m\Delta v}{\Delta t} = \frac{(1100)(22)}{20} = 1210N$

49. $F\Delta t = m\Delta v \Rightarrow \Delta t = \frac{m\Delta v}{F} = \frac{(1000)(200)}{2000} = 100 \sec$

50. a) $\Delta P = m\Delta v = (600)(44 - 10) = 20400$ kgm/s

b) $F\Delta t = \Delta P \Rightarrow F = \frac{\Delta P}{\Delta t} = \frac{20400}{60} = 300N$

51. $m_1u_1 = (m_1 + m_2)v \Rightarrow v = m_1u_1/m_1 + m_2 = \frac{(700)(20)}{700+1400} = 6.67 m/s$

52. a) $s = \left(\frac{v+u}{2}\right)t \Rightarrow t = \frac{2s}{u} = \frac{2(3)}{20} = 0.3 \sec$

b) $F\Delta t = m\Delta v \Rightarrow F = m\Delta v/\Delta t = 900\ \frac{(20)}{0.3} = 60000N$

53. $m_1u_1 + m_2u_2 = (m_1 + m_2)v \Rightarrow 8 \times 3.2 - 2 \times 6.8 = 10v \Rightarrow v = 1.2$ m/s

54. $P_x = m_cv_c = (5)(3) = 15, P_y = m_dv_c = (20)(2.5) = 50$

$P = \sqrt{P_x^2 + P_y^2} = \sqrt{15^2 + 50^2} = 52.2 = m_mv_m \Rightarrow v_m = \frac{52.2}{m_m} = 0.7$ m/s

Unit Four

Work Energy and Power

Different types of machines and vehicles operate and do useful **work** because of the **energy** they get from fuel, wind or electricity. Plants grow because of the energy they get from the sun. We walk, talk, work and engaged in variety of activities because of the chemical energy we get from the food we eat. The term **power** is also used to indicate how fast a given work is done by a man or machine. Therefore, the concepts, work, energy and power are all round us every single day of our lives.

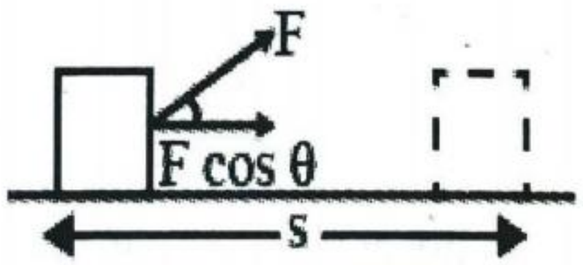
4.1 Mechanical work

In everyday language, any activity that requires muscular or mental effort is termed as work. But in physics, this is not true. For instance, a daily laborer who carried a quantal of cement and travelled a long distance horizontally, a man pushing a building as hard as he could untile he is tired and sweating, a manager sitting in his office and checking hundred and thousands of documents, according to physics, all the three men mentioned above did No work at all.

In physics, the term work has a special meaning.

> **Work**: Is said to be done when a force acts on a body and the body moves in the direction of the force.

That is, if a body is displaced horizontally on a smooth surface, then work is done on it by the horizontal component of the applied force ($F_x = F\cos\theta$).



The work done on the body by the applied force is then obtained by:

$$\boxed{W = Fs\cos\theta}$$

Where, θ is the angle between the force and the displacement of the body, note also that *work is a scalar quantity and its SI unit is Joule (J).*

Hence $\boxed{1J = 1Nm}$

☛ Important points to remember

Work is the product of displacement and the force in the direction of the displacement or.

Work is the amount of energy transferred when an object is moved through a distance by a force.

☛ The two necessary conditions for work to be done are:

i) Force must be applied

ii) The applied force must produce displacement in any direction except perpendicular to the direction of the force

Energy is the ability to do work thus an object may gain or lose energy in the process of doing work.

When a body is displaced horizontally, the vertical component of the force ($F \sin \theta$) does no work because, in this case, the displacement and the vertical component of the force are perpendicular that is, $\theta = 90°$, and $\cos 90° = 0 \Rightarrow W = 0$.

Conceptual Examples

1. A man carried a 100 kg box from a truck and travelled horizontally for 500m. Calculate the work done by this man.

   Explanation: $W = Fs \cos \theta$, but $F = mg = 100 \times 10 = 1000N$

   $s = 500m$. In this case, the man supported the box upward with a force of 1000N and he is moving horizontally. So, the angle between the force and the displacement is 90°. $\cos 90° = 0$

   $\Rightarrow W = Fs \cos \theta = (1000)(500)(0) = 0$

   The man did No work!

2. Can work done by a body be ever be:

   a) zero
   b) negative
   c) positive

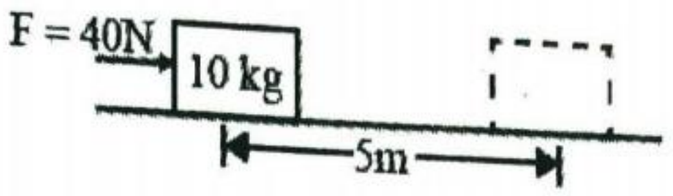
   Explanation: a) Yes, work done can be zero when a force is applied on a body but the body did not displaced or when the force and the displacement are at right angle to each other.

   b) Yes, work done can be negative when the applied force and the displacement are opposite to each other. Here, $\theta = 180°$, $\cos 180° = -1$ when the work done is negative, it means, the object loses energy.

   c) Yes, work done can be positive when the applied force and the displacement are in the same direction to each other. Here, $\theta = 0°$, $\cos 0° = 1$, $W = Fs$ when the work done is positive, it means, the object gains energy.

Illustrative Examples

3. A constant force of 40N is applied on a mass of 10 kg through a distance of 5m. What is the work done by this force?



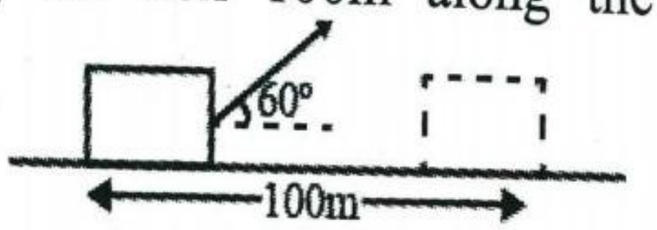
Solution: $W = F\,s\cos\theta$

$F = 40N$, $s = 5m$ and $\theta = 0°$

$W = (40)(5)(\cos 0°)$, $\cos 0° = 1$

$W = 200J$

4. A man pulled a box using a rope that makes an angle of 60° form the horizontal. If the man exerts a force of 200N, what is the work done by him in displacing the box 100m along the horizontal?



Solution: $W = Fs\cos\theta$

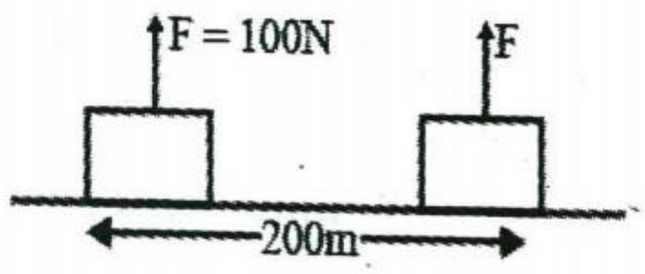
$F = 200N$, $s = 100m$, $\theta = 60°$

$W = (200)(100)(\cos 60°)$, $\cos 60° = 0.5$

$W = (200)(100)(0.5)$

$W = 10{,}000\ J$

5. When an object is moving along a horizontal direction for 200m, a boy tried to lift the object vertically upward by applying a force of 100N. How much work is done by the boy?



Solution: $W = Fs\cos\theta$

$F = 100N$, $s = 200m$, $\theta = 90°$

$W = (100)(200)(\cos 90°)$, $\cos 90° = 0$

$W = (100)(200)(0)$

$W = 0J$

6. What will be the nature of the work done when the angle between the force and the displacement is:

a) $\theta < 90°$ b) $\theta = 90°$ c) $\theta > 90°$

Solution: a) $W = Fs\cos\theta$, for $\theta < 90°$, $\cos\theta$ is positive

$\Rightarrow$Work done is positive

b) $W = Fs\cos\theta$, for $\theta = 90°$, $\cos\theta = \cos 90° = 0$

$\Rightarrow$ Work done is zero

c) $W = Fs\cos\theta$, for $\theta > 90°$, $\cos\theta$ is negative

$\Rightarrow$Work done is negative

7. When a force of 150N is made to act on a body at rest, its velocity increases to 10 m/s in 5 seconds. Calculate the work done by this force.

Solution: $W = Fs\cos\theta$, $\theta = 0°$ and $\cos 0° = 1$

$$W = Fs,\ s = \frac{(v+u)}{2}t$$

$$W = F\left(\frac{v+u}{2}\right)t$$

$$W = 150\left(\frac{10+0}{2}\right)5 = 3750J$$

8. a) What is the work done by a force of 20N acting through a parallel distance of 8m?
   b) What force will do the same work through a distance of 4m?

   Solution:

   a) $W = Fs\cos\theta$
      $F = 20N,\ s = 8m,\ \theta = 0°$
      $W = (20)(8)(\cos 0°),\ \cos 0° = 1$
      $W = 160J$

   b) $W = Fs\cos\theta$
      $F = ?,\ s = 4m,\ W = 160J,\ \theta = 0°$
      $160 = (F)(4)(1) \Rightarrow F = 40N$

9. How much work do you do as you exert a 95N force to push a 300N shopping cart through a 14m long super market floor?

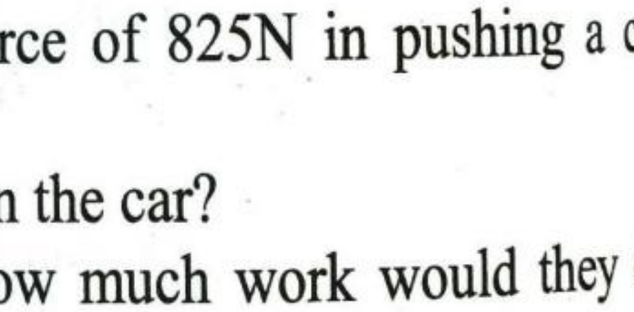
   Solution: $W = Fs\cos\theta$, $\theta = 0° \Rightarrow \cos\theta° = 1$
   $W = (95)(14)(1) = 1300\ J = 1.3\ kJ$

10. Two students together exert a force of 825N in pushing a car 35m.
    a) How much work do they do on the car?
    b) If the force were doubled, how much work would they do pushing the car the same distance?

    Solution:

    a) $W = Fs\cos\theta$, $\theta = 0° \Rightarrow \cos 0° = 1$
       $W = (825)(35) = 28875J = 28.875kJ$
    b) $W = Fs\cos\theta$, but $F = 2(825) = 1650N$
       $W = (1650)(35) = 57750J = 57.750kJ$

11. A block is dragged a distance of 10m by a rope under a constant force of 120N. If a 720J of work is done by the force, what angle does the rope make with the ground?

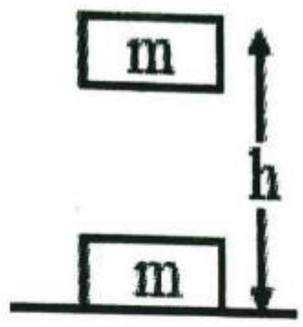


Solution: $W = Fs\cos\theta$

$$\Rightarrow \cos\theta = \frac{W}{Fs} = \frac{720}{(120)(10)} = 0.6\ \theta = \cos^{-1}(0.6) = 53°$$

Work Done Against Gravity

When an object of mass m is lifted to a height h from the ground, work is done in lifting the object against gravity. In this case, the minimum force required to lift the object is equal to its weight, mg.



Thus, $W = Fs\cos\theta$ but here, $F = mg$, $s = h$, $\theta = 0°$

$\Rightarrow \boxed{W = mgh}$

Where; g is acceleration due to gravity.

Illustrative Examples

12. How much work is done against gravity in lifting a 4 kg object through a distance of 50cm? (take, g= 10 m/s^2)

    Solution: $W = mgh = (4)(10)(0.5) = 20J$

13. The third floor of a building is 8m above the ground. How much work is needed to move a 150 kg refrigerator to the third floor?

    Solution: $W = mgh = (150)(10)(8) = 12000J = 12\ kJ$

14. The World's highest waterfall, the Cherun- Meru in Venezuela, has a height of 980m. How much work does gravity do on a cubic meter of water dropping down the Cherun – Meru?

    Solution: $W = mgh$ but, $1m^3 = 1000$ litre and 1 kg = 1 litre

    $\Rightarrow 1\ m^3 = 1000\ kg$

    $W = (1000)(10)(980) = 9800000J = 9.8\ MJ$

15. A 50 kg boy carrying a 10 kg bag walks upstairs of height 7m. What is the work by the boy?

    Solution: $W = mgh = (50 + 10)(10)(7) = 4200J = 4.2kJ$

16. A body of mass 0.4kg falls freely, what is the work done by the earth during the first 5 second?

    Solution: $W = mgh$, but $h = \frac{1}{2}gt^2$

    $$W = mg\left(\frac{1}{2}gt^2\right)$$

    $W = (0.4)(10)(0.5)(10)(5)^2$

    $W = 500J$

7. A pump pumps 3000 liters of water from a well of 20m below the ground. What is the work done by the pump (take $g = 10\ m/s^2$ and $\rho_w = 1000\ kg/m^3$)

Solution: *Method 1*, $3000\ lt = 3\ m^3$

$W = mgh$, but $m = v\rho = (3)(1000)$

$W = (3)(1000)(10)(20)$

$W = 600000J = 600kJ$

Method 2, take $1\ \ell t = 1\ kg$

$W = mgh = (3000)(10)(20)$

$W = 600000J = 600kJ$

18. A crane lifts a 10 kg object 4m vertically upward with an acceleration of $0.5\ m/s^2$. What is the work done by the crane? ($g = 10\ m/s^2$)

Solution: $W = Fs$, but $F - mg = ma \Rightarrow F = mg + ma$

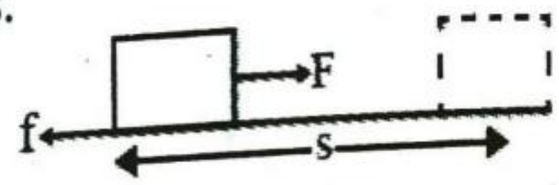
$W = (mg + ma)\ h$

$W = [(10)(10) + (10)(0.5)]\ 4$

$W = 420J$

Work done against friction

To push or pull a body along the horizontal a minimum force that is equal to the kinetic friction between the surfaces is required. Thus, the work done to overcome friction is: $W_f = f_k s$.



$$\boxed{W = \mu_k F_N s}$$

Where, μ_k: coefficient of kinetic friction

F_N: Normal force

f_k: kinetic friction

Illustrative Examples

19. A 200N force acts on a 10 kg box along a horizontal. If the box moves horizontally for 20m. **Calculate**

a) Work done by the applied force.

b) Work done by friction if the coefficient of kinetic friction between the box and the surface is 0.4 (take $g = 10\ m/s^2$)

Solution:

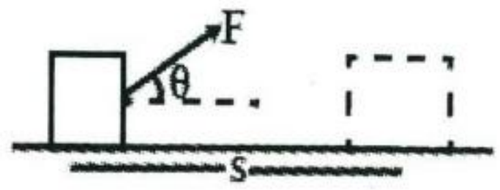
a) $W = Fs = (200)(20) = 4000J$

b) $W_f = \mu F_N s = \mu_k mgs = (0.4)(10)(10)(20)$

$W_f = 800J$

20. A 300N force acts on a 40 kg box at an angle of 37° with the horizontal surface. If the box moves horizontally for 20m. **Calculate**
   a) The work done in the absence of friction.
   b) The work done to overcome friction if the coefficient of friction between the box and the surface is 0.2 (take g = 10 m/s^2)
   c) The net work done

Solution:



a) $W = Fs\cos\theta$
$W = (30)(20)(0.8) = 4800J$

b) $W_f = f_k s = \mu_k F_N s$
$= (\mu_k)(mg - F\sin\theta)(s)$
$W_f = (0.2)(40 \times 10 - 300 \times 0.6)(20) = 880J$

c) $W_{net} = W_{app} - W_f = 4800 - 880 = 3920J$

21. A man pushes a 50 kg box on a level ground with a force of 1.75×10^2N horizontally through a distance of 6m. If the net work done on the box is 1.5×10^2J, find the coefficient of kinetic friction.

Solution: $W_{net} = W_{app} - W_f$
$W_{net} = Fs - W_{net}$
$\Rightarrow \mu mgs = Fs - \mu mgs$

$$\mu = \frac{Fs - W_{net}}{mgs} = \frac{1.75\times10^2 \times 6 - 1.5\times10^2}{(50)(10)(6)} = 0.3$$

22. A physics book is pushed 1.5m along a horizontal table top by a horizontal force of 2N. The opposing force of friction is 0.4N.
   a) How much work is done on the book by the 2N force?
   b) What is the work done on the book by the friction force?

Solution:
a) $W = Fs = (2)(1.5) = 3J$
b) $W = fs = (0.4)(1.5) = -0.6J$

23. A boy pushes a 35 kg crate a distance of 5m along a level floor at constant speed by pushing horizontally on it. The coefficient of kinetic friction between the crate and floor is 0.25.
   a) What magnitude of force must the boy apply?
   b) How much work is done on the crate by this force?
   c) How much work is done on the crate by friction?
   d) How much work is done by the normal force? By gravity?

Solution:

a) $F = f = \mu mg = (0.25)(35)(10) = 87.5N$

b) $W = Fs = (87.5)(5) = 437.5J$

c) $W = -fs = -(\mu)(mg)(s) = -(0.25)(35)(10)(5) = -437.5J$

d) Work done by the normal force = work done by gravity, which is zero.

Work done against both gravity and friction

If you were to push or pull on an object up on inclined plane, then you end up doing work against both friction and gravity. In this case, the total work done could be found as follows.

Total work done

= work done against gravity + work done against friction

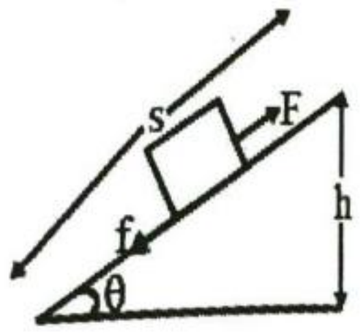
Work done against gravity = (weight) × (vertical distance moved)

$\Rightarrow W_{gravity} = mgh$

Work done against friction = (force of kinetic friction) × (distance moved up the plane)

$\Rightarrow W_{friction} = \mu_k F_N s$

So we have $W_{total} = W_{gravity} + W_{friction}$

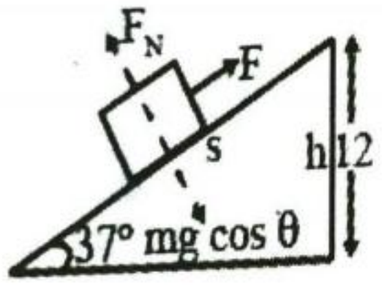


$$\boxed{W_T = mgh + \mu_k F_N s}$$

Illustrative Examples

24. What is the minimum work done by the force F to move an 8 kg block to the top of the incline if $\mu_k = 0.4$? (Take g = 10 m/s)

Solution: $W_T = mgh + (\mu_k F_N)\, s$



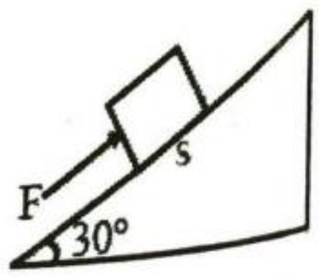
$$\sin\theta = \frac{h}{s} \Rightarrow s = \frac{h}{\sin\theta}$$

$$W_T = mgh + \mu_k mg\cos\theta\left(\frac{h}{\sin\theta}\right)$$

$$W_T = (8)(10)(12) + (0.4)(8)(10)\cos 37\left(\frac{12}{\sin 37}\right)$$

$$W_T = (8)(10)(12) + (0.4)(8)(10)(0.8)\frac{(12)}{(0.6)} = 1472J$$

25. A push of 80N moves a 5 kg block up a 30° inclined plane. The coefficient of kinetic friction is 0.25 and the length of the plane is 20m. **Calculate** the net work on the block.



Solution: Method I

$W_{net} = W_{app} - W_{friction} - W_{gravity}$

$W_{net} = (F)(s) - \mu_k(mg\cos\theta)(s) - mgh,\ h = (s)(\sin\theta)$

$W_{net} = Fs - \mu_k mg\cos\theta s - mg\sin\theta s$

$= (80)(20) - (0.25)(5 \times 10 \times \cos 30°)(20) - (5 \times 10 \times \sin 30°)(20)$

$= (80 \times 20) - (0.25 \times 5 \times 10 \times 0.87 \times 20) - (5 \times 10 \times 0.5 \times 20)$

$= 882.5J$

Method II

$F_R = F - f_k - mg\sin\theta = F - \mu mg\cos\theta - mg\sin\theta$

$F_R = 80 - (0.25 \times 5 \times 10 \times 0.87) - (5 \times 10 \times 0.5) = 44.125N$

$W_{net} = (F_R)(s) = (44.125)(20) = 882.5J$

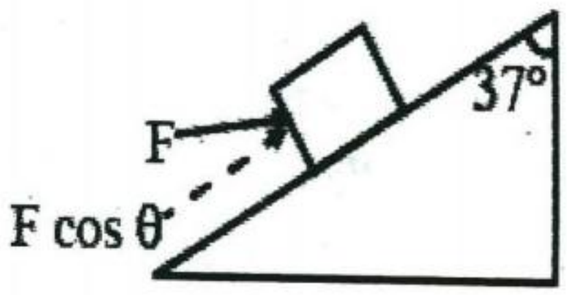
26. A block is pushed up an inclined plane of angle 37° with the vertically by a horizontal force of 80N. Ignoring friction, what is the work done by this force in moving the block 3m along the incline?

Solution: $W = Fs\cos\theta$

but $\theta = 53°$, and $\cos 53° = 0.6$

$\Rightarrow W = Fs\cos 53° = (80)(3)(0.6)$

$W = 144J$



Work done by a variable force

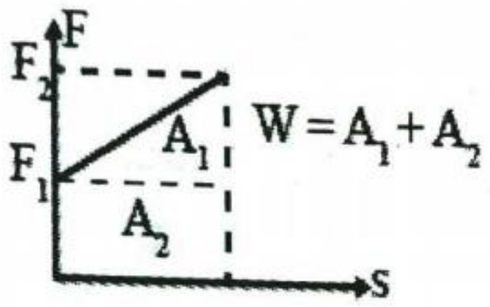
So far, we have calculated work only for constant force. Yet, most force in nature vary with positions. For example, the force exerted by a spring depends on how far the spring is stretched, and the force of gravity between planets depends on their separation. To calculate the work done by a force that varies with position, we cannot use the formula $W = Fs\cos\theta$.

☛ Work done by a variable force can be obtained by:

a) Calculating the area under the force v_s displacement graph or by

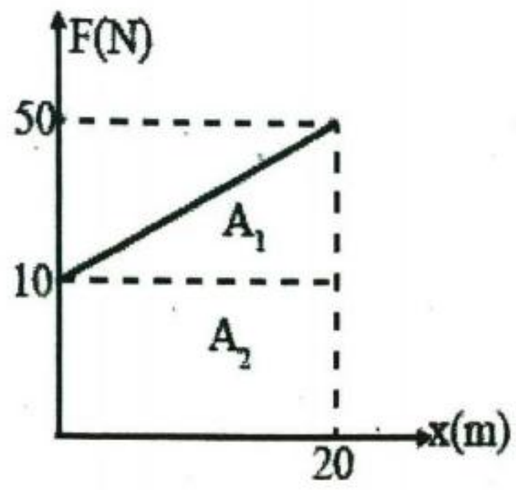
b) Using the relation, $\boxed{W = (F_{av})(s)}$

Where, $F_{av} = \dfrac{F_1 + F_2}{2}$



Illustrative Examples

27. The force acting on a body varies with the displacement of the body and is given by $F = (10 + 2x)N/m$. What is the work done by this force as the body displaces



from x = 0 to x = 20m?

Solution*: *Method I

W = Area under F – x graph

When x = 0, F = (10 + (2)(0)) = 10N

x = 20, F = (10 + (2)(20)) = 50N

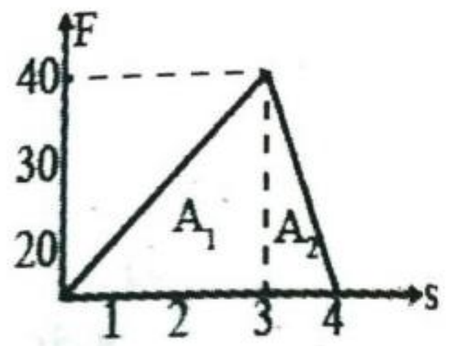
$$W = A_1 + A_2 = \frac{1}{2}(20)(40) + (20)(10) = 600J$$

Method II

$$W = (F_{av})(s) = \frac{(F_1 + F_2)}{2}(s) = \frac{(10+50)}{2}(20) = 600J$$

28. Find the total work done by the force shown as the object on which it acts moves.
   a) From x = 0 to x = 3m
   b) From x = 3m to x = 4m
   c) From x = 0 to x = 4m

***Solution*:**



a) $W = A_1 = \frac{1}{2}(3)(40) = 60J$

b) $W = A_2 = \frac{1}{2}(4-3)(40) = 20J$

c) $W = A_1 + A_2 = 60J + 20J = 80J$

4.2 Energy

When we say that an object has energy, we mean it is able to do work. Conversely, if you do work on an object, you have added to it an amount of energy that is equal to the work you did on it.

> **Energy**: Is the ability of a body to do work.

It is a scalar quantity and its SI unit is Joule (J).

Forms of Energy

Regardless of its different forms, energy is always either kinetic or potential.

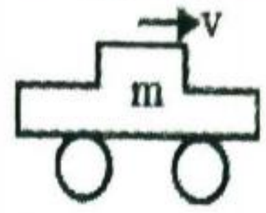
Kinetic energy

In order to knock a nail into a piece of wood with a hammer, the distance the nail can be driven depends on the speed of the hammer head. This energy which the hammer posses by virtue of its motion is known as kinetic energy. A moving car, a fired bullet and a rotating wheel. etc possess kinetic energy.

Kinetic energy: Is the energy a body possess because of its motion.

A body of mass m moving at a speed v posses, kinetic energy given by:

$$kE = \frac{1}{2}mv^2$$



Kinetic energy is a scalar quantity and its SI unit is Joule.

Conceptual Examples

29. Can the kinetic energy of a body ever be negative? Explain

Explanation: Kinetic energy can never be negative since m and v^2 are always positive or zero.

30. By what factor does an object's kinetic energy change if its speed is doubled?

Explanation: $kE_1 = \frac{1}{2}mv_1^2$

$kE_2 = \frac{1}{2}mv_2^2$, but now $v_2 = 2v_1$

$$\Rightarrow kE_2 = \frac{1}{2}m(2v_2)^2 = \frac{1}{2}(4mv_1^2) = 4\left(\frac{1}{2}mv_1^2\right)$$

$\Rightarrow kE_2 = 4kE_1$, it become four times

31. Two bodies are known to have the same momentum, does this mean that these two bodies necessarily have the same kinetic energy? Explain

Explanation: No suppose one of the bodies has mass m and speed v and the second has mass $\left(\frac{m}{2}\right)$ and speed (2v). The momenta are the same but the second body has twice as much kinetic energy. $\frac{1}{2}\left(\frac{m}{2}\right)(2v)^2 = 2\left(\frac{1}{2}mv^2\right)$

32. A body is known to have zero kinetic energy. What can you say about the momentum of the body?

Explanation: If the kinetic energy is zero, the speed must be zero as well. This means that the momentum is zero.

Illustrative Examples

33. A car is traveling at a speed of 20 m/s. If the mass of the car is 400 kg. What is the kinetic energy of the car?

Solution: $kE = \frac{1}{2}mv^2 = \frac{1}{2}(400)(20)^2 = (200)(400)$

$kE = 80000J = 80\ kJ$

34. An object with mass 1000 kg revolves around the earth having a kinetic energy of 2.45×10^{10}J. What is the speed of the revolving object?

Solution: $kE = \frac{1}{2}mv^2 \Rightarrow 2kE = mv^2$

$$v = \sqrt{\frac{2kE}{m}} = \sqrt{\frac{2(2.45\times10^{10})}{1000}} = \sqrt{\frac{4.9\times10}{1000}} = 7 \times 10^3 m/s$$

Work Energy theorem

This theorem relates work and kinetic energy. Since work done is energy transfer, this theorem says, if an external force acts on an object it will cause its kinetic energy to change.

Work Energy theorem: states that "the work done on an object is equal to the change in its kinetic energy."

$$\boxed{W = \Delta kE = kE_f - kE_i}$$

This can be verified as follows:

When the applied force and the displacement are parallel, then the work done;

$W = Fs$, but $F = ma$

$$W = (ma)s, \text{ from } v^2 = u^2 + 2as \Rightarrow as = \frac{v^2 - u^2}{2}$$

$$W = m\left(\frac{v^2 - u^2}{2}\right) = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = k_Ef - k_Ei = \Delta kE$$

$\boxed{\boxed{W = \Delta kE}}$ ← work energy theorem

Conceptual Examples

35. If the net work done on a body is zero. What can you say about its speed?

Explanation: From work energy theorem,

$$W = \Delta kE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

When $W = 0$, $\frac{1}{2}mv^2 = \frac{1}{2}mv^2 \Rightarrow v = u$

It means, the body moves with constant speed.

36. A 100J of work accelerates a car from 0 to 5 m/s. How much work is needed to accelerate the car from 5 m/s to 15 m/s.

Explanation: From work energy theorem,

$W = \Delta kE$

$$W_1 = 100 = \frac{1}{2}m(v_1^2 - u_1^2) = \frac{1}{2}m(5^2 - 0^2) = \frac{25}{2}m = 100 \Rightarrow m = 8kg$$

$$W_2 = \frac{1}{2}m(v_2^2 - u_2^2) = \frac{1}{2}m(15^2 - 5^2)$$

$$W_2 = \frac{1}{2}m(225 - 25) = \frac{200}{2}m = 100\ m, \text{ but } m = 8$$

$W_2 = (100)(m) = (100)(8) = 800J$

37. Whenever network is done on a body, will the body necessarily undergo acceleration? Explain

Explanation: From work energy theorem, $W = \Delta kE = kE_f - kE_i$

$\Rightarrow W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$, when W is none zero, the body speed changes. Yes it does.

Illustrative Examples

38. What is the work done by a constant force in accelerating a 2 kg block from rest to 15 m/s in a distance of 100m?

Solution: $W = \Delta kE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$$W = \frac{1}{2}(2)(15)^2 - \frac{1}{2}(2)(0)^2 = 225J$$

39. If a 4 kg body initially at rest travels a distance of 5m in 2 seconds under the action of a constant force, what work is done on the body during this time?

Solution: Method I

$$W = \Delta kE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$W = \frac{1}{2}mv^2,\ u = 0 \text{ and } s = \left(\frac{v+u}{2}\right)t \Rightarrow v = \frac{2(s)}{t}$$

$$W = \frac{1}{2}(4)(5)^2,\ v = \frac{2(5)}{2} = 5\,m/s$$

$$W = 50J$$

Method II

$$W = Fs,\ F = ma, \text{ and } a = \frac{v}{t} = \frac{5}{2} = 2.5m/s^2$$

$$W = (ma)s = (4)(2.5)(5) = 50J$$

40. What average force is necessarily to stop a 16g bullet travelling at 260 m/s as it penetrates into wood a distance of 12 cm?

Solution: $W = \Delta kE = \frac{1}{2}m\,v^2 - \frac{1}{2}mu^2$, but $v = 0$

$$Fs = -\frac{1}{2}mv^2$$

$$F(0.12) = -\frac{1}{2}(0.16)(260)^2$$

$$F = \frac{-(0.016)(260)^2}{(2)(0.12)} = -4510N$$

The negative sign indicates that the force is opposing force.

41. The driver of a 2000 kg car traveling at 20 m/s applies the brakes of the car. If the brakes provide a frictional force of 4000N. Determine the stopping distance of the car.

Solution: $W = \Delta kE_f - kE_i$

$$W = \frac{1}{2}m\,v^2 - \frac{1}{2}mu^2,\ v = 0$$

$$-fs = -\frac{1}{2}mu^2 \Rightarrow s = \frac{mu}{2f} = \frac{(2000)(20)}{2(4000)} = 100m$$

42. How large a force is needed to accelerate a 2000 kg car from rest to 15 m/s in 80m if an average friction force of 500N opposes the motion?

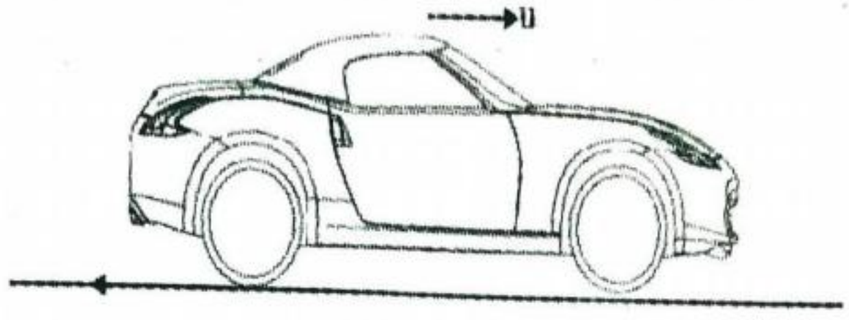
Solution: $W = \Delta kE$

$= kE_f - kE_i$

$(F - f)(s)$

$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$,

$u = 0$



$(F - 500)(80) = \frac{1}{2}(2000)(15)^2$, solving this gives

$F = 3300N$

Exercise: left for you

How much heat energy is generated as the car travels the 80m?

Answer: 40,000J

43. A body of mass 2 kg initially at rest moves under the action of applied horizontal force of 7N on a table with coefficient of kinetic friction 0.1. **Calculate**:
    a) Work done by the applied force in 10 sec.
    b) Work done by friction in 10 sec.
    c) Work done by the net force on the body in 10 sec.
    d) Change in kinetic energy of the body in 10 sec.

    Solution:

    a) Friction force, $f_k = \mu_k mg = (0.1)(2)(10) = 2N$ and, $F - f = ma$

    $\Rightarrow a = \frac{F-f}{m} = \frac{7-2}{2} = \frac{5}{2} = 2.5m/s^2$, distance moved in 10 seconds,

    $s = ut + \frac{1}{2}at^2 = (0)(10) + \frac{1}{2}(2.5)(10)^2 = 125m$

    Work done by the applied force in 10 seconds;

    $W_{app} = Fs\cos\theta = (7)(125)(\cos 0°) = 875J$

    b) Work done by friction in 10 seconds

    $W_f = fs\cos\theta = (2)(125)(\cos 180°) = -250J$

    c) Work done by the net force on the body in 10 sec.

    Method I

    $W = (F - f)s\cos\theta = (7 - 2)\,125\cos 0° = 625\ J$

    Method II

    $W = W_{app} + W_f = 875 + (-250) = 625J$

    d) Change in kinetic energy of the body in 10 sec.

    Method I

    $\Delta kE = kE_f - kE_i$, but $kE_i = 0$

$\Delta kE = \frac{1}{2}mv^2$, velocity of the body after 10 seconds is,

$v = u + at = 0 + (2.5)(10) = 25$ m/s

$\Delta kE = \frac{1}{2}mv^2 = \frac{1}{2}(2)(25)^2 = 625J$

Method II From work energy theorem

$W_{net} = \Delta kE$, but $W_{net} = 625J$

$\Rightarrow \Delta kE = 625J$

Potential energy

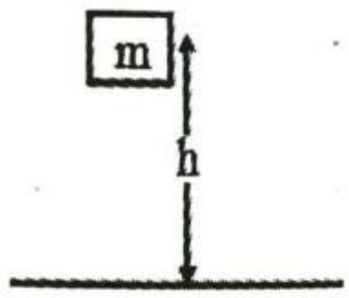
The energy possessed by a body because of its position or configuration, that is particle arrangement of the body is known as *potential energy*. Potential energy is of two types. Gravitational and elastic.

1. Gravitational potential energy

When water stored behind a dam is released it rotate a turbine that is placed at some height below the stored water. The water do work on the turbine because it possessed an energy called *gravitational potential energy*.

Gravitational potential energy: Is the energy possessed by a body because of its height from some reference surface.

The gravitational potential energy of a body of mass *m* placed some height *h* above the ground is given by:



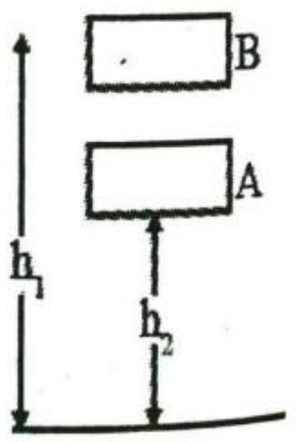
$$GPE = mgh$$

It is a scalar quantity and its SI unit is Joule (J).

Gravitational potential energy and work

☞ Let's consider two cases

a) When a body is displaced upward (from A to B) $W = mg\Delta h = mgh_1 - mgh_2$, it is *positive* since $h_1 > h_2$. Thus the work done against gravity is equal to the rise in gravitational potential energy of the body.



b) When a body is released downward (from B to A). In this case, work is done by the body.

$\boxed{W = mg\Delta h = mgh_2 - mgh_1}$, which is *negative* since $h_2 < h_1$.

Thus, the work done by the body is equal the fall in its gravitational potential energy.

Conceptual Examples

44. Can the gravitational potential energy of a body be ever be negative? Explain

Explanation: Yes. In gravitational potential energy, GPE = mgh. We can take h as negative or positive depending on our sign convention for upward and downward motion. Hence there is a chance for the GPE to be negative.

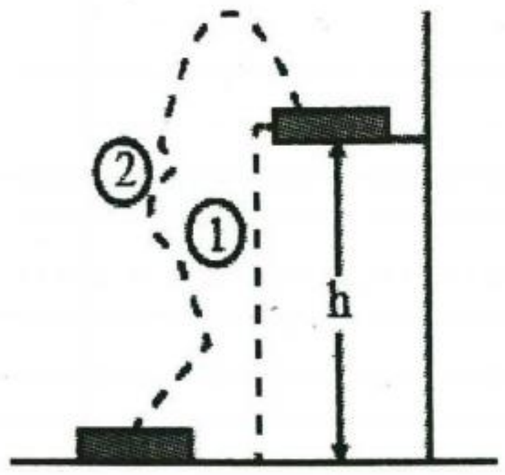
45. Is the change in gravitational potential energy of a ball that is dropped to the ground different from the same type of ball thrown to the ground from the same height? Explain

Explanation: No the change in gravitational potential energy depends only on the change in height, not on the change in speed.

46. If the change in gravitational potential energy between two points is zero, does this necessarily mean that the object experiences no force?

Explanation: No. It means, there is no change in elevation of the object between these two points.

47. A book is moved from the floor to the shelf in two ways. In which case is more work done against gravity? Explain



Explanation: The work done against gravity is the same in both cases. Work done against gravity is path independent.

Illustrative Examples

48. A weight lifter raises a 180 kg rock to a height of 2m. What is the increase in the potential energy of the rock? (take g =10 m/s^2)

Solution: GPE = mgh = (180)(10)(2) = 3600J = 3.6kJ

49. A 2 kg block rests on top of a table 80 cm from the floor. Find the potential energy of the block relative to:

a) The floor
b) The seat of a chair 40 cm from the floor.
c) The ceiling 3m from the floor.

Solution:

a) GPE = mgh = (2)(10)(0.8) = 16J

b) GPE = mgh = (2)(10)(0.8 – 0.4) = (2)(10)(0.4) = 8J

c) GPE = mgh = (2)(10)(0.8 – 3) = (2)(10)(-2.2) = -44J the negative sign indicates the fall in potential energy.

50. You lift a 2 kg text book from the floor to a shelf 2.1m above the floor. What is

a) The book gravitational potential energy relative to the floor.

b) Its gravitational potential energy relative to your head assuming that you are 1.65m tall.

Solution:

a) GPE = mgh = (2)(10)(2.1)

GPE = 42J

b) GPE = mgh = (2)(10)(2.1 – 6.5)

GPE = 9J

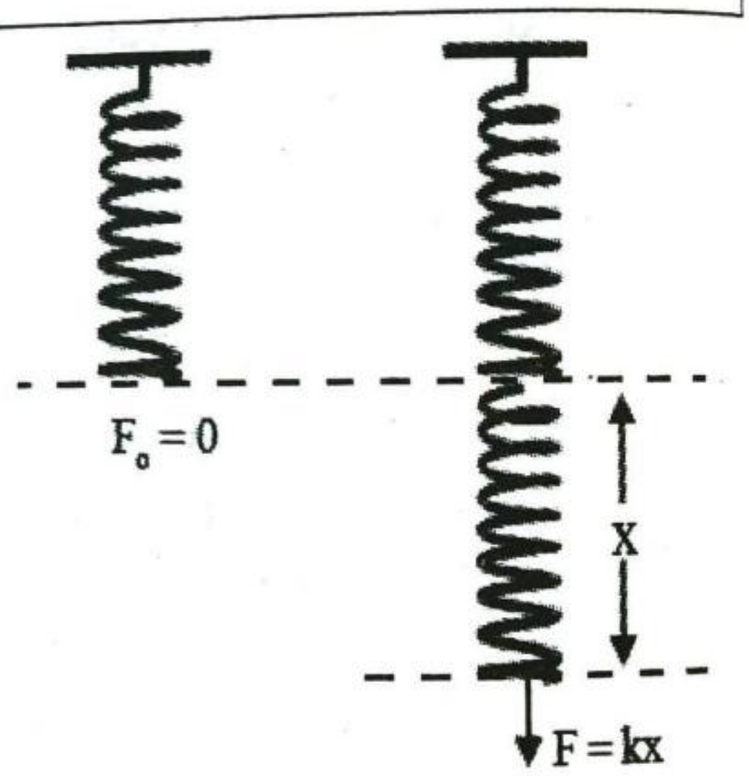
Elastic Potential Energy

A stone in a stretched catapult, an arrow in a stretched bow, and a block in a stretched or compressed spring, all possessed a stored energy called elastic potential energy.

Elastic potential energy: Is the energy possessed by a body because of its configuration.

The elastic potential energy of a spring of force constant k that is stretched or compressed by a distance x from its equilibrium position is given by:

$$G_{PE} = \frac{1}{2}kx^2$$



This can be derived as follows;

from Hook's law, the force required to stretch the spring to an extension x.

$$F = kx$$

The work done in stretching the spring an amount x is,

$$W = (F_{av})(x),\ F_{av} = \frac{F_o + F}{2} = \frac{0 + kx}{2} = \frac{kx}{2}$$

$$W = \left(\frac{kx}{2}\right)(x) \qquad \boxed{W = E_{PE} = \frac{1}{2}kx^2}$$

Conceptual Examples

51. If the stretch of a spring is doubled, by what factor does the potential energy increase?

Explanation: $E_{PE_1} = \frac{1}{2}kx_1^2$, when $x_2 = 2x_1$

$$E_{PE_2} = \frac{1}{2}kx_2^2, = \frac{1}{2}k(2x_1)^2 = 4\left(\frac{1}{2}kx_1^2\right), \text{ by four}$$

Illustrative Examples

52. A spring with a force constant of 500 N/m is compressed so that the elastic potential energy is 0.625J. Find the maximum distance the spring will be compressed.

Solution: $E_{PE} = \frac{1}{2}kx^2 \Rightarrow x = \sqrt{2E_{PE}/k}$

$$x = \sqrt{\frac{(2)(0.625)}{500}} = 0.05m$$

53. A spring with a spring constant of 400 N/m will have elastic potential energy of 800J. **Calculate** the average force acted on the spring.

Solution: $E_{PE} = \frac{1}{2}kx^2 \Rightarrow 800 = \frac{1}{2}(400)x^2 \Rightarrow x = 2m$

$$E_{PE} = F_{av}x \Rightarrow F_{av} = E_{PE}/x = \frac{800}{2} = 400N$$

54. When a force of 120N is applied on a spring, it causes a stretch of 2.25cm. What is the potential energy of this spring when it is compressed by 3.5 cm?

Solution: $F = kx \Rightarrow k = \frac{F}{x} = \frac{120}{0.0225} = 5330N/m$ then,

$$E_{PE} = \frac{1}{2}kx^2 = \frac{1}{2}(5330)(0.0350)^2 = 3.26J$$

Conservation of energy

The total mechanical energy of a system which is the sum of all possible kinetic and potential energies is constant if the system is closed system.

Law of conservation of energy states that "energy cannot be created or destroyed but it transformed from one form to the other."

Which means, the change in mechanical energy is zero for a closed system. $\Rightarrow \Delta M_E = 0$

Hence, $M_{Ei} = M_{Ef}$

$$P_{Ei} + K_{Ei} = P_{Ef} + K_{Ef}$$

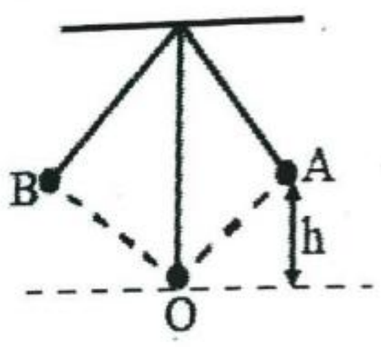
Conceptual Examples

55. A leaf falls to the ground with constant speed. Is there any difference in the mechanical energy of the leaf? Explain

Explanation: As the leaf falls, air resistance exerts a force on it opposite to its direction of motion. Hence, a negative work is done on it and its final mechanical energy is less than its initial mechanical energy.

Illustrative Examples

56. A pendulum is raised from its rest position 0 to point A that is at a height of 20 cm above point 0. What is the speed of the bob as it pass point 0?



Solution: $M_{Ei} = M_{Ef}$

$$P_{EA} + k_{EA} = P_{Eo} + k_{Eo}$$

$$mgh + 0 = 0 + \frac{1}{2}mv^2$$

$$v = \sqrt{2gh} = \sqrt{(2)(10)(0.2)} = 2m/s$$

57. A ball of mass 300g is released from a height of 5m above the ground. **Calculate**

a) The speed of the ball just as it reached the ground.

b) The speed of the ball after it fell 3m.

Solution:

a) $M_{Ei} = M_{Ef}$

$$P_{Ei} + k_{Ei} = P_{Ef} + K_{Ef}$$

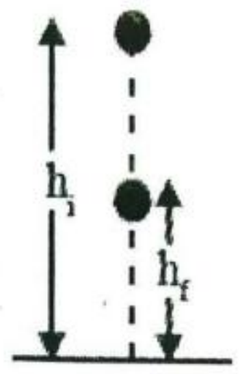
$$mgh + 0 = 0 + \frac{1}{2}mv^2$$

$$\Rightarrow v = \sqrt{(2)(g)(h)} = \sqrt{(2)(10)(5)} = 10m/s$$

b) $M_{Ei} = M_{Ef}$

$$P_{Ei} + k_{Ei} = P_{Ef} + k_{Ef}$$

$$Mghi + 0 = mgh_f + \frac{1}{2}mv^2$$



$$gh_i - gh_f = \frac{1}{2}v^2$$

$$10(5-2) = \frac{1}{2}v^2 \Rightarrow v = 7.7 \text{ m/s}$$

58. Consider the situation shown. What will be the maximum elongation of the spring when the block is released from rest?

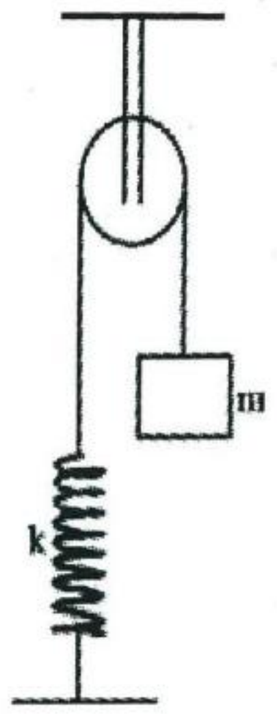
Solution: $M_{Ei} = M_{Ef}$

$$mgh = \frac{1}{2}kx^2$$

but here, $h = x$

$$mgx = \frac{1}{2}kx^2$$

$$\Rightarrow x = \frac{2mg}{k}$$



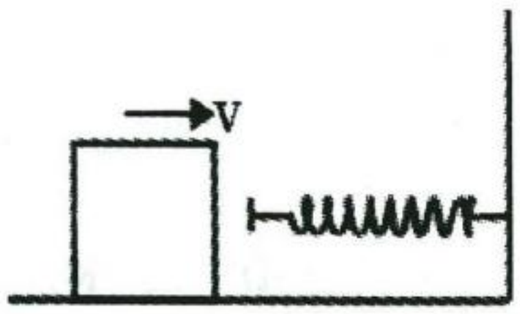
59. A 1 kg block collides with a horizontal spring of force constant 2 N/m. The block compresses the spring 4m from the rest position. If $\mu_k = 0.25$. What was the speed of the block at the instant of collision?

Solution: $KE = E_{PE} + W_f$

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 + \mu mgx$$

$$\frac{1}{2}(1)(v^2) = \frac{1}{2}(2)(4)^2 + (0.25)(1)(10)(4)$$

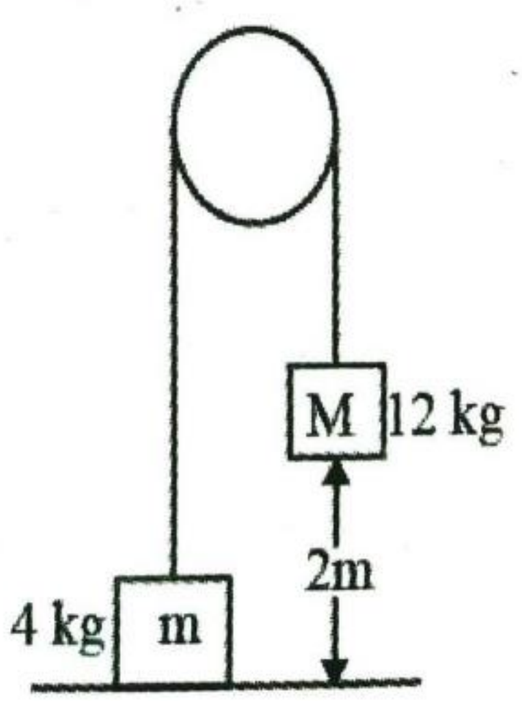
$$v = 5 \text{ m/s}$$



60. A block of mass 12 kg is released from rest from a height of 2m above the floor. Find the speed with which this block strikes the floor.

Solution: $M_{Ei} = M_{Ef}$

$$Mgh = \frac{1}{2}mv^2 + mgh$$



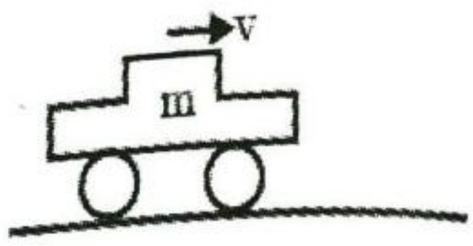
$$\frac{1}{2}mv^2 = Mgh - mgh$$

$$\frac{1}{2}(12)v^2 = (12)(10)(2) - (4)(10)(2)$$

$$v^2 = 26.7 \approx 5.1 \text{ m/s}$$

Kinetic energy and Momentum

Any object of mass **m** moving at a speed **v** is said to have both, momentum and kinetic energy. For instance, a car of mass **m** travelling at a speed **v** in a straight line do have;



→ Momentum, $p = mv \ldots (1)$ and

→ Kinetic energy, $KE = \frac{1}{2}mv^2 \ldots (2)$

From equation (1) and equation (2)

$$KE = \frac{1}{2}mv^2, p = mv \Rightarrow v = \frac{p}{m}$$

$$KE = \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m}$$

$$\Rightarrow \boxed{kE = \frac{P^2}{2m}} \text{ or } \boxed{P = \sqrt{(2m)(KE)}}$$

Illustrative Examples

61. If the momentum of a body increases by 20%. What will be the percentage increase in its kinetic energy?

Solution: $KE_1 = \frac{P_1^2}{2m} \ldots (1)$

$$KE_2 = \frac{(P_1 + 0.2P_1)^2}{2m} = \frac{P_1^2(1.2)^2}{2m} = \frac{1.44P_1^2}{2m} \ldots (2)$$

Taking ratio,

$$\frac{KE_2}{KE_1} = \frac{1.44P_1^2}{2m} \div \frac{P_1^2}{2m} = 1.44$$

$$\Delta KE = KE_2 - KE_1 = 1.44K_{E1} - K_{E1} = 0.44$$

$$\Delta KE = 44\%$$

62. If the kinetic energy of a body increases by 0.1% what will be the percentage increase in its momentum.

Solution: $P = \sqrt{2mkE}$

$$\%P_{increase} = \frac{1}{2}(0.1\%) = 0.05\%$$

Elastic and Inelastic Collisions

By a collision we mean a situation in which two objects strike one another, and in which the net external force is zero. For example, if two cars moving in a level road hit one another, this is a collision. Since the net external force is zero, the momentum of the system is conserved in the processes of collision.

☛ The fact that momentum of a system is conserved during a collision does not necessarily mean that the system's kinetic energy is conserved.

On the bases of what happens to the kinetic energy of the system after collision, collisions are either elastic or inelastic.

Elastic Collision: Is a collision in which both momentum and kinetic energy are conserved. That is, $\vec{P}_i = \vec{P}_f$ and $K_{Ei} = K_{Ef}$.

Collision of billiard balls, and collisions of subatomic particles are best examples of elastic collision.

Inelastic collision: Is a collision in which momentum is conserved but kinetic energy is not conserved. That is, $\vec{P}_i = \vec{P}_f$ and $K_{Ei} \neq K_{Ef}$.

☛ When objects stick together after collision, the collision is completely inelastic.

Conceptual Example

63. In an inelastic collision, can we conclude that the final kinetic energy is always less than the initial.

Explanation: No. When the collision sets off explosion this is not true.

Mechanical Power

Suppose a man takes one day to carry 200 bricks to a height of 5m. If a crane takes only one minute to do the same work, we say the crane is more powerful than the man.

Power is a measure of how quickly work is done.

Power: Is the rate of doing work.

$$\text{Power} = \frac{\text{work done}}{\text{time taken}} \Rightarrow \boxed{P = \frac{w}{t}}$$

It's a scalar quantity and its SI unit is Watt (W)

from $P = \frac{w}{t}$, $W = Fs$

$$P = \frac{Fs}{t} \Rightarrow \boxed{P = Fv}$$

Conceptual Examples

64. In a recent News, a reporter declared, "this hydro – power could yield 50 Megawatts of electricity an hour." Do you think this phrase makes sénse?

Explanation: Megawatt per an hour, doesn't make sense since there is no such quantity described by the unit watt/hour.

65. Force F_1 does 5J of work in 10 sec. Force F_2 does 3J of work in 5 sec. Which force produces the greater power?

Explanation: $P_1 = \frac{W_1}{t_1} = \frac{5}{10} = 0.5W$ and $P_2 = \frac{W_2}{t_2} = \frac{3}{5} = 0.6W$

Thus, F_2 produce more power.

66. Engine 1 produces twice the power of engine 2. Is it correct to conclude that engine 1 does as much work as engine 2? Explain

Explanation: No. Engine 1 may do the same amount of work as engine 2 in half the time.

Illustrative Examples

67. A kettle uses 180,000J of electrical energy in 3 minutes. What is the power of the Kettle?

Solution: $P = \frac{W}{t} = \frac{180,000}{(3)(60)} = 100\text{Watt}$

68. A 1000 kg car accelerates uniformly form 10 m/s to 30 m/s in 2 sec. **Calculate**:

a) The work done on the car.

b) The average power developed.

Solution:

a) $W = \Delta KE = K_{Ei} - K_{Ef} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$

$$W = \frac{1}{2}m(v^2 - u^2) = \frac{1}{2}(1000)(30^2 - 10^2)$$

$$W = \frac{1}{2}(1000)[900 - 100] = 400{,}000\text{J}$$

b) $P = \frac{W}{t} = \frac{400{,}000}{2} = 2\times10^5\,\text{Watt}$

69. A car is traveling along a straight level road at 20 m/s against a force of 3000N. What power form its engine is needed?

Solution: $P = Fv = (3000)(20) = 60{,}000\ \text{Watt} = 60\ \text{KW}$

70. If 50000 kg of water pour every second over the Tisaabbay fall of height 50m. What power could be available?

Solution: $P = \frac{W}{t} = \frac{mgh}{t} = \frac{(50000)(10)(50)}{1} = 0.25\times10^8\,\text{W}$

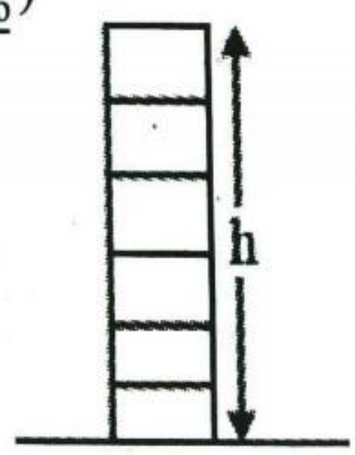
71. An elastic motor lifts a 100N object to a height of 20m in 8 sec. At what rate does this motor convert electrical energy to mechanical energy?

Solution: $P = \frac{W}{t} = \frac{\Delta E}{t} = \frac{mgh}{t} = \frac{(100)(20)}{8} = 250\,\text{Watt}$

72. How much power is required to construct vertically 6 boxes in 2 min if each has a mass of 100 kg and 30 cm thickness?

Solution:

$$P = \frac{W}{t} = \frac{mgh}{t} = \frac{mg(h_1 + h_2 + h_3 + h_4 + h_5 + h_6)}{t}$$

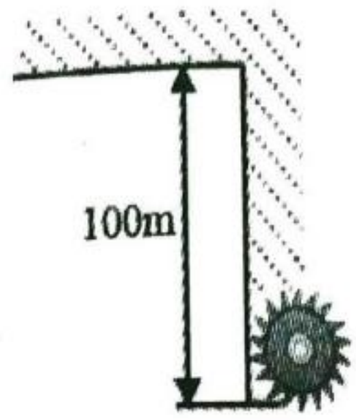


$$P = \frac{(100)(10)[0 + 0.3 + 0.6 + 0.9 + 1.2 + 1.5]}{(2)(60)}$$

$$P = \frac{4500}{120} = 37.5\,\text{Watt}$$

73. What should be the rate of flow of water that falls from a height of 100m on to a turbine in order for the turbine to produce a power of 2MW?

Solution: $P = \frac{mgh}{t} \Rightarrow \frac{P}{gh} = \frac{m}{t}$ thus,



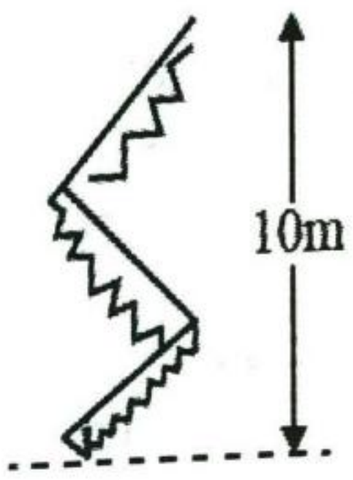
$$\frac{m}{t} = \frac{2\times10^6}{(10)(100)} = 2\times10^3 \text{ kg/s}$$

But, $m = v\rho$, and $\rho_w = 10^3 \text{ kg/m}^3$

$$\Rightarrow \frac{v\rho}{t} = 2\times10^3$$

$$\Rightarrow \frac{v}{t} = 2\text{m}^3/\text{s}$$

74. What average power can be developed by a man of mass 70 kg as he moved himself up stairs to a height of 10m in 20 seconds?



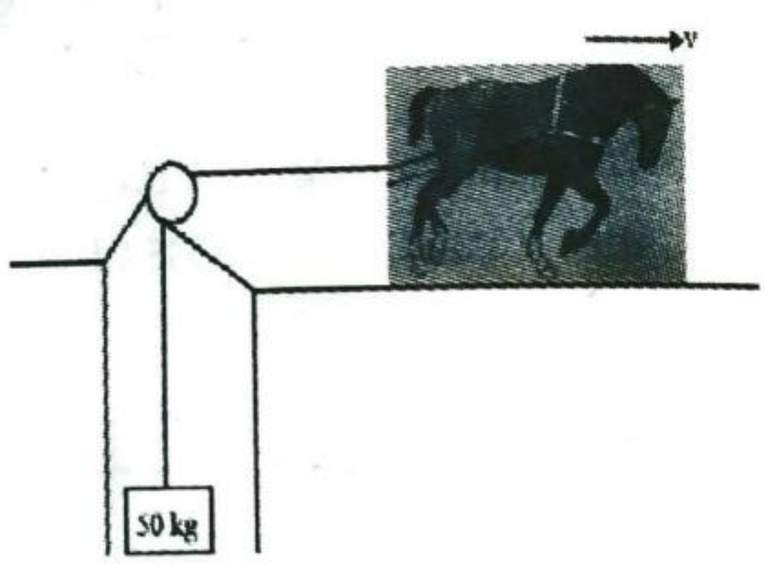
Solution:

$$P = \frac{W}{t} = \frac{mgh}{t}$$

$$P = \frac{(70)(10)(10)}{20}$$

$P = 350$ Watt

75. Once upon a time, James Watt let his horse pull up a mass of 50 kg from a well. He found that it was able to walk comfortably at 1 m/s. What was the power of his horse? ($g = 9.8 \text{ m/s}^2$)



Solution: $P = \frac{W}{t} = Fv$

$P = mgv$

$\Rightarrow P = (50)(9.8)(1)$

$P = 490$ Watt

General Secondary Education Certificate Exam

76. A nurse pushes a patient in a wheelchair for a 100m doing 400J of work in the process. What average force did the nurse exert in moving the patient?

A. 10N B. 4N C. 40N D. ¼ N

Solution: $W = Fs \Rightarrow F = W/s = 400/100 = 4\text{N}$ **Answer: B**

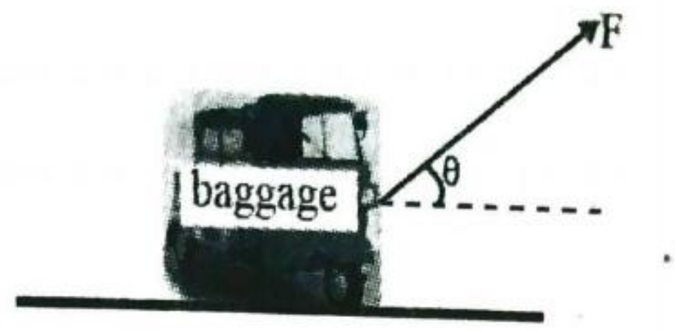
77. A typist uses a force of 1N to push a key down to 1 cm, and does this each in 0.5 seconds. How much work does she do on the keys each second?

A. 0.02J B. 0.01J C. 0.2J D. 0.1J

Solution: $W = Fs = (1)(0.01) = 0.01J$

Answer: B

78. In the diagram below, the direction of the force your need to exert on the baggage is indicated. What can you say about the work you do on the baggage as you move it through the horizontal displacement s?



A. It is simply zero.
B. It is greater than Fs.
C. It is less than Fs.
D. It is equal to Fs.

Explanation: $W = Fs\cos\theta$, here $\theta > 0°$ thus $\cos\theta < 1$ hence, $Fs\cos\theta < Fs$.

Answer: C

79. A body of mass 10 kg is pulled on a horizontal surface by applying on it a force of 50N parallel to the surface. If $\mu_k = 0.1$, what is the total work done when the body is pulled 5m?

A. 250J B. 5J C. 200J D. 50J

Solution: $W = F_{net}s\cos\theta$, $F_{net} = F - f = F - \mu mg = 50 - (0.1)(10)(10)$

$F_{net} = 40N, \Rightarrow W = Fs = (40)(5) = 200J$

Answer: C

80. A force does a negative work when:

A. The force is opposite to the displacement.
B. The force is perpendicular to the displacement.
C. Both the force and the displacement are positive.
D. Both the force and the displacement are negative.

Explanation: $W = fs\cos\theta = -fs$ when $\theta = 180°$, which is when the force and the displacement are opposite to each other.

Answer: A

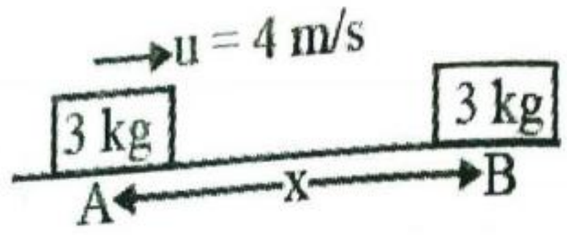
81. The block in the figure, started with 4 m/s at point A. After sliding over the frictional surface $\mu = 0.5$, it come to rest at point B. How large is x? (take $g = 10\ m/s^2$)

A. 0.8m B. 1.6m C. 3.2m D. 2.4m

Solution: $W = \Delta kE \Rightarrow -fs = -\frac{1}{2}mu^2$, $f = \mu mg$

$$\Rightarrow \mu mgx = -\frac{1}{2}mu^2 \Rightarrow x = \frac{u^2}{2\mu g} = \frac{(4)^2}{(2)(0.5)(10)}$$

$$x = \frac{16}{10} = 1.6\text{m}$$



Answer: B

82. Which of the following energy lies change/s by work of friction, when a block slides across a rough horizontal table top?
A. Kinetic energy only
B. Internal energy only
C. Kinetic and internal energy
D. Kinetic and potential energy

Explanation: Friction produces heat (internal energy) and the kinetic energy changes.

Answer: C

83. The system shown is released from rest. What is the kinetic energy of the body of mass 2m when it reaches the floor?

A. mgh
B. 2mgh
C. $\frac{1}{3}$mgh
D. $\frac{2}{3}$mgh

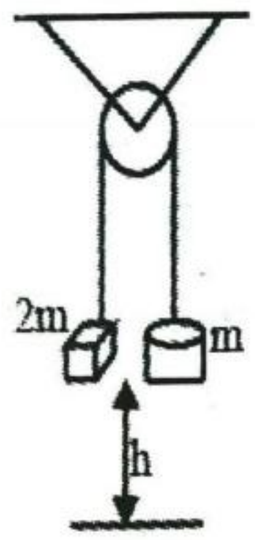
Solution: $2mg - T = 2ma \ldots (1)$

$T - mg = ma \ldots (2)$

$\Rightarrow a = \frac{9}{3}$ and $v^2 = u^2 + 2ah = 2ah,\ u = 0$

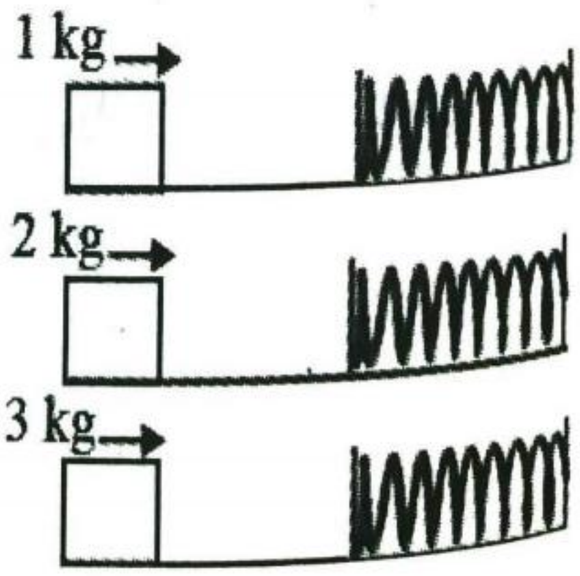
$v^2 = 2\left(\frac{g}{3}h\right) = \frac{2}{3}gh$

$kE = \frac{1}{2}(2m)v^2 \Rightarrow kE = \frac{2}{3}mgh$



Answer: D

84. Three blocks of masses 1 kg, 2kg and 3 kg are sliding a long a frictionless surface with the same momentum towards three identical springs held against a wall. Which block does the largest amount of work in compressing the spring from the equilibrium position?
A. They do equal amount of work.
B. The 2 kg block



C. The 3 kg block
D. The 1 kg block

Solution: Momentum, $P = mv$, here $P_1 = P_2 = P_3 = P$

$\Rightarrow m_1v_1 = m_2v_2 = m_3v_3$, the 1 kg block has the highest speed. Thus, from work energy theorem, $W = \Delta kE$

$$W_1 = \frac{1}{2}(m_1)(v_1)^2 = 0.5v_1^2$$

$$W_2 = \frac{1}{2}(m_2)(v_2)^2 = v_2^2 = \left(\frac{1}{2}v_1\right)^2 = 0.25v_1^2$$

$$W_3 = \frac{1}{2}(m_3)(v_3)^2 = 1.2v_3^2 = 1.5\left(\frac{v_1}{2}\right)^2 = 0.17v_1^2$$

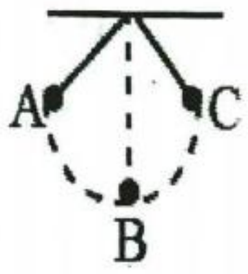
Thus, clearly $W_1 > W_2 > W_3$ **Answer: D**

85. Two paths lead to the top of a big hill. One is steep and direct while the other is twice as long but less steep. Compared to the potential energy you gain by taking the direct path, how much potential energy would you gain if you take the longer path?

A. Half as much
B. Twice as much
C. Four times as much
D. The same

Explanation: Gravitational potential energy depends only on the height from the given surface, not on the path followed to reach that height. **Answer: D**

86. All of the following about the transformation of energy by the work done on the pendulum bob are correct. Except

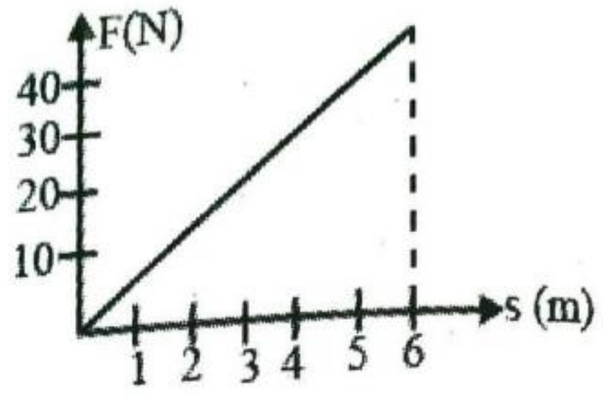


A. Gravitational work done increases the kinetic energy by decreasing the potential energy as the bob moves from A to B.
B. Gravitational work done increases the potential energy by decreasing the kinetic energy as the bob moves from B to C.
C. The tension force decrease the kinetic energy while it increases the potential energy by moving the bob from B to C.
D. The tension force does no work on the bob as its kinetic energy is transformed into potential energy.

Answer: C

87. The graph below shows the force acting on a body when it is at different points from its starting point. How much is the total work done on the body when it is 6m from the starting point?

A. 240J
B. 120J
C. 720J
D. 1440J



Solution: $W = \text{Area} = \frac{1}{2}Fs$

$$W = \frac{1}{2}(40)(6) = 120J$$

Answer: B

88. An object of mass m moving with velocity v across a level surface comes to rest after a distance d. What is the work done by friction?

A. –mvd B. $-\frac{1}{2}mu^2$ C. $-\frac{1}{2}mv^2$ D. zero

Solution: $W_f = \Delta kE$,

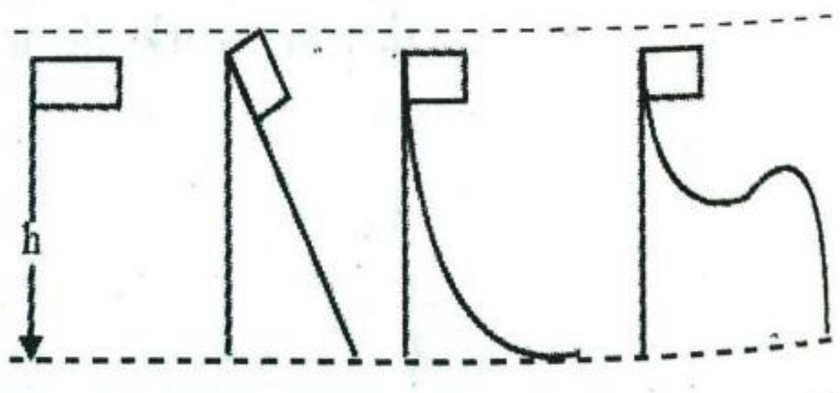
$fd\cos\theta = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$, here $v = 0$, $\theta = 180°$

$-fd = -\frac{1}{2}mu^2 = -\frac{1}{2}mv^2$

Answer: B

89. Four blocks are released from the same height h as shown in the figure. If friction is ignored, which one is correct about the speeds of the blocks when they reach the bottom?

A. $V_1 < V_2 < V_3 < V_4$
B. $V_1 = V_2 = V_3 = V_4$
C. $V_1 < V_2 < V_3 < V_4$
D. $V_4 > V_1$ and $V_2 = V_3$



Explanation: In all cases, since they are the same height they posses the same potential energy and which is transferred to kinetic energy at the bottom. Thus, $V_1 = V_2 = V_3 = V_4$ at the bottom. **Answer: B**

90. A ball, thrown vertically upward, gains gravitational potential energy u at the maximum height. If it is thrown up at half the speed, its gain in gravitational potential energy will be:

A. u B. u/2 C. u/9 D. u/4

Solution: $u_1 = mgh_1$, here $mgh_1 = \frac{1}{2}mv_1^2$

Thus, $u_2 = mgh_2$ $\qquad h_1 = \frac{v_1^2}{2g}$

$u_2 = mg\left(\frac{1}{4}h_1\right)$ $\qquad h_2 = \frac{v_2^2}{2g}, v_2 = \frac{v_1}{2}$

$u_2 = \frac{1}{4}mgh_1$ $\qquad h_2 = \frac{\left(v_1/2\right)^2}{2g}, v_2 = \frac{v_1^2}{4(2g)} = \frac{1}{4}h_1$

$\Rightarrow u_2 = \frac{1}{4}u_1$ **Answer: D**

91. Which one of the following can do work?
   A. static friction
   B. kinetic friction
   C. the tension force in the string of an oscillating simple pendulum
   D. the gravitational force on a satellite orbiting the earth in a circular path

   Explanation: Work is done when there is force component in the direction of motion. $W = Fs\cos\theta$
   In choice A, there is no motion, thus no work in choice C and D, the force is perpendicular to the displacement, hence no work.
   Answer: B

92. If a resultant horizontal force of 15N acts on a body of mass 2 kg and displaces it from rest through a displacement of 5m, the kinetic energy of the body becomes
   A. 75J B. 30J C. 0J D. 150J

   Solution: $W = \Delta kE = K_{Ef} - K_{Ei}$
   $(F)(s) = \Delta kE = (15)(5) = 75J$ **Answer: A**

93. A particle of mass 30 g attached to a spring of force constant 200 N/m is oscillating along the x – axis about the equilibrium position x = 0. If the amplitude x_{max} = 10 cm, what is the kinetic energy of the particle when it reaches the equilibrium position?
   A. 1000J B. 100J C. 1J D. 10J

   Solution: From the transformation of energy.

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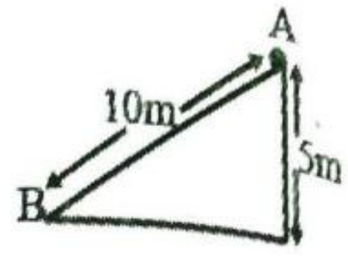
$$KE = PE = \frac{1}{2}kx^2 = \frac{1}{2}(200)(10^{-1})^2 = 1J$$ **Answer: C**

94. In the diagram below, a ball of 1 kg is rolling down the inclined plane starting from rest at A. What will be the kinetic energy of the ball at B at the bottom of the inclined plane? ($g = 10\ m/s^2$)



A. 250J B. 150J B. 100J D. 50J

Solution: $M_{EA} = M_{EB}$

$$mgh = \frac{1}{2}mv^2 = (1)(10)(5) = 50J$$ **Answer: D**

95. A 60 kg man jumps down form a 0.8m table. What is the speed when he hits the ground ($g = 10\ m/s^2$)?

A. 16m/s B. 8 m/s C. 4 m/s D. $\sqrt{8}$ m/s

Solution: From conservation of energy

$$P_E = K_E \Rightarrow mgh = \frac{1}{2}mv^2, gh = \frac{1}{2}v^2 \Rightarrow (10)(0.8) = \frac{1}{2}v^2$$

$\Rightarrow v^2 = 16, v = 4m/s$ **Answer: C**

96. A ball is thrown straight up. It rises for a while and then drops back to the ground. Which of the following energy transformations best describe the process?

A. Chemical energy → kinetic energy → potential energy
B. kinetic energy → chemical energy → potential energy
C. potential energy → kinetic energy → potential energy
D. kinetic energy →potential energy →kinetic energy

Answer: D

97. Suppose you lifted a load from the ground and carried it then you walked for some distance on a level floor at a constant velocity. The work you did on the load is:

A. when you were lifting it up
B. when you were walking
C. proportional to the distance you walked
D. proportional to the time you carried it

Answer: A

98. A car driver accelerated a car of mass 40,000 kg from rest to 25 m/s. What is the work done on the car?

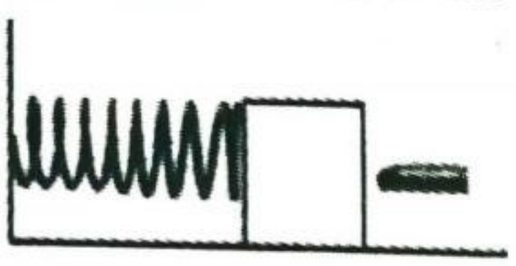
A. $15 \times 10^5 J$ C. $10 \times 10^5 J$

B. 125×10^5J D. 250×10^5J

Solution: $W = \Delta kE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$, $u = 0$

$$W = \frac{1}{2}mv^2 = \frac{1}{2}(40{,}000)(25)^2 = 125 \times 10^5 J$$ **Answer: B**

99. The mass spring system rests on frictionless surface. How far will the system be displaced from its equilibrium position if a bullet of 40 g mass is shut into it at a speed of 450 m/s ($k = 10^4$N/m)



A. 13.5m B. 9m C. 0.135m D. 0.9m

Solution: $K_E = E_{PE}$

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow x = \sqrt{\frac{mv^2}{k}} = \sqrt{\frac{(0.04)(450)}{10000}} = \sqrt{0.81} = 0.9m$$

Answer: D

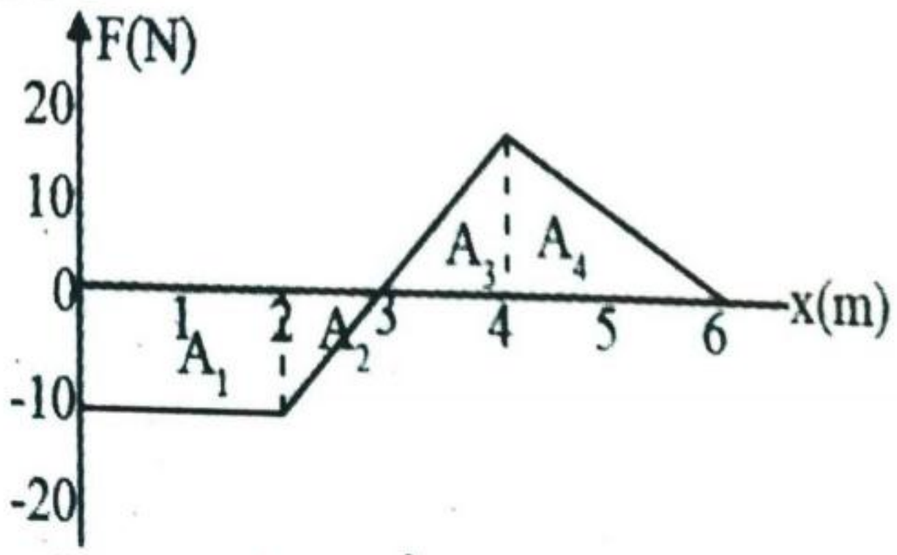
100. The force acting on a particle varies with position as shown. What is the work done as the particle moves from x = 0 to x = 6m?

A. 10J B. 40J C. -10J D. -15J

Solution: $W = A_1 + A_2 + A_3 + A_4$

$$W = (-10)(2) + \left(\frac{1}{2}\right)(-10)(1) + \frac{1}{2}(10)(1) + \frac{1}{2}(10)(2) = -10J$$

Answer: C



101. Car A with mass 2×10^3 kg moving at speed of 10 m/s collides and sticks with car B of mass 3×10^3 kg initially at rest. How much kinetic energy is lost during the collision?

A. 4×10^3J C. 1×10^5J
B. 6×10^3J D. 6×10^4J

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Solution: $k_{Ei} = \frac{1}{2}m_A v_A^2 = \frac{1}{2}(2\times10^3)(10)^2 = 1\times10^5 J$

Now, from $m_A u_A + m_B u_B = m_A v_A + m_B v_B$, $v_A = v_B = v$

$$m_A u_A = (m_A + m_B)v \Rightarrow \frac{m_A u_A}{m_A + m_B} = \frac{(2\times10^3)(10)}{2\times10^3 + 3\times10^3} = 4 m/s$$

$$k_{Ef} = \frac{1}{2}mv^2 = \frac{1}{2}(m_A + m_B)v^2 = \frac{1}{2}(2\times10^3 + 3\times10^3)(4)^2$$

$k_{Ef} = 4\times10^4 J$ then;

$\Delta k_E = |k_{Ef} - k_{Ei}| = |4\times10^4 - 10^5| = 6\times10^4 J$

Answer: D

102. Which one of the following statement is true about the change in gravitational potential energy?
   A. It depends only on the initial and final height.
   B. It is positive because mass and gravity are positive.
   C. It depends on the height were the body is placed.
   D. It depends on path of motion of the body.

Explanation: $\Delta G_{PE} = G_{PEf} - G_{PEi} = m_g h_f - m_g h_i$

$\Rightarrow \Delta G_{PE} = mgh(hf - hi)$

Answer: A

103. The impulse experienced by an object is equivalent to change in its
   A. Momentum
   B. Velocity
   C. Kinetic energy
   D. Potential energy

Answer: A

Unit Four: Conceptual Questions

1. Is it possible to do work on an object that remains at rest?
2. Only a net force acting on an object can do work. Is this true or false? Explain
3. A car and a truck are moving at the same speed. Which one has more kinetic energy?
4. If a positive work is done on a body and the potential energy increase, what can you say about its kinetic energy?
5. If you drop a tennis ball onto a concrete floor, it will bounce back. But it will not if you drop it on to a mud. Where does its energy goes?

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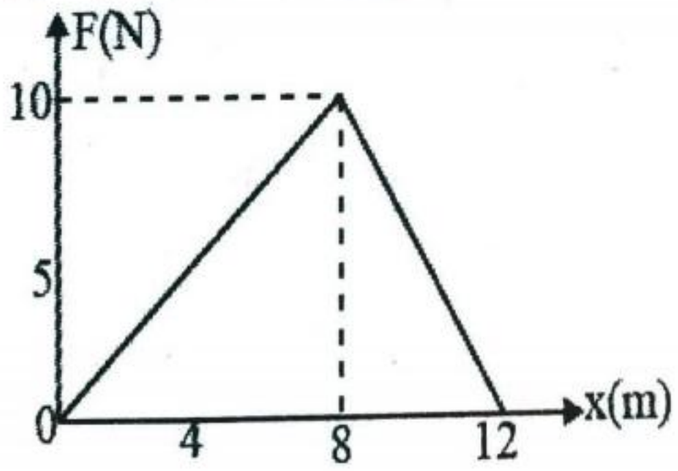
6. If two identical balls are raised to the same height, one on earth, the other on the moon, which one will have the larger potential energy?
7. An earth quake can release energy to destroy a city. Where does this energy reside moments before the earth quake takes place?
8. What will be the largest possible kinetic energy of an arrow shot from a bow that has been pulled so that it stores 50J of elastic potential energy?

Unit Four: Problems

Work done

9. Calculate total work done in the following cases.
   a) A 20 kg log lifted 2m into the air.
   b) Thirty 6 kg boxes lifted on to a shelf 1.5m high.
   c) A car of mass 1400 kg pushed 50m along a rod $\mu = 0.3$.
   d) A concrete slab of mass 200 kg pulled 10m up a slope at an angle of 30° to the horizontal. $\mu = 0.6$
10. Calculate the work done if a boulder of mass 100 kg is rolled 40m up a slope at an angle of 20°. Assume there is no friction.
11. A force varies with the displacement as shown below. Calculate the work done by the variable force as the body is displaced from x = 0 to x = 12m.



Work – Energy Theorem

12. Calculate the kinetic energy of a ball of mass 50 g travelling at 30 m/s. How much work will need to be done to stop the ball?
13. How much work is needed for a 65 kg runner to accelerate from rest to 10 m/s?
14. A 65 kg bicyclist rides his 10 kg bicycle with a speed of 12 m/s.
    a) How much work must be done to stop the bicycle?
    b) How far it travel if it takes 4 sec to come to rest?
    c) What is the magnitude of the stopping force?

Energy and Power

15. If a pendulum of mass 50 g is lifted so that it has a GPE of 0.1 J. Calculate

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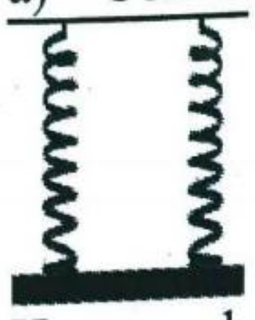
a) Its increase in height.
b) The velocity of the bob at the bottom.

16. A spring is used to launch a ball. The spring has a spring constant of 200 N/m and is compressed by 5 cm. A ball of mass 10 g is placed just above the spring. Calculate
a) The energy stored in the spring.
b) The velocity of the ball as it launches.
c) The height reached by the ball.

17. Two springs with force constants k_1 and k_2 are:
a) Connected in parallel b) Connected in series



How much work is needed to stretch the system in each case a distance x?

18. A man raises 100 kg from the floor to a height of 2m in 2.5 sec. What is the work done and the power developed?

19. It is proposed to use a small waterfall to turn an electricity generator $10m^3$ of water fall 50m per minute. Only one fifth of its energy can be obtained usefully. Calculate the power developed.

20. Calculate the resistive force acting on a sport car if it is travelling at a steady speed of 25 m/s when the engine is providing 200 kw.

Unit Four: Explanation to Questions

1. No. Work done is zero if there is no displacement.
2. It is false. There might be work done by friction etc.
3. The truck. Because of its bigger mass.
4. If mechanical energy is conserved, as G_{PE} increases, its k_E decreases with the same amount.
5. It change its form, such as vibration.
6. The one on the Earth. Because $g_E > g_m$.
7. It was stored energy in the core of the earth.
8. From the conservation of mechanical energy, the elastic potential energy is converted to kinetic energy in the absence of friction or drage force. Thus, $k_E = 50J$.

Unit Four: Solution to Problems

9. a) $W = mgh = (20)(10)(2) = 400J$
   b) $W = mgh = (6)(30)(10)(1.5) = 2700J$

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c) $W = \mu mgs = (0.3)(1400)(10)(50) = 210000J = 210\ kJ$

d) $W = \mu mg \sin \theta s = (0.6)(200(\sin 30)(10)(10) = 6000J$

10. $W = (mg \sin \theta)s = (100)(10)(\sin 20)(40) = (40000)(0.34) = 13680.8J$

11. $W = A_1 + A_2 = \frac{1}{2}(8)(10) + \frac{1}{2}(4)(10) = 60J$

12. $kE = \frac{1}{2}mv^2 = \frac{1}{2}(0.05)(30)^2 = 22.5J,\ W = \Delta kE,\ W = 22.5J$

13. $W = \Delta kE = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}(65)(10)^2 = 3250J$

14. a) $W = \Delta kE = \frac{1}{2}(mv^2) - \frac{1}{2}mu^2 = \frac{1}{2}(65 + 10)(12)^2 = 5400J$

b) $s = \left(\frac{v+u}{2}\right)t = \left(\frac{0+12}{2}\right)(4) = 24m$

c) $fs = W \Rightarrow f = \frac{W}{s} = \frac{5400}{24} = 225N$

15. a) $mgh = 0.1 \Rightarrow h = 0.1\ mg = \frac{0.1}{(0.05)(10)} \Rightarrow h = 0.2m$

b) $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{(2)(10)(0.2)} = 2m/s$

16. a) $E_{PE} = \frac{1}{2}kx^2 = \frac{1}{2}(200)(0.05)^2 = 0.25J$

b) $\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \Rightarrow v = \sqrt{\frac{0.5}{0.01}} = 7m/s$

c) $mgh = \frac{1}{2}mv^2 \Rightarrow h = 2.5m$

17. a) $W = E_{PE} = \frac{1}{2}kx^2$ but for parallel spring, $k = k_1 + k_2$

$$W = \frac{1}{2}(k_1 + k_2)x^2$$

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b) $W = E_{PE} = \frac{1}{2}kx^2$, but for springs in series, $k = \frac{k_1 k_2}{k_1 + k_2}$

$$W = \frac{1}{2}\left(\frac{k_1 k_2}{k_1 + k_2}\right)x^2$$

18. $W = mgh = (100)(10)(2) = 200J,\ P = \frac{W}{t} = \frac{2000}{2.5} = 800Watt$

19. $W = mgh$ but $1m^3 = 1000$ kg thus, $10m^3 = 10000$ kg

$W = (1000)(10)(50) = 5000000J$ but only $\frac{1}{5}W$ is useful

$$\Rightarrow W_{out} = \frac{1}{5}W = \frac{1}{5}(5000000) = 1000000J, P = \frac{W_{out}}{t} = \frac{10^6}{60} = 16.7KW$$

20. $P = Fv \Rightarrow F = \frac{P}{v} = \frac{200000}{25} = 8000N$

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Unit Five

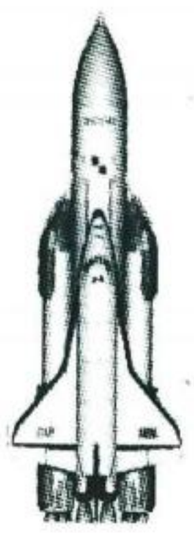
Simple Machines

Machines have made it possible for mankind to accomplish amazing things, from building the ancient pyramids of Egypt to landing on the moon from opening a bottle and cutting food to sharpening a pencil. In sum machines are parts of our daily lifes, and without them, life would become so complex.

5.1 Purposes of Machines

Machines might range from the very complex space shuttle, down to a simple pair of scissors.

Simple Machine



Complex Machine

Weather operated by engine or people, machines help to do work easier.

Machine: Is a device that is designed to do work easier.

☞ A machine makes it easier to do work by performing one or more of the following.

- Increase the magnitude of the applied force.
- Increase the distance or speed against the applied force.
- Change the direction of the force.

The term simple machine, refers to a machine that is well simple! That means;

- A device that only requires a single force to do work.
- A device for doing work that has only one part.
- A device that uses a single effort to do work against a single load force.

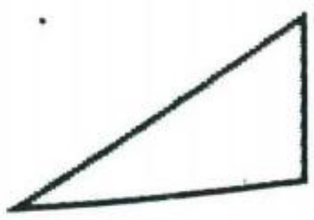
Simple machine: Is a device which requires a single effort to do work against a single force.

Simple machines are often described as the elementary building blocks from which all other machines are made.

Types of Simple machines

There are **six** different types of simple machines.

1. Inclined plane



2. Wedge

3. Screw

4. Lever

5. Pulley

6. Wheel and axle

☞ Important terms to remember

Effort (E): Is the force applied to a machine.

Load (L): Is the force applied on the machine.

Work input (W_{in}): Is the work done by the effort

W_{in} = (Effort) × (distance moved by effort)

Work out put (W_{out}): is the work done by the machine

W_{out} = (load) × (distance moved by load)

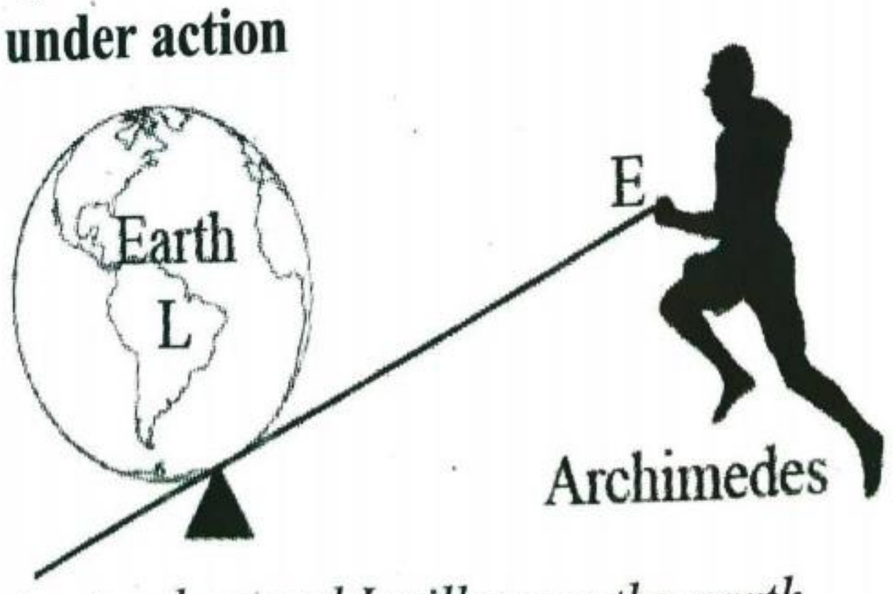
- No matter which type of simple machine we deal with, it will fall under one or more of the following categories.

1. Force multipliers machines

These are machines designed so that, *the load is greater than the effort.* This is only possible if the **load** moves through **a smaller distance** than **the effort**.

Example: bottle opener, scissors, pillars, are force multipliers.

Example, lever under action

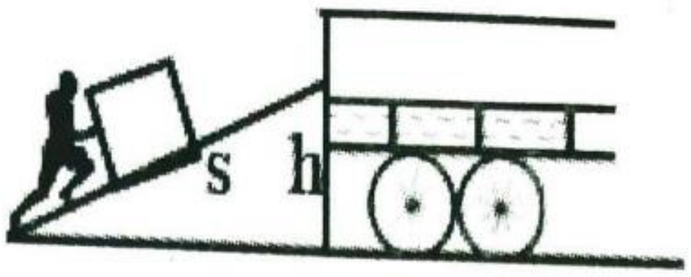


Give me a place to stand on and I will move the earth.

2. Speed multipliers machines

These are machines designed so that, *the distance moved by the load is greater than the distance move by the effort in the same time*. This is only possible if the load is **smaller** than the **effort**.

Example: the rear wheel of a bicycle.



Force multiplier

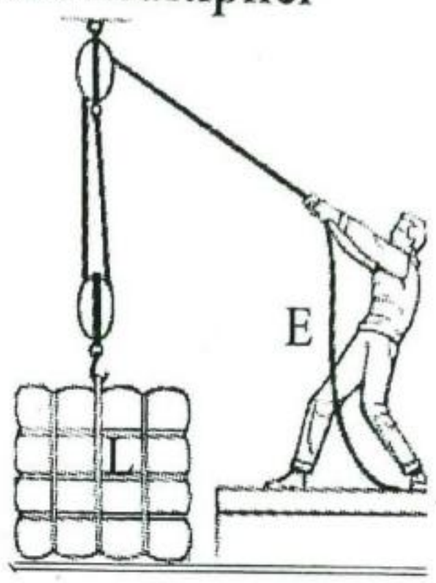
3. Direction changer machines

These are machines designed so that the load is moved in a different direction to the effort.

Example: single fixed pulley

⇒L = upward

E = downward



Conceptual Examples

1. Can a machine multiply energy? Explain

   Explanation: No. that would break the law of conservation of energy.

2. Can a machine multiply force and speed at the same time? Explain

   Explanation: No. This would mean the work output is greater than the work input.

☞ **Important terms to remember**

Actual mechanical Advantage (AMA): Is the ratio of load and effort taking into account energy losses due to friction etc,

$$\text{Mechanical advantage} = \frac{\text{Load}}{\text{Effort}} \Rightarrow \boxed{MA = \frac{L}{E}}$$

Ideal Mechanical Advantage (IMA): is the ratio of load and effort, assuming no energy loss due to friction etc.

Velocity Ratio (VR): is the ratio between the distance moved by the effort and the distance moved by the load.

$$\text{Velocity ratio} = \frac{\text{distance moved by effort}}{\text{distance moved by load}} \Rightarrow \boxed{VR = \frac{S_E}{E_L}}$$

both MA and VR are unit less.

Efficiency (η) of a machine: Is the ratio between work output and work input.

Efficiency $= \dfrac{\text{workout put}}{\text{work in pu}} \Rightarrow \boxed{\eta = \dfrac{W_{out}}{W_{in}}}$ efficiency of a machine can also be expressed in terms of MA and VR.

From $\eta = \dfrac{W_{out}}{W_{in}} = \dfrac{(L)(S_L)}{(E)(S_L)} = \dfrac{\left(L/E\right)}{\left(\dfrac{SE}{SL}\right)} = \dfrac{MA}{VR} \Rightarrow \boxed{\eta = \dfrac{MA}{VR}}$

Moreover, efficiency can be expressed in percentage (%) such as $\eta = 50\%$ or using a decimal point such as $\eta = 0.5$.

Conceptual Examples

3. Can the efficiency of a machine be ever be greater than 100% or 1? Explain

   Explanation: Never. If it is greater than 100% or one, it means energy can be created. Which is not the case in nature.

4. Can the efficiency of a real machine be over be equal to 100% or 1? Explain

   Explanation: Never. In real machine, there is a loss of energy due to external forces such as friction, hence, the workout is always less than the work input, which means, the efficiency of real machine is always less than 100% or unity.

Illustrative Example

5. A simple machine is able to move a 400N load a distance of 20cm when a force of 20N is moved through a distance of 5 m. Calculate:
   a) The work input
   b) The workout
   c) The actual mechanical advantage
   d) The velocity ratio
   e) The ideal mechanical advantage
   f) The efficiency of the machine

   Solution:

   a) $W_{in} = (E)(S_E) = (20)(5) = 100J$

   b) $W_{out} = (L)(S_L) = (400)(0.2) = 80J$

   c) $AMA = \dfrac{L}{E} = \dfrac{400}{20} = 20$

d) $VR = \frac{S_E}{S_L} = \frac{5}{0.2} = 25$

e) $IMA = VR = 25$

f) $\eta = \frac{W_{out}}{W_{in}} = \frac{80}{100} = 0.8$ or 80%

Or $\eta = \frac{AMA}{VR} = \frac{20}{25} = 0.8$ or 80%

Conceptual Example

6. Can we categorize a machine as force multiplier, speed multiplier or direction changer merely by observing its value of MA or VR? Explain

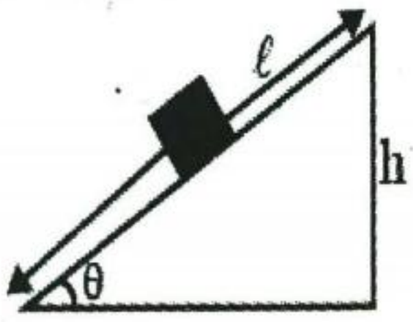
Explanation: Yes. A simple machine is said to be force multiplier when MA > 1 or VR >1. It is speed multiplier if MA < 1 or VR < 1 and it is a direction changer when MA = 1 or VR = 1.

The six simple machines mentioned before can be split into two groups. Wedges and screws can be thought of as a special kinds of inclined planes. Pulleys and wheels & axles can be considered to be special kinds of levers. Now, let's see them one by one.

5.2 Inclined plane, Wedge and Screw

The inclined plane

One of the simplest devices for raising heavy loads is the inclined plane (ramp). It is thought that Egyptians used it for raising large stones in the building of pyramids. It is still in use today to load and unload small cars onto or from a truck. Workmen use it on building sites so that wheelbarrows may be easily raised. Now, consider an object that is lifted to a height (h) by sliding it up the length of the slope (ℓ).



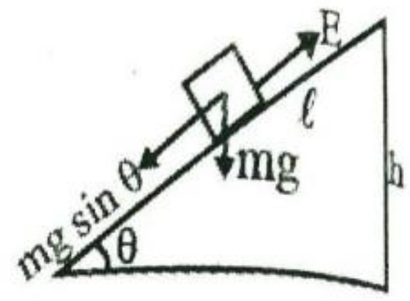
The mechanical advantage, velocity ratio and efficiency of an inclined plane can then be calculated as follows:

1. ***When there is no friction***

$E = mg \sin\theta$ and $L = mg$

$$MA = \frac{L}{E} \Rightarrow \boxed{MA = \frac{mg}{mg\sin\theta} = \frac{1}{\sin\theta}}$$

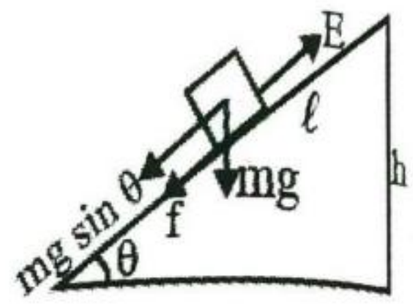
$$VR = \frac{S_E}{S_L} \Rightarrow \boxed{VR = \frac{\ell}{h} = \frac{1}{\sin\theta}}$$



2. ***When there is friction***

$E = mg \sin\theta + f$ and $L = mg$

$$MA = \frac{L}{E} \Rightarrow \boxed{MA = \frac{mg}{mg\sin\theta + f}}$$



$$VR = \frac{S_E}{S_L} \Rightarrow \boxed{VR = \frac{\ell}{h} = \frac{1}{\sin\theta}}$$

and $f = \mu F_N$, $F_N = mg\cos\theta$; $f = \mu mg\cos\theta$

$$\text{and } \eta = \frac{MA}{VR} = \frac{W_{out}}{W_{in}} \Rightarrow \boxed{\eta = \frac{mgh}{(mg\sin\theta + f)\ell}}$$

Illustrative Examples

7. A 40m length ramp is used to lift up a 1000N object through a height of 10m. If the ramp is frictionless. Calculate
   a) The work done in lifting the box.
   b) The efficiency of the machine.

Solution:

a) $W_{in} = (E)(S_E)$, but

$$E = mg\sin\theta = mg\left(\frac{h}{\ell}\right) = (1000)\left(\frac{10}{40}\right) = 250N$$

$S_E = \ell = 40m$

b) $$h = \frac{W_{out}}{W_{in}} = \frac{mgh}{(mg\sin\theta)\ell} = \frac{(1000)(10)}{(250)(40)}$$

$$\eta = \frac{10000}{10000} = 1$$

$\eta = 1$ or 100% because the machine is ideal, there is no friction.

8. An inclined plane of 60% efficiency is used to raise a load of 6000N through a height of 6m while sliding 20m along the incline. Calculate:
   a) the work done on the load
   b) the effort applied

Solution:

a) $W_{out} = (L)(S_L) = (mg)(h) = (6000)(6) = 36000J$

b) $\eta = \dfrac{W_{out}}{W_{in}} = \dfrac{mgh}{E\ell} \Rightarrow E = \dfrac{mgh}{\ell\eta}$

$$E = \frac{36000}{(20)(0.6)} = 3000N$$

9. A block of weight 5000N is pushed up a ramp of height 10m by a force of 250N. Calculate the length of the lamp. Assume there is no friction.

Solution: In the absence of friction, MA = VR

$$\Rightarrow \frac{\ell}{h} = \frac{L}{E} \Rightarrow \ell = \left(\frac{L}{E}\right)(h) = \left(\frac{5000}{250}\right)(10) = 200m$$

10. A certain machine raises a car of mass 2000 kg through a height of 3m in one minute. If the efficiency of the machine is 50%, what is the machine's power intake ($g = 10\ m/s^2$).

Solution: $W = mgh = (2000)(10)(3) = 60000J$

$$P_{out} = \frac{W}{t} = \frac{60000}{60} = 1000Watt$$

$$\eta = \frac{P_{out}}{P_{in}},\ P_{in} = \frac{P_{out}}{\eta} = \frac{1000}{0.5}$$

$P_{in} = 2000$ Watt

11. A 200N block is to be raised to a platform of 6m high. A ramp 12m long is used to slide the crate from the ground to the plat form. Assume that the coefficient of friction is 0.3. Calculate

a) The ideal mechanical advantage of the ramp.

b) The actual mechanical advantage of the ramp.

c) The efficiency of the ramp.

Solution:

a) $IMA = VR = \dfrac{\ell}{h}$

$$IMA = \frac{12}{6} = 2$$

b) $$AMA = \frac{L}{E} = \frac{mg}{mg\sin\theta + f} = \frac{mg}{mg\sin\theta + \mu mg\cos\theta}$$

$$AMA = \frac{200}{200\sin 30 + (0.3)(200)(\cos 30)}$$

$$AMA = \frac{200}{152} = 1.32$$

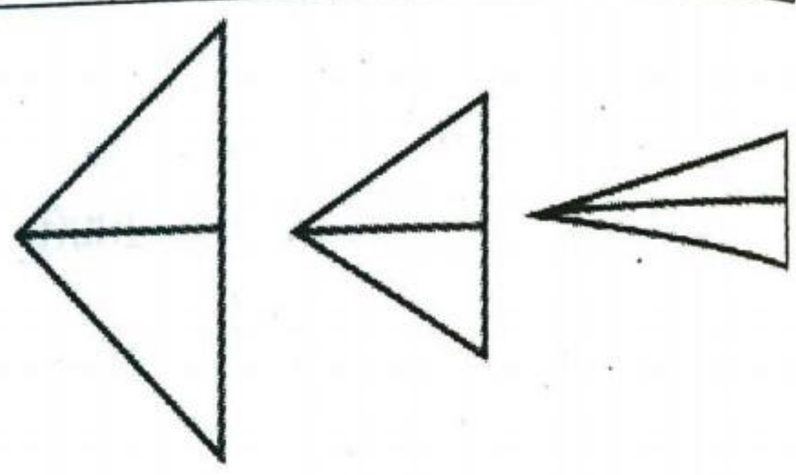
c) $$\eta = \frac{MA}{VR} = \frac{1.32}{2} = 0.66$$

or $\eta = 66\%$

The Wedge

Wedge: Is a piece of material such as metal or wood thick at one edge and tapered to a thin edge at the other.

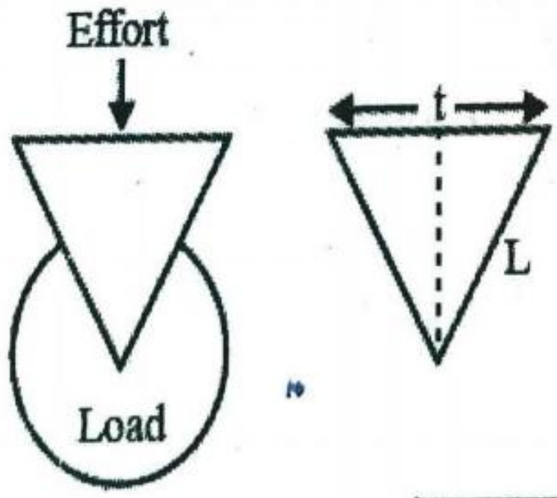
☛ A wedge can either be composed of one or more inclined planes. A double wedge can be thought of as two inclined planes joined together with their sloping surface outward.



Example of wedge types

Wedges are used to separate two object or split object apart. *Knives, nails, spears, axes and arrow heads and razor blades are all examples of wedges.*

Mechanical advantage, velocity ratio and efficiency of a wedge, consider a wedge of length (L) and thickness (t) is used to split a log g of wood by an effort (E).



Then, the mechanical advantage is; $\boxed{MA = \frac{L}{E}}$

the velocity ratio is; $\boxed{VR = \frac{S_E}{S_L} = \frac{L}{t}}$

the efficiency is;

$$\eta = \frac{MA}{VR} = \frac{L/E}{L/t} = \frac{(load)(thickness)}{(effort)(length)}$$

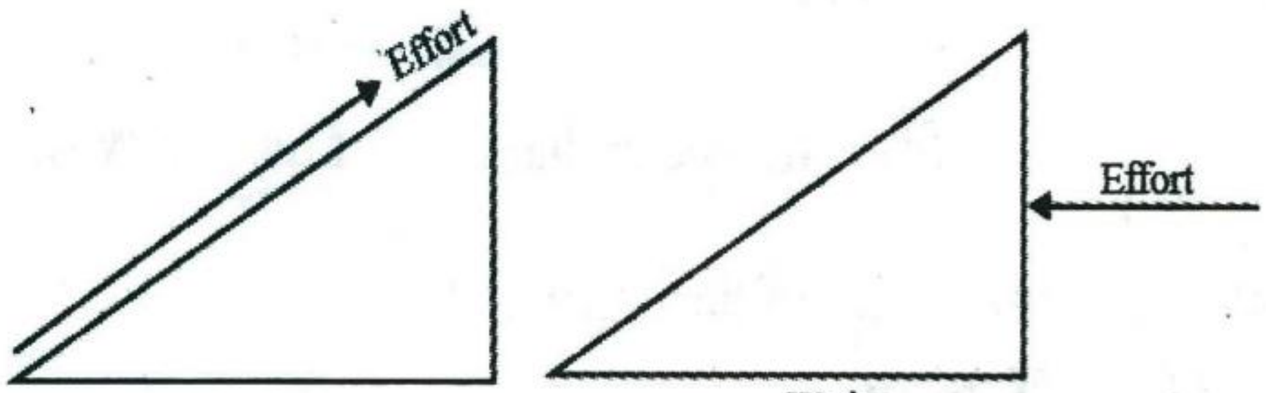
$$\boxed{\eta = \frac{Lt}{EL}}$$

Conceptual Example

12. Is there any difference between inclined plane and wedge? Explain

Explanation: Yes. Firstly, when in use an inclined plane remains stationary whereas the wedge moves. Secondly, the effort is applied parallel to the slope of an inclined plane but when using a wedge the effort is applied to the top of the wedge.

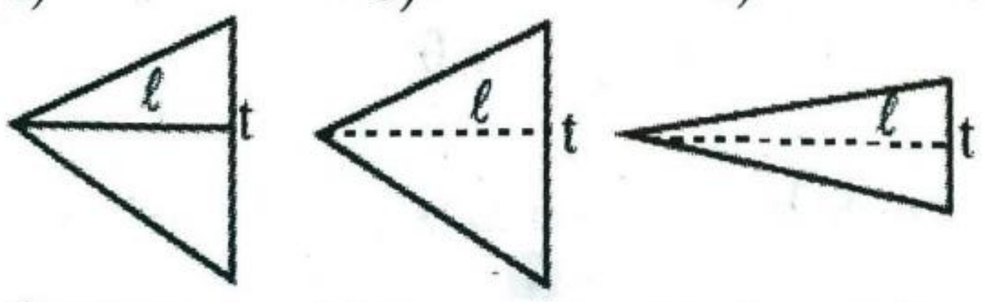
That is,



but both, inclined plane and wedge can be used as force multiplier and direction changer.

13. Which one of the following wedges do have the greater ideal mechanical advantage? Explain

a) b) c)



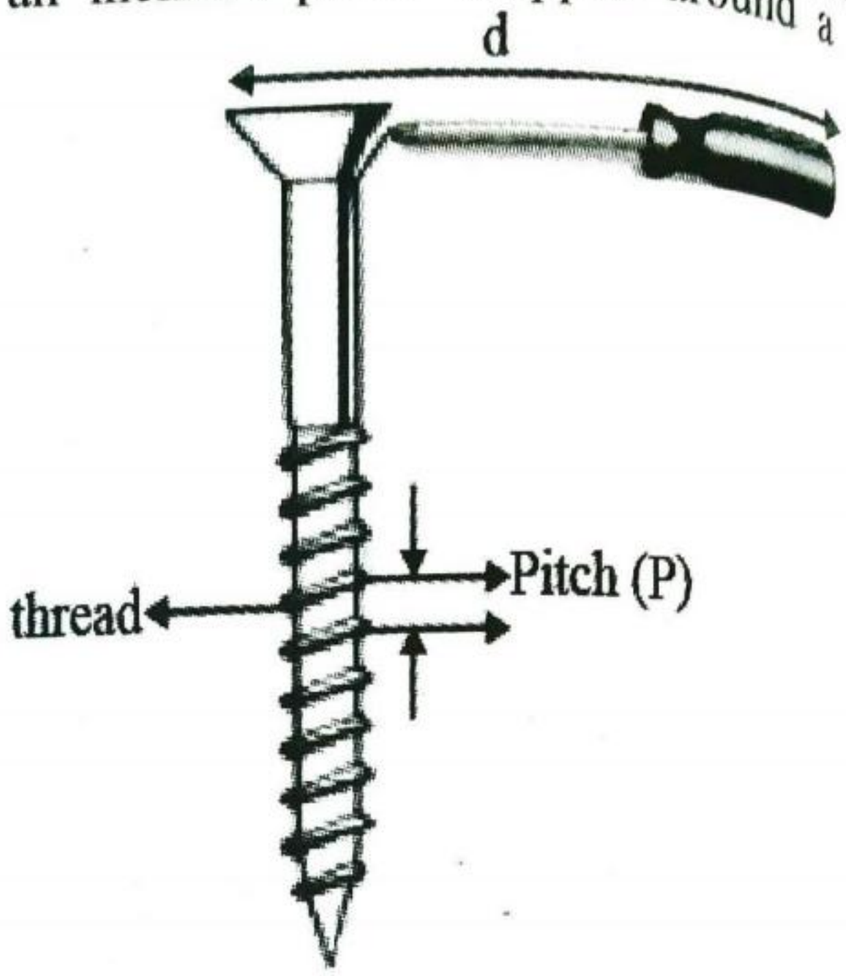
Explanation: The more narrow the wedge, the greater the IMA.

Answer: C

The Screw

Screw: Is a cylinder of material with a helical thread around it.

You can think of a screw like an inclined plane wrapped around a cylinder. This means it includes nuts and bolts etc and used to hold objects together, to dig into the ground and to bore through materials.



In one turn, the screw moves a distance equal to the separation between the threads. This distance is called *pitch* (P) of the screw and it is analogous to the height of an inclined plane.

The maximum theoretical mechanical advantage (IMA) for a screw can be obtained by:

$\boxed{IMA = \frac{\pi d}{P}}$ where (d) is the mean diameter of the screw shaft

(πd) is the circumference of the screw shaft

(P) is pitch of the screw

Illustrative Example

14. Suppose the pitch of the screw is 0.1cm and that the effort applied at the end of the spanner describes a circle of radius 16 cm. Calculate the V.R of the screw.

Solution: $VR = IMA = \frac{\pi d}{P} = \frac{2\pi r}{P} = \frac{(2)(3.14)(16)}{0.1} = 1000$

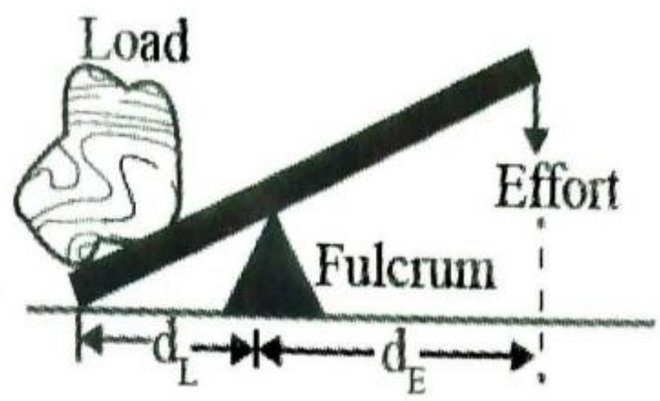
5.3 Levers

Probably the oldest device used by men and most general useful machine is the lever.

Lever: Is a rigid bar which is free to turn about a fixed pivot point called the fulcrum.

☛ It is important to notice that the effort distance (d_E) and the load distance (d_L) are always perpendicular to the load and effort forces. And that, if you had a really long lever you could lift almost anything! That is why Archimedes once said, . . ."*Give me a place to stand on, and I will move the Earth.*"

The mechanical advantage, velocity ration and efficiency of lever can be calculated as follows.



$$\boxed{MA = \frac{L}{E}}, \boxed{VR = \frac{d_E}{d_L}} \text{ and } \boxed{\eta = \frac{AMA}{VR}}$$

Note that;

d_E: is the distance from the effort to the fulcrum

d_L: is the distance from the load to the fulcrum

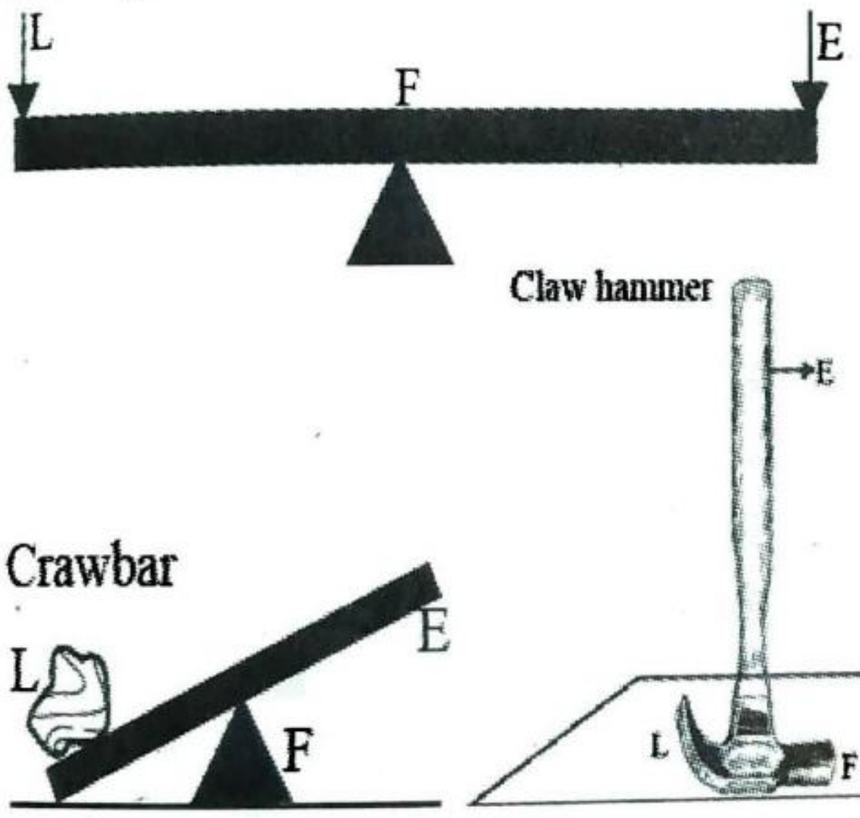
☛ Depending on the relative distances, levers can be force multiplier, speed multiplier or direction changer.

Different classes of lever

Depending on the relative position of the *load*, *fulcrum* and *effort*.

1st class: Fulcrum is between the load and the effort.

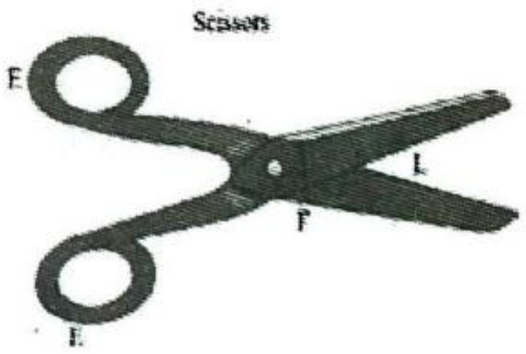
Examples



Claw hammer

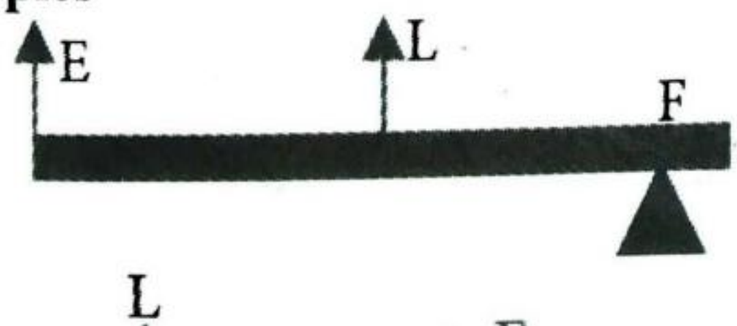
Crawbar

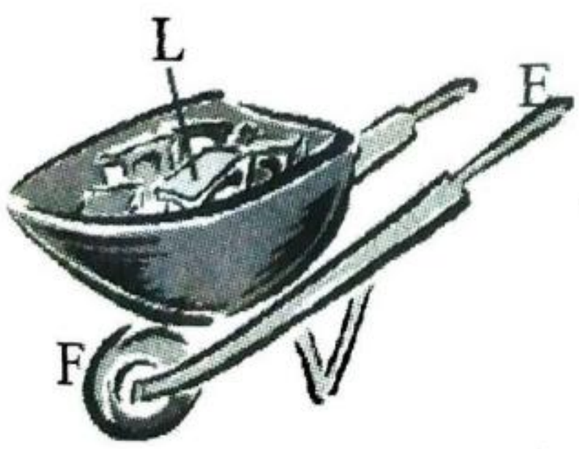
Scissors

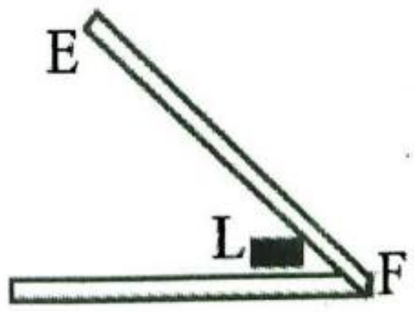


2nd class: Load is between the effort and the fulcrum.

Examples

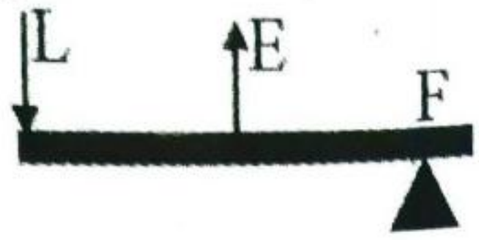


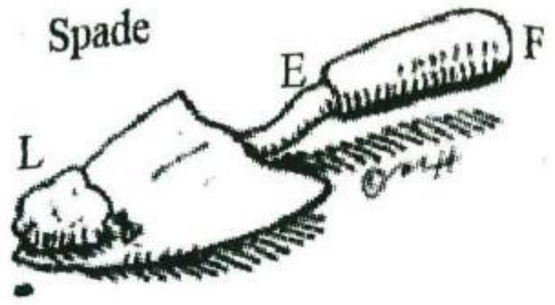


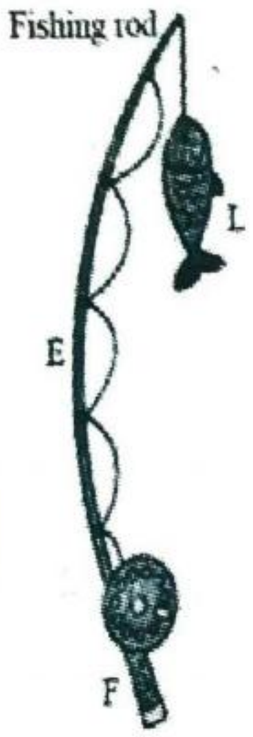


3rd class: Effort is between the load and the fulcrum.

Examples







Illustrative Examples

15. In a nut cracker, a nut is located 2 cm from the fulcrum and an input force of 20N is applied at the handles, which are 10 cm from the fulcrum. What is the force applied to crack the nut?

Solution: In ideal case, $MA = VR \Rightarrow \frac{L}{E} = \frac{S_E}{S_L}$

$$\Rightarrow L = \left(\frac{S_E}{S_L}\right)(E) = \left(\frac{10}{2}\right)(20) = 100N$$

16. Find the ideal mechanical advantage of a claw hammer if the input force is applied 30cm from the nail and the fulcrum is located 2 cm from the nail.

Solution: In this case,

$d_E = 30 - 2 = 28$ cm

$d_L = 2$ cm

$$VR = IMA = \frac{d_E}{d_L} = \frac{28}{2} = 14$$

17. An iron bar 3m long is used to lift a 60 kg block. The fulcrum is placed 80 cm from the block.
    a) What is the IMA of the bar?
    b) What input force is required?

Solution: Now we have

$d_E = 2.2m$ and $L = mg = (60)(10) = 600$

$d_L = 0.8m$

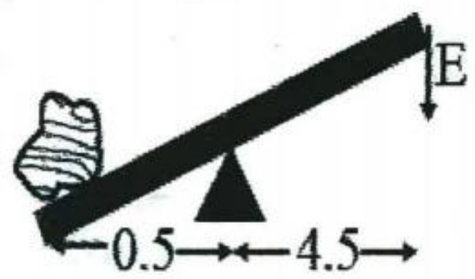
a) $IMA = VR = \frac{d_E}{d_L} = \frac{2.2}{0.8} = 2.75$

b) $IMA = VR$

$$\frac{L}{E} = \frac{d_E}{d_L} \Rightarrow E = \frac{(L)(d_L)}{(d_E)} = \frac{(600)(0.8)}{2.2}$$

$E = 218N$

18. A man raises a 5000N stone by means of a 5m long lever. If the fulcrum is 0.5m from the end that is in contact with the stone. Determine the VR of the lever.



Solution: $VR = \frac{d_E}{d_L} = \frac{4.5}{0.5}$

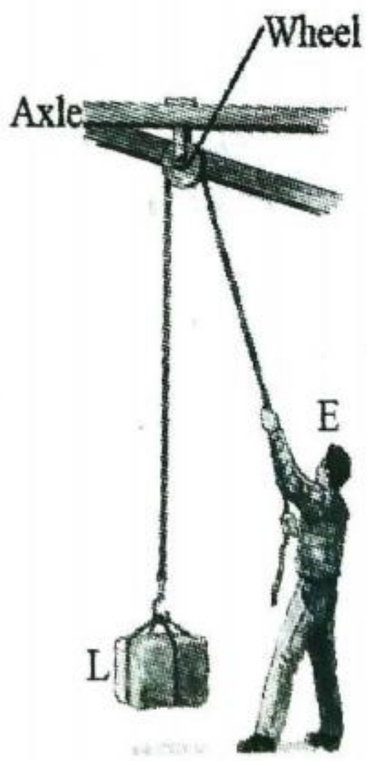
$VR = 9$

The Wheel and axle

> **The wheel and axle**: Is a large wheel secured to a smaller wheel called an axle.

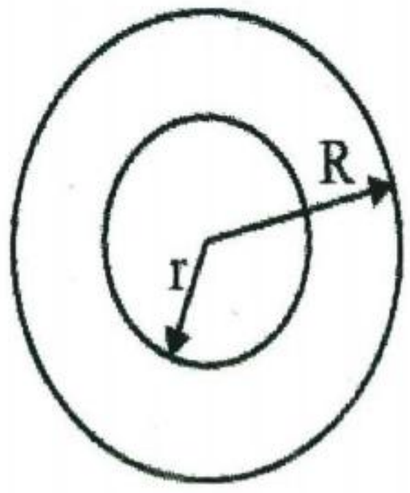
EXAMPLES: A windlass, a door knob, steering wheel and even screw driver and starting handle of a car are all examples of wheel and axle.

When an effort is applied at the end of the rope, in one single revolution, the effort moves a distance equal to the circumference of the wheel ($2\pi r$). So, the mechanical advantage and the velocity ratio can be calculated as follows.



$\boxed{MA = \frac{L}{E}}$ and $VR = \frac{d_E}{d_L}$,

but $d_E = 2\pi R$; $d_L = 2\pi r$

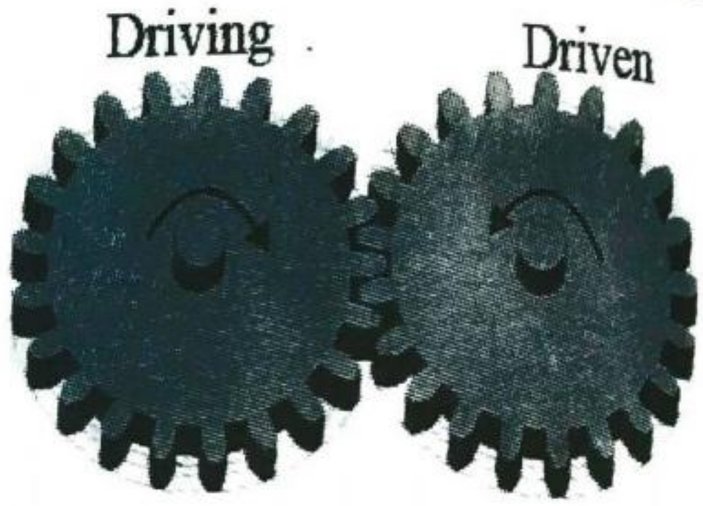


$$\Rightarrow \boxed{VR = \frac{R}{r}}$$

Depending on the relative radii, wheel and axle can be thought of a force multiplier, speed multiplier or direction changer.

Gears

Gear: Is a notched wheel that can transmit turning effect by meshing with another notched wheel.



Gears are used in many machines including the analogue clock where when the minute hand moves 60 minutes the hour hand moves 5 minutes thus the system has a velocity ration 12.

When the driven gear turns, its teeth lock into the driving gear and force it to rotate.

The velocity ratio of the gear is then obtained by:

$$VR = \frac{\text{No of teeth on the driven gear}}{\text{No of teeth on the driving gear}} = \frac{\text{diameter of driven}}{\text{diameter of driving}}$$

$$\Rightarrow VR = \boxed{\frac{N_{driven}}{N_{driving}} = \frac{D_{driven}}{D_{driving}}}$$

Illustrative Examples

19. Using a wheel and axle, a 400N load can be lifted by a force of 50N applied to the rim of the wheel. The radii of the wheel and axle are 85 cm and 6 cm respectively. Determine

a) MA b) VR c) η

Solution:

a) $MA = \frac{L}{E} = \frac{400}{50} = 8$

b) $VR = \frac{R}{r} = \frac{85}{6} = 14.2$

c) $\eta = \frac{MA}{VR} = \frac{8}{14.2} = 0.56 = 56\%$

20. Find the velocity ratio of a gear when the number of teeth on the driving is 8 and the number of teeth on the driven is 16.

Solution: $VR = \frac{N_{driven}}{N_{driving}}$

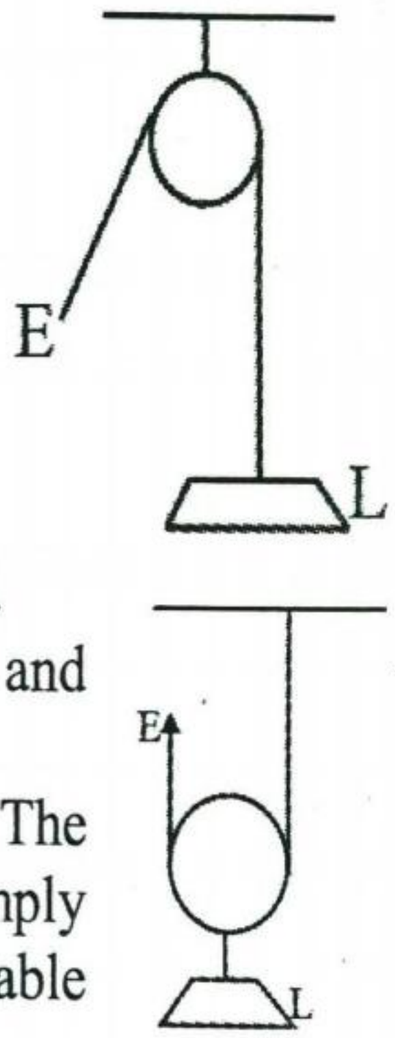
$VR = \frac{16}{8}$

$VR = 2$

The Pulley system

Pulley: Is a wheel with a grooved rim over which a rope or a chain is passed.

There are different kinds of pulleys, but the most common ones are; fixed pulley, movable pulley and compound pulleys.

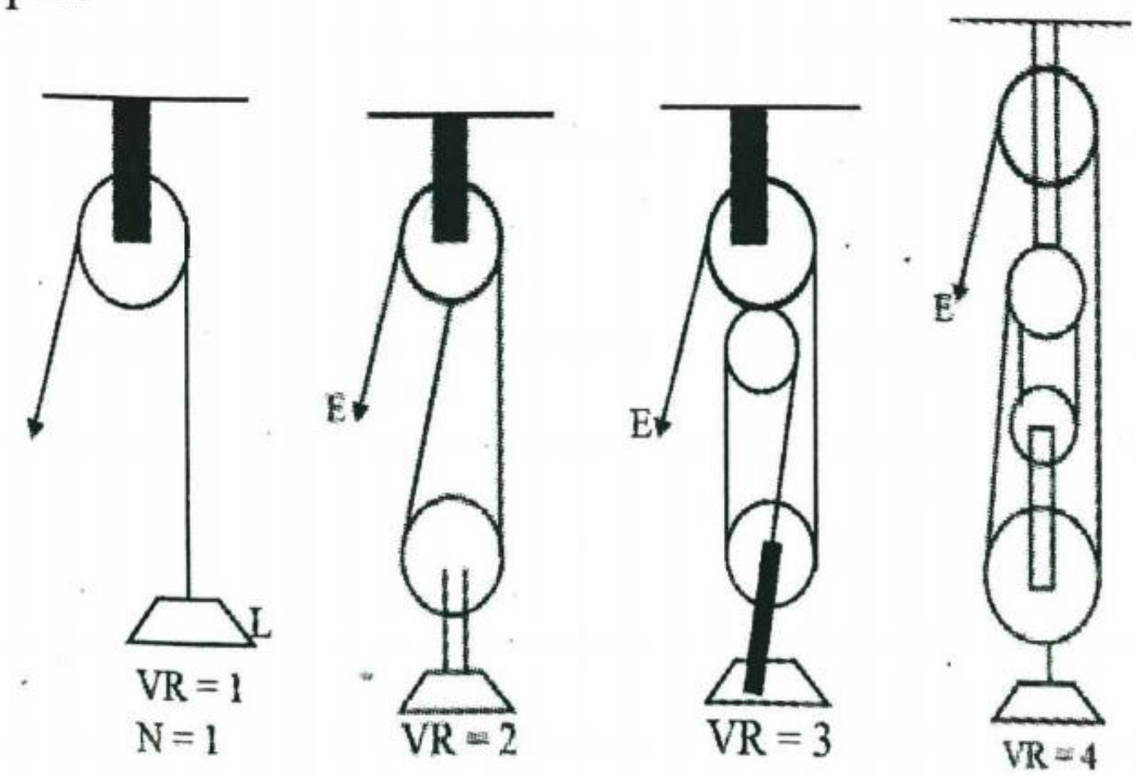


1. **Fixed Pulley**: Has a fixed axle with a rope passing over the top
   fixed pulley is a direction changer only and has a velocity ration, VR = 1.
2. **Moveable pulley**: Has an axle that is free to move up and down.
   Movable pulley has a velocity ratio of two VR = 2.
3. **Compound pulley**: Is a combination of fixed and movable pulleys.

☞ A compound pulley also called a block and tackle. The velocity ration of a block and tackle is obtained by simply counting the number of ropes supporting the movable pulley.

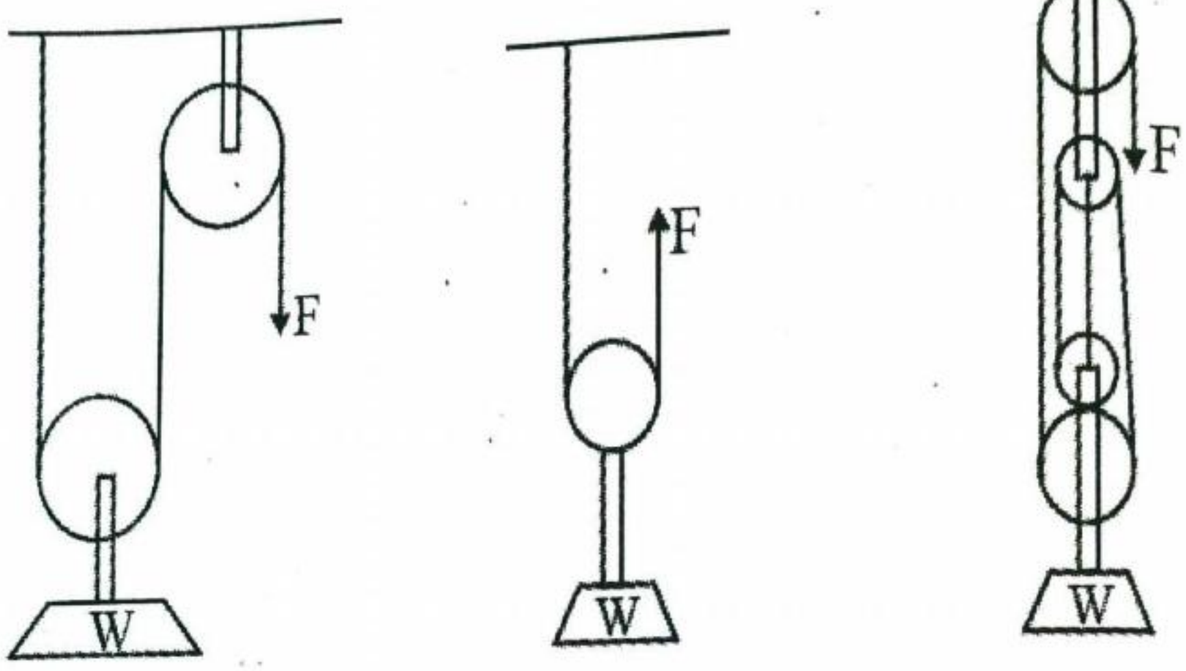
VR = number of the rope sections that support the load
VR = N

Examples



Illustrative Examples

21. For each of the pulley systems. Determine
    a) The VR
    b) The force needed to lift the 100N load.



Solution:

i, a) VR = 2

b) $2F = W \Rightarrow F = \frac{W}{2} = \frac{100}{2} = 50N$

ii, a) VR = 2

b) $2F = W \Rightarrow F = \frac{W}{2} = \frac{100}{2} = 50N$

iii, a) VR = 4

b) $4F = W \Rightarrow F = \frac{W}{4} = \frac{100}{4} = 25N$

Complex Machines

Complex machines: Is a device where two or more simple machines are combined to make a single mechanism.

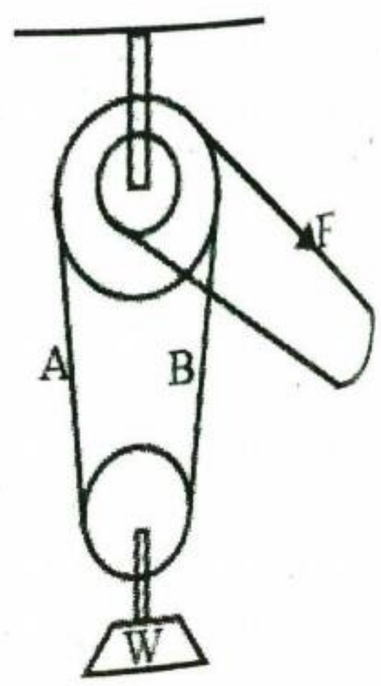
Examples includes, bicycle, differential pulley, jackscrew, etc.

The differential pulley

Differential pulley: Is a pulley with combination with a wheel and axle.

It is used to lift extremely large masses over a short distance. A differential composed of two fixed pulleys at the top with radii R and r.

When a force F is applied, the part of the rope marked A moves up a distance of $2\pi R$ and the part marked B moves down a distance $2\pi R$. And the net result is $2\pi R - 2\pi r = 2\pi (R - r)$ and the load moves



up half this distance thus, $d_L = \pi(R - r)$ and the effort moves $2\pi R$ for one revolution. The mechanical advantage and the velocity ratio can be obtained by:

$$\boxed{MA = \frac{L}{E} = \frac{W}{F}} \text{ and}$$

$$VR = \frac{d_E}{d_L}, \text{ but } d_E = 2\pi R \text{ and } d_L = \pi(R - r)$$

$$VR = \frac{2\pi R}{\pi(R - r)} \Rightarrow \boxed{VR = \frac{2R}{R - r}}$$

Illustrative Example

22. A differential pulley with radii, r = 10 cm and R = 11 cm are fastened together and turn on the same axle. If a force of F = 50N is applied on the rope to lift a load of W = 700N, calculate:
    a) VR b) η

Solution:

a) $$VR = \frac{2R}{R - r} = \frac{(2)(11)}{(11 - 10)} = 22$$

b) $$MA = \frac{L}{E} = \frac{W}{F} = \frac{700}{50} = 14$$

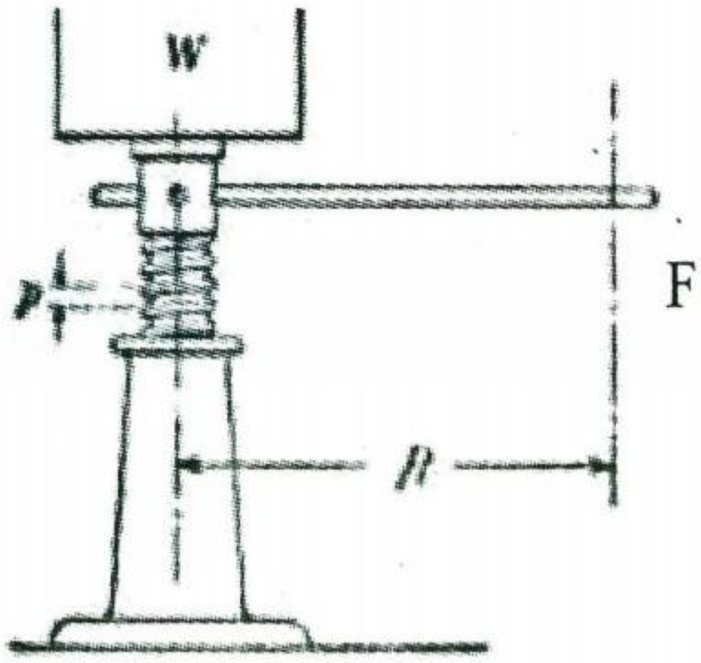
$$\eta = 64\%$$

The Jack Screw

Jack screw: is a screw in combination with a lever.

It helps to lift large loads such as a car to give it service or change tires. The mechanical advantage and the velocity ratio is obtained by:

$$\boxed{MA = \frac{L}{E} = \frac{W}{F}}, \text{ and } \boxed{VR = \frac{d_E}{d_L} = \frac{2\pi R}{P}},$$



where p is called pitch which is the distance between threads.

☛ Note that;

The longer the handle (R) and the smaller the pitch (P) the greater the VR, but it would take more turns to lift the car.

Illustrative Example

23. A jackscrew has a lever arm of 40 cm and a pitch of 5 mm. If the efficiency is 30 percent, what force F is required to lift a load W = 2700N.

Solution: $d_E = 2\pi R$, $d_L = P$

$$VR = \frac{d_E}{d_L} = \frac{2\pi R}{P}$$

$$VR = \frac{2\pi(0.4)}{0.005} = \frac{(2)(3.14)(0.4)}{0.005} = 502$$

$$\eta = \frac{MA}{VR} \Rightarrow MA = (\eta)(VR) = (0.3)(502) = 151$$

$$\text{But } MA = \frac{L}{E} = \frac{W}{F} \Rightarrow F = \frac{W}{MA} = \frac{2700}{151}$$

$F = 18N$

General Secondary Education Certificate Exam

24. A 7 cm think and 15 cm long wedge is used to pierce an object 2m long with diameter 20 cm. What is the VR of the wedge?

A. 10 B. 2.14 C. 0.1 D. 0.47

Solution: $VR = \frac{L}{t} = \frac{15}{7} = 2.14$ **Answer: B**

25. The efficiency of an inclined plane is 50%. If the inclined plane has a length of 4m and a height of 1m. What is its MA?

A. 3 B. 2 C. 5 D. 4

Solution: $VR = \frac{L}{h} = \frac{4}{1} = 4$, and $\eta = \frac{MA}{VR} \Rightarrow MA = (\eta)(VR)$

$MA = (0.5)(4) = 2$ **Answer: B**

26. What is the purpose of using an inclined plane?

A. Multiplying force C. Multiplying energy

B. Multiplying distance D. Multiplying speed

Answer: A

27. A boy weighs 400N. If he lifts a car of weight 20,000N by applying his weight on a lever. What must be the mechanical advantage of the lever?

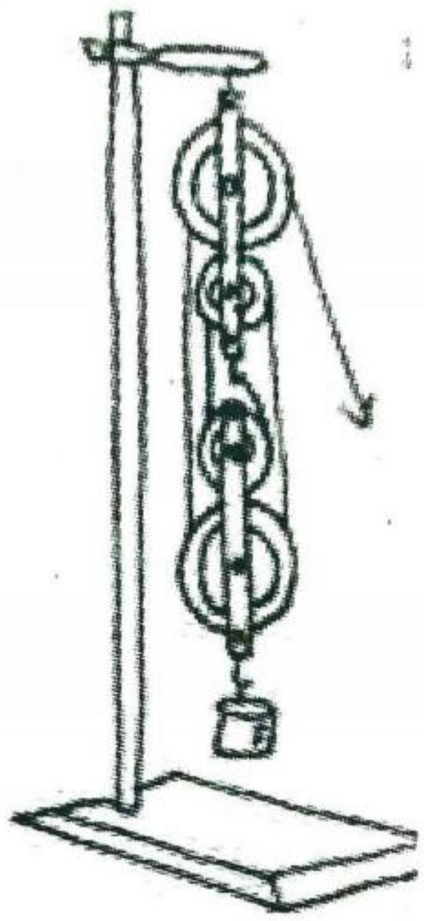
A. 20 B. 50 C. 0.02 D. 0.5

Solution: $MA = \frac{L}{E} = \frac{20,000}{400} = 50$

Answer: B

28. When the pulley system below is used to lift a block by1m, how far will the effort moves?

A. 2m B. 5m C. 0.2m D. 1m



Solution: $VR = \frac{dE}{dL} = \frac{dE}{1} = 5$

$dE = 5m$ or $VR = N = 5 = \frac{dE}{1} \Rightarrow dE = 5m$

Answer: B

29. What do we mean when we say a certain machine is 98 percent efficient?

A. The machine wastes 2 Joules of an input energy.
B. The machine wastes 2 percent of an input energy.
C. The machine multiplying an input force by a factor of 98.
D. The machine multiplying energy by a factor of 98.

Answer: B

30. If MA of a simple machine is greater than one, the machine is

A. Energy multiplier C. speed multiplier
B. Force multiplier D. distance multiplier

Answer: B

31. Which one of the following equation correctly defined the efficiency of a simple machine?

A. $\mu = \frac{VR}{MA}$ B. $\mu = \frac{IMA}{VR}$ C. $\mu = \frac{IMA}{AMA}$ D. $\mu = \frac{AMA}{IMA}$

Answer: D

32. If it were possible to get rid of friction, we would have a machine:

A. That does more work than the work done on it.
B. Whose efficiency is greater than 100%.
C. With a MA equal to VR.
D. Whose power output is greater than power input?

Solution: If friction is zero,

$\eta = 1, \frac{MA}{VR} \Rightarrow MA = VR$ **Answer: C**

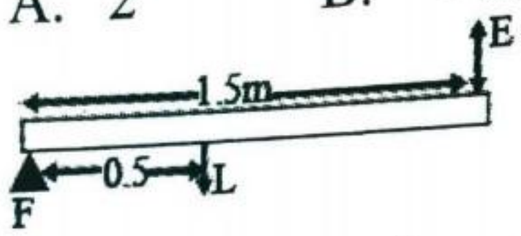
33. In raising an object to a given height h by means of an inclined plane as compared with raising the object vertically to the same height h, there is a reduction in:

A. Work required
B. Force required
C. Mechanical advantage
D. Distance pushed

Solution: Inclined plane multiplies force. **Answer: B**

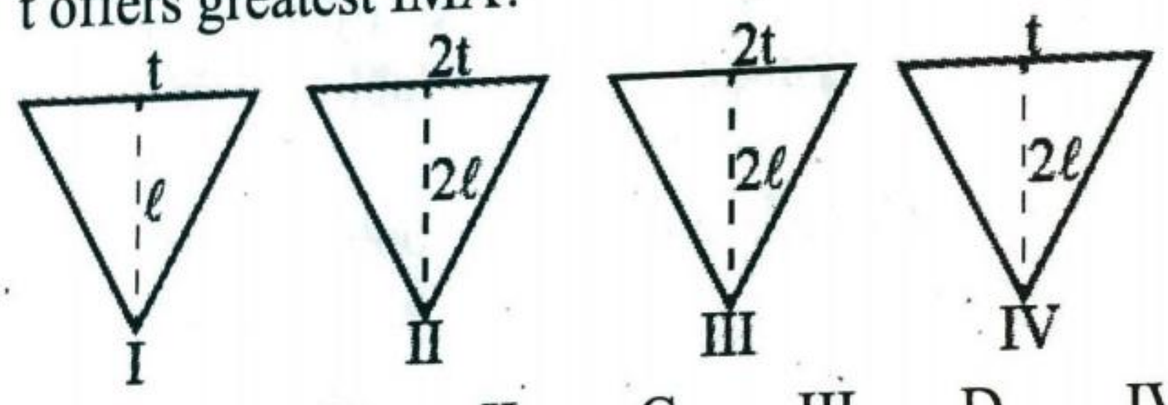
34. On a 1.5m long 2nd class lever, a load of 450N is placed 0.5m away from the fulcrum. How large is the velocity ratio of the lever?

A. 2 B. 0.33 C. 3 D. 0.5



Solution: $VR = \frac{d_E}{d_L} = \frac{1.5}{0.5} = 3$ **Answer: C**

35. Which of the following double wedges with length l and thickness t offers greatest IMA?



A. I B. II C. III D. IV

Solution: $IMA = VR = \frac{L}{t}$ **Answer: D**

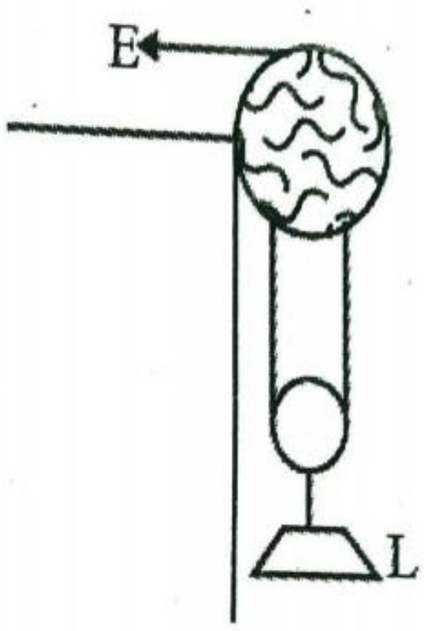
36. The diagram shows a pair of gear wheels, the number of teeth on gear A is 32 and on B is 16. Which statement is correct if B is driving gear?

A. The load is greater than the effort and A rotates slower than B.
B. The load is less than the effort A rotates faster than B.
C. The load is less than the effort A rotates slower than B.
D. The load is greater than the effort A rotates faster than B.

Solution: $VR = \frac{N_{driven}}{N_{driving}} = \frac{32}{16} = 2 \Rightarrow$ force multiplier

Answer: A

37. If the pulley system is frictionless, the machine is:

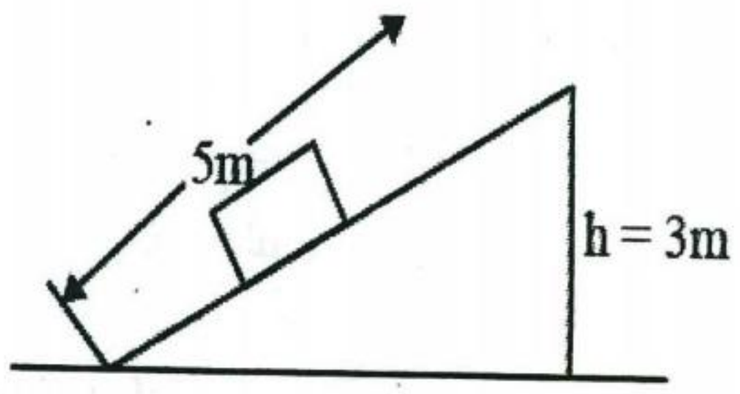


A. Force multiplier
B. Speed multiplier
C. Both speed and force multiplier
D. Neither speed nor force multiplier

Solution: VR = 2 = number of ropes supporting the load.

Answer: A

38. Due to the presence of friction, the effort needed to slide a 600N load along the inclined plane is 600N. What is the efficiency of the plane in lifting the land to the 3m height?



A. 100% B. 50% C. 75% D. 60%

Solution: $MA = \frac{L}{E} = \frac{600}{600} = 1$

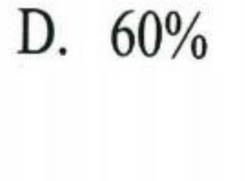
$VR = \frac{d_E}{d_L} = \frac{5}{3}$

$\eta = \frac{MA}{VR} = \frac{1}{\frac{5}{3}} = \frac{3}{5} = 0.6$

$\eta = 60\%$

Answer: D

39. A block is sliding down on a surface inclined at angle θ. If the coefficient of friction is μ, what is the MA of the plane?



A. $\frac{1}{\cos\theta}$ B. $\frac{1}{\sin\theta}$ C. $\frac{1}{\sin\theta + \mu\cos\theta}$ D. $\frac{1}{\mu\cos\theta}$

Answer: C

Unit Five: Conceptual Questions

1. Is it possible to get more work out of a machine than you put in?
2. How could you increase the ideal mechanical advantage of a machine?
3. How could you increase the mechanical advantage of a wedge without changing the ideal mechanical advantage?
4. In what condition will the lever serve as a force multiplier machine.
5. The ratio of output speed to input speed for a gear is 2: 1. What is the mechanical advantage?

Unit Five: problems

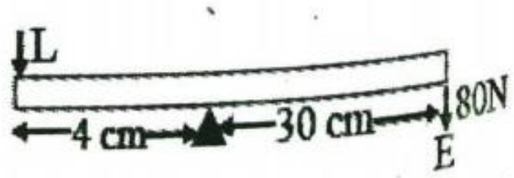
6. A simple machine has an efficiency of 0.75 and a VR of 12. Determine the MA and the load that can be moved if an effort of 100N is applied.
7. A 25 percent of efficient machine performs a work of 200J. What input work is required?
8. A 60% efficient machine lifts a 10 kg mass at a constant speed of 3 m/s. What is the required input power?

Inclined plane, wedge and screw

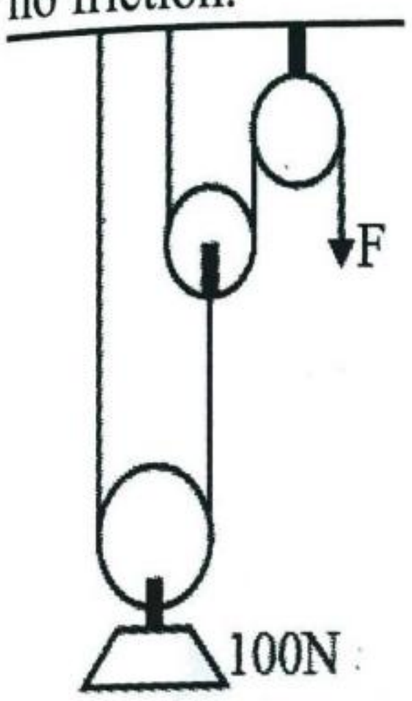
9. A block of weight 5000N is pushed up a slope by a force of 250N. Assume there is no friction. Calculate
   A. AMA B. VR
   C. The length of the slope if the height of the slope is 10m.
10. An inclined plane is 100m long and at an angle of 20° to the horizontal. The AMA is two. Calculate
    a) The effort required to push a 7200N block up the slope.
    b) The IMA
    c) The efficiency of the incline plane
11. What must be the thickness of the base if a wedge is 20 cm long and it is desired that the input force be one – length of the output force?
12. The handle of a screw jack is 50 cm long. If the screw has five threads per centimeter. What is the ideal mechanical advantage?

Lever, Wheel and axle and pulley

13. For the see – saw shown, calculate
    a) The load
    b) The MA if y = 100%



14. A wheel and axle is used to lift a bucket of water out of a well. The radii of the wheel and axle are 20 cm and 4 cm respectively. Determine
    a) The VR
    b) The theoretical effort needed to lift a load of 30N assuming no energy loss.
    c) The efficiency of the actual effort required is 10N
15. Determine the force F required to lift a 100N load. Assume there is no friction.



16. A 10 cm long, 2 cm wide wooden wedge is pushed into a soft wood block. Calculate
    a) The VR of the wedge.
    b) The load on the soft wood if the effort applied is 30N. Assume the wedge is 100% efficient.

Unit Five: Explanation to Questions

1. NO! it is not possible due to the conservation of energy.
2. By increasing the distance moved by the effort.
3. By reducing the effect of friction.
4. When the lever is in the 2nd class and the load is very close to the fulcrum.
5. MA = ½

Unit five: Solutions to problems

6. $\eta = \dfrac{MA}{VR} \Rightarrow MA = (\eta)(VR) = (0.75)(12) = 9$ and

$$MA = \frac{L}{E} \Rightarrow L = (E)\frac{MA}{M} = (9)(100) = 900N$$

7. $$\eta = \frac{W_o}{W_i} \Rightarrow W_i = \frac{W_o}{\eta} = \frac{200}{0.25} = 800J$$

8. $\eta = \frac{P_o}{P_i} \Rightarrow P_i = \frac{P_o}{\eta} = \frac{(mg)(v)}{\eta} = \frac{(10)(10)(3)}{0.6} = 500W$

9. a) $MA = \frac{L}{E} = \frac{5000}{250} = 20$

b) $VR = MA = 20$

c) $VR = \frac{L}{h} \Rightarrow L = (VR)(h) = (20)(10) = 200m$

10. a) $AMA = \frac{L}{E} \Rightarrow E = \frac{L}{AMA} = \frac{7200}{2} = 3600N$

b) $IMA = VR = \frac{L}{h} = \frac{1}{L\sin\theta} = \frac{1}{\sin\theta} = \frac{1}{\sin 20} = 2.9$

c) $\eta = \frac{MA}{VR} = \frac{2}{2.9} = 0.69 = 69\%$

11. $IMA = VR = \frac{\ell}{t} = \frac{L}{E} \Rightarrow \ell = \left(\frac{E}{L}\right)(t) = \left(\frac{1}{10}\right)(20) = 2cm$

12. $VR = \frac{2\pi R}{P} = \frac{2(3.14)(50)}{0.2} = 1570$

13. a) $MA = VR \Rightarrow \frac{L}{E} = \frac{30}{4} \Rightarrow L = 80\left(\frac{30}{4}\right) = 600N$

b) $MA = \frac{L}{E} = \frac{600}{80} = 7.5$

14. a) $VR = \frac{R}{r} = \frac{20}{4} = 5$ b) $\frac{L}{E} = 5 \Rightarrow E = \frac{(30)}{5} = 6N$

c) $MA = \frac{L}{E} = \frac{30}{10} = 3$ and $\eta = \frac{MA}{VR} = \frac{3}{5} = 0.6 = 60\%$

15. $F = \frac{W}{4} = \frac{100}{4} = 25N$

16. a) $VR = \frac{L}{t} = \frac{10}{2} = 5$ b) $\frac{L}{E} = 5 \Rightarrow L = (30)(5) = 150N$

Unit Six

Fluid Statics

Fluids are extremely important in many aspects of everyday life. We drink them, breath them, swim in them, airplane fly through them, ships float in them etc.

Fluids: Are substances that can flow. Gases and liquids are fluids.

And, the study of fluids at rest is called *fluid statics*.

Pressure

The ratio of force to the contact area that is perpendicular to the force is known as pressure.

Pressure: Is force per unit area.

$$P = \frac{F}{A}$$

Pressure is a scalar quantity and its SI unit is Pascal (Pa)

thus, $1\ pa = 1\ N/m^2$.

☛ **Note that;**

* The greater the area over which the force acts, the less the pressure. And,
* The smaller the area over which the force acts, the greater the pressure.

Conceptual Examples

1. Why do large vehicles, such as a tractor fitted with large tyres? Explain

Explanation: It is to reduce the pressure on the ground, so that the wheels will not sink deeply into soft mud and become struck.

2. A girl who wears a sharp pointed heels is more likely to destroy a wooden floor than when she wear a flat shoes. Why?

Explanation: It is because, the heel has a smaller area and hence the pressure is larger. Thus the floor will be penetrated and destroy by the heel.

3. A physics professor sleeps on a bed of nails suffering no injury and only moderate discomfort. How is this possible?

Explanation: If you try to sleep on a vertically upward single nails, it will penetrate your body and thus, you will feel a sever pain but, if thousand nails are used and form a bed then you can safely sleep on the bad of nails because, in this case your weight is distributed over large area and hence, the pressure on your body is smaller.

Illustrative Examples

4. Calculate the pressure exerted on a level ground by a man whose mass is 75 kg and the area of both feet is 280 cm^2.

Solution: $P = \frac{F}{A} = \frac{mg}{A} = \frac{(75)(10)}{(280)(10^{-4})} = 26800Pa$

Note: $1\ cm^2 = 10^{-4}m^2$. Thus, $280\ cm^2 = 280 \times 10^{-4}\ m^2$

5. The pressure exerted by the point of the needle on the body when a patient is being injected is about $1 \times 10^8\ N/m^2$. If the area of the needle tip is 0.1 mm^2, calculate the force the doctor needs to exert to produce the necessary pressure.

Solution: $P = \frac{F}{A} \Rightarrow F = PA = (1 \times 10^8)(1 \times 10^{-7}),\ 0.1mm^2$

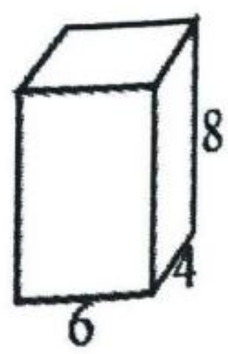
$= 0.1 \times 10^{-6}\ m^2$

$F = 10N$

6. A rectangular box 8m by 4m has weight 48N. Calculate
   a) the maximum pressure it exerts
   b) the minimum pressure it exerts

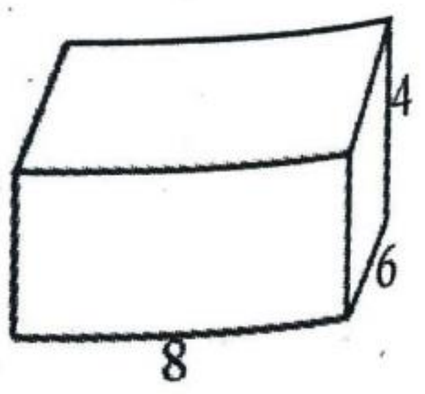
Solution:

a) pressure is maximum when the box stands on its ends



$P = \frac{F}{A} = \frac{48.}{(6)(4)} = 2N/m^2$

b) pressure is minimum when the box is lying flat



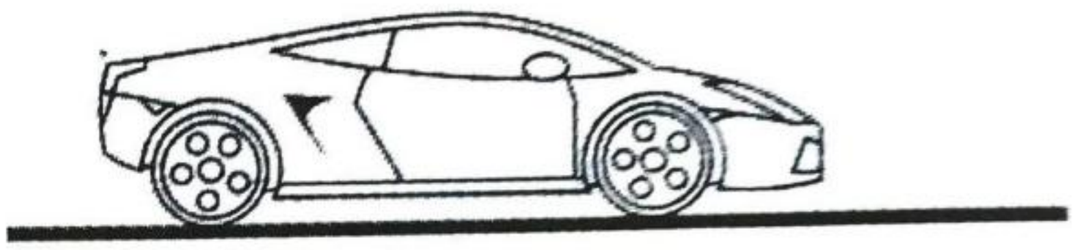
$$P = \frac{F}{A} = \frac{48}{(8)(6)} = 1N/m^2$$

7. A woman weighs 495N and wears shoes that touch the ground over an area of 412 cm^2. What is the pressure she exerts on the ground when she stands on one foot?

Solution: $P = \frac{F}{A}, A = \frac{412}{2} = 206cm^2 = 206 \times 10^{-4}\ m^2$

$$P = \frac{495}{206 \times 10^{-4}} = \frac{495}{0.0206} = 24kPa$$

8. A car tire makes contact with the ground on a rectangular area of 12 cm by 18 cm. If the mass of the car is 925 kg. What pressure does the car exert on the ground?



Solution:

$$P = \frac{F}{A} = \frac{mg}{4(W \times \ell)}$$

$$P = \frac{(925)(10)}{4(0.12 \times 0.18)} = 1 \times 10^5 Pa$$

Fluid Pressure

From the air moving in and out in our lung to the blood pumped by the heart throughout our body, pressure in fluids plays an important role in our lives. Without atmospheric pressure as an example, our blood would simply boil and life on earth would become impossible.

Atmospheric pressure

The atmosphere is a layer of air that surrounds the earth. Although we can't feel it, air has mass, this means, it also has a weight.

> **Atmosphere Pressure**: Is the pressure exerted by the weight of air at any point on the earth's surface.

The weight of the column of air on $1m^2$ surface at sea level is found to be 101000N. This means the atmospheric pressure at sea level is around 101 kPa. This is often known as one atmosphere or 1 atm.

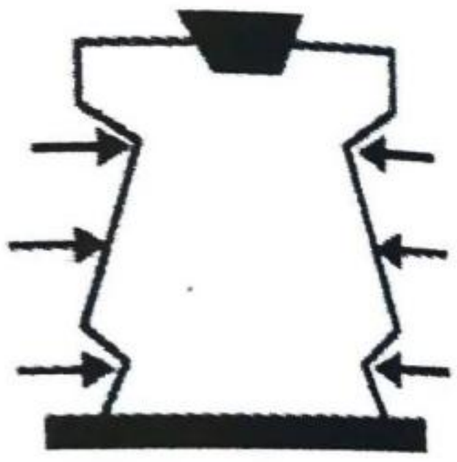
Thus, $1\ atm = 101\ kPa = 1.01 \times 10^5\ Pa.$

Conceptual Examples

9. Even if the atmospheric pressure is so huge and assumed as if an elephant is sitting on top of our head, we didn't feel its existence, why?

Explanation: Because, as the atmosphere pushes on our body equally from all directions, our body also pushes back out equally. Thus the atmospheric pressure and the pressure in our body cancel out and we feel nothing.

10. What will happen to our body when we go somewhere where the pressure is greater than the atmospheric pressure?
Explanation: Our body will be crushed.
11. If you went somewhere where the pressure was very low (in to space) what will happen to your body.
Explanation: If you went into space without pressurized suit, the pressure inside your body will push outward with some nasty effects, like explosion.
12. If you remove the air inside a can, or if you add some water and boil it and then close the can tightly and then add cold water on it, what will happen to it? Explain



Explanation: It will collapse. This is because, the pressure inside will be smaller than the atmospheric pressure. Hence the can collapses!

Factors that affect atmospheric pressure

The most dominant factor that affects atmospheric pressure is altitude, that is the height above the ground. And, it is observed that; "*As altitude increases, the atmospheric pressure decreases*."

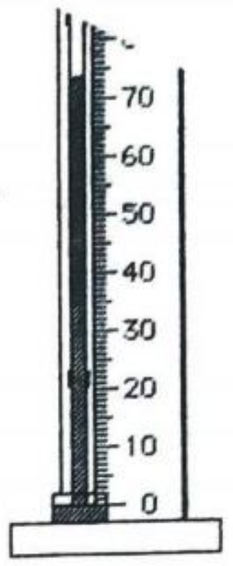
Conceptual Examples

13. As altitude increases, the atmospheric pressure decreases. Why?
Explanation: Because, as the height above the ground increases, the air mass decrease, so less weight is pushing down and thus a decreased atmospheric pressure is acting.
14. At sea level, water boils at 100°C. At the top of tall mountains, it will boil at 75°C. Why is this?
Explanation: At sea level, the atmospheric pressure is about 101000Pa, which is large. But at the top of tall mountains, the atmospheric pressure is much less than 101000Pa. Hence as water is heated at mountains, the pressure pressing the molecules is small hence the water boils at lower temperature than 100°C. Thus, as altitude increases, boiling point decreases.

Measuring atmospheric pressure

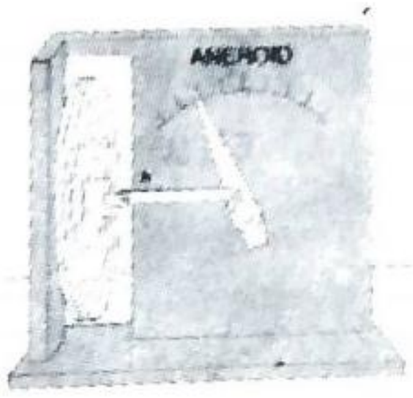
There are several instruments used to measure atmospheric pressure. The most common is a barometer.

Mercury Barometer: when a tube is immersed in a vessel that contains mercury, for 1 atm or 101000Pa atmospheric pressure, the mercury is pushed up through the tube up to 76 cm or 760 mm. Thus, 1 atm = 760 mm of mercury.



1 atm = 101000Pa = 76 cm Hg = 760 mmHg

Aneroid Barometer: Is a device for measuring atmospheric pressure using a corrugated metal box and not a liquid. As the pressure changes, the upper face of the box moves and this in turn moves a pointer in centimeter scale.



Conceptual Examples

15. Why do we commonly use Aneroid barometer than Mercury barometer. Explain

Explanation: This is because mercury barometer is long and inconvenient, heavy and contains a liquid that is hazardous and easily spoilt. But aneroid barometer is compact and portable, thus we can manage it easily.

16. Mercury is quite toxic and needs to be handled very carefully. Why not use water barometer instead? Explain

Explanation: This is because water has a much lower density (1000 kg/m^2) than mercury (13600 kg/m^3). This means when mercury is pushed 760mm in the tube, water will be pushed up the tube about 10000mm = 10m. This will make the water barometer very tall and impractical. **Note**: you can check h = 10m, using the relation $p = \rho gh$. Thus, $h = \frac{P}{\rho g} = \frac{101000}{(1000)(10)} = 10.1\text{m} \simeq 10\text{m}$.

Some uses of air pressure

There are several uses of air pressure. Most rely on creating a pressure difference by pumping air into or out of a container (chamber).

Common pump: a pump that relies on atmospheric pressure to move water.

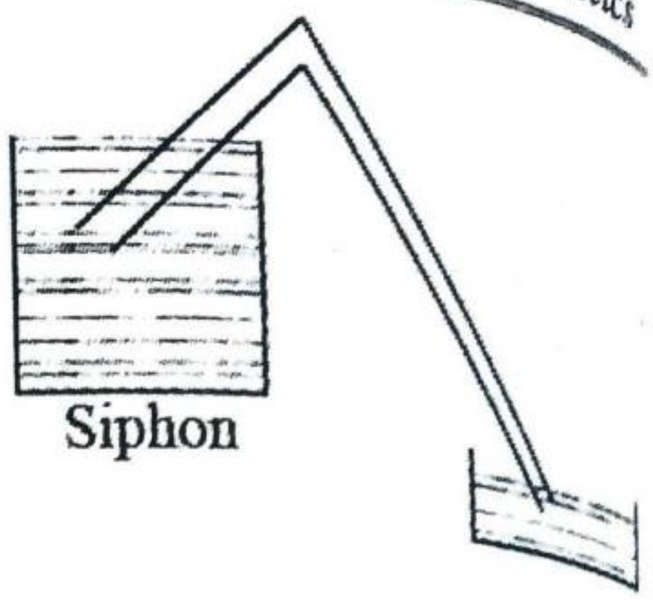
Force pump: a pump that relies on atmospheric pressure and compressed air to move water, often to a great height.

Lift pump: a pump that relies on atmospheric pressure to move water.

Suction pad: a round rubber pad that relies on atmospheric pressure to stick to smooth surfaces.

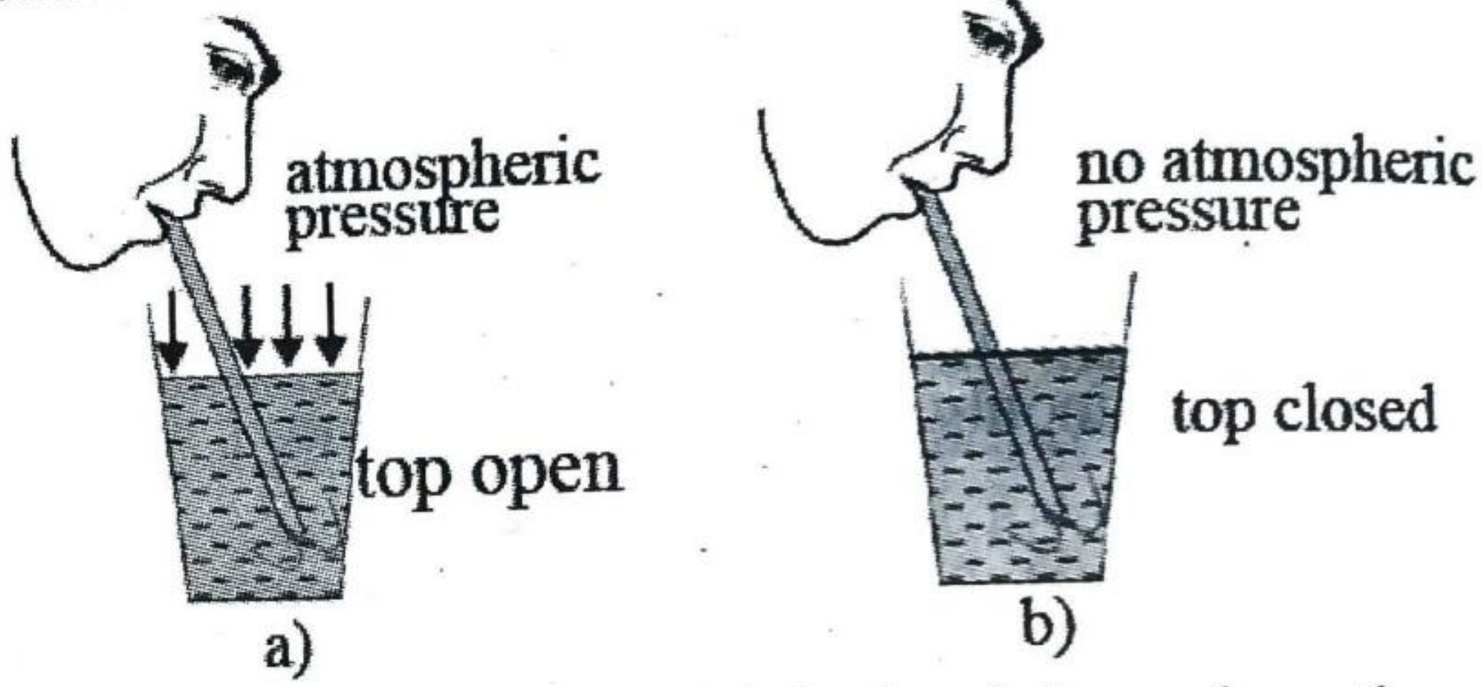
Siphon: a tube which can move liquid using the difference between pressure of the liquid and the atmosphere.

Drinking straw: a thin tube used to suck liquids into the mouth.


Siphon

Conceptual Example

17. In which case will you be successful in drinking the juice using a straw.



Explanation: You will drink the juice when the can is open. Because when you suck, you increase the volume of your lung and this reduces the pressure in your mouth. Thus, the juice pushed up the straw by the atmosphere. But, when the can is closed, there is no pressure difference to push the juice up the straw.

Fluid Density

Fluid is a substance that can freely flow. All gases as well as liquids are fluids. Fluids always take the shape of the container they are put in. However, there are still some differences between liquids and gases. For instance, gases are compressible and liquids are incompressible. This means the volume of a liquid stays the same when a force is applied. And, the other difference is that, liquids are denser than gases.

Density: Is mass of the substance divided by its volume.

$$\rho = \frac{m}{v}$$

The Greek letter ρ (rho) stands for density.

Density is a scalar quantity and its SI unit is kg/m^3.

Note also that; $1\ g/cm^3 = 1000\ kg/m^3$

List of some common fluids density;

Fluid	Density (kg/m^3)
Mercury	13600
Honey	1400
Water	1000
Alcohol	800
Air	1.2

Relative density

It is the density of a substance relative to any other substance.

In most cases, relative density involves comparing the density of a fluid to that of water.

$$\text{Relative density} = \frac{\text{density of substance}}{\text{density of water}}$$

For example, the relative density of alcohol is,

$$\text{Relative density} = \frac{\text{density of alcohol}}{\text{density of water}} = \frac{800}{1000} = 0.8$$

Relative density can also be calculated as the ratio of mass of fluids of the same volume.

$$\text{Relative density} = \frac{\text{mass of substance}}{\text{mass of equal volume of water}}$$

☞ **Notice that;** relative density has no unit and that; relative density is also called specific gravity.

Pressure and Depth

In any fluid, the pressure increases with depth. Thus, the deeper the depth within in a given fluid, the greater the fluid pressure. In general, the fluid pressure at a depth (h) due to a fluid of density (ρ) is given by

$p = \rho gh$, where g is acceleration due to gravity. This relation can be derived as follows;

Consider a fluid in a container. The fluid pressure at a depth h from the fluid surface can be obtained by $P = \frac{F}{A}$, but $F = mg = \rho vg$ and $v = Ah$

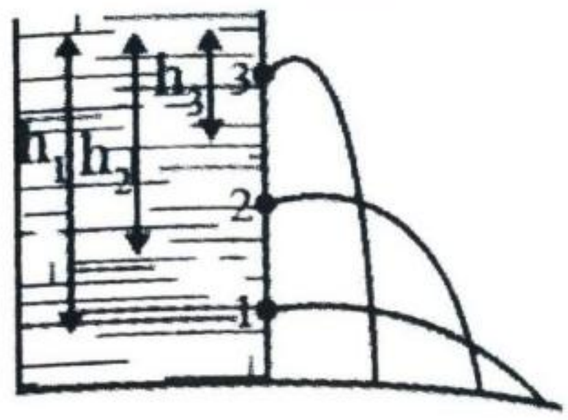
$\Rightarrow F = \rho Ahg$, where A is base area and v is volume.

$P = \frac{F}{A} = \frac{\rho Ahg}{A} = \rho gh$. Thus, $P = \rho gh$, h is the depth from the surface to the point of interest from the relation, $p = \rho gh$, we can see that;

- pressure increases with depth, and
- at any given depth, the pressure is equal in all directions

Conceptual Examples

18. Through which hole will water moves out fastest and furthest and why? Explain



Explanation: Through hole 1, because, the depth of hole 1 is greater than the other two. That is, $h_1 > h_2 > h_3$. Thus, according to the relation $p = \rho gh$, the pressure is highest at point (hole) 1.

19. If you pour water into a set of connected tubes of various shapes, which tube do you think will contain the maximum height of water column. Explain

Explanation: All the tubes have the same height of water. Because, the liquid pressure depends only on the height or the liquid and not on the shape of width of the tube. In this case, we can say, a liquid finds its own level.

20. What is the pressure 100m below the surface of sea water of density 1150 kg/m^2?

Solution: $P = \rho gh$, ignoring air pressure

$P = (1150)(10)(100) = 1150000$ Pa

Pascal's Principle

When a force is applied on any part of a confined fluid, it creates pressure inside the fluid. This pressure is transferred throughout the fluid and is the same value everywhere. This phenomenon is referred to as Pascal's principle and is stated as follows;

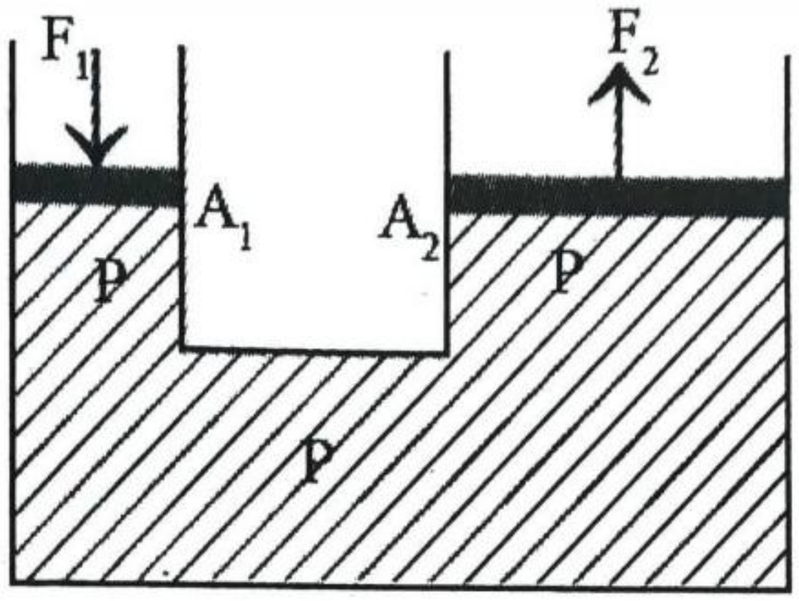
> **Pascal's principle:** sates that
> "The pressure applied to an enclosed fluid is transmitted to every part of the fluid, as well as to the walls of container without reducing in value."

Pascal's principle is applicable in the design of hydraulic press, automobile brake system, aircraft wing, bulldozer, cranes and robots.

Now, let's consider a hydraulic lift with piston areas of A_1 and A_2.

If a force F_1 is applied to piston 1, it creates a pressure inside the fluid. This pressure is transmitted throughout the liquid. It is the same

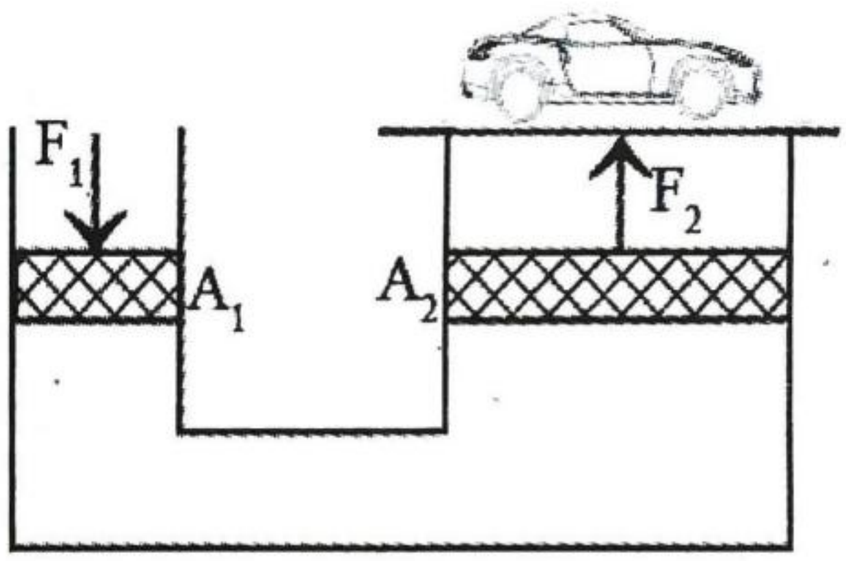
everywhere and exerts a force F_2 on piston 2. And according to the Pascal's principle, P = constant $\Rightarrow P_1 = P_2$.



So we have; $\boxed{\frac{F_1}{A_1} = \frac{F_2}{A_2}}$ ← Pascal's Principle

Illustrative Example

21. To examine a 14500N car, it is raised with a hydraulic lift. If the radius of the small piston is 4 cm and the radius of the larger piston is 17 cm, find the force that must be exerted on the smaller piston to lift the car.



Solution: From Pascal's principle,

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}, \text{ where } A_1 = \pi r_1^2 \text{ and } A_2 = \pi r_2^2$$

$$\Rightarrow F_1 = \left(\frac{A_1}{A_2}\right)(F_2) = \frac{\pi(0.04)^2}{\pi(0.17)^2}(14,500) = 800\text{N}$$

Absolute, Atmospheric and Gauge pressure

Absolute Pressure (P_s)

Is the actual pressure at a given point. That is, it is a pressure exerted at a point by a fluid including atmospheric pressure.

Atmospheric Pressure (P_{atm})

Is the pressure of the surrounding air when measured at the surface of the earth. It has a value of 101 kPa and varies depending on the temperature, the altitude above sea level and the impact of weather system.

Gausge Pressure (P_g)

Is a pressure exerted by a fluid. That is by a liquid or a gas without including the atmospheric pressure. Gauge pressure can be calculated as;

$$\boxed{P_g = P_s - P_{atm}}$$

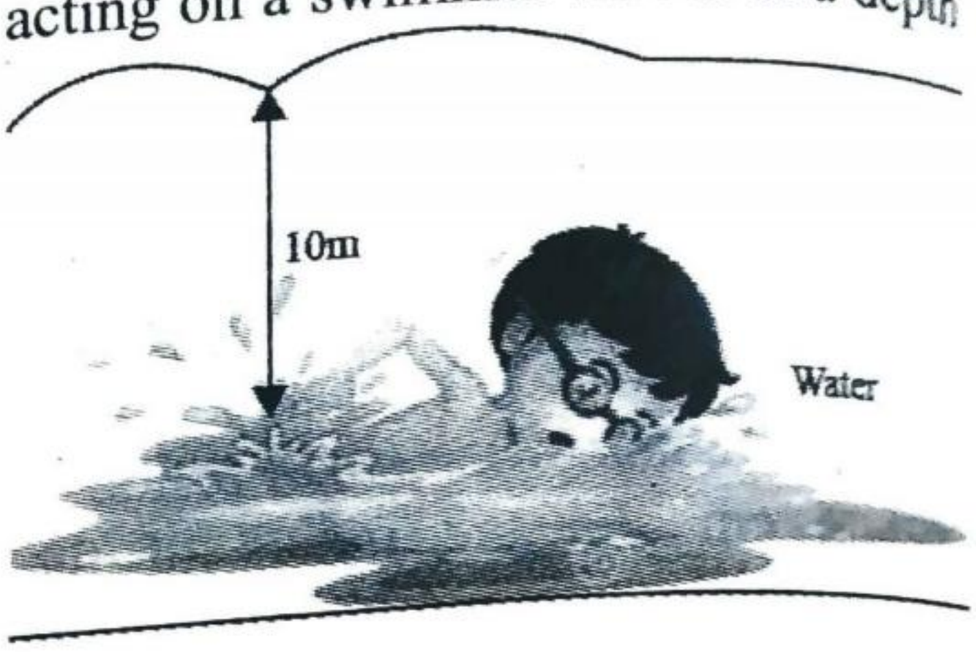
Conceptual Example

22. Can absolute pressure be ever be zero or negative? What about gauge pressure?

Explanation: Absolute pressure can never be negative but a gauge pressure can be zero or negative. For example, a gauge pressure of 0 Pa would mean an absolute pressure of 101 kPa, a gauge reading of -10 kPa would mean 10 kPa below atmospheric pressure.

Illustrative Example

23. Calculate the total pressure acting on a swimmer who is at a depth of 10m from the surface of the swimming pool? (take, $g = 10\ m/s^2$ and $\rho = 1000\ kg/m^3$).



Solution:

$P = P_{atm} + P_{fluid}$

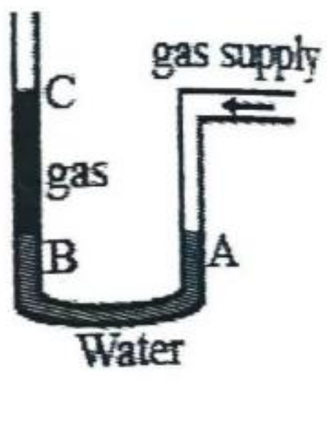
$P = 101\ kPa + \rho gh$

$P = 101000 + (1000)(10)(10)$

$P = 111000 = 111\ kPa$

Manometer

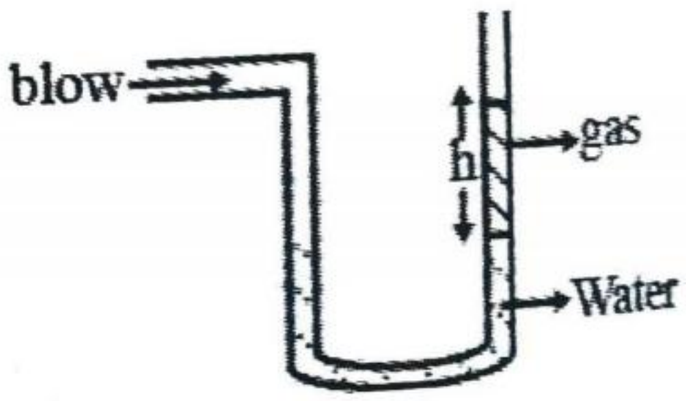
Is a simple instrument often used to measure the pressure of a gas supply. It comprises a U shaped tube open at both ends. For example, the height difference between B and C can be used to find the pressure of the gas supply. Thus, pressure of gas = atmospheric pressure + pressure due to the column of liquid BC.



$$\boxed{\text{Pressure of gas} = P_{atm} + \rho gh_{BC}}$$

Illustrative Example

24. A man blows into one end of a water u tube manometer until the levels differ by 40 cm. Calculate his lung pressure.



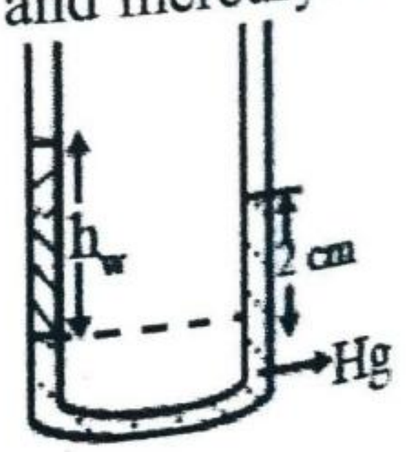
Solution: $P = P_{atm} + \rho gh$

$= 101\ kPa + (1000)(10)(0.4)$

$= 101000 + 4000 = 105000\ Pa$

$P = 1.05 \times 10^5\ N/m^2$

25. The liquids in the u tube manometer are water and mercury. If the column of mercury is 2 cm, calculate the column of water. (take, $\rho_w = 1000\ g/m^3$, $\rho_{Hg} = 13600\ kg/m^3$)



Solution: $P_{atm} + \rho_w gh_w = P_{atm} + \rho_{Hg} gh_{Hg}$

$\Rightarrow \rho_w gh_w = \rho_{Hg} gh_{Hg}$

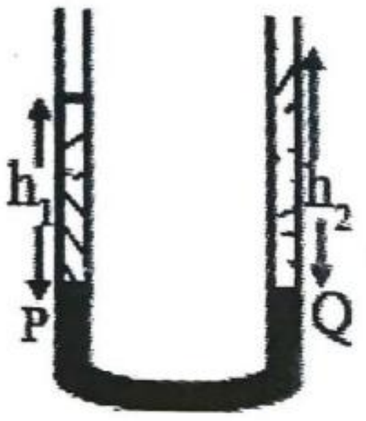
$$h_w = \frac{\rho_{Hg} h_{Hg}}{\rho_w} = \frac{(13600)(2)}{1000} = 27.2\text{cm}$$

☛ **Important points to remember**

- If liquid of density (ρ_1) has height (h_1) and liquid of density (ρ_2) has height (h_2) from points P and Q respectively, and the points are at the same level. Then the pressure at P and Q are the same

  $P_1 = P_2 \Rightarrow P_{atm} + \rho g h_1 = P_{atm} + \rho_2 g h_2$

  $\Rightarrow \rho_1 g h_1 = \rho_2 g h_2 \Rightarrow \boxed{\rho_1 h_1 = \rho_2 h_2}$



- If the density of one liquid is known, the density of the unknown liquid can be determined.
- If only one type of liquid is poured into the manometer, then the levels at both ends would be equal.

Forces in fluids

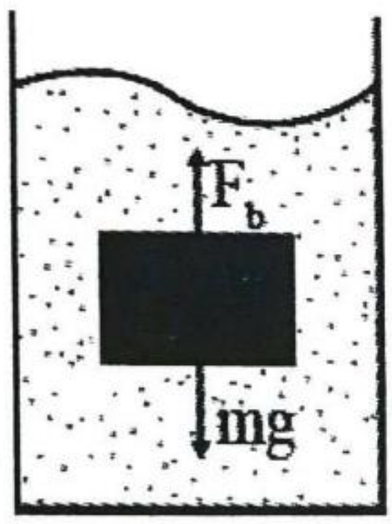
If you try to lift a heavy object which is under water, you will find it surprisingly light and much easier to lift than when it is out of water. This is due to the force from the water that pushes it up. This force of the liquid against gravity is called buoyant force or sometimes up thrust.

Buoyant force: Is a force from the water which pushes a body upwards against gravity.

☛ The size of the buoyant force (F_b) depends on factors such as the density of the fluid and the volume of the object.

Apparent weight

Objects seem less heavy in water, this is so because, the buoyant force pushes upwards, acting against the objects weight and so the weight appears to drop. And the weight of the body immersed in liquid, we call apparent weight. The apparent weight of a body immersed in a liquid can be found as,



apparent weight = Weight – buoyant force

$\Rightarrow \boxed{\text{apparent weight} = mg - F_b}$

Illustrative Example

26. A body that weighs 6N in air is found to weigh 4N when immersed in water. What is the buoyant force of the water?

Solution: Apparent weight = $mg - F_b$

$\Rightarrow F_b$ = Weight in air – Weight in water

$F_b = 6 - 4 = 2\text{N}$

Archimedes's Principle

Archimedes's realized that when an object is immersed in a liquid it displaces a certain volume of the liquid. He determined that, the weight of the displaced fluid was equal to the buoyant force. Or in his own words;

> **Archimedes's Principle:** states that;
> "Any object, wholly or particularly immersed in a fluid, is buoyed up by a force equal to the weight of the fluid displaced by the object."

In other words, the buoyant force acting on an object is equal to the weight of the displaced liquid.

buoyant force = weight of displaced fluid

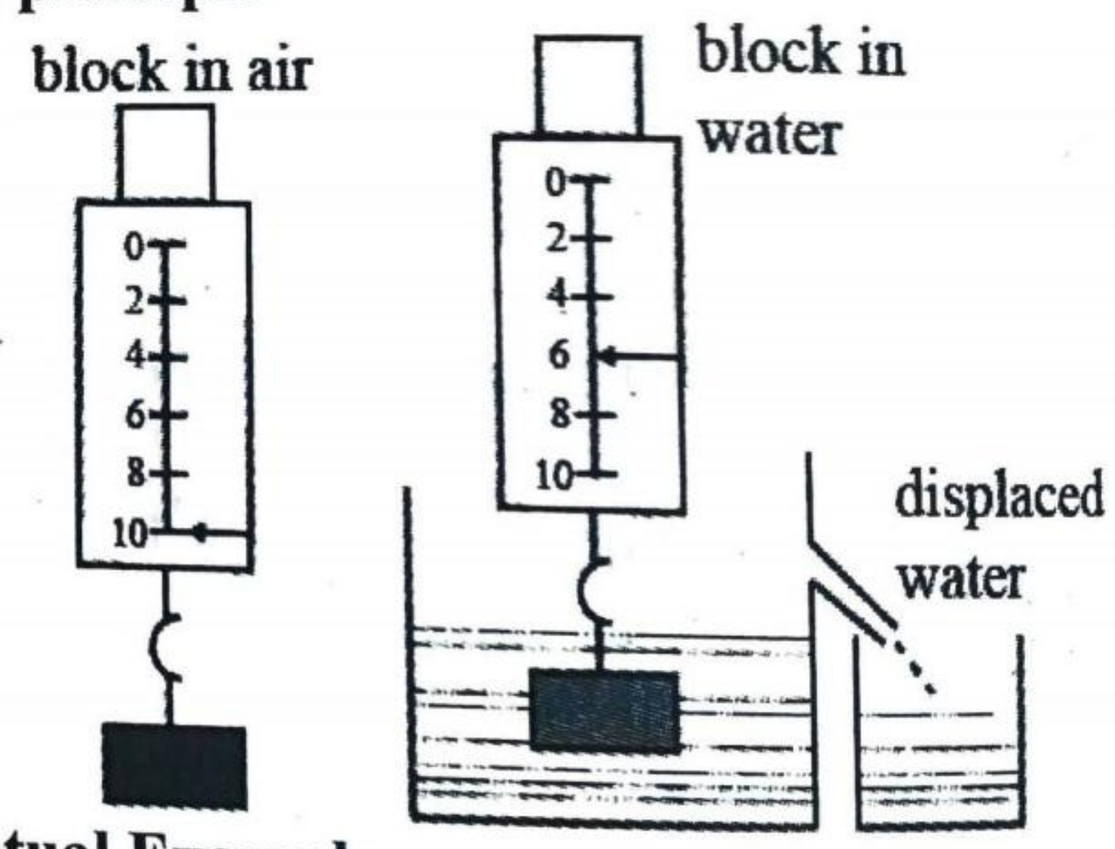
☛ The greater the volume of liquid displaced, the greater the buoyant force.

The apparent weight in light of Archimedes's principle is

- apparent weight = weight – buoyant force
- buoyant force = weight of displaced fluid
- apparent weight = weight – weight of displaced fluid

Investigating Archimedes's principle

Assume that a block weighs 10N in air and 6N when completely immersed in water. The drop in weight of the block is then 10N – 6N = 4N. The buoyant force on the block is therefore 4N. Which is also equals to the weight of the displaced water.



Conceptual Example

27. Lead is more dense than aluminum. What can you say about the buoyant force if equal volume of each material is immersed in water.

Explanation: The buoyant force is the same for both since they have equal size.

Illustrative Examples

28. A solid block is attached to a spring scale. When the block is suspended in air, the scale reads 20N, when it is completely immersed in water, the scale reads 17.7N. What is
    a) the volume and

b) the density of the block

Solution:

a) buoyant force = weight in air – weight in water

$F_b = 20 - 17.7 = 2.3N$

But buoyant force = weight of displaced water

$F_b = mg = \rho_w Vg$

$$\Rightarrow V = \frac{F_b}{\rho_w g} = \frac{2.3}{(1000)(10)} = 2.3\times10^{-4}m^3$$

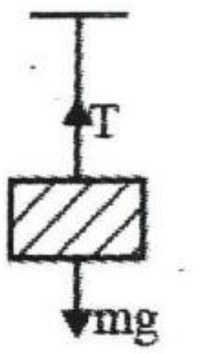
b) $W = mg \Rightarrow m = \frac{W}{g} = \frac{20}{10} = 2kg$ then

$$\rho = \frac{m}{v} = \frac{2}{2.3\times10^{-4}} = 8.7\times10^{3}kg/m^3$$

29. A brass block with a mass of 500g and density of 8×10^3 kg/m^3 is suspended from a string.

a) What is the tension in the string if the block is in air?

b) What is the tension if it is completely immersed in water?



Solution:

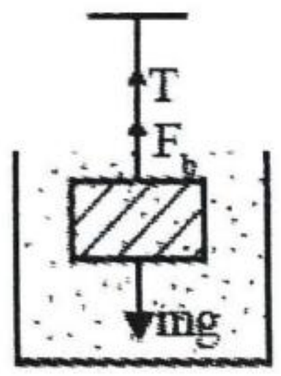
a) $T = mg = (0.5)(10)$

$T = 5N$

b) When the block is immersed in water,

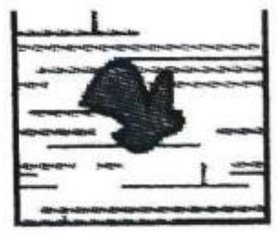
$T + F_b - mg = 0$

$T = mg - F_b = mg - \rho_w Vg$



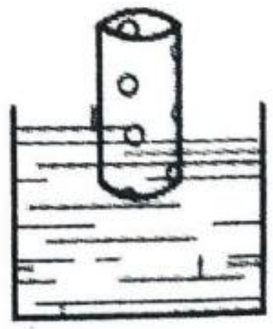
$$T = (0.5)(10) - (1000)\left(\frac{0.5}{8\times10^3}\right)(10)$$

$T = 4.375N$



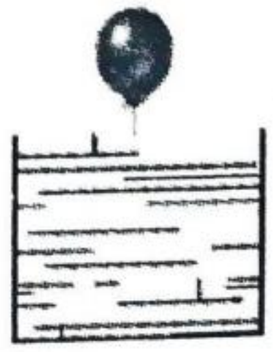
Floating and sinking

Whether or not an object floats or sinks depends on the weight of the object, the size of the buoyant force acting on the object and the relative density between the object and the fluid. For instance,



The stone sinks because its weight is greater than the buoyant force.

The cork floats because its weight is equal to the buoyant force, and



The balloon rises because its weight is less than the buoyant force.

Law of floatation

In order to float an object must displace a volume of fluid equal to its own weight.

> **Law of floatation:** states that;
> "An object floats when it displaces an amount of fluid equal to its weight."

In other words, if the buoyant force is equal to the weight of the object then the object will float. Moreover, we can think of the following

- If the density of the object is greater than the density of the fluid it will sink (relative density > 1).
- If the density of the object is less than the density of the fluid it will float (relative density < 1).

If the object floats, then;

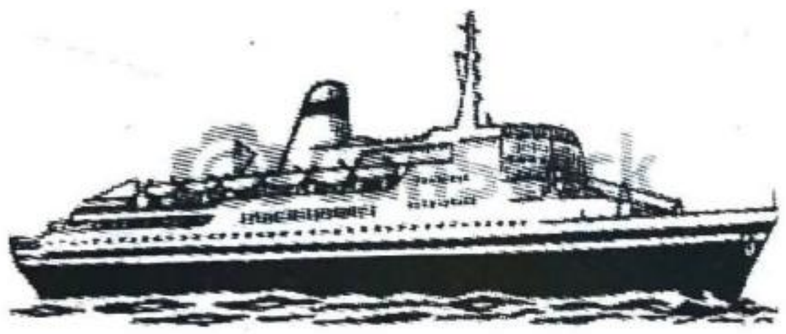
Buoyant force = weight of displaced liquid = weight of object

$\rho_{object} V_{object} g = \rho_{fluid} V_{fluid} g$

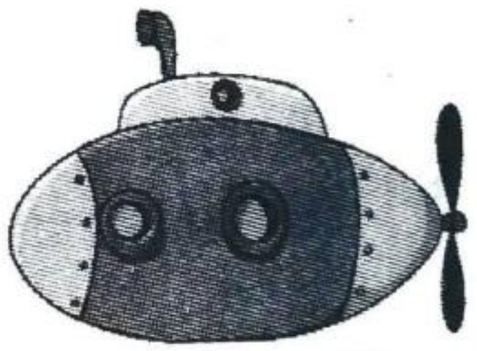
$\boxed{\rho_{object} V_{object} = \rho_{fluid} V_{fluid}}$ → for a floating body

Some practical examples in fluid statics.

(i) **Ships**: A floating ship displaces a weight of water equal to its own weight including that of the cargo.



(ii) **Submarines:** A submarine sinks by taking water into its buoyancy tanks. To come out, compressed air is used to blow the water out of the tanks.



(iii) **Balloons**: A balloon filled with hydrogen weighs less than the weight of air it displaces. The up thrust is therefore greater than its weight and resultant upwards force on the balloon causes it to rise.

(iv) **Flight and lift**

Airplanes, helicopters, birds are supported in flight by aerodynamic forces resulting from their dynamic interaction with the air.

What lifts an Airplane?

If you hold a sheet of paper and blow over the top of it, the paper will rise. This is because of the decrease in pressure above the paper caused by the stream of moving air. This and other shape and design factors are the working principles of an airplane. Remember, fluid dynamics is about fluid in motion and you will learn about it in later grades.

Conceptual Examples

30. A large ship made of steel floats while a piece of stone sinks in an ocean. Why?

Explanation: Because, the ship displaces such a large volume of water. This volume of water has the same weight as the ship. And hence, the ship floats.

31. An ice cube is floating in a cup of water. What will happen to the water level when the ice melts? Explain

Explanation: When it melts, its weight is the same. Hence, the melted ice fills exactly the same volume that the ice cube displaced when floating. As a result, the water level is unchanged!

32. Why is it easier to float in the sea (salty water) than in fresh water? Explain

Explanation: Because, the density of sea water is larger than that of the fresh water.

33. If a small cube of steel is dropped into water, it will sink. But if you hammer it into a bowl shape, it will float. Why? Explain

Explanation: Because, the steel in cube form does not displace enough water. But when it is hammered into bowl or sheet form, it displaces a greater volume of water and so will float.

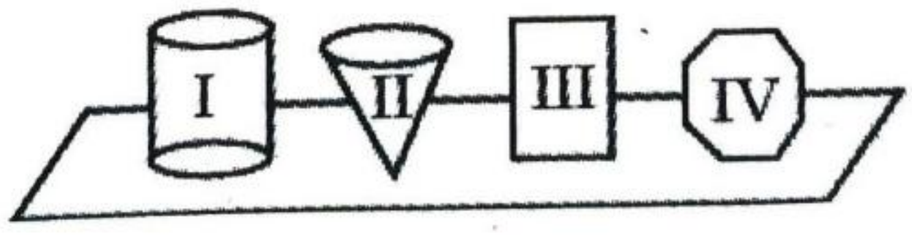
34. On a planet where the acceleration due to gravity is greater than that on the earth, is it easier to float in water?

Explanation: As far as floating takes place, there is no difference to float on earth or on the planet. Because, your weight and the weight of water increase by the same amount as g increases.

General School Education Certificate Exam

35. In which of the following cases is the pressure on the supporting surface relatively high? Assume the same mass in all cases.

A. I and IV
B. II and III
C. II and IV
D. I and II



Solution: From $P = \frac{F}{A}, F = mg \Rightarrow P = \frac{mg}{A}$ that is $P \sim \frac{1}{A}$, as contact area decreases pressure increases. **Answer: C**

36. A woman weighing 600N lies on a bed of 5000 nails. What is the pressure on her body if each nail has an area of $1mm^2$?

A. $8.22\ N/mm^2$
B. $0.12\ N/mm^2$
C. $3\ N/mm^2$
D. $1.2\ N/mm^2$

Solution: $P = \frac{F}{A}$, $F = W = 600$, and total contact area is

$A = (5000)(1mm^2) = 5000\ mm^2$

$\Rightarrow P = \frac{F}{A} = \frac{600N}{5000mm} = 0.12N/mm$ **Answer: B**

37. The atmospheric pressure of the earth is about 100,000 Pascal near the ground. By how much force is the atmosphere pressing against each square centimeter of your body?

A. zero B. 10N C. 1000N D. 100,000N

Solution: $P = \frac{F}{A} \Rightarrow F = (P)(A) = (10^5)(10^{-4}) = 10N$ **Answer: B**

Note that: $1\ cm^2 = 10^{-4}\ m^2$

38. A 3600 kg circus elephant balances on one foot. If the foot is circular and has an area of $3 \times 10^{-2}\ N/m^2$, what is the pressure it exert on the ground ($g = 10\ m/s^2$)

A. $3 \times 10^6\ N/m^2$
B. $6 \times 10^5\ N/m^2$
C. $9 \times 10^5\ N/m^2$
D. $1.2 \times 10^6\ N/m^2$

Solution: $P = \frac{F}{A} = \frac{mg}{A} = \frac{(3600)(10)}{3 \times 10^{-2}} = 1.2 \times 10^6 N/m^2$

Answer: D

39. Atmospheric pressure increases as

A. latitude decrease
B. air density decrease
C. air temperature increase
D. altitude decrease

Solution: As altitude increases, the atmospheric pressure decreases. **Answer: D**

40. It is easier to swim in salty sea water than in pure pool water. What is the reason for this?

A. The density of sea water is larger than that of pure water.

B. The volume of sea water is larger than that of a pure pool water.
C. The depth of sea water is smaller than the depth of pool water.
D. The density of pure water is larger than the density of sea water.

Explanation:
It is water density difference. **Answer: A**

41. If the molecules of a substance move freely and occupy all the space available to them, then which of the following is true about the substance?
A. It must be in liquid state
B. It must be I the gaseous state
C. It must have a definite shape
D. It must have a definite volume

Answer: B

42. Suppose that body A is three times as dense as body B. For equal volumes of A and B, how do the masses compare?
A. $M_A = M_B$ C. $M_A = 3M_B$
B. $M_A = \frac{1}{3}M_B$ D. $M_A = \frac{1}{9}M_B$

Solution: $\rho_A = \frac{M_A}{V_A} \ldots (1)$ but, $\rho_A = 3\rho_B$ and $V_A = V_B = V$

$$\rho_B = \frac{M_B}{V_B} \ldots (2) \Rightarrow \frac{\rho_A}{\rho_B} = \frac{M_A}{M_B}, \rho_A = 3\rho_B$$

$$\frac{3\rho_B}{\rho_B} = \frac{M_A}{M_B} \Rightarrow MA = 3M_B$$ **Answer: C**

43. A cylindrical piston of mass m exerts a pressure P on the gas in a container. What is the radius of the cross section of the piston?
A. $\left(\frac{m}{\pi P}\right)^2$ B. $\left(\frac{mg}{\pi P}\right)^2$ C. $\frac{mg}{\pi P}$ D. $\left(\frac{mg}{\pi P}\right)^{1/2}$

Solution: $P = \frac{F}{A}$ but $F = mg$ and $A = \pi r^2$

$$\Rightarrow P = \frac{mg}{\pi r^2} \Rightarrow r^2 = \frac{mg}{\pi P} \Rightarrow r = \sqrt{\frac{mg}{\pi P}} = \left(\frac{mg}{\pi P}\right)^{1/2}$$ **Answer: D**

44. Which of the following describes Pascal's principle?
A. In a given liquid, the greater the depth the greater is the pressure.
B. A body immersed in a liquid is pushed up by a force equal to the weight of water it displaces.
C. Pressure at the same depth in a given liquid is the same everywhere.
D. Pressure change in an enclosed liquid is transmitted to every part equally.

Answer: D

45. The mass of a rock is 8.2 g and its volume is 2.35 cm^3. What si the most accurate value of the density of the rock?
A. 3.49 g/cm^3 C. 3.489 g/cm^3
B. 3.4894 g/cm^3 D. 3.5 g/cm^3

Solution: $\rho = \frac{m}{v} = \frac{8.2}{2.35} = 3.489 g/cm^3 = 3.5 g/cm^3$

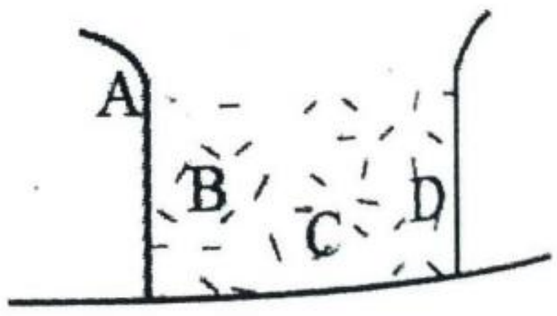
In this case, the significant figure is two. **Answer: D**

46. A driver can determine the depth at which she is swimming by measuring pressure. If the measuring device reads 248.4 kPa at certain depth in the sea water (ρ = 1080 kg/m^3) what is the depth of the water at which she is swimming?
A. 43.5m B. 435m C. 230m D. 23m

Solution: $P = \rho gh \Rightarrow h = \frac{\rho}{\rho g} = \frac{248.4 \times 10^3}{(1080)(10)} = 23m$

Answer: D

47. The figure below shows a water filled tank in the atmosphere. At which of the labeled points must the pressure always be equal.



A. B and D C. A and B
B. A, B, C and D D. B and C

Solution: Points at the same depth have the same pressure. $P = \rho gh$

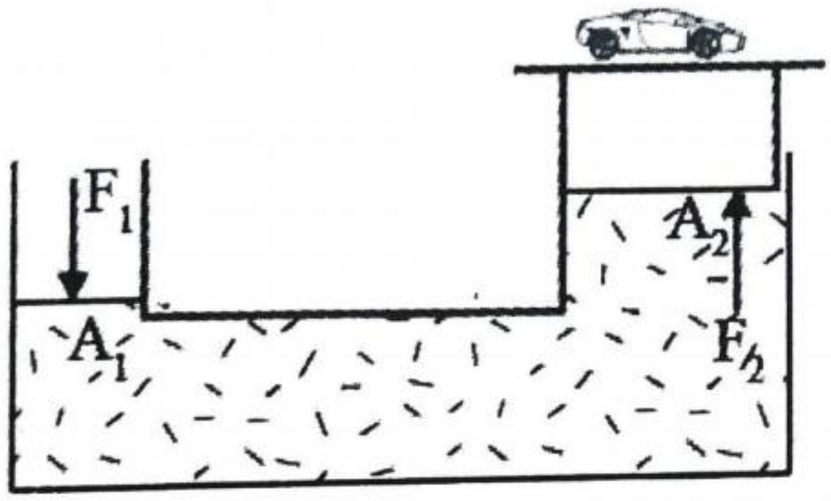
Answer: A

48. Hydraulic car lifts make use of the principle of a hydraulic press in which the smaller circular piston has a diameter 8 cm. What force

must be applied to the smaller piston to obtain a force of 1600N at the larger piston?

A. 6.25N B. 25N C. 400N D. 100N

Solution: From Pascal's principle,



$$\frac{F_1}{A_1} = \frac{F_2}{A_2},$$

$$D_1 = 2\text{ cm},\ r_1 = \frac{D_1}{2} = 1\text{cm}$$

$$D_2 = 8\text{ cm},\ r_2 = \frac{D_2}{2} = 4\text{cm}$$

$$F_2 = 1600\text{N}$$

$$\Rightarrow F_1 = \left(\frac{A_1}{A_2}\right)(F_2) = \left(\frac{\pi r_1^2}{\pi r_2^2}\right)(F_2) = \frac{\pi(1\times10^{-4})}{\pi(16\times10^{-4})}\times1600 = 100\text{N}$$

Answer: D

49. Assuming that atmospheric pressure at sea level is $10^5 N/m^2$. At what depth below sea level in the red sea is the total hydrostatic pressure twice the atmospheric pressure (take g = 10 m/s^2, $\rho_w = 10^3\ kg/m^3$)

A. 1000m B. 1m C. 100m D. 10m

Solution: Hydrostatic pressure below sea level is $P = P_o + \rho gh$. Where P_o is the atmospheric pressure $P = 2P_o$.

$$\Rightarrow 2P_o = P_o + \rho gh \Rightarrow h = \frac{P_o}{\rho g} = \frac{10^5}{(1000)(10)} = 10\text{m}$$ **Answer: D**

50. A certain object floats in three fluids F_1, F_2 and F_3. Of densities 0.9ρ, ρ and 1.1ρ respectively. The volume of the displaced fluid by the object is:

A. Least for F_1 C. Least for F_2

B. Least for F_3 D. Greatest for F_2

Solution: From the law of floatation, force of buoyancy = Weight of displaced fluid = Weight of object

$\Rightarrow$Weight of displaced fluid = weigh of object

$Wd_f = W = mg\ Md_fg = mg$

$V_{df}\delta f = gm$, Hence,

$$Vdf_1 = \frac{m}{\rho_1} = \frac{m}{0.9\rho} \qquad Vdf_2 = \frac{m}{\rho_2} = \frac{m}{\rho}$$

$$Vdf_3 = \frac{m}{\rho_3} = \frac{m}{1.1\rho}, \text{ thus}$$

Answer B

51. One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by ΔP, the pressure at the larger piston increases by

A. ΔP B. $\frac{\Delta P}{2}$ C. $2\Delta P$ D. $4\Delta P$

Explanation: Pascal's principle says, any change in the pressure of a confined fluid is transmitted equally to all parts of the fluid.

Answer: A

52. An air pressure of 101 kPa supports a 76 cm of mercury in a barometer column. What would be the height of the column if another liquid of density 10 g/cm^3 were used ($\rho_{mercury}$ = 13.6 g/cm^3

A. 55.88 cm B. 103.36 cm C. 79.6cm D. 76 cm

Solution: $P = \rho gh \Rightarrow h = \frac{P}{\rho g} = \frac{101000}{(10)(9.8)} = 103.36\text{cm}$ **Answer: B**

53. An object hangs from a spring balance. The balance indicates 30N in air and 20N when totally submerged in water with density 10^3 kg/m^3. What is the volume of the object? (g = 10 m/s^2)

A. 10^3 m^3 B. 10^{-1}m^3 C. 10^{-6}m^3 D. 10^{-3}m^3

Solution: From Archimedes's principle, buoyant force = apparent loss of weight and, buoyant force = weight of displaced fluid

$$\Rightarrow W_{loss} = W_{df} \Rightarrow 30 - 20 = \rho g v_{d\ell} \Rightarrow V_{d\ell} = \frac{W_{loss}}{\rho g}$$

$$\Rightarrow V_{d\ell} = \frac{10}{(1000)(10)} = 10^{-3}\,\text{m}^3$$

Answer: D

54. A block of lead has a greater density than a block of iron, and both are denser than water. Compared, to buoyant force on a block of iron, the buoyant force on a block of lead is:

A. The same if they have equal volumes.
B. The same if they have equal masses.
C. Greater if they have equal masses.
D. Greater if they have equal volumes.

Solution: Buoyancy = weight of displaced liquid

$B_{(Lead)}$ = Weight of displaced water = $V_{dw}\rho_w g$

$B_{(iron)}$ = Weight of displaced water = $V_{dw}\rho_{wg}$

Thus, $B_{(lead)} = B_{(iron)}$ **Answer: A**

55. Glucose solution of density 1025 kg/m^3 is given to a patient. If the average gauge pressure in the artery is 12300 Pa, what must be the minimum height of the bag in order to infuse glucose into the artery (g = 10 m/s^2).

A. 1.6 m B. 11.05m C. 8.6m D. 1.2m

Solution: Hydrostatic pressure = gauge pressures

$$\rho_e gh = P \Rightarrow h = \frac{P}{\rho g} = \frac{12300}{1025} = 1.2m$$ **Answer: D**

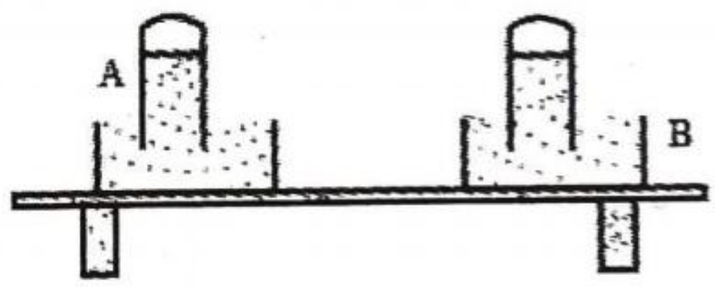
56. The highest of the liquid in the inverted tube of barometer A is less than the height of the liquid in barometer B. If the liquid are at the same temperature, what does this height difference indicate?

A. The diameter of the inverted tube in barometer A is greater than the one in barometer B.

B. The density of the liquid in barometer A is less than the one in barometer B.

C. The atmospheric pressure on barometer A is greater than that on barometer B.

D. The density of the liquid in barometer A is greater than the one in barometer B.



Solution: In both cases, the hydrostatic pressure equals to the atmospheric pressure $P_A = P_{atm}$ and $P_B = P_{atm}$

$P_A gh_A = P_{atm}$(*) $\rho_B gh_B = P_{atm}$ (**)

Taking ratio, $\rho_A h_A = \rho_B h_B$, but $h_A < h_B \Rightarrow \rho_A > \rho_B$ **Answer: D**

57. Liquids and gases are collectively called fluids because

A. Both can flow

B. Both can be compressed

C. Both have no fixed volume

D. Both fill the volume of their container **Answer: A**

Unit Six: Conceptual Questions

1. Why are dams made thicker at the bottom than at the top?
2. Suppose you drink a liquid through a straw. Explain why the liquid moves upward, against gravity into your mouth.

3. Why it is more practice to use Mercury as a barometer liquid than water? Explain
4. A person floats in a boat in a small swimming pool. Inside the boat with the person are some bricks. If the person throws the bricks into the pool, what will happen to the water level in the pool? Explain

Unit Six: Problems

Pressure

1. A wooden block of mass 2 kg is 20 cm thick, 10 cm wide and 30 cm tall. Calculate
   a) The minimum
   b) The maximum pressure the block exert on the surface.
2. Calculate the pressure, if a force of 250N is acting on an area of $2m^2$.
3. An elephant has a mass of 3200 kg. Each of its feet covers an area equal to 0.08 m^2. Calculate the pressure from each foot.

Fluid pressure

4. a) Show that at 1 atm, the height of a column of mercury equals 760 mm..
   b) Calculate the pressure in Pa if the reading from a barometer is 820 mmHg
5. Calculate the pressure caused by sea water when diving to a depth of 100m. What is the total pressure acting on the driver?
6. The pressure of water in a pipe in the ground floor of a building is 4×10^5 Pa. What is the height between the ground floor and the third floor?

Pascal's principle

7. Two pistons are connected together to make a hydraulic lift. The smaller piston has an area of 0.05 m^2 and the larger piston has an area of $2m^2$. Calculate
   a) The pressure in the fluid and the force at the larger piston if the force on the smaller piston is 50N.
   b) The pressure in the fluid and the force from the smaller piston required to lift a car of mass 1200 kg.

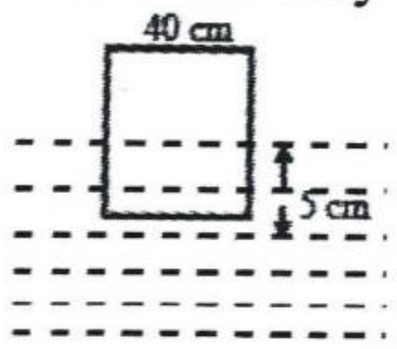
Manometer

8. Calculate the pressure of a gas supply that causes a column of water in a manometer 15 cm high.

9. If a column of oil 0.85m high is balanced by a column of mercury 0.6m high, what is the density of the oil? The density of mercury is 13600 kg/m^3?

Archimedes's principle and floatation

10. The weight of an object is measured in air to be 7N. The object is then immersed in water and its apparent weight is measured to be 4N. Determine the buoyant force. Will the object float?
11. A large ship floating in the sea has a volume of 375000 m^3 of water. Determine the density and mass of the ship.
12. A block of wood weighs 4N in air and 3N when immersed in a liquid. What is the buoyant force?
13. A 900g cubical block of iron 5 cm on each side is floating on mercury in a vessel. What is the height of the block above mercury level?
14. A cubical block of wood, 40 cm along each side floats in water with its lower surface 5 cm below the free surface of the water. Calculate the mass of the block.



Unit Six: Explanations to Questions

1. Because, fluid pressure increases with depth, thus the hydrostatic pressure is largest at the base of the dam. And to prevent the dam from sliding and overturning, it is designed thicker at the base.
2. To draw a liquid up a straw, you expand your lung, which in turn reduce the pressure in your mouth to less than the atmospheric pressure. The resulting difference in pressure produces a net upward force on the liquid in the straw.
3. Because, Mercury is much denser than water and it rises only up to 76 cm while water rises to 10m due to 101 kPa along a capillary tube.
4. When a brick is in the boat, it displaces a volume of water equal to its weight. When the brick is at the bottom of the pool it displaces a volume of water equal to its own volume. Thus, the brick displaces a greater volume of water when it is in the boat. As a result, the water level decreases when the brick is thrown.

Unit Six: Solutions to problems

5. a) $$P = \frac{F}{A} = \frac{mg}{A} = \frac{(2)(10)}{(0.3)(0.2)} = 333\text{Pa}$$

b) $P = \frac{F}{A} = \frac{mg}{A} = \frac{(2)(10)}{(0.1)(0.2)} = 1000\text{Pa}$

6. $P = \frac{F}{A} = \frac{mg}{A} = \frac{(3200)(10)}{(0.08)} = 4\times10^5\,\text{Pa}$

7. a) $P = \rho gh \Rightarrow h = \frac{P}{\rho g} = \frac{101000}{(13600)(10)} = 0.760\text{m} = 760\text{mm}$

b) $P = \rho gh = (13600)(10)(0.82) = 111520\ \text{Pa}$

8. $P = \rho gh = (1020)(10)(100) = 1020000\ \text{Pa}$
$P_T = P + P_{atm} = 1020000 + 101000 = 1121000\ \text{Pa}$

9. $\Delta P = \rho gh \Rightarrow h = \frac{\Delta P}{\rho g} = \frac{(4-2)\times10^5}{(1000)(10)} = 20\text{m}$

10. a) $P = \frac{F_1}{A_1} = \frac{50}{0.05} = 1000\text{Pa}$ and $F_2 = (P)(A_2) = (1000)(2) = 2000\text{N}$

b) $F_2 = mg = (1200)(10) = 12000\text{N}$, but from pascal,

$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_1 = \frac{A_1}{A_2}(F_2) = \frac{0.05}{2}(1200) = 300\text{N}$ and

$P = \frac{F_1}{A_1} = \frac{F_2}{A_2} = \frac{12000}{2} = 6000\text{Pa}$

11. $P_{gas} = P_{atm} + \rho gh = 101000 + (1000)(10)(0.15) = 102500\text{Pa}$

12. $\rho_o h_o = \rho_{Hg} h_{Hg} \Rightarrow \rho_o = \frac{h_{Hg}}{h_o}(\rho_{Hg}) = \frac{0.6}{0.85}(13600) = 9600\text{kg/m}^3$

13. $F_b = W_{air} - W_{water} = 7 - 4 = 3\text{N}$. No it will not float.

14. For a floating object, $\rho_{object}\, V_{object} = \rho_{fluid} V_{fluid}$

$\Rightarrow \rho_s V_s = \rho_w V_w \Rightarrow \rho_s = \frac{V_w}{V_s}(\rho_w) = \frac{50000}{375000}(1000) \simeq 134\text{kg/m}^3$

$m_s = V_s \rho_s = (37500)(134) \simeq 5\times10^7\ \text{kg}$

15. $F_b = W_{air} - W_{water} = 4 - 3 = 1\text{N}$

16. Let the height above mercury level be h, then the volume of mercury displaced is $(5\times5\times(5-h))\text{cm}^3$ and by floatation, $W_{block} = W_{Hg}$ displaced $900 = (5\times5\times(5-h)(13.6)) \Rightarrow h = 2.4\ \text{cm}$

17. Volume of block immersed, $v = 40\times40\times5 = 0.008\ \text{cm}^3$ by floatation, $m = 8\ \text{kg}$.

Unit Seven

Temperature and Heat

We feel heat when the sun shines and when we sit near a fire. We cook food using a hot stove, heat helps us to prepare hot drinks and to boil water, and it causes many things to melt. Moreover, we often say the room temperature is 20°C and the temperature of a normal human being as 37°C. Thus, heat and temperature are the two concepts that we experience in our everyday lives.

7.1 Heat (Q)

When we heat a body, we are transferring energy to the body.

☞ And this means

→ The particles of the body gain kinetic energy and/ or

→ The bonds between the particles in the body are broken and the potential energy of the particles increases. When this happens, the body changes state.

In symbols,

$$Q = \sum k_E + \sum P_E$$

Where, the Greek letter sigma (Σ) means "sum of".

Heat: Is a form of energy that flows from a hotter body to a colder body.

Heat is a scalar quantity and its SI unit is Joule (J).

Heat can flow from one region to the other in three different ways. That is by *conduction, convection* and *radiation*.

a) Conduction

If you hold one end of a metal rod and put the other end in fire, you will feel warm after a time. The heat you feel is transported along the rod by conduction.

Conduction: Is the flow of heat through a material without any flow of the particles of the material.

☞ In conduction heat transfer takes place by the collision of neighboring molecules.

b) Convection

If you lit fire wood in a room, then gradually the room gets warmer. This is due to the transport of heat due to convection.

> **Convection**: Is the flow of heat due to the actual movement of particles of liquids and gases.

c) Radiation

☞ In radiation, heat flows through empty space, that is vacuum. The heat energy of the sun reaches the ground due to radiation.

> **Radiation**: Is the flow of heat by means of electromagnetic wave.

In sum, the sun heats the earth by radiation and the ground level air gets warmer and begins to move by convection and finally, if you walk across the ground in bare foot, you will feel the heat under your feet due to conduction.

Temperature (T)

Temperature is a measure of hotness or coldness of a body. The higher the temperature, the hotter the body.

> **Temperature:** Is a measure of the average kinetic energy of the particles within a body.

☞ Temperature is a scalar quantity and its SI unit is kelvin (k).

- Temperature indicates the direction of heat flow. Heat flows from a higher temperature region to a lower temperature region.
- Particles at higher temperature move faster. That is, particles at 50°C move faster than those at 5°C.

Conceptual Examples

1. Is there any difference between heat and temperature? Explain

   Explanation: Yes! There is, heat is an energy that flows from a hotter object to a colder object. But, temperature is the measure of the average kinetic energy of particles of a given body.

2. Is it valid to say a hot object contains more heat than a cold object? Explain

   Explanation: No. the only thing we can say is that the two bodies are at different temperature.

Temperature scales

The temperature scales that are often used in measuring temperature are the Celsius scale (°C), the Fahrenheit scale (°F) and the kelvin scale (K).

The Celsius scale (°C) named in honor of the Swedish astronomer Anders Celsius (1701 – 1744).

In Celsius scale, the freezing (ice) point of water is 0°C and the boiling point of water is 100°C.

The Fahrenheit scale (°F) named after Gabriel Fahrenheit (1686 – 1736). In Fahrenheit scale, water freezes at 32°F and boils at 212°F.

The Kelvin scale (K) named for the Scottish physicist William Thomson, Lord Kelvin (1824 – 1907). In Kelvin scale, the freezing point of water is 273.15k and the boiling point of water is 373.15K.

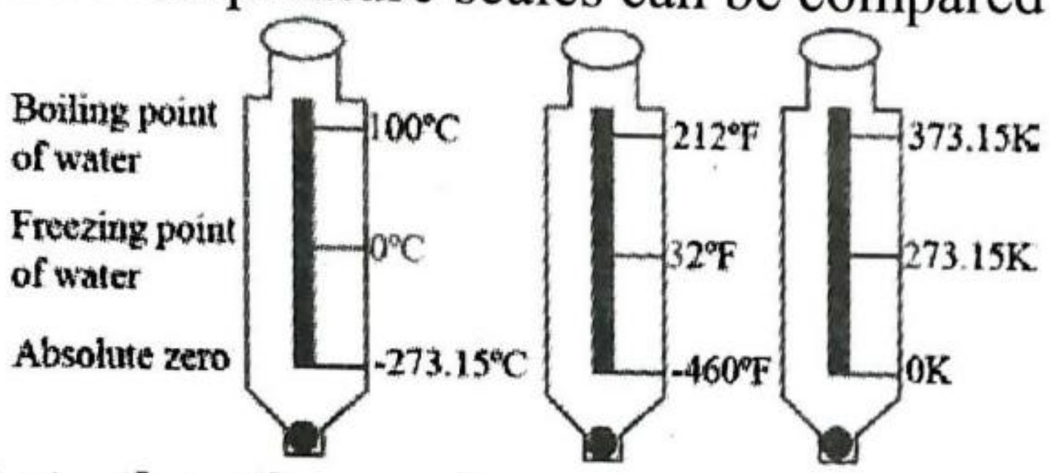
Absolute zero: Is the lowest temperature at which particles stop moving and it is impossible to cool an object to a temperature lower than the absolute zero, which is 0K = –273.15°C.

Conceptual Example

3. Do you think a body at absolute zero possesses thermal energy? Explain

Explanation: A body has no thermal energy at the absolute zero. This is so because, at 0K the particles in the body ceases motion.

In sum, the three temperature scales can be compared as follows.



Changing one scale to the other scale

To convert temperature scale x to temperature scale y, use the following general relation.

$$\frac{T_x - \text{ice point of x}}{\text{boiling point of x - ice point of x}} = \frac{T_y - \text{ice point of y}}{\text{boiling point of y - ice point of y}}$$

So we can have the following;

a) To convert *Celsius scale* to *Fahrenheit scale* or vice versa ($°C \rightleftarrows °F$)

$$\frac{T_C - 0}{100 - 0} = \frac{T_F - 32}{212 - 32} \Rightarrow \boxed{\frac{T_C}{100} = \frac{T_F - 32}{180}}$$

b) To convert *Celsius scale* to *Kelvin scale* or vice ver

(°C $\rightleftarrows$ K)

$$\frac{T_C - 0}{100 - 0} = \frac{T_K - 273}{373 - 273} \Rightarrow \boxed{\frac{T_C}{100} = \frac{T_K - 273}{100}}$$

c) To convert *Fahrenheit scale* to *Kelvin scale* or vice versa

(°F $\rightleftarrows$ K)

$$\frac{T_F - 32}{212 - 32} = \frac{T_K - 273}{373 - 273} \Rightarrow \boxed{\frac{T_F - 32}{180} = \frac{T_K - 273}{100}}$$

Conceptual Examples

4. A mercury in a thermometer may not be used at temperature below –40°C. Why?

Explanation: This is because, Mercury freezes at this temperature.

5. Which way will heat transfer fastest from one region to the other? Explain

Explanation: Radiation. This is because, when heat flows by radiation, the thermal energy is transported by electromagnetic waves at a speed of light through vacuum.

6. Can a vacuum be heated so that its temperature be changed? Explain

Explanation: No. Since there is not particle in vacuum it cannot be heated and its temperature cannot be changed since temperature is the measure of the average kinetic energy of particles of a body.

7. Which one is greatest, 0°C, 0°F or 0K?

Explanation: 0°C = 273K, and 0°F = 268K

Thus, 0°C is the largest temperature measurement.

Illustrative Examples

8. At what temperature will the Celsius scale and Fahrenheit scale have the same reading?

Solution: We know, $\frac{T_C}{100} = \frac{T_F - 32}{180}$, let $T_C = T_F = T$

$$\Rightarrow \frac{T}{100} = \frac{T - 32}{180} \Rightarrow 18T = 10T - 320 \Rightarrow 8T = -320$$

T = –40°. Thus at –40°, both °C and °F scales read the same.

9. Body temperature is normal at 98.6°F. What is the corresponding temperature on the Celsius scale?

Solution: From, $\frac{T_C}{100} = \frac{T_F - 32}{180}$

$$\frac{T_C}{100} = \frac{98.6 - 32}{180} \Rightarrow T_C = 37°C$$

10. The boiling point of oxygen is –297.35°F. Express this temperature
    a) In Kelvin b) In Celsius

Solution: a) From $\frac{T_F - 32}{180} = \frac{T_K - 273}{100}$ and $T_F = -297.35$

$$= \frac{-297.35 - 32}{180} = \frac{T_K - 273}{100} \Rightarrow T_K = 90K$$

a) Using $\frac{T_C}{100} = \frac{T_F - 32}{180}$

$$\frac{T_C}{100} = \frac{-297.35 - 32}{180} \Rightarrow T_C = -183°C$$

11. Gold melts at 1336K. What is the corresponding temperature in degree Celsius?

Solution: $\frac{T_C}{100} = \frac{T_K - 273}{100} \Rightarrow \frac{T_C}{100} = \frac{1336 - 273}{100}$

$T_C = 1063°C$

12. At what temperature is the Fahrenheit scale reading equal to twice of Celsius?

Solution: From, $\frac{T_C}{100} = \frac{T_F - 32}{180}$, but $T_F = 2T_C$

$$\Rightarrow \frac{T_C}{100} = \frac{2T_C - 32}{180} \Rightarrow T_C = 160°C$$

13. A sample of gas cools from –120°C to –180°C. Express the change of temperature.
    a) In kelvin
    b) In Fahrenheit scale

Solution:

a) From $\frac{T_C}{100} = \frac{T_K - 273}{100}$

When $T_C = -120°C$, then T_K is;

$$\frac{-120}{100} = \frac{T_{K_1} - 273}{100} \Rightarrow T_{K_1} = 153K$$

When $T_C = -180°C$, then T_K is;

$$\frac{-180}{100} = \frac{T_{K_2} - 273}{100} \Rightarrow T_{K_2} = 93K$$

Then, $\Delta T_K = T_{K_2} - T_{K_1} = 93 - 153 = -60K$

b) From $\frac{T_C}{100} = \frac{T_F - 32}{180}$

When $T_C = -120°C$, then T_F is;

$$\frac{-120}{100} = \frac{T_{F_1} - 32}{180} \Rightarrow T_{F_1} = -184°F$$

When $T_C = -180°C$, then T_F is;

$$\frac{-180}{100} = \frac{T_{F_2} - 32}{180} \Rightarrow T_{F_2} = -292°F$$

Then, $\Delta T_F = T_{F_2} - T_{F_1} = -292 - (-184) = -108°F$

Thermal equilibrium

Place your hand near an oven and you can feel the heat energy flowing into your hand. It feels hot. Place your hand inside a fridge and the heat energy flows from you into the fridge, it feels cold.

> When there is a movement of heat energy from a hotter object to a colder object, we say that the two objects are in *thermal contact*.

Imagine that we have two bodies A at 90°C and B at 50°C. When A and B are in thermal contact, there will be a net flow of heat energy from A to B. In the process, the temperature of A will decrease to 70°C and that of B will increase to 70°C. This process of heat exchange between the bodies will go on until *thermal equilibrium* is reached.

> **Thermal equilibrium**: Is a situation where there is no net movement of heat energy between bodies.

☞ If two bodies are in thermal equilibrium, they will also be at the *same temperature*.

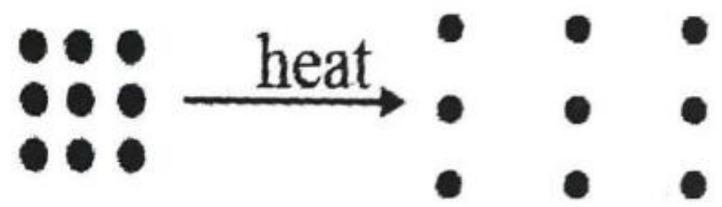
Conceptual Example

14. Can heat flow spontaneously from a colder object to a hotter one? Explain

Explanation: No. Heat energy will not flow from a colder object to a hotter one spontaneously unless work is done. Energy must be used to reverse the usual flow of heat energy. This principle is used in refrigeration, freezers and air conditioning units.

7.3 Expansion of solids, liquids and gases

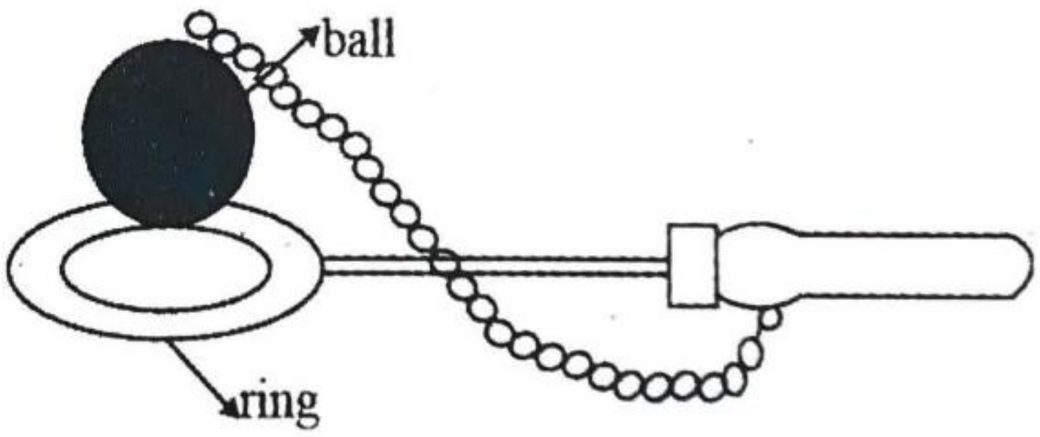
When a body is heated, the particles vibrates more in a solid and move faster in a liquid or gas. As the particles gain more energy, we can see that they move further apart from each other, which means the substance will expand, that is increase in size.

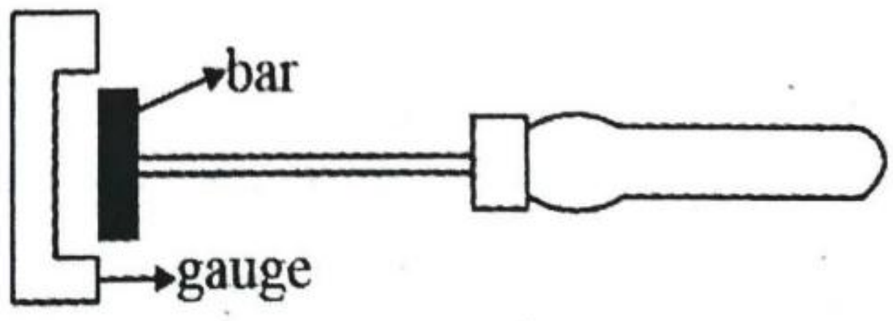


Thermal expansion: is the increase in size of a substance as a result of heating.

The expansion of solids

When a solid is heated, its particles move further apart and hence the solid expands, that is it increase in size. The ball and ring experiment is a good demonstration of the expansion of a solid. The cold metal ball easily passes through the ring. After heating, the ball expands and it is no longer able to pass through the ring. In the same manner, the metal bar will just fit into the gap in the gauge when both the bar and the gauge are cold. In general, when a solid is heated, its expansion could be *one dimensional* **(linear)**, *two dimensional* **(Area)** or *three dimensional* **(volume) expansion**. Now, let's see each of them one by one.



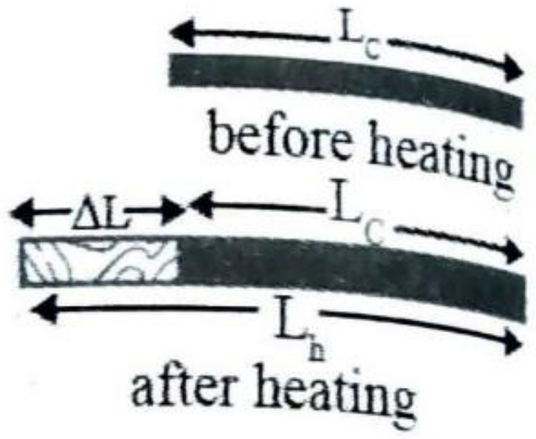


A. Linear expansion of solids

When a metal rod is heated, it expands and increases in length. This expansion is referred to as *linear expansion*.

Linear expansion: Is the increase in length of a substance due to heating.

If a metal rod or original length (L_C) is heated to a temperature increase (ΔT), then the increase or change in length (ΔL) of the rod after heating is given by;



$$\Delta L = \alpha L_C \Delta T$$

Where, $L_C \rightarrow$ length of the rod when it is cold

$\Delta T = T_2 - T_1$ or $T_f - T_i$

α (alpha), is called *coefficient of linear expansion.*

And its unit is $\frac{1}{k} = k^{-1}$

Coefficient of linear expansion: Is the increase in length of a 1m rod of a given substance when its temperature increases by 1k.

The total length of the rod after heating (L_h) is given by:

$$L_h = L_C + \Delta L = L_C + \alpha L_C \Delta T$$

$$\Rightarrow \boxed{L_h = L_C (1 - \alpha \Delta T)}$$

Observe from the relation, $L_h = L_C + \Delta L$ that,

$$\boxed{\Delta L = L_h - L_C}$$

Conceptual Example

15. The coefficient of linear expansion of iron is found to be $1.1 \times 10^{-5} k^{-1}$. What does this mean?

Explanation: This means that a 1m iron rod will expand by 1.1×10^{-5}m for every 1k rise in temperature.

Illustrative Examples

16. The Eiffel Tower in Paris is made of iron. If the tower is 301m high at 22°C, how much does its height increase when the temperature rises to 45°C?

Solution: $\Delta L = \alpha L_C \Delta T = (1.1 \times 10^{-5})(301)(45 - 22) = 7.6$ cm

17. The world's longest suspension bridge is the Akashi Kaikyo Bridge in Japan. The bridge is 3910m long and is constructed of steel. How much will its length change as the temperature changes from -5°C to 30°C (as the season changes from summer to winter)

Solution: $\Delta L = \alpha L_C \Delta T = (1.1 \times 10^{-5})(3910)(30 - (-51)) = 1.5$m

18. An iron pipe is 300m long at 20°C. If the pipe is to be used as a steam pipe, calculate

a) The change in length

b) The new length of the pipe

Solution: At boiling point, T = 100°C

a) $\Delta L = \alpha L_C \Delta T = 1.1 \times 10^{-5}(300)(100 - 20) = 0.3m$

b) $L = L_C + \Delta L = 300 + 0.29$

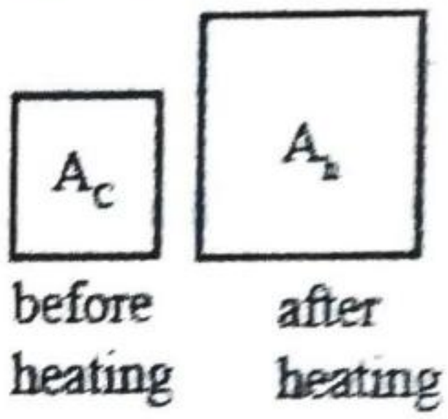
$L = 300.3m$

B) Area expansion of solids

When a metal plate is heated, it expands and increases in size. This expansion is referred to as *surface* or *area expansion.*

Area expansion: is the increase in length and width or height of a substance due to heating.

If a metal plate of original area (A_C) is heated to a temperature increase of (ΔT), then the increase or change in area (ΔA) of the plate after heating is given by:



$$\boxed{\Delta A = \beta A_C \Delta T}$$

Where, β (beta) is coefficient of area expansion and is equal to,

$$\boxed{\beta = 2\alpha}$$

The total area of the plate after heating (A_h) is givne by,

$A_h = A_C + \Delta A = A_C + \beta A_C \Delta T$

$\boxed{A_h = A_C(1 + \beta\Delta T) = A_C(1 + 2\alpha\Delta T)}$

Illustrative Examples

19. Calculate the increase in surface area of an iron plate with an area of $0.75m^2$ at 20°C when it is heated to a temperature of 100°C ($\beta = 2.2 \times 10^{-5}k^{-1}$)

Solution: $\Delta A = \beta A_C \Delta T = (2.2 \times 10^{-5})(0.75)(100 - 20)$

$= 1.3 \times 10^{-3}m^2$

20. A 6m long square of iron sheet is heated from 20°C to 80°C calculate ($\alpha = 1.1 \times 10^{-5}k^{-1}$)

a) The change in area

b) The final area of the sheet

Solution: a) $\Delta A = \beta A_C \Delta T$

$\beta = 2\alpha = 2(1.1 \times 10^{-5}) = 2.2 \times 10^{-5}$

$A_C = (6)(6) = 36$, and $\Delta T = 80 - 20 = 60°C$

$\Rightarrow \Delta A = (2.2 \times 10^{-5})(36)(60) = 0.05m^2$

b) $A_n = A_C + \Delta A$

$A_n = 36 + 0.05 = 36.05m^2$

21. The coefficient of area expansion of an unknown container is $4.6 \times 10^{-5}k^{-1}$. Determine from what material the container is made from.

Solution: $\beta = 2\alpha \Rightarrow \alpha = \frac{\beta}{2} = \frac{4.6\times10^{-5}}{2} = 2.3\times10^{-5}k^{-1}$

So, the container is made from aluminum.

22. A sheet of copper has an area of $500cm^2$ at 0°C. Find the area of this sheet at 80°C ($\alpha = 1.7 \times 10^{-5}k^{-1}$)

Solution: $A_h = A_C + \Delta A$,

But $\Delta A = \beta A_C \Delta T = 2\alpha A_C \Delta T = (2 \times 1.7 \times 10^{-5})\ (500)\ (80 - 0)$

$\Delta A = 1.3cm^2$

$\Rightarrow A_h = A_C + \Delta A = 500 + 1.3 = 501.3cm^2$

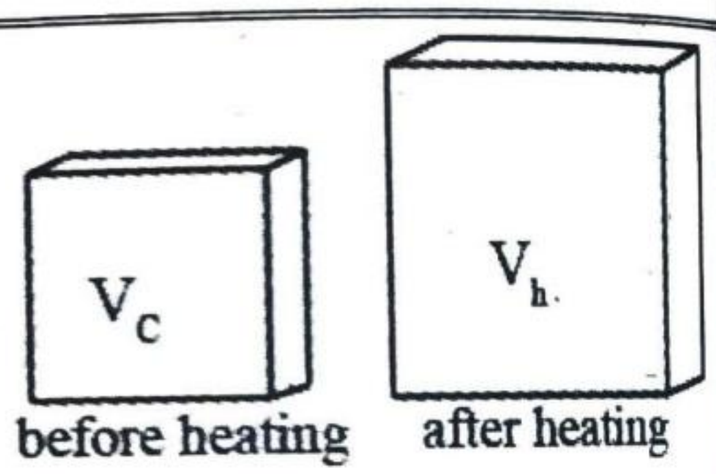
C) Volume expansion of solids

When a metal block is heated, it expands and increases in size. This three dimensional expansion of a body is referred to as *volume expansion*.

Volume expansion: is the increase in length, width and height of a substance due to heating.

If a metal block of original volume (V_C) is heated to a temperature increase of (ΔT), then the increase or change in volume (ΔV) of the block after heating is given by:

$$\boxed{\Delta V = \gamma V_C \Delta T}$$



Where, γ (gamma) is coefficient of volume expansion and is equal to,

$$\boxed{\gamma = 3\alpha}$$

The total volume of the block after heating (V_h) is givne by,

$$V_h = V_C + \Delta V = V_C + \gamma V_C \Delta T$$

$$\boxed{V_h = V_C(1 + \gamma\Delta T) = V_C(1 + 3\alpha\Delta T)}$$

Illustrative Examples

23. Calculate the increase in volume of an aluminium block with a volume of $0.008m^2$ at 25°C when it is heated to a temperature of 90°C ($\alpha = 2.3 \times 10^{-5}k^{-1}$)

Solution: $\Delta V = \gamma V_C \Delta T = 3\alpha V_C \Delta T = 3\alpha V_C(T_2 - T_1)$

$\Delta V = 3(2.3 \times 10^{-5})\ (0.008)\ (90 - 25)$

$\Delta V = 3.6 \times 10^{-5}m^3$, this is the increase in volume.

The total volume of the block after heating is;

$V_h = V_C + \Delta V = 0.008 + 0.000036$

$V_h = 0.00804m^3$

24. A 10m long iron cube is heated from 20°C to 100°C.
Determine a) the change in volume
b) the final volume of the cube

Solution: a) $\Delta V = \gamma V_C \Delta T = 3\alpha V_C \Delta T$

$\gamma = 3\alpha = 3(1.1 \times 10^{-5}) = 3.3 \times 10^{-5}k^{-1}$, $V_C = (10)(10)(10)$
$= 1000m^3$

$\Delta T = (T_2 - T_1) = 100 - 20 = 80°C$

$\Rightarrow \Delta V = (3.3 \times 10^{-5})(1000)(80) = 2.64m^3$

b) $V_h = V_C + \Delta V = 1000 + 2.64 = 1002.64m^3$

25. When a block is heated from 10°C to 50°C to change the volume of a block by $1000cm^3$ from $2m^3$ of original volume. What is the coefficient of volume expansion of the block.

Solution: a) $\Delta V = \gamma V_C \Delta T = \Rightarrow \gamma = \dfrac{\Delta V}{V_C \Delta T} = \dfrac{0.001}{2(50-10)}$

$\gamma = 1.25 \times 10^{-5}k^{-1}$

26. Determine the change in volume of a block of iron 5cm × 10cm × 6cm when the temperature changes from 15°C to 47°C. ($\alpha = 1.1 \times 10^{-5}k^{-1}$)

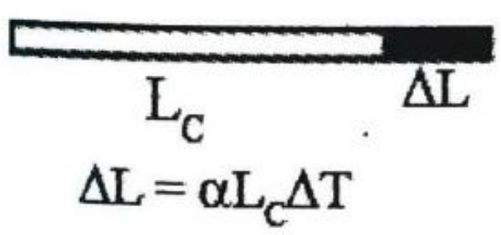
Solution: $\Delta V = \gamma V_C \Delta T$

$\gamma = 3\alpha = 3(1.1 \times 10^{-5}) = 3.3 \times 10^{-5}k^{-1}$

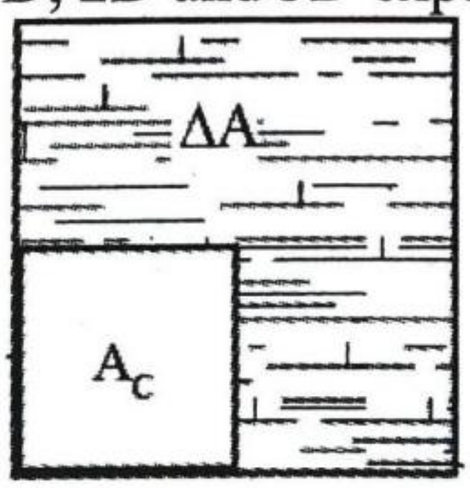
$V_C = (5)(10)(6) = 300cm^3$ and $\Delta T = 47°C - 15°C = 32°C$

$\Rightarrow \Delta V = (3.3 \times 10^{-5})(300)(32) = 0.35cm^3$

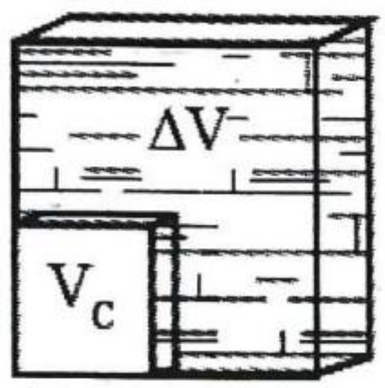
1D, 2D and 3D expansion **summary**



$\Delta L = \alpha L_C \Delta T$



$\Delta A = 2\alpha A_C \Delta T$



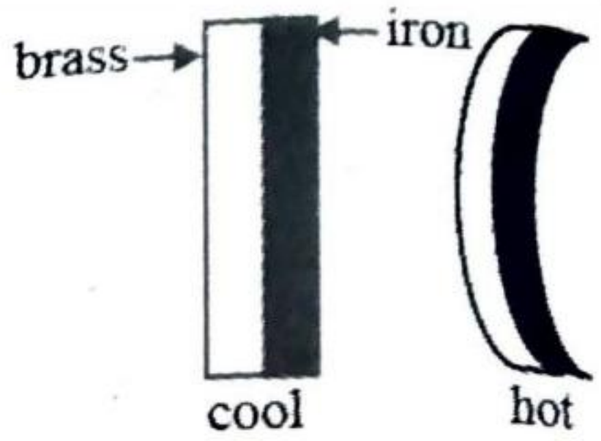
$\Delta V = 3\alpha V_C \Delta T$

The bimetallic strip

A bimetallic strip is made out of two metals.

Bimetallic Strip: Is a strip made of two different metals bonded together along their length.

For examples, if iron and brass are bonded together, since coefficient of linear expansion of iron ($1.1 \times 10^{-5}k^{-1}$) is less than that of brass ($1.9 \times 10^{-5}k^{-1}$). When the strip is heated, the brass expands more than the iron and the strip bends.



Expansion of liquids

The expansion of a liquid is somehow related with the expansion of its container. To understand this well, notice the following points.

→ In the case of liquids, we consider only volume expansion.

→ When a liquid is heated, its container is also heated and it expands.

→ Liquids have larger coefficient of volume expansion than solids. ($\gamma_{liquid} > \gamma_{solid}$), and

☞ The observed expansion of a liquid is always less than its real or actual expansion. Thus, the observed expansion of a liquid is equals to the apparent expansion of the liquid.

Real expansion: Is the actual expansion of liquid.
Apparent expansion: Is the observed expansion of a liquid which may be affected by the expansion of its container.

Due to the expansion of the vessel holding a liquid the real expansion of the liquid is computed as follow.

$$\Delta V_{real} = \Delta V_{apparent} + \Delta V_{container}$$

$$\Rightarrow \gamma_{real} V_C \Delta T = \gamma_{apparent} V_C \Delta T + \gamma_{container} V_C \Delta T$$

$$\Rightarrow \gamma_{real} = \gamma_{apparent} + \gamma_{container}$$

Illustrative Examples

27. Calculate the real and apparent expansion of a $1000cm^3$ of water in glass when it is heated from 20°C to 80°C ($\gamma_{glass} = 9.9 \times 10^{-6}k^{-1}$, $\gamma_{water} = 2.07 \times 10^{-4}k^{-1}$

Solution: $\rightarrow \Delta V_{real} = \gamma_{real} V_C \Delta T$, $V_C = \dfrac{1000}{1000000} m^3 = 0.001m^3$

$= (2.07 \times 10^{-4})\ (0.001)\ (60)$, $\Delta T = 80 - 20 = 60°C$

$\Delta V_{real} = 1.24 \times 10^{-5} m^3$

$\rightarrow \Delta V_{apparent} = \gamma_{apparent} V_C \Delta T$

But $\gamma_{apparent} = \gamma_{real} - \gamma_{container}$

$= 2.07 \times 10^{-4} - 9.9 \times 10^{-6}$

$\gamma_{apparent} = 1.97 \times 10^{-4} k^{-1}$

$\Delta V_{apparent} = (1.97 \times 10^{-4})(0.001)\ (60)$

$\Delta V_{apparent} = 1.18 \times 10^{-5} m^3$

28. Calculate the increase in the volume of $0.0025m^3$ of mercury at 5°C when it is heated to a temperature of 55°C. ($\gamma_{mercury} = 1.8 \times 10^{-4} k^{-1}$).

Solution: $\Delta V_{real} = \gamma_{real} V_C \Delta T$

$= (1.8 \times 10^{-4})\ (0.0025)\ (50)$, $\Delta T = (55 - 5) = 50°C$

$= 0.225 \times 10^{-4} m^3$

Conceptual Examples

29. The density of solids is greater than the density of liquids. Is this statement always true? Explain.

Explanation: No! ice, the solid state of water, has a lower density and a higher volume than warmer water. This is why ice floats on water. The abnormal molecular structure when it is below 4°C. The expansion of water on freezing can cause problems.

For example, when it gets cold, water in pipes can freeze, expand and then break the pipe.

7.4 Quantity of heat, specific heat capacity and heat capacity

Quantity of heat

The amount of heat energy required to raise the temperature of a substance depends on three things;

→ The **substance** being heated **(c)**: A given mass of aluminium will require more energy to raise its temperature by lk than the same mass of wood.

→ The **mass** of the substance **(m)**: The greater the mass of the substance, the more heat energy required to raise its temperature.

→ The **temperature rise** required (ΔT): For a given mass of a substance, a large temperature increase will require a large amount of heat.

In general, the quantity of heat energy (Q) required to increase the temperature of a substance of mass (m) by (ΔT) is obtained by.

$$\boxed{Q = mC\,\Delta T}$$

Where, C is specific heat capacity, which is defined as

Specific Heat Capacity (c): is the heat energy required to raise the temperature of 1kg of a given substance by 1k. $\boxed{c = \frac{Q}{m\Delta T}}$

Every substance has its own specific heat capacity and its unit is J/kg k.

☞ **Remember,**

- ✓ A change in temperature of 1°C is the same as a change in temperature of 1k. So °C or k can be used to express the change in temperature.
- ✓ The SI unit of heat is Joule (J). But, calorie (Cal) can also be used to measure heat.

One calorie: Is the quantity of heat energy required to increase the temperature of 1g of water by1°C. $\boxed{1cal = 4.2J}$

Heat capacity of a body

Specific heat capacity is the heat energy required to increase the temperature of 1kg of a substance by 1k. But the heat energy required to raise the temperature of the whole body is referred to as *heat capacity*.

Heat capacity: Is the heat energy required to raise the temperature of a body by 1k $\boxed{C = \frac{Q}{\Delta T}}$

Heat capacity has a unit of J/k.

Specific heat capacity (c) and heat capacity (C) can be related as follows;

$$\text{From } c = \frac{Q}{m\Delta T} = \frac{1}{m}\frac{Q}{\Delta T}, \text{ but } \frac{Q}{\Delta T} = C$$

$$\Rightarrow c = \frac{C}{m} \Rightarrow \boxed{C = cm}$$

That is, heat capacity = (specific heat capacity) (mass of body)

Conceptual Examples

30. When you touch a metal and a wood that are both at room temperature, the metal feels cooler. Why?

    Explanation: One possible reason is that, metal is a good conductor of heat than wood.

31. Explain why land is warmer than the sea during the day and the sea is warmer than the land at night?

Explanation: During the day, the sun warms the land than the water. This is because, the land has lower specific heat than the water. The warm land heats the air above it, which becomes less dense and cooler air flows from the water "sea breeze".
At night, the land cools off more rapidly than the water because of its lower specific heat. Here, the air above the water became less dense and cooler air flows from the land "land breeze".

Illustrative Examples

32. Determine the quantity of heat energy required to raise the temperature of a 4kg iron from 20°C to 40°C ($C_{iron} = 470J/kg\ k$)

Solution: $Q = mC\Delta T = (4)(470)(40 - 20) = 37600J$

33. Find out the specific heat capacity of a 4kg block if a 67200J of heat is required to change its temperature by 40°C.

Solution: From, $Q = mC\Delta T \Rightarrow C = \dfrac{Q}{m\Delta T}$.

$$C = \frac{67200}{(4)(40)} = 420J/kg\ k.$$

34. Suppose 63J of heat is added to a 128g piece of aluminum at 25°C. What is the final temperature of the aluminum? ($C_{Al} = 910J/kg\ k$).

Solution: From, $Q = mC\Delta T \Rightarrow \Delta T = \dfrac{Q}{mC}, m = 0.128kg$

$$\Delta T = \frac{63}{(0.128)(910)} = 0.5$$

$$\text{But, } \Delta T = T_f - T_i \Rightarrow T_f = T_i + \Delta T = 25 + 0.5$$
$$T_f = 25.5°C$$

35. If 2200J of heat is added to a 190g object its temperature increases by 12°C. What is the

a) Specific heat capacity

b) Heat capacity of the object.

Solution:

a) $Q = mC\Delta T \Rightarrow C = \dfrac{Q}{m\Delta T} = \dfrac{2200}{(0.19)(12)} = 965J/kg\ k.$

b) $C = \frac{Q}{\Delta T} = \frac{2200}{12} = 183J/k$

or $C = cm = (965)(0.19) = 183J/k$

36. Hot coffee with a specific heat of 880J/kg k is poured into a 0.5kg ceramic cup. How much heat is absorbed by the cup if its temperature increases from 20°C to 80°C.

Solution: $Q = mC\Delta T$

$Q = (0.5)(880)(80 - 20)$

$Q = 26400J = 26.4kg$

37. A 70kg woman eat a chocolate having 1000Cal. To what height should she run up stair inorder to burn the chocolate in her body.

Solution: $Q = 1000Cal$

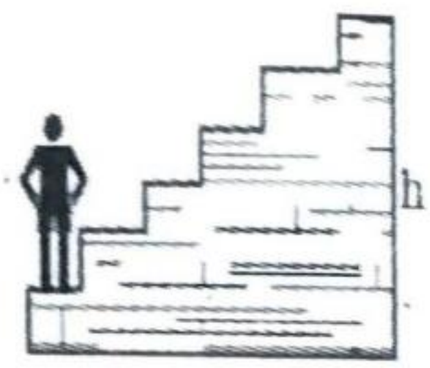
But, $1cal = 4.2J$

$\Rightarrow Q = (1000)(4.2) = 4200J$

Now, the heat energy is assumed to be totally converted into the work done against gravity.

$Q = mgh$

$\Rightarrow h = \frac{Q}{mg} = \frac{4200}{(70)(10)} = 6m$



38. How many grams of iron at 20°C must be heated to 100°C so that it releases 1800Cal of heat ($C_{iron} = 470J/kg\ k$).

Solution: $Q = mC\Delta T$, $Q = 1800Cal = (1800)(4.2) = 7560J$

$$m = \frac{Q}{C\Delta T}$$

$$m = \frac{7560}{(470)(100 - 20)}$$

$m = 0.2kg = 200g$

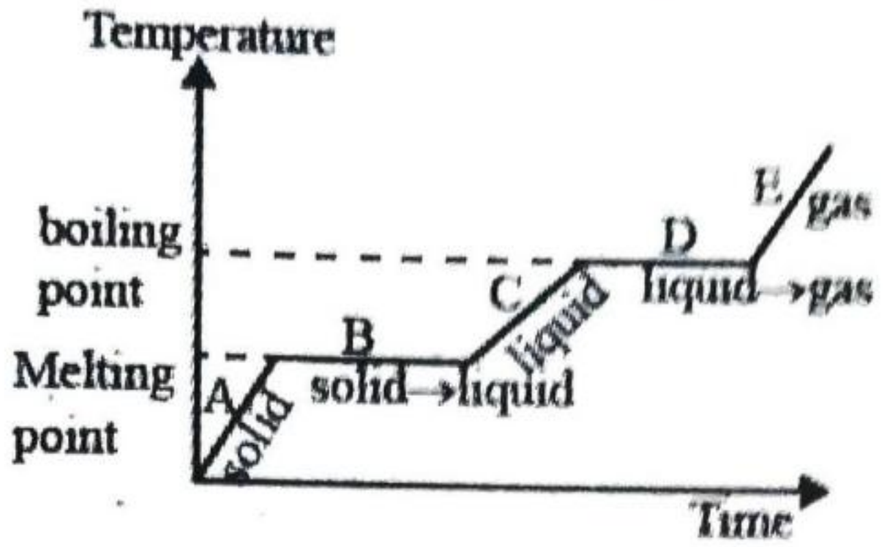
7.5 Changes of state

Phase or state, is the distinct of a substance under different condition example solid, liquid and gas.

Heating and cooling curves

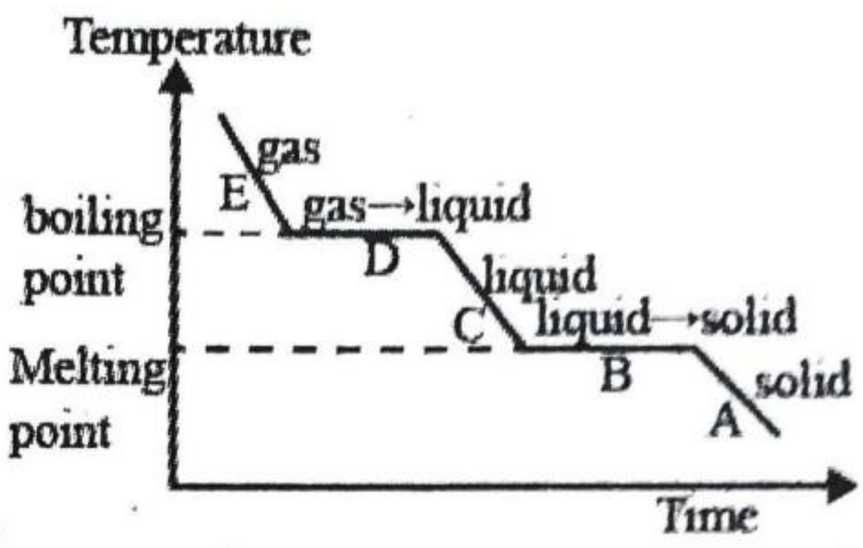
Heating curve: is a graph showing the temperature of a substance against time as heat energy is applied and it changes state.

The temperature of a solid (A) varies with time as it is heated until it melts (B) and finally forms a liquids (C). the liquid is heated until it boils (D) and until all the liquid changes into a gas (E).



Cooling curve: Is a graph showing the temperature of a substance against time as it loses heat energy and changes state.

The cooling curve for a gas cooling to eventually become a solid looks this.



Latent heat: Is the energy required to change a substance from one state to another at constant temperature.

It is given by:

$$Q = mL$$

Where: Q, heat

L, specific latent heat, its unit is J/kg

m, mass

Since there are two changes of state to consider, liquid to gas and solid to liquid, we use two versions of latent heat, the *latent heat of fision (melting)* and *latent of vaporization (boiling)*.

Latent heat of fuslon: Is the amount of heat energy required to change the state of a substance from a solid to a liquid at constant temperature.

Latent heat of Vaporisation: Is the amount of heat energy required to change the state of a substance from a liquid to a gas at constant temperature.

The specific latent heat (L) of a substance is defined as

Specific latent heat (L): Is the quantity of heat energy required to change 1kg of a substance from one state to another at constant temperature.

Specific latent heat of fusion (L_f): Is the quantity of heat energy required to change 1kg of a substance from *a solid* to a *liquid* at constant temperature.

Specific latent heat of vaporisation (L_v): Is the quantity of heat energy required to change 1kg of a substance from a *liquid* to a *gas* at constant temperature.

The heat energy required when substances are heated, taking into account changes in state is given by;

☞ Total heat energy = energy required to increase temperature
+ energy required to change state

$$\Rightarrow \boxed{Q_{Total} = mC\Delta T + mL}$$

Calorimetry: Is the experimental approach to measure heat capacities and heat changes during chemical and physical processes.

Calorimetry problems involve the sharing of heat among initially hot and cold objects. Since energy is conserved, heat lost by the hot body is equal to the heat gained by the cold body. That is,

$$\Delta Q = 0 \Rightarrow \boxed{\Sigma Q_{gaind} = \Sigma Q_{lost}}$$

Conceptual Examples

39. Assume a solid is heated continuously. Explain what will happen to the temperature in the process.

Explanation: The temperature will not increase until the entire solid has melted.

40. Which one will cause a serious burn, a water at 100°C or a steam at 100°C? Explain.

Explanation: When a water at 100°C comes in contact with the skin, it cools to a skin temperature. This means that a certain amount of heat is transferred to the skin, which can cause a burn. The steam, on the other hand, must first give off the heat required for it to condense to water at 100°C. After that, it cools to the body temperature. Thus, the steam causes sever burn.

Illustrative Examples

41. Calculate the heat energy required to melt a 2g gold at its melting point (L_f= 63700J/kg)

Solution: $Q = mL_f = (0.002)\,(63700) = 127.4J$

42. The amount of heat that raises the temperature of 1kg of a certain metal by 10°C raises the temperature of 1kg of water by 2°C. What is the specific heat of the metal ($C_{water} = 4200J/kg\ k$).

Solution: $Q_{metal} = Q_{water}$

$$mC_{metal}\Delta T_m = mC_{water}\Delta T_w$$

$$C_{metal} = \frac{\Delta T_w}{\Delta T_m} C_{water}$$

$$C_{metal} = \left(\frac{2}{10}\right)(4200) = 840J/kg\ k.$$

43. How much energy is needed to change 2kg of ice at 0°C to water of 50°C. ($C = 4200J/kg\ k$), ($L_f = 3.336 \times 10^5 J/kg$)

Solution: $Q = mL_f + mC\Delta T$

$$Q = (2)(3.336 \times 10^5) + (2)(4200)(50)$$

$$Q = 1.09 \times 10^6 J$$

44. Calculate the heat energy required to increase the temperature of 400 g of water from 30°C to 130°C. (take, $C_{water} = 4200J/kg\ k$, $C_{steam} = 2080J/kg\ k$, $L_V = 2.5 \times 10^6 J/kg$).

Solution: There are three stages

→ Heat energy required to heat the water from 30°C to 100°C.

$$Q_1 = mC_w\Delta T_1 = (0.4)(4200)(100 - 30) = 117600J$$

→ Heat energy required to boil the water

$$Q_2 = mL_V = (0.4)(2.5 \times 10^6) = 1000000J$$

→ Heat energy required to heat the stem from 100°C to 130°C.

$$Q_3 = mC_S\Delta T_2 = (0.4)(2080)(130 - 100) = 24960J$$

Thus, the total amount of heat required is

$$Q = Q_1 + Q_2 + Q_3 = 117600 + 1000000 + 24960$$

$$Q = 1142560J \simeq 1.1 \times 10^6 J$$

45. A 20g aluminum at 90°C is dropped into a block of ice at 0°C. How much ice does the aluminum melt?
($C_{Al} = 910J/kg\ k$, $L_{f\,ice} = 304000J/kg$).

Solution: From heat exchange,

Heat gained = heat lost

→ Heat energy of ice + heat change of aluminum = 0

$$L_f m_{ice} + m_{Al}C_{Al}\Delta T_{Al} = 0$$

$$(304000)m_{ice} + (0.02)(910)(0 - 90) = 0$$

→ $m_{ice} = 5.4g$

46. A certain amount of heat is added to a mass of aluminum and its temperature is raised by 57°C. If the same amount of heat is added to the same mass of copper, how much does the temperature of the copper rise? ($C_{Al} = 910J/kg\ k$, $C_{Cu} = 420J/kg\ k$)

Solution: The heat supplied is the same for both

$$mC_{Al}\Delta T_{Al} = mC_{Cu}\Delta T_{Cu}$$

$$C_{Al}\Delta T_{Al} = C_{Cu}\Delta T_{Cu}$$

$$\Rightarrow \Delta T_{Cu} = \frac{C_{Al}\Delta T_{Al}}{C_{cu}} = \frac{(910)(57)}{(420)}$$

$$\Delta T_{Cu} \simeq 124°C$$

47. Determine the temperature T that results when 150g of ice at 0°C is mixed with 300g of water at 50°C. ($L_f = 304000J/kg$, $C_w = 4200J/kg\ k$)

Solution: From energy conservation;

→ Heat energy of ice + heat change of water = 0

$$mL_f + mC\Delta T_{ice} + mC\Delta T_{water} = 0$$

$$(0.15)(304000) + (0.15)(4200)(T - 0) + (0.3)(4200)(T - 50) = 0$$

$$45600 + 630T + 1260T - 63000T = 0$$

$$\Rightarrow T \simeq 9°C$$

48. A 3g bullet (C = 128J/kg °C) moving at 180m/s enters a bag of sand and stops. By what amount does the temperature of the bullet change if all its kinetic energy is changed to heat.

Solution: From transformation of energy,

Kinetic energy = thermal energy

$$\frac{1}{2}mV^2 = mC\Delta T$$

$$\Rightarrow \Delta T = \frac{V^2}{2C} = \frac{(180)^2}{2(128)} \Rightarrow \Delta T \simeq 127°C$$

49. Suppose a 60kg person consumes 2.5×10^6Cal of food in one day. If the entire heat of this food is in his body, how large a temperature change would it cause? (take $C_{body} = 0.83Cal/g\ °C$)

Solution: Using the relation;

$$Q = mC\Delta T$$

$$\Rightarrow \Delta T = \frac{Q}{mC} = \frac{2.5\times10^{6}}{(60000)(0.83)}, 60kg = 60000g$$

$$\Delta T = 50°C$$

50. A 50g metal at 95°C is dropped into 250g of water at 17°C and warms it to 19.4°C. What is the specific heat of the metal?

Solution: From heat exchange;

Heat change of metal + heat change of water = 0

$(m_{metal})(C_{metal})(\Delta T_{metal}) + (m_{water})(C_{water})(\Delta T_{water})$

$(0.05)(C_{metal})(19.4 - 95) + (0.25)(4200)(19.4 - 17) = 0$

$-3.8C_{metal} + 2520 = 0$

$\Rightarrow C_{metal} \simeq 663 J/kg\ k$

General Secondary Education Certificate exam

51. Which of the statements is correct?

A. For all ways of heat transfer, conduction, convection and radiation, a material medium is necessary

B. In conduction and convection, heat transfer is brought by the motion of matter

C. In conduction and radiation, heat transfer occurs without the motion of matter

D. Heat is transferred to the earth from the sun mainly by convection

Explanation: The radiant energy travels with the velocity of light and transmitted through a vacuum as well as a medium.

Answer: C

52. The temperature of a gas is most closely related to

A) The kinetic energy of translation of its molecules

B) The size of its molecules

C) The potential energy of the molecules

D) The total energy of its molecules

Explanation: The temperature of a substance is a measure of the average kinetic energy of the particles within the substance.

Answer: A

53. A 2kg block is made of a metal that has higher specific heat capacity than the metal out of which a 1kg block is made. The 2kg block was originally at 50°C and the 1kg block at 20°C. When the two blocks are placed in contact, what will the final equilibrium temperature be?

A) 35°C C) Between 40°C and 50°C

B) 40°C D) Between 35°C and 40°C

Solution: From conservation of heat

$$Q_{lost} = Q_{gained} = m_1C_1\Delta T_1 = m_2C_2\Delta T_2$$

$$(2)(C_1)(50 - T) = (1)(C_2)(T - 20)$$

$$100C_1 - 2C_1T = C_2T - 20C_2$$

$$C_2T + 2C_1T = 20C_2 + 100C_1$$

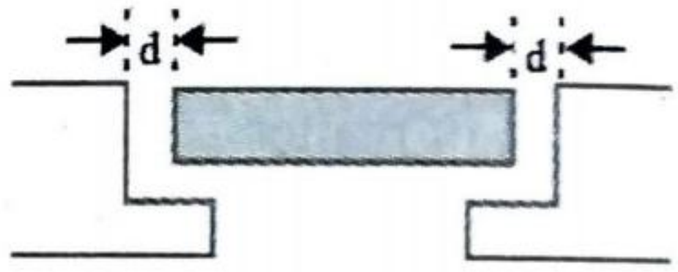
$$T = \frac{20C_2 + 100C_1}{C_2 + 2C_1}, \text{ for } C_1 = C_2,\ T = 40°C$$

But since $C_1 > C_2$

Answer: C

54. A 10m steel bridge at 5°C is constructed with some gap at its ends. The range of temperature in the surrounding of the bridge is from 5°C to 30°C. What must be the gap d at one end for the bridge to expand and construct without disturbing the road? ($\alpha = 1.2 \times 10^{-5}/°C$).

A) 0.15cm
B) 0.3cm
C) 0.11cm
D) 0.4cm



Solution: $\Delta L = L_C\alpha\Delta T = (10)(1.2 \times 10^{-5})(30 - 5) = 0.3cm$

But $\Delta L = 2d \Rightarrow d = \frac{\Delta L}{2} = \frac{0.3}{2} = 0.15cm$

Answer: A

55. A railroad is laid at 20°C with rail length of 10m. What should be the minimum gap left between the rails in order to avoid buckling if the maximum expected temperature is 40°C. ($\alpha = 1.2 \times 10^{-5}k^{-1}$).

A) 0.24 cm C) 4.8×10^{-4}m
B) 0.0006cm D) 3×10^{-5}m

Solution: $\Delta L = L_C\alpha\Delta T = (10)(1.2 \times 10^{-5})(40 - 20) = 0.24cm$

Answer: A

56. A steel beam is 25m long in winter at 0°C. How much does the length change in the summer at 30°C. ($\alpha = 1.2 \times 10^{-5}k^{-1}$).

A) 6mm B) 9mm C) 25mm D) 5mm

Solution: $\Delta L = L_C\alpha\Delta T = (25)(1.2 \times 10^{-5})(30 - 0) = 9mm$

Answer: B

57. Specific heat capacity.

A) Is different for different phase of the same material

B) Determines the rate at which heat flows
C) Is the heat energy needed to raise the body temperature by 1°C.
D) Is measured in J/kg

Answer: A

58. The temperature of a body is directly proportional to.
A) The average speed of the body in motion.
B) The average kinetic energy of the whole body
C) The average speed of the molecules of the body
D) The average kinetic energy of the molecules of the body.

Answer: D

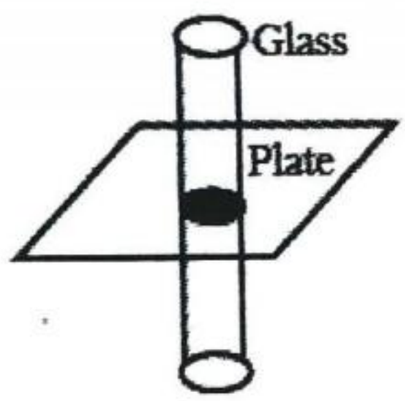
59. When two bodies are in thermal equilibrium, they will have equal.
A) Thermal conductivity C) temperature
B) Specific heat D) heat energy

Answer: C

60. A mixture of gas is made of two types of molecules. The molecules of A are lighter than the molecules of B. The gas mixture has the same temperature everywhere. Which one is correct about the speed of the molecules.
A) The molecule of B move faster than those of A
B) The molecules of A move faster than those of B
C) The molecules of A move as fast as those of B
D) The mass of the molecules and their speed are unrelated

Answer: B lighter molecules move faster

61. A glass rod ($\alpha = 0.9 \times 10^{-5}k^{-1}$) is fitted into a hole of the same diameter in copper plate ($\alpha = 1.7 \times 10^{-5}/k$). What will happen if the temperature of the glass and the copper increase?

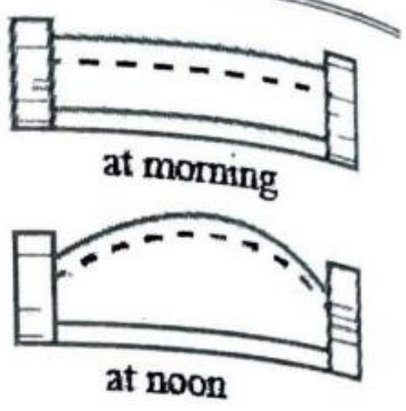


A) The glass rod will be dropped
B) The glass will be crack
C) There will be no change
D) The plate will be folded

Explanation: since the copper has a greater coefficient of expansion than the glass, the copper plate will expand more than the glass. Thus, the glass will fall/drop through the hole.

Answer: A

62. If the strips are heated, which one is the right explanation?
A) The strips have equal linear expansion



B) The strips have same coefficient of expansion but different length
C) The top strip expands faster
D) The bottom strip expands faster

Explanation: The top strip expands faster than the bottom strip and the system bulges upwards.

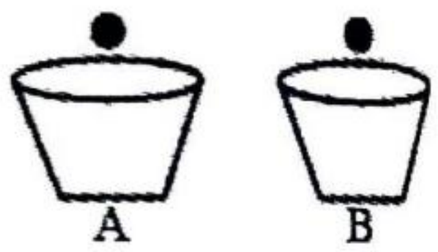
Answer: C

63. When heated from room temperature to higher temperature, aluminum bar expands faster than a bar of steel. What is the reason for this?

A) Aluminum will have greater temperature
B) Molecule of Aluminum vibrate more
C) Steel has greater temperature coefficient
D) Aluminum has less temperature coefficient

Answer: D

64. Two hot copper spheres are dropped into buckets contained water at the same temperature. If it is found that, the temperature of water in bucket A increased more than bucket. B, which one is a possible reason for this temperature difference.



A) The mass of sphere A is smaller than sphere B
B) There is more water in bucket B than in bucket A
C) The specific heat of the spheres is different
D) The initial temperature of water in bucket A was greater.

Answer: B

65. A technician noticed that a round steel rod ($\alpha = 1.2 \times 10^{-5}$) is fitted into a circular brass ring ($\alpha = 1.9 \times 10^{-5}/°C$). What should the technician do in order to easily pull out the rod out of the ring?

A) Increase the temperature of the rod while keeping that of the ring constant
B) Decrease the temperature of the ring while keeping that o the rod constant.
C) Decrease the temperature of both the ring and the rod
D) Increase the temperature of both the ring and the rod.

Explanation: If both heated, the ring will expand more than the rod.

Answer: D

66. Thin strips of iron and zinc are riveted together to form a bimetallic strip that bends when heated. The iron is on the inside of the bend because,

A) It has a higher specific heat
B) It has a lower specific heat
C) It has a lower coefficient of linear expansion
D) It has a higher coefficient of linear expansion

Answer: C, $\alpha_{iron} = 1.1 \times 10^{-5}/k$ an $\alpha_{zinc} = 2.6 \times 10^{-5}/k$.

67. Which one of the following is a factor on which specific heat capacity of a conductor depends

A) Heat B) Substance C) Mass D) Temperature

Answer: B

68. When you are taking a bath, how many kilogram of hot water at 48°C must you mix with cold water at 12°C so that the temperature of the bath is 36°C? The total mass of water (hot plus cold) is 90kg.

A) 40kg B) 50kg C) 20kg D) 60kg

Solution: From heat exchange,

Heat lost by hot water = heat gained by cold water

$m_hC_h\Delta T_h = m_CC_C\Delta T_C$, $m_h + m_C = 90 \Rightarrow m_C = 90 - m_h$

$\Rightarrow m_hC_h\Delta T_h = (90 - m_h)\, C_C\Delta T_C$, $C_h = C_C = C$

$m_h\Delta T_h = (90 - m_h)\,(\Delta T_C) = 90\Delta T_C - m_h\Delta T_C$

$m_h\,(48 - 36) + m_h\,(36 - 12) = 90(36 - 12)$

$12m_h + 24m_h = 2160$

$36m_h = 2160 \Rightarrow m_h = 60kg$

Answer: D

69. Which one of the following is constant during change of state?

A) temperature C) volume
B) molecular speed D) density

Answer: A

70. In an effort to stay awake for an all night study session, a student makes a cup of coffee using an immersion heater in 0.3kg of water. How much heat must be added to raise it temperature from 20°C to 80°C (C = 4200J/kg k)

A) 2.6×10^4J C) 8.4×10^4J
B) 2.1×10^2J D) 7.56×10^4J

Solution: $Q = mC\Delta T = (0.3)\,(4200)\,(80 - 20)$

$Q = 7.56 \times 10^4$J

Answer: D

71. The temperature of water suitable for hot bath is about 60°C. If you have 5kg hot water at 90°C room temperature 20°C should you mix to reduce its temperature to 60°C.

A) 3.75kg B) 2.25kg C) 5kg D) 6.7kg

Solution: From the law of heat exchange,

Heat lost by hot water = heat gained by cold water

$$Q_n = Q_C$$

$$m_n C_n \Delta T_n = m_C C_C \Delta T_C$$

$$m_C = \frac{m_n \Delta T_n}{\Delta T_C} = \frac{5(90-60)}{60-20} = 3.75\text{kg}$$

Answer: A

72. Water at 50°C can be obtained by mixing

A) 1kg of ice at 0°C and 1kg of water at 100°C

B) 1kg of water at 0°C and 1kg of water at 100°C

C) 1kg of ice at 0°C and 1kg of steam at 100°C

D) 1kg of water at 0°C and 1kg of steam at 100°C

Solution: from the law of heat exchange,

Answer: B

73. Which one of the following is a heat converter?

A) Refrigerator C) Heat pump

B) Car engine D) radiator

Answer: B, A car engine converts heat energy into mechanical energy.

74. A refrigerator converts 500g of water at 0°C into ice 0°C. What amount of heat is extracted from the water? ($L_f = 3.36 \times 10^5$J/kg).

A) 168000J C) 1.68J

B) 16800J D) 1.68kJ

Solution: $Q = mL_f = (0.5)(3.36 \times 10^5) = 168000\text{J}$

Answer: A

75. A tourist guide wanted to make tea in a cold morning at the top of Ras Dashen. Instead of water he got ice at -10°C. He knows that water boils at 90°C and freezes at 0°C. How much electrical energy should he use to get boiling water of 0.1kg to make tea?

($L_f = 334$k J/kg, $C_{ice} = 2.1$k J/kg °C , $C_w = 4.2$k J/kg °C).

A) 76.82kJ C) 73.3kJ

B) 73.3J D) 170.9J

Solution: $Q = Q_1 + Q_3 + Q_3$

$Q = m_{ice}C_{ice}\Delta T_{ice} + m_{ice}L_f + m_wC_w\Delta T_w$

$Q = (0.1)(2.1)(0-(-10)) + (0.1)(334) + (0.1)(4.2)(90 - 0)$

$Q = 73.3k\ J$

Answer: C

76. If 4200J of heat is required to raise the temperature of 1kg of water by 1°C, what is the amount of heat required to raise the temperature of 500g of water by 20°C.

A) $4.2 \times 10^2 J$ C) $4.2 \times 10^4 J$

B) $1.05 \times 10^2 J$ D) $1.68 \times 10^5 J$

Solution: $Q = mC\Delta T \Rightarrow C = 4200\ J/kg\ k$

$Q = (0.5)(4200)(20) = 42000J$

Answer: C

77. Which of the following is correct

A) Temperature and heat are two names of the same thing

B) Temperature measures motion of atoms and molecules

C) Temperature and heat have the same SI unit

D) Heat is not a form of energy.

Answer: B

78. When a substance changes state from gas to liquid state, which one of the following occurs

A) Heat is absorbed by the substance

B) Heat is given off by the substance

C) The temperature of the substance increases

D) The temperature of the substance decreases

Answer: B

79. What is the dominant way of heat transfer by which the engine of a car is cooled when the car is in motion

A) Conduction C) Convection

B) Radiation D) Evaporation

Answer: B

80. The heat energy required to change state is
A) Latent heat of vaporization
B) Latent heat of fusion
C) Specific heat
D) Latent heat

Answer: D

81. When heat is felt from a burning fire this is due to
A) Conduction
B) Convection
C) Radiation
D) Evaporation

Answer: C

82. The heat capacity of object B is twice that of object A initially A is at 300k and B is at 450k. They are placed in thermal contact and the combination is isolated. What is the final temperature of the objects?
A) 450k B) 300k C) 150k D) 400k

Solution: By the law of heat exchange,

$$m_A C_A \Delta T_A = m_B C_B \Delta T_B$$

But $C_B = 2C_A \Rightarrow C_B m_B = 2C_A m_A$

$$\Rightarrow m_A C_A \Delta T_A = 2m_A C_A \Delta T_B$$
$$\Delta T_A = 2\Delta T_B$$
$$T - T_A = 2(T_B - T)$$
$$T - T_A = 2T_B - 2T \Rightarrow 3T = 2T_B + T_A$$
$$3T = 2(450) + 300 = 1200$$
$$\Rightarrow T = 400k$$

Answer: D

83. A copper calorimeter of mass 100g contains 200g of water at 0°C. How much heat is required to convert the water to steam at 100°C ($C_{Cu} = 390J/kg\ k$, $C_w = 4200J/kg\ k$, $L_v = 2.26 \times 10^6 J/kg$)
A) 87900J
B) 452000J
C) 226000J
D) 539900J

Solution: $Q = Q_1 + Q_2 + Q_3$

$$Q = m_{Cu} C_{Cu} \Delta T_{Cu} + m_w C_w \Delta T_w + m_w L_V$$
$$Q = (0.1)(390)(100 - 0) + (0.2)(4200) + (100 - 0)(0.2)(2.26 \times 10^6 J)$$
$$Q = 539900J$$

Answer: D

84. A gas in a cylinder expanded and does 150J of work while 500J of heat is supplied to it. What is the change in the internal energy of the gas in the process.
A) 350J B) 150J C) 650J D) 500J

Solution: $\Delta u = \Delta Q + \Delta W = \Delta Q - \Delta W$, since it expands
$\Delta u = 500 - 150 = 350J$

Answer: A

85. Why is an ice at 0°C more effective than the same mass of water at 0°C in cooling a drink?
A) The ice has move average kinetic energy than water
B) The ice absorbs heat from the drink to melt to water
C) The water transfers heat to the drink to freeze to ice
D) The water particles move faster than the ice

Answer: B

86. Which one of the following systems can absorb some amount of heat without any change in temperature
A) Water at 10°C C) Water at 10°C
B) Ice at -10°C D) Ice at 0°C

Explanation: ice at 0°C melts due to the latent heat of fusion.

Answer: D

87. What is the ratio of the coefficient of linear expansion to coefficient of volume expansion?
A) 1:2 B) 1:4 C) 1:3 D) 1:2

Explanation: coefficient of linear expansion is α and
coefficient of volume expansion is $\gamma = 3\alpha$

Now, taking the ratio, $\frac{\alpha}{\gamma} = \frac{\alpha}{3\alpha} = \frac{1}{3} = 1:3$

Answer: C

Additional examples

88. When the door of an operating refrigerator is opened, the temperature of the room will
A) Increase
B) Decrease
C) Unchanged
D) Decrease in summer increases in water

Explanation: from first law of thermodynamics,

Answer: A

89. Metals are good conductors of heat because
A) They contain free electrons
B) They are relatively far apart
C) Their atoms collide very frequently
D) They have a reflecting surface

Explanation: free electrons are responsible for conduction

Answer: A

90. Wollen clothes are used in winter because they;
   A) are good sources of heat
   B) absorb heat from surrounding
   C) are bad conductors of heat
   D) provide heat to the body

Answer: C

91. The most suitable material for cooking is the one having
   A) high specific heat and low conductivity
   B) low specific heat and low conductivity
   C) low specific heat and high conductivity
   D) high specific heat and low conductivity

Answer: C

92. The upper and lower fixed points of a faulty thermometer are 5°C and 105°C. If the thermometer reads 25°C, what is the actual temperature?
   A) 15°C B) 30°C C) 20°C D) 35°C

Solution: $\frac{25-5}{105-5} = \frac{x-0}{100-0} \Rightarrow x = 20°C$

Answer: C

93. The normal temperature of a human body is x degrees above absolute zero. What is the value of x?
   A) 100 B) 200 C) 310 D) 340

Solution: $T_C = 37°C$. Since $T_k = T_C + 273 = 37 + 273 = 310K$

Answer: C

94. At what temperature is the Fahrenheit scale reading equal to twice of Celsius?
   A) 160°C B) 320°C C) 175°C D) 273°C

Solution: $\frac{T_F - 32}{180} = \frac{T_C}{100}$, but $T_k = 2T_C$

$$\frac{2T_C - 32}{180} = \frac{T_C}{100} \Rightarrow 20T_C - 320 = 18T_C$$

$$2T_C = 320 \Rightarrow T_C = 160°C$$

Answer: A

95. Two thermometers, one Celsius scale and the other of Fahrenheit scale are kept in a bath. If the temperature of the bath is thrice that of Celsius reading, the temperature of bath is?

A) 80°C B) 80°F C) 32°C D) 12°C

Solution: $\frac{T_F - 32}{180} = \frac{T_C}{100}$, but $T_F = 3T_C \Rightarrow T_C = \frac{T_F}{3}$

$$\frac{T_F - 32}{180} = \frac{T_F}{300} \Rightarrow 30T_F - 960 = 18T_F$$

$$\Rightarrow T_F = 80°F$$

Answer: B

96. Which of the following temperatures is highest?

A) 100k B) -13°F C) -20°C D) -23°C

Solution:, change all into the same scale. Thus,

$100k \Rightarrow -273 + 100 = -173°C$ and $\frac{-13-32}{180} = \frac{T_C}{100} \Rightarrow$

-13°F = -25°C. Now compare all the four, that is -173°C,-25°C, -20°C, -23°C.

Answer: C

97. If the coefficient of linear expansion of a solid is 0.00009, its coefficient of volume expansion is;

A) 0.00027 C) 0.018

B) 0.00018 D) 0.27

Solution:, $\gamma = 3\alpha = 3(0.00009) = 0.00027$.

Answer: A

98. On a hypothetical scale x, the ice point is 40° and the steam point is 120°. For another scale Y, the ice point and the steam point are -30° and 130° respectively. If X reads 50°, the Y would read

A) -5° B) -10° C) -80° D) -12°

Solution: $\frac{50-40}{120-40} = \frac{Y-(-30)}{130-(-30)} \Rightarrow Y = -10°$

Answer: B

99. When a Celsius thermometer reads 90°C, a faulty Fahrenheit thermometry reads 190°F. The correction to be made in the Fahrenheit scale is,

A) +2°F B) -4°F C) -20°F D) +4°F

Solution: $\frac{T_F - 32}{180} = \frac{90 - 0}{100} \Rightarrow T_F = 194°F$

The correct reading is 194°F. Thus, the correction that should be made on 190°F is +4°F

Answer: D

100. 5kg of water at 10°C is added to 10kg of water at 40°C. The equilibrium temperature will be

A) 30°C B) 25°C C) 35°C D) 33°C

Solution: $m_1C\Delta T_1 = m_2C\Delta T_2$

$(5)(T - 10) = (10)(40 - T) \Rightarrow 5T - 50 = 400 - 10T$

$\Rightarrow 15T = 450 \Rightarrow T = 30°C$

Answer: A

101. 100g of ice is mixed with 100g of water at 100°C. The final temperature of the mixture is? (L_f= 80cal/g and C_w = 1Cal/g)

A) 10°C B) 20°C C) 30°C D) 40°C

Solution: $m_1L_f + m_1C_w\Delta T_1 = m_2C_w\Delta T_2$

$(100)(80) + (100)(1)(T - 0) = (100)(1)(100 - T)$

$80 + T = 100 - T \Rightarrow 2T = 20 \Rightarrow T = 10°C$

Answer: A

102. One gram of ice is mixed with one gram of steam. After thermal equilibrium, the temperature of the mix is

A) 0°C B) 100°C C) 55°C D) 80°C

Explanation: → Heat required to melt 1g of ice at 0°C into 1g of water at 0°C is 80cal.

→ Heat required to convert 1g of water at 0°C into 1g of water at 100°C is $1 \times 1 \times 100 = 100$cal.

→ Heat required to condense 1g of steam is 1×540cal = 540cal.

Since the whole of steam is not condensed, the temperature of the mixture is 100°C

Answer: B

103. A change in temperature of 25°C is equivalent to

A) 45°F B) 72°F C) 32°F D) 25°F

Solution: $\Delta F = \frac{9}{5}\Delta C = \frac{9}{5}(25) = 45°F$

Answer: A

104. An electric heater rated as 2KW is used to heat 200kg of water from 10°C to 70°C. What time it takes to do so?
($C = 4200 J/kg\ k$)

A) 25.2 sec
B) 6×10^3 sec
C) 25.2×10^3 sec
D) 25.2×10^6 sec

Solution: $P = \frac{W}{t} = \frac{Q}{t} \Rightarrow t = \frac{Q}{P} = \frac{mC\Delta T}{P}$

$$t = \frac{(200)(4200)(70-10)}{2000} = 25200 \text{ sec}$$

$$t = 25.2 \times 10^3 \text{ sec}$$

Answer: C

105. 540g of ice at 0°C is mixed with 540g of water at 80°C. The final temperature of the mix is

A) 0°C B) 80°C C) 40°C D) 20°C

Solution: $m_1 L_f + m_1 C(T - 0) = m_2 C\ (80 - T)$

$(540)\ (80) + (540)\ (1)\ (T) = (540)\ (1)\ (80 - T)$

$80 + T = 80 - T \Rightarrow T = 0°C$

Answer: A

106. The height of a water fall is 84m. If the potential energy of the falling water is converted into heat, the rise in temperature of the water is?

A) 0.2°C B) 0.96°C C) 1.960°C D) 0.0196°C

Solution: $mgh = mC\Delta T = \Delta T = \frac{gh}{C} = \frac{(10)(84)}{4200}$

$$\Delta T = 0.2°C$$

Answer: A

107. Morning sun is not so hot as the mid day sun because?

A) Heat rays travel slowly in the morning
B) Sun is cooler in the morning
C) Earth is farther away from the sun in the morning
D) The sun ray travel longer in the morning

Explanation: In the morning, the sun rays are not vertical.

Answer: D

108. On a cold morning, a metal surface feel colder to touch than a wooden surface, because

A) Metal has high specific heat
B) Metal has high thermal conductivity

C) Metal has low specific heat
D) Metal has low thermal conductivity.

Explanation: Due to high thermal conductivity of the metal, heat flows quickly from the body into the metal. This gives a cold feeling.

Answer: B

109. 4200J of work is required for
A) Increasing the temperature of 10g of water to 10°C
B) Increasing the temperature of 100g of water to 10°C
C) Increasing the temperature of 1kg of water to 10°C
D) Increasing the temperature of 10kg of water to 10°C

Solution:

$$Q = mC\Delta T = m = \frac{Q}{C\Delta T} = \frac{4200}{(4200)(10)} = 0.1kg = 100g$$

Answer: B

110. A piece of glass is heated to a high temperature and then allowed to cool. If it cracks, the probable reason for this is
A) Low thermal conductivity
B) High thermal conductivity
C) High specific heat
D) High melting point

Explanation: The transmission of heat is small due to low thermal conductivity.

Answer: A

111. As compared to the person with white skin, the person with black skin will experience
A) Less heat and more cold
B) More heat and less cold
C) More heat and more cold
D) Less heat and less cold

Answer: A

112. Body x and body y are separately in thermal equilibrium with body z what can you conclude about body x and body y.
A) They have the same temperature
B) They have the same heat capacity
C) They are in thermal contact
D) They are in thermal equilibrium with other body

Answer: A

Unit Seven: Conceptual Questions

1. What will be the key difference in the energy of the particles in iron at 250k and 500k?
2. Explain why a solid expands on heating?
3. Which one is largest 1°C, 1k or 1°F
4. Which one is denser, water at 2°C or at 8°C
5. Is it possible to change phase of matter without changing its temperature? Explain

Unit Seven: problems

Expansion of solids.

6. Calculate the increase in length of a 2m brass rod that is heated from 0°C, to 150°C. ($\alpha = 1.9 \times 10^{-5}k^{-1}$)
7. Calculate the surface area of an iron cover with a surface area of $0.67m^2$ at 10°C, when it is heated to a temperature of 105°C. ($\beta_{iron} = 2.2 \times 10^{-5}k^{-1}$).
8. Calculate the increase in the volume of a $0.1m^3$ sample of water at 10°C, when it is heated to a temperature ($\gamma_{water} = 2.1 \times 10^{-4}k^{-1}$).

Heat, heat capacity and specific heat capacity

9. A metal bar of mass 100g is warmed from 20°C to 80°C. How much heat is absorbed by the bar. (C = 450J/kg k).
10. Use the data given and find specific heat capacity
    mass of metal = 300g — power of electrical heater = 100w
    start temperature = 20°C — time of heating = 150sec
    end temperature = 75°C
11. A hot metal is immersed into a 50g of water. The water increases in temperature from 20°C to 32°C. Find the heat supplied by the metal (C = 4200J/kg k).
12. Use the experimental data given and find the specific heat capacity of the metal.
    mass of metal = 200g — end temperature = 105°C
    start temperature = 20°C — electrical energy supplied = 200J
13. A metal block increases in temperature from 15°C to 60°C when supplied with 13500J of heat energy
    A) Calculate the heat capacity of the metal
    B) Calculate the specific heat capacity is the mass is 0.75kg

Latent heat and heat exchange

14. Calculate the heat energy required to melt 1.2kg of gold at its melting point (L_f= 63700J/kg k).

15. Calculate the heat liberated when 75g of iron freezes at its freezing point (L_f= 245000J/kg k).
16. 20g of water at 42°C was placed in copper calorimeter with mass 27g at 20°C. Determine the final temperature of the water (C_w = 4200J/kg k, C_{Cu} = 420J/kg)
17. Calculate the heat energy required to increase the temperature of 0.1kg of water from 10°C to 150°C? (C_w = 4200J/kg k, C_{steam} = 2080J/kg k, L_v = 2500000J/kg)
18. Calculate the heat energy required to increase the temperature of 10kg ice from -5°C to 5°C? (C_{ice} = 2100J/kg k, L_f = 334000J/kg, C_{water} = 4200J/kg k)

Unit Seven: Explanation to Questions

1. At 500k, the particles posses larger kinetic energy than at 250k.
2. Because, when a solid is heated, the bond between particles break and they move far apart, which results in expansion
3. 1°C is largest than 1k and 1°F.
4. Water at 8°C is more dense than water at 2°C!
5. Yes. Latent heat can do that

Unit Seven: Solution to Problems

6. $\Delta L = L_C \alpha \Delta T = (2)(1.9\times10^{-5})(150) = 5.7\text{mm}$

7. $\Delta A = \beta A_C \Delta T = (2.2\times10^{-5})(0.67)(105-10) = 1.4\times10^{-3}\text{m}^2$

8. $\Delta V = \gamma V_C \Delta T = (2.1\times10^{-4})(0.1)(80-10) = 1.47\times10^{-3}\text{m}^3$

9. $Q = mC\Delta T = (0.1)(450)(80-20) = 2700\text{J}$

10. $$P = \frac{W}{t} = \frac{Q}{t} \Rightarrow \frac{mC\Delta T}{t} = P$$

$$\Rightarrow C = \frac{(P)(t)}{m\Delta T} = \frac{(100)(150)}{(0.3)(75-20)} = 909\text{J/kg k}$$

11. $Q = mC\Delta T = (0.05)(4200)(32-20) = 2520\text{J}$

12. $$Q = mC\Delta T = C = \frac{Q}{m\Delta T} = \frac{2000}{(0.2)(105-20)} = 118\text{J/kg k}$$

13. a) $$C = \frac{Q}{\Delta T} = \frac{13500}{(60-15)} = 300\text{J/kg}$$

b) $c = \frac{C}{m} = \frac{300}{0.75} = 400 J/kg\ k$

14. $Q = mL_f = (1.2)\ (63700) = 76440J$

15. $Q = mL_f = (0.075)\ (245000) = 18375J$

16. $m_w C_w \Delta T_w = m_{Cu} C_{Cu} \Delta T_{Cu} = m_w C_w (42 - T) = m_{Cu} C_{Cu} (T - 20)$

$(0.02)(4200)(42 - T) = (0.027)(420)(T - 20) \Rightarrow T = 39°C$

17. $Q = Q_1 + Q_2 + Q_3 = mC_w \Delta T_1 + mL_v + mC_5 \Delta T_2$

$Q = (0.1)\ (4200)\ (100 - 10) + (0.1)\ (250000) + (0.1)\ (2080)$
$(150 - 100)$

$Q = 73200J$

18. $Q = Q_1 + Q_2 + Q_3 = mC_{ice} \Delta T_1 + mL_f + mC_w \Delta T_2$

$Q = (0.1)\ (2100)\ (0-(-5)) + (10)\ (334000) + (10)\ (4200)\ (5 - 0)$

$Q = 3655000J$

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Unit Eight

Wave Motion and Sound

Waves make possible all communications in radio, Tv, Phones...etc. They help us to see the world around us. Waves enable doctors to make operations without surgery, they help them to examine the internal condition of a patient. Doctors use waves even to know the general condition of unborn baby. Moreover waves help us to control devices such as TV using remote control. Knowledge of waves is also essential in the design of bridges and many other structures.

8.1 Wave Propagation

All waves carry energy as they propagate (move) from one place to the other. For example, light waves travelling from a bulb transfer energy from the bulb to your eye. Sound waves transfer energy from a speaker to your ear.

Conceptual Example

1. As a wave travels from place to place in a given material, is there any transfer (transportation) of particles of the material from place to place? Explain

   Explanation: No. the particles of the material do not be transported. They only vibrate in their position. As they vibrate, they kick each other and the wave energy move from one particle to the next. In simple terms consider a wave that propagates through a field of wheat. The individual wheat stalks sway back and forth as a wave passes, but they do not change their location. That is how wave passes from place to place without carrying particles.

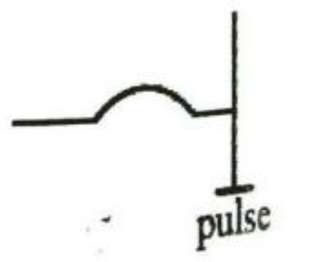
Wave Pulses and continuous waves

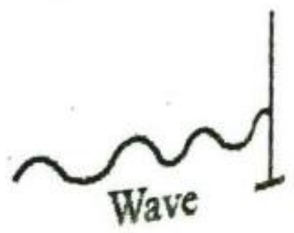
Wave Pulse: Is a wave with single distrurbance.

Wave pulse, is a wave with no repeated vibration

Wave: Is a series of disturbances.

Wave is resulted due to repeated vibrations.





8.2 Type of Waves

In one category, wave's are *mechanical* or *electromagnetic* and in the other category they are *transverse* and *longitudinal*.

A. **Based on the requirement of material medium** for their transmition, waves are classified as mechanical or electromagnetic.

Mechanical waves: Are waves that require material medium (solid, liquid, or gas) for their propagation.

Examples of mechanical waves include, sound wave, water wave, string wave, aspring wave etc.

☞ **Remember;**

Medium: is a material through which a wave moves. A medium may be solid, liquid or gas.

Electromagnetic Waves: Are waves that can move through medium or through empty space (Vacuum).

Examples of electromagnetic waves include, light wave, microwave, infrared wave, radio wave, ultraviolet, x ray, Gamma ray… etc.

☞ **Comparing Mechanical and electromagnetic Waves**

- All mechanical waves require a material to travel through. Mechanical waves travel through the material due to the vibration of the particles in the material. But,
- Electromagnetic waves do not require a medium to travel through. They are comprised of vibrating electric and magnetic fields. There are no particle vibration at all. Electromagnetic waves are able to travel through a vacuum.

Conceptual Example

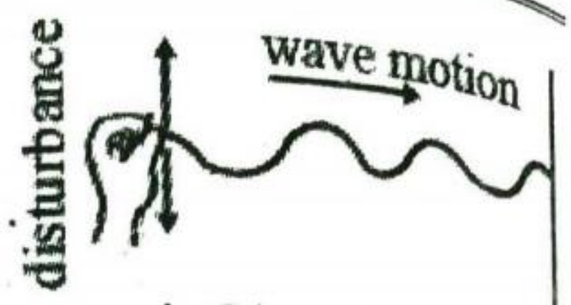
2. When electromagnetic waves travel through a material, can we say the waves travel because of the vibration of the particle in the materials? Explain

   Explanation: No. when electromagnetic waves travel through a medium, there are no particle vibration inside that medium, the wave propagate, because of the electric and magnetic fields in the region.

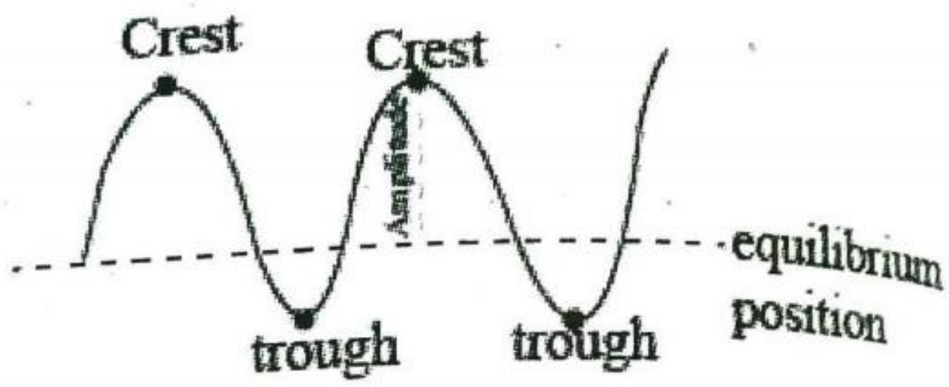
B. **Based on the direction of the disturbance and the direction of the wave motion**, waves are classified as transverse or longitudinal.

Transverse waves: Are waves where the direction of the disturbance and the direction of the wave motion are perpendicular to each other.

In transverse waves, the direction of the vibration and the direction of the wave propagation are at right angles.



Examples of transverse waves include, all electromagnetic waves, that is light wave microwave, infrared, Uv light, x-ray etc and some mechanical waves such as string waves, earth quake s-waves, water waves, etc.



☞ **Terms in transverse waves**

The following are some elements found in transverse wave.

Crest: Is the maximum point on a transverse wave.

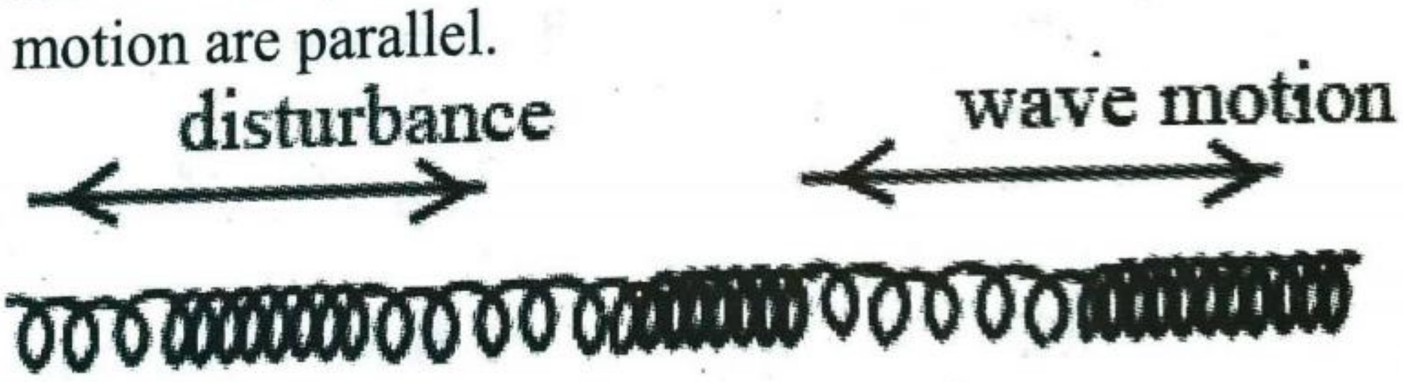
Through: Is the minimum point on a transverse wave

Amplitude: Is the maximum displacement from the equilibrium position.

Equilibrium position: Is the central point about which vibration occurs.

Longitudinal waves: Are waves where the direction of the disturbance and the direction of the wave motion are parallel to each other.

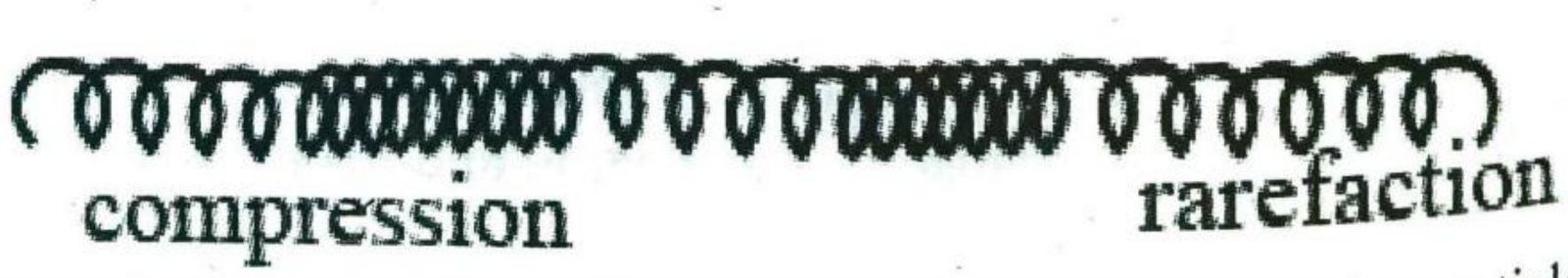
In longitudinal waves, the direction of the vibration and the direction of the wave motion are parallel.



Examples of longitudinal waves include, sound wave, spring wave, p waves in earth quake.

☞ **Terms in longitudinal waves**

The following are some elements found in longitudinal waves.



Compression: Are regions of a longitudinal wave where the particles are close together.

Rarefactions: Are regions of a longitudinal wave where the particles are spread out.

☞ **Comparing transverse and longitudinal waves**

- In transverse wave, the vibrations are perpendicular to the direction of the wave motion. Transverse waves comprises a series of crests and troughs.
- In longitudinal wave, the vibrations are parallel to the direction of the wave motion. Longitudinal waves comprises a series of compressions and rarefactions.

Wave characteristics

The following terms and equations are used to describe any type of wave.

Wave speed (V): Is the speed at which a wave moves.

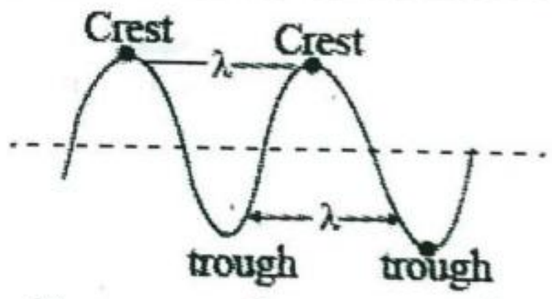
Wave speed tells us the distance the wave travels in one second. Its SI unit is m/s.

Amplitude (A): Is the maximum displacement from the equilibrium position.

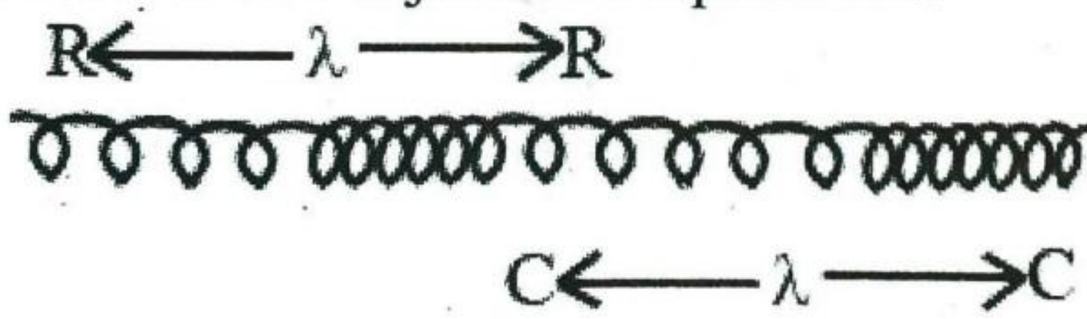
Amplitude is the height of a crest or the height of a trough measured from the undisturbed position. (Equilibrium position)

Wavelength (λ ; lambda): Is the distance between two adjacent points which are identical.

In a transverse wave, the distance between successive crests (crest to crest) or the distance between successive throughs (trough to trough) is a wave length.



In a longitudinal wave, one wave length is the distance between two adjacent rarefactions or two adjacent compressions.



☞ **Remembers**, the unit of wavelength (λ) is meter (m).

Frequently (f): Is the number of complete cycles passing a given point per second.

The SI unit of frequency is Hertz (H_z). A frequency of $10H_z$ would mean 10 waves per second.

Period (T): Is the time taken for a wave to complete one cycle.

The SI unit of period is second (s).

☞ **Remember,** period and frequently of a wave are reciprocal to each other.

$$\textbf{frequency} = \frac{\mathbf{1}}{\textbf{period}}$$

$\boxed{f = \frac{1}{T}}$, thus, $H_z = \frac{1}{sec} = sec^{-1}$

☞ If you consider a wave with frequency of $4H_z$, this would mean, four waves pass a given point in one second. Each would therefore take 0.25 second to pass that point. Thus, the period of the wave is, $T = \frac{1}{f} = \frac{1}{4} = 0.25\,sec$. Likewise, a wave of frequency $10H_z$ would have a period of 0.1 sec.

The wave equation

The equation that relates wave speed, *wave length* and frequency of a wave is known as the *wave equation*. And it is given by,

Wave speed = (frequency) (wavelength)

In symbols,

$\boxed{V = \lambda f}$ ⟵ The wave equation

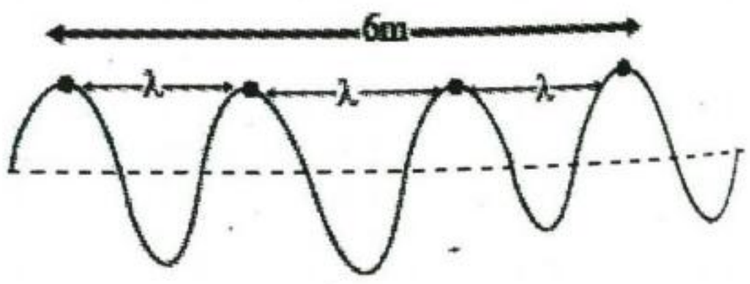
Illustrative Examples

3. The total distance between 4 consecutive crests of a transverse wave is 6m. What is the wavelength of the wave?

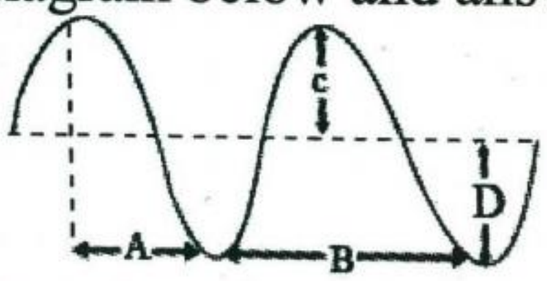
Solution:

There are three wavelengths between four crests. Thus,

$3\lambda = 6 \Rightarrow \lambda = {}^{6}\!/_{3} = 2m$



4. Consider the diagram below and answer the questions that follow.



a) The wavelength of the wave is shown by letter _____
b) The amplitude of the wave is shown by letter _____
c) What does letter A represent in the diagram?

Solution: a) The wavelength is represented by letter B

b) The amplitude of the wave is shown by both letters C and D

c) Letters A is at a distance of crest to trough. Thus, the distance from crest to the next adjacent through is $\frac{\lambda}{2}$.

Hence, letter A represents half wavelength.

5. A student counts his heartbeats and finds that it makes 75 beats in one minute. Calculate

a) The frequency of his heartbeat

b) The period of his heartbeat

Solution: a) $\text{frequency} = \frac{\text{Number of oscillation}}{\text{time taken}}$

$$f = \frac{75}{60} = 1.25H_z$$

b) $\text{Periods} = \frac{\text{time taken}}{\text{Number of oscillation}}$

$$T = \frac{60}{75} = 0.8 \text{ sec}$$

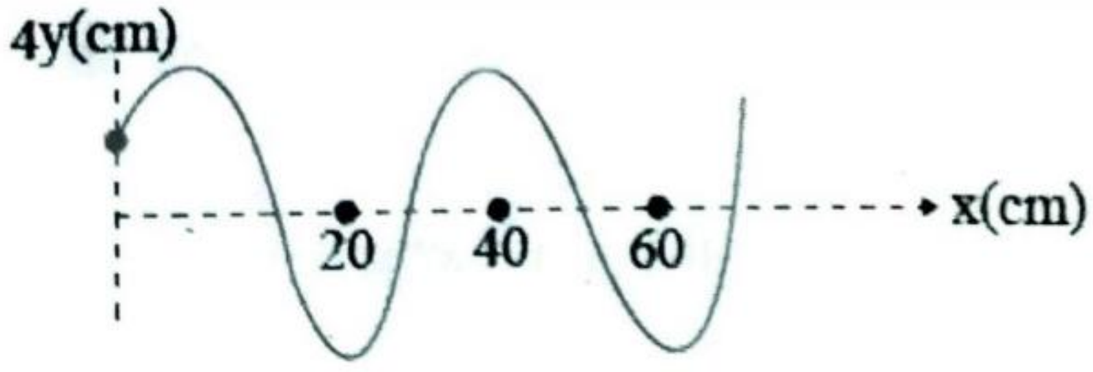
$$\text{Or } T = \frac{1}{f} = \frac{1}{1.25} = 0.8\,\text{sec}.$$

6. Use the following diagram and find?

a) The amplitude

b) The wavelength

c) If the frequency of the wave is 8Hz, calculate the speed of the wave



Solution: a) $A = 4cm$

b) $\lambda = 60 - 20 = 40cm$

c) $v = \lambda f = (40)(8) = 320cm/s = 32m/s$

7. If the speed of sound in air is 330m/s and its frequency is $750H_z$. What is?

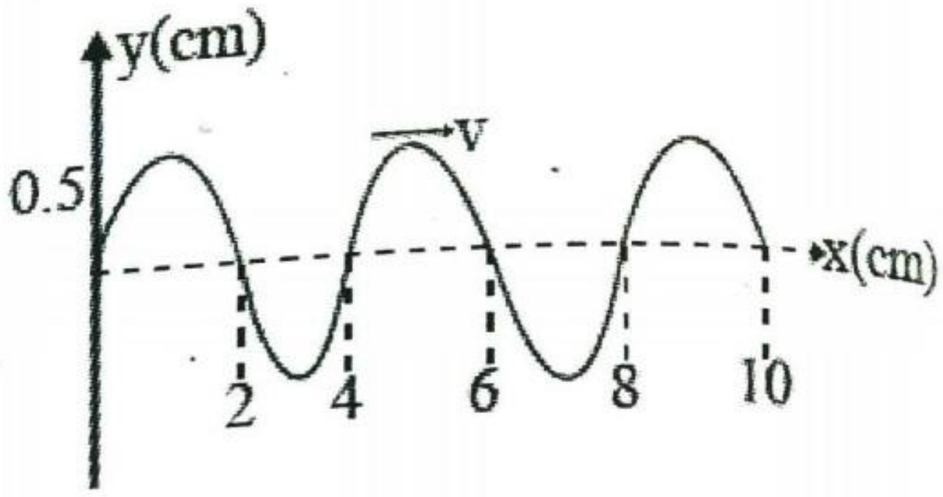
a) Its wavelength

b) Its period

Solution: a) $v = \lambda f = \lambda = {}^{v}\!/_{f} = \frac{330}{750} = 0.44m$

b) $T = \frac{1}{f} = \frac{1}{750} = 1.3 \times 10^{-3} sec.$

8. The wave shown below is moving to the right at 20cm/s. Find its
   a) Wavelength
   b) Amplitude
   c) Frequency, and
   d) Period



Solution:

a) $\lambda = 6 - 2 = 4cm$

b) $A = 0.5cm$

c) $v = \lambda f = f = {}^{v}\!/_{\lambda} = \frac{20}{4} = 5H_z$

d) $T = \frac{1}{f} = \frac{1}{5} = 0.2sec$

Conceptual Examples

9. Does a wave pulse have a period? A frequency? A speed? An amplitude? Explain

Explanation: A wave pulse is a single disturbance which we can visualize it as. Thus, pulse has a speed and an amplitude.

10. What happens to the period of a wave as the frequency increases?

Explanation: Period and frequency of a wave are reciprocal to each other. That is, as one increases, the other decreases. $T = \frac{1}{f}$.

Thus, as the frequency increases, the period decreases.

11. What happens to the wavelength of a wave as the frequency increases?

Explanation: From the wave equation, we have; $v = \lambda f = \lambda = {}^{v}\!/_{f}$.

Thus keeping the speed of the wave constant, as the frequency increases, the wavelength decreases.

12. Consider a light wave and a sound wave with the same wavelength. Which has the higher frequency?

Explanation: From the wave equation, we have; $v = \lambda f = f = {}^{v}/_{\lambda}$.

Now, for the same wavelength, as the speed increases, the frequency increases. Since the speed of light is much larger than the speed of sound, light has a higher frequency.

8.3 Wave properties

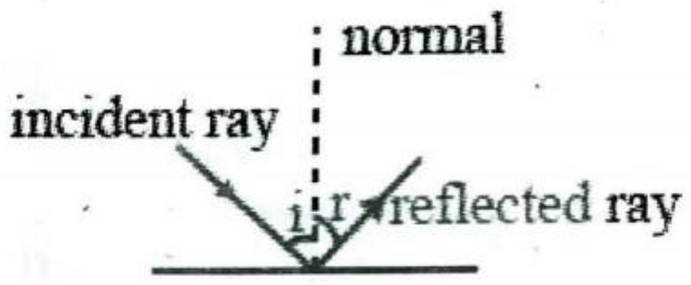
All types of waves do have four common properties. There are *reflection*, *refraction*, *diffraction* and *interference*.

a) Reflection

It occurs when a wave meets a fixed surface on its way and returns back.

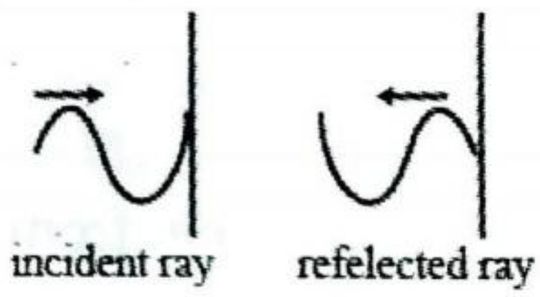
Reflection: Is the bouncing off and change in direction of a wave as it strikes a fixed surface.

When a light wave meets a reflecting surface, the incident ray bounces off the surface in such a way that, the angle of incidence $\left(\hat{i}\right)$ is



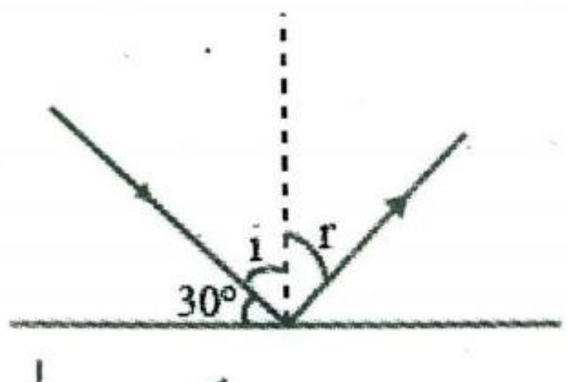
equal to the angle of reflection $\left(\hat{r}\right)$. This relation is what we usually call the first law of reflection $\left(\hat{i} = \hat{r}\right)$.

☞ **Remember,** when a wave is reflected from a fixed surface, it changes its direction. In the process, its crest changes to trough and its through will became crest.



Illustrative Examples

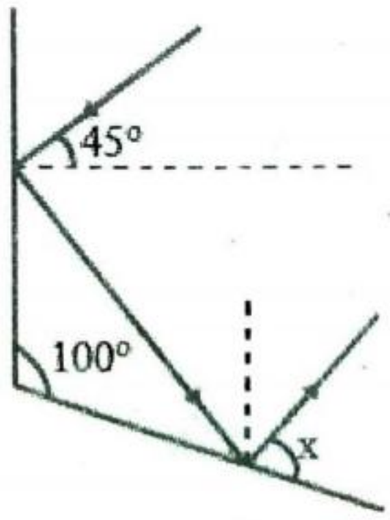
13. Light wave is reflected from a glass as shown. Find the angle of incidence (i) and the angle of reflection (r)?



Solution: Since, $i + 30° = 90° \Rightarrow i = 60°$ and from the first law of reflection $i = r$

$\Rightarrow \hat{r} = 60°$

14. Self test, find the angle marked x ?



Answer: $x = 35°$

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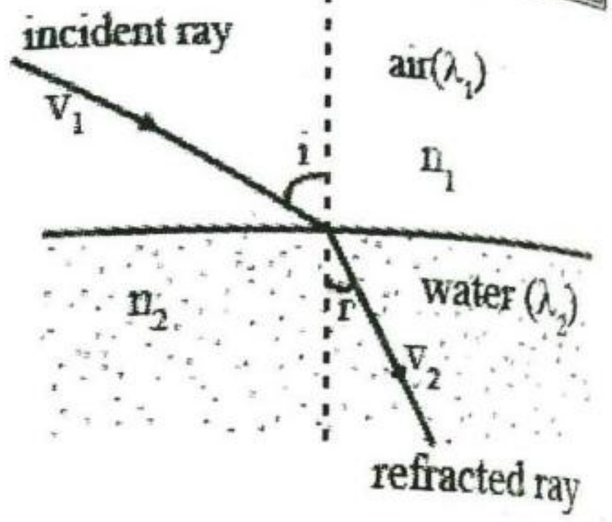
b) Refraction

It occurs when waves change speed as they move from one medium to another.

Refraction: Is the bending of wave due to the change in speed as it moves from one medium to the other.

When a light wave moves from air into water, it bends as shown due to the difference in its speed in the two media.



Here, we have **snell's law**, which states that "*the ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant.*"

$$\frac{\sin(i)}{\sin(r)} = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$ ← snell's law

Where, n → index of refraction
v → speed of the wave
λ → wave length
} both in medium 1 and medium 2.

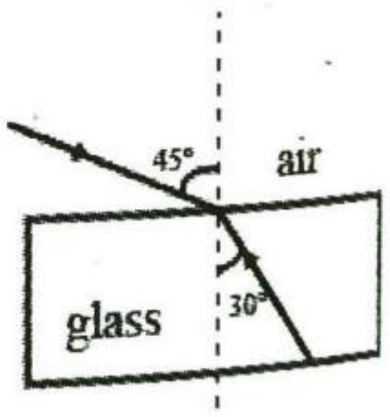
Illustrative Example

15. A ray of light moves from air into a glass as shown below. Calculate the refractive index of the glass (take refractive index of air = 1).

Solution: from snell's law

$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$

$$\Rightarrow n_2 = \left(\frac{\text{Sin } i}{\text{Sin } r}\right)(n_1) = \left(\frac{\text{Sin } 45°}{\text{Sin } 30°}\right)(1) = 1.4$$



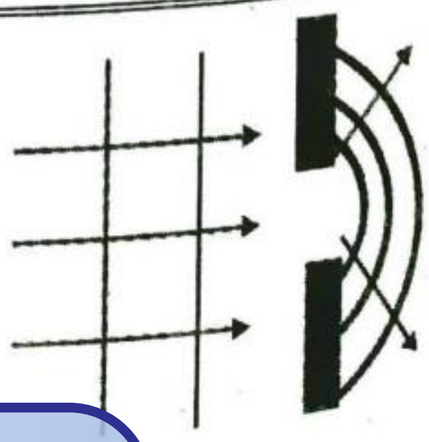
Diffraction

Occurs when a wave passes the edge of an obstacle or through a gap.

Diffraction: Is the spreading out of a wave when it travel through a narrow gap or around sharp edges or corners.

A water wave spreads itself through the narrow gap.

Or around the obstacle as shown here



Interference

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It occurs when two waves of the same kind join together.

Interference: Is the mixing up of two waves either to add up or cancel each other out.

Interference is of two types, so we have *constructive* or *destructive* interference patterns.

Constructive Interference: Is an interference where two waves that are inphase combine to make a bigger wave.

→ When two waves are in constructive interference, they give rise to a wave of larger amplitude.

Destructive interference: Is an interference where two waves that are out of phase combine to cancel each other out.

→ When two waves are under destructive interference, they result to wave of smaller amplitude.

8.4 Sound Waves

A laughing baby, a singing bird, a barking dog, etc are sources of sound. Sound wave is a longitudinal mechanical wave that is produced when an object vibrates. For example, when we speak, our vocal cords in the throat vibrate as the air is pushed over them. In the same manner, different musical instruments produce sound by making part of the instrument vibrate.

Sound: Is a longitudinal mechanical wave that is produced due to vibrating objects.

☞ As sound waves are mechanical waves, they require a medium to travel through. Thus, sound travels through air, liquid and solids. But, it can never travel through vacuum.

Conceptual Examples

16. It is said that, sound wave do not move through vacuum. Why?
Explanation: It is because, in vacuum, there is no particle. Thus, nothing is responsible to carry over the sound wave from one point to the other.

17. How do we hear sound? Explain
Explanation: When vibrations from different sources reach our ears, they make our ear drums vibrate. These vibrations are transmitted via special cells to our brain and it interpret the message as sound.

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18. It is said, we cannot hear even an explosion in vacuum. But how could Astronauts speak to each other on the moon, assuming that there is no air on the moon.

Explanation: Astronauts can speak to each other on the moon using radio waves. Radio waves are electromagnetic waves that can travel through vacuum.

Hearing

A normal human ear can detect a range of frequencies from $20H_z$ to $20{,}000H_z$. This is referred to as *audible range*.

Audible range: Is the range of sound frequency that can be detected by the human ear.

The speed of sound

Sound travels at different speeds through different materials. For example, the speed of sound through water is around five times faster than in air and it is about fifteen times faster in solids.

And in general, sound travels faster in solids and faster and faster in liquids than in liquids than in gases.

Conceptual Example

19. The denser the material, the faster the speed of sound. This is usually the case. Why? Explain

Explanation: This is so because, in denser medium, the particles are closer together and so in such a medium, the sound wave pass from particle to particle much quickly.

→ The speed of sound in any gas increases the as temperature of the gas increases, and the speed of sound through the gas at any temperature (T) is given by.

$$V = \sqrt{(\gamma R^* T)} = \sqrt{(KT)}$$

Where γ is a constant

$$R^* = \left(\frac{\text{molar gas constant (R)}}{\text{The molar mass (M)}} \right), R^* = \frac{R}{M}$$

$$K = \gamma R^*$$

For air, $\gamma = 1.4$ and $R^* = 286 \ \frac{m^2}{s^2 K}$

Thus, $K = \gamma R^* = (1.4)\ (286) = 400 \frac{m^2}{s^2 K}$

Thus, at any temperature (T), the speed of sound in air is given by,

$$V = \sqrt{KT}\ , \text{ but } K = 400$$

$$\Rightarrow \boxed{V = \sqrt{400T} = 20\sqrt{T}}, \text{ T is in Kelvin}$$

☞ The speed of sound in air at any temperature (T) can also be calculated using the relation

$$\boxed{V = 331 + 0.6T}$$

Conceptual Example

20. As air temperature increases, the speed of sound in air increases why? Explain

Explanation: As temperature increases, the particles of the air get higher kinetic energy and move faster. Thus, the faster the particles are moving, the faster the sound wave carried over from particle to particle. .

Illustrative Examples

21. Calculate the speed of sound in air at 25°C?

Solution: **Method I** using the relation:

$$V = 20\sqrt{T}\ .\ \text{Where T is in Kelvin}$$

Since $T = 25°C \Rightarrow T_K = 273 + 25 = 298K$

$$V = 20\sqrt{298} = 345m/s\ .$$

Method II using the relation:

$$V = 331 + 0.6T,\ T = 25°C.$$

$$V = 331 + 0.6(25) = 346m/s.$$

22. If the speed of sound in air at 324K is 340m/s. Calculate its speed at 441K?

Solution: using the relation, $V = 20\sqrt{T}$, thus for

$$V_1 = 20\sqrt{T_1} \ldots\ldots\ldots\ldots (*)$$

$$V_2 = 20\sqrt{T_2} \ldots\ldots\ldots\ldots (**)\ \text{, taking ratio,}$$

$$\frac{V_2}{V_1} = \frac{\sqrt{T_2}}{\sqrt{T_1}} \Rightarrow V_2 = (V_1)\left(\sqrt{\frac{T_2}{T_1}}\right) = (340)\left(\sqrt{\frac{441}{324}}\right) = 397m/s$$

Hence, the speed of sound at 441K is 397m/s.

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23. The human ear can detect sounds with frequencies between $20H_z$ and $20{,}000H_z$. Find the largest and smallest wavelengths the ear can detect while sound is travelling in air at 25°C?

Solution: at 25°C, $T = 273 + 25 = 298K$

$$\text{Then, } V = 20\sqrt{T} = (20)\left(\sqrt{298}\right) = 345\text{m/s}$$

and from the wave equation, $V = \lambda f$

$$V_1 = \lambda_1 f_1 \Rightarrow \lambda_1 = \frac{V_1}{f_1} = \frac{345}{20} = 17.25\text{m}$$

$$\text{and } V_2 = \lambda_2 f_2 \Rightarrow \lambda_2 = \frac{V_2}{f_2} = \frac{345}{20{,}000} = 0.01725\text{m}$$

Note that, $V_1 = V_2 = 345\text{m/s}$

Describing sound

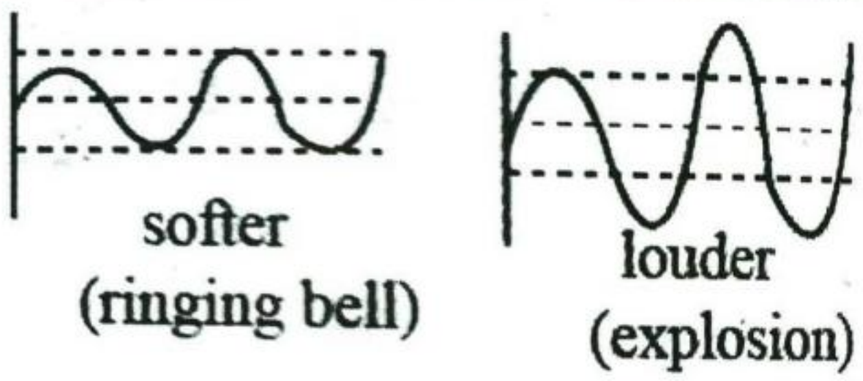
Terms such as *loudness*, *pitch*, *timbre* etc are used to describe sound.

Loudness

The loudness of a sound depends on the **amplitude** of the sound wave. The larger the amplitude, the louder the sound.

Loudness: Is the audible strength of a sound which depends on the amplitudes of the sound wave.

An explosion has a louder sound than a ringing bell.

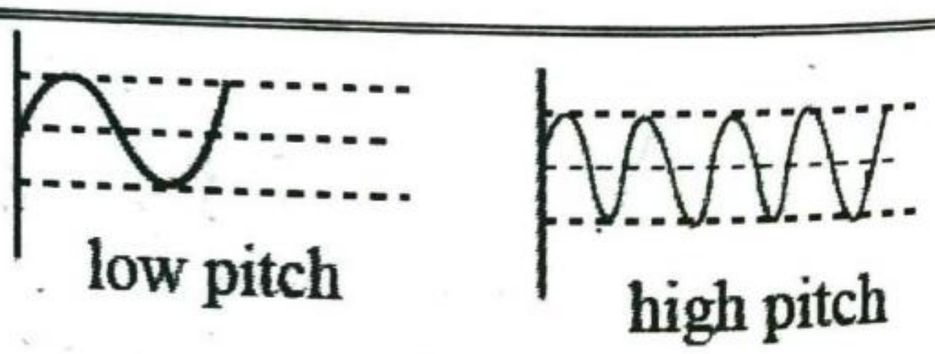


Pitch

The pitch of sound depends on the **frequency** of the sound wave. The higher the frequency, the higher the pitch.

Pitch: Is the highness or lowness of a sound which depends on the frequency of the sound wave.

The chirp of a bird has a high pitch and the roar of a lion has a low pitch.



Timber

The timber of a sound tells the **quality** of the sound. Quality does not mean good or bad. It just refers to the difference in sound.

Timber: Is the quality of sound (tone).

Trumpet and violin do have different sounds.

Echoe

Sound, like all waves is able to reflect off surfaces. This reflection of sound is known as *echoe*.

Echoe: Is the reflection of sound from surfaces.

☞ Solid surfaces such as metals and rocks are better sound reflecting surfaces than soft surface such as cloth.

Conceptual Example

24. Why are hard surfaces good sound reflectors than soft surfaces? Explain

Explanation: Because, softer surfaces tend to absorb part of the sound wave and hence, the sound reflected off from such surfaces is less.

Intensity of sound

The intensity of sound wave is the energy received by each square meter per second. A higher intensity would mean more energy per second falling on each square meter.

Intensity: Is the energy incident on each square meter of a surface per second.

$$\text{That is; Intensity} = \frac{\text{energy /time}}{\text{Area}}$$

$$\text{But} = \frac{\text{energy}}{\text{time}} = \frac{\text{Work}}{\text{time}} = \text{Power}$$

$$\Rightarrow \text{Intensity} = \frac{\text{Power}}{\text{Area}}$$

$$\Rightarrow \boxed{I = \frac{P}{A}}, \text{ its unit is } w/m^2.$$

☞ Since sound is transmitted in all directions, then the area cover by the sound wave is spherical.

Thus, at a distance (r) from the source, the sound covers a spherical area of $A = 4\pi r^2$. So we have,

$$I = \frac{P}{A} = \frac{P}{4\pi r^2}$$

→ From this relation, we can observe that, the larger the distance from the source, the smaller the intensity

Conceptual Example

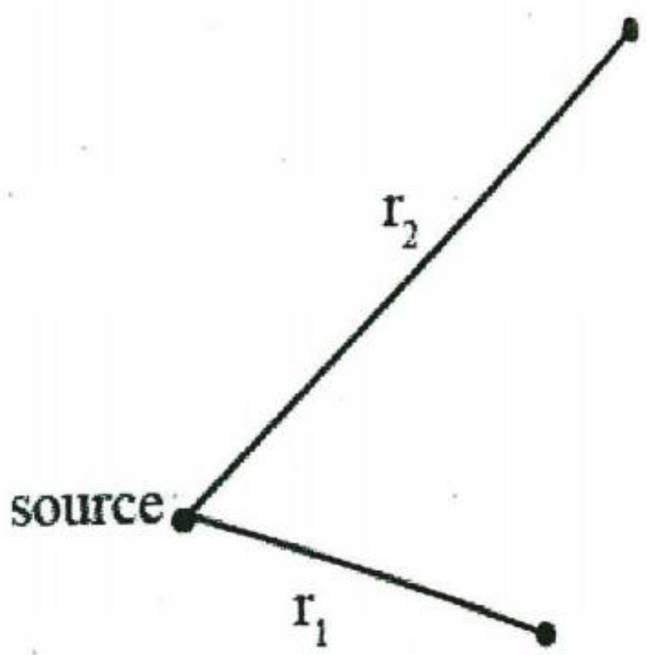
25. The further the source of sound from you, the quiter the sound. Why?

Explanation: This is because, the energy is spread out over a much wider area.

The intensity of sound at two different distances r_1 and r_2 from a given source can be related as follows;

$$I_1 = \frac{P}{4\pi r_1^2} \ldots\ldots\ldots (*)$$

And $$I_2 = \frac{P}{4\pi r_2^2} \ldots\ldots\ldots (**)$$



Taking ratio, we get

$$\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$$

Illustrative Examples

26. A speaker produces $440H_z$ sound with a power of 1.2w. Calculate the intensity of the sound at a distance of 5m from the speaker.

Solution: $I = \frac{P}{A} = \frac{P}{4\pi r^2} = \frac{440}{(4)(3.14)(5)^2}$

$$I = 1.4 w/m^2$$

27. Romio and Juliet once sitting underneath of tree shadows in a garden are impressed by the song of a singing bird. If Juliet is 1m from the bird and hears an intensity of $2.8 \times 10^{-6} w/m^2$. Calculate
   a) The intensity heard by Romio, if he is 4.25m from the bird.
   b) The power output of the bird's song?

Solution: a) Using $\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2} \Rightarrow I_2 = I_1\left(\frac{r_1^2}{r_2^2}\right)$

$$I_2 \Rightarrow \left(2.8\times10^{-6}\right)\left(\frac{(1)^2}{(4.25)^2}\right)=1.55\times10^{-7}\,\text{w/m}^2$$

b) $I_1=\dfrac{P}{A_1}\Rightarrow P=(I_1)(A_1)=\left(2.8\times10^{-6}\right)\left(4\pi(1)^2\right)$

$P = 3.52 \times 10^{-5}$w, or $P = (I_2)\,(A_2) = 3.52 \times 10^{-5}$w

28. By what factor will the intensity of a sound wave change as the distance from the source is doubled.

Solution: Using $\dfrac{I_1}{I_2}=\dfrac{r_2^{\,2}}{r_1^{\,2}}$

Here, $r_2 = 2r_1$

$$\Rightarrow I_2 = I_1\frac{r_1^{\,2}}{r_2^{\,2}} = I_1\frac{r_1^{\,2}}{(2r_1)^2} = \frac{I_1}{4}$$

$$I_2=\frac{1}{4}I_1$$

Thus, as the distance from the sound source is doubled, the intensity decreases by a factor of $\dfrac{1}{4}$.

☞ Applications of sound waves

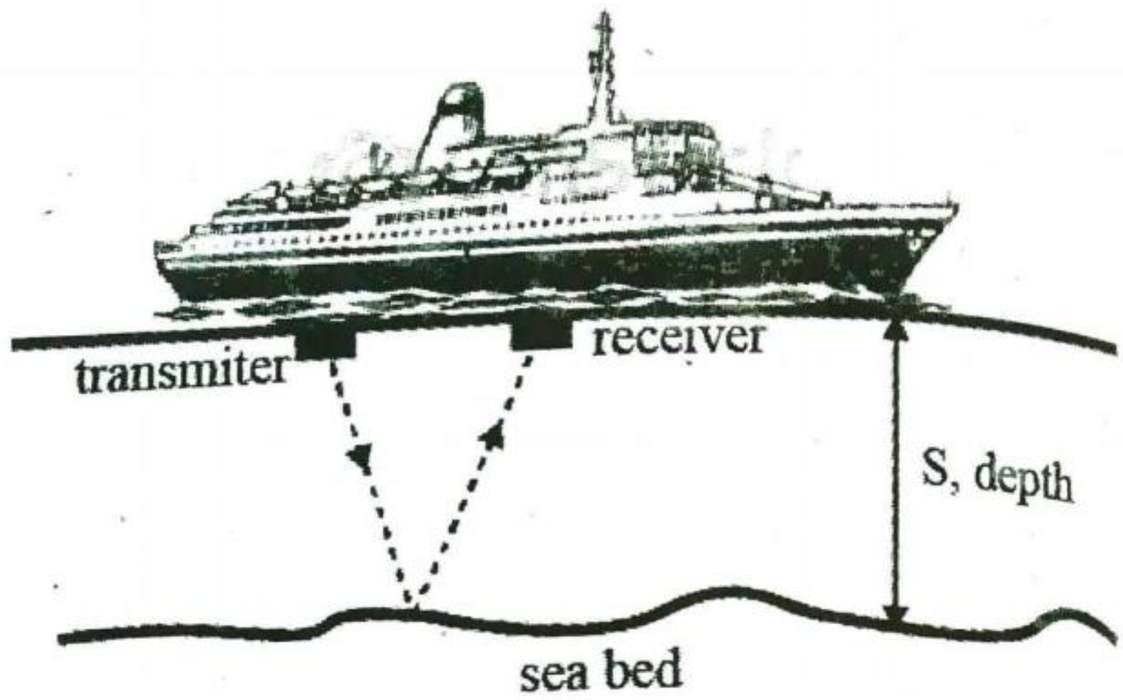
Sound waves have many uses, in addition to their uses in communication and music.

Most of these uses depend on the reflection or refraction of sound.

For example, dogs, dolphins and bats produce ultrasound that is sound of frequency above 20,000H_z and use the echoe to explore their environment and detect their prey-insects or fishs etc. this, we call *echolocation*. Ultrasound is also used in medicine to investigate the inside condition of a human body. Moreover, its reflection from the unborn baby allow doctors to monitor the progress and general condition of a developing baby.

Infrasound, that is sound of frequency below 20H_z, which humans feel as vibration rather than hear as sound, is used to elephants to communicate each other. What is more, ships on the ocean make use of the reflection of sound waves to determine the depth of the ocean. A sound wave is transmitted and the reflected sound is recoded in a receiver. Then, since

the speed of sound in a sea is known, and the time elapse between sending and receiving the sound is recorded, the distance from the ship to the seabed is calculated using the relation.



$$\boxed{S = \frac{vt}{2}}$$

Illustrative Examples

29. A ship sends a signal to the bottom of the ocean to determine the depth of the ocean. If the speed of sound in sea water is 1450m/s and the signal is received 1.5 seconds later, how deep is the ocean.

Solution: Using $S = \frac{vt}{2} = \frac{(1450)(1.5)}{2} = 1087.5\text{m}$

30. What is the distance of a thunder storm when you observe a 5 seconds delay between the flash of light and the sound of thunder. Take the speed of sound as 340m/s.

Solution: Now, $S = (v)\,(t) = (340)\,(5) = 1700\text{m}$

31. A boy says "hello" infront of a mountain and the echoe reaches him after 3 seconds. If the air temperature was 25°C at that moment. How far is the mountain from the boy.

Solution: Using the relation;

$$v = 20\sqrt{T}, \qquad T = 25 + 273 = 298\text{K}$$

$$v = 20\sqrt{298} = 345\text{m/s}$$

Next using the relation;

$$S = \frac{vt}{2} = \frac{(345)(3)}{2} = 414.5\text{m}$$

32. A person shouts at a cliff and hears an echo 1 sec later. If the cliff is 172m away, what was the speed of sound at that time?

Solution: from $S = \frac{vt}{2} \Rightarrow v = \frac{(2)(S)}{t}$

$$V = \frac{(2)(172)}{1} = 344\text{m/s}$$

General Secondary Education Certificate exam

33. What is the basic difference between mechanical and electromagnetic waves?
    A. Mechanical waves require material medium while electromagnetic waves do not
    B. Mechanical waves are longitudinal waves while electromagnetic waves are transversal
    C. Mechanical waves are compressions while electromagnetic waves are rarefactions
    D. Mechanical waves travel in one direction while electromagnetic waves do in all direction.

    Answer: A

34. When water wave is traveling from deeper to shallower water, what can you observe?
    A. The wave speed reduces
    B. The wavelength decreases
    C. The direction of the wave changes
    D. The frequency changes.

    Answer: All except D.

35. Which one of the following statement is correct?
    A. Waves of the same frequency always interfere constructively
    B. Waves of different frequency always interfere destructively
    C. The phase of waves affect the interference pattern
    D. The interference of waves does not depend on phase kind.

    Answer: C

36. Which one of the following is a term used to tell the maximum displacement of a body from equilibrium position.
    A. Wavelength
    B. Period
    C. Pulse
    D. Amplitude

    Answer: D

37. A student standing by the side of a pond, counted the number of water ripples passing a given point in 10 seconds. The student is trying to measure.
    A. Wavelength
    B. Amplitude
    C. Period
    D. Frequency

Explanation: $\frac{\text{No of oscillation}}{\text{Time taken}}$ = frequency.

Answer: D

38. As a wind blows across a field of grain, a wave can be seen to move across the field as the tops of the crops sway back and forth. What type of wave is this?
    A. Spherical
    B. Electromagnetic
    C. Transverse
    D. Longitudinal

Answer: D

39. A series of water waves pass an observer. He measured that, there is a distance of 6m between crests. What is the observer measuring?
    A. Displacement
    B. Height
    C. Amplitude
    D. Wavelength

Answer: D

40. As a wave moves along the strings of a guitar, the particles of the string:
    A. Remain stationary as the wave moves.
    B. Move antiparallel to the wave motion
    C. Move parallel to the direction of wave motion
    D. Move perpendicular to the direction of wave motion.

Answer: D

41. The following is true except
    A. Waves do interfere
    B. Waves need medium to move
    C. Wave speed = (wavelength) × (frequency)
    D. Wave carries energy as it propagates.

Answer: B. Not always the case, consider electromagnetic waves for example.

42. What is the purpose of the receiver antenna in radio communication?
    A. Separating the carried wave from modulation wave
    B. Converting electromagnetic wave to electrical oscillation
    C. Converting sound wave to electromagnetic wave
    D. Increasing amplitude of the radio wave.

Answer: B

43. Which quantities decrease as sound travels from water to air
    A. Speed and frequency
    B. Wavelength and speed

C. Frequency and wavelength
D. Frequency and period

Answer: B

44. An electromagnetic wave propagates in vacuum at speed C and wavelength λ. As the wave eneters some unknown medium, its speed decrease to 0.6c. What is the wavelength of the wave in the unknown medium?

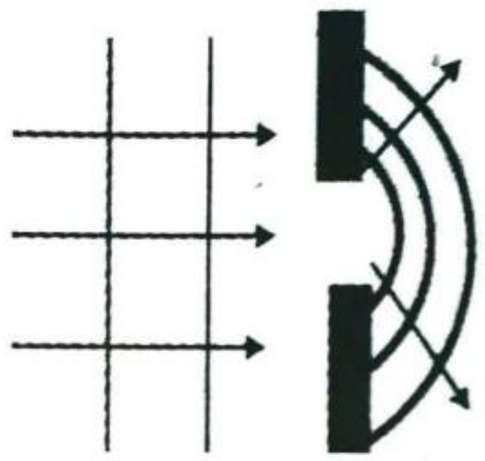
A. 0.6λ B. 0.8λ C. λ D. 1.25λ

Solution: $v = \lambda f \Rightarrow \lambda^1 = \frac{v^1}{f} = \frac{0.6C}{f} = 0.6\left(\frac{C}{f}\right)$

$$\lambda^1 = 0.6\lambda$$

Answer: A

45. The diagram below shows straight wave front passing through an opening in a barrier. The wave phenomena is called



A. Reflection
B. Refraction
C. Diffraction
D. Interference

Answer: C

46. An echo was received after 4 seconds by a sonar system to find a wrecked ship on the sea bed. If the speed of sound in sea water is 1445m/s. How deep is the ship?

A. 2890m B. 381.2m C. 5780m D. 1445m

Solution: $S = \frac{vt}{2} = \frac{(1445)(4)}{2} = 2890\text{m}.$

Answer: A

47. Sound travels fastest in materials that are

A. Highly compressible and very dense
B. Least compressible and very dense
C. Least compressible and less dense
D. Highly compressible and less dense.

Answer: B

48. If you are four times closer to an explosion than your friend, the intensity you detect as compared to your friend is

A. 4 times greater C. 10 times greater
B. 2 times greater D. 16 times greater

Solution: $\frac{I_1}{I_2} = \frac{r_2^{\,2}}{r_1^{\,2}}$, $r_2 = 4r_1$

$$I_1 = I_2 \frac{(4r_1)^2}{r_1^{\,2}} = 16I_2$$

Answer: D

49. The intensity of sound 1m away from a speaker is $8w/m^2$. What will be the intensity at a distance of 10m from the speaker.
    A. $8w/m^2$
    B. $0.8w/m^2$
    C. $2.5w/m^2$
    D. $0.08w/m^2$

Solution: $\frac{I_1}{I_2} = \frac{r_2^{\,2}}{r_1^{\,2}}, I_2 = I_1\left(\frac{r_1^{\,2}}{r_2^{\,2}}\right) = (8)\frac{(1)^2}{(10)^2} = 0.08^2/m^2$

Answer: D

50. What happens to a wave that passes obliquely from one medium to another?
    A. It changes wavelength
    B. It continuous to propagate in the same direction
    C. It continuous to propagate with the same speed
    D. It change from spherical to plane wave.

Answer: A

51. The property of a sound wave that is related to the loudness is.
    A. Wave length
    B. Pitch
    C. Intensity
    D. Speed

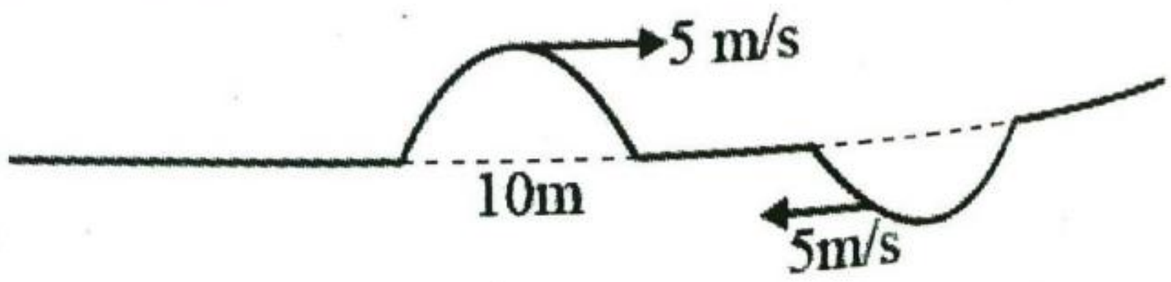
Answer: C

52. A bat flying in a cave emits sound and receives the echo 2 sec latter. How far is the cave wall. Take the speed of sound as 340m/s.
    A. 100m B. 680m C. 340m D. 170m

Solution: $S = \frac{vt}{2} = \frac{(340)(2)}{2} = 340m.$

Answer: C

53. The figure shows two identical transverse waves pulses each travelling at 5m/s toward, each other. At what time will completely destructive interference occur?
    A. 1 sec
    B. 4 sec
    C. 2 sec



D. 10 sec

Solution: $S = vt \Rightarrow t = \frac{s}{v} = \frac{10}{5} = 2\,\text{sec}.$

Answer: C

54. A rhinocer is calling her mate using infrasonic sound with a frequency of $5H_z$. If her mate is 476m away and the speed of sound is 340m/s. how many cycles of sound wave are there between the two animals?

A. 7 B. 6 C. 9 D. 5

Solution: $\lambda = \frac{v}{f} = \frac{340}{5} = 68\text{m}$, and the number of cycles is,

$$No = \frac{S}{\lambda} = \frac{476}{68} = 7.$$

Answer: A

55. A sound is moving towards a stationary observer with a constant speed. What happens to the frequency and intensity of the sound heard by the observe?

A. Both frequency and intensity increase
B. Both frequency and intensity decrease
C. Intensity increases and frequency decreases,
D. Intensity decreases and frequency increases

Answer: A

56. A girl measured the time between her claps and the echoes in a relatively warmer day of temperature 25°C and found that it is 0.68sec. What would be the time interval if she did the measurement in a cold day of temperature 15°C.

A. 0.346sec C. 0.580sec
B. 0.692sec D. 0.480sec

Solution: $v_1 = 20\sqrt{T_1} = 20\sqrt{298} = 345\text{m/s}$ and,

$$S = \frac{v_1 t_1}{2} = \frac{(345)(0.68)}{2} = (172.5)(0.68)$$

$$v_2 = 20\sqrt{T_2} = 20\sqrt{298} = 340\text{m/s}$$

$$S = \frac{v_2 t_2}{2} \Rightarrow 2(S) = v_2 t_2 \Rightarrow t_2 = \frac{2(S)}{v_2}$$

$$t_2 = \frac{2(172.5)(0.68)}{340} = 0.692 \text{ sec}.$$

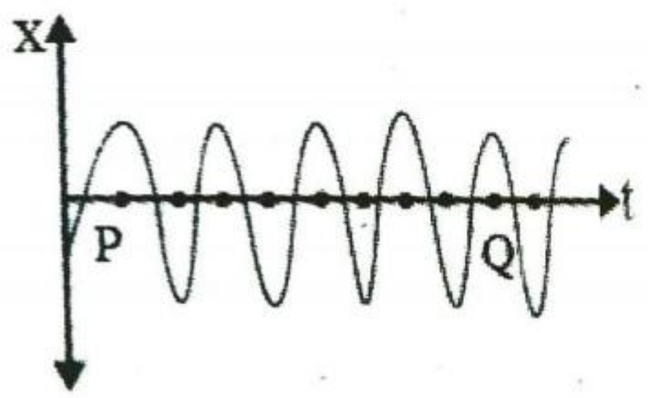
Answer: B

57. If the period of a transverse wave is 0.1 sec and its wavelength is 40cm..which one of the following interpretation is correct?
   A. The distance between two successive crests is 20cm
   B. In 0.1sec, the wave travels a distance of 40cm
   C. In one second the wave travels a distance of 40 cm
   D. In one second 20 crests pass a given point

   Explanation: Wavelength is the distance moved by a wave in one period.

   Answer: B

58. The diagram below shows displacement time graph of a wave generated by a certain source. The time interval between the crests P and Q is one second. What is the frequency of the source?
   A. $4H_z$
   B. $5H_z$
   C. $0.25H_z$
   D. $0.2H_z$



Solution: There are 5 crests between P and Q and hence there are 4 wave lengths (4λ). And from $S = vt \Rightarrow V = \frac{s}{t} = \frac{4\lambda}{1} = 4\lambda$ and $v = \lambda f \Rightarrow 4\lambda = \lambda f \Rightarrow f = 4H_z$

Answer: A

59. For a sound wave travelling in air, rarefaction occurs in the regions where the air pressure is
   A. higher than the atmospheric pressure
   B. lower than the atmospheric pressure
   C. equal to the atmospheric pressure
   D. higher or lower than the atmosphere.

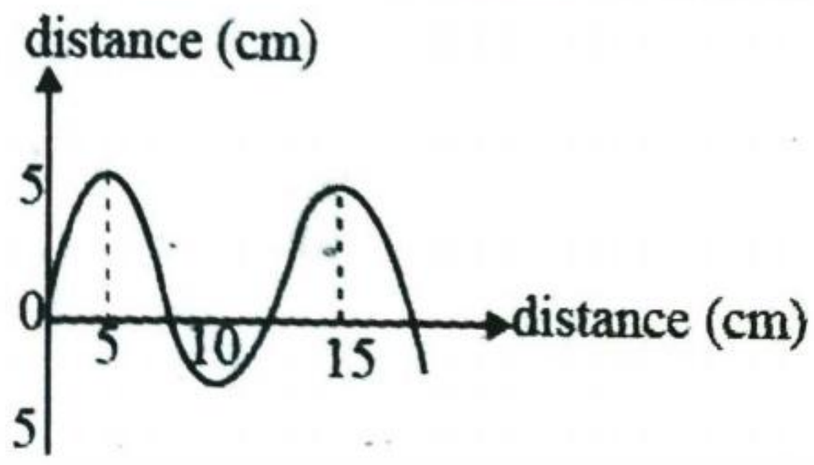
   Answer: B

Unit Eight – Conceptual Questions

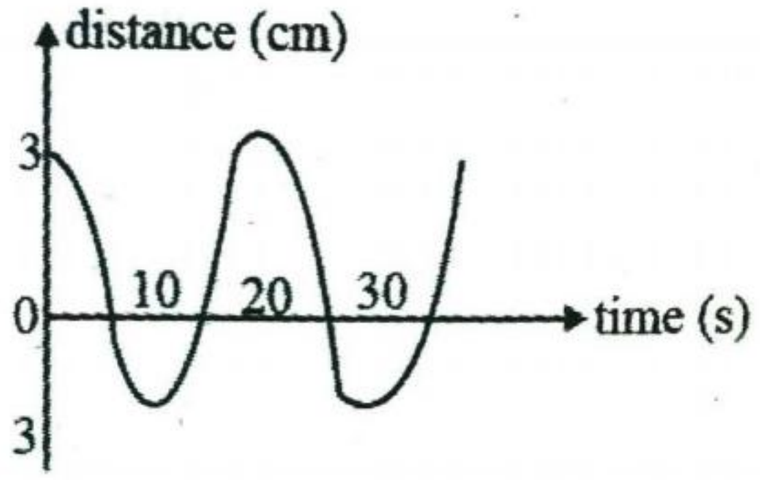
1. Sound travels faster when the medium is hot. Is this statement true or false?
2. If the frequency of a sound is an indication of the pitch, then the amplitude determines _____.
3. Where will sound travel faster in air, at sea level or at high mountains? Explain
4. X and Y are waves. In air, X travels faster than Y but has a much shorter wavelength. What can you say about the wave type?
5. Astronauts in a spaceship seen explosion on the surface of the moon. Why can they not hear it?
6. What will happen to the loudness and pitch of a wave when the amplitude and frequency of a wave are both increased.
7. Explain the basic difference between a sound wave and a light wave.
8. Does halving the frequency of a wave source halfed or double the speed of the waves?
9. Explain why a flash of lightning is usually seen before the clap of thunder is heard.
10. Two identical waves of amplitude 5cm meet. What will be the amplitude of the combined wave at a point where they interfere.
    a) Constructively
    b) Destructively

Unit Eight: problems

11. What are the values of the amplitude and the wavelength of the wave shown



12. Look at the wave shown and find its amplitude and period.



13. A) A wave has a frequency of 400Hz. What is its period?
    B) A wave has a period of 20 μ s. What is its frequency?

The wave equation

14. A guitarist plays a high note of frequency 2000H_z. The sound wave produced has a wavelength of 0.17m. What is the speed of sound in air?

15. A drummer plays a note with a frequency of 85H_z. What is the wavelength of this sound wave in air if its speed is 340m/s.

16. A radio station broad casts an FM signal with a wavelength of 2.85m. If the speed of radio waves is 3×10^8m/s, what is the frequency of the FM signal?

17. An electromagnetic wave has a wavelength of 10nm. Calculate its frequency.

18. Two sound waves are traveling through a container of nitrogen gas. The first wave has a wavelength of 1.5m, while the second wave has a wavelength of 4.5m. What can be said about their velocities?

19. Calculate the frequency and wavelength of the annoying sound made by a mosquito when it beats its wings at the rate of 600 wing beats per second. Assume the speed of sound is 344m/s.

20. The speed of a certain wave is 300m/s. If the frequency of the wave is 1500cycle/second. What is the wavelength of the wave.

Intensity of Sound

21. A speaker produces a sound at a power of 500w. Determine the intensity at:

    A. 2m B. 4m

22. The intensity of a sound source at a distance of 3m is 4w/m^2. Calculate the intensity received at a distance of 1m from the source.

Echo

23. A child claps her hands together whilst facing a tall building. The echo reaches her ears after 0.6sec. How far is she from the building? (v = 340m/s)

24. A ship is sailing in a part of the sea where the bed is 600m below the ship. The ship uses sonar to detect the seabed. How long will it take a pulse of sound to travel to the seabed and return to the ship? (Speed of sound in sea water is 1500m/s).

Unit Eight: explanation to Questions

1. True 2. Loudness 3. At sea level
4. Wave X is an electromagnetic wave
5. Because sound can not travel in vacuum
6. Both will increase.
7. Sound is a mechanical wave but light is an electromagnetic wave.
8. It halves the speed, since $V = \lambda f$, when f is halfed v is also decrease by half.
9. Because the speed of light is much larger than the speed of sound.
10. a) 10cm b) 0

Unit Eight: solutions to Problems

11. Amplitude = 5cm and wavelength = 15 – 5 = 10cm
12. Amplitude = 3cm and period = 30 – 10 = 20 seconds
13. a) $T = \frac{1}{f} = \frac{1}{400} = 0.0025\,\text{sec}$

    b) $f = \frac{1}{T} = \frac{1}{20\times10^{-6}} = 5000H_z$

14. $v = \lambda f = (0.17)(200) = 34\text{m/s}$
15. $\lambda = v/f = \frac{340}{85} = 4\text{m}$
16. $f = v/\lambda = \frac{3\times10^8}{2.85} \simeq 10^8 H_z$
17. $f = v/\lambda = \frac{3\times10^8}{10\times10^{-9}} = 3\times10^{16}H_z$
18. The wave speed is the same because the two sound waves are travelling through the same gas, $v = \sqrt{KT}$.
19. $f = 600H_z$ and $\lambda = v/f = \frac{344}{600} \simeq 0.6\text{m}$

Unit Nine

Motion in two Dimensions

The motion of a body on xy plane can be considered as two dimensional motion. In this case, the motion has x and y components. A detailed understanding of 2D motion is essential for physicists and engineers in the design of complex machines, rockets, roads etc. projectile motion and uniform circular motion can be taken as examples of 2D motion

9.1 Projectile Motion

A projectile can be a football kicked into the air, a stone thrown from a cliff, a drop of water, a package dropped from an airplane, a bullet shot from a rifle, a rocket launched into space or any object that moves through the air only under the influence of the force of gravity.

Projectile: Is any object that moves through the air without an engine or other motive force.

☞ **Note:** The path followed by a projectile is called a *trajectory*.

Conceptual Example

1. Can we consider the motion of an airplane as a projectile motion? Explain?

   Explanation: No, the motion of an airplane is due to an engine, which is a deriving force for the plane. In the same manner, a flying bird cannot also be considered as projectile motion.

   Depending on the direction of the initial velocity, a projectile motion can be *horizontal* or *inclined*.

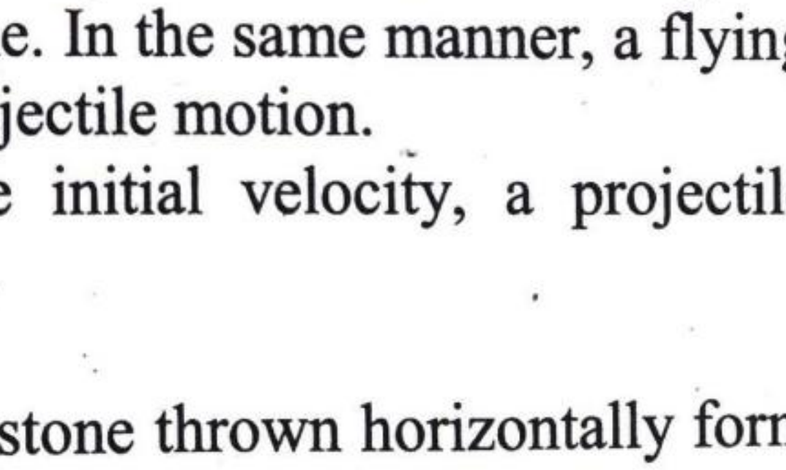
A) Horizontal projection

A metal ball that is rolling off a table or a stone thrown horizontally form a cliff, can be taken as horizontal projections.

Horizontal Projection: Is where the initial velocity of the projectile is along the horizontal.

Suppose a body is thrown horizontally with a velocity (u) from a height (h). Then,

→ The horizontal velocity of the body at any time is;



$$\boxed{V_x = u}$$

→ The vertical velocity of the body at any time is;

$V_y = U_y + gt$, but

$U = U_x$, $U_y = 0$

$$\boxed{V_y = gt}$$

→ The horizontal distance moved by the body at any time is,

$$\boxed{X = V_x t = ut}$$

→ The vertical distance moved by the body at any time is,

$$Y = U_y t + \frac{1}{2}gt^2, \text{ but } U_y = 0$$

$$\boxed{Y = \frac{1}{2}gt^2}$$

→ The time to reach the ground, can be calculated from:

$$X = Vt \Rightarrow t = \frac{X}{u}, \text{ and}$$

$$Y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2y}{g}}, \text{ so we have,}$$

$$\boxed{\frac{X}{u} = \sqrt{\frac{2y}{g}}}$$

→ The speed of the body at any time is obtained by,

$$\boxed{V = \sqrt{(V_x)^2 + (V_y)^2}}, \text{ where, } V_x = u \text{ and,}$$

$$V_y = gt$$

→ The angle with which the body hits the ground from the horizontal is given by,

$$\boxed{\tan\theta = \frac{V_y}{V_x}}$$

Conceptual Examples

2. If you throw a ball through a window horizontally from the second floor of an apartment, and at the same time a friend of you drop

another all from the same place, which ball will hit the ground first? Explain

Explanation: Both balls will hit the ground at the same time. This time is, $t = \frac{x}{u} = \sqrt{\frac{2y}{g}}$.

3. What can you conclude about the nature of the horizontal and the vertical velocities of a projectile.

Explanation: In the absence of air resistance there is no horizontal force acting on the body. Hence, the horizontal velocity is constant. But the vertical velocity does change, because gravity acts on the body vertically.

4. What can you say about the horizontal and vertical component motions of a projectile. Are they dependent or independent? Explain

Explanation: They are independent of each other because, the horizontal component remain unchanged but the vertical component is affected by the force of gravity. Thus, while the horizontal motion is a uniform motion, the vertical motion is a uniformly accelerated motion.

Illustrative Examples

5. A stone is thrown horizontally from the top of a building with an initial speed of 25m/s. The stone strikes the ground after covering a horizontal distance of 75m. ($g = 10m/s^2$).
   a) How high is the building
   b) What is the speed of the stone just when it hit the ground?

Solution: a) $y = \frac{1}{2}gt^2 = \frac{1}{2}(10)(t^2)$, but, from

$$x = ut,\ t = \frac{x}{u} = \frac{75}{25} = 3sec$$

$$\text{Thus, } y = \frac{1}{2}(10)(3)^2 = 45m$$

b) $v = \sqrt{v_x{}^2 + v_y{}^2}$, where

$v_x = u = 25m/s$ and $v_y = gt = (10)(3) = 30m$

$$v = \sqrt{(25)^2 + (30)^2} = 39m/s$$

6. A stone is thrown horizontally at 15m/s from the top of a cliff 44m high.
   a) How far from the base of the cliff does the stone hit the ground?
   b) How fast is it moving the instant before it hits the ground?

Solution: a) $x = ut$, but $t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2(44)}{10}} = 3\,\text{sec}$

$\Rightarrow x = (15)(3) = 45\text{m}$

b) $v = \sqrt{v_x^2 + v_y^2}$, $v_x = u = 15\text{m/s}$ and

$v_y = gt = 30\text{m/s}$

$v = \sqrt{(15)^2 + (30)^2} = 33.5\text{m/s}$

7. A stone is thrown horizontally at a speed of 10m/s from the top of a cliff 80m high.
   a) How long does it take the stone to reach the bottom of the cliff?
   b) How far from the base of the cliff does the stone strike the ground?

Solution: a) $y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2y}{g}} = \sqrt{\frac{2(80)}{10}} = 4\,\text{sec}$

b) $x = ut = (10)(4) = 40\text{m}$

8. A steel ball rolls with constant velocity across a tabletop 0.95m high. It rolls off and hits the ground 0.352m from the edge of the table.
   a) How fast was the ball rolling?
   b) What are the horizontal and vertical component velocities of the ball just before it hits the ground.
   c) What are the horizontal and vertical displacements of the ball at $t = 0.2\text{sec}$?

Solution: a) $\frac{x}{u} = \sqrt{\frac{2y}{g}} \Rightarrow \frac{0.352}{u} = \sqrt{\frac{2(0.95)}{10}} \Rightarrow u = 0.8\text{m/s}$

b) $v_x = u = 0.8$ and $v_y = gt = g\left(\frac{x}{u}\right) = 10\left(\frac{0.352}{0.8}\right) \Rightarrow v_y = 4.4\text{m/s}$

c) $x = ut = (0.8)(0.2) = 0.16m$

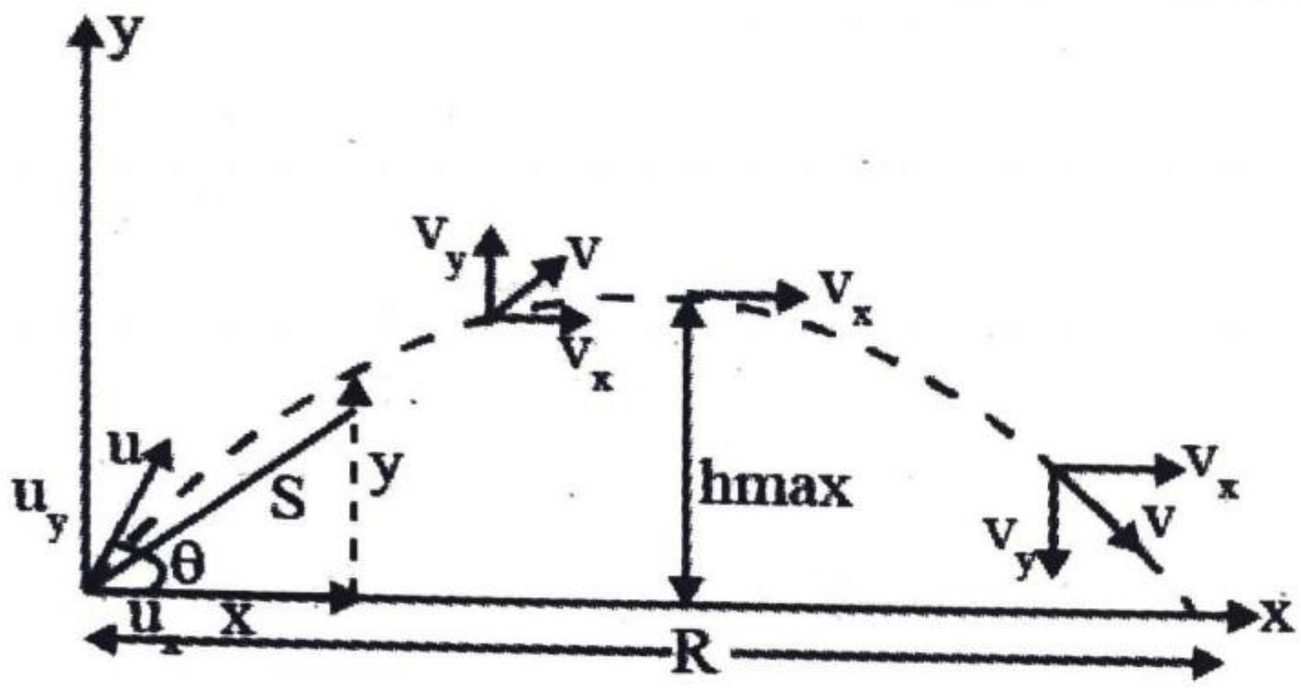
$$y = \frac{1}{2}gt^2 = \frac{1}{2}(10)(0.2)^2 = 0.2m$$

B) Inclined projection

If a body is thrown, shot, kicked or launched at some angle from the horizontal, then it is said to be a *projectile at an angle* or *inclined projection.*

Inclined projection: Is where, the initial velocity of the body is at an angle.

Suppose a body is project at an angle θ from the horizontal. Then, we have;

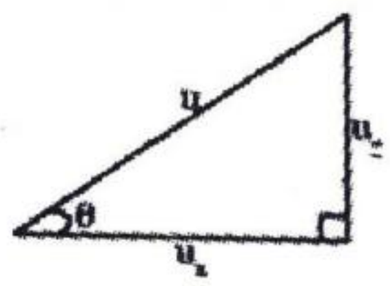


→ The horizontal component of the initial velocity.

$$u_x = u\cos\theta$$

→ The vertical component of the initial velocity

$$u_y = u\sin\theta$$



→ The horizontal velocity at any time is,

$$v_x = u_x = u\cos\theta \leftarrow \text{constant}$$

→ The vertical velocity at anytime is

$$v_y = u_y - gt = u\sin\theta - gt$$

→ The horizontal displacement at any time.

$$x = (v_x)(t) = (u\cos\theta)(t)$$

→ The vertical displacement at any time is,

$$y = u_y t - \frac{1}{2}gt^2 = u\sin\theta t - \frac{1}{2}gt^2$$

→ The displacement of the body at any time is

$$S = \sqrt{x^2 + y^2}$$

→ The maximum height reached by the body, is from,
$v_y^2 = u_y^2 - 2gh_{max}$, but $v_y = 0$ at h_{max}.

$$\Rightarrow \boxed{h_{max} = \frac{u_y^2}{2g} = \frac{u^2 \sin^2\theta}{2g}}$$

→ The time taken to reach the maximum height is obtained from,
$v_y = u_y - gt$, $v_y = 0$ at h_{max}.

$$\Rightarrow \boxed{t = \frac{u_y}{g} = \frac{u\sin\theta}{g}}$$

→ The total time taken to reach the ground (time of flight) is found by;

$y = u_y t_T - \frac{1}{2} g t_T^2$, but now $y = 0$

$$\boxed{t_T = \frac{2u_y}{g} = \frac{2u\sin\theta}{g}}$$

→ The maximum horizontal distance (Range) is calculated as;
$x = (u\cos\theta)(t)$, here $t = t_T$ and $x = R$

$$R = (u\cos\theta)\left(\frac{2u\sin\theta}{g}\right) = \frac{2(u_x)(u_y)}{g}$$

$$\Rightarrow \boxed{R = \frac{2u^2\sin\theta\cos\theta}{g} = \frac{u^2\sin(2\theta)}{g}}$$

→ The speed of the body at any time is then

$$\boxed{v = \sqrt{v_x^2 + v_y^2}}$$

From, $y = u\sin\theta t - \frac{1}{2} g t^2$, and

$x = u\cos\theta t \Rightarrow t = \frac{x}{u\cos\theta}$, substituting for t in y.

We have, $y = u\sin\theta\left(\frac{x}{u\cos\theta}\right) - \frac{1}{2} g\left(\frac{x}{u\cos\theta}\right)^2$

$$y = x\frac{\sin\theta}{\cos\theta} - \frac{gx^2}{2v^2\cos^2\theta}$$

$$\boxed{y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}} \leftarrow \text{equation of trajectory}$$

Conceptual Examples

9. For what angle of projection is the range maximum? Explain.

Explanation: $R = \dfrac{u^2\sin(2\theta)}{g}$, R is the maximum when $\sin 2\theta = 1 \Rightarrow 2\theta = 90°$. Thus, the range is maximum when $\theta = 45°$. In this case, $R = \dfrac{u^2}{g}$.

10. For what angle of projection is the height maximum? Explain.

Explanation: $h = \dfrac{u^2\sin^2\theta}{2g}$, h is maximum when $\sin^2\theta = 1$

$\Rightarrow \sin\theta = 1 \Rightarrow \theta = 90°$. In this case, $H = \dfrac{u^2}{2g}$

11. What are the two possible angles for which the range of a projectile is the same. Explain?

Explanation: the two possible angles for which the range of a projectile is the same for the same initial velocity are θ and $90 - \theta$.

12. Express the maximum height of a body interms of the range R, when both are maximum.

Explanation: R is maximum when $\theta = 45°$ and is given by, $R = \dfrac{u^2}{g}$, and the height is maximum at $90°$ and is given by,

$h = \dfrac{u^2}{2g} \Rightarrow h = \dfrac{1}{2}\dfrac{u^2}{g} = \dfrac{1}{2}R$, it is half of R

13. For what angle of projection is the range equal to the maximum height? Reason?

Explanation: $R = \dfrac{2u^2\sin\theta\cos\theta}{g}$ and $h = \dfrac{u^2\sin^2\theta}{2g}$,

In our case, $R = h$

$$\frac{2u^2 \sin\theta \cos\theta}{g} = h = \frac{u^2 \sin^2\theta}{2g}$$

$$\Rightarrow \tan\theta = 4$$

Illustrative Examples

14. A projectile is fired at an angle of 53° with the horizontal. The speed of the projectile is 200m/s. Calculate

a) time of flight

b) the range

Solution: a) $t = \frac{2u\sin\theta}{g} = \frac{(2)(200)(\sin 53°)}{10} = \frac{(2)(200)(0.8)}{10}$

$t = 32\text{sec}$

b) $R = \frac{2u^2 \sin\theta\cos\theta}{g} = \frac{(2)(200)^2 \sin 53°\cos 53°}{10}$

$$= \frac{(2)(40000)(0.8)(0.6)}{10}$$

$R = 3840\text{m}$

15. A projectile is fired at such an angle that the vertical component of its velocity is 49m/s. The horizontal component of its velocity is 60m/s.

a) How long does the projectile remain in air?

b) What horizontal distance does it travel?

Solution: a) $t_T = \frac{2u\sin\theta}{g} = \frac{2u_y}{g} = \frac{(2)(49)}{10} = 9.8\text{sec}$

b) $R = \frac{2u^2 \sin\theta\cos\theta}{g} = (u\cos\theta)\frac{(2u\sin\theta)}{g}$

$$R = \frac{(u_x)(2u_y)}{g} = \frac{(49)(2)(60)}{10} = 588\text{m}$$

16. A boy made a fascinating shot into the goal. The ball strikes inside part of the horizontal bar at its highest point of trajectory.

a) At what distance x did he shot the ball?

b) With what speed has the ball left his foot?

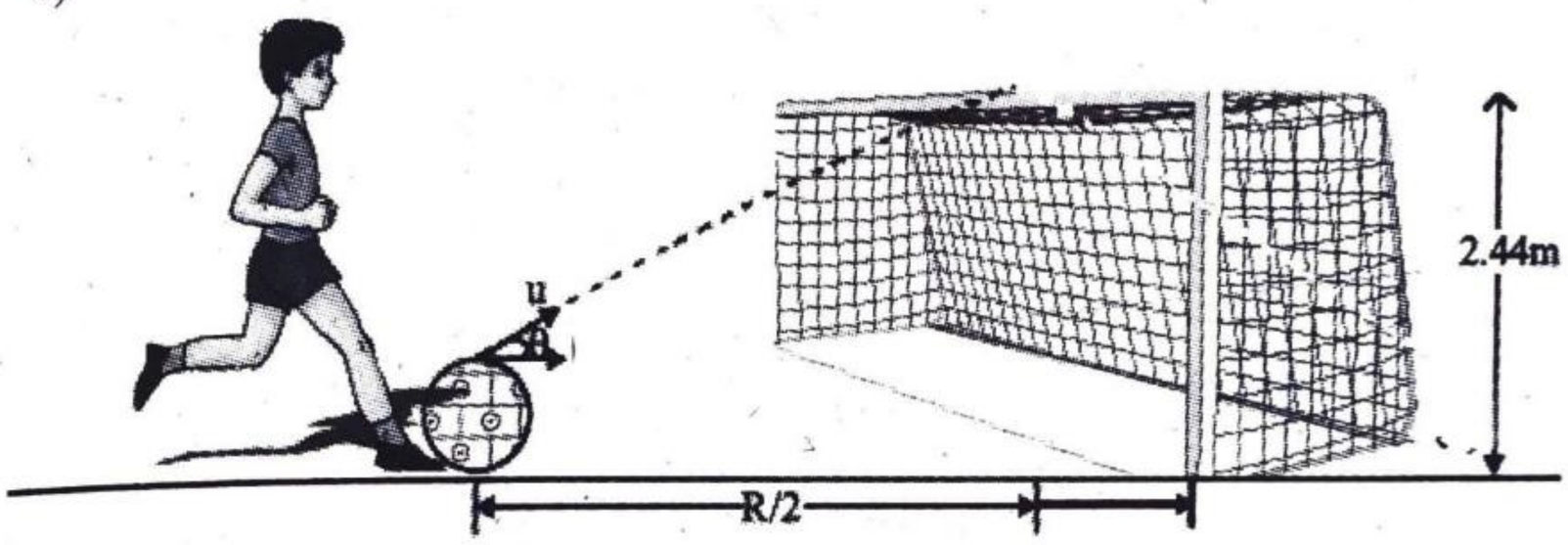
Solution: $h_{max} = \dfrac{u^2\sin^2\theta}{2g} = 2.44 = \dfrac{(u^2)(\sin^2 37)}{(2)(10)} = \dfrac{u^2(0.6)^2}{20}$

$$\Rightarrow u^2 = \frac{(2.44)(20)}{(0.6)^2}, u = 11.64\text{m/s}$$

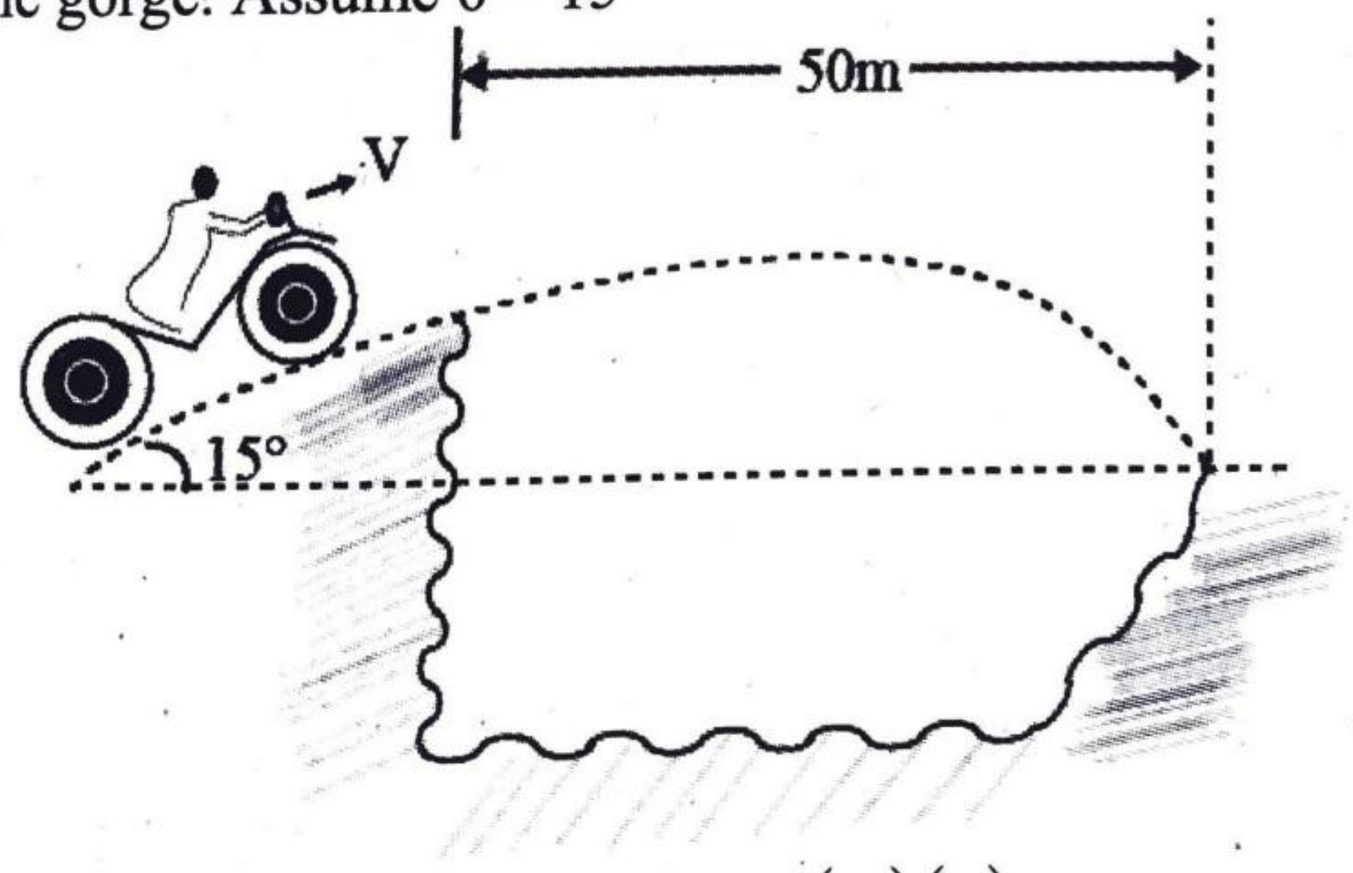
$$x = \frac{R}{2} = \frac{-u^2\sin 2\theta}{2g} = \frac{u^2\sin\theta\cos\theta}{g} = \frac{(11.64)^2(0.6)(0.8)}{10}$$

$x = 6.5\text{m}$

a) $x = 6.5\text{m}$, b) $u = 11.64\text{m/s}$



17. Vandam, the film star, attempts to jump a 50m wide gorge with a motorcycle. What is the minimum speed necessary for the motor to clear the gorge. Assume $\theta = 15°$

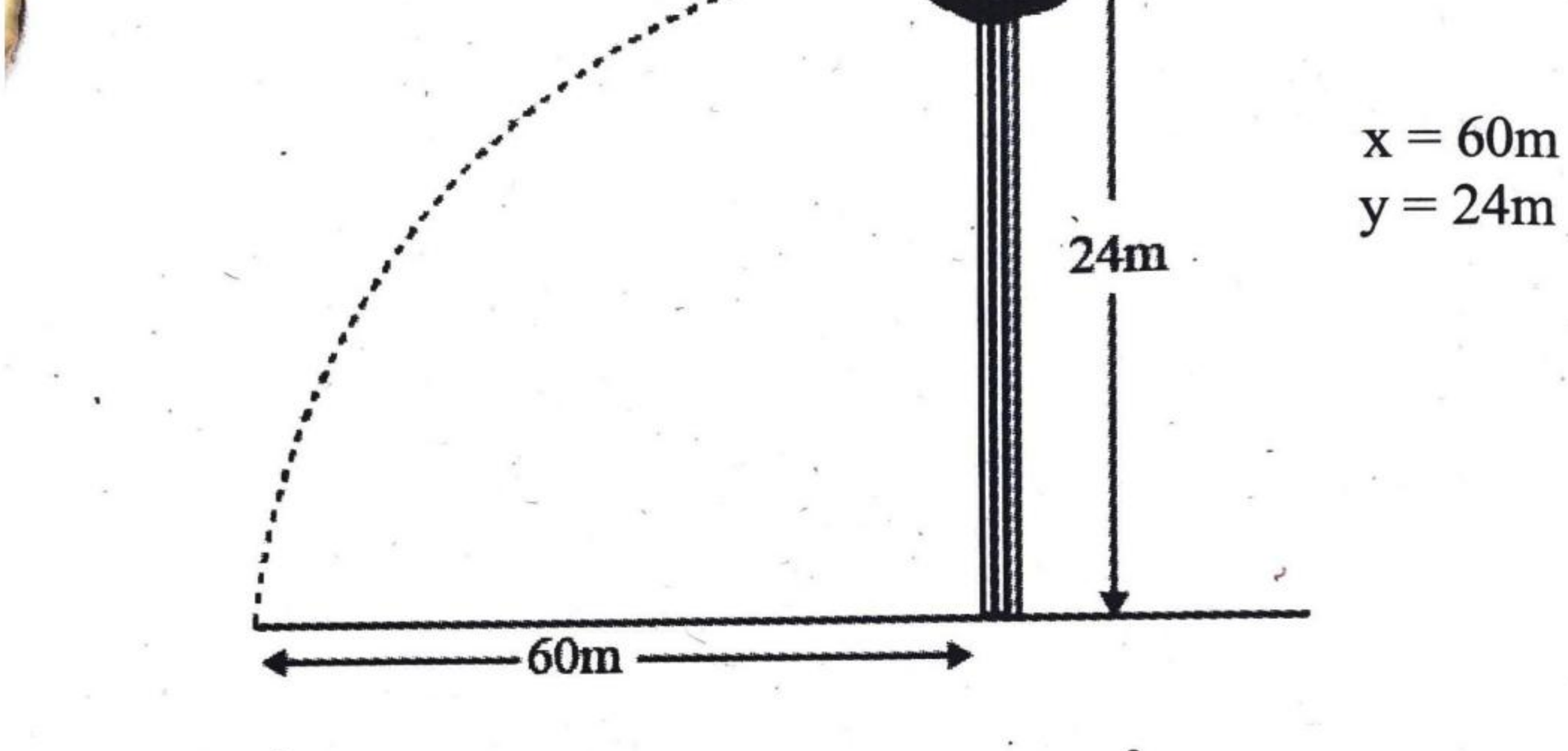


Solution: $R = \dfrac{u^2 2\sin\theta\cos\theta}{2g} \Rightarrow u^2 = \dfrac{(R)(g)}{2\sin\theta\cos\theta}$

$$u^2 = \frac{(50)(10)}{(2)(\sin 15)(\cos 15)} = \frac{500}{(2)(0.26)(0.97)}$$

$$u = 31.5 m/s$$

18. The Olympic Flame at the 1992 Barcelona Olympics was lit by a flaming arrow shot from a point located 60m horizontally and 24m vertically from the flame. Find the initial speed necessary for the arrow to reach the flame at the peak of its trajectory.



x = 60m
y = 24m

Solution: Now, $x = 60$, and $R = 2x = \frac{u^2}{g} \sin 2\theta$

$$\Rightarrow u^2 = \frac{2gx}{\sin 2\theta} = \frac{gx}{\sin\theta\cos\theta} \text{ and } y = x\tan\theta - \frac{gx^2}{2\frac{gx}{\sin\theta\cos\theta}\cos^2\theta}$$

$$\Rightarrow y = x\tan\theta - \frac{1}{2}x\tan\theta \Rightarrow \theta = 39°, \text{ for } x = 60m,\ y = 24m$$

and $u = 35 m/s$.

19. A ball is kicked at an angle of 37° from the horizontal giving it an initial speed of 30m/s.
   a) The horizontal and vertical components of its initial velocity
   b) The horizontal and vertical velocities at t = 1.5 sec
   c) The horizontal and vertical displacements at t = 1.5 sec
   d) The time it takes to reach the maximum height
   e) The speed of the ball just before it hits the ground.

f) The maximum height reached by the ball
g) The range of the ball
h) The displacement of the ball at t = 1.5 sec.

Solution: a) $u_x = u\cos\theta = (30)(\cos 37°) = (30)(0.8) = 24\text{m/s}$

$u_y = u\sin\theta = (30)(\sin 37°) = (30)(0.6) = 18\text{m/s}$

b) $v_x = u_x = 24\text{m/s}$ and $v_y = u_y - gt = 18 - (10)(1.5) = 3\text{m/s}$

c) $x = (u_x)(t) = (24)(1.5) = 36\text{m}$,

$$y = u_y t - \frac{1}{2}gt^2 = (18)(1.5) - \frac{1}{2}(10)(1.5) = 15.75\text{m}.$$

d) $$t = \frac{u_y}{g} = \frac{18}{10} = 1.8\text{sec and } t_T = 2t = 3.6\text{sec}$$

e) $v = \sqrt{v_x^2 + v_y^2}$, $v_x = 24\text{m/s}$ and $v_y = v_y - gt = 18 - 36 = -18\text{m/s}$

$$v = \sqrt{(24)^2 + (-18)^2} = 30\text{m/s}$$

f) $$h_{max} = \frac{u_y^2}{2g} = \frac{(18)^2}{2(10)} = 16.2\text{m}$$

g) $$R = \frac{2(u_x)(u_y)}{g} = \frac{(2)(24)(18)}{10} = 86.4\text{m}$$

h) $$S = \sqrt{x^2 + y^2}, \sqrt{(36)^2 + (15.75)^2} = 39.3\text{m}$$

Uniform Circular Motion

A car moving around a curve, a planet orbiting the sun, an electron orbiting an atom, a ball tied on a string and whirled around are examples of circular motion.

Uniform Circular Motion: Is the motion of a body along a curved path with constant radius at constant speed.

Conceptual Examples

20. When a body is in uniform circular motion, what can you say about its

a) Speed b) Velocity

Explanation: a) Speed is constant

b) Velocity is changing, because there is a continuous change in direction.

21. Can a projectile motion be considered as uniform circular motion? Explain

Explanation: No, the path of a projectile is a parabola hence, its radius is changing with time.

Centripetal Acceleration (a_c)

The acceleration of an object in a uniform circular motion always points towards the center of the circle and for that reason, it is called *center – seeking* or *centripetal acceleration*.

Suppose a body is in a uniform circular motion with position and velocity vectors shown. Since triangle of the position vector and triangle of velocity vector and similar, then the ratio of the length of corresponding sides are equal.

$\frac{\Delta r}{r} = \frac{\Delta v}{v}$, dividing both sides by Δt

$\frac{\Delta r}{r\Delta t} = \frac{\Delta v}{v\Delta t}$, but $\frac{\Delta r}{\Delta t} = v$ and $\frac{\Delta v}{\Delta t} = a$

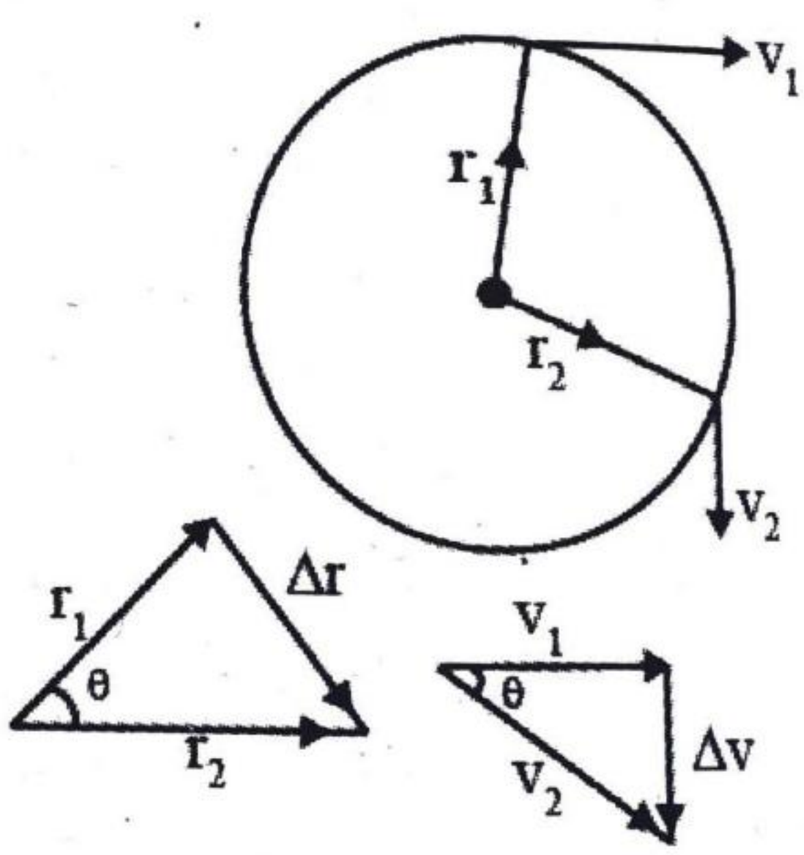
$\Rightarrow \frac{v}{r} = \frac{a}{v} \Rightarrow \boxed{a = \frac{v^2}{r}}$ ←centripetal acceleration for one complete cycle,

$S = 2\pi r$ and $t = T$

$\Rightarrow \boxed{v = \frac{s}{t} = \frac{2\pi r}{T}}$ substitute it into a_c,

$$a_c = \frac{\left(2\pi\, r/T\right)^2}{r}$$

$$\Rightarrow \boxed{a_c = \frac{4\pi^2 r}{T^2}}$$



Illustrative Examples

22. A stone tied to a string is whirled in a horizontal circle with a constant tangential speed of 10m/s. If the length of the string is 15cm. Find the centripetal acceleration.

Solution: $a = \frac{v^2}{r} = \frac{(10)^2}{(0.15)} = \frac{100}{0.15} = 666.7 m/s^2$

23. A rubber stopper is attached to a 0.93m string. The stopper is swung in a horizontal circle, making one revolution in 1.18sec. Find the centripetal acceleration of the stopper .

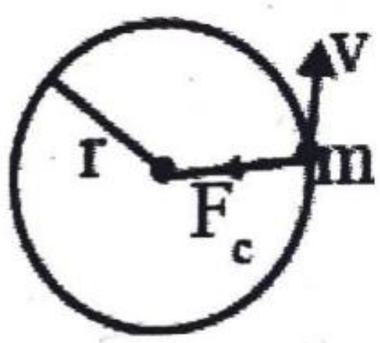
Solution: $a = \frac{4\pi^2 r}{T^2} = \frac{(4)(3.14)^2(0.93)}{(0.18)^2} = 26 m/s^2$

Centripetal Force (F_c)

The force that is responsible to help a body follow a circular path is known as centripetal force.

Centripetal force (F_c): Is the net force needed to make a body follow a circular path.

The central force on a body of mass (m) moving at a velocity (v) along a curved path of radius (r) is given by, $\boxed{F_c = \frac{mv^2}{r}}$.



☞ **Remember**, the centripetal acceleration is produced due to the centripetal force. And from Newton's second law, $\boxed{F_c = ma_c}$

Conceptual Examples

24. The gravitational force between the sun and the earth helps the earth to follow a circular path. What is the centripetal force for a car that turns around a curve. Explain?

Explanation: The frictional force between the tire and the road is the cause for the centripetal force on a car moving around a curve.

25. What will happen if the tension of a string on a rotating stone is larger than the centripetal force? Explain?

Explanation: The string will snap (break) and the stone starts moving in a straight line along the direction of the tangential velocity.

26. What will happen for a car moving along a curve if the friction between the road and the tyre is low. Explain?

Explanation: The car will skid or slide outward.

Illustrative Examples

27. The mass of the earth is 6×10^{24}kg. It travels at a steady speed of 30,000m/s around the sun at a radius of 1.5×10^{9}m. What is the centripetal force on the earth.

Solution: $F = \dfrac{mv^2}{r} = \dfrac{\left(6\times10^{24}\right)\left(30{,}000\right)^2}{1.5\times10^{9}}$

$F = 3.6 \times 10^{24}{}_{N}$

28. Suppose the maximum frictional force between the road and the tyres of a car is 6500N. If the mass of the car is 1200kg and the curve has radius of 85m, what is the maximum speed at which the car can safely bend.

Solution: from $F = \dfrac{mv^2}{r} \Rightarrow v^2 = \dfrac{Fr}{m}$, $v = \sqrt{\dfrac{Fr}{m}}$

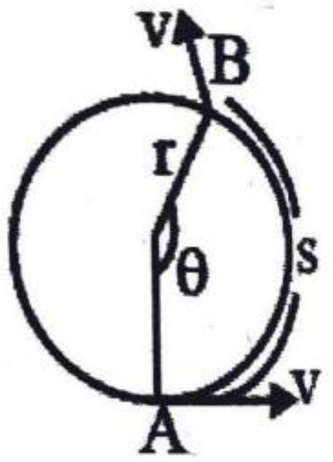
$$v = \sqrt{\frac{(6500)(85)}{1200}} = 21\text{m/s}$$

9.2 Rotational Kinematics

The angular displacement, angular velocity and the angular acceleration of a body in circular motion are discussed here.

Angular displacement (θ)

Consider a body that rotates around a circle of radius (r) with a tangential speed (v).



As the body moves from A to B, an arc length (S) on the circular path, a central angle (θ) is formed inside the circle and this angle is called *angular displacement*.

Angular displacement (θ): Is the central angle formed as a body moves from one point to the other on the circle.

The arc length (tangential displacement, S), the angular displacement (θ) and the radius of the circle (r) are related by:

$\boxed{S = r\theta}$, where $\theta = \dfrac{s}{r}$

The angular displacement is measured in revolution, degree or radian. But the SI unti of angular displacement is the *radian* (rad).

Radian: is the central angle for which the radius is equal to the tangential displacement.

That is, one radian is formed when r = s, and we have the following relations.

$\boxed{1\text{rev} = 360° = 2\pi\text{rad}}$, from which

$$1\text{rad} = \frac{360°}{2\pi} = \frac{360°}{(2)(3.14)} = 57.3°$$

Illustrative Examples

29. Convert a) 2rev into Degrees, Radians
b) 45° into rev, Radians
c) $\frac{\pi}{2}$ rad into Degrees, rev.

Solution: a) $2\text{rev} = 2(360°) = 720°$

$2\text{rev} = 2(2\pi\text{ rad}) = 4\pi\text{ rad}$

b) $45° = \left(\frac{45°}{360°}\right)\text{rev} = \frac{1}{8}\text{ rev}$

$$45° = \frac{(2\pi)(45°)}{360°}\text{ rad} = \frac{\pi}{4}\text{ rad}$$

c) $\frac{\pi}{2}\text{rad} = \frac{(360°)\left(\frac{\pi}{2}\right)}{4} = 90°$

$$\frac{\pi}{2}\text{rad} = \frac{(1)\left(\frac{\pi}{2}\right)}{2\pi} = \frac{1}{4}\text{rev}$$

30. A stone tied to a tring is whirled in a circular path of radius 25cm. **What is**
a) the angular displacement in radian when it covers 3rev
b) the tangential displacement of the stone in 3 rev.

Solution: a) $\theta = 3\text{rev} = (3)\ (2\pi\text{ rad}) = 6\pi\text{ rad}$

b) $S = r\theta = (0.25)\ (2\pi) = (0.25)\ (2)\ (3.14) = 4.7\text{m}$

Angular Velocity (ω): Is the rate of change of angular displacement.

$$\text{Angular velocity} = \frac{\text{angular displacement}}{\text{time taken}}$$

$\Rightarrow \boxed{\omega = \frac{\theta}{t}}$, where ω(omega) is a Greek letter.

The SI unit of angular velocity (ω) is rad/sec.

Remember; for one complete revolution, θ is equal to 2πrad and t become period (T). So for one complete cycle,

$$\boxed{\omega = \frac{\theta}{t} = \frac{2\pi}{T}}$$

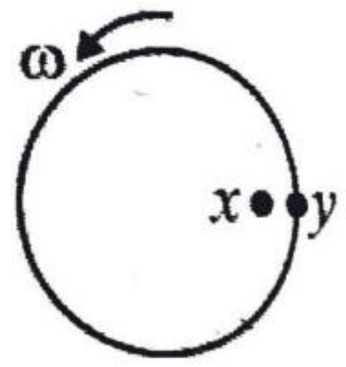
Note also that; the angular velocity (ω), the tangential velocity (v) and the radius (r) are related by,

$$\boxed{v = r\omega}$$

Conceptual Examples

31. Consider a spinning disk, shown. What can you say about the angular and tangential velocities of points x and y on the disk?

Explanation: both points x and y do have the same angular velocity as they both cover the same angle in the same time. However, point *y* has a higher tangential velocity than *x* as it covers a greater distance in the same time



Illustrative Examples

32. A bicycle wheel turns at an angular velocity of 20rad/se. If the radius of the wheel is 30cm. Calculate
    a) the tangential speed of the wheel
    b) the distance travelled by the bicycle in 30 minutes.

Solution: a) $v = r\omega = (0.3)\ (20) = 6\text{m/s}$

b) $S = vt = (6)\ (30)(60) = 10{,}800\text{m} = 10.8\text{km}$

33. The angular velocity of a belt of grindstone is 30rad/s. To what angle does the belt rotate in one minute?

Solution: $\theta = (\omega)\ (t) = (30)\ (60) = 1800\text{rad}$

Angular Acceleration (α): Is the rate at which angular velocity changes.

$$\text{Angular acceleration} = \frac{\text{change in angular velocity}}{\text{time taken}}$$

$\Rightarrow \boxed{\alpha = \frac{\Delta\omega}{t}}$, where α(alpha) is a Greek letter.

The SI unit of angular acceleration (α) is rad/s^2.

In the general case, a body in rotational motion can have different accelerations. Those are

a) **Tangential acceleration (a_t):** which is due to the change in tangential velocity through time and is given by;

$$\boxed{a_t = \frac{\Delta v_t}{t}}, \quad \Delta v = r\Delta\omega \Rightarrow \boxed{a_t = \alpha r}$$

b) **Centripetal (radial) acceleration (a_c):** which arises due to a net centripetal force and is given by;

$$\boxed{a_c = \frac{v^2}{r}}, \text{ but } v = r\omega \Rightarrow \boxed{a_c = r\omega^2}$$

c) **Angular acceleration (α):** is resulted due to the change of angular velocity, and is obtained by:

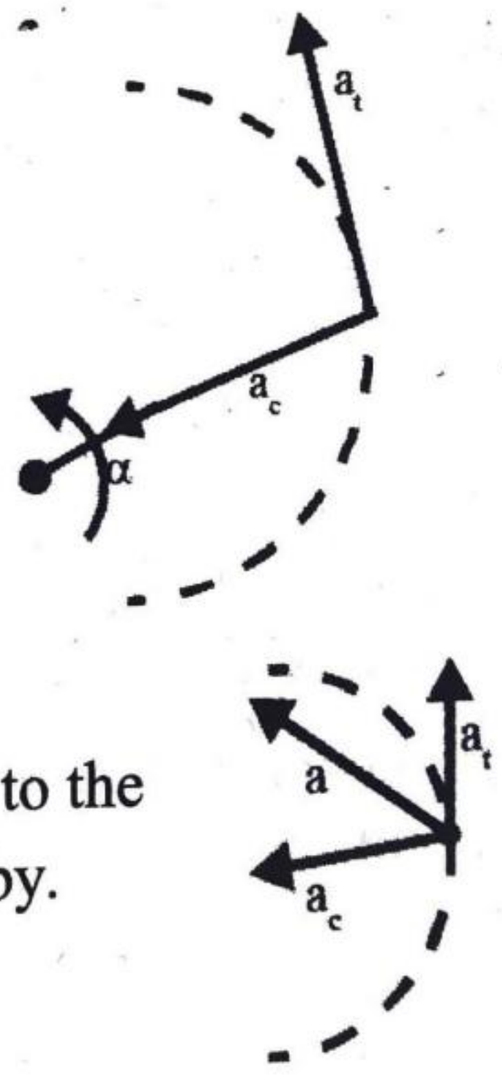
$$\boxed{\alpha = \frac{\Delta\omega}{t}}$$

Graphically, the three accelerations can be summarized as follows

$\alpha \rightarrow$ Angular acceleration

$a_t \rightarrow$ tangential acceleration

$a_c \rightarrow$ centripetal acceleration



Moreover, the effective acceleration of a body due to the centripetal and tangential accelerations is obtained by.

$$\boxed{a = \sqrt{(a_t)^2 + (a_c)^2}}$$

Conceptual Examples

34. What can you say about the tangential, centripetal and effective accelerations of a body that is in uniform circular motion.

Explanation: The tangential acceleration is zero, because the tangential speed is constant, $a_t = \frac{\Delta v}{t} = 0$. But, the centripetal acceleration is with a non zero constant value of $a_c = \frac{v^2}{r}$ and in this case the effective acceleration equals the centripetal acceleration; $a = \sqrt{a_c^{\,2} + a_t^{\,2}}$, $a_t = 0 \Rightarrow a = a_c$.

Illustrative Examples

35. A wheel starts from rest and rotates with uniform angular acceleration to attain an angular velocity of 2rev/sec in one second. The radius of the wheel is 50cm.

Find a) the angular acceleration
b) the tangential acceleration
c) the total acceleration of a point on the rim.

Solution: a) $\alpha = \frac{\Delta\omega}{t} = \frac{\omega - \omega_o}{t} = \frac{2(2\pi) - 0}{1} = 4\pi \text{ rad/s}^2$

b) $a_t = \alpha r = (4\pi)(0.5) = 2\pi \text{ m/s}^2 = 6.3 \text{ m/s}^2$

c) $a = \sqrt{a_t^{\,2} + a_c^{\,2}}$, $a_c = \frac{v^2}{r} = r\omega^2 = 8\pi^2 \text{m/s}^2$

$$\Rightarrow a = \sqrt{(2\pi)^2 + (8\pi^2)^2} = \sqrt{(2\times 3.14)^2 + (8\times(3.14)\pi^2)^2}$$

$$a = 79 \text{m/s}^2$$

Equation of constant angular acceleration motion

The equations used for linear uniformly accelerated motion can be used for uniformly accelerated angular motion with some symbol modifications. Here we use, (ω) in place of (v), (α) instead of (a) and (θ) rather than (s).

Correspondence between uniformly accelerated linear motion and circular motion with uniformly angular acceleration.

Linear Motion	**Angular Motion**
$v = u + at$	$\omega = \omega_o + \alpha t$

$$v_{av} = \frac{u+v}{2} \qquad \omega_{av} = \frac{\omega_o + \omega}{2}$$

$$S = \left(\frac{u+v}{2}\right)t \qquad \theta = \left(\frac{\omega_o + \omega}{2}\right)t$$

$$S = ut + \frac{1}{2}at^2 \qquad \theta = \omega_o t - \frac{1}{2}\alpha t^2$$

$$S = vt - \frac{1}{2}at^2 \qquad \theta = \omega t - \frac{1}{2}\alpha t^2$$

$$S = v^2 = u^2 2as \qquad \omega^2 = \omega_o{}^2 + 2\alpha\theta$$

Illustrative Examples

36. A ball is attached to a string of length 2m and is rotated with uniform angular acceleration on a horizontal circle. If the ball attains an angular velocity of 2rad/s in 2sec. starting from rest. Calculate
    a) the angular displacement
    b) the angular acceleration and the linear velocity
    c) the tangential distance the ball covers in 2sec.
    d) the total acceleration of the ball

Solution: a) $\theta = \frac{(\omega - \omega_o)t}{2} = \frac{(2+0)\,2}{2} = 2\text{rad}$

b) $\alpha = \frac{\omega - \omega_o}{t} = \frac{2-0}{2} = 1\text{rad/sec}^2$

and, $v = r\omega = (2)(2) = 4\text{m/s}$

c) $S = r\theta = (2)(2) = 4\text{m}$

d) $a = \sqrt{a_t{}^2 + a_c{}^2}$, $a_c = \frac{v^2}{r} = \frac{(4)^2}{2} = 8\text{m/s}^2$

and $a_t = r\alpha = (2)(1) = 2\text{m/s}^2$

then, $a = \sqrt{(2)^2 + (8)^2} = 8.25\text{m/s}^2$

37. A belt runs on a wheel of radius 50cm. during the time the wheel retards from an initial angular velocity of 2rad/s to rest, 25m of the belt passes over the wheel. Find
    a) the angular acceleration

b) the angle turned through before it stops.

Solution: a) $\omega^2 = \omega_o{}^2 + 2\alpha\theta$, but $\omega = 0$ and $\theta = \frac{s}{r} = \frac{25}{0.5} = 50\text{rad}$

$$\Rightarrow \quad \alpha = \frac{-\omega_o{}^2}{2\theta} = \frac{-(2)^2}{(2)(50)} = -0.04\text{rad/s}^2$$

b) $\theta = \frac{s}{r} = \frac{25}{0.5} = 50\text{rad}$

38. How large an angular acceleration must a wheel have if it is to accelerate from rest to 460rad/s after 5 revolutions?

Solution: $\omega^2 = \omega_o{}^2 + 2\alpha\theta,\ \omega_o = 0$

$$\alpha = \frac{\omega^2}{2\theta},\ \theta = 5\text{rev} = 5(2\pi)\text{rad} = 10\pi\text{rad}$$

$$\alpha = \frac{(60)^2}{(2)(10\times 3.14)} = 57.3\text{rad/s}^2$$

39. The angular acceleration of a wheel is 2rev/s^2. To what angle does it rotate starting form rest in 20 seconds?

Solution: $\theta = \omega_o t + \frac{1}{2}\alpha t^2, \omega_o = 0$ and $\alpha = 2(2\pi) = 4\pi\text{rad/s}^2$.

$$\theta = \frac{1}{2}(4\pi)(20)^2 = 800\pi\text{rad}$$

40. The wheel of an engine is initially rotating at an angular speed of 24 rad/sec. When the engine is turned off, the fly wheel decelerates at constant rate and comes to rest after 3 sec. what is the angle though which the fly wheel rotates before combing to rest.

Solution: $\theta = \frac{1}{2}(\omega + \omega_o)t, \omega = 0$ and

$$\theta = \frac{1}{2}(0 + 24)3 = 36\text{rad}$$

9.3 Rotational dynamics

Rotational kinetic energy, torque, angular momentum and work done by a torque are discussed here under.

Moment of inertia

The manner in which the mass of a body is distributed and the tendency of the rotating body to oppose the agent that attempts to change its angular state of motion is referred to as moment (rotational) inertia.

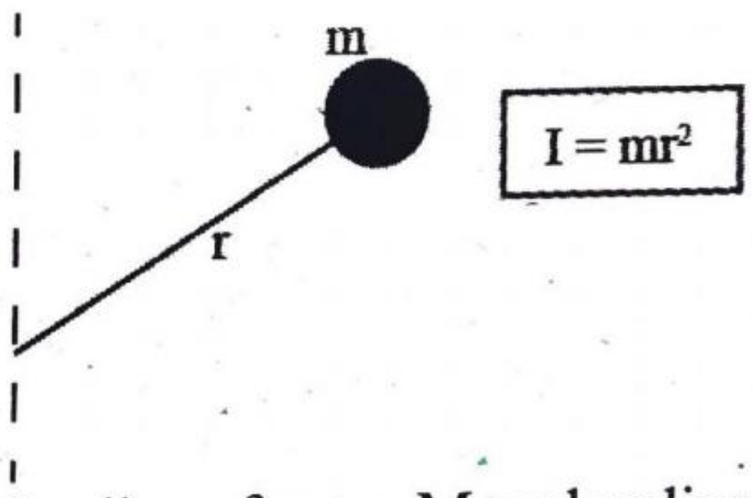
Moment of inertia (I): Is a measure of the distribution of the mass of a body in relation to its axis of rotation.

☞ The moment of inertia of a body depends on;

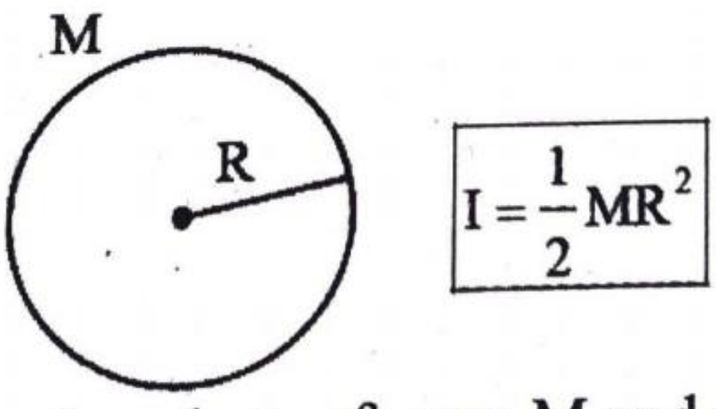
→ The mass of the body
→ The size of the body
→ The shape of the body, and
→ The axis of rotation of the body

Bodies with different shapes do have different moments of inertial. As an example, consider the following.

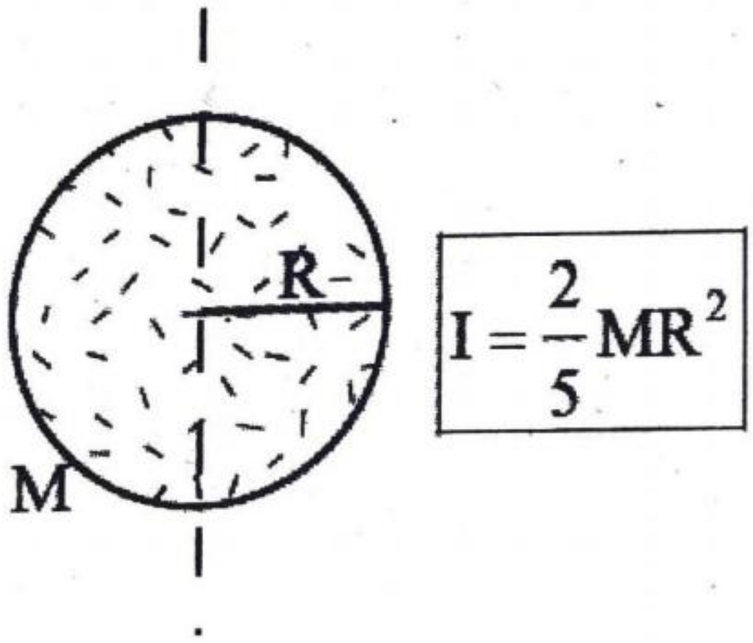
→ Moment of inertia of a point mass m at a distance r from the axis of rotation is



→ Moment of inertia of a disc of mass M and radius R about its center is



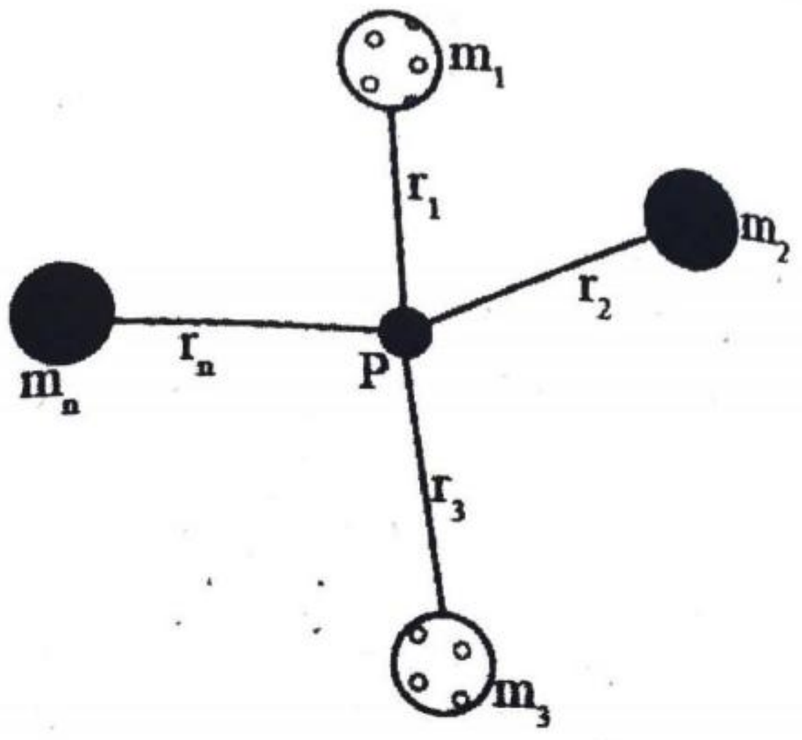
→ Moment of inertia of a sphere of mass M and radius R about its center is



Note that: the SI unit of moment of inertia is kgm^2. And,

The total moment of inertia of several point masses $m_1, m_2, m_3 \ldots m_n$ that are at distances of $r_1, r_2, r_3 \ldots r_n$ from the axis of rotation, is simply the sum of the moment of inertia of each mass about that axis.

That is, if we have, the arrangement shown;



Then the total moment of inertia of the system of the point mass about point p is

$I = I_1, I_2, I_3 + \ldots I_n$

$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + \ldots\ldots\ldots\ldots m_nr_n^2$$

Conceptual Examples

41. Can we say that the moment of inertia of a body is constant about any axis of rotation?

Explanation: No. the moment of inertia of a body changes as the axis of rotation for that body changes.

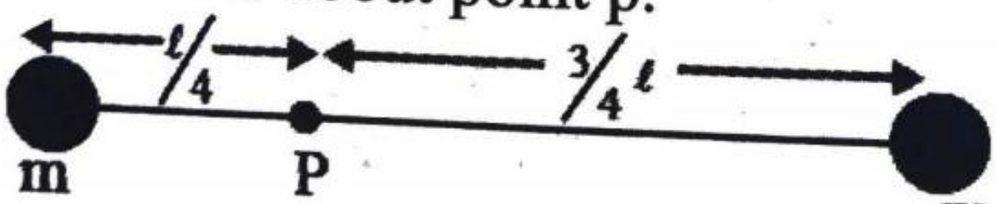
Illustrative Examples

42. Calculate the moment of inertia about the center of
    a) a disc of mass 5kg and radius 10cm
    b) a sphere of mass 10kg and radius 2cm

Solution: a) $I = \frac{1}{2}MR^2 = \frac{1}{2}(5)(0.1)^2 = 0.25kgm^2$

b) $I = \frac{2}{5}MR^2 = \frac{2}{5}(10)(0.02)^2 = 0.0016kgm^2$

43. Calculate the moment of inertia of two equal masses of m = 0.64kg attached on a massless rod of length ℓ = 85cm when the axis of rotation is about point p.



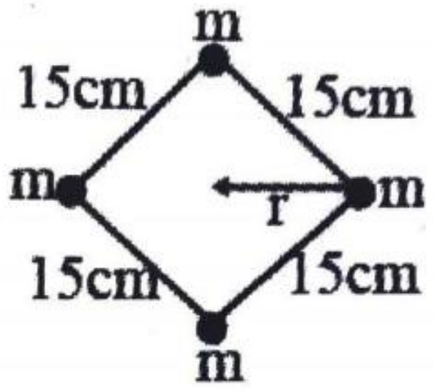
Solution: $I = m_1r_1^2 + m_2r_2^2$

$$= m\left(\frac{1}{4}\ell\right)^2 + m\left(\frac{3}{4}\ell\right)^2, \; r_1 = \frac{1}{4}\ell, \; r_2 = \frac{3}{4}\ell$$

$$I = \frac{1}{16}m\ell^2 + \frac{9}{16}m\ell^2 = \frac{10}{16}m\ell^2 = \frac{5}{8}m\ell^2$$

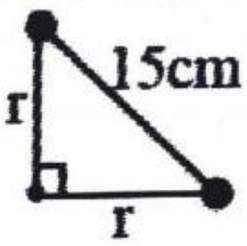
$$I = \frac{5}{8}(0.64)(0.85)^2 = 0.29\text{kgm}^2$$

44. What is the moment of inertia of four equal masses 10g each situated at the corners of a square of side 15cm, about an axis passing through the center of the square perpendicular to the plane of the square.



Solution: The distance r form the axis of rotation to each mass is

$$r^2 + r^2 = (15)^2 \Rightarrow 2r^2 = (15)^2 \Rightarrow r = \frac{15}{\sqrt{2}} = 7.5\sqrt{2}$$



$\Rightarrow r = 10.6\text{cm}$

Then, $I = m_1r_1^2 + m_2r_2^2 + m_3r_2^2 + m_4r_4^2$, but $m_1 = m_2 = m_3 = m_4$ and

$r_1 = r_2 = r_3 = r_4$

$\Rightarrow I = mr^2 + mr^2 + mr^2 + mr^2,$

$I = 4mr^2 = 4(0.01)(0.106)^2$

$I \simeq 4.5 \times 10^{-4}\text{kgm}^2$

Rotational Kinetic energy

The rotational kinetic energy of a body of moment of inertia (I) rotating at an angular speed of (ω) is given by;

$$\boxed{K_{Er} = \frac{1}{2}I\omega^2}$$

The SI unit of rotation kinetic energy is Joule(J).

Illustrative Examples

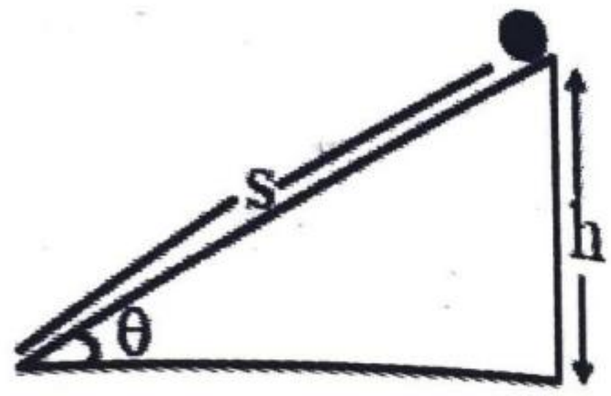
45. A grindstone with a radius of 0.61m is being used to sharpen an ax. If the linear speed of the stone is 15m/s and its rotational kinetic energy is 13J what is its moment of inertia?

Solution: $K_E = \frac{1}{2}I\omega^2$, but $v = \omega r \Rightarrow \omega = \frac{v}{r} = \frac{1.5}{0.61} = 2.46$ rad/s

$$\Rightarrow I = \frac{2K_E}{\omega^2} = \frac{2(13)}{(2.46)^2} = 4.3\text{kgm}^2$$

46. What is the acceleration of a point mass rolling down a plane inclined at angle θ with the horizontal?

Solution: from conservation of energy



$$M_{Ei} = M_{Ei}$$

$$P_E = K_E + K_{Er}$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2, \text{ but } I = mr^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\left(mr^2\right)\omega^2, \ \omega = \frac{v}{r}$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\left(mr^2\right)\left(\frac{v}{r}\right)^2$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}mv^2$$

$$gh = \frac{1}{2}v^2 + \frac{1}{2}v^2 = v^2$$

And from, $v^2 = u^2 + 2as$, $u = 0$, here

$$v^2 = 2as$$

$$gh = 2as \Rightarrow a = \frac{1}{2}g\frac{h}{s}, \text{ but } h/s = \sin\theta$$

$$a = \frac{1}{2}g\sin\theta$$

Torque

Torque is the tendency of a force to rotate a body about a given pivot point.

Torque (τ): Is the turning effect of a force

Torque = (Force) × (perpendicular distance)

$\Rightarrow \boxed{\tau = Fr_{\perp}}$, where τ (taw) is a Greek letter.

Torque is a vector quantity and its SI unit is Newton meter (N.m). Applying the concept of Newton's second law, torque can also be calculated as,

Torque = (moment of inertia) × (Angular acceleration)

$$\boxed{\tau = I\alpha}$$

Conceptual Examples

47. The SI unit of torque is N.m, can we then conclude that torque and work (energy) do have the same SI unit? Explain?

Explanation: No. Torque has SI unit of N.m and work has SI unit of Nm which is usually called joule (J) Hence N.m is different from Joule (J).

48. If the force and the distance of the force from the pivot point are both nonzero. Can the torque acting on the body ever be zero? Explain?

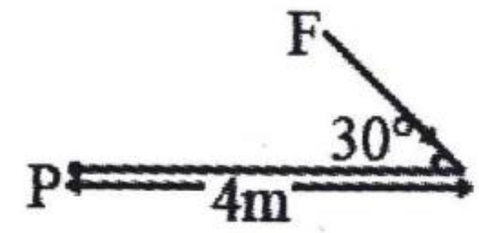
Explanation: Yes, even if both the force and lever arm are both nonzero, the torque on the body can be zero whenever the force and the lever arm are parallel or anti parallel to each other.

Illustrative Examples

49. A force of 20N is acting at a perpendicular distance of 2m from a point of rotation. Calculate the torque acting on the body?

Solution: $\tau = Fr_{\perp} = (20)(2) = 40\text{N.M}$

50. Find the magnitude and direction of the torque due to the 100N force about point p?



Solution: $\tau = Fr_{\perp} = Fr\sin\theta$

$\tau = (100)\,(4)\,(\sin 30°) = (100)\,(4)\,(0.5)$

$\tau = 200\text{N.m}$ clockwise

51. Find the torque of a wheel of moment of inertia 30kgm^2 rotating with angular acceleration of 0.2rad/s^2?

Solution: $\tau = I\alpha = (30)\,(0.2) = 6\text{N.m}$

52. A ball of mass 50g rotates in a circular path of radius 20cm if a force of 10N acts on the ball for 5 seconds in the direction of rotation. By how much does its angular velocity increase?

Solution: $I = mr^2 = (0.05)\,(0.2)^2 = 0.002\text{kgm}^2$

$$\tau = Fr = I\alpha\,,\ \alpha = \frac{\Delta\omega}{\Delta t}$$

$$\Rightarrow Fr = I\frac{\Delta\omega}{\Delta t}$$

$$\Delta\omega = \frac{Frt}{I} = \frac{(10)(0.2)(5)}{(0.002)} = 5000\text{rad/sec}$$

Work done by torque

The work done by a torque is the torque multiplied by the angle it turn through.

$$\text{Work} = (\text{torque}) \times (\text{angular displacement}) \Rightarrow \boxed{W = \tau\theta}$$

Illustrative Examples

53. A torque of 100Nm rotates a door by 45° about its hinges. What is the work done on the door?

Solution: $W = \tau\theta \Rightarrow \theta = 45° = \dfrac{\pi}{4}\text{rad}$

$$W = (100)\left(\frac{\pi}{4}\right) = (100)\left(\frac{3.14}{4}\right) = 78.5\text{J}$$

54. Find the torque when the work done to move through an angle of $\dfrac{\pi}{2}$ rad is 50J.

Solution: $W = \tau\theta \Rightarrow \tau = \dfrac{\omega}{\theta} = \dfrac{50}{\pi/2} = \dfrac{100}{\pi} = \dfrac{100}{3.14} \simeq 32\text{Nm}$

55. A ball of mass 25 g rotates in a circular path of radius 40cm. If a force of 10N acts on the ball for 10 seconds in the direction of rotation. Calculate

a) The moment of inertia
b) The angular acceleration
c) The angular velocity
d) The work done on the ball

Solution: a) $I = mr^2 = (0.025)\,(0.4)^2 = 4\times 10^{-3}\text{kgm}^2$

b) $\tau = I\alpha = Fr \Rightarrow \alpha = \dfrac{Fr}{I} = \dfrac{(10)(0.4)}{(4\times 10^{-3})} = 1000\text{rad/s}^2$

c) $\tau = Fr = I\alpha = I\dfrac{\Delta\omega}{\Delta t} \Rightarrow \Delta\omega = \dfrac{Fr\Delta t}{I} = \dfrac{(10)(0.4)(10)}{4\times 10^{-3}} = 10^{-4}\text{rad/s}$

d) $W = \tau\theta = \Delta K_{Er} = \dfrac{1}{2}I\Delta\omega^2$

$$= \left(\frac{1}{2}\right)\left(4\times10^{-3}\right)\left(4\times10^{-4}\right)^2 = 3.2\times10^{-4}\,\text{J}$$

Angular Momentum

The angular momentum (L) of a body of moment of inertia (I) rotating at an angular velocity (ω) is given by

$$\boxed{L = I\omega}$$

Angular momentum is a vector quantity and its SI unit is kgm^2/s.

Note that: the angular momentum of a body can also be calculated using the relation,

$\boxed{L = pr}$, where p is linear momentum and

r is perpendicular distance.

So we can have, a maximum angular momentum

$\boxed{L = pr = mvr}$, $p = mv$

And, we can see that, $1kgm^2/s = 1Nms$

The law of conservation of angular momentum; It states that,
"If no net torque is acting, the angular momentum of the system is conserved."

That is, if, $\sum\tau = 0$, then

Initial angular Momentum = Final angular momentum

$$L_i = L_f$$

$$\boxed{I_i\omega_i = I_f\omega_f}$$

Conceptual Examples

56. An ice skater holds her arms out stretched to either side and starts to spin. If she holds her arms in, what will happen to
    a) Her rotational speed
    b) Her moment of inertia

Explanation: a) her rotational speed increases
b) her moment of inertia decrease.

Illustrative Examples

57. An ice skater with her arms outstretched has a moment of inertia of $5.4kgm^2$ and spins at 5rad/s. If her moment of inertia is changed to $3.8kgm^2$ when she pulls her arms in, what is her new angular velocity.

Solution: from conservation of angular momentum

$$L_i = L_f$$

$$I_i\omega_i = I_f\omega_f$$

$$\Rightarrow \omega_f = \frac{I_i\omega_i}{I_f} = \frac{(5.4)(5)}{3.8} = 7.1\text{rad/s}$$

58. Calculate the angular momentum of a 95kg man running with a speed of 5.1m/s on a circular track of radius 25m.

Solution: $L = rmv = (25)(95)(5.1) = 12000\text{kgm}^2/\text{s}$.

Comparison between Linear and angular quantities.

Linear	**Angular**
$F = ma$	$\tau = I\alpha$
$P = mv$	$L = I\omega$
$W = FS$	$W = \tau\theta$
$K_E = \frac{1}{2}mv^2$	$K_{Er} = \frac{1}{2}I\omega^2$

Equilibrium

Equilibrium is a stable situation in which forces or torques acting on the body cancel one another out.

Equilibrium: Is a state of balance of a system.

The two conditions of equilibrium

For a body to become in equilibrium, two conditions must be satisfied.

1. The first condition (translational) equilibrium

"The sum of forces acting on a body must be zero."

$$\Rightarrow \boxed{\sum F = 0 \;\; ; \quad \sum F_x = 0 \text{ and } \sum F_y = 0}$$

2. The second condition (rotational) equilibrium

"The sum of torques acting on a body must be zero."

$$\Rightarrow \boxed{\sum \tau = 0, \quad \sum \tau_{\text{clockwise}} = \sum \tau_{\text{anticlockwise}}}$$

Conceptual Examples

59. Can an object in motion be ever in equilibrium

Explanation: Yes. If the body is moving at constant velocity.

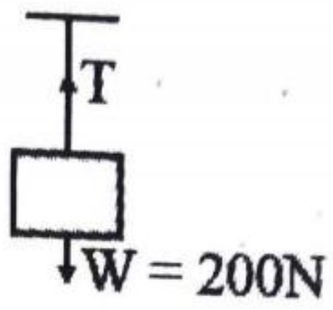
60. Is it sufficient for a body to be in equilibrium if it experiences only a net zero force? Explain.

Explanation: No. When a body is under a zero net force, it is said to be in translational equilibrium only. To be completely in

equilibrium both the force and the torque acting on the body must add to give zero.

Illustrative Examples

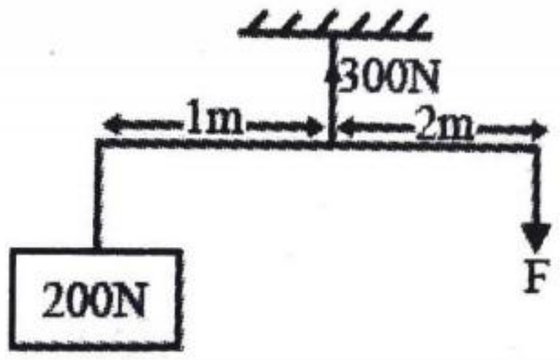
61. What should be the minimum value of the tension in the string so that it can carry the load safely without snap.



Solution: from the first condition of equilibrium,

$\sum F = 0 \Rightarrow T - W = 0 \Rightarrow T = W = 200N$

62. What should be the value of F so that the system becomes in equilibrium.



Solution: $\sum F = 0 \rightarrow 1^{st}$ condition

$\Rightarrow 300 - 200 - F = 0$

$F = 100N$

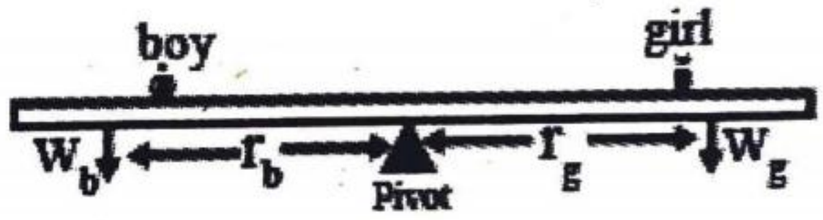
Check 2^{nd} condition, (200) (1) = (F) (2)

$\Rightarrow F = 100N$

63. A boy of weight 500N sits on the left of a see saw a distance of 2.4m from its pivot. If a girl can balance the see saw by sitting 3m from the pivot on the right side, what is her weight.

Solution: from the second condition of equilibrium,

$\sum \tau = 0$,



$\tau_{boy} = \tau_{girl}$

$$W_b r_b = W_g r_g \Rightarrow W_g = \frac{W_b r_b}{r_g} = \frac{(500)(2.4)}{3} = 400N$$

Stability

Stability is of three forms, those are, *stable*, *unstable* and *neutral*.

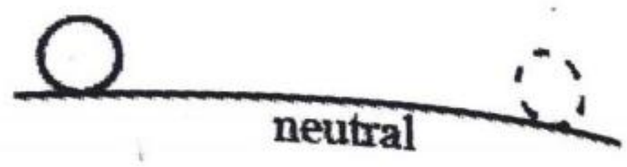
A) Stable Equilibrium

A body is said to be in stable equilibrium if when given a small displacement and then released it *returns to its original position*.

B) Unstable Equilibrium

A body is said to be in unstable equilibrium if when given a small displacement and then released it *moves farther from its original position.*

C) Neutral Equilibrium



A body is said to be in neutral equilibrium if when given a small displacement and then released it *stays in its new position*.

Center of mass

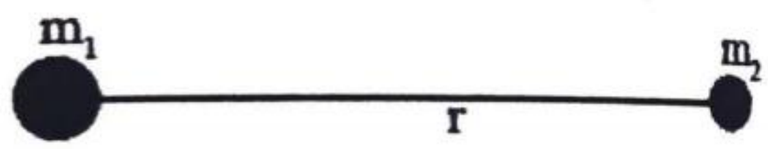
The center of mass of a body is the point at which the body's whole mass is considered to be concentrated. The center of mass is often called the *center of gravity* but this is only true in a system where the gravitational forces are uniform such as on Earth.

9.4 Newton's Law of universal gravitation

Gravitation is a force of attraction that exists between any two lumps of matter.

Newton's law of universal Gravitation: It states that;
"Every mass in the universe attracts every other mass with a force that is directly proportional to the product of the masses and inversely proportional to the square of the distance separating their centers."

That is, if we have two masses (m_1) and (m_2) separated by a distance (r), then the gravitational force between them is,



$\boxed{F = \frac{Gm_1m_2}{r^2}}$ where G is a gravitational constant and is equal to,

$G = 6.67 \times 10^{-11} Nm^2/kg^2$

Conceptual Examples

64. What is the nature of the gravitational force between any two bodies, is it attractive or repulsive.

Explanation: Gravitational force is always attractive force.

Illustrative Examples

65. Estimate the gravitational force between a 55kg woman and a 75kg man when they are 1.6m a part.

Solution: $F = \frac{Gm_1m_2}{r^2} = \frac{(6.67\times10^{-11})(55)(75)}{(1.6)^2} = 10^{-7}\,N$

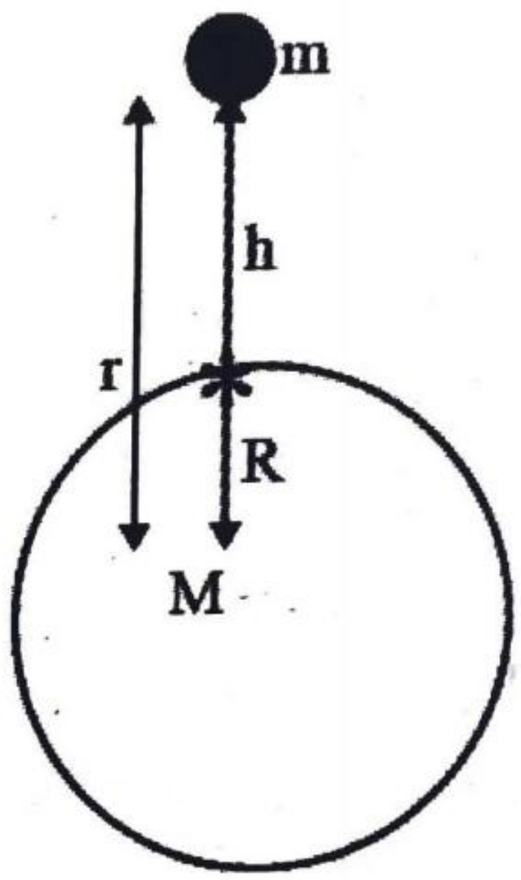
66. The force of attraction between two point masses is 4N. If one of the masses is doubled, the other mass is made four times greater and the distance between them is haved, what will be the new attractive force between the masses.

Solution: $F_1 = \frac{Gm_1m_2}{r^2} = 4N$. And $F_2 = \frac{G(2m_1)(4m_2)}{\left(\frac{r_1}{2}\right)^2}$

$$F_2 = 32\frac{Gm_1m_2}{r^2} = 32F_1 = (32)(4) = 128N.$$

Acceleration due to gravity (g) at any distance above the surface of the earth.

If a body of mass (m) is placed a distance (h) above the surface of the earth, then the acceleration due to gravity at that point is found by equating the weight of the body and the gravitational force between the body and the earth as follows,



Weight = gravitational force

$$mg = \frac{GmM}{r^2}$$

$$\Rightarrow \boxed{g = \frac{GM}{r^2}}, \text{ where } r = R + h, \text{ and}$$

R is radius of the earth

M is mass of the earth.

Conceptual Examples

67. As altitude increases, does the gravitational acceleration decrease or increase? Explain

Explanation: As height above the earth's surface increases, the gravitational acceleration decreases, we can see this form the relation $g = \frac{GM}{r^2}$.

Illustrative Examples

68. If the mass of the earth is 6×10^{24}kg and its radius is 6400km. What will be the acceleration due to gravity at an altitude of 300km above the earth's surface

Solution: $g = \frac{GM}{r^2}$, $r = h + R = 300 + 6400 = 6700km = 6.7 \times 10^6$ m

$$g = \frac{\left(6.67\times10^{-11}\right)\left(6\times10^{24}\right)}{\left(6.67\times10^{6}\right)^2} = 8.94\text{m/s}^2$$

69. At what altitude above the surface of the earth will the acceleration due to gravity becomes 9.71m/s^2

Solution: $g = \frac{GM}{r^2} \Rightarrow r^2 = \frac{GM}{g} = \frac{\left(6.67\times10^{-11}\right)\left(6\times10^{24}\right)}{9.71}$

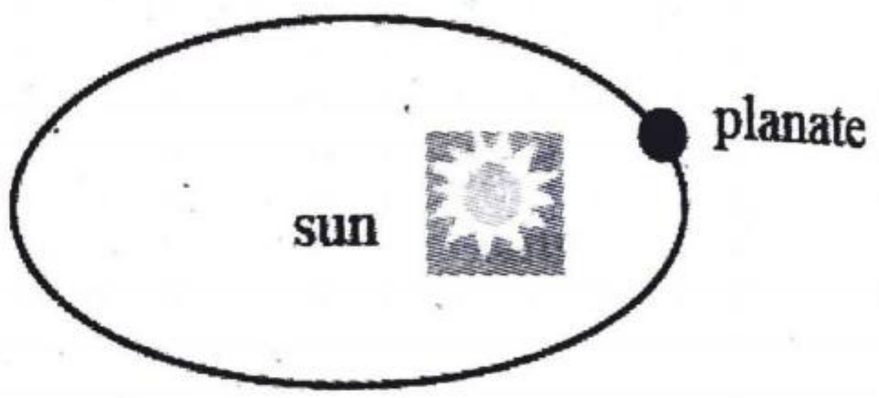
$r = 6420000\text{m}$

but, $r = R + h \Rightarrow h = r - R = 6420000 - 640000$

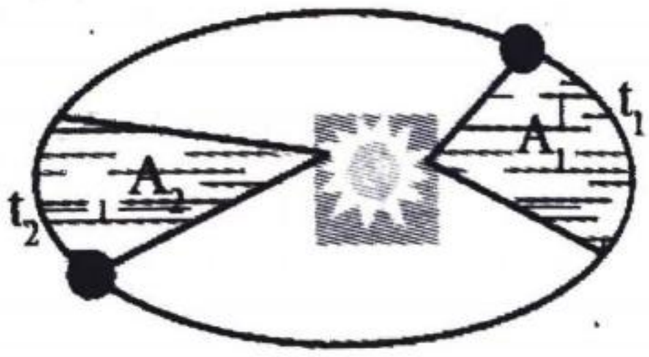
$h = 20000\text{m} = 20\text{km}.$

Kepler's law of planetary motion

Kepler's laws describe the motion of a planet around the sun. There are three laws of kepler.



1. **Kepler's first law;** state that;
"The planets orbit the sun in an elliptical path with the sun at one focus."



2. **Kepler's second law;** states that;
"The line joining the sun with the planet sweeps out equal areas in equal time intervals." That is for $t_1 = t_2$, $A_1 = A_2$.

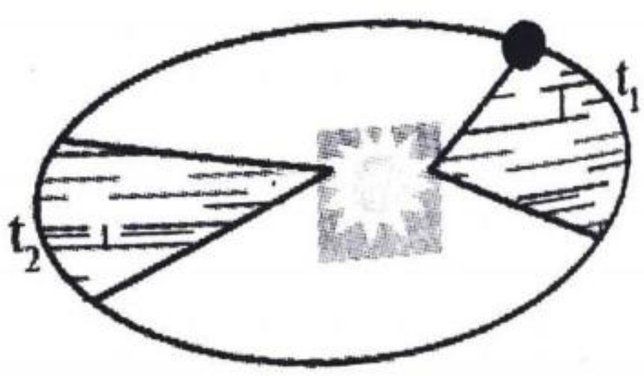
3. **Kepler's third law;** states that;
"The squares of the periods of revolution of the planets are proportional to the cubes of their mean distances from the sun."

$$\frac{T^2}{r^3} = \text{constant} = \frac{4\pi^2}{GM}$$

$$\Rightarrow \boxed{\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}}$$

Conceptual Example

70. According to Kepler's law, in which case is the planet moving faster, is it when it is closer or further away from the sun? Explain?



Explanation: It moves faster when it is closer to the sun. Because,

according to the second law, at equal times, the planet covers equal areas, so, as can be see, it moves long distance in short time when it is close to the sun. Because $s_2 > s_1$.

Illustrative Examples

71. The third law of kepler is given by, $\frac{T^2}{r^3} = \frac{4\pi^2}{GM}$. Derive this relation.

Solution: we know, $mg = \frac{GmM}{r^2} = \frac{mv^2}{r}$

$$\Rightarrow \frac{GM}{r} = v^2, \text{ but for one revolution.}$$

$$v = \frac{2\pi r}{T} \text{ and } v^2 = \frac{4\pi^2 r^2}{T^2},$$

$$\Rightarrow \frac{GM}{r} = \frac{4\pi^2 r^2}{T^2} = \text{criss crossing,}$$

$$T^2 GM = 4\pi^2 r^3$$

$$\Rightarrow \boxed{\frac{T^2}{r^3} = \frac{4\pi^2}{GM}} \leftarrow \text{kepler's } 3^{rd} \text{ law.}$$

72. The earth revolves around the sun once a year at an average distance of 1.5×10^{11}m . Calculate the mass of the sun.

Solution: from, $\frac{T^2}{r^3} = \frac{4\pi^2}{GM}, M = \frac{4\pi^2 r^3}{GT^2}$,

$$M = \frac{(4)(3.14)^2 (1.5\times10^{11})^3}{(6.67\times10^{-11})(3.16\times10^{7})^2} = 2\times10^{30}\text{kg}$$

73. Planet x revolves about the sun once every 88 days while planet y takes 365 days. If the average distance of planet y is 150×10^6km from the sun, what is the average distance of planet x from the sun.

Solution: from kepler's third law,

$$\frac{T_x^{\,2}}{r_x^{\,3}} = \frac{T_y^{\,2}}{r_y^{\,3}} \Rightarrow r_x^{\,3} = \frac{(T_x^{\,2})}{(T_y^{\,3})}(r_y^{\,3}),$$

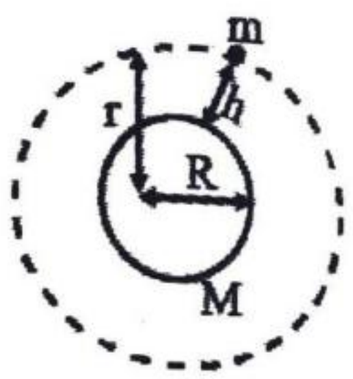
$$r_x^{\,3} = \frac{(88)^2}{(365)^2}\left(1.5\times10^{11}\right)^3 = 1.96\times10^{32},$$

$$r_x = 5.8\times10^{10}\text{m} = 5.8\times10^{7}\text{km}$$

Orbital velocity of a satellite

The earth has a large number of satellites in orbit around it. Most made by humans but the biggest of them all is natural – the moon.

Suppose a satellite of mass (m) is moving at constant speed of (v) around the earth, situated at a distance (r) from the center of the earth. The orbital speed of the satellite is then obtained from the relation of centripetal force and gravitational force.



Centripetal force = gravitational force

$$\frac{mv^2}{r} = \frac{GmM}{r^2}$$

Orbital velocity $\rightarrow \boxed{v = \sqrt{\frac{GM}{r}}}$, $r = h + R$

Geostationary Satellite: a satellite that orbits the equator with the same angular velocity as the earth.

Illustrative Examples

74. A satellite is in a circular orbit 250km above earth's surface. Calculate
    a) The orbital speed
    b) The period of the satellite.

Solution: a) $$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{\left(6.67\times10^{-11}\times6\times10^{24}\right)}{\left(6.4\times10^{6}+2.5\times10^{5}\right)}}$$

$$v = 7.8\text{km/s}$$

b) $$\frac{T^2}{r^3} = \frac{4\pi^2}{GM} \Rightarrow T = \left(\frac{4\pi^2 r^3}{GM}\right)^{1/2}$$

$$T = \left(\frac{(4\pi^2)(6.4\times10^6 + 2.50\times10^5)^3}{(6.67\times10^{-11})(6\times10^{24})} \right)^{\frac{1}{2}}$$

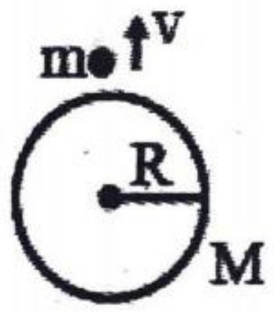
$$T = 5.4\times10^3 \text{ sec}$$

Escape velocity of a satellite

If a satellite is launched upward, it will be able to climb out of the earth's potential and escape. If the satellite is launched with a speed v, it will have a kinetic energy that will eventually stored as potential energy. The minimum escape velocity of the satellite is obtained from.

$$K_E = P_E$$

$$\frac{1}{2}mv^2 = \frac{GMm}{R}$$



$$\Rightarrow v = \sqrt{\frac{2GM}{R}}, \text{ but } g = \frac{GM}{R^2} \Rightarrow \frac{GM}{R} = gR$$

$$\boxed{v = \sqrt{2gR}} \leftarrow \text{escape velocity}$$

Conceptual Examples

75. Is the escape speed of a 10N rocket different from the escape speed of a 1000N rocket? Explain.

Explanation: The escape velocity is given by $v = \sqrt{2gR}$, Thus it is independent of mass. Hence the escape velocity of a 10N rocket is the same as that of 1000N rocket or any other body.

Illustrative Examples

76. Calculate the escape velocity for the moon.

Solution: $$v = \sqrt{\frac{2GM_m}{R_m}} = \left(\frac{(2\times6.67\times10^{-11}\times7.35\times10^{22})}{(1.74\times10^6)} \right)^{\frac{1}{2}}$$

$$v = 2370 \text{m/s}.$$

Additional Examples

77. A particle moves in a circle of radius 0.25m at two revolutions per second the acceleration of the particle is

A. π^2 B. $8\pi^2$ C. $4\pi^2$ D. $2\pi^2$

Solution: $a = r\omega^2 = (0.25)\left[(2)(2\pi)\right]^2 = 4\pi^2$.

Answer: C

78. A stone of mass 0.5kg tied with a string of length 1m is moving in a circular path with a speed of 4m/s. the tension on the string is

A. 2N B. 8N C. 0.2N D. 0.8N

Solution: $T = \dfrac{mv^2}{r} = \dfrac{(0.5)(4)^2}{1} = 8N$

Answer: B

79. In uniform circular motion, the factor that remains constant is

A. Linear velocity C. acceleration
B. Centripetal force D. speed

Answer: D

80. When a particle moves in a circle with uniform speed.

A. Its velocity and acceleration are both constant
B. Its velocity and acceleration both change
C. Its velocity is constant but not its acceleration
D. Its acceleration is constant but not its velocity

Answer: B

81. A stone of mass m tied to a string of length ℓ is rotated in a circle with the a speed v. if the string breaks, the stone will move.

A. Towards the center C. away from the center
B. Along the tangent D. will stop

Answer: B

82. The angular momentum changes from 2 units to 6 units in 4sec. the torque is

A. 1 unit B. $\frac{1}{2}$ unit C. $\frac{3}{2}$ unit D. 4 unit

Solution: $\tau = \dfrac{\Delta L}{t} = \dfrac{(6-2)}{4} = 1unit$

Answer: A

83. A fan of moment of inertia 0.6kgm^2 is turned up to a working speed of 0.5rev/s. The angular momentum is

A. 0.6π B. 3 C. 6 D. $\frac{\pi}{6}$

Solution: $L = \omega I = (0.6)(0.5)(2\pi) = 0.6\pi$

Answer: A

84. The total kinetic energy of a rolling sphere is

A. $\frac{1}{2}mv^2$ C. $\frac{7}{10}mv^2$

B. $\frac{1}{5}mv^2$ D. $\frac{7}{5}mv^2$

Solution:

$$K_E = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{2}{5}MR^2\right)\left(\frac{v^2}{R^2}\right) = \frac{7}{10}Mv^2$$

Answer: C

85. A flywheel of mass 25kg has a radius of 0.2m. What force should be applied tangentially to the wheel so that it acquires an angular acceleration of 2rad/s^2.

A. 1N B. 3N C. 5N D. 8N

Solution:

$$\tau = I\alpha = Fr = F = \frac{I\alpha}{r} = \frac{\frac{1}{2}MR^2\alpha}{R} = \frac{1}{2}MR\alpha$$

$$F = \frac{1}{2}(0.2)(25)(2) = 5\text{N}.$$

Answer: C

86. If the angular momentum of a body is increased by 50%, its kinetic energy of rotational increased by

a) 50% B. 125%N C. 200% D. 150%

Solution: $K_{E1} = \frac{1}{2}I\omega_1^2,\ \omega_2 = 1.5\omega_1 \Rightarrow K_{E2} = \frac{1}{2}I(1.5\omega_1)^2$

$K_{E2} = 2.25K_{E1} \Rightarrow \Delta K_E = (1.25)100\% = 125\%$

Answer: B

87. Two bodies with moment of inertia I_1 and I_2 with $(I_1 > I_2)$ have equal angular momenta. If E_1 and E_2 are the rotational kinetic energies then;

A. $E_1 = E_2$ C. $E_1 < E_2$

B. $E_1 > E_2$ D. $E_1 \geq E_2$

Solution: $I_1\omega_1 = I_2\omega_2 \Rightarrow \omega_1 < \omega_2$.

Answer: C

88. The torque required to stop a wheel of moment of inertia $5 \times 10^{-3}kgm^2$ from a speed of 20rad/s in 10sec is

A. 1Nm B. 0.1Nm C. 0.01Nm D. 0.001Nm

Solution: $\tau = I\alpha = \left(5\times10^{-3}\right)\left(\frac{20}{10}\right) = 0.01Nm$

Answer: C

89. When a sphere rolls without slipping, the ratio of its translational kinetic energy to its total energy is

A. 1:7 B. 1:1 C. 1:2 D. 5:7

Solution: $\frac{\frac{1}{2}mv^2}{\frac{7}{10}mv^2} = \frac{10}{14} = 5:7$

Answer: D

90. If the distance between two point masses is doubled, the gravitational attraction between them

A. Doubled C. increase four times

B. Halved D. decrease four times

Solution: $F_1 = \frac{Gm_1m_1}{r_1^2}, r_2 = 2r_1 \Rightarrow F_2 = \frac{1}{4}, \left(\frac{Gm_1m_2}{r^2}\right)$.

Answer: D

91. Orbital velocity of a satellite does not depend on

A. Mass of earth C. radius of earth

B. Mass of satellite D. acceleration due to gravity

Solution: $\frac{mv^2}{r} = \frac{GmM_1}{r^2} \Rightarrow v = \sqrt{\frac{GM}{r}}$

Answer: B

92. The acceleration due to gravity on earth is $10m/s^2$. What is its value on a planet $\frac{1}{5}$ the mass and $\frac{1}{2}$ radius of the earth.

A. $4m/s^2$ B. $6\ m/s^2$ C. $8m/s^2$ D. $12m/s^2$

Solution: $mg = \frac{GmM}{R^2} \Rightarrow g = \frac{GM}{R^2}$, $g_p = \frac{G\frac{1}{5}(M)}{\left(\frac{1}{2}R\right)^2}$

$$g_p = \frac{4}{5}\frac{GM}{R^2} = \frac{4}{5}(10) = 8m/s^2$$

Answer: C

93. What keeps an earth satellite moving in its orbit
A. Gravitational force between earth and satellite
B. Ejection of gas from the satellite
C. Burning of fuel
D. Gravitational attraction of sun

Answer: A

94. When a planet moves around the sun
A. The angular momentum remains conserved
B. The angular speed remains constant
C. The linear velocity remains constant
D. The linear momentum remains constant

Answer: A

95. A body is detached gently from the outer wall of a satellite circling round earth. The body will
A. Continue to move along with the satellite
B. Fall to the earth
C. Fall to the moon
D. Follow irregular path

Answer: A

96. If the earth shrinks, what will be the effect on the value of gravitational acceleration.
A. decrease C. become zero
B. increase D. become infinite

Solution: $g = \frac{GM}{R^2}$, when R decrease, g increases.

Answer: B

97. Orbital velocity of a satellite moving around a planet.
   A. Is proportional to density of plant
   B. Is proportional to the square root of density of plant
   C. Is proportional to the square of density of planet
   D. Is inversely proportional to density of planet

Answer: B

98. If earth orbits the sun at double of its present radius, the year on earth will be
   A. 365 days
   B. $\frac{365}{2}$ days
   C. $365\sqrt{2}\times 2$days
   D. 365×4days

Solution: $T^2 \sim r^3$, when r is doubled, $T = 365\times 2\sqrt{2}$

Answer: C

99. The gravitational force of attraction between two bodies each of mass 1kg and separated by 1m is
   A. 6.67×10^{-11}N
   B. 6.67×10^{-8}N
   C. 6.67N
   D. 1N

Solution: $F = \frac{Gm_1m_2}{r_1^2} = \frac{\left(6.67\times10^{-11}\right)(1)(1)}{r_1^2} = 6.67\times10^{-11}$N

Answer: A

100. The escape velocity from the earth for a rocket is 11.2km/s. The escape velocity of 10^{-2}g of sand from the earth will be
   A. 0.112km/s
   B. 11.2km/s
   C. 1.12km/s
   D. 22.4km/s

Solution: $v = \sqrt{2gR}$, thus it is the same.

Answer: B

101. A system consisting of two masses m_1 and m_2 separated by distance r is immersed in water. What is the effect on gravitational force?
   A. No effect
   B. It is weakened
   C. It become stronger
   D. In become zero

Answer: A

102. Earth exerts a force of 2.5N on an apple. What is the force exerted by apple on earth?
   A. 2.5N B. zero C. 10N D. 0.5

Answer: A

103. An earth satellite 5 has an orbit radius which is 4 times that of a communication satellite C. The period of revolution of 5 is?

A. 4 days B. 8 days C. 16 days D. 32 days

Solution: $\frac{T_s^2}{r_s^3} = \frac{T_c^2}{r_c^3}$, $r_s = 4r_c$

Answer: B

104. Acceleration due to gravity increases if

A. We go up from the surface of the earth towards its center
B. We go up from the surface of the earth
C. We go from the equator towards the poles
D. The rotational velocity of eh earth increased.

Answer: C

105. Escape velocity on a planet is v_e. If radius of the planet remains same and mass becomes 4 times, the escape velocity becomes.

A. $4v_e$ B. v_e C. $2v_e$ D. $\frac{1}{2}v_e$

Solution: $v = \sqrt{2\frac{GM}{R}}$, when $M_2 = 4M$, $v_e = 2v_e$.

Answer: C

106. The kinetic energy of a satellite in its orbit around the earth is e. what should be the kinetic energy of the satellite so as to enable it to escape from the gravitational pull of the earth?

a) 4E b) 2E c) $\sqrt{2}E$ d) E

Solution: escape kinetic energy, $KE = \frac{1}{2}Mv^2$, $v = \sqrt{\frac{2GM}{R}}$

$$KE_e = \frac{1}{2}\left(\frac{2GM}{R}\right) = \frac{GMm}{R}$$

Orbital kinetic energy, $KE = \frac{1}{2}Mv_e^2$, $v = \sqrt{\frac{GM}{R}}$

$$KE_o = E = \frac{1}{2}m\frac{GM}{R} = \frac{1}{2}\frac{GMm}{R} \Rightarrow KE_e = 2KE_o$$

Answer: B

107. Height of geostationary satellite above the earth is about
A. 3600km C. 30000km
B. 42000km D. 36000km
Answer: D

108. The gravitational potential at a point on the surface of earth is
A. g B. gR C. $\frac{1}{2}$gR D. zero
Answer: B

109. Kepler's law of area is a consequence of
A. Conservation of energy
B. Conservation of mass
C. Conservation of linear momentum
D. Conservation of angular momentum
Answer: D

General Secondary Education Certificate exam

110. A girl can throw a stone to a maximum horizontal distance of 64m. If she throws the stone vertically upward with the same speed, what is the maximum height reached by the stone.
A. 64m B. 32m C. 16m D. 8m

Solution: from, $R = \frac{u^2 \sin(2\theta)}{g}$, R is maximum

when $\theta = 45° \Rightarrow 2\theta = 2(45) = 90°$ and $\sin 90° = 1$

$$R = \frac{u^2}{g} \Rightarrow u = \sqrt{(R)(g)} = \sqrt{(64)(10)}$$

$$u = 8\sqrt{10}\text{m/s}$$

and $h = \frac{u^2 \sin^2\theta}{2g}$, h is maximum

when $\theta = 90° \Rightarrow \sin 90° = 1$ and $\sin^2 90° = 1$

$$h = \frac{\left(8\sqrt{10}\right)^2}{2(10)} = \frac{(64)(10)}{(2)(10)} = 32\text{m}$$

Answer: B

111. A certain body is thrown upwards at an angle of 30° to the horizontal with a speed of 20m/s. What is the time the body takes to reach the maximum height?

$\left(\sin 30° = \frac{1}{2}, \cos 30° = \frac{\sqrt{3}}{2}, g = 10m/s^2\right)$

A. 2 sec B. 5 sec C. 0.5 sec D. 1 sec

Solution: from, $v_y = u_y - g_t$, $v_y = 0$ at h_{max}.

$$\Rightarrow t = \frac{u_y}{g} = \frac{u\sin\theta}{g} = \frac{(2)(\sin 30)}{10} = \frac{(20)\left(\frac{1}{2}\right)}{10} = 1 \text{ sec}$$

Answer: D

112. An Olympic long jumper leaves the ground at an angle of 45° and travels through the air for a horizontal distance of 8.1m before landing. What is the take off speed f the jumper (take $g = 10m/s^2$)?

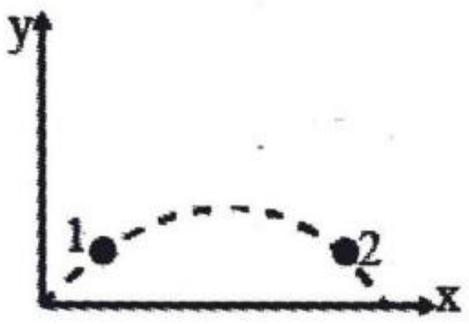
A. 9m/s B. 11m/s C. 8.1m/s D. 10m/s

Solution: $R = \frac{u^2\sin(2\theta)}{g} \Rightarrow 8.1 = \frac{u^2\sin(2\times 45)}{10}$

$u^2 = 81 \Rightarrow 9m/s$ **Answer: A**

113. A football is kicked into the air at some angle ignoring air friction, the horizontal component of the velocity.

A. Is zero at the maximum point
B. Remains constant
C. Increases with time
D. Decreases with time



Explanation: if air resistance is neglected, then there is no force that affects the horizontal component of the velocity. **Answer: B**

114. Consider two position 1 and 2 of a projectile at the same height as it moves up and down respectively. Which one of the following is the same at those points.

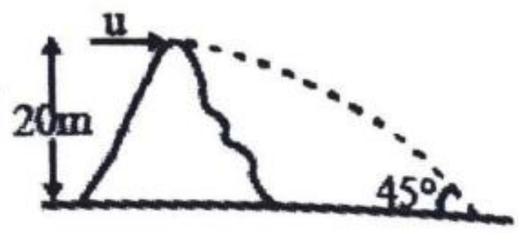
A. Speed C. Displacement
B. Velocity D. Distance

Explanation: the speed is the same.

Answer: A

115. A ball is thrown horizontally from the top of a 20m tall hill. It strikes the ground at an angle of 45°, with what speed was it thrown? (take $g = 10m/s^2$)

A. 40m/s C. 32m/s
B. 28m/s D. 20m/s



Solution:

$$h = \frac{u^2\sin^2\theta}{2g} \Rightarrow 20 = \frac{u^2\sin^2(45)}{(2)(10)}$$

$$\Rightarrow u^2 = \frac{(20)(2)(10)}{(\sin 45)(\sin 45)} = \frac{400}{(0.7)(0.7)} \Rightarrow u = \frac{20}{0.7} = 28m/s$$

Answer: B

116. What is the magnitude of the centripetal force that is necessary to keep a body of mass 2kg in a circular motion of radius 2m with a speed of 10m/s.

A. 10N B. 40N C. 100N D. 25N

Solution: $F_C = \frac{mv^2}{r} = (2)\frac{(10)^2}{(2)} = 100N$

Answer: C

117. Which one of the following statement is incorrect?

A. Uniform circular motion cannot be maintained with out radial force.
B. The centripetal and centrifugal forces associated with a body in circular motion are always of the same magnitude.
C. A body in uniform circular motion has a non zero radial acceleration
D. Uniform circular motion can be maintained by applying a constant tangential force.

Answer: C

118. Which one of the following statement is incorrect?

A. Uniform circular motion cannot be maintained with out radial force.
B. The centripetal and centrifugal forces associated with a body in circular motion are always of the same magnitude.
C. A body in uniform circular motion has a non zero radial acceleration

D. Uniform circular motion can be maintained by applying a constant tangential force.

Answer: D

119. A ball tied to the end of a sting is allowed to move in a circular path at a uniform speed. What keeps the ball along the circular path?

A. Torque B. Gravity C. Tension D. Inertia

Explanation: The centripetal force, that is the tension in the string keeps the ball in circular path.

Answer: C

120. Which of the following is NOT true about a uniform circular motion?

A. The direction of the velocity continuously changes
B. The magnitude of the velocity continuously changes
C. The direction of the acceleration continuously changes.
D. The magnitude of the acceleration remains constant the speed is constant here.

Answer: B

121. A body in rotational equilibrium has a constant

A. Acceleration C. Velocity
B. Angular acceleration D. angular velocity

Solution: now, $\tau = 0 \Rightarrow \tau = I\alpha = 0 \Rightarrow \alpha = 0$

$$\text{but } \alpha = \frac{\Delta\omega}{t} = \frac{\omega - \omega_o}{t} = 0 \Rightarrow \omega = \omega_o$$

Answer: D

122. A wheel of radius 20cm is accelerating at $5\pi rad/s^2$. What is the tangential acceleration of a point at the rim of the wheel?

A. 2π rad/s^2 C. π m/s^2
B. 1.5π rad/s^2 D. 0.5π m/s^2

Solution: $a_t = r\alpha = (0.2)(5\pi) = \pi m/s^2$

Answer: C

123. A body tide to a string of length 1.5m completes 3 rev in one second. What is the tangential displacement of the body in 5 seconds?

A. 60πm B. 45πm C. 30πm D. 15πm

Solution: $S = r\theta = (1.5)(3)(2\pi)(S) = 45\pi m.$

Since $\theta = \omega t = (3)(2\pi)(5) = 30\pi$ rad, $S = r\theta = 45\pi m$.

Answer: B

124. How much work is needed to be done on a 1kg disk of rotational inertia $0.03kgm^2$ to accelerate it from rest to an angular speed of 200 rad/sec in 1sec.

A. 600J B. 30J C. 1200J D. 2000J

Solution: $W = \tau\theta = I\alpha\theta$. but $\alpha = \dfrac{\omega - \omega_o}{t} = \dfrac{200-0}{1}$,

$\alpha = 200rad/s^2$, $\theta = \dfrac{1}{2}(\omega - \omega_o)t = \dfrac{1}{2}(200+0)(1) = 100rad$

$\Rightarrow W = I\alpha\theta = (0.03)(200)(100) = 600J$

Answer: A

125. The wheel of a bicycle rotates with a constant angular acceleration of $2rad/s^2$. It attains an angular velocity of 100rad/s in 30 sec. what was its initial angular velocity.

A. Zero B. 60rad/s C. 160rad/s D. 40rad/s

Solution: $\alpha = \dfrac{\omega - \omega_o}{t} \Rightarrow \omega_o = \omega - \alpha t = 100 - 2 * 30$

$\omega_o = 40rad/s.$

Answer: D

126. A uniformly accelerating wheel increased its angular velocity from 2rad/s to 5rad/s with in an angular displacement of 3.5 rad. What is the angular acceleration of the wheel?

A. $0.43rad/s^2$
B. $6rad/s^2$
C. $0.86rad/s^2$
D. $3rad/s^2$

Solution: $\alpha = \dfrac{\omega^2 - \omega_o{}^2}{2\theta} = \dfrac{(5)^2 - (2)^2}{2(3.5)} = \dfrac{25-4}{7} = 3rad/s^2$

Answer: D

127. Two people start at the same point and walk around a circular lake in opposite direction. One has an angular speed of $\pi/120$ rad/s, while the other has an angular speed of $\pi/240$ rad/s, how long will it be when they meet?

A. 160 sec B. 80 sec C. 120 sec D. 60 sec

Solution: $\theta = \theta_1 + \theta_2 = 2\pi$ rad

$\Rightarrow \omega_1 t_1 + \omega_2 t_2 = 2\pi$, but $t_1 = t_2 = t$

$$\left(\frac{\pi}{120}\right)t + \left(\frac{\pi}{240}\right)t = 2\pi \Rightarrow \frac{t}{120} + \frac{t}{240} = 2 \Rightarrow \frac{2t+t}{240} = 2$$

$$\frac{3t}{240} = 2 \Rightarrow 3t = 480$$

$t = 160$ sec

Answer: A

128. A particle of mass 5gm is attached to the end of a rigid rod of length 8cm and negligible mass. The other end of the rod is pivoted. If the particle acceleration from rest to 2m/s in 10 sec. How large is the torque that accelerated the particle.

A. 6.4×10^{-6}Nm
B. 4×10^{-5}N/m
C. 8×10^{-5}Nm
D. 1×10^{-5}Nm

Solution: $\tau = I\alpha$, $\alpha = \frac{a}{r}$ and $I = mr^2$, $a = \frac{v-u}{t} = \frac{2-0}{10} = 0.2$

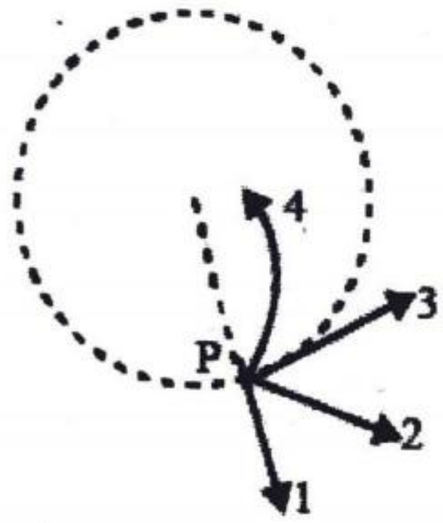
$$\tau = \left(mr^2\right)\left(\frac{a}{r}\right) = mra$$

$$\tau = \left(5*10^{-3}\right)\left(8*10^{-2}\right)\left(0.2\right) = 8 \times 10^{-5}\text{Nm}$$

Answer: C

129. Stone attached to a string is moved at constant speed in a circular path. What path would the stone take if the string broke at point P?

A. 1 B. 2 C. 3 D. 4



Explanation: the stone will move in the direction of the tangential speed.

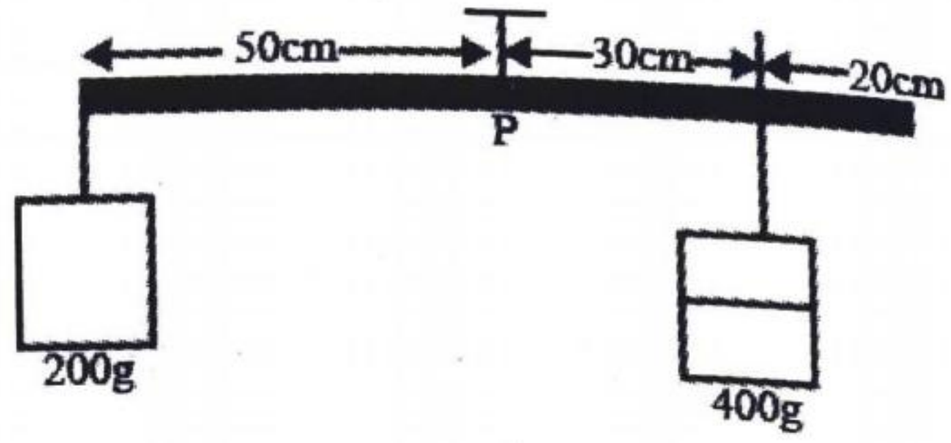
Answer: C

130. The uniform rod in the diagram is pivoted at its center, and 200g and 400g masses are suspended from the rod. But, the system is not balanced. The best strategy to balance system is moving the

A. 200g mass by 20cm towards the pivot
B. 200g mass by 25cm away from the pivot
C. 400g mass to the end of the rod
D. 400g mass to the pivot by 5cm

Solution: $\tau_{\text{clockwise}} = \tau_{\text{Anticlockwise}}$

$m_1r_1 = m_2r_2$, (200) (50) = (400) $(r_2) \Rightarrow r_2 = 25cm$.
Thus, the 400g should be moved to the pivot by 5cm
Answer: D



131. The average distance of the earth form the sun is 1.49×10^8km. What would be the period of revolution of the earth if it were located twice as far form the sun?

A. 2 year
B. $2\sqrt{2}$ years
C. 4 years
D. $3\sqrt{4}$ years

Solution: $\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}$, $T_1 = 1$ year and $r_2 = 2r_1$

$$\Rightarrow T_2^2 = \left(T_1^2\right)\left(\frac{r_2^3}{r_1^3}\right) = (1)\frac{(2r_1)^3}{(r_1)^3} = \frac{8r_1^3}{r_1^3}$$

$$T_2^2 = 8 \Rightarrow T_2 = 2\sqrt{2}\text{years}$$

Answer: B

132. Two satellites are monitored as they orbit the earth's satellite y is four times as for from the earth's center as is satellite x. If the period of revolution of x is T, what is the period of revolution of y.

A. 4T B. 8T C. 2T D. 16T

Solution: $\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3} \Rightarrow \frac{T_x^2}{r_x^2} = \frac{T_y^2}{r_y^2}$, but $r_y = 4r_x$ and

$$\Rightarrow \frac{T^2}{r_x^3} = \frac{T_y^2}{(4r_x)^3} \Rightarrow T_y^2 = \frac{(4r_x)^3}{r_x^3}T^2 = 64T^2$$

$$T_y^2 = 64T^2,\ T_y = 8T$$

Answer: B

133. A satellite of mass m is orbiting the earth in a circular path of radius R. if the mass of the earth is M, what is the speed of the satellite?

A. $\sqrt{GM/R}$
B. $\sqrt{GM/R}$
C. GM/R
D. GM/R

Solution: Centripetal force = Gravitational force

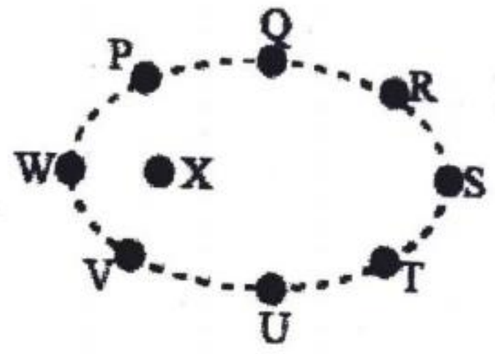
$$\frac{mv^2}{R}=\frac{GMm}{R^2}\Rightarrow v^2=\frac{GM}{R},\ v=\sqrt{\frac{GM}{R}}$$

Answer: A

134. A planet travels in an elliptical orbit about star x. The kinetic energy of the planet is greatest at point?

A. S B. J C. Q D. W

Explanation: When a planet moves closer to the sun, its speed is greater and hence its kinetic energy.



Answer: D

135. What initial minimum velocity must a projectile have to reach a target at a distance of 500m away ($g = 9.8m/s^2$)?

A. 70m/s B. 700m/s C. 140m/s D. 1400m/s

Solution: $R=\dfrac{u^2\sin 2\theta}{g}$, let $\sin 2\theta = 1$

$$R=\frac{u^2}{g},\ u=\sqrt{gR}\ =\sqrt{(9.8)(500)}=70m/s$$

Answer: A

136. If a point on a spinning wheel changes its angular speed by a factor of 3 and its angular acceleration by a factor of 9. The magnitude of the total acceleration changes by a factor of ?

A. $\dfrac{1}{3}$ B. 3 C. 9 D. $\dfrac{1}{9}$

Explanation: $a=\sqrt{a_c^2+a_t^2}$, $a_c=r\omega^2$ and $a_t=r\alpha$

$$\Rightarrow a_1=\sqrt{a_{c1}^2+a_{t1}^2},\ a_{c1}=r\omega_1^2 \text{ and } a_{t1}=r\alpha_1$$

$$a_2=\sqrt{a_{c2}^2+a_{t2}^2},\ a_{c2}=r(3\omega_1)^2=9r\omega_1^2$$

$$a_{t2}=r(9\alpha_1)=9r\alpha_1$$

$$=\sqrt{9^2(a_c^2)+9^2(a_t^2)}$$

$$=9\sqrt{a_c^2+a_t^2}=9a_1$$

Answer: C

137. The escape speed at the surface of a certain planet is twice that of earth. If the planet has radius twice that of earth, what is its mass in unit of Earth's mass?

A. 2 B. 4 C. 8 D. 0.5

Solution: $v = \sqrt{\frac{GM}{R}} = v_p = 2v_e,\ R_p = 2R_e$

$$v_{planet}, (v_p) = \sqrt{\frac{GM_p}{R_p}}, v_{earth}\ (v_e) = \sqrt{\frac{GM_e}{R_e}}$$

$$\text{Take ratio, } \frac{v_p}{v_e} = \frac{\sqrt{\frac{GM_p}{R_p}}}{\sqrt{\frac{GM_e}{R_e}}} \Rightarrow \frac{2v_e}{v_e} = \sqrt{\frac{M_p}{2M_e}}$$

$$\Rightarrow 2 = \sqrt{\frac{M_p}{2M_e}}, \text{ squaring both sides, } 4 = \frac{M_p}{2M_e}$$

$$\Rightarrow M_p = 8M_e$$

Answer: C

138. Which one of the following is true about Kepler's laws of planetary motion?

A. the path of each planet about the sun is an ellipse with the sun at one focus

B. the square of the time it takes the planet to go round the sun is proportional to its average distance from the sun.

C. the period of revolution of the planet around the sun is proportional to its average distance from the sun

D. the line that joins the sun to the orbiting planet sweeps out different area in equal times.

Answer: A

139. A package is dropped from an airplane flying parallel to the ground with constant velocity. How would two observers, one in the plane and the other standing an the ground, describe the motion of the package?

A. To the observer on the ground the package moves vertically down
B. To the observer on the plane, the package moves vertically down
C. To the observer on the plane the package moves horizontally down
D. To the observer on the plane the package moves along a curved path.

Explanation: because of relative motion

Answer: B

140. What is the measure of distribution of the mass of a body in relation to its axis of rotation?

A. Torque
B. Moment of inertia.
C. Angular momentum
D. Angular displacement

Answer: B

141. A rock and a ball are initially at the same height from the ground. The rock is dropped at the same instant the ball is thrown horizontally, if air resistance is ignored?

A. The ball strikes the ground earlier than the rock
B. The rock strikes the ground earlier than the ball
C. The rock and the ball strike the ground simultaneously
D. The rock and the ball strikes the ground differently.

Answer: C

142. If there were no gravity, which would still happen?

A. An apple on a tree would fall to the ground
B. The moon would revolve around the earth
C. An object thrown vertically down would strike the ground
D. The earth would revolve the sun.

Answer: D

143. An object is initially in stable equilibrium. What happens when the object is lightly displaced form its original position?

A. It does not necessarily return to its original position
B. It moves further away from its original position
C. It turns back to its original position.
D. It remains stable in the new position.

Answer: C

144. A constant torque of 100Nm rotates a door by 90° about its hinges. What is the work done on the door?

A. 9000J B. 10J C. 157J D. 0.7J

Solution: $W = \tau\theta = (100)\left(\frac{\pi}{2}\right) = (100)\left(3.14/2\right) = 157J$

Answer: C

Unit Nine: Conceptual Questions

1. Can an object have an acceleration in a direction other than its direction of motion? Explain
2. Can a body have a constant speed even though it is accelerated?
3. Can a body have a constant acceleration even when its velocity is zero.
4. Can an object moving in a circle have a centripetal acceleration whose magnitude changes? Explain
5. Do you attract a wall? If so, why not the wall move? Does the wall attract you? If Yes, why don't you move?
6. A solid sphere and a hollow sphere of the same mass and radius roll without slipping at the same speed. What can you say about their rotational kinetic energy
7. Can the centripetal acceleration be non zero while the tangential acceleration is zero? Explain
8. Can the centripetal acceleration be zero while the tangential acceleration is nonzero? Explain
9. Two forces produce the same torque. Does it follow that they have the same magnitude? Explain
10. Can an object have zero translational acceleration and at the same time have non zero angular acceleration.
11. Is it possible to change the angular momentum of an object without changing its linear momentum? Explain
12. If the earth were to magically expand, keeping its mass constant, would the length of the day change?
13. Suppose the earth were to suddenly shrink keeping its mass constant. Would its escape speed change?
14. It is possible for a satellite to have any desired speed in a given orbit?

Unit Nine: Problems

Projectile

15. A football is kicked at a velocity of 15m/s at an angle of 25° to the horizontal. Calculate
    a) The total flight time
    b) The maximum height

c) The range of the ball

16. A car accidently rolls off a cliff. As it leaves the cliff it has a horizontal velocity of 13m/s. if it hits the ground 60m from the shoreline. Calculate the height of the cliff.

17. Find the range of a projectile launched at an angle of 45° with an initial velocity of 25m/s.

Rotational kinematics

18. Find the tangential velocity of an object moving in a path of radius 2m with an angular velocity of 3rad/s.

19. Find the final angular velocity when the initial angular velocity is 5rad/s, the angular acceleration is 2rad/s^2 and the time of motion is 10 sec.

20. Find the distance travelled when the final angular velocity is 20rad/s, the angular acceleration is 2rad/s^2 and the time of motion is 5 sec.

21. A satellite is in orbit 35600km above the surface of the earth. Its angular velocity is 7.27×10^{-5}rad/s. What is the velocity of the satellite (R = 6400 km)

22. Calculate the centripetal force on an astronaut of mass 80kg if a centrifuge of radius 5m rotates once every 2 sec.

Rotational dynamics

23. a) Find the moment of inertia of a point mass of 0.005g at a perpendicular distance of 3m from its axis

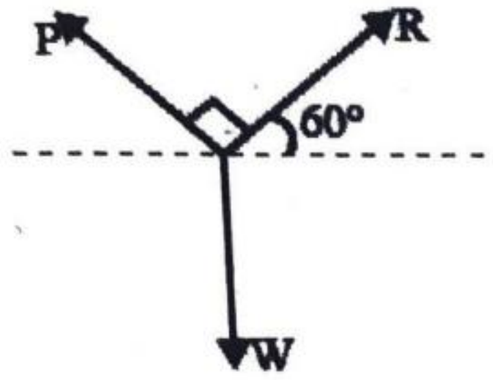
b) Find the moment of inertia of a sphere of mass 0.3kg and radius 60cm

c) Find the kinetic energy of a rotating body with moment of inertia 0.003kgm^2 and angular velocity of 0.6rad S^{-1}.

24. a) Find the work done when a torque of 3.5Nm moves through an angle of $\frac{1}{2}\pi$ radians

b) Find the torque when the work done to move through an angle of $\frac{1}{4}\pi$ radians in 3J.

25. The system is in equilibrium. Write an expression linking P, R and W.



Newton's law of universal gravitation

26. Calculate the value of g at 1000m above the surface of the earth (R = 6400km)
27. Determine the magnitude of the force of attraction between the moon and the earth. They are about 4×10^8m apart. The mass of the moon is 2×10^{24}kg and the mass of the earth is 6×10^{24}kg.
28. Geostationary satellites are placed in orbits of radius 4.2×10^4km. Deduce g at that height.

Unit Nine: Explanation to Questions

1. Yes, consider a body thrown upward, the acceleration due to gravity is downward but the motion is upward.
2. Yes, a horizontal projection can be example.
3. Yes, a body thrown upward has zero velocity at the peak but the acceleration is stile there $g = 10m/s^2$.
4. No! This doesn't happen for uniform circular motion.
5. Yes, but the force is so small to feel.
6. The hollow sphere has higher rotational kinetic energy as it acquires larger moment of inertia
7. Yes, this is so when the body is in uniform circular motion
8. Yes, when the body is moving in a straight line.
9. No, not necessarily. If $F_1 = 10N$, $r_1 = 5m$ and $F_2 = 5N$ and $r_1 = 10m$. Then $\tau_1 = \tau_2 = 50N.m$ but $F_1 \neq F_2$.
10. Yes, for uniform circular motion
11. Yes, by changing the radius.
12. Yes. It's moment of inertia increase and the speed of rotation decreases hence the length of the day increases.
13. $v = \sqrt{2gR}$. Yes because the radius decreases the escape speed decreases
14. No. it is impossible to adjust the speed at any orbit

Unit Nine: Solution to problems

15. a) $t = \dfrac{2u\sin\theta}{g} = \dfrac{(2)(15)(\sin 25°)(2)}{10} = 1.3 \text{ sec}$

b) $h_{max} = \dfrac{u^2 \sin^2\theta}{2g} = \dfrac{(15)^2 \sin^2 25°}{2(10)} = 2m$

c) $R = \dfrac{(2u\sin\theta)(u\cos\theta)}{g} = \dfrac{(2)(15)^2(\sin 25°)(\cos 25°)}{10}$

$= 17.2\ m$

16. $\dfrac{x}{u} = \sqrt{\dfrac{2y}{g}} \Rightarrow \dfrac{x^2}{u^2} = \dfrac{2y}{g} \Rightarrow y = \dfrac{x^2 g}{2u^2} = \dfrac{(60)^2(10)}{(2)(13)^2} = 106.5m$

17. a) $R = \dfrac{u^2 \sin(2\theta)}{g} = \dfrac{(25)^2(\sin(2\times 45))}{10} = 62.5m$

18. $v = r\omega = (3)(2) = 6m/s$

19. $\omega = \omega_o + \alpha t = 5 + (2)(10) = 25rad/s$

20. $\theta = \omega t - \dfrac{1}{2}\alpha t = (20)(5) - \dfrac{1}{2}(2)(5)^2 = 75rad/s$

21. $v = \omega r = \left(7.27\times 10^{-7}\right)(6400000 + 35600000) = 30.5m/s$

22. $a_c = r\omega^2 = (5)\left(\dfrac{2\pi}{2}\right)^2 = 49.3m/s^2,\ F_c = ma_c = (80)(49.3) = 3944N$

23. a) $I = mr^2 = \left(0.005\times 10^{-3}\right)(3) = 1.5\times 10^{-5} kgm^2$

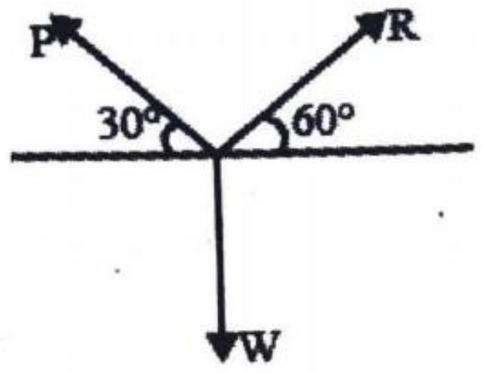
b) $I = \dfrac{2}{5}mR^2 = \dfrac{2}{5}(0.3)(0.6)^2 = 0.0432kgm^2$

c) $K_{Er} = \dfrac{1}{2}I\omega^2 = \dfrac{1}{2}(0.003)(0.6) = 9\times 10^{-4} J$

24. a) $W = \tau\theta = (3.5)\left(\dfrac{1}{2}\times 3.14\right) = 5.5J$

b) $W = \tau\theta = \tau = \dfrac{W}{\theta} = \dfrac{3}{\frac{3.14}{4}} = \dfrac{12}{3.14} = 3.82N.m$

25.



$$\Rightarrow \sum F_x = 0,\ R\cos 60 = P\cos 30$$

$$\frac{1}{2}R = \frac{\sqrt{3}}{2}P \Rightarrow R = \sqrt{3}P$$

$$\Rightarrow \sum F_y = 0,\ R\sin 60 + P\sin 30° = W$$

$$\frac{\sqrt{3}}{2}R + \frac{1}{2}P = W$$

$$\Rightarrow W = \frac{3}{2}P + \frac{1}{2}P = 2P$$

26. $$g = \frac{GM}{r^2} = \frac{\left(6.67\times10^{-11}\right)\left(6\times10^{24}\right)}{\left(6400000+1000\right)^2} = 9.77 m/s^2$$

27. $$F = \frac{Gm_1m_2}{r^2} = \frac{\left(6.67\times10^{-11}\right)\left(2\times10^{24}\right)\left(6\times10^{24}\right)}{\left(4\times10^{8}\right)^2} = 5\times10^{21}N$$

28. $$g = \frac{GM}{r^2} = \frac{\left(6.67\times10^{-11}\right)\left(6\times10^{24}\right)}{\left(42000000\right)^2} = 0.23 m/s^2$$

Unit Ten

Electrostatics

We are all made up of electrical charges. Moreover, the technology we are using today depends on the manipulation of electric charges. Starting from the fascinating natural phenomena of lighting and thunderstorm to the use of spray to paint car body, chimeny to trap dust, smoke or soot that escape from cement or textile plant. Photo copy machines, printers all do work on the principle of electrostatics.

Electrostatics: Is the study of charges at rest.

10.1 Electric charge

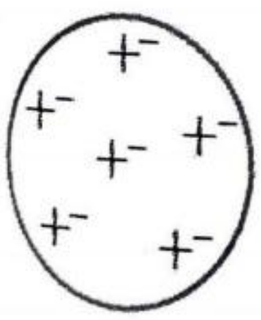
All objects are made up from tiny particles called *atoms*. Atoms are inturn made up of subatomic particles named, neutron, proton and electron. Conventionally, neutrons carry no charge, protons carry positive charge and electrons carry negative charge.

Under normal condition an atom is electrically neutral since it contains equal number of protons and electrons. An atom is said to be *charged* when it *gained* or *lost* electrons.

☞ An atom is said to be positively charged when it lost electrons and negatively charged when it gained electrons.

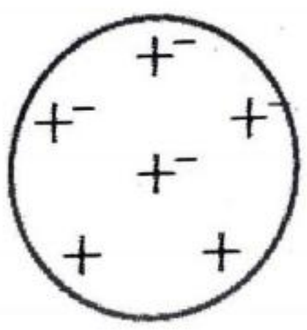
As an example, consider the following

$6 + (-6) = 0$



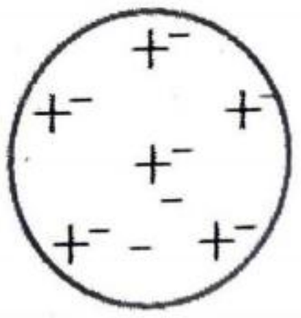
The net charge is zero
→ The body is neutral

$6 + (-4) = 2$



The net charge is +2
→ This body is positively charged

$6 + (-8) = -2$



The net charge is -2
→ This body is negatively charged

Conceptual Example

1. Can an object gain or lose proton under normal condition? Explain

   Explanation: No. Protons are inside the nucleus bonded with the neutron and can not be readily displaced with out applying huge energy.

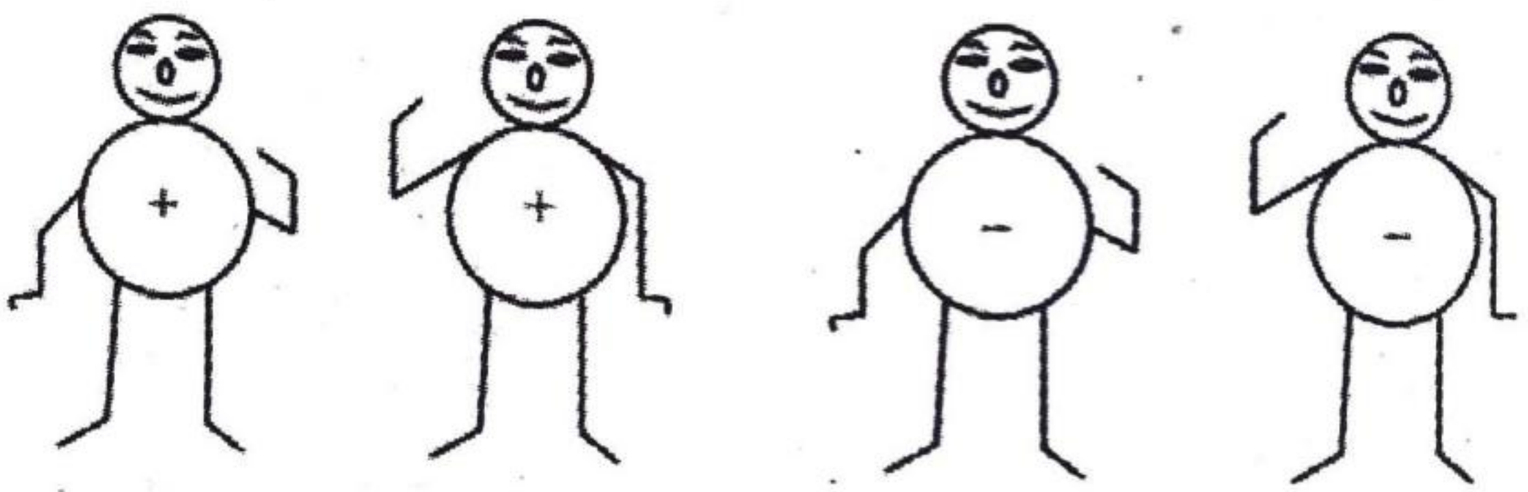
Law of electrostatics

Two positive charges repel each other, two negative charges also repel each other. But, a positive charge attracts a negative charge and a negative charge attracts a positive charge. This is what we call *law of electrostatics*.

> **Law of electrostatics** states that;
>
> "Like charges repel each other and unlike charge attract each other."

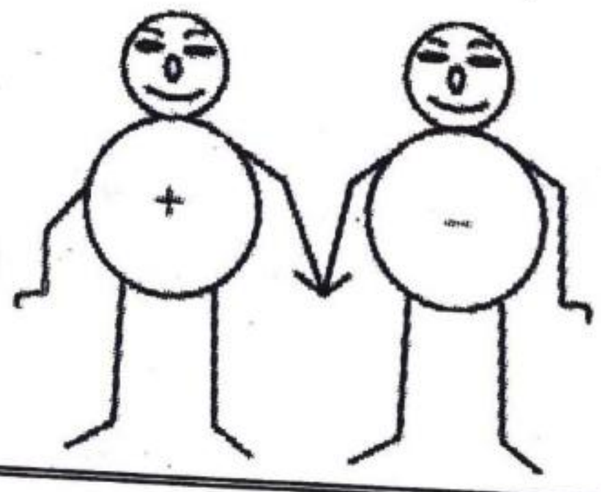
Remember, always that

Like charges repel



And

Unlike charges attract



> **Law of conservation of charge** states that
>
> "The total electric charge of a system remains constant despite changes inside the system".

Amount of charge

The amount of charge carried by a body is obtained by,

$$Q = ne$$

Where Q, is amount of charge n, is number of electrons lost or gained by the body and e, is the charge carried by an electron which is called

elementary charge $\left(e=-1.6\times10^{-19}C\right)$. Coulomb (C), is the SI unit of electric charge.

Illustrative Examples

2. How many electrons are there in 1 coulomb of charge?

Solution: $Q=ne \Rightarrow n=\frac{Q}{e}=\frac{1}{1.6\times10^{-19}}=6.25\times10^{18}$ electrons.

3. If 1.875×10^{19} electrons are transferred from a cloth to a plastic rod, how much charge did the rod acquire?

Solution: $Q=ne=\left(1.875\times10^{19}\right)\left(1.6\times10^{-19}\right)=3C$

4. Two spheres carried charges of $5\times10^{-11}C$ and $-11\times10^{-11}C$. If they made contact,

a) Calculate the final charge on each sphere

b) How many excess electrons are there on each sphere?

Solution:

a) $Q=\frac{Q_1+Q_2}{2}=\frac{5\times10^{-11}+\left(-11\times10^{-11}\right)}{2}=-3\times10^{11}C$

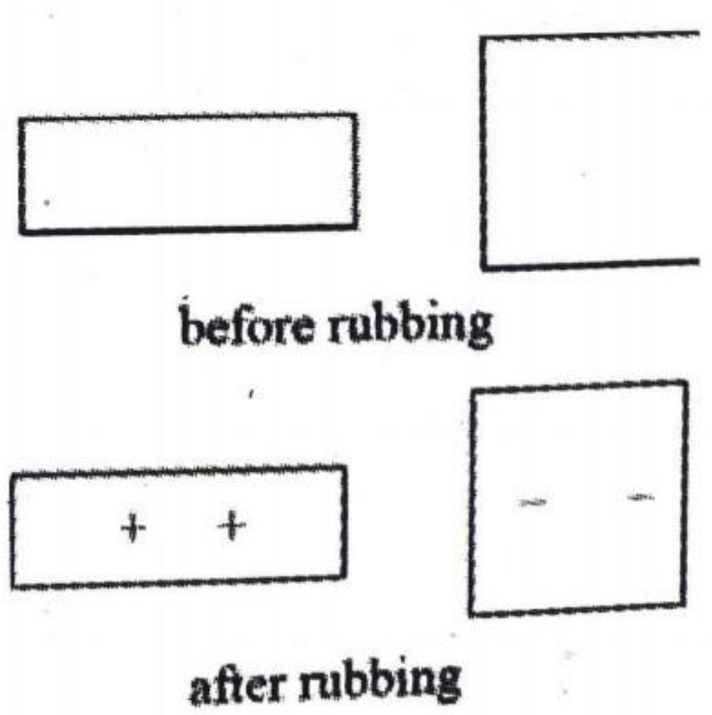
b) $Q=ne=n=\frac{Q}{e}=\frac{-3\times10^{11}C}{1.6\times10^{-19}}=1.875\times10^{8}$ electron

Process of Charging

The process of producing charge on a neutral body is called *charging*. The most common methods of charging are *rubbing*, *conduction* and *induction*. Let's see them one by one.

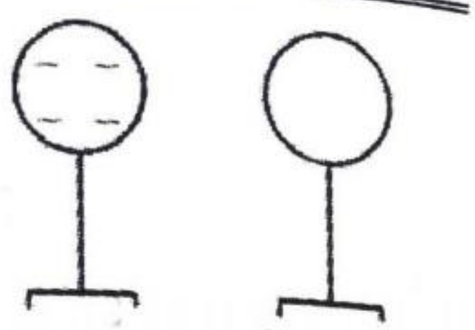
A. Charging by rubbing (friction)

Two neutral bodies are rubbed together in the process, electrons transfer from one body to the other. And after the process, the bodies became *oppositely* charged.

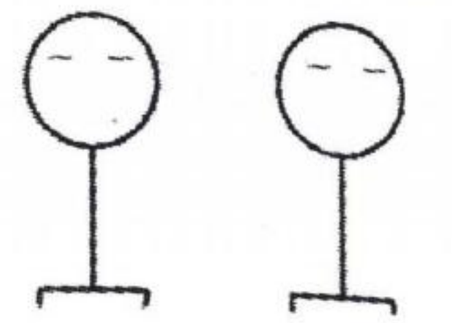


B. Charging by conduction (sharing)

A *charged body* and a *neutral body* are brought in physical contact. In the process charges flow from the charged body to the neutral body and the bodies finally carry the *same type* of charge.



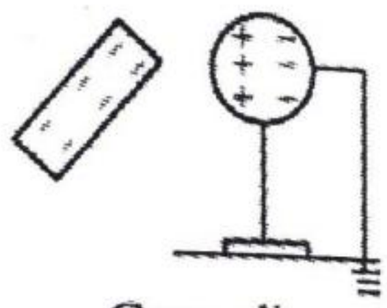
before conduction



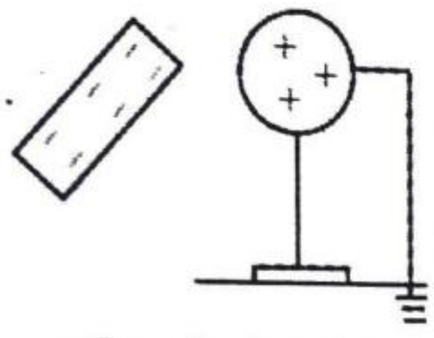
after conduction

C. Charging by Induction

A *charged body,* say a negatively charged rod is brought near to but not with physical contact with a neutral metal sphere on an insulating stand. The electrons in the sphere are pushed to one side while the protons are attracted to the other side. If you touch the negative side with your hand or connect it with the earth using a wire, the sphere will attain positive charge. And after the process the bodies become *oppositely* charged.



Grounding



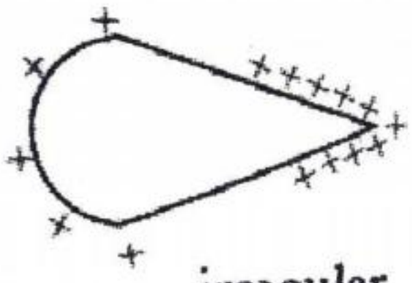
after induction

Distribution of charge on conductors

☞ Static charge distribute itself on the outer surface of a conductor.

- Charges distribute *uniformly* on the outer surface of a sphere.
- On an irregular conductor, the charge distribution is *denser* at the curved or *sharp edge*.

sphere



irregular conductor

Conceptual Examples

5. Why is that, the static charges distribute them selves on the outer surface of a conductor and not in the inside?

   Explanation: The charges cannot stay inside because of the repulsion force between them.

6. Where will the static charge distribute itself if it gets a hollow conductor?

343

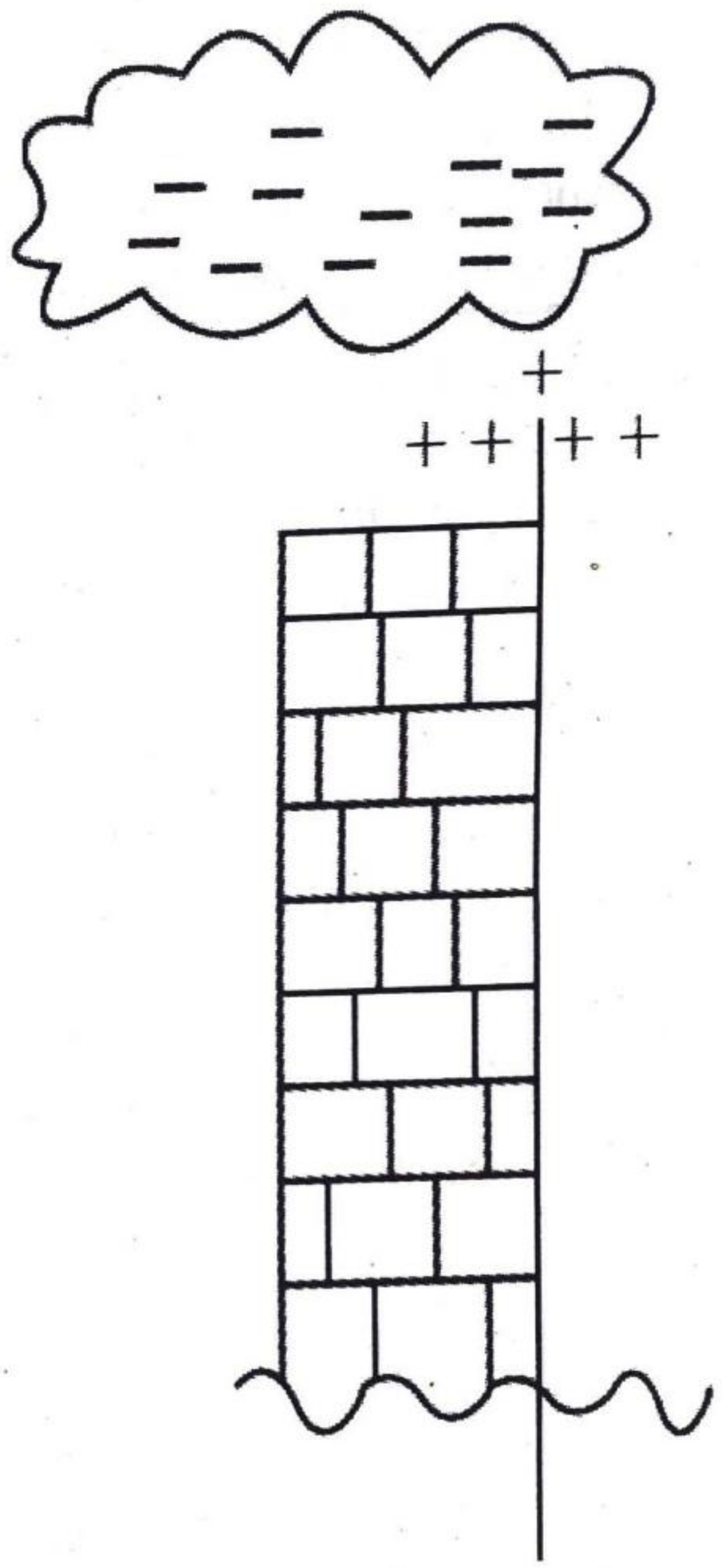
Explanation: Static charge distributes itself on the outer surface of a solid or a hollow conductor.

7. Why is the charge concentration high at the sharp edge of a conductor?

Explanation: Because, the charge density = $\frac{\text{charge}}{\text{Area}}$ hence as area decreases charge density increases.

Thunder and lightning

During rainy season, clouds are formed in different places and move from place to place. as a result of the relative motion between layers of clouds, the clouds become charged by rubbing. When these charged clouds meet, they neutralize each other and in the process a spark of lightning is formed. This lighting releases enormous amount of energy which rapidly heat and expand the air through which the spark passes, it produces a huge and roaring sound which is called thunder.



Lightning rod

It is a metal bar erected at the top of buildings to protect the building form lightning.

- The rod is pointed up with the other end of it buried in the ground. The rod saves the building form lightning either by fighting via shoting opposite charges to neutralize the coming charge or by simply receiving the coming charge and sending it to the ground.

Conceptual Example

8. You will not be struck by lightning when you are inside a car or an aeroplane. Why?

Explanation: Because, the metal shell will divert the charge away.

10.2 Electric forces and fields

Force between charges

The force between charges that are at rest is called *electrostatic force*. And the amount of force with which two charges are repelling or attracting each other is determined by *coulombs law*.

> **Coulomb's law** states that;
>
> "The force between two point charges is directly proportional to the product of the two charges and inversely proportional to the square of the distance between them'.

That is, if two charges (Q_1) and (Q_2) are a distance (r) apart, then the electrostatic force between them is given by

$$\boxed{F = K\frac{Q_1 Q_2}{r^2}}$$

Where, K is called the *electrostatic constant* and is equal to,

$$K = \frac{1}{4\pi\varepsilon_o}, \text{ where (epsilon) is permittivity of free space}$$

$$\varepsilon_o = 8.85 \times 10^{-12} c^2/\mu m^2$$

$$K = \frac{1}{(4)(3.14)\left(8.85\times10^{-12}\right)} = 9\times10^9 Nm^2/c^2$$

Conceptual Example

9. What is the physical interpretation if the force between two point charges is negative?

Explanation: The force is negative when the charges are opposite and attract each other.

Illustrative Examples

10. Two point charges, $Q_1 = 3\times10^{-9}$ C and $Q_2 = -5\times10^{-9}$ C are 2m apart. Determine the magnitude of the force between them and state whether it is attractive or repulsive.

Solution:

$$F = K\frac{Q_1Q_2}{r^2} = \frac{(9\times10^9)(3\times10^{-9})(5\times10^{-9})}{(2)^2} = 3.37\times10^{-8}\,N$$

The force is *attractive*.

11. Two charges 6nC and 3nC are placed 30mm apart. What is the magnitude and nature of the force acting on the charges?

Solution:

$$F = K\frac{Q_1Q_2}{r^2} = \frac{(9\times10^9)(6\times10^{-9})(3\times10^{-9})}{(3\times10^{-2})^2} = 1.8\times10^{-4}\,N$$

The nature of the force is *repulsive*.

12. Compare the magnitude of the electric force between two charges of each charge 1C separated by a distance of 1m with the magnitude of the gravitational force if each have a mass of 1kg and placed 1m apart.

Solution: $F_e = K\frac{Q_1Q_2}{r^2} = \frac{(9\times10^9)(1)(1)}{(1)^2} = 9\times10^9\,N$

$$F_g = \frac{Gm_1m_2}{r^2} = \frac{(6.67\times10^{-11})(1)(1)}{(1)^2} = 6.67\times10^{-11}\,N$$

Now, taking ratio, $\frac{F_e}{F_g} = \frac{9\times10^9}{6.67\times10^{-11}} = 1.3\times10^{20}$

Hence, the electrostatic force is about 10^{20} times the gravitational force in this example.

3. Calculate the electrostatic force between two charges of 3×10^{-7}C each placed 10mm apart.

Solution: $F_e = \frac{KQ_1Q_2}{r^2} = \frac{KQ^2}{r^2} = \frac{(9\times10^9)(3\times10^{-7})}{(10^{-2})^2} = 8.1N$

14. The force of electrostic between two equal charges separated by a distance of 3cm is 3.6N. Find out the magnitude of the charges.

Solution: $F = \frac{KQ_1Q_2}{r^2}$, but $Q_1 = Q_2 = Q$

$$\Rightarrow F = \frac{KQ^2}{r^2} = Q^2 = \frac{Fr^2}{k}$$

$$Q = \sqrt{\frac{Fr^2}{k}} = \sqrt{\frac{(3.6)(0.03)^2}{9\times10^9}} = \sqrt{4\times10^{-14}} = 2\times10^{-7}C$$

$$Q = Q_1 = Q_2 = 2\times10^{-7}C$$

15. When charges $Q_1 = 6\mu c$ and $Q_2 = 15\mu c$ are separated at some distance, they experience an electric force of 900N. what is the separation between the charges.

Solution:

$$F = \frac{KQ_1Q_2}{r^2} \Rightarrow r^2 = \frac{KQ_1Q_2}{F}$$

$$r = \sqrt{\frac{KQ_1Q_2}{F}} = \sqrt{\frac{(9\times10^9)(6\times10^{-6})(15\times10^{-6})}{900}} = 3\times10^{-2}m$$

$r = 3cm$

16. Two small metal spheres carry charges of $6\times10^{-5}C$ and $-4\times10^{-5}C$. If they are brought into contact. Calculate

a) The charge on each sphere after contact

b) The number of electrons lost or gained by each sphere on contact.

c) The magnitude of the force between each sphere if they are placed 40mm apart after contact.

Solution:

a) $$Q = \frac{Q_1 + Q_2}{2} = \frac{6\times10^{-5} + \left(-4\times10^{-5}\right)}{2} = 1\times10^{-5}\text{C}$$

b) $$\Delta Q = ne \Rightarrow n = \frac{\Delta Q}{e} = \Delta Q = Q_f - Q_o = 5\times10^{-5}\text{C}$$

$$\Rightarrow n = \frac{5\times10^{-5}}{1.6\times10^{-19}} = 3.125\times10^{14} \text{ electrons}$$

c) $$F = \frac{KQ_1Q_2}{r^2} = \frac{\left(9\times10^{9}\right)\left(1\times10^{-5}\right)^2}{\left(1\times10^{-2}\right)^2} = 562.5\text{N}$$

17. The electrostatic force between two equal point charges when they are 50mm apart is 80N. What will be the force when they are 5mm apart.

Solution:

$$F_1 = \frac{KQ_1Q_2}{r_1^{\,2}} = \frac{KQ^2}{r_1^{\,2}} = KQ^2 = \left(F_1\right)\left(r_1^{\,2}\right) = \left(80\right)\left(5\times10^{-2}\right)^2$$

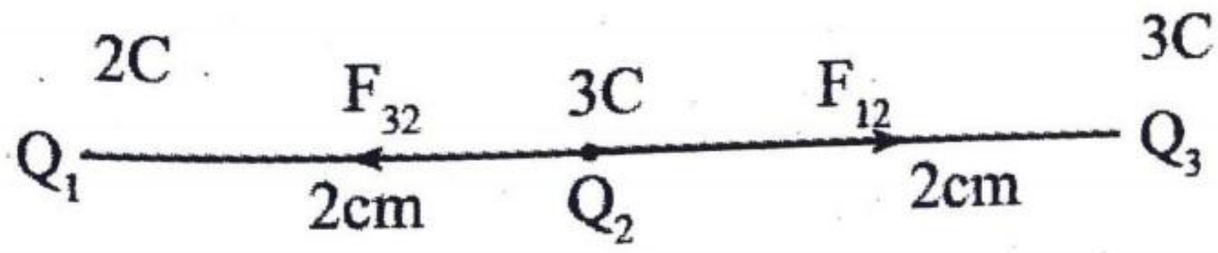
$$KQ^2 = 0.2$$

$$F_2 = \frac{KQ_1Q_2}{r_2^{\,2}} = \frac{KQ^2}{r_2^{\,2}} = \frac{0.2}{\left(5\times10^{-3}\right)^2} = 8000\text{N}$$

Coulomb's law when there are more than two charges.

18. Three charges are arranged in a straight line with 2cm separation as shown. Find the force exerted on Q_2 by the other two charges.

Solution:



→ The force between Q_1 and Q_2 is repulsive: $Q_2 \longrightarrow F_{12}$

$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{\left(9\times10^9\right)(2)(3)}{\left(2\times10^{-2}\right)^2} = 13.5\times10^{13}\text{N}$$

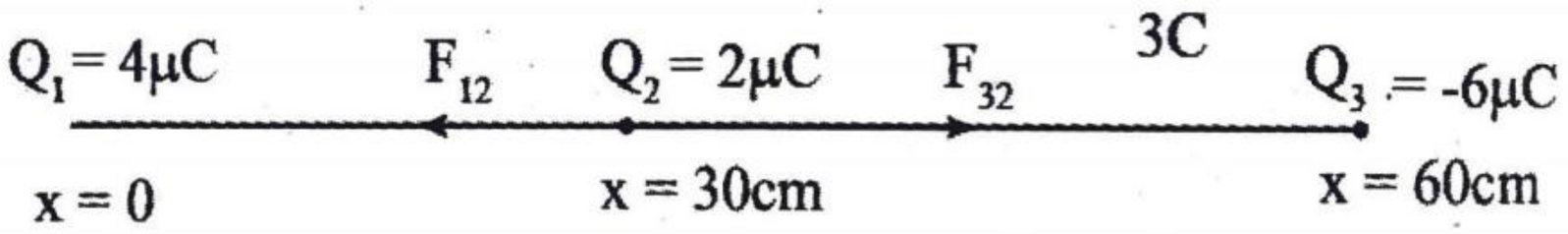
→ The force between Q_2 and Q_3 is repulsive $F_{32} \longleftarrow Q_2$

$$F_{23} = \frac{KQ_2Q_3}{r_2^2} = \frac{\left(9\times10^9\right)(3)(3)}{\left(2\times10^{-2}\right)^2} = 20.25\times10^{13}\text{N left}$$

$$F_{net} = F_{23} - F_{12} = 20.25\times10^{13} - 13.5\times10^{13}\text{N} = 6.75\times10^{13}\text{N}$$

19. If $-4\mu c$ is at $x = 0$, $+2\mu c$ at $x = 30cm$ and $-6\mu c$ at $x = 60cm$. Find the force on the $+2\mu c$ charge

Solution:



→ The force between Q_1 and Q_2 is attractive: $F_{12} \longleftarrow Q_2$

$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{\left(9\times10^9\right)\left(4\times10^{-6}\right)\left(2\times10^{-6}\right)}{\left(0.3\right)^2}$$

$= 0.8\text{N}$ to the left

→ The force between Q_1 and Q_2 is also attractive:

$$F_{23} = \frac{KQ_2Q_3}{r^2} = \frac{\left(9\times10^9\right)\left(6\times10^{-6}\right)\left(2\times10^{-6}\right)}{\left(0.3\right)^2}$$

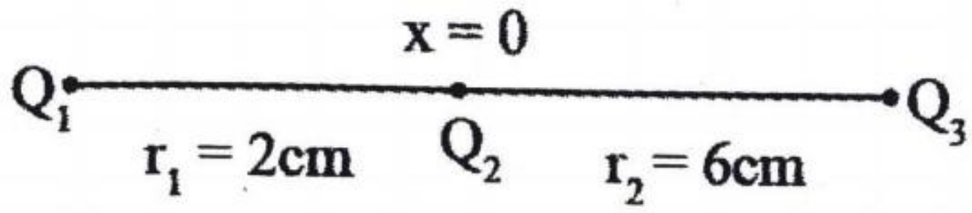
$= 1.2\text{N}$ to the right

Hence, the net force on Q_2 is,

$$F_{net} = F_{32} - F_{12} = 1.2N - 0.8N = 0.4N \text{ to the right}$$

20. Three charges are located on the x axis. A charge $Q_1 = 40nc$ at $x = -2cm$ an unknow positive charge Q_2 at the origin. If a net force of 0.015N acts on Q2, what is the charge carried by it? and a third charge $Q_3 = -60nc$ at $x = 6cm$

Solution:



→ The force between Q_1 and Q_2 is attractive:

$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{\left(9\times10^9\right)\left(40\times10^{-9}\right)}{\left(2\times10^{-2}0.3\right)^2}Q_2 = 90\times10^4Q_2$$

→ The force between Q_2 and Q_3 is attractive:

$$F_{32} = \frac{KQ_1Q_2}{r_2^2} = \frac{\left(9\times10^9\right)\left(60\times10^{-6}\right)}{\left(6\times10^{-2}\right)^2}Q_2 = 15\times10^4Q_2$$

The net force is then

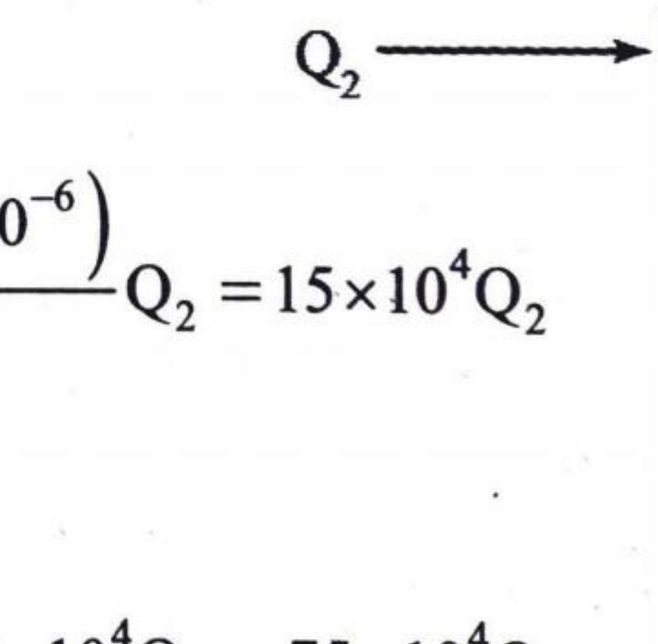
$$F_{net} = F_{12} - F_{23} = 90\times10^4Q_2 - 5\times10^4Q_2 = 75\times10^4Q_2$$

$$\Rightarrow\ 75\times10^4Q_2 = 0.015N \Rightarrow Q_2\frac{15\times10^{-3}}{75\times10^4} = 20\times10^{-9}C$$

Thus, $Q_2 = 20nc$

21. Find the resultant force on the 10μc

Solution:



→ The force between Q_1 and Q_2 is

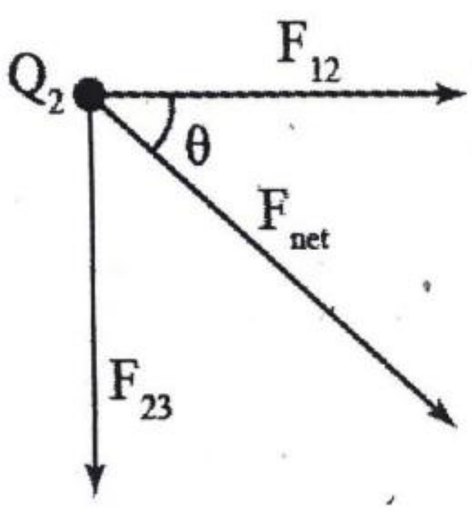
$$F_{12} = \frac{KQ_1Q_2}{r_1^2} = \frac{\left(9\times10^{9}\right)\left(6\times10^{-6}\right)\left(10\times10^{-6}\right)}{\left(30\times10^{-2}\right)^2}$$

$$F_{12} = 6N$$

→ The force between Q_2 and Q_3 is attractive:

$$F_{23} = \frac{KQ_2Q_3}{r_2} = \frac{\left(9\times10^{9}\right)\left(10\times10^{-6}\right)\left(8\times10^{-6}\right)}{\left(20\times10^{-2}\right)^2} = 18N$$

The forces on Q_2 due to Q_1 and Q_3 are then.



the net force is then

$$F_{net} = \sqrt{F_{12}^2 + F_{23}^2} = \sqrt{(6)^2 + (18)^2}$$

$$F_{net} = 19N$$

and its direction is,

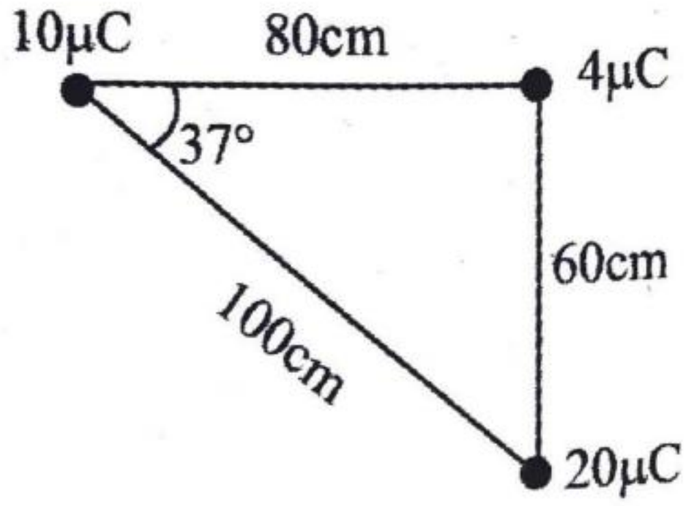
$$\tan\theta = \frac{F_{23}}{F_{12}} = \frac{18}{6} = 3$$

$$\Rightarrow \theta = 71.5°$$

22. Find the resultant force on the 20μc

Solution:

→ The force between the 4μc and the 20μc is,



$$F_1 = \frac{KQ_1Q_2}{r_1^{\,2}}$$

$$= \frac{\left(9\times10^{9}\right)\left(4\times10^{-6}\right)\left(20\times10^{-6}\right)}{\left(60\times10^{-2}\right)^2}$$

$$F_1 = 2N$$

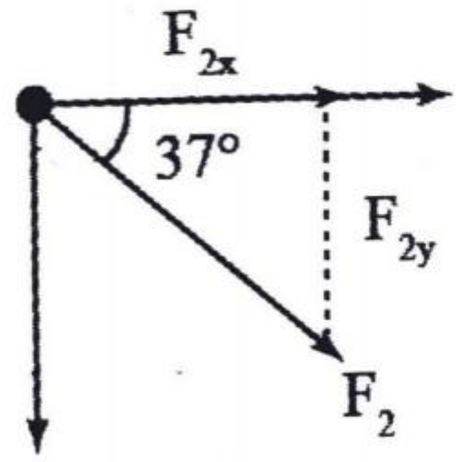
→ The force between the 10μc and the 20μc is, Q_2 and Q_3 is attractive:

$$F_2 = \frac{KQ_2Q_3}{r_2^{\,2}} = \frac{\left(9\times10^{9}\right)\left(10\times10^{-6}\right)\left(20\times10^{-6}\right)}{\left(100\times10^{-2}\right)^2} = 1.8N$$

Resolving F_2 on to x and y

$$F_{2x} = F_x\cos 37 = \left(1.8\right)\left(0.8\right) = 1$$

$$F_{2y} = F_2\sin 37 = \left(1.8\right)\left(0.6\right) = 1.08N$$



the net force is then

$$F_x = 14 \text{ and } F_y = F_1 + F_{2y} = 2+1.08 = 3.08$$

$$F_{net} = \sqrt{F_x^{\,2} + F_y^{\,2}} = \sqrt{\left(1.44\right)^2 + \left(3.8\right)^2}$$

$$= 3.4N \text{ at } 65°$$

Electric field

The region where an electric charge will experience a repulsive or attractive force due to the presence of other electric charge is known as *electric field*.

Electric field: Is a region around a point charge

Plotting electric field lines

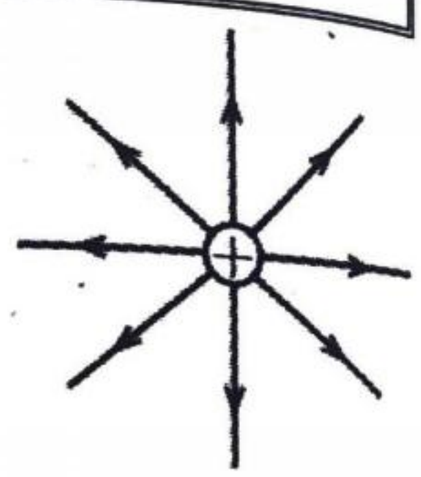
Imaginary lines that are used to visualize electric field are called *electric field lines* or *electric lines of force*.

Electric filed lines: Are imaginary lines drawn to visualize electric field in space produced due to a point charge.

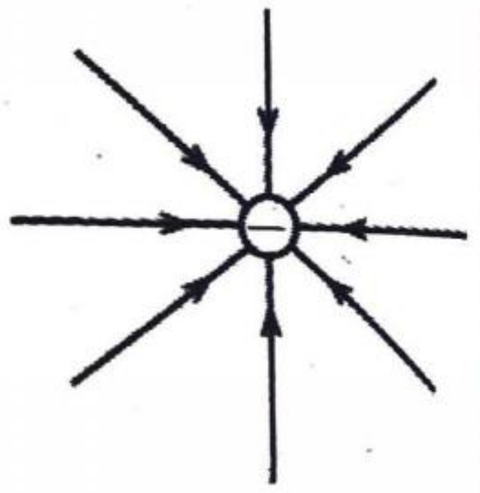
To plot electric field lines around a point charge or any charged body, we place a positive test charge (q^+) near the charge.

The direction of the electric field due to the point charge is the same as the direction of the electric force on a positive test charge placed at that point.

The electric field lines around a *positive* point charge are *radially outward* showing that the point charge is repelling the test charge.

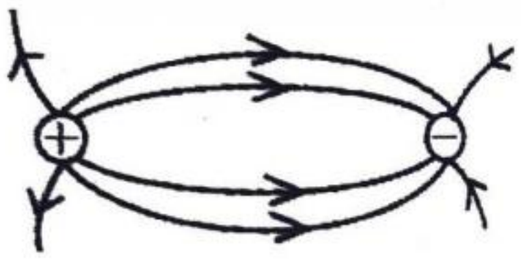


The electric field lines around a *negative* point charge are *radially inward* indicating that the point charge is attracting the test charge.

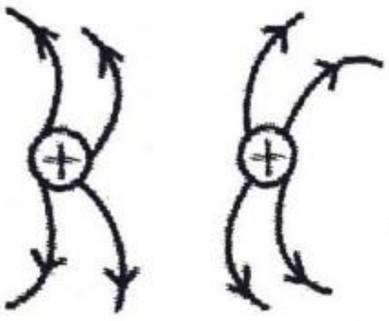


In general, we can have the following types of electric field line patterns.

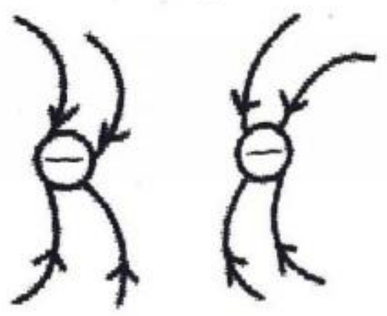
→ Electric field lines between a positive and a negative charge.



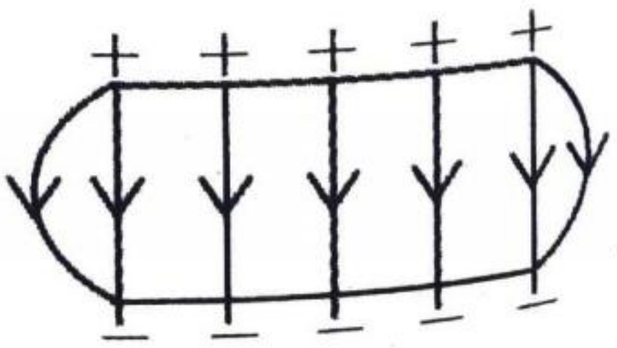
→ Electric field lines between two positive point charges



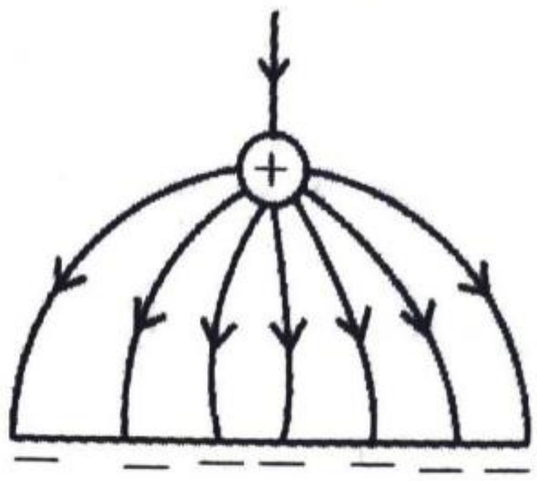
→ Electric field lines between two negative point charges.



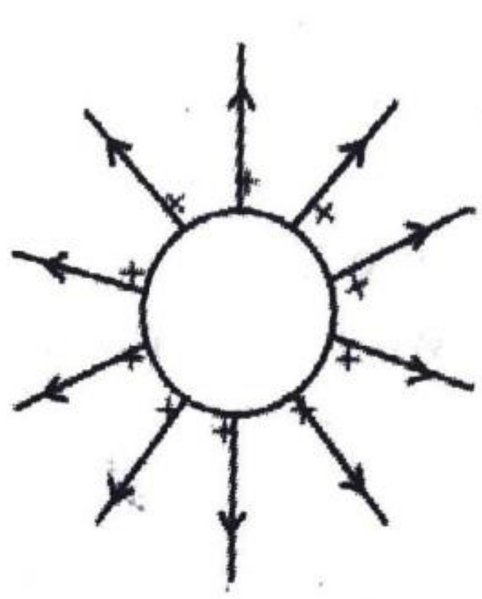
→ Electric field lines between two oppositely charged parallel plates.



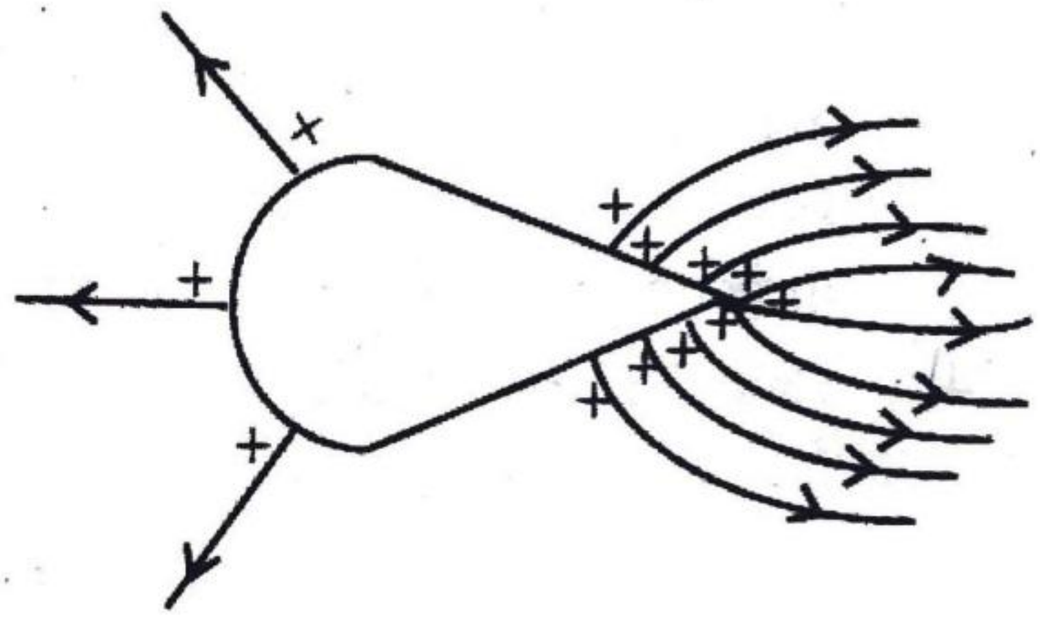
→ Electric field lines between a point charge and charged plat



→ Electric field lines due to a charged sphere.



→ Electric field lines due to an irregularly shaped charged conductor.



☞ **Some important points to remember about electric fields**

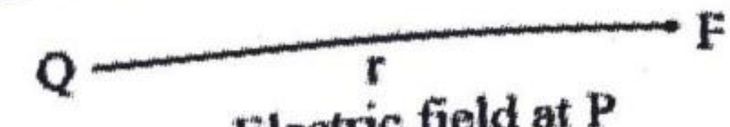
- Electric field lines never cross each other
- Electric field lines start form the positive charge and goes to a negative charge
- Electric field lines are close together where the electric force is stronger
- Electric field is uniform in side a charged parallel plate.
- Electric field due to a point charge is not uniform. It varies from point to point.
- Electric field lines always start at right angle to the charged object causing the field.
- The direction of the electric field is the direction of the force on a positive test charge.

Electric field strength

The strength or intensity of the electric field (E) is defined as.

Electric filed strength: Is the force experienced per unit charge by a positive test charge placed at the point.

Mathematically, $\vec{E} = \frac{F}{q^+} = \frac{KQq^+}{r^2 / q^+}$



$$\boxed{E = \frac{KQ}{r^2}}$$

Electric field strength is a vector quantity and its SI unit is N/c (Newton/coulombs)

Illustrative Examples

23. A positive charge of 30c is in an electric field of 5N/C. what is the magnitude and direction of the force?

Solution: $E = F/Q \Rightarrow F = EQ = (5)(30) = 150N$

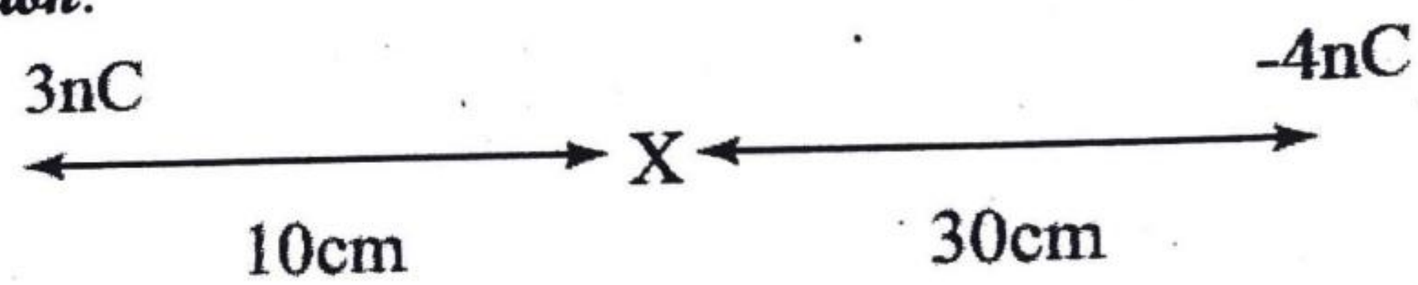
since the charge is positive, the direction of the force is in the direction of the electric field.

24. Calculate the electric field strength at a distance of 30cm from a 5nc charge.

Solution: $E = \frac{KQ}{r^2} = \frac{\left(9\times10^{9}\right)\left(5\times10^{-9}\right)}{\left(30\times10^{-2}\right)^2} = 5\times10^{2}N/c$

25. Calculate the electric field at point x between the two charges.

Solution:

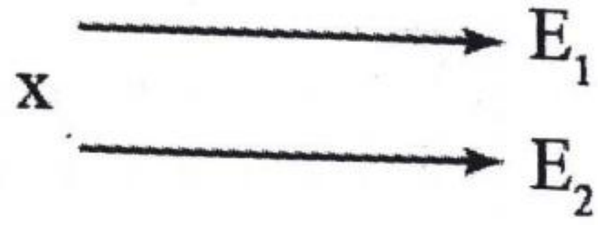


→ The electric field due to the 3nc charge at point x is,

$$E_1 = \frac{KQ_1}{r_1^2} = \frac{\left(9\times10^{9}\right)\left(3\times10^{-9}\right)}{\left(10\times10^{-2}\right)^2} = 0.27N/c$$

→ The electric field strength due to the –4nc charge at point x is,

$$E_2 = \frac{KQ_2}{r_2^{\,2}} = \frac{\left(9\times10^{9}\right)\left(4\times10^{-9}\right)}{\left(30\times10^{-2}\right)^2} = 0.04\text{N/c}$$



The electric field at point x due to the two point charge is,

$$E_{total} = E_1 + E_2 = 0.27 + 0.04 = 0.31\text{N/c}$$

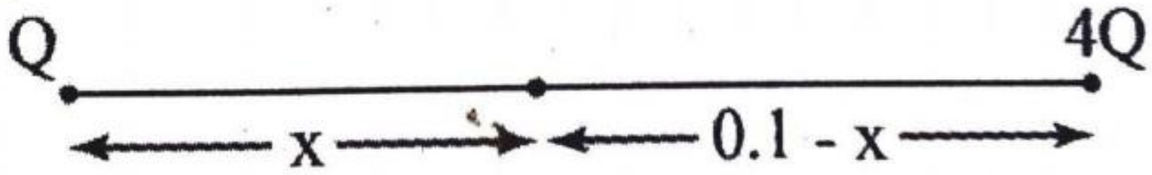
26. What must be the charge on a particle of mass 2gm for it to remain stationary in a downward electric field of 500N/c.

Solution:

$$E = \frac{F}{Q} = \frac{mg}{Q} \Rightarrow Q = \frac{mg}{E} = \frac{\left(2\times10^{-3}\right)(10)}{500} = 4\times10^{-5}\text{C}$$

27. Two positive charges Q and 4Q are placed 10cm apart. Calculate the distance from the charge Q where the electric field strength is zero between these charges.

Solution:



→ At distance x from the charge Q, the field is,

$$E_1 = \frac{KQ}{x^2}$$

→ At distance (0.1 – x) from the charge 4Q, the field is,

$$E_2 = \frac{4KQ}{\left(0.1-x\right)^2}$$

$$E_{net} = E_1 - E_2 = 0$$

$$\Rightarrow E_1 = E_2 \Rightarrow \frac{KQ}{x^2} = \frac{K4Q}{(0.1-x)^2} \therefore 4x^2 = (0.1-x)^2$$

$$\Rightarrow 4x^2 = (0.01 - 0.2x + x^2) \Rightarrow 3x^2 + 0.2x - 0.01 = 0$$

Using the general quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-0.2 \pm \sqrt{(0.2)^2 - 4(3)(-0.01)}}{2(3)}$$

$$x = 0.033m = 3.3cm$$

28. An electron is situated in an electric field of 1500N/c pointing in an easterly direction. Calculate the magnitude and direction of the electron at this point in the electric field ($m_e = 9.1\times10^{-31}kg$).

Solution:

$$E = F/Q = EQ = (1500^2)(-1.6\times10^{-9})$$

$$F = -2.4 \times 10^{-6}N$$

And from Newton's second law,

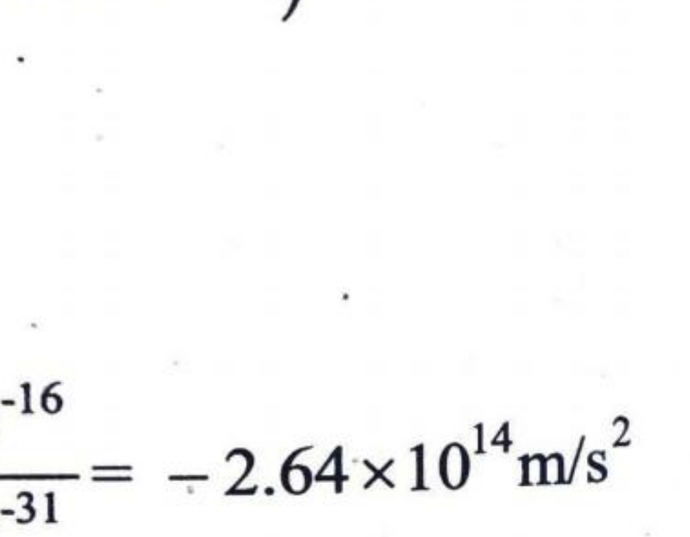
$$F = ma \Rightarrow a = \frac{F}{m} = \frac{2.4\times10^{-16}}{9.1\times10^{-31}} = -2.64\times10^{14}m/s^2$$

The negative sign tells us the electron accelerate, in the opposite direction to the electric field

$$\Rightarrow a = 2.64\times10^{14}m/s^2 \text{ west.}$$

29. A spherical conductor of radius 50cm carries a charge of 10c. calculate the electric field strength at a distance of

a) 10 cm, from the center
b) 50 cm, from the center



c) 100 cm, from the center

Solution:

a) $E = \dfrac{KQ}{r^2}$,

but Q = 0 inside the sphere E = 0

b) $E = \dfrac{KQ}{R^2} = \dfrac{(9\times10^9)(10)}{(50\times10^{-2})^2} = 36\times10^{10}\text{N/c}$

c) $E = \dfrac{KQ}{r^2} = \dfrac{(9\times10^9)(10)}{(1)^2} = 9\times10^{10}\text{N/c}$

Electric potential energy

The electrical potential energy of a charge is the energy it has because of its position relative to other charges that it interact with.

The potential energy of a charge (Q_1) relative to a charge (Q_2) a distance (r) away is calculated by,

$$\boxed{U = \frac{KQ_1Q_2}{r}}$$

Illustrative Examples

30. What is the electric potential energy of a 7nc charge that is 2cm from a 20nc

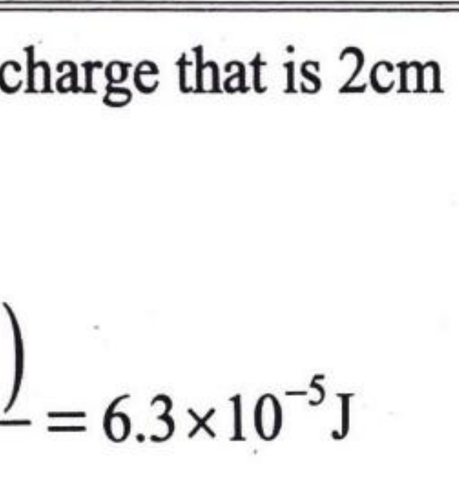
Solution:

$$U = \frac{KQ_1Q_2}{r} = \frac{(9\times10^9)(7\times10^{-9})(20\times10^{-9})}{(2\times10^{-2})^2} = 6.3\times10^{-5}\text{J}$$

Electric Potential

The electric potential (Absolute potential) at a point is the electric potential energy divided by charge.

The absolute potential at point A is the work done in carrying a test charge from infinity up to point A and is given by



$$V = \frac{KQ}{r}$$

The SI unit of electric potential in volt (v)

Illustrative Examples

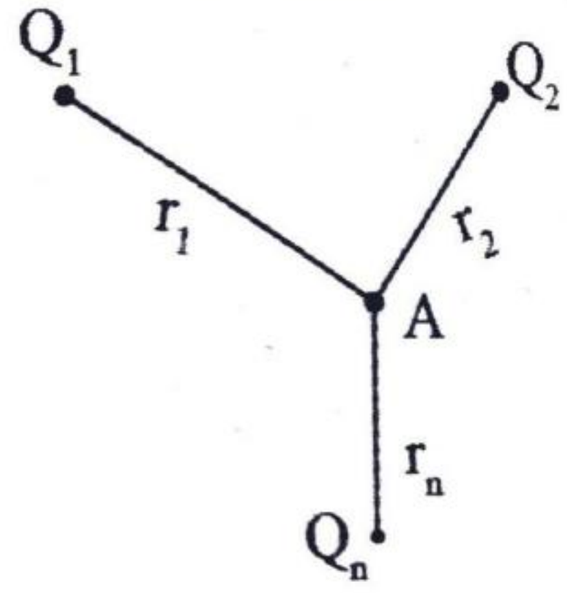
31. Show that electric potential is $V = \frac{KQ}{r}$

Solution: $V = \frac{\text{work}}{\text{charge}} = \frac{\text{Potential energy}}{\text{charge}} = \frac{KQq^{+}}{r/q^{+}} = KQ/r$

Electric potential due to a collection of charges

The electric potential at a point due to charges Q_1, Q_2, Q_n that are at distance r_1, r_2, r_n from the given point is the algebraic sum of potential of each charge at the point.

That is, the total potential of the charges Q_1, Q_2,... Q_n at distance r_1, r_2,... r_n from point A is given by



$$V = V_1 + V_2 + \ldots + V_n$$

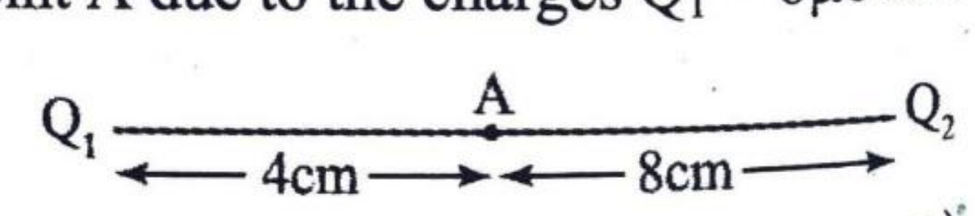
$$V = \frac{KQ}{r} + \frac{KQ_2}{r_2} + \ldots \frac{KQ_n}{r_n}$$

Illustrative Examples

32. Calculate the absolute potential of a point charge $Q = 6\mu c$ at a distance of $r = 30cm$

Solution: $V = K\frac{Q}{r} = \frac{\left(9\times10^{9}\right)\left(6\times10^{-6}\right)}{30\times10^{-2}} = 18000v$

33. Calculate the potential at point A due to the charges $Q_1 = 6\mu c$ and $Q_2 = -6\mu c$ shown below.



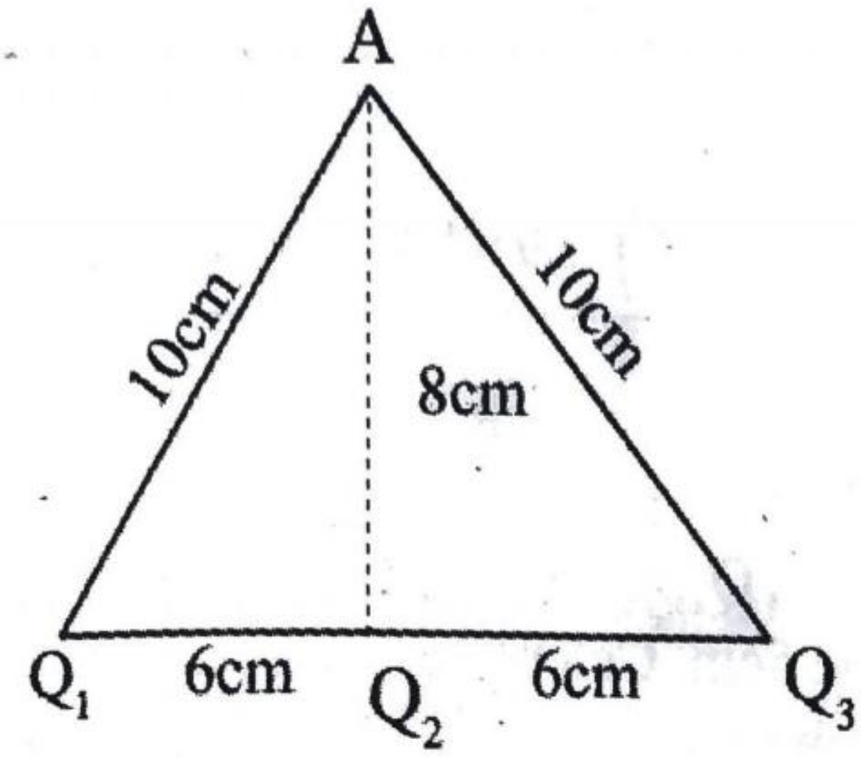
Solution: $V_A = V_1 + V_2$

$$V_A = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} = \frac{\left(9\times10^{9}\right)\left(6\times10^{-6}\right)}{4\times10^{-2}} + \frac{\left(9\times10^{9}\right)\left(-6\times10^{-6}\right)}{8\times10^{-2}}$$

$$= 13.5\times10^{5} - 6.75\times10^{5}$$

$$V_A = 6.75\times10^{5}$$

34. Compute the absolute potential at point A due to the three point charges.



Solution: $V_a = V_1 + V_2 + V_3$

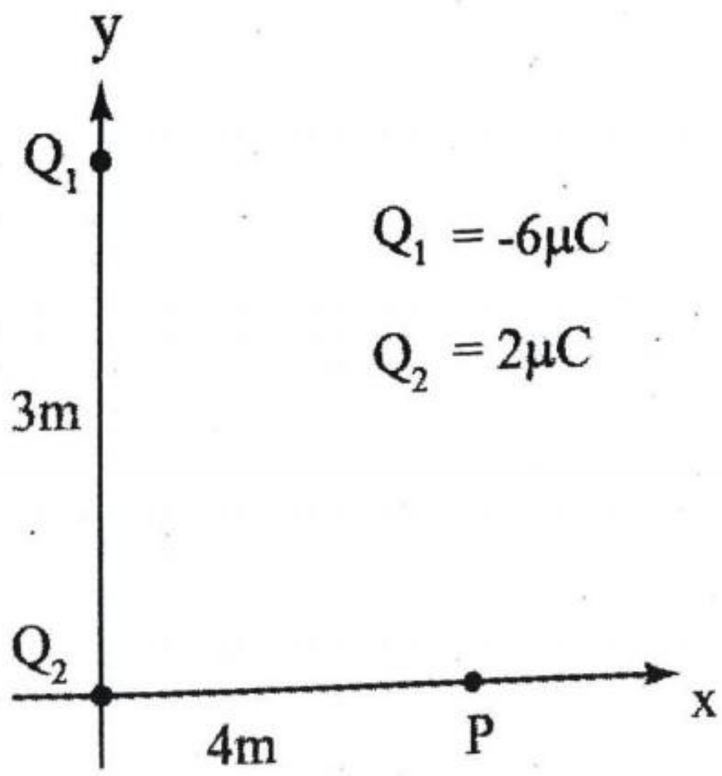
$$V_1 = K\frac{Q_1}{r_1} = \frac{\left(9\times10^{9}\right)\left(-2\times10^{-6}\right)}{10\times10^{-2}} = -18{,}0000\text{v}$$

$$V_2 = K\frac{Q_2}{r_2} = \frac{\left(9\times10^{9}\right)\left(8\times10^{-6}\right)}{8\times10^{-2}} = 90{,}0000\text{v}$$

$$V_3 = K\frac{Q_3}{r_3} = \frac{\left(9\times10^{9}\right)\left(4\times10^{-6}\right)}{10\times10^{-2}} = 36{,}0000\text{v}$$

$$\Rightarrow V_A = V_1 + V_2 + V_3 = -180000 + 900000 + 360000 = 1080000\text{V}$$

35. Calculate the electric potentail at point p



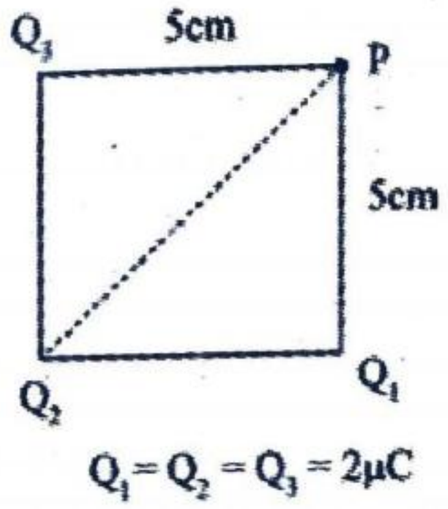
Solution: $V_p = V_1 + V_2$

$$V_p = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2}, \; r_1 = \sqrt{3^2 + 4^2} = 5\text{m, and } r_2 = 4\text{m}$$

$$= 9\times10\frac{9\times\left(-6\times10^{-6}\right)}{5} + \frac{9\times10^{9}\left(2\times10^{-6}\right)}{4}$$

$$\Rightarrow V_p = -6.3\times10^{3}\text{V}$$

36. Three equal charges Q_1, Q_2 and $Q_3$are placed at the three corners of a square of side 5cm. Calculate the electric potential at the fourth corner (point P).



Solution: $V_p = V_1 + V_2 + V_3$

$$V_p = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2} + K\frac{Q_3}{r_3},$$

$$r_2 = \sqrt{5^2 + 5^2} = 5\sqrt{2}$$

$$= 9\times10^{9}\times2\times10^{-6}\left(\frac{1}{5\times10^{-2}} + \frac{1}{5\times\sqrt{2}\times10^{-2}} + \frac{1}{5\times10^{-2}}\right)$$

$$= 9.7\times10^{5}\text{V}$$

Electric potential difference

The electric potential difference is the difference of the absolute potential of a point charge at two different point.

> The potential difference V_{AB} between two points A and B is defined as the work done per unit charge in moving a charge from A to B in electric field.

That is

$$\boxed{V_{AB} = \frac{W_{AB}}{Q}}$$

The unit of potential difference is volt (v)

Thus, $\boxed{1\text{V} = 1\text{J/C}}$

→ The potential difference, $V = \frac{W}{Q}$, but $W = Fd$ and $F = EQ$ substituting these,

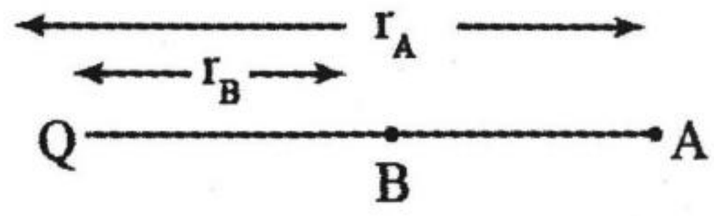
$$V = \frac{EQd}{Q}$$

$$= \boxed{V = Ed} \quad E \text{ is electric field}$$

d is distance moved

→ The potential difference between two points say A and B is written as,

$$\boxed{V_{AB} = V_B - V_A = KQ\left(\frac{1}{r_B} - \frac{1}{r_A}\right)}$$



Conceptual Example

37. Explain the difference between electric potential energy, electric potential and potential difference.

Explanation: Electric potential energy is energy stored and is given by $U = K\frac{Q_1Q_2}{r}$. Electric potential is energy per unit charge, $V = K\,Q/r$ And potential difference is the difference between electric potential $V_{AB} = V_B - V_A = KQ\left(\frac{1}{r_B} - 1/r_A\right)$.

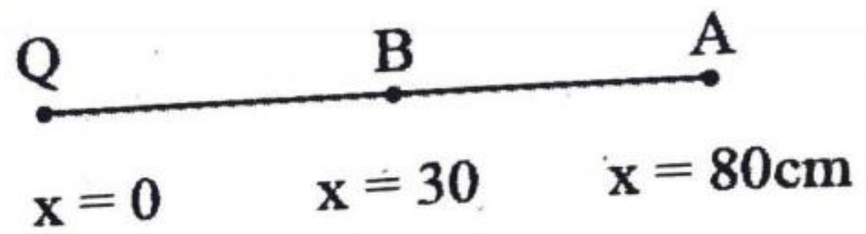
Illustrative Examples

38. What is the potential difference between two points in an electric field if it takes 600J of energy to move a charge of 2c between these points?

Solution: $V = \frac{W}{Q} = \frac{600}{2} = 300v$

39. A 40nc charge is located at x = 0. Consider two points A and B located at x = 80cm and x = 30cm respectively. Calculate the electrical potential difference V_{AB} between the points A and B.

Solution: $V_{AB} = V_B - V_A$



$$V_B = K\frac{Q}{r_B} = 9\times10^9\frac{\left(40\times10^{-9}\right)}{0.3} = 1200V$$

$$V_A = K\frac{Q}{r_A} = 9\times10^9\frac{\left(40\times10^{-9}\right)}{0.8} = 450V$$

$$V_{AB} = V_B - V_A = \left(1200-450\right) = 750V$$

40. The potential difference between two plates 5mm apart is 10,000V. Determine the electric field intensity between the plates.

Solution: $V = Ed \Rightarrow E = {}^{V}\!/_{d} = \frac{10{,}000}{5\times10^{-3}} = 2\times10^6 N/m$

41. A 20nc charge is moved between two points A and B that are 30mm apart and have a potential difference of 600v. Calculate

a) The electric field between the points

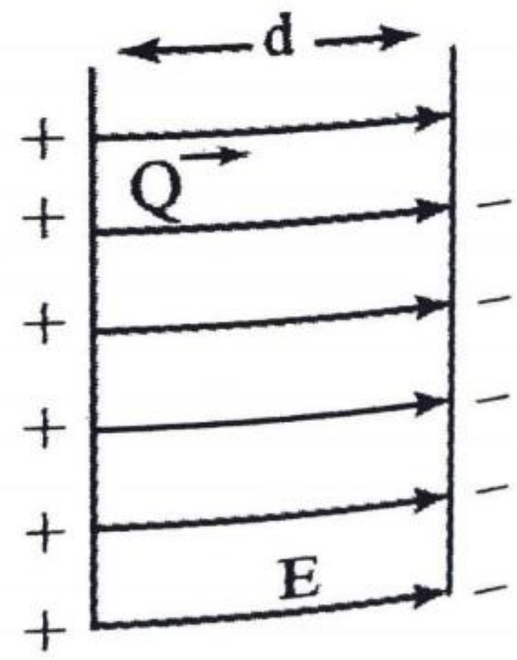
b) The work done on the charge

Solution: a) $V = Ed \Rightarrow E. = {}^{V}\!/_{d} = \frac{600}{30\times10^{-3}} = 20000v/m$

b) $V = \frac{W}{Q} \Rightarrow W = VQ = \left(600\right)\left(20\times10^{-9}\right) = 1.2\times10^{-5}J$

→ When a charge (Q) is moved against a constant electric force for a distance (d), the charge stored potential energy, given by

$$\boxed{P_E = QEd}$$



And if the charge is released, it acquires a kinetic energy.

$$\boxed{K_E = \frac{1}{2}mv^2}$$ m, mass

v, speed

Now, from conservation of energy, $P_E = K_E$

$$\Rightarrow \boxed{QEd = \frac{1}{2}mv^2}$$

☞ **Remember that:** The potential energy increases as a positive charge moved against the electric field and the potential energy decreases as a negative charge is moved against the same field.

Illustrative Examples

42. A proton is accelerated from rest to a speed of 2×10^5m/s over a distance of 8cm in a uniform electric field. Calculate the strength of the electric field ($m_p = 1.67 \times 10^{-27}$kg)

Solution: $P_E = K_E \Rightarrow EQd = \frac{1}{2}mv^2$

$$E = \frac{mv^2}{2Qd}$$

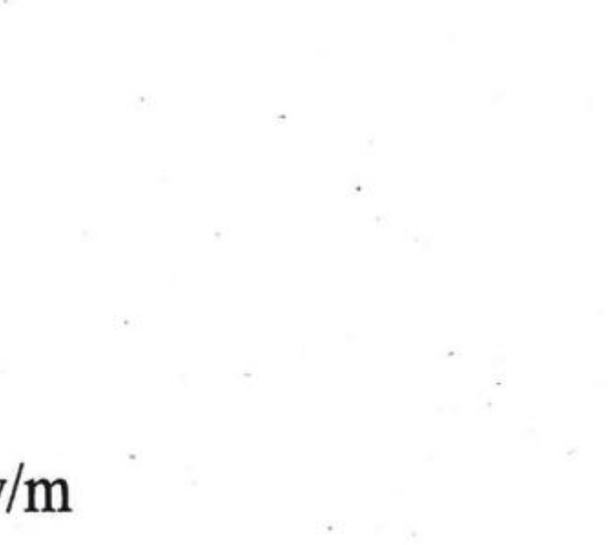
$$E = \frac{\left(1.67\times10^{-27}\right)\left(2\times10^{5}\right)^2}{(2)\left(1.6\times10^{-19}\right)\left(8\times10^{-2}\right)} = 2609\text{v/m}$$

43. Two charges $Q_1 = -50$nc and $Q_2 = 70$nc are located on the x axis at x = –30cm and x = 80cm respectively. Calculate the total electric potential at point p which is 40cm on the y axis.

Solution: $V_p = V_1 + V_2 = K\frac{Q_1}{r_1} + K\frac{Q_2}{r_2}$

$r_1 = \sqrt{0.3^2 + 0.4^2} = 0.5$

and $r_2 = \sqrt{0.8^2 + 0.4^2} = 0.9$

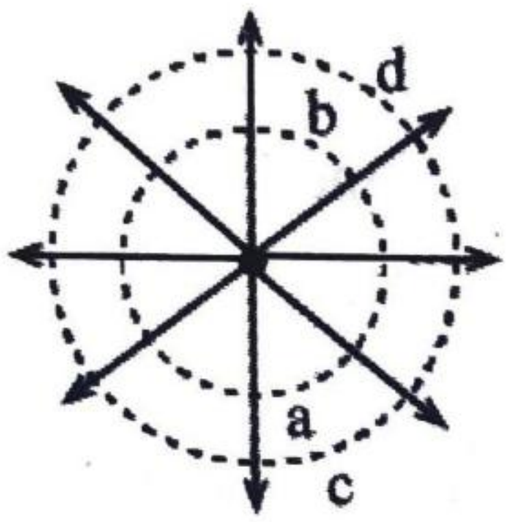


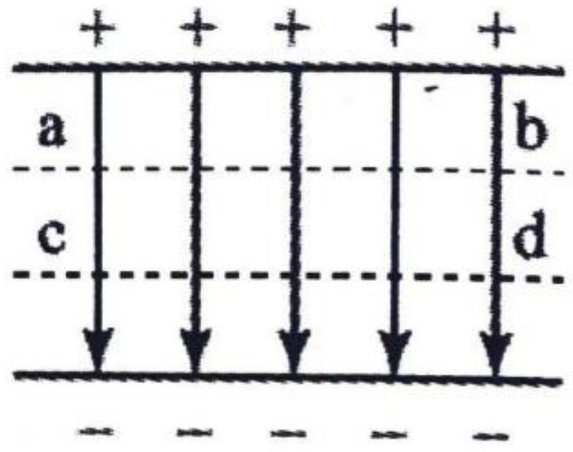
$$V_p = 195V$$

Equipotential lines and surfaces

The term "equi" means equal. Equipotential therefore means 'equal potential'. Thus, equipotential lines are lines that join points having the same electric potential. And equipotential surfaces are surfaces that contain equipotential lines.

In both cases, points a, b and c, d are equipotential points and the lines containing a, b and c, d are equipotential lines and the surfaces are equipotential surfaces.





Conceptual Example

44. Work done in moving a charge along an equipotential line or surface is zero. True or false?

Explanation: True. Because $W = (\Delta V)(Q)$. But $\Delta V = 0$ for equipotential line. Hence $W = 0$

10.4 Capacitors

Capacitor: Is a device used to store charge and energy.

The electric circuit symbol of a capacitor is (parallel plates joined by straight lines in opposite directions), ——||——.

The charge (Q) stored on a capacitor is given by the capacitor equation $Q = CV$ Where V is potential difference and

C is capacitance of the capacitor

Capacitance: Is the ability or capacity of a capacitor to store charge.

$$\text{Capacitance} = \frac{\text{charge stored}}{\text{potential difference}}$$

$$C = \frac{Q}{V}$$

The SI unit of capacitance is Farad (F)

Thus, $\boxed{1F = 1\frac{C}{V}}$

Conceptual Example

45. What is the net charge stored on a parallel plate capacitor?

Explanation: It is zero, because equal charges of opposite sign reside on each plate.

46. Capacitance of a capacitor is usually expressed in microfarad (10^{-6}F), nano Farad (10^{-9}F) or pico Farad (10^{-12}F) and not in Farad (F) why?

Explanation: Because, a capacitor to have 1F, its size would be around 2000 times radius of the Earth.

Illustrative Example

47. When a potential difference of 4.5V is supplied in a capacitor, a 5.4μc charge is stored. What is the capacitance of this capacitor?

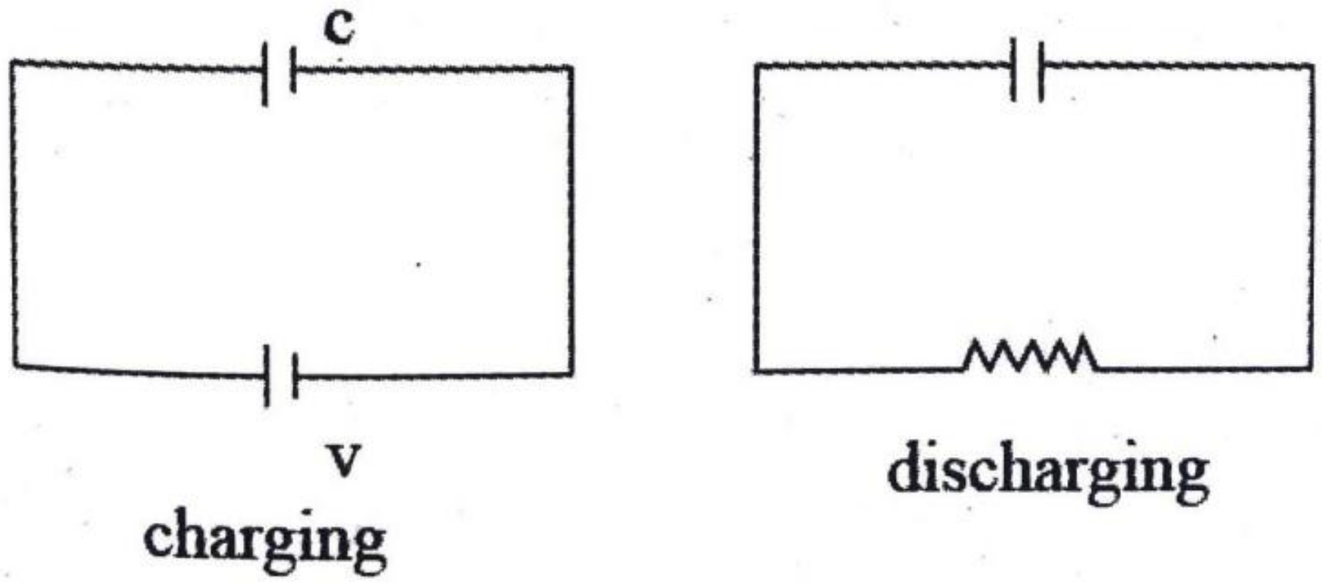
Solution: $C = \frac{Q}{V} = \frac{5.4\times10^{-6}}{4.5} = \left(1.2\times10^{-6}F\right) = 1.2\mu F$

48. A charge of 15μc exists on each of the capacitance of 5μF. Calculate the potential difference across the conductors.

Solution: $Q = CV \Rightarrow V = \frac{Q}{C} = \frac{1.5\times10^{-6}}{5\times10^{-6}} = 3V$

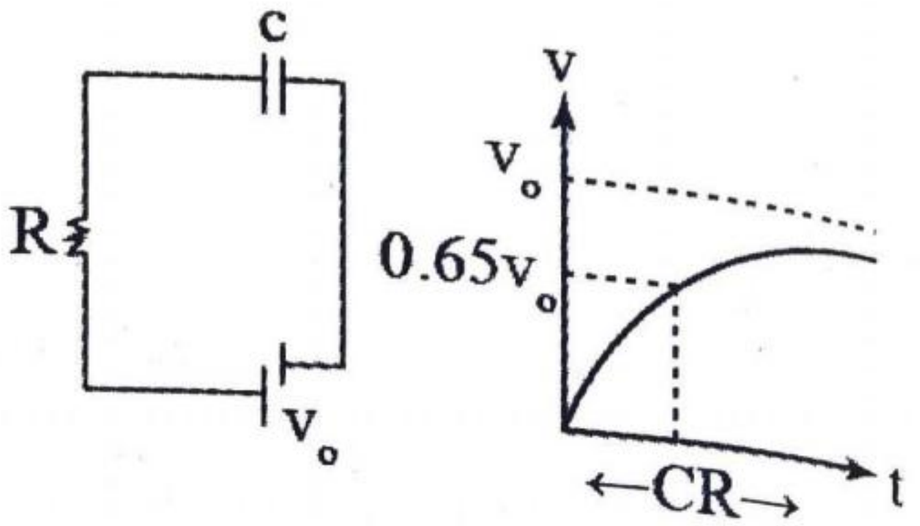
The Circuit symbol for a capacitor

The usual circuit symbols for a capacitor are



Charging a capacitor

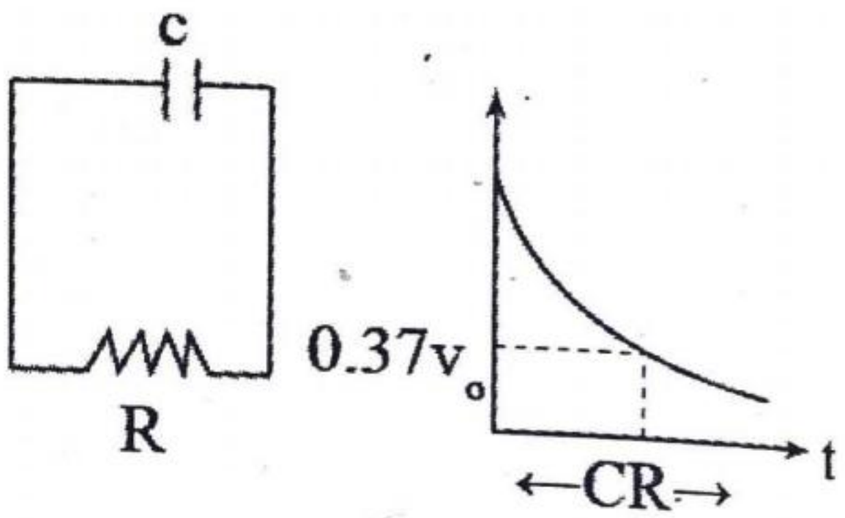
When a capacitor is charged through a resistor, it charges rapidly at the start, but then continues at an ever declining rate. When t = CR (time constant), the capacitor is almost two third full, ($V = 0.63V_o$)



Charging

Discharging a capacitor

When a capacitor discharges through a resistor, it acts like a battery. Its voltage drops rapidly as its charge drains away – exponential decay. When t = CR (time constant), the voltage drops to 0.37 of its starting value.



Discharging

Conceptual Examples

49. Show that t = CR

Explanation: $CR = \left(\frac{Q}{V}\right)\left(\frac{V}{I}\right) = \frac{Q}{I}$, but $I = \frac{Q}{t}$ hence

$$t = \frac{Q}{I} = CR$$

50. When will the capacitor became full (stop charging)

Explanation: When the voltage between the plates of the capacitor equals that of the battery, the charging ceases.

Combination of capacitors

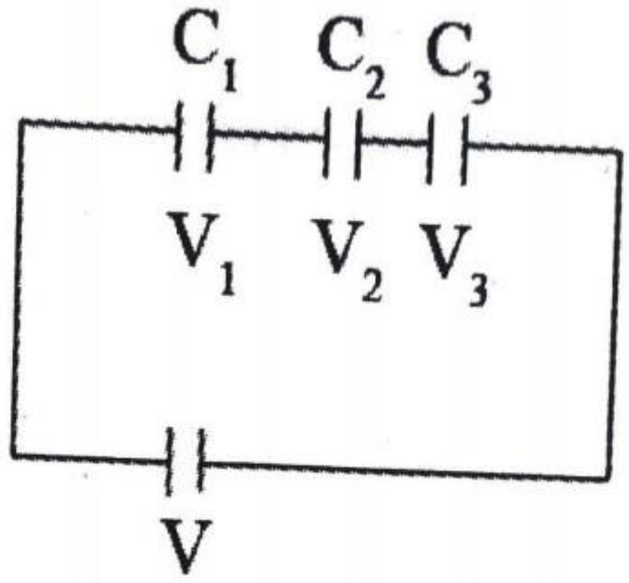
In practice, combination of several capacitors is used in electronic devices such as radio, television, computer etc to get the desired value of capacitance.

Capacitors in series

When capacitors are connected in series, they are connected one after the other.

→ For a series connection, the charge stored in each capacitor is the same as the charge stored in the network

$$Q_1 = Q_2 = Q_n = Q_T = Q$$



→ The total potential drop in the network is equal to the sum of the individual potential drops across each capacitor.

$$V_T = V_1 + V_2 + \ldots\ldots\ldots + V_n$$

$$\Rightarrow \frac{Q_T}{C_T} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \ldots\ldots\ldots + \frac{Q_n}{C_n}$$

But charge is constant, $Q_1 = Q_2 = Q_n = Q_T = Q$

Thus, the equivalent capacitance of the network when capacitors are connected in series is given by

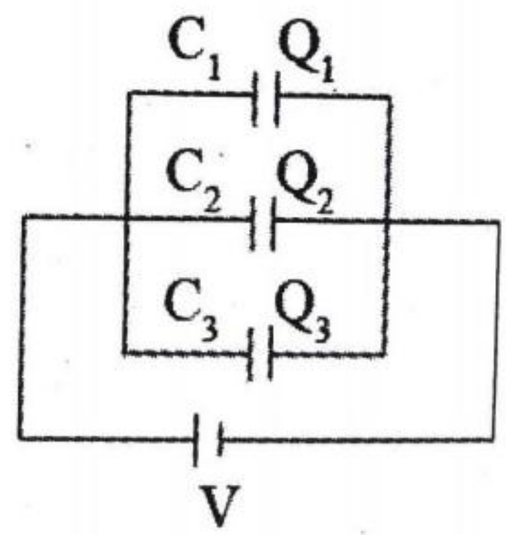
$$\boxed{\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots\ldots\ldots + \frac{1}{C_n}}$$

Capacitors in parallel

When capacitors are connected in parallel, they are connected face to face.

→ For a parallel connection, the total potential drop in the system is the same as the potential drop across each capacitor

$$V_1 = V_2 = V_n = V_T = V$$



→ The total charge stored in the sysem is equal to the sum of charges stored in each individual cpacitors

$$Q_T = Q_1 + Q_2 + \ldots\ldots\ldots + Q_n$$

$$\Rightarrow C_T V_T = C_1 V_1 + C_2 V_2 Q_1 + \ldots\ldots\ldots + C_n V_n$$

But potential drop is constant, $V_1 = V_2 = V_n = V_T = V$

This, the equivalent capacitance of the network when capacitors are connected in parallel is given by

$$C_T = C_1 + C_2 + \ldots\ldots + C_n$$

Conceptual Example

51. Which capacitor connection results in an equivalent capacitance of

a) smallest capacitance

b) largest capacitance

Explanation: a) the total capacitance has a smallest value than any one of the individual capacitances when capacitors are connected in series.

b) the total capacitance became largest when the capacitors are connected in parallel.

Illustrative Examples

52. Determine the effective capacitance of two capacitors $C_1 = 5.4\mu F$ and $C_2 = 15\mu F$. When connected in

a) Series b) parallel

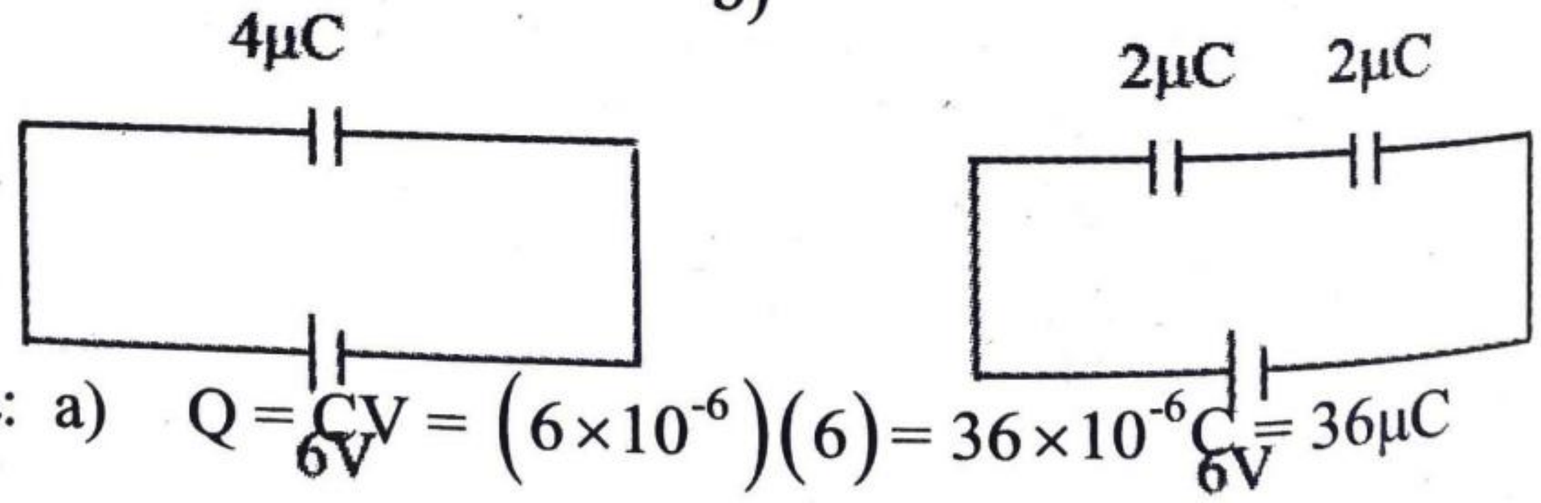
Solution: a) $\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6} + \frac{1}{15} = \frac{5+2}{30} = \frac{7}{30}$

$$C_T = \frac{30}{7} = 4.3\mu F$$

b) $C_T = C_1 + C_2 = 6 + 15 = 21\mu F$

53. In which of the two circuits is more charge stored?

a) b)



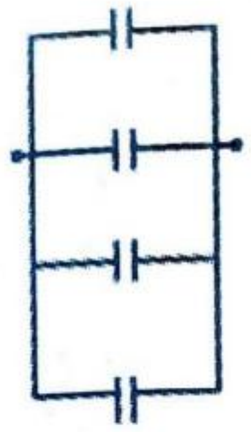
Solution: a) $Q = CV = \left(6\times10^{-6}\right)(6) = 36\times10^{-6}C = 36\mu C$

b) $\frac{1}{C_T} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} = 1\mu F \Rightarrow C_T = 1\mu F$

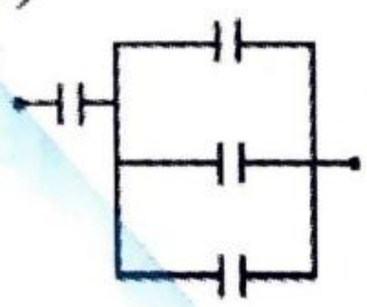
$Q = CV = \left(1\times10^{-6}\right)(6) = 6\times10^{-6}C = 6\mu C$, in (a).

54. Four identical capacitors are connected as shown. Organize these circuits into order of increasing effective capacitance.

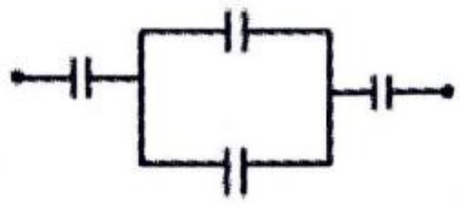
a)



b)



c)



d)

Solution:

a) $C_T = C + C + C + C = 4C$

b) $C^1 = C + C + C = 3C, \quad \frac{1}{C_T} = \frac{1}{C} + \frac{1}{3C} = \frac{4}{3C}$

$$C_T = \frac{3}{4}C = 0.75C$$

c) $C^1 = C + C = 2C, \quad \frac{1}{C_T} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C^1} = \frac{1}{C} + \frac{1}{C} + \frac{1}{2C}$

$$\frac{1}{C_T} = \frac{2+2+1}{2C} = \frac{5}{2C} \Rightarrow 0.4C$$

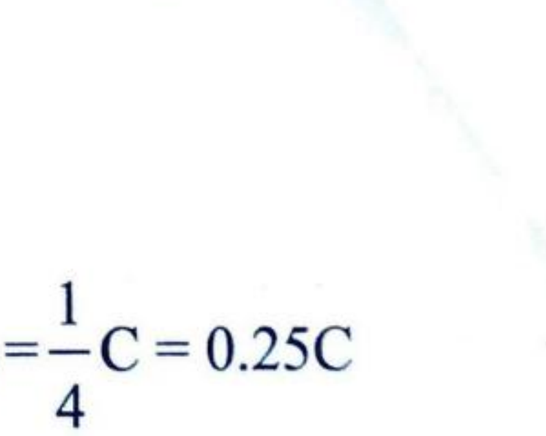
d) $\frac{1}{C_T} = \frac{1}{C} + \frac{1}{C} + \frac{1}{C} + \frac{1}{C} = \frac{4}{C} \Rightarrow C_T = \frac{1}{4}C = 0.25C$

So we have, 0.25C (d), 0.4C (c), 0.75C (b) 4C (a)

55. Consider a capacitor network and calculate

a) the equivalent capacitance

b) the potential difference across each capacitors

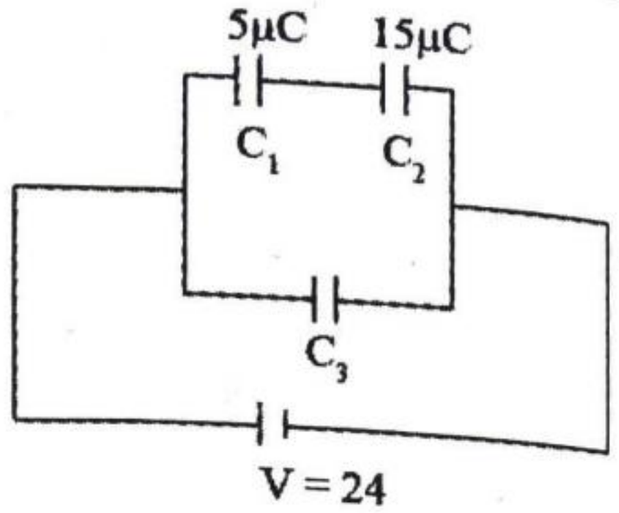


Solution: a) C_1 and C_2 are in series,

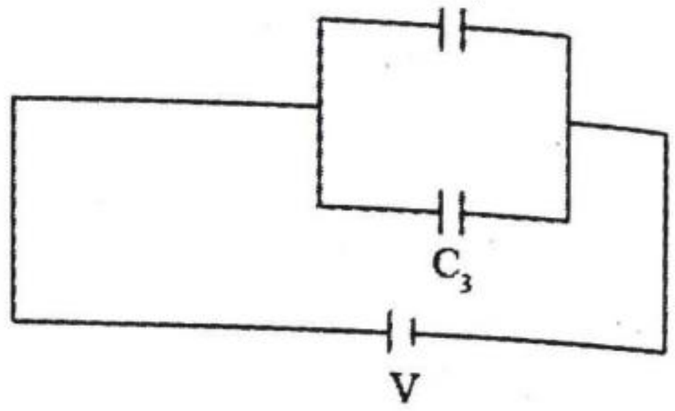
$$\frac{1}{C^1} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{5} + \frac{1}{15}$$

$$= \frac{3+2}{15} + \frac{4}{15} = \frac{15}{4}\mu F$$

$$\Rightarrow C^1 = 3.75\mu F$$



The circuit modified to,



$$C_T = C^1 + C_3 = 3.75 + 10$$

$$C_T = 13.75\mu F$$

b) The potential difference across C^1 and C_3 is the same since they are in parallel.

$$V_3 = V^1 = 24V$$

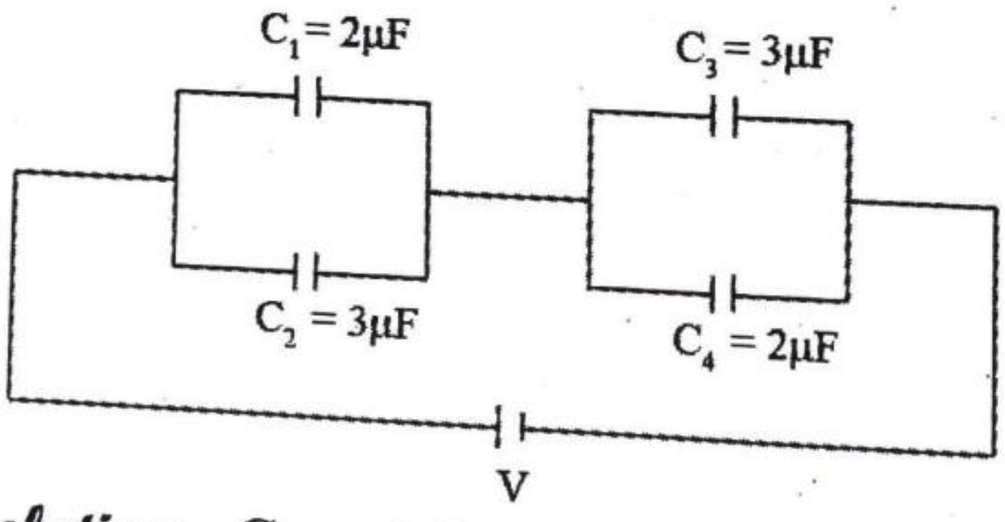
Charge stored in the C^1 capacitor is,

$$Q^1 = C^1V^1 = (3.75)(24) = 90\mu C$$

This charge is the same for both C_1 and C_2 since they are in series. There fore,

$$V_1 = \frac{Q_1}{C_1} = \frac{90}{5} = 18V \text{ and } V_2 = \frac{Q_2}{C_2} = \frac{90}{15} = 6V$$

56. What is the resultant capacitance of a capacitor shown.

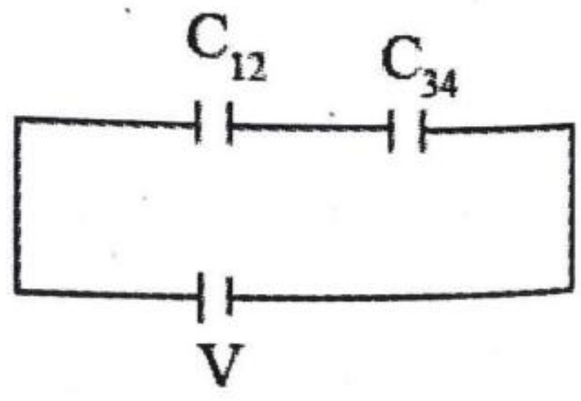


Solution: C_1 and $C_2$are in parallel,

$$C_{12} = C_1 + C_2 = 2 + 3 = 5\mu F$$

C_3 and $C_4$are in parallel

$$C_{34} = C_3 + C_4 = 3 + 2 = 5\mu F$$

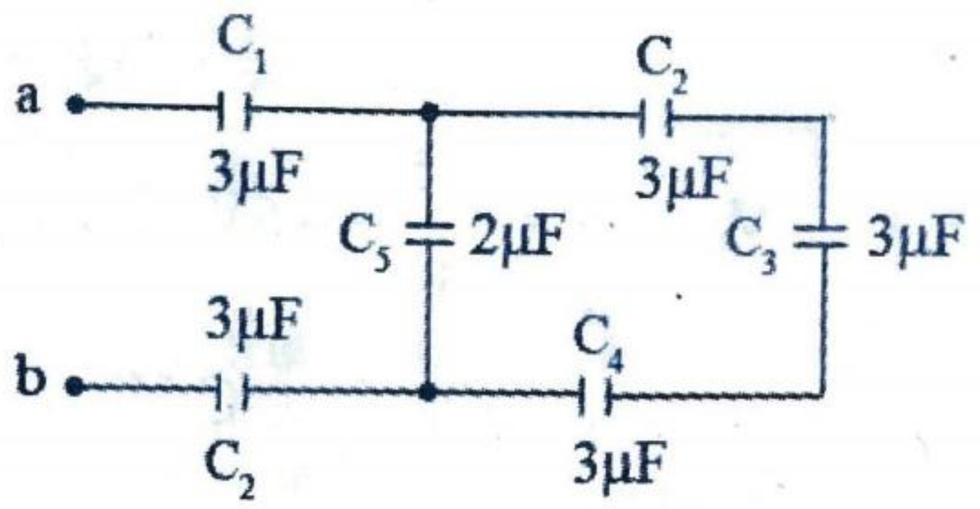


Now, the circuit modified to C_{12} and C_{34} which are in series.

$$\frac{1}{C_T}=\frac{1}{C_{12}}+\frac{1}{C_{34}}=\frac{1}{5}+\frac{1}{5}=\frac{2}{5}\Rightarrow C_T=\frac{5}{2}\mu F=2.5\mu F$$

57. Find the equivalent capacitance of the network shown?

Solution: C_2, $C_3$and $C_4$are in series.



$$\frac{1}{C_{234}}=\frac{1}{C_2}+\frac{1}{C_3}+\frac{1}{C_4}$$

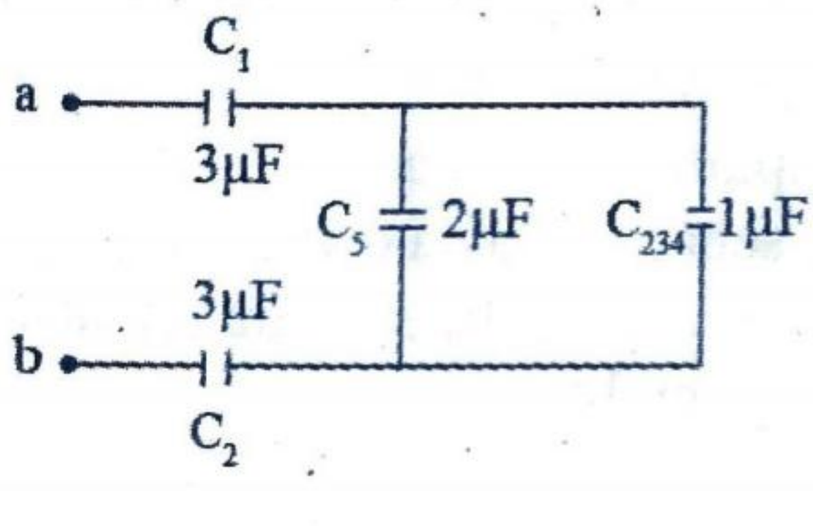
$$\frac{1}{C_{234}}=\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=\frac{3}{3}$$

$$\Rightarrow C_{234}=1\mu F$$

The network modified to;

C_{234}andC_5 are in parallel.

$C^1 = C_{234} + C_5 = 1 + 2 = 3\mu F$



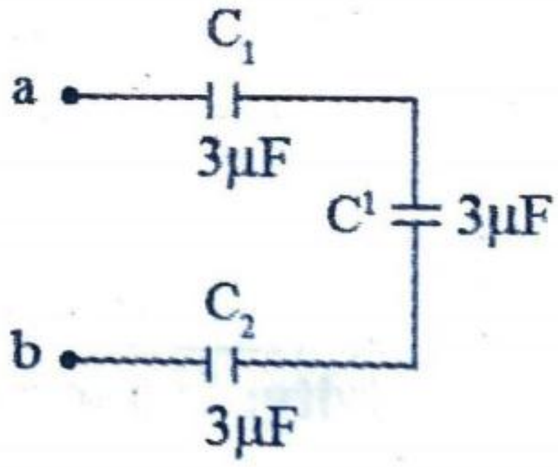
The network modified to;

Now, C_1, C^1 and $C_2$are in series.

$$\frac{1}{C_T}=\frac{1}{C_1}+\frac{1}{C^1}+\frac{1}{C_2}$$

$$\frac{1}{C_T}=\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=\frac{3}{3}$$

$$\Rightarrow \frac{1}{C_T}=\frac{1}{1},\ C_T=1\mu F$$

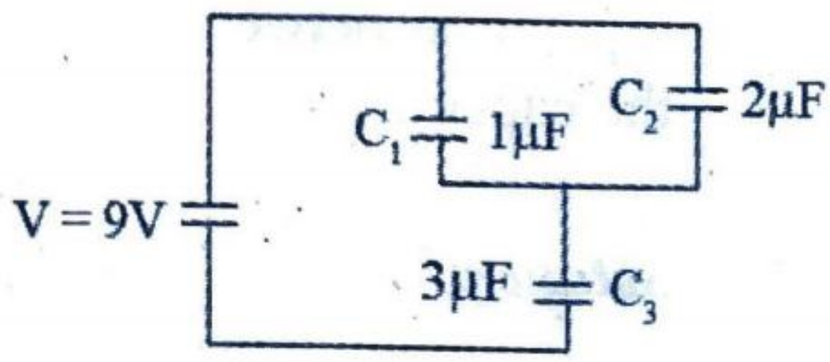


58. From the network shown, find

a) the total capacitance

b) the potential drop across each capacitor

c) the charge on each capacitor

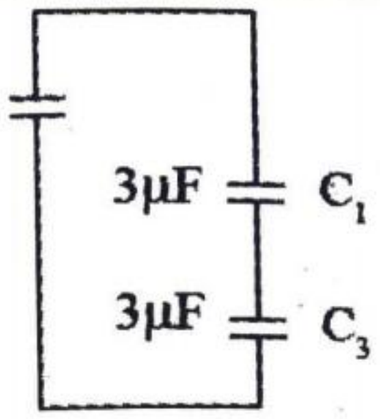


Solution: a) C_1 and C_2 are in parallel, $C_{12}=C_1+C_2 = 1+2 = 3\mu F$

Now the circuit modified to

$$\frac{1}{C_T}=\frac{1}{C_{12}}+\frac{1}{C_3}=\frac{1}{3}+\frac{1}{3}=\frac{2}{3}$$

$$C_T=\frac{3}{2}=1.5\mu F$$



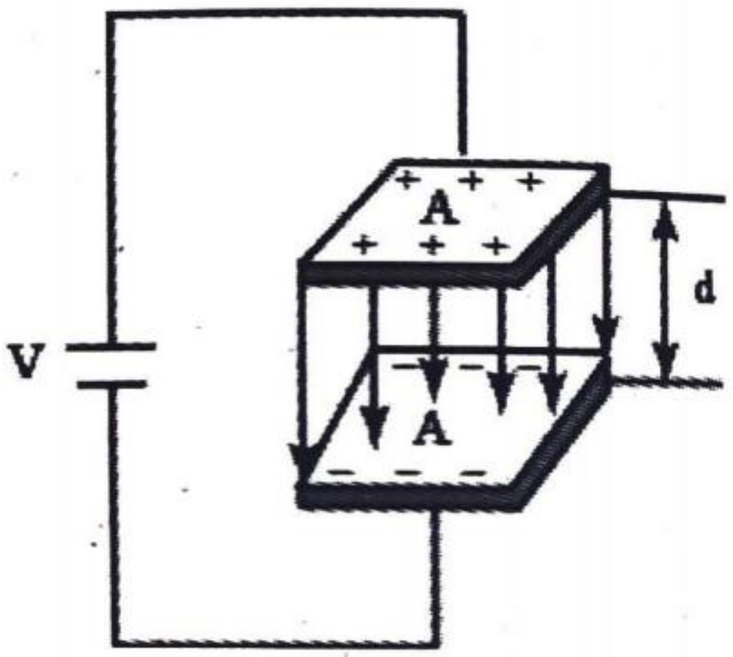
b) $Q_T = Q_{12} = Q_3 = CV = (1.5)(9) = 13\mu C$

$V_3=\frac{Q_3}{C_3}=\frac{13.5}{3}=4.5V$, and $V_1=V_2=V-V_3=9-4.5=4.5V$

c) $Q_1 = C_1V_1=(1)(4.5)=4.5\mu C$, $Q_2=C_2V_2=(2)(4.5)=9\mu C$ and $Q_3 = 13.5\mu C$

Parallel plate Capacitors

A parallel plate capacitor has two parallel plates of each having area (A) and separated by a distance (d). The capacitance of an air filled parallel plate capacitor is given by:



$$\boxed{C=\epsilon_o\frac{A}{d}}$$ where ϵ_o (Epsilon) is permittivity of free space and is equal to $\epsilon_o = 8.85\times10^{-12}F/m$.

Permittivity: Is a measure of how a material is able to transmit (permit) an electric field.

59. The area of the plates of a certain capacitor is $0.1m^2$ and the separation between the plates is 10cm. what is the capacitance of the capacitor?

Solution: $C=\epsilon_o\frac{A}{d}=\left(8.85\times10^{-12}\right)\frac{(0.1)}{(0.1)}=8.85\times10^{-12}F$

60. An air capacitor is made from two flat parallel plates 0.5mm apart. The magnitude of the charge on each plates is $0.01\mu C$ when the potential difference is 200V. Calculate

a) The capacitance b) the area of each plate.

Solution: a) $C = \frac{Q}{V} = \frac{0.01\times10^{-6}}{200} = 5\times10^{-11}F$

b) $C = \epsilon_o \frac{A}{d} \Rightarrow A = \frac{Cd}{\epsilon_o} = \frac{\left(5\times10^{-11}\right)\left(0.5\times10^{-3}\right)}{8.85\times10^{-12}}$

$A = 0.28\times10^{-2}m^2$

Dielectric Material

Dielectric: is an electrically insulating material such as, *mica*, *rubber*, *paper*, *air* etc inserted between the plates of a capacitor to increase its capacitance.

In the presence of dielectric, the capacitance of the parallel plate capacitor is given by

$$C = \epsilon \frac{A}{d} = \epsilon r \epsilon_o \frac{A}{d}$$

Where: ϵ is permittivity of the dielectric, and $\epsilon r = \frac{\epsilon}{\epsilon_o}$ is the relative permittivity of the dielectric.

☞ **Remember:** the factors that affect the capacitance of a parallel plate capacitor are
- distance between plates
- area of plates, and
- material between plates

Conceptual Example

61. Why is a dielectric an insulator? Explain

Explanation: If it were a conductor, charges will soon leak off the capacitor. Hence since a dielectric is an insulator, it stops the charged plates coming into contact with each other and neutralize.

Illustrative Examples

62. You wanted to make a 5μF capacitor. For this purpose, you used a waxed paper of thickness 0.4mm with dielectric constant of $\epsilon = 7.1\times10^{-2}F/m$. What areas of plates is needed ?

Solution: $C = \epsilon \frac{A}{d} = A = \frac{Cd}{\epsilon} = \frac{\left(5\times10^{-6}\right)\left(4\times10^{-3}\right)}{7.1\times10^{-12}} = 281.7m^2$

63. The plates of a capacitor have sides of 5cm and 8cm. If a paraffin paper of 0.001cm thickness and relative permittivity of 2.1 is introduced between the plates, calculate the capacitance of the capacitor.

Solution:

$$C = \epsilon_r \epsilon_o \frac{A}{d} = (2.1)\left(8.85\times10^{-12}\right)\frac{(0.08)(0.05)}{10^{-5}} = 281.7m^2$$

$$C = 7.4\times10^{-9}F = 7.4nF$$

Energy stored in a capacitors

If a capacitor carried a charge (Q) at a potential (V), the energy it stored is given by

$$\text{Energy} = \frac{(\text{charge})\ (\text{potential difference})}{2}$$

$\rightarrow$ $\boxed{\text{Energy} = U = \frac{1}{2}QV}$, but $Q = CV \leftarrow$ Capacitor equation

$$U = \frac{1}{2}QV = \frac{1}{2}(CV)(V)$$

$\boxed{\text{Energy} = U = \frac{1}{2}CV^2}$, also $V = Q/C$

$$U = \frac{1}{2}Q\left(Q/C\right)$$

$$\boxed{\text{Energy} = U = \frac{1}{2}\frac{Q^2}{C}}$$

Conceptual Example

64. Why is the energy stored in a capacitor not given as, energy = QV?

Explanation: Because, the energy stored starts from zero level and goes up to its final value. Hence in actual sense, the energy stored at any time is an average value.

Illustrative Examples

65. A 60μF capacitor is charged to a potential difference of 3000V. Calculate the energy stored in this capacitor.

Solution: $U = \frac{1}{2}CV^2 = \frac{1}{2}\left(60\times10^{-6}\right)\left(3000\right)^2 = 270J$

66. What capacitance is necessary in order to store energy of 8Kwh at a potential difference of 950V?

Solution: $U = \frac{1}{2}CV^2 \Rightarrow C = \frac{2U}{V^2} = \frac{2(1000)(8)(3600)}{(950)^2} = 63.83F$

67. A parallel plate capacitor with plates area of $50cm^2$ and mica dielectric of thickness 0.002cm with relative permittivity of 6 is charged to source of 6000V. Calculate the energy stored in this capacitor.

Solution:

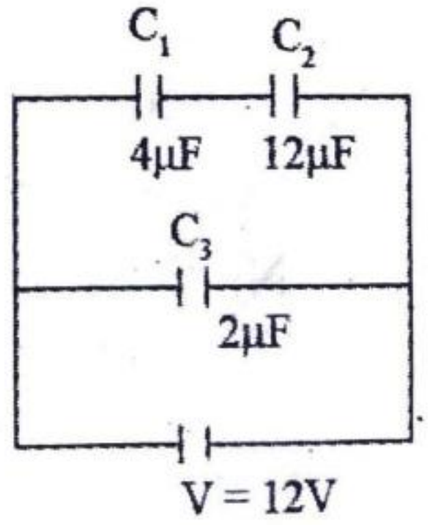
$$C = \epsilon_r\epsilon_o\frac{A}{d} = (6)\left(8.85\times10^{-12}\right)\frac{\left(50\times10^{-4}\right)}{2\times10^{-5}} = 1.33\times10^{-8}F$$

$$U = \frac{1}{2}CV^2 = \frac{1}{2}\left(1.33\times10^{-8}\right)\left(6000\right)^2$$

$$= 0.24J$$

68. For the circuit shown, calculate

a) The effective capacitance
b) The energy stored in each capacitor
c) The total energy stored



Solution: a) C_1 and C_2 are in serious,